Introduction to Psychology (A critical approach)

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The current version of the textbook was written, remixed and edited by Jill Grose-Fifer, Ph.D., Professor of Psychology at John Jay College, CUNY, in 2023. It is a work in progress. This book was adapted from a clone of the 2019 BCcampus Open Education's clone of the first edition of OpenStax's *Psychology* into Pressbooks.

In chapters 1-8 much of the content has been rewritten or replaced to reflect a more critical approach to the discipline of psychology, but these do not have interactive activities yet. At present, chapters 9 through 15 are in their original format so they represent a more traditional view of psychology (but have interactive activities). This edited work has been sponsored by John Jay College and was inspired by many people, including the students of John Jay College and CUNY more broadly. However, I particularly want to acknowledge the role of Dr. Della Mosely and Pearis Bellamy, founders of Academics for Black Survival and Wellness, as major forces for the impetus for the edits to this book.

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CHAPTER 1. INTRODUCTION TO PSYCHOLOGY

Figure 1.1. Psychologists study thinking, feeling, and doing.

In this first chapter, we provide a general outline for the themes that we will cover in this book. We also provide some basic information about the field of psychology, including what kind of work psychologists do and what it takes to become a psychologist. You will also learn about what it means to major in Psychology in terms of skills learned and the kinds of career opportunities that are available to you after graduation. Finally, we take a critical look at the history of psychology.

LEARNING OBJECTIVES:

- Define psychology and identify the main focus of the field.

- Explain the importance of context in influencing behavior and mental processes.
- Describe potential biases in traditional Western psychology and the need for a more diverse and representative approach to research.
- Demonstrate knowledge of skills gained by psychology majors and potential entry-level jobs.
- Explore different subfields and career paths in psychology.
- Describe the history and evolution of psychology as a field of study
- Discuss historical and current barriers faced by marginalized groups in psychology and their contributions to the field.
- Describe the role of psychology in promoting social justice and fighting against oppression.

1.1. What Is Psychology?

JILL GROSE-FIFER

Psychology is the study of the mind (thoughts and feelings) and behavior (what we do). Unequal distribution of wealth and resources across the United States and in many other countries in the world leads to vastly different opportunities and experiences for people, depending on who they are, and where they live. Not surprisingly, these lived experiences interact with biological (and other individual factors) determining how people think, feel, and behave in different situations.

Figure 1.2. Opportunities and experiences affect what we do, think, and feel

Figure 1.3. People across the world have different experiences and opportunities, which affects how we think, feel, and behave

Psychologists often use a **biopsychosocial** lens (see Figure 1.4) to try to better understand people's behaviors, and their thoughts and feelings. In other words, they look at the influence of biological, psychological, and social factors. **Biological** factors include looking at the impact of genes, nervous system function, and physical health. **Psychological** factors include significant life events relating to mental health, emotion, distress, beliefs, and expectations. Finally, **social** influences include interpersonal factors, such as social interactions, community experiences, and culture. Clearly, what we do, think, and feel at any given moment is highly context-dependent and psychologists are increasingly recognizing the importance of studying contextual influences such as ethnicity, race, ability, gender, sexual orientation, religion, and other cultural norms. However, it is also important to recognize that modern-day Western psychology has been dominated by research in the United States and Europe with foundational studies by White middle-class males.

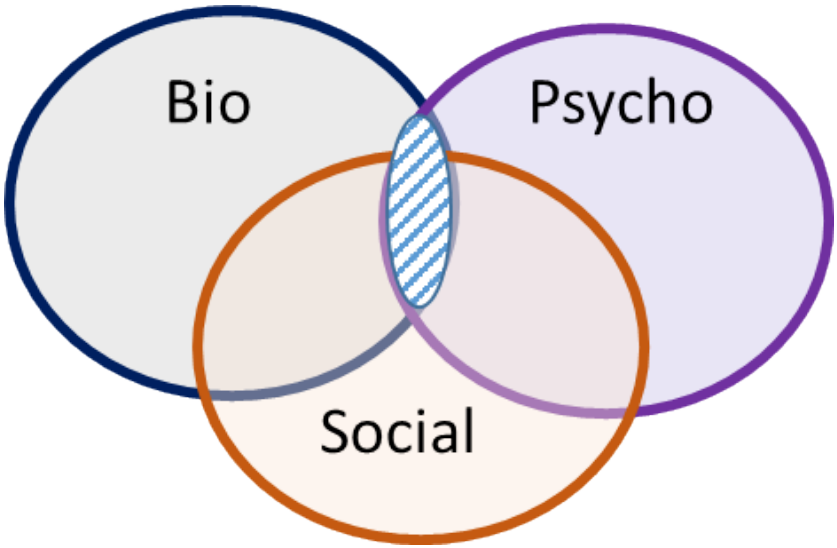

Figure 1.4. Psychology often takes a biopsychosocial perspective in trying to understand the mind and behavior

At first glance, the research conducted by Western psychologists appears to be highly objective. Psychologists frequently employ the **scientific method** using statistics to test their predictions. Also, psychologists typically publish their findings in scientific journals after review by other experts in the field, a process known as peer review. This is done to try and ensure quality control. However, Western Psychology has also tended to make universal claims about *all* humans based on studies of a very small percentage of the world population – often referred to as **WEIRD** participants, i.e., mostly **W**hite undergraduates from **e**ducated, **i**ndustrialized, **ri**ch **d**emocracies, such as the United States and Europe (Apicella et al., 2020). Furthermore, if other groups of people behaved, thought, or felt differently from the published "norms", Western psychologists have often considered them atypical or deficient. However, psychologists much less frequently questioned whether they (and the studies they conducted) were culturally biased. In addition, more recent attempts to reproduce many of the original research

studies in Western psychology have not always succeeded – leading to what has been termed the **replication crisis** in psychology (and science more broadly). When you read about the many different ways that modern psychologists carry out research in the upcoming chapter on *Research Methods*, you will see that there are many reasons for this lack of replication. All of these issues call into question the validity of some "foundational" studies in traditional Western Psychology. However, the field of psychology is becoming more critical of its roots. As more women and people of color have broken through barriers and became psychologists, priorities have shifted and the diversity of the human experience has become increasingly important to the field of psychology (you will read more about this in the section on *Social Justice and Intelligence*). Many psychologists are now acknowledging that Western Psychology has been highly **ethnocentric**, that is evaluating people from the context of one's own culture, and not necessarily relevant to the global majority.

Figure 1.5. Mamie Phipps Clark and Kenneth Clark, two Black United States psychologists whose research helped to desegregate schools in the United States.

All of this might sound confusing and maybe even slightly worrying to a new student of psychology, but we encourage you to think of yourself as standing at an exciting crossroad in the discipline. Psychologists and students of psychology are becoming

increasingly aware and critical of shortcomings in the field, including the role that Western psychology has played in oppressing people historically considered to be at the margins of society. But, we also have the chance to reimagine the discipline, to move away from outdated theories that fail to capture the breadth of the human experience, and instead to embrace research that offers a more contextualized, nuanced, and representative view of people's thoughts, feelings, and behavior. Therefore, in this book, although we have reused open source materials, we also provide some pointed critiques about the **"traditional canon of knowledge"** in Western psychology, and we strongly encourage you to think of more. We have also tried to broaden our coverage beyond traditional textbooks, to include and uplift perspectives from Black, Latinx, and Asian-American psychology, and to highlight the importance of critical race, intersectionality, feminist, and queer theories in understanding humankind. Thirdly, we have tried to infuse a social justice focus throughout this text to show how psychology can promote equality and fight against oppression (see Figures 1.5 and 1.6). In short, we have tried hard to introduce you to the good, bad, and the ugly in the field of psychology in the hope that you will think deeply about how to use this knowledge to improve your life and the lives of other people. Finally, we acknowledge that despite our lofty goals, this book (and its associated materials) are a work in progress and we welcome your feedback and any suggestions for improvements. **Please email us at jgrose-fifer@jjay.cuny.edu**

Figure 1.6. Protestors at a rally after George Floyd is murdered by a police officer in Minnesota.

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1.2. Careers and Psychology

JILL GROSE-FIFER

SHOULD I MAJOR IN PSYCHOLOGY?

You might be thinking about becoming a Psychology major because you want to pursue a career in psychology. Bear in mind that nearly all undergraduate degrees in the United States are general – they will provide you with a broad education in the **liberal arts** and hone the skills that you will need for a variety of jobs after you graduate. But, to become a psychologist, you will need a **graduate-level degree** where you will study the field of psychology in greater depth. Even if you are not planning to go to **graduate school**, being a psychology major can prepare you for a wide range of jobs. In fact, Psychology is one of the most popular majors for college students (Irwin et al., 2022), so, if you choose this path, you will be in good company! Most psychology majors do NOT go to graduate school (Lin et al., 2017). Figure 1.7 shows you the kind of work they end up doing. As you can see, psychology majors work in a wide variety of jobs after they graduate – but about one-third work in health or human service-related areas. A further 15% work in managerial roles, 14% in teaching, 10% in office/finance work, 8% in sales and 5% in Science, Technology, Engineering and Math fields, often referred to as STEM (National Center for Science and Engineering Statistics [NCSES], 2021b).

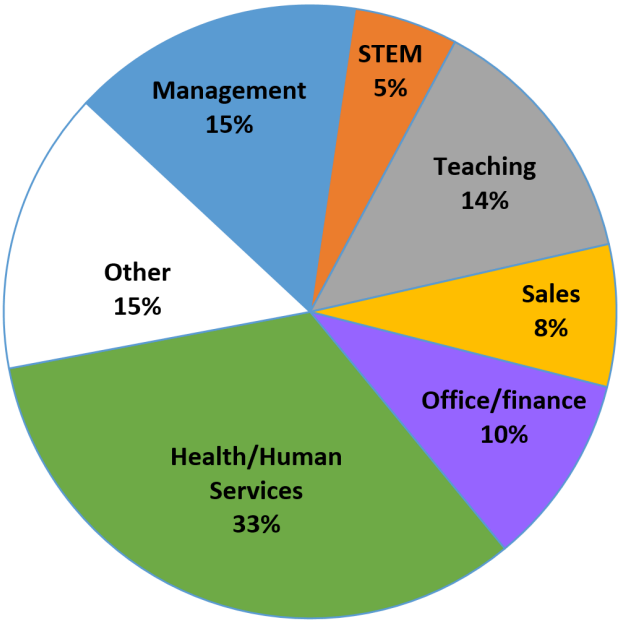

Figure 1.7. Workplace settings for people who graduated as Psychology Majors. Figure created from National Center for Science and Engineering Statistics data (NCSES; 2021b)

Courses in the Psychology major help to cultivate the kinds of skills that employers particularly look for in college graduates (Landrum, 2018). Table 1.1 lists the skills that employers value in order of importance (Gray, 2021). Psychology majors learn about the ways that other people (and animals) think, feel, and behave, but they also learn about their own strengths and weaknesses, which can help them to work more effectively with others. Psychology courses help to cultivate good listening and other communication skills, as well as the ability to evaluate and organize information from multiple sources (information literacy). The Psychology major has a strong emphasis on psychology as a science and so students learn about the scientific method and quantitative reasoning – the

ability to use math and information to solve real-world problems. Psychology courses encourage students to think critically and analytically, and to learn different ways of collecting, organizing, and analyzing data as well as communicating their findings effectively. Psychology is a "hub science" because it has strong connections to the medical sciences, social sciences, and education; psychology majors sometimes choose to go into these fields after graduation too (Boyack et al., 2005).

Figure 1.8. Psychology majors often have good interpersonal skills

Table 1.1. *The Top 12 Most Desirable Skills Employers Look For in College Graduates (Gray*

DESIRABLE QUALITIES EMPLOYERS LOOK FOR IN GRADUATES
1. Problem-solving skills
2. Analytical/quantitative skills
3. Ability to work in a team
4. Communication skills (written)
5. Initiative
6. Strong work ethic
7. Technical skills
8. Flexibility/adaptability
9. Detail-oriented
10. Leadership
11. Communication skills (verbal)
12. Interpersonal skills (relates well to others)

Link to Learning

WATCH THIS VIDEO ABOUT BEING A PSYCHOLOGY MAJOR TO HELP YOU TO DECIDE WHETHER THIS IS THE RIGHT MAJOR FOR YOU (Watch at least until the 10:00 min point – or longer if you are interested).

Your professors and your College Career Advisement Center will be able to help explore your options further, but Table 1.2 below gives you more details about the kinds of jobs that psychology majors go on to do. The Occupational Information Network (O*net) is a helpful web-based career resource that can help you explore

your options further. You can type a job title from the table (or another job you are curious about) into the search bar, you will then see the kinds of duties expected, as well as the skills and qualifications that you will need. For each job, O*net provides information about average salaries within each US state and the likelihood of new jobs being available in the near future.

Many of the positions in Table 1.2 require that you have relevant work experience; you might be able to gain this through internships, fieldwork courses, and doing other types of volunteer work while you are an undergraduate. Your College Career Advisement Center can guide you with your applications for work experience positions. Some employers might provide on-the job training for new employees too. Working with one of your professors on their research projects is a great way to gain research experience, which is especially important if you are thinking of applying to graduate school. Working in a professor's lab builds research-related skills, but it also allows your professor to get to know you really well. This is important when you ask them to write letters of recommendation for jobs or graduate school. Other ways to get to know your professors well, is to talk to them before and after class and to visit them during their office hours. This can feel awkward at first, but remember that we are here to help you and we love to talk with students.

Table 1.2. *Entry Level Jobs for Psychology Majors (after Landrum, 2018)*

Activities Director	Labor Relations Manager
Admissions Evaluator	Loan Officer
Advertising Sales Representative	Management Analyst
Alumni Director	Market Research Analyst
Animal Trainer	Occupational Analyst
Benefits Manager	Patient Resources Reimbursement Agent
Career/Employment Counselor	Personnel Recruiter
Career Information Specialist	Police Officer
Caseworker	Polygraph Examiner
Child Development Specialist	Preschool Teacher
Child Welfare/Placement Caseworker	Probation/Parole Officer
Claims Supervisor	Project Evaluator
Coach	Psychiatric Aide/Attendant
Community Organization Worker	Psychiatric Technician
Community Worker	Psychological Stress Evaluator
Computer Programmer	Psychosocial Rehabilitation Specialist (PSR)
Conservation Officer	Public Relations Representative
Correctional Treatment Specialist	Purchasing Agent
Corrections Officer	Real Estate Agent
Criminal Investigator (FBI & other)	Recreation Leader
Customer Service Representative Supervisor	Recreation Supervisor
Data Base Administrator	Research Assistant
Data Base Design Analyst	Retail Salesperson
Department Manager	Sales Clerk
Disability Policy Worker	Social Services Aide
Disability Case Manager	Substance Abuse Counselor
Employee Health Maintenance Program Specialist	Systems Analyst
Employee Relations Specialist	Technical Writer
Employment Counselor	Veterans Contact Representative
Employment Interviewer	Veterans Counselor
Financial Aid Counselor	Victims' Advocate
Fund Raiser	Vocational Training Teacher
Health Care Facility Administrator	Volunteer Coordinator
Information Specialist	Human Resource Advisor
Job Analyst	Writer

Figure 1.9. Psychology majors often work in an office after graduation.

Figure 1.10. Psychology majors sometimes become preschool teachers after graduation.

BECOMING A PSYCHOLOGIST

People often assume that all psychologists are mental health providers**.** This is not true, but **health service psychology** is a popular career choice for psychologists. In fact, about two thirds of psychologists work as clinical, counseling, or school psychologists (Bureau of Labor Statistics, 2020). Nevertheless, health service psychology is only *one* area of psychology. As you will discover in your Introduction to Psychology course, psychology is a very broad discipline – consisting of numerous different specializations, and many psychologists work outside of health service psychology. Figure 1.11 shows the different settings that psychologists work in. Ask your professor about the kind of graduate degree they earned and the different jobs they have had!

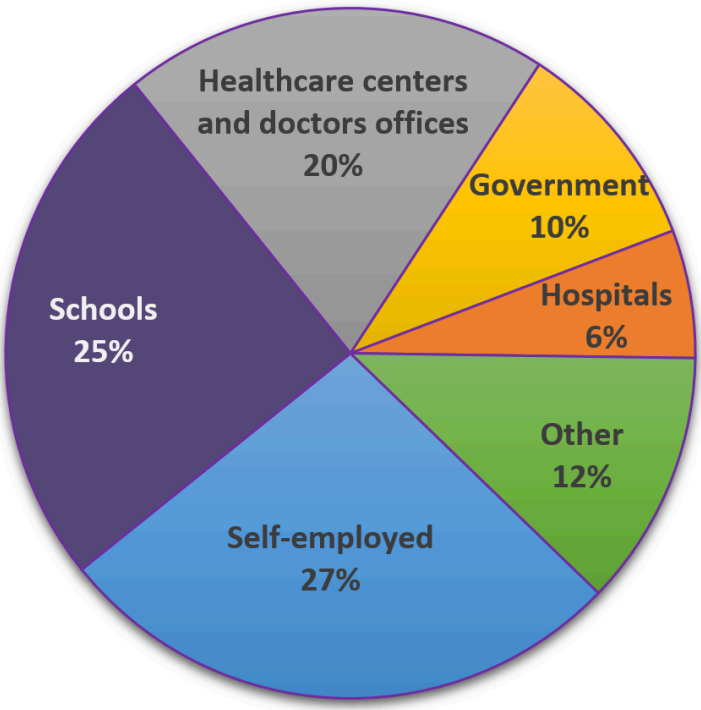

Figure 1.11. Where Do Psychologists Work? Created from data from the Bureau of Labor Statistics, (2020)

DOCTORATES IN PSYCHOLOGY

Figure 1.12. Doctoral degrees are generally conferred in formal ceremonies involving special attire and rites.

To become a psychologist you will need a doctoral degree in psychology. There are three types of doctorate degrees in psychology, the most common of which is a **Doctor of Philosophy (PhD)**. A PhD is a research-based degree where you learn about a specific subfield of psychology in great depth, as well as writing a dissertation based on your own original research. In contrast, **Doctor of Psychology (PsyD)** degrees are only available in applied fields – such as health service psychology. A much less common doctoral degree in psychology is the **Doctorate of Education (EdD)**, which focuses on both practice and research in areas such as educational psychology and counseling psychology.

Health Service Psychology

Many students want to be health service providers (e.g., a clinical psychologist, counseling psychologist, or school psychologist). If you decide on this kind of career path, you will also need to pass licensing/certification exams that are specific to the State where you will work. It is much easier to become certified/licensed and find a job, if you graduate from a doctoral program that is accredited by the **American Psychological Association (APA),** or in the case of school psychology—the **National Association of School Psychology** (NASP).

All doctoral students in health services complete a year-long internship (usually at the end of the program) where they gain intensive hands-on experience in their chosen field. Health service students typically graduate more quickly from PsyD programs than PhD programs because they do not have to carry out extensive research projects (American Psychological Association [APA], 2021a).

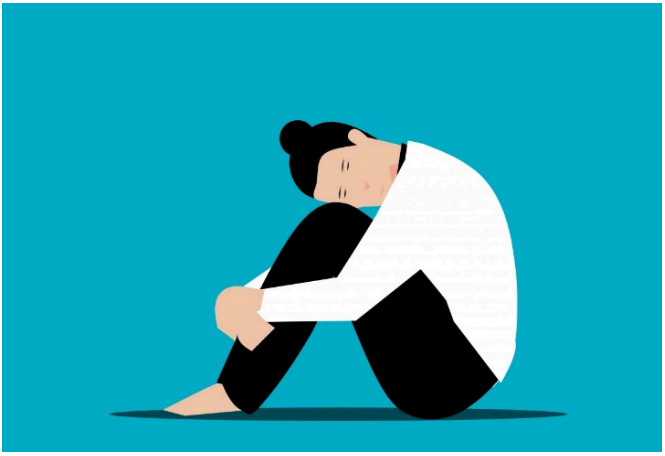

Figure 1.13. Many people who study psychology become health service providers

How do you decide which doctoral program is best for you?

Deciding on the best program for you will depend in part on the type of work that you want to do. In health service psychology, PsyD students train to become practitioners. Health service PhD students also train to be clinicians, but like other PhD students, they learn how to be teachers and researchers as well. PhDs have more flexibility in terms of the kind of work that they do, they might work in a University or College or other research settings. Health service PhDs might also work in the same clinical or other professional settings that PsyDs work in.

All doctoral programs are long and demanding, and admission is very competitive. In general, PhD programs in psychology accept fewer than 15% of all applicants (click here for the latest statistics for different subfields in psychology. In contrast, PsyD programs accept 40 to 50% of applicants (Norcross & Hogan, 2016). Admissions committees use multiple criteria to evaluate doctoral applicants, including relevant work experience. Students applying to PhD programs typically have worked in a research lab (usually at their college or university), and those applying to health service doctoral programs typically have some clinical or professional experience. Admissions committees also consider grades earned as an undergraduate (known as grade point average [GPA]); application essay(s), letters of recommendation and Graduate Record Entrance (GRE) exam scores (though some programs are phasing out the GRE). Most PhD psychology programs require a GPA of at least a 3.0 on the 4.0 scale, which is the equivalent of a B average. However, the average GPA of students who are actually admitted is much higher at 3.6 (between a B+ and A- average) overall and 3.7 (A- average) in psychology courses (Norcross & Hogan, 2016). Admitted students at very competitive programs often have even higher average GPAs (Wegenek & Buskist, 2010).

As we already mentioned, PsyD programs are somewhat easier to get into than PhD programs, however, there are some

disadvantages to consider. PsyD students receive substantially less funding than PhD students; ~1% of PsyD students receive full funding (tuition waiver and a stipend high enough to live on) compared to ~85% of PhD students (Norcross & Hogan, 2016). This means that total student debt after graduation is typically much greater for PsyD compared to PhD students (Norcross & Hogan, 2016; Wilcox et al. 2021). Average student debt is substantially higher for PhDs in health service psychology (~$72,000) compared to other psychology subfields (~$58,000), but it is still considerably less than PsyDs (~$120, 000; Wilcox et al. 2021). Also, it is often more difficult for PsyD graduates to obtain an APA-accredited internship and fewer PsyDs pass the licensing exams on their first try (Norcross & Hogan, 2016). You will find specific information about fees, financial aid, application criteria, and the qualifications of current and recent students on each program's website.

Figure 1.14. Studying for a doctorate to become a psychologist is time-consuming and takes a lot of effort

OTHER SUBFIELDS OF PSYCHOLOGY

Although we have talked a lot about health service psychology in this module, as you can see from Table 1.3, people earn doctorates in many other areas of psychology. Slightly more than half of the psychology doctorates awarded in 2018-2020 were in health-service areas, and about one quarter were in general psychology, with the remainder split across multiple other subfields (American Association of Psychology, 2021b). In general, job prospects for psychology doctorates are good. About two-thirds of psychology students accept a job offer (or decide to work for themselves) in the final year of their doctoral program (National Center for Science and Engineering Statistics (NCSES), 2021a). Most new doctorates in 2018-2020 went directly into post-doctoral positions, which are paid opportunities for further training and specialization in a field. A further 10% went directly into academic jobs, most likely as assistant professors, and the others were self-employed or worked in industry, business or other settings (NCSES, 2021a).

Table 1.3. ***Most popular areas of study for psychology doctorates awarded in 2018-20.***

Health service psychology fields are shown in yellow.
Data from American Psychological Association (2021b).

PSYCHOLOGY SUBFIELD	PERCENTAGE DOCTORATES AWARDED IN 2018-20
Clinical Psychology*	35.9%
Counseling Psychology*	6.5%
Community Psychology	0.6%
Developmental Psychology	0.8%
Educational Psychology	5.3%
Experimental Psychology	3.0%
Forensic Psychology	1.0%
General Psychology	26.0%
Industrial/Organizational Psychology	3.1%
School Psychology*	5.4%
Social Psychology	0.6%
Other Research Subfields	7.2%
Other Health Service Psychology Subfields	4.7%

To learn more about the kinds of work that different kinds of psychologists do, click on the links below.

Clinical psychologists assess and treat mental, emotional and behavioral problems. NOTE: Psychologists use therapy, not drugs for treatment. Psychiatrists, who have medical degrees, can prescribe drugs for treatment and often work alongside psychologists.

Cognitive psychologists study human perception, thinking, language, and memory.

Community psychologists work with communities to meet people's needs.

Counseling psychologists use culturally-informed and culturally-sensitive practices to help people prevent and alleviate distress and increase their ability to function better in their lives.

Developmental psychologists study psychological development across the lifespan (although many study a specific period, e.g., infancy, adolescence or old age).

Educational psychologists focus on how effective teaching and learning take place in a variety of settings.

Experimental psychologists are interested in using the scientific method to investigate thinking, feeling, and behavior in humans and non-humans. Many other psychologists also conduct research in this way – learning about research methods (experimental psychology) is an important aspect of most doctoral programs.

Forensic psychologists focus on the intersection between psychology and the law. Some may have clinical training too.

Health psychologists specialize in how biological, psychological, and social factors affect our health.

Industrial/organizational (I/O) psychologists work with businesses and organizations to improve productivity and health of workers and the quality of work life.

Neuroscientists study the relationship between biology (especially the brain) and thinking, feeling, and behavior in humans and/or animals. They might use EEG recording, PET scans, and MRIs to image the nervous system, or they might record the activity from individual nerve cells (neurons).

Neuropsychologists are clinical psychologists who have undergone extensive additional training to learn more about the relationship between the brain and behavior. They often assess and help treat people with brain damage and related disorders. They use behavioral testing in their assessments.

School psychologists help children, adolescents and families to succeed in schools.

Social psychologists study how a person's thoughts, feelings, and behavior are affected by interactions with other people.

Sport psychologists use psychological principles to help athletes perform better.

MASTERS AND SPECIALIST DEGREES IN PSYCHOLOGY

Figure 1.15. Graduating with a Masters or a specialized degree in Psychology is another way to gain access to psychology-related careers

You might be wondering whether it is possible to work in a psychology-related field without a doctorate? The answer is – YES! Increasing numbers of students are graduating with masters or specialized degrees in psychology and related fields, which you can complete in much less time than a doctorate. Table 1.4 shows the most popular areas for masters' degrees in Psychology. Health service psychology degrees are the most popular (55%). It is easier to get admitted to a masters' program in psychology than a doctoral program. Like doctoral programs, admissions to Masters' programs are also typically based on letters of recommendation, your application essay(s), relevant experience, GRE scores (though these are becoming less frequently required) and GPA. Master's programs often have the same minimum GPA as doctoral programs (3.0), but the GPA of admitted students tends to be a bit

lower for Masters compared to doctoral students; 3.4 (overall) and 3.5 in psychology courses (Norcross & Hogan, 2016).

Table 1.4. Most popular areas of study for psychology Masters degrees awarded in 2018-20.
Health service psychology fields are shown in yellow.
Table created from data from American Psychological Association (2021b).

PSYCHOLOGY SUBFIELD	PERCENTAGE MASTERS AWARDED IN 2018
Clinical Psychology*	
Counseling Psychology*	
Community Psychology	
Developmental Psychology	
Educational Psychology	
Experimental Psychology	
Forensic Psychology	
General Psychology	
Industrial/Organizational Psychology	
School Psychology*	
Social Psychology	
Other Research Subfields	
Other Health Service Psychology Subfields	

Let's take a look at some of the things you can do with a Masters degree in psychology. In many states, you can become an entry-level **licensed school psychologist** after graduating with a specialized three-year degree from an accredited program (this includes a one-year internship); you will also have to sit licensing exams at the end of this period (National Association of School Psychologists, 2022). Similarly, you can become a **licensed/certified counselor** after finishing a Master's degree in

a specialized field of counseling (e.g., *Addiction Counseling; Career Counseling; Clinical Mental Health; Community Agency Counseling; Forensic Mental Health Counseling; Marriage, Couple and Family Counseling; School Counseling; Student Affairs and College Counseling; Gerontological Counseling; or Counselor Education & Supervision). T*hese programs are typically 2 years, but you will also need to complete an extensive period of post-graduate supervised practical experience (American Counseling Association, 2022) before you can take your professional licensing exams. While you cannot become a licensed clinical psychologist with only a masters' degree in clinical psychology, you can work as a clinician or a caseworker for mental health agencies under the supervision of a licensed mental health practitioner.

If you are considering becoming a social worker – you will need to complete a **Masters in Social Work** (MSW), which qualifies you to sit the associated licensing exam (Association of Social Work Boards, 2022). **Industrial/organizational** (I/O) psychology is currently a rapidly growing field and helps businesses and organizations resolve workplace issues, such as problems with employee engagement, efficiency and retention. Graduates with an Masters in I/O psychology often work in consulting, marketing and human resources, click on this link to see more specific job titles). Other graduates from MA Psychology programs go into similar areas to those shown in Table 1.2, but at a higher level than graduates with a Bachelors degree. They might become research project coordinators or senior research assistants within a lab setting, or might even teach undergraduate courses at a college.

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1.3. The good, bad, and ugly aspects of the history of US Psychology

JILL GROSE-FIFER

Modern-day Western psychology has its roots in philosophy, medicine, theology, and the natural sciences. However, Western historians often fail to acknowledge that psychology has actually been "studied and practiced" around the world in different ways for centuries. Egyptian, Greek, Roman, Indian, Hebrew, Tibetan, and Chinese texts show us how ancient civilizations across the globe theorized about the mind and behavior. Many texts claim that Freud was the "father of psychoanalysis", but we know that ancient Egyptians interpreted dreams and practiced hypnotherapy to relieve psychological distress more than two thousand years before Freud was born. Similarly, indigenous and folk practices to ensure physical, mental, and spiritual well-being have passed from generation to generation via word of mouth and through hands-on teaching in many cultures. However, psychology as an **academic discipline**—a subject studied in a college or university—is relatively new.

European (especially German) psychologists dominated the academic discipline of psychology when it emerged during the 19th

century. In 1879, **Wilhelm Wundt** established the first "modern" psychology research lab in Leipzig, Germany. Wundt was particularly influential because he trained nearly 200 students in psychology; most were from Germany and other European countries, but a few (like Stanley Hall and Edward Titchener) worked in the United States after studying with Wundt (Walsh et al., 2014). White men dominated both in society and in psychology at this time; among Wundt's many doctoral students, only one, Anna Berliner, was a woman!

Wundt focused on understanding the structure of mental processes by breaking them down into small components, thus, creating the **school of structuralism** in psychology**.** Wundt tried to use the scientific method to study basic psychological functions, like vision, and so was one of the first **experimental psychologists** in the West. Wundt trained his graduate students to be participants in his studies. He asked them to **introspect** (look inside themselves) to describe their experiences of a stimulus (e.g., a ticking clock). Wundt's students also went on to train many other influential psychologists. Wundt was influenced by two earlier scientists, Gustav Fechner (a physicist) and Ernst Weber (a physiologist), who established the field of psychophysics. Psychophysical experiments investigate how changing the physical properties of a stimulus (like the brightness of a light) affected people's perception. Psychophysics is very important in the field of Sensation and Perception, which focuses on understanding how our five senses work. However, it was probably King Psamtik I of Egypt in the 7th century BCE, who conducted one of the first known psychology experiments. **Psamtik** believed that everyone's natural language was Egyptian, so to test his hypothesis, he arranged for two newborn babies to be taken from their mother and raised without hearing anyone speak. This cruel study did not support his prediction. The children's first words were not Egyptian. In fact they only ever babbled in meaningless sounds, which is not surprising

given we now know that children need to hear language in order to speak actual words.

Wundt also formally established the field of **cultural psychology**—since he believed that complex psychological functions like language, traditions, and beliefs needed to be studied within the context of culture. Despite this, critics of Wundt's cultural work suggest that it is highly ethnocentric, in other words, Wundt was unaware that his own worldview dominated his theories, he apparently believed that his own culture and beliefs were the norm (Walsh et al., 2014). This problem has plagued mainstream psychology for decades.

Early psychologists, both in Europe and in the United States, focused on the study of the mind, rather than behavior. **William James** taught the first psychology course in the USA in 1875 at Harvard; he taught students about experimental psychology focusing predominantly on sensation and perception. James also found the phenomena of consciousness and metaphysics intriguing, and he acknowledged they were difficult to study within the constraints of the scientific method. His thoughts and theories were published in a very influential two-volume book on psychology in 1890, and helped to establish the **school of functionalism.** Functionalist psychologists focused on understanding the meaning (or function) of people's thoughts, feelings, and behaviors, and how they helped a person to adapt to their environment. One of William James's doctoral students, G. **Stanley Hall** (who had studied briefly in Germany with Wundt), was the first person in the US to receive a PhD in psychology from Harvard. Hall set up the first psychology research lab in the United States in 1892 at Johns Hopkins University and was a strong proponent of using the scientific method. In contrast, in 1895, **Sigmund Freud** (an Austrian neurologist) wrote an influential text about using psychoanalysis for the treatment of mental illness. **Psychoanalytic theory** relates mental health problems to unconscious thoughts (of which people are unaware) and their

early childhood experiences. This was a popular perspective in clinical psychology for several decades (Thorne & Henley, 2005).

In the 20th century, the United States expanded its economic power and became the largest producer of psychological research in the world. Mainstream psychologists in the USA believed that psychology needed to be more objective for it to become a "serious discipline", and so embraced a **reductionist** approach focusing on using the scientific method. They largely redefined psychology as **Behaviorism**–"the scientific study of observable behavior". Their philosophy stemmed from Russian physiologist's **Ivan Pavlov'**s work on learning in animals. Leading behaviorists in the USA, like **John Watson** and **BF Skinner** believed that children (and animals) were "blank slates" who could be taught almost anything through a program of rewards and punishments. In contrast, most of European psychology took a different direction during this time. In Europe, psychoanalytical theory continued to be popular and other psychological perspectives emerged. During the 1920s, **Lev Vygotsky,** a highly influential developmental psychologist in Russia, studied development in children using a new **sociocultural framework.** This framework hinged on cultural-historical influences on the biological roots of the mind and behavior. In particular, Vygotsky stressed that learning strongly depended on the environment, especially social interactions among peers. He also warned against studying psychology in an overly simplistic way, urging psychologists to consider behavior and the mind as systems, rather than reducing them to individual functions. Somewhat similarly, in the 1940s, Russian-American psychologist, **Urie Bronfenbrenner,** also emphasized the importance of societal influences, social interactions, and environment on child development.

With the advance of computers and the advent of new technologies to study the brain, US psychologists in the 1960s finally turned their attention again to study the mind. Cognitive psychology, linguistics, and cognitive neuroscience became popular

areas of study and this turning point became known as the **cognitive revolution,** championed in part by psychologists, such as Noam Chomsky. German **Gestalt** psychologists, such as Wertheimer, Koffka, and Köhler, who immigrated to the United States in the 1930s to escape persecution in Nazi Germany, also contributed to this renewed study of the mind-brain connection. Gestalt psychologists believed that structuralism oversimplified the study of the mind. In contrast, Gestalt psychologists emphasize that the brain interprets information received from the senses in a holistic way, rather than simply adding together the bits of information it receives. For example, if we focus on structure – we could describe a jigsaw puzzle as being composed of multiple different shaped pieces, but our general perception (also known as the gestalt) is the overall "big picture" that we see when the puzzle is completed, such as an animal, or a landscape. Other US critics of behaviorism, like **Abraham Maslow** and **Carl Rogers** formed a new school of thought in Psychology in the 1960s, known as **Humanism**. Humanists emphasized human potential and the capacity to change, with a focus on the "good" in human behavior and often used qualitative methods (rather than the scientific method) in their studies.

PSYCHOLOGY AND EUGENICS

During the 20th century, Western Psychology placed an emphasis on empirical or experimental psychology. Theories based on studies of participants from **Western, Educated, Industrialized, Rich Democracies,** swept through the world's universities and became the hegemonic norm, with little regard for individual differences among people. During this time, White upper class men dominated Western Psychology, and members of the discipline often espoused sexist, racist, and homophobic views about humans, condoning the oppression of women and non-White

people, as well as other groups traditionally pushed to the margins of society. During the first half of the 20th century, many Western psychologists embraced evolutionary theories promoted by two British scientists, Charles Darwin and Francis Galton. Darwin and Galton believed that genes determined certain traits, such as intelligence, and that environment had little influence on their development. Galton went on to create the very ugly field of **Eugenics**—promoting the idea that people with "superior" genes should be encouraged to reproduce, while advocating for the sterilization of others with "inferior" genes. Others in this context, referred to people of color, people with mental illness, and people in poverty.

The eugenics movement, championed by highly influential people (including American President, Theodore Roosevelt), spread like a virus throughout the world. It was taught in schools and colleges. Notably, between 1842 and 1947, 31 presidents of the American Psychological Association were members of eugenics movements. For decades, US and UK psychologists including **Terman, Yerkes, Spearman** (among many others), actively promoted the idea that non-Whites, Jews, and Eastern and Southern European immigrants were intellectually inferior to Whites. Similarly, eugenicists expected women of all races and ethnicities to engage in traditional gender roles, such as housework and child rearing, because they believed that women lacked the capacity for more intellectual pursuits. Chapter 8's section on Intelligence delves deeper into these issues.

The research that was conducted by eugenicist psychologists was used to advocate for, and justify social and economic inequalities in the USA, the UK, and the countries they colonized. In the USA, eugenics contributed to the ban on interracial marriages, racial segregation, and increased wealth and resources for Whites. The government passed laws that restricted who could immigrate to the USA, dramatically limiting the admittance of Jews, Eastern and Southern Europeans, and people who were mentally ill. From 1907

to 1932, 30 states in the USA passed laws allowing forced sterilization of people deemed to be "defective" in some way. In many states, this included people living in institutions for the mentally ill, those accused of crimes, and those with disabilities or chronic illnesses, such as epilepsy. By 1944, 40,000 people were subjected to eugenic sterilizations across these 30 US states. Poor Black women and those born in foreign countries were the most likely targets. During the pre-Nazi period, Adolf Hitler and German eugenicists publicly admired the USA for its success in perpetuating white superiority, and its immigration and sterilization laws, which they used as models for their own laws. Germany took eugenics to even greater extremes by mandating the murders of young children with physical or mental disabilities. Eugenics was also the basis for the persecution and mass murders of six million Jews in Nazi Germany during the holocaust. Subsequently, eugenics lost much of its popularity. Over time, psychologists and scientists from other fields discredited the field of eugenics by conclusively demonstrating that it has no scientific basis. DNA studies show us that there is no genetic basis to race, and that the environment strongly influences intelligence and psychological functioning. However, despite public outcry against the inhumanity of the holocaust in Nazi Germany, the USA still carried out about 22,000 forced sterilizations across 27 states between 1943 and 1963. It was not until the 60s and 70s, that USA sterilization laws were largely repealed (Sofair and Kaldjian, 2000; Miller, 2020).

PSYCHOLOGY AND SOCIAL JUSTICE

The biases of the White Christian men who dominated the early years of psychology in the USA barred other groups from becoming psychologists. Women could not apply to most graduate psychology programs and those who earned a PhD had difficulty securing an academic job. Even by 1970, women accounted for only

20% of psychology PhDs. Similarly, in the 1940s, there were only four Black colleges offering psychology majors and Black students at predominantly White institutions faced exclusion and racism at every step of their studies. Jews were also generally unwelcome in the field of psychology. In the 1920s, Harvard, Yale, and Princeton all had tight quotas to restrict the entry of Jews into their psychology programs. Similarly, psychologists with "Jewish-sounding" names (regardless of faith) were frequently unsuccessful in securing academic positions until they changed their names. Harry Harlow, a well-known psychologist who studied attachment was a protestant, but he changed his name from Harry Israel because he was concerned that people would assume he was Jewish and would discriminate against him.

Very gradually, however, women and members of other marginalized groups began to make inroads into society, and, in turn, the discipline of psychology. The earliest pioneers in particular, fought hard to counter prejudice and discrimination as they struggled to gain admission into the field. Increased diversity among psychologists has forced modern-day psychology to evolve from its early roots, which focused on the study of White men by White men, into a far richer discipline where studies of gender, sexuality, and multiculturalism in general, are not only subfields in their own rights, but are also becoming infused more regularly throughout the discipline of psychology. *Please note, there are relatively few copyright-free images of the social justice champions that we are showcasing in the next section. However, if you click on the link on their names you will see a picture of the psychologist.*

Mary Whiton Calkins was the first woman in the United States to complete her doctoral studies in psychology. She worked under William James at Harvard University in the 1890s and passed all the requirements for the Ph.D. with distinction. Although she had permission to attend classes, she could not formally enroll in the PhD program because Harvard was an all-male institution. Harvard offered her a degree from Radcliffe College (a women's college

affiliated with Harvard) but, Calkins, a strong advocate for women's rights, refused it as a form of protest against Harvard's sexist policies. Even without a Ph.D., Calkins went on to be a professor at Wellesley College and was the first to set up a psychology laboratory in a women's college in the United States. In 1904, she became the first woman president of the American Psychological Association (APA). Calkins had a distinguished career and published over 100 papers and 4 books on dream analysis, memory, and "self-psychology"—the study of the conscious self (Young, 2010).

Figure 1.16. Mary Whiton Calkins was the first female psychologist in the United States

In 1894, **Margaret Floy Washburn** was the first woman to receive a Ph.D. in psychology in the United States. After her undergraduate studies at Vassar College (an all-women's college at the time), Washburn began her doctoral studies under James Cattell at Columbia University. Like Calkins, Washburn's educational journey was hampered because she was a woman. At the time, Columbia was an all-male university and although they allowed Calkins to audit classes, she could not enroll in the doctoral program. Instead,

Cattell persuaded Calkins to enroll at Cornell University where she studied under Titchener and was awarded a Ph.D. Washburn became a professor at Vassar College and many of undergraduates worked in her lab. Washburn studied consciousness in humans and animals. In 1921, Washburn became the second woman president of the APA (Rodkey, 2010).

Figure 1.17. Margaret Floyd Washburn was the first woman in the United States to receive a Ph.D. in psychology.

Helen Thompson Woolley was awarded a PhD in psychology (with the highest distinction) from the University of Chicago. Woolley's PhD dissertation research provided evidence that went against the sexist views that were prevalent in society at that time. Woolley found that men and women performed similarly on sensory, motor, and intelligence tasks, and that there were greater differences within each sex than between the sexes. Her research went against the popular idea of the time that women were less

logical than men. Her later work showed the environment and education strongly influenced intelligence scores in adolescents, and that children who left school to go to work were at an intellectual disadvantage. Her work helped to influence labor laws to protect children. Woolley's work was in direct opposition to the views of the eugenicists, who believed that intelligence was innate, that is, fixed from birth (Helen Thompson Woolley, 2022; Rodkey, 2010).

Leta S. Hollingworth also helped to counteract negative stereotypes about women with her doctoral dissertation (PhD awarded in 1916) from Columbia University in Educational Psychology. Her dissertation showed that menstruation had no effect on women's decision making abilities, interestingly, her advisor Edward Thorndike, was one of the very psychologists who had helped to popularize the opposite view. Hollingworth taught at Teacher's College, Columbia University and also worked on intelligence testing and with gifted children for much of her career (Benjamin 1990; Leta Stetter Hollingworth, 2023).

Figure 1.18. Leta S. Hollingworth

.In 1920, **Francis Sumner** (click on link for photo) became the first African American to be awarded a Ph.D. in Psychology in the United States. Sumner did not attend high school because in Virginia, where he lived, the schools that were available to Black students were of a very low quality. Recognizing this, his father homeschooled him and when he was 15, he passed the entrance exam to Lincoln University. Four years later, he graduated with a degree in English and was awarded a Masters' degree two years later. His doctoral studies at Clark University were interrupted when he was drafted into the US Army, but he completed his Ph.D. after finishing his service. After working as a professor at two other colleges, Sumner became a professor and Chair of Psychology at Howard University (a historically Black University) where he helped to start and build the reputation of the psychology department.

Sumner was no stranger to discrimination, when he was a PhD student at Clark University, he was given his own separate table in the dining hall, with the expectation that no one would want to eat with him. Perhaps, not surprisingly his research focused on trying to reduce racism and bias toward African Americans. Kenneth Clark, another psychologist who fought for social justice, was one of his most well-known students.

In 1933, **Inez Beverly Prosser** was the first African American woman to be awarded a doctorate in psychology in the United States. In 1912, Prosser attained a teaching certificate from the Prairie View A&M University in Texas, and then taught at various schools for Black children. However, at that time, Texas did not award graduate degrees to Black students and so Prosser had to apply out of state to continue her studies. She completed a Masters' in Education at the University of Colorado, and then a doctorate in Educational Psychology at the University of Cincinnati. Prosser's work highlighted the psychological difficulties that Black children experienced in desegregated schools, were they were predominantly taught by White teachers. Prosser taught in small colleges in both Austin, Texas and Jacksonville, Mississippi. Tragically, Prosser died in a car accident, a year after she was awarded her doctorate.

Figure 1.19. Inez Prosser was the first African American woman to earn a doctorate in psychology in the United States

In 1940, **Kenneth Clark** was the first African American to earn a Ph.D. in psychology from Columbia University. After graduating from Columbia, Clark became a professor at City College (CUNY) and worked there for many years. In 1960, he was the first Black person in New York City to be promoted to the rank of tenured full professor, and he later became a distinguished professor at the college. He was also the first African American president of APA.

Figure 1.20. Kenneth Clark's work with his wife, Mamie Phipps Clark, helped to desegregate schools in the United States

In 1943, **Mamie Phipps Clark** became the first African American woman to earn a Ph.D. in psychology from Columbia University. Phipps Clark's earlier research for her Masters' degree at Howard University examined race consciousness in children. This then led to the development of her doctoral research and another set of highly influential research studies she conducted with her husband, Kenneth Clark. After she graduated from Columbia, Phipps Clark found that universities and

colleges were unwilling to hire a young Black woman as a professor. Eventually, she became a counselor at the Riverdale Home for Children in New York, where she worked with African American girls who were homeless. Phipps Clark and her husband, Kenneth Clark, later founded the Northside Center for Child Development in Harlem, which provided psychological services to minority children, and she served as the director here for many years. The Clark's collaborative research studies on racial consciousness in children (often referred to as the doll studies) were used as evidence in Brown vs. Board of Education of Topeka, Kansas (1954). Their work contributed to the decision to desegregate schools by persuading the Supreme Court that segregation was unconstitutional and led to low self-esteem among children of color.

Figure 1.21. Mamie Phipps Clark was the first African American woman to earn a PhD in psychology from Columbia University. Her research was influential in de-segregating schools in the United States

In 1943, **Robert Chin** (who changed his name from Chen Yuli) was the first Chinese American to be awarded a Ph.D. in psychology. After Chin graduated from Columbia University with a

doctorate in social psychology, he joined the U.S. army and worked in counterintelligence in both China and Washington. Chin later joined the faculty at Boston University where his research focused on organization theory, and ethnic and minority group relationships. His work on intelligence testing helped to negate the prevalent idea of the time that non-Caucasians were intellectually inferior to Whites. This evidence was used in the Brown vs. Board of Education of Topeka, Kansas (1954) decision that helped to desegregate schools. Chin was a cofounder of the Humans Relations Center at Boston University, and also helped to bring organization theory to China.

Kurt Lewin was a social psychologist and German-American Jew who sought refuge in the USA during the 1930s. He worked as a professor at the University of IOWA. Lewin advocated that psychological research should result in improvements in the lives of people who are often marginalized in society. Some of his earlier work in the USA focused on productivity in factories. He showed that when workers were involved in decision-making and were giving opportunities to collaborate with managers about their training, they outperformed workers who only received orders from management. Lewin also worked with community members to reduce gang violence towards Jews. He believed that **action research** (as he termed it) always needed to be collaborative and could bring about meaningful change if community members were intimately involved in designing and carrying out research that related to the social problems in their lives. After his death in 1947, some psychologists, including his graduate students, continued to expand on his social justice legacy, which spread to other fields

such as education and business. However, many social psychologists felt that action research was too "messy" and unscientific and so it was not fully embraced by mainstream psychology at the time.

In 1953, **Efrain Sanchez Hidalgo** (click name for photo**)** was the first Puerto Rican to receive a Ph.D. in Psychology in the United States. He studied Social Psychology at Columbia University under Edward Thorndike. He had previously earned a Masters' degree from Columbia University in Educational Psychology and his textbook (*Psicología educativa*) was widely used throughout Latin America. He also served as Secretary of Education for Puerto Rico. Sanchez Hidalgo worked for many years at the University of Puerto Rico as a professor of psychology.

Evelyn Hooker was one of the first psychologists to receive a grant from the National Institute of Health (NIH) to study homosexuality. Hooker was awarded a PhD in psychology from Johns Hopkins University in 1932, and went onto be a professor at UCLA. At the time, homosexuality was considered to be a mental illness by psychologists and members of the medical profession. In the 1950s, one of her gay students at UCLA asked Hooker to do a study to help fight against those beliefs and Hooker took action. After applying for and being awarded an NIH grant, Hooker collaborated with a gay-rights organization who helped her to find

participants for her research study. Participants responded to three personality tests (including the famous Rorschach inkblot test). Hooker reported that experts found no differences in the personality test results of homosexual and heterosexual men. In 1956, Hooker presented her research findings to the annual convention of the American Psychological Association. Her findings helped to pave the way forward to normalizing homosexuality. In 1973, homosexuality was removed as a mental health diagnosis from the next edition of the American Psychiatric Association's Diagnostic and Statistical Manual (DSM II), which helped to reduce some of the stigma associated with being queer. However, the DSM II had a new diagnosis—same sex orientation disorder—which applied if a person was attracted to people of the same sex and was distressed by it. This loophole meant that sex conversion therapy continued to be practiced by psychologists and psychiatrists after 1973. It was not until 1987, that homosexuality was properly removed from the DSM. In 1990, the World Health Organization also removed homosexuality from its list of illnesses. These changes helped to decriminalize homosexual acts (sodomy laws) and create laws protecting LGBTQ+ rights. Psychologists moved away from research that questioned what caused

queerness and how to "cure" it, to investigations focusing on wellness in LGBTQ+ populations.

In 1962, **Martha Bernal** was the first Latina to be awarded a Ph.D. in psychology in the United States. Her doctorate was in clinical psychology from Indiana University. Bernal's parents were originally from Mexico, and Bernal was born and grew up in Texas, but she was not allowed to speak Spanish in the segregated schools she attended. She also struggled to persuade her father to allow her to go to graduate school because he believed she should get married and stay home to raise children. Bernal's mother and sister helped to persuade him. Bernal faced a lot of sexism and racism, especially in her early career and she had difficulty finding a faculty position after she graduated. However, Bernal did not give up and was given a post-doctoral research position at UCLA. She later held academic appointments at the University of Denver and at Arizona State University. Bernal's research focused on multiculturalism using learning theories to help children with behavioral issues. Bernal was deeply committed to helping other women, especially Latinas and women of color, to succeed in graduate school. She was a founding member of the National Hispanic Psychological Association (NHPA).

In 1964, **Marigold Linton** became the first Native American woman (Cahuilla-Cupeño nations) to earn a PhD in psychology.[1] Linton grew up in poverty on the Morongo Indian Reservation in Southern California and graduated from UCLA as a cognitive psychologist. Her first job as a professor was at San Diego State University, where she campaigned for equal pay for women professors. She later worked at Arizona State University as the Director of American Indian Programs, and the University of Kansas as the Director of American Indian outreach. In addition to her cognitive psychology research, where she used diaries to study how her own memory changed over time, Dr. Linton received millions of

1. Carolyn Attneave earned a PhD in psychology from Stanford in 1952. Attneave's mother was Lenni-Lenape and her father from Scandinavia.

dollars in grants to promote the advancement of American Indians in higher education.

In 1975, indigenous psychologists in the United States formed the Native American Psychological Association. **Logan Wright** became the first Native American (Osage Nation) President of APA in 1986. Wright was awarded a Ph.D. in clinical psychology from Vanderbilt University in 1964. He began his academic career at Purdue University and then moved to the University of Oklahoma Health Sciences Center, where he worked for many years in the department of pediatrics. Indeed, Wright has often been credited with being the "father of pediatric psychology"; he helped to found the Society for Pediatric Psychology and his research focused on behavioral interventions for children.

In 1999, **Richard M. Suinn** was the first Asian American president of the APA. Suinn earned his doctorate in counseling psychology from Stanford University in 1959. Suinn is an emeritus professor at Colorado State University where he worked tirelessly to increase recruitment and retention of ethnic minority students at the university. Suinn's research focused on ethnic minorities, behavior therapy, and sport psychology. Before he became the president of APA, Suinn served on APA's committee on ethnic minority affairs. He also helped to establish the Society for the Psychological Study of Cultural, Race and Ethnicity (APA Division 45).

In 2009, **Melba Vasquez** became the first Latina president of APA. She was awarded a doctorate in counseling psychology from the University of Texas in 1978. Like Bernal before her, Vasquez was Mexican-American and grew up in Texas where her early public school experiences were fraught with racism. Vasquez's life improved dramatically when she started Catholic school in 4th grade. Vasquez was an English major as an undergraduate and then went on to earn a teaching certificate, but she soon realized that teaching children was not her passion. She wanted to learn more about human behavior, especially prejudice and discrimination. Vasquez met Martha Bernal when she was a

graduate student, and became one of the many students that Bernal mentored. Vasquez was one of the first people to receive an APA Minority Fellowship to support minority students. Vasquez works as an independent counseling psychologist in Austin, Texas. Her work focuses on multicultural issues in psychology and **feminist psychology**.

In 2018, **Jessica Henderson Daniel** was the first African American woman to become an APA president. Daniel earned her doctorate in Educational Psychology from the University of Illinois-Urbana. She works as a professor of psychology in the Department of Psychiatry at Harvard Medical School. Daniel has dedicated her life to instruction and mentoring, especially helping students of color to succeed. She developed and garnered funding for a very successful mentoring program for early-career Black women psychologists in academia with other more senior Black women serving as mentors. Although funding for this project was short-lived, Dr. Daniel's insights about effective mentoring strategies have been widely disseminated.

Lillian **Comas-Diaz** was the first person of color to receive a gold medal for lifetime achievement in the practice of psychology from the American Psychological Association. Comas-Diaz was born in Chicago to Puerto Rican parents; she then grew up and went to college in Puerto Rico. She moved to the USA in her twenties to go to graduate school and was awarded a PhD in clinical psychology from the University of Massachussetts, Amherst, in 1979. Comas-Diaz pioneered the use of ethnocultural approaches to mental health and is a specialist in racial trauma and the use of liberatory, decolonial therapeutic practices. **Liberatory psychology** stems from work by **Frantz Fanon**, a psychiatrist from the French colony of Martinique and by Brazilian educator, **Paulo Friere**, and later, **Ignacio Martin-Baro**, a Spanish-born psychologist and Jesuit priest, who worked most of his life in El Salvador. **Liberatory psychology** focuses on practices that fight against systems of oppression and bring about social improvements. Comas-Díaz

worked for the American Psychological Association and played a critical role in creating Division 45, the Society for the Psychological Study of Ethnic Minority Issues. She is a practicing psychologist and a professor at George Washington University.

Figure 1.22. Lillian Comas Dias

HOW PSYCHOLOGISTS HAVE HELPED TO PROMOTE SOCIAL JUSTICE IN THE UNITED STATES (CLICK HERE FOR MORE INFORMATION)

This is a brief summary of some of the events that psychologists have played a role in helping to promote social justice in the United States. This list is by no means exhaustive (it does not, for instance, include any mention of multiple participatory action research projects that have brought about social change in many communities across the country). Many of the events described below focus on the role of APA, including the formation of special

interest groups that came about because APA itself was not addressing the needs of its members.

In 1936, the Society for Psychological Study of Social Justice Issues (SPPSI) was formed. In 1945, SPSSI became part of APA (division 9)

In 1954, evidence from psychologists, especially Dr. Kenneth Clark, was used in reaching the decision that segregated schools were unconstitutional and needed to be dismantled as quickly as possible (Brown vs. Board of Education of Topeka, Kansas).

In 1962, APA submitted an amicus brief (a document that is designed to educate legal decision makers) in the case of Jenkins vs. United States, which contributed to the ruling psychologists were competent to serve as expert witnesses and to testify about the nature of mental illnesses in court cases. Psychologists also testify in court cases on other issues, including factors that impact the reliability of eyewitness testimony or lead to false confessions.

In 1968, shortly after the assassination of Martin Luther King, the Association of Black Psychologists (ABPsi) was formed at the annual APA conference amid frustration that APA was not addressing the issues facing Black psychologists and the Black community ingeneral. ABPsi asked APA to stop exploiting the Black community by viewing them only as research subjects and to instead use their power to address social problems, such as poverty and racism. They also asked APA to commit to increasing the number of Black psychologists. Influential figures in the development of the Black Psychology movement included **Dr. Joseph White** and **Dr. Robert Williams**. Williams developed the Black Intelligence Test of Cultural Homogeneity (B.I.T.C.H.) to demonstrate racial biases in testing. The test asked for definitions of words that were predominantly used in Black communities and so Blacks outperformed Whites, this helped to highlight the fact that most standard IQ tests were biased to favor White middle class people. Leaders of the ABPsi helped to highlight the importance of research on race in psychology, and established new academic journals devoted to

publishing work on racial issues. They also promoted increased access for Black students wanting to become psychologists.

In 1970, the National Hispanic Psychological Association was formed. This was later renamed the National Latina/o Psychological Association (NLPA). The mission of NLPA is to advance the field of psychology to benefit Latina/os.

In 1972, the Asian American Psychological Association was formed with the goal of using psychological knowledge to advance the mental health and wellbeing of Asian Americans.

In 1974, APA established a Minority Fellowship to providing funding for doctoral psychology students who were members of minority groups.

In 1975, the Society of Indian (American Indian and Alaska Native) Psychologists was formed to promote the mental health of Native Americans.

In 1975, Division 35 of APA, which later became the Society for the Psychology of Women, was finally formed. Attempts to form a women's division within APA had been turned down three times previously (in 1948, 1958, and 1959). The Feminist Psychology movement critiqued traditional psychology studies as male-centric and lacking insights into the psychology of women. Pioneers, like **Karen Horney**, paved the way for research looking at gender differences and promoting women's rights. Horney critiqued Freud's psychoanalytical theories; Freud suggested that women's issues were often centered on the fact that they were not male (e.g., penis envy). Instead Horney put forward the idea that men may experience feelings of inferiority because they could not give birth (i.e., womb envy).

In 1975, APA released an update to its publication manual stating that sexist language should not be used in psychology articles and papers.

In 1978, the Office of Ethnic Minority affairs was established at APA with the goal of working toward diversity, inclusion and empowerment

In 1979, a U.S. district court in California used research evidence provided by psychologists to rule that standard IQ tests were discriminatory (Larry P. v. Wilson Riles; 1979). Large numbers of Black children had been placed in remedial programs on the basis of these tests.

In 1986, psychologist **Susan Fiske** provided evidence as an expert witness in a U.S. Supreme Court case that ruled that gender-stereotyping had prevented a woman gaining promotion to partner in the law firm in which she worked (Price Waterhouse v. Hopkins).

In 1990, APA Society for Peace Psychology was formed (now Society for the study of Peace, Conflict and Violence)

In 1991, SPSSI became consultants to the United Nations

In 2000, APA released treatment guidelines for LGB clients.

In 2001, New Jersey was the first state to control the way in which police lineups for eyewitnesses of crimes were conducted. This was informed by research on eye witness testimony by psychologists.

In 2003, APA provided an amicus brief to the U.S. Supreme Court to provide support for the University of Michigan's affirmative action admissions policies to increase recruitment of minority students.

In 2004, APA adopted a policy in support of same sex marriage.

In 2005, Roper vs. Simmons abolished the juvenile death penalty in the United States (APA provided an **amicus brief** with neuroscience evidence that documented the immaturity of adolescent brains, which was presented to court as supporting evidence for reduced culpability).

In 2007, APA taskforce published a report showing that sexualization of girls in the media leads to mental health problems.

In 2010, Graham vs. Florida banned life without parole for all juveniles (except those found guilty of homicides), APA provided an amicus brief with neuroscience evidence about the immaturity of adolescent brains.

In 2012, Jackson versus Hobbs banned life without parole for all juveniles (*regardless of type of conviction*), APA provided an amicus

brief with neuroscience evidence about the immaturity of adolescent brains.

In 2015, psychologists contributed to evidence used in the President's Taskforce on 21st Century Policing, which urged police departments to implement implicit bias training

In 2016, SPPSI members gave a congressional seminar on the impacts of climate change on people around the world, particularly those who are poor and/or socially marginalized. They gave another congressional seminar on LGBTQ discrimination.

In 2017, SPPSI members gave congressional seminars on criminal justice reform as well as health disparities research. The American Arab, Middle Eastern, and North African Psychological Association (AMENAPsy) was formed.

In 2018, SPSSI members hosted a congressional seminar on the psychological effects of genocide, and gave a congressional briefing on human trafficking.

In 2021, APA issued an apology to people of color in the United States for APA's role in promoting, perpetuating, and failing to challenge racism, racial discrimination, and human hierarchy in the U.S.

In 2022, APA issued a resolution on policing advocating that police departments should expand training programs for officers to learn about de-escalation techniques and how to reduce the use of force, as well as learning how to minimize targeting people of lower socioeconomic status. APA also highlighted the need for the police force to build stronger relations with mental health service agencies.

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Chapter 1. Conclusions

In conclusion, the first chapter of this textbook has provided you with a glimpse into the multifaceted nature of the field of psychology. We have explored the fundamental question of "What is Psychology?" and have discussed its diverse subfields and applications. Moreover, we have delved into the rich history of psychology, acknowledging both its achievements and its ugliness, such as psychology's contribution to eugenics and its failure to recognize the intellectual potential of BIPOC people and women. We have highlighted the transformative efforts of social justice activists within the discipline. As you consider becoming a psychology major or potential careers in psychology, it is important to recognize the profound impact psychologists can have on individuals and society as a whole. By studying the human mind and behavior, we can contribute to understanding, supporting, and advocating for the well-being of others. With this knowledge, we hope that you will embrace opportunities to make a positive difference and foster social change.

Chapter 1. Personal Application Questions

- Why are you taking this course? What do you hope to learn about during this course?
- Now that you've been briefly introduced to some of the major areas within psychology, which are you most interested in learning more about? Why?
- Which of the career options in the field of psychology is most appealing to you?
- What are some of the most surprising things that you learned about the history of psychology?
- Of all the psychologists that you read about in this chapter, who do you admire the most and why?

Chapter 1. Critical Thinking Questions

- Why do you think psychology courses like this one are often requirements of so many different programs of study?
- Why do you think many people might be skeptical about psychology being a science?
- Given the incredible diversity among the various areas of psychology that were described in this chapter, how do they all fit together?
- Why is an undergraduate education in psychology so helpful in a number of different lines of work?
- Other than a potentially greater salary, what would be the reasons an individual would continue on to get a graduate degree in psychology?

Chapter 1. Key Terms

action research
a research method in which researchers and participants collaborate to identify and address practical problems in a particular setting or community

American Psychological Association (APA)
professional organization representing psychologists in the United States

behaviorism
focus on observing and controlling behavior

biopsychology
study of how biology influences behavior

biopsychosocial model
perspective that asserts that biology, psychology, and social factors interact to determine an individual's health, actions, thoughts and feelings

Cognitive Revolution
period in the 1950s and 1960s where psychologists shifted their focus from behaviorism to the study of mental processes, such as perception, memory, and problem-solving.

cultural psychology
A subfield of psychology that studies how culture shapes and influences human behavior and mental processes, including perception, cognition, and emotion.

dissertation

long research paper about research that was conducted as a part of the candidate's doctoral training

empirical method

method for acquiring knowledge based on observation, including experimentation, rather than a method based only on forms of logical argument or previous authorities

ethnocentric

refers to the tendency of people to view and judge other cultures based on the standards and values of their own culture, often resulting in a biased or narrow perspective.

eugenics

The pseudoscientific belief in improving the human race by selectively breeding people with desirable traits, condemned today as unethical and discriminatory.

feminist psychology

focuses on the ways in which gender and gender-related issues impact behavior and mental processes, and aims to promote gender equality and challenge gender-based biases and stereotypes.

functionalism

focused on how mental activities helped an organism adapt to its environment

Gestalt Psychology

Perspective in psychology that emphasizes the holistic nature of perception, where individuals perceive objects as whole entities rather than a collection of parts. It emphasizes the role of context and organization in shaping perception and understanding.

graduate-level degree

An advanced academic degree beyond the undergraduate level, usually requiring 2+ years of coursework and research at the Masters level, and 5+ years at the Doctoral level

graduate school

A school or department within a university that offers advanced

academic programs leading to graduate-level degrees, such as master's and doctoral degrees.

health service psychology

A branch of psychology that focuses on the integration of psychological principles and practices in the provision of healthcare services, includes Clinical Psychology, Counseling Psychology, and School Psychology.

humanism

perspective within psychology that emphasizes the potential for good that is innate to all humans

introspection

process by which someone examines their own conscious experience in an attempt to break it into its component parts

liberal arts education

emphasizes a broad education, including humanities, social sciences, natural sciences, and mathematics, to develop critical thinking and problem-solving skills.

Liberatory psychology

emphasizes social justice and the liberation of marginalized and oppressed individuals and groups from systemic forms of oppression and discrimination, through the use of critical consciousness and collective action.

personality psychology

study of patterns of thoughts and behaviors that make each individual unique

personality trait

consistent pattern of thought and behavior

PhD

(doctor of philosophy) doctoral degree conferred in many disciplinary perspectives housed in a traditional college of liberal arts and sciences

postdoctoral training program

allows young scientists to further develop their research programs

and broaden their research skills under the supervision of other professionals in the field

psychoanalytic theory

focus on the role of the unconscious in affecting conscious behavior

psychology

study of the mind and behavior

PsyD

(doctor of psychology) doctoral degree that places less emphasis on research-oriented skills and focuses more on application of psychological principles in the clinical context

reductionist approach

breaks complex phenomena into simpler components, often oversimplifying complex phenomena, leading to an incomplete understanding of human behavior and experience. Typically neglects the role of contextual factors.

replication crisis

difficulty in repeating and confirming previously published research, particularly in psychology and social sciences. This has raised concerns about biased or weak research methods, leading to calls for more rigorous and transparent scientific practices.

scientific method

A systematic approach to investigating phenomena based on empirical evidence and logical reasoning.

sociocultural framework

An approach to understanding human behavior and experience that emphasizes the influence of social and cultural factors, such as language, values, and cultural norms.

structuralism

understanding the conscious experience through introspection

traditional canon of knowledge

refers to a set of standard works or concepts that are considered to be important and essential to a particular field of study or culture.

WEIRD participants

People from Western, educated, industrialized, rich, and democratic societies used in social science research that may not generalize to other cultures.

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CHAPTER 2. PSYCHOLOGICAL RESEARCH

Figure 2.1. Psychologists use empirical studies to better understand the mind and behavior in animals and humans.

The discipline of psychology is based on evidence that is gathered

in research studies. This research plays a pivotal role in our understanding of human behavior, cognition (thinking), and emotions. By conducting rigorous studies, researchers uncover valuable insights into the complexities of the human mind and its underlying mechanisms. Psychology research helps to shed light on issues such as mental health, social interactions, and learning processes. It can provide evidence-based solutions for real-world problems, aiding in the development of effective therapeutic interventions, educational strategies, and organizational practices. There are many different approaches to conducting research in psychology — let's dive in and take a look.

HOW DO PSYCHOLOGISTS CREATE KNOWLEDGE?

We often make assumptions that certain groups of people behave differently from others, but, we are not always correct in our intuitions. Consider a common stereotype about gender differences. Many people believe that women are more talkative than men. *What do you think? What information are you using to form your opinion?* Psychologists use **empirical methods** to answer research questions like this one. Empirical methods involve collecting and organizing evidence (data) and using it to draw logical conclusions. In 2007, a group of psychologists led by Matthias Mehl conducted an empirical study to investigate whether women did actually talk more than men. Nearly 400 college students wore voice recorders for a few days, and the researchers counted how many words the research **participants** spoke in the recordings. They found that both men and women spoke about 16,000 words per day on average. Thus, they concluded that there was no basis to the stereotype (Mehl et al. 2007).

There are many different ways to conduct research in Psychology, depending on the research question. Each method has its pros and cons and no one method is perfect. Using different

methods provides different types of information and all contribute to a large body of knowledge about the mind and behavior.

LEARNING OBJECTIVES:

- Describe the different types of research methods used by psychologists
- Describe the steps of the scientific process in psychology
- Describe the key elements of an experiment and how it differs from a quasi-experiment
- Discuss the strengths and weaknesses of case studies, naturalistic observation, surveys, and archival research
- Describe the various ways that researchers collect data
- Compare and contrast correlation and causation
- Explain what a correlation coefficient tells us about the relationship between variables
- Identify independent and dependent variables
- Define the concept of replication crisis in psychology and explain its potential consequences for scientific research.
- Compare and contrast qualitative research with quantitative research in psychology, highlighting the strengths and limitations of each approach.
- Describe how research involving human subjects and animals is regulated
- Describe how psychology studies can help fight for social justice

The text in this chapter has been edited by Jill Grose-Fifer. Most of the material has come from the OERs mentioned in the REFERENCE section at the end of this chapter.

2.1. The scientific approach and quantitative research

Western Psychology has relied heavily on **quantitative research (using numbers)** and the **scientific approach** to build its theoretical foundations. The study by Mehl and colleagues exploring gender differences in talkativeness (Mehl et al. 2007) that we just described is a good example of this. There are three basic features to using the scientific approach.

1. Research strives to be **systematic** – Mehl and colleagues were careful and organized in their study design, data collection, and analysis. Scientific studies follow specific structures, which we will discuss in more detail below.
2. Research is focused on an **empirical question** – In the study by Mehl and colleagues, the question was *"Do women talk more than men?"* Not all questions relevant to psychology are empirical, for example, consider the question: *is there such a thing as free wi*ll? It would not be possible to collect data to answer this question.
3. **Research creates public knowledge**. Psychologists, like other scientists, publish their work. This usually means writing an article in a **peer-reviewed** professional journal targeted for other scholars. The article explains their research question in the context of previous research,

describes the methods used to answer their question, and clearly presents their results and conclusions.

Psychological knowledge is mostly based on **theories**. Theories are a well developed set of ideas that propose an explanation for observed behaviors we see in the world. For example, the research question that Mehl and his team were interested in relates to gender communication theories that attempt to explain how status affects the way men and women communicate in different contexts. Theories are typically based on multiple studies published by different researchers over many years. Despite researchers' best efforts, sometimes methods are flawed and/or their conclusions incorrect. Publication allows others in the scientific community to detect and correct these errors, and so over time, theories become more refined and accurate. As we can see in Figure 2.2, knowledge that is built through the scientific process often changes over time.

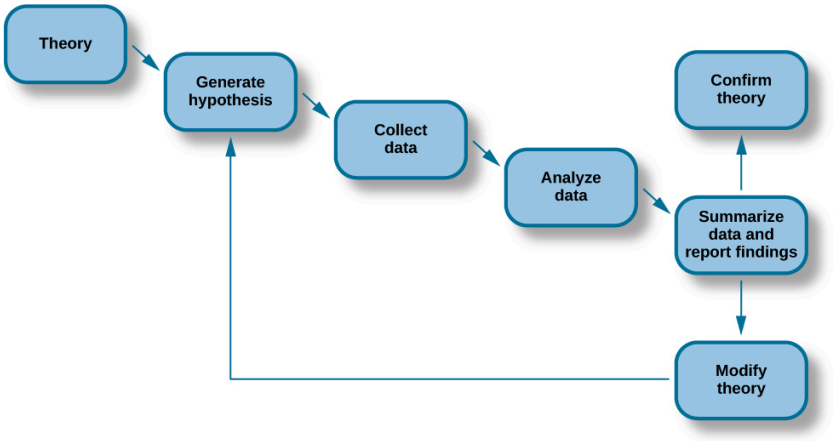

Figure 2.2. The scientific method involves deriving hypotheses from theories and then testing those hypotheses. If the results are consistent with the theory, then the theory is supported. If the results are not consistent, then the theory should be modified and new hypotheses will be generated (Source: Openstax.org)

Theories are too complex to be tested all at once; instead, researchers create **hypotheses** to test specific aspects of a theory. A **hypothesis** is a testable prediction. For example, Mehl and colleagues hypothesized that women talk more than men. Then they set out to test their prediction by collecting and analyzing data (see Figure 2.2). They did not find support for their hypothesis, and so this in turn, contributes to gender communication theories. Interestingly, other studies *have* found evidence for gender differences in talkativeness. For example, male students tend to talk more than female students in college classrooms (Lee & McCabe, 2021) and women tend to talk more in public, than private settings (Crawford 1995; Tannen, 1990). Clearly, context has to be taken into account. Gender communication theories acknowledge that how much men and women talk depends on the situation. In psychology, studies based on the scientific method consist of three main types: **experiments, quasi-experiments** and **correlational** studies. Let's take a look at experimental research first.

EXPERIMENTAL RESEARCH STUDIES

Figure 2.3. Psychologists have studied whether making donations to charity make people feel happier.

If somebody gave you $20 that you *had* to spend today, how would you spend it? Would you spend it on an item you've been wanting for weeks, or would you donate the money to charity? If you're like most people, you'd choose to spend the money on yourself. But, which choice do you think would make you most happy? Professor Elizabeth Dunn's research team at the University of British Columbia conducted TWO studies on whether giving to charities is connected to feeling happy (Dunn et al., 2008). One was a correlational study and the other was an experiment. In the experiment, they gave each of the participants $20 and then told half of the people they had to spend it on themselves. They told the rest of the people that they had to donate the money to charity. At the end of that day, both groups answered a question about how

happy they felt. Those who made the donation reported feeling happier. Dunn and her colleagues concluded that giving money to others contributed to overall happiness. This research study used the scientific method. It was systematic and it asked an empirical question – *Does giving money away make you happy?* You can ask many different kinds of questions with experiments, but they all have the same basic design features. Let's take a look at the features. Firstly, researchers always systematically change one variable, and then measure the impact of those changes on another variable. A variable is just something we can vary or measure in some way. The variable that is under the experimenters' control is called the **independent variable**. In the case of Dunn et al.'s experiment, the independent variable was how the participants spent the money. There were two conditions (options) – spend it on themselves or on others. The researchers (not the participants) decide on the options. This is what we mean by manipulating a variable. Experiments investigate whether the independent variable causes a change in the **dependent variable**. In this example, the participants' happiness is the dependent variable, because it depends on how the participants spend their money (independent variable). The independent variable is often thought of as the "cause" and the dependent variable – "the effect" (see Figure 2.4). Dr. Dunn was able to conclude that group differences in happiness were caused by how the money was spent.

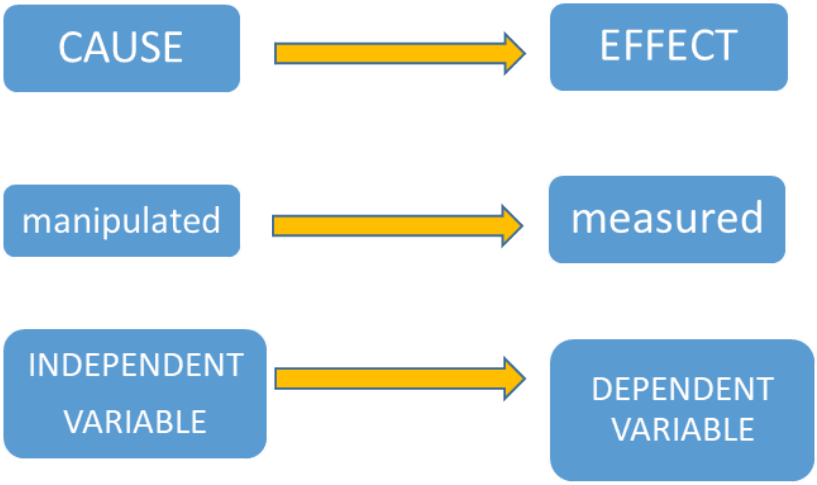

Figure 2.4. The relationship between the independent and dependent variables.

Did the results of Dr. Dunn's study surprise you? If they did, you might be feeling a little bit skeptical about the results of this study. You might be thinking—*doesn't happiness depend on a lot of different factors—including a person's upbringing or life circumstances? What if some people had happy childhoods and that's why they're happier? Wouldn't people who are having a bad day feel less happy?* You are absolutely right and you are thinking like a scientist! All of these factors (and many more) can easily affect a person's level of happiness. So, how can we take all of these different variables into account? A second feature of experiments that helps to compensate for these individual differences across people, is that they all use **random assignment.** In other words, participants cannot pick which group they are in, the experimenter assigns them to a particular condition or group randomly. This might be based on the flip of a coin, the roll of a die, or the generation of a random number by a computer. Random assignment makes it so the groups are relatively similar on all characteristics, except the

one that the experimenter wants to manipulate. In Dunn's study, there should be equal numbers of people with happy childhoods and equal numbers of people who were having a bad day in each group.

Some experiments are conducted to see whether an intervention has an effect on behavior, for example, psychologists might be interested to know whether a certain type of drug or therapy improves symptoms of depression more than another. Researchers manipulate the independent variable by randomly assigning participants to an **experimental group** (who are given the drug being tested) and the **control group** (who are given a sugar pill—also known as a placebo). Researchers often conduct studies like this using "blinding". A **single–blind** design is one where participants do not know which group they are in and so their expectations are less likely to influence the results. Nevertheless, participants in the control group sometimes believe that they received the actual drug and show signs of improvement or even side effects—this is known as the **placebo effect.** That is why a control group is so important, the drug must have significantly greater benefits than the placebo in order to be considered effective. In **double-blind** studies, the researchers are also unaware of the group assignments until after the study is completely finished. This reduces experimenter expectations and biases when analyzing the data. In experiments, there always has to be at least comparison group/condition in order to know whether the manipulation of the independent variable had any effect. In the case, of Dunn's study, she hypothesized that prosocial spending, i.e., giving money to others would lead to greater happiness than self-spending, so we can consider those who were in the donation group as the experimental group while the other participants were in the control group.

Another aspect of experiments is that the dependent variable must be quantifiable, in other words, it must be measurable in some way using numbers. In the experiment on happiness and

giving to charity you just read about, Dunn's team asked participants a single question: *Do you feel happy on this day?* Participants used a rating scale from 1 to 5 (like the one shown in Figure 2. 5) to indicate how happy they felt (5 = most happy).

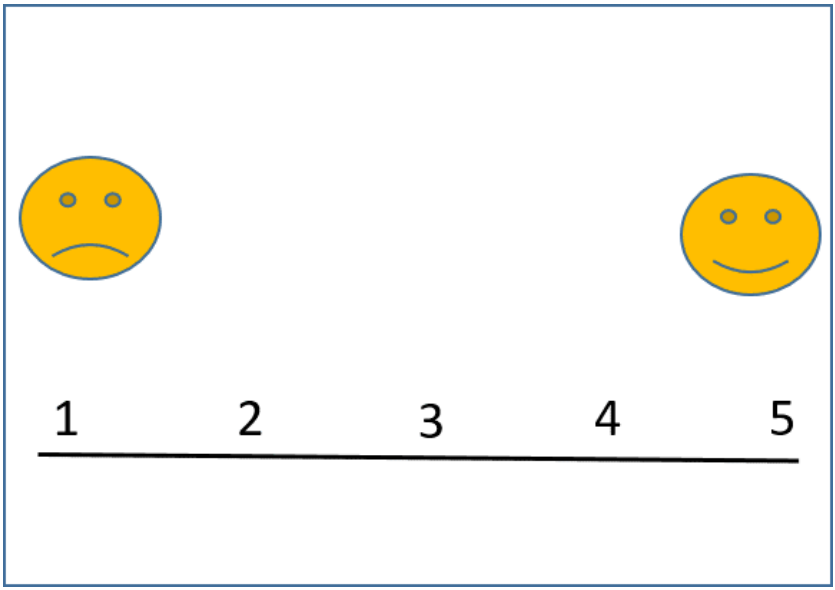

Figure 2.5. Using a 5 -point scale to measure happiness

Psychologists use **statistics in quantitative studies** to assess whether groups differences in the dependent variable are meaningful and reliable, or merely due to chance. In experimental research, there is a tremendous amount of control over the variables of interest. This is a powerful approach. However, psychology experiments are often conducted in artificial settings, like labs, and complex phenomena are reduced to simple numbers. It is sometimes difficult to know whether experimental findings would apply in real-world settings. This is one of the reasons why it is important that psychology as a discipline includes multiple types of research methods to build knowledge about the mind and behavior.

QUASI-EXPERIMENTAL DESIGNS

Remember the study that you read about in the beginning of this chapter where Mehl and colleagues looked at gender differences in talkativeness (Mehl et al., 2007)? This study also has an independent variable—gender, and a dependent variable—talkativeness. But, psychologists cannot manipulate gender or use random assignment for their male and female groups, gender is already determined. The study design that Dr. Mehl used is known as a **quasi-experiment**; but, it is similar to a regular experiment. Let's look at another example to distinguish between these two different types of design. Imagine you want to know who is a better professor: Dr. Smith or Dr. Khan? To judge their ability, you're going to look at their students' final grades. Letter grades can easily be turned into numbers. Here, the independent variable is the professor (Dr. Smith vs. Dr. Khan) and the dependent variable is the students' grades. In an experimental design, you would randomly assign students to one of the two professors and then compare the students' final grades. However, in real life, researchers can't randomly force students to take one professor's class over another. Instead, the researchers would just have to study the classes as they stand (quasi-experimental design). Students' choice of professor may seem random, but it's most likely not. Students often hear things about professors that affect their enrollment. Maybe Dr. Khan has a reputation for being kind and so more students want to take her course. But, because some schools allow Honors students (who typically have better grades than other students) to register first, Dr. Khan's class could have more honors students than Dr. Smith's and this might affect the average grades in the class. So, even though a quasi-experimental design is similar to an experimental design (i.e., it has independent and dependent variables), because there's no random assignment you cannot reasonably draw the same conclusions that you would with an experimental design. However, quasi-

experiments are still very useful in providing information about different groups of people, based on various aspects of their identity, or their demographics.

CORRELATIONAL DESIGNS

In correlational research, scientists do not intervene and measure changes in behavior. Instead, they passively observe and measure variables and identify patterns of relationships between them. So, they can make predictions about relationships between two variables, but they **cannot** make conclusions about **cause and effect**. Dr. Dunn's team first conducted a correlational study with a different set of participants than those in her experiment. She asked participants in the correlational study how much of their income they donated to charity in the past month; she later asked them how happy they were (using the 5-point scale). Then they looked at the relationship between the two variables. Dunn found that the more donations people made to charity; the happier they reported feeling. We call this a correlation and we measure the strength of the relationship using statistics. To find out how well two variables relate to each other, we create a scatterplot (see Figures 2.6 and 2.7). In the scatterplot, each dot represents a data point. In Figure 2.6, each point represents a participant in the study. Importantly, each dot provides us with two pieces of information—in this case, donations in the past month (x-axis) and how happy the person felt (y- axis).

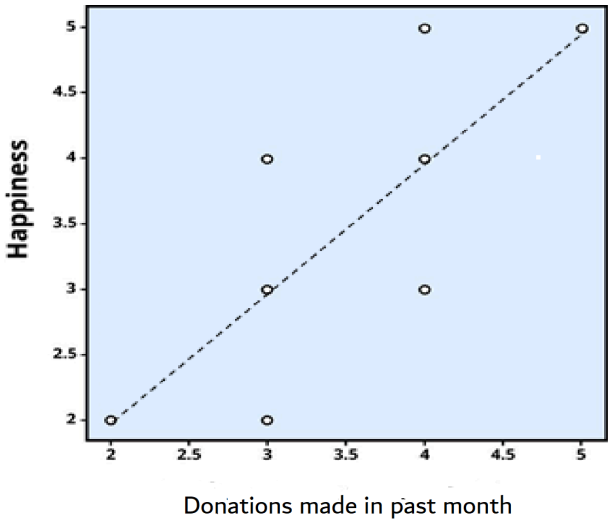

Figure 2.6. Scatter plot of happiness and donation ratings for each participant based on Dunn et al. (2008). Correlation coefficient (r) =+ 0.81.

We have added a best fit (dotted) line through the data in Figure 2.6, the line represents the average relationship between the two variables. It has equal numbers of points above and below it. The relationship between two variables can also be summarized statistically using the **correlation coefficient** (abbreviated as *r*). The sign of *r* (positive or negative) indicates the direction of **correlation**. In Figure 2.6, the line slopes upwards, therefore, *r* is a positive number (+0.81) indicating that the data show a **positive correlation**. In other words, the variables move in the same direction – as one goes up so does the other. Low donations correlate with low happiness, and high donations with high happiness ratings.

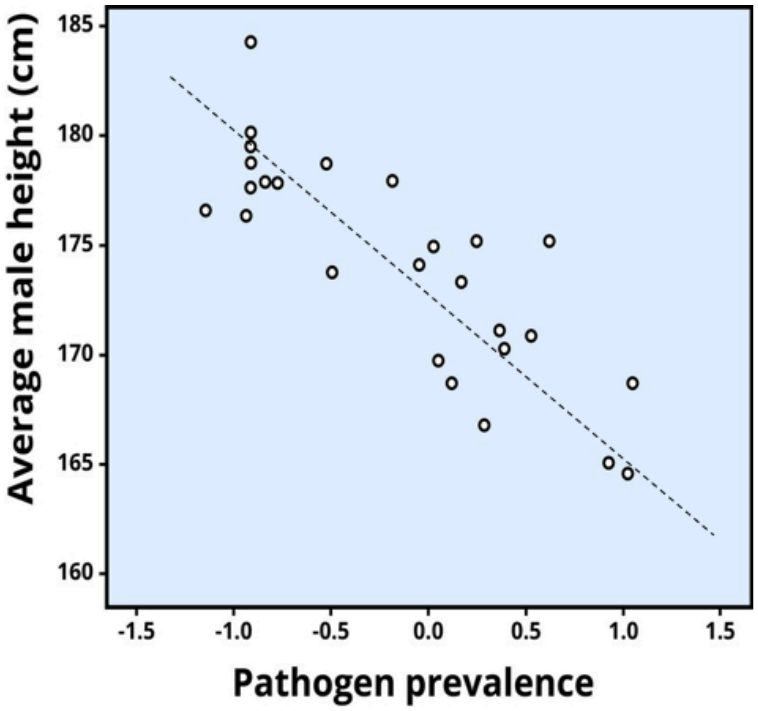

Figure 2.7. Scatter plot of average male height and pathogen prevalence for different countries. Correlation coefficient (r) =- 0.83.

Figure 2.7 shows the data from a different research study. We see the relationship between average height of men in different countries on the y-axis and the prevalence of a pathogen (something that causes disease) in each country on the x-axis. Each dot represents a different country. The line of best fit slopes downwards because there is a **negative correlation** between the two variables. We can see that countries with taller men have lower incidence of the pathogen and vice versa. In other words, as one variable (height) goes up, the other one (pathogen prevalence) goes down. In Figure 2.7, the correlation coefficient (r = -0.83) is a negative number.

The **strength** of a correlation tells us about the closeness of

the relationship between the two variables. If all the points sat on the best fit lines, this would be the very strong relationship and *r* would be 1 or -1 (depending on the direction of the association). The higher the **absolute** value of r, the stronger the relationship. Absolute value means that we ignore the +/- sign. Perhaps you can see that the dots are a little closer to the line in Figure 2.7 compared to Figure 2.6? This is because the absolute value of *r* is slightly larger (a stronger correlation) for Figure 2.7 (-0.83) compared to Figure 2.6 (+0.81). The relationship between variables can vary from strong to weak. Sometimes, *r* is so low (close to zero), that we say there is no correlation, or the variables are uncorrelated, and the best fit line would be horizontal or flat.

Correlation is NOT causation

Correlational studies like the ones we see in Figures 2.6 and 2.7are very useful for making predictions about the relationships between two variables. They are usually cheaper and less time consuming to run than experiments. However, correlational studies cannot tell us whether one variable causes the other. In the case of Dr. Dunn's study on charitable giving and happiness, she ran the correlational study first and found a significant correlation between the two variables. However, she could not conclude that giving to charity caused happiness. It is possible that a third variable (that she did not measure) such as wealth, might be linked to both happiness and generosity. So, correlation is NOT the same as causation. However, the results of the correlational study, gave Dr. Dunn the motivation to conduct a follow-up experiment to explore cause and effect. She was able to randomly assign participants to give away or keep the money she gave them. Only then was she able to draw the conclusion that donating to charity made people happier.

RESEARCH PARTICIPANTS

As we mentioned at the beginning of this chapter, psychologists using the scientific approach have to design their studies in a very systematic way. We have seen some evidence of this already in our discussion of experimental design. In addition, researchers must determine the number of participants they need in each study, as well as their characteristics (e.g., age, gender, socioeconomic status, etc.). The characteristics of the research participants should be described in detail. The research question often determines who the researchers recruit for a particular study. For example, if researchers are investigating causes of depression in the elderly in New York City, then it would not be prudent to recruit college students, or people who live outside of New York City. However, it is not practical to test every elderly person in the area. Instead, psychologists try to recruit a **sample** (a subset) of elderly people. Ideally, the sample should be random and large enough to be representative of the population of New York City. This means that the percentages of specific characteristics in the sample (sex, ethnicity, socioeconomic level, and any other characteristics that might affect the results)are close to those percentages in the **population** of elderly people in NYC.

OPERATIONAL DEFINITIONS

Researchers must also decide on very precise definitions of how they will measure the variables. We call these **operational definitions**—because they determine how the study will be run (operated). When a study is published, these methods should be described in detail—just like a cooking recipe, so that another researcher should be able to duplicate the study exactly. For example, *if you wanted to measure happiness – what measure would you use?* We know that Dunn and colleagues used a rating scale, like

the one shown in Figure 2.5. Notice that she asked the participants to rate how they felt "on this day", rather than "right now" or "during this week, month or year". She also had to operationally define the independent variable – by deciding how much money to give each person ($20), how long participants had to spend the money (1 day), and what they could spend the money on (spend on yourself, give to a charity). There are many different ways to measure psychological variables, but remember that scientific studies require that we quantify (use numbers to measure) the data in some way. Each answer needs to be turned into a number for analysis. This allows psychologists to display data in graphs and use statistics to draw meaningful conclusions. Let's take a look at some of the different ways that psychologists gather data from participants.

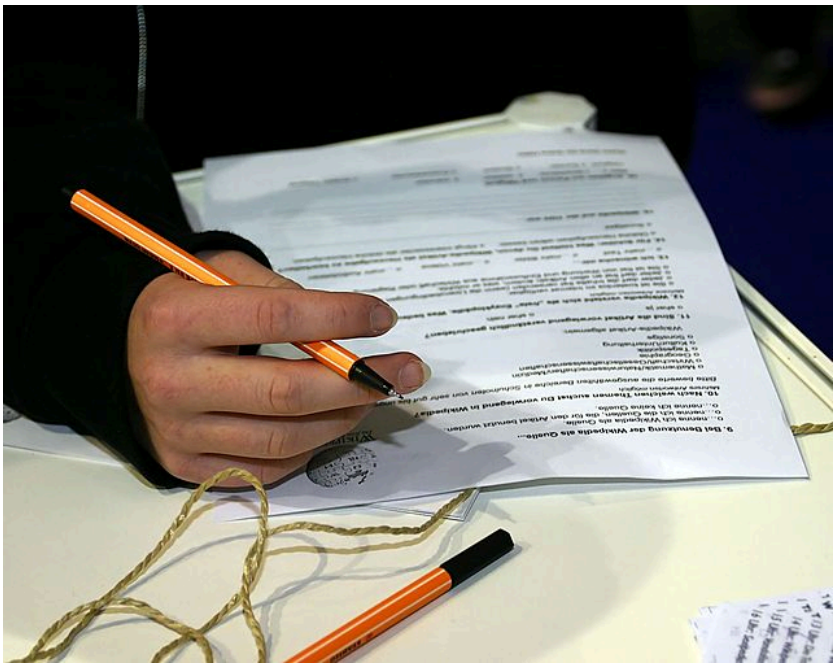

FIGURE 2.8. Research studies often use surveys to ask participants questions about themselves

DATA COLLECTION METHODS

Surveys are among the most common way to collect psychological data. Participants complete questionnaires and the scores from each question are aggregated (summed together). Surveys can be administered in person or sent by mail/email; they can also be administered online. Alternatively, psychologists may collect other kinds of data, such as counting the number of words a person speaks over the course of a few days (Mehl et al., 2007), or measuring how quickly they walk, or whom they sit next to in the cafeteria. Psychologists might measure participants' brain responses to various sounds or pictures, or how quickly they

respond to them by pressing a button. Throughout this book, you will read about many of the different measures that psychologists use to draw conclusions about the mind and behavior.

CRITICISMS OF QUANTITATIVE PSYCHOLOGICAL RESEARCH

Although most quantitative studies seek to generate generalizable findings, the vast majority of study participants are not representative of the general population of the USA, much less the world. Most psychology research has been conducted with participants from WEIRD (White, Educated, from Rich, Industrialized, Democratic) cultures, and so are not representative of the global majority. A very large percentage of studies use college students as research participants, because they are a convenient sample (Sears, 1986; Arnett, 2008). Relying on such a limited pool of research participants can be problematic, college students tend to be younger, more educated, more liberal, and less diverse than the general population.

Scientific research is often upheld as being highly objective. Indeed, it is very systematic in the way data are measured and analyses conducted. As we have already mentioned, **double-blind** studies can be used to improve objectivity by reducing **participant** and **experimenter bias** (the influence the participants and researcher has on the study). However, in general, much less attention has been paid to the issue of researcher bias in terms of how our lived experiences affect the research we conduct. For example, if a research team is conducting a quasi-experiment on how intelligence varies across race and/or gender, they might require that participants come into their lab to be tested. From the very beginning, an experimenters' behavior toward the participants (welcoming, smiling, friendly) might vary depending on the participant's characteristics (such as gender, race, socioeconomic

status) which in turn could affect how the participants perform on the test. As you will read in the upcoming chapter on Intelligence, in the 1930s, Black researchers like Horace Mann Bond and George Herman Canady found that White psychologists often took time to develop rapport with White, but not Black children, when conducting psychological testing. They also found that Black children scored higher on intelligence tests if they were tested by a Black, rather than a White, examiner. Among more recent studies, not all have found that the examiner's race affects performance on psychological tests (Sattler & Gwynne, 1982). However, a significant number of studies have found that when the participants were feeling stressed or threatened by the test, they performed better when tested by someone of the same race/ethnicity as themselves (Moore & Retish, 1974; Nagra et al., 2007; Thames et al., 2013).

Similarly, a researcher's own personal identity (including but not limited to age, race, gender, sexual orientation, socioeconomic status, nationality and language, health history, and dis/ability) and lived experiences are likely to influence the kinds of research questions they ask, the measures they use, and how they interpret their data. For example, increased diversity among psychologists over time has resulted in a greater focus on social justice issues in the field in general. However, the field of psychology continues to be dominated by White psychologists. In 2021, about 80% of psychologists and 87% of neuropsychologists were White (American Psychological Association, 2022; Klipfel et al., 2023). Regardless of race, it is of paramount importance for all psychologists to consider whether the measures they use are culturally appropriate for all research participants. Ethnocentric behaviors among psychologists can negatively impact the people they are meant to serve. Recently, a group of Black psychologists have drawn attention to the cultural inappropriateness of an item within the Boston Naming Test—a commonly used test for assessing language abilities, especially among people who may have brain damage. In the test, people are asked to name what

they see when shown various simple black and white drawings. One of the 60 pictures is a drawing of a noose, which Black and Brown clients find highly distressing because of its strong association with hatred and violence toward African Americans. This distress can affect performance on the naming task and other subsequent tests. Professor Desiree Byrd and colleagues have shone a spotlight on the fact that this picture has been included in the test for over 40 years, but the field of psychology has failed to recognize the violence caused to people of color by the inclusion of this image. Byrd and colleagues have called for psychologists to stop administering the Boston Naming Test until the offensive picture has been removed (Byrd et al., 2021). You will see a greater discussion of racism in psychology in the upcoming section on Intelligence.

REPLICATION CRISIS

Successful **replication**s of published research studies make psychologists (and scientists in general) more apt to adopt those findings, but repeated failures tend to cast doubt on the legitimacy of the original article and lead scientists in different directions. For example, we might be eager to take a new drug if a study shows it improves peoples' memories. But if other scientists cannot replicate the results, we would question the original study's claims.

In recent years, there has been increasing concern about a *"replication crisis"* that has affected a number of scientific fields, including psychology. Some of the most well-known scientists have produced research that has failed to be replicated by others (as discussed in Shrout & Rodgers, 2018). In some cases, this might be explained in part by a lack of transparency in how the research was carried out; there may be important details missing from the original article that make the "recipe" hard to follow precisely, or participants may not be adequately described. Also, sample sizes

(numbers of participants) in the original studies were often too small and not representative of the general population. There is also a publication bias in psychology, where journals typically only publish significant results that confirm hypotheses, whereas studies that fail to show significant differences between conditions are not made public. Finally, although psychologists are bound by a code of ethics, a very small number of researchers **falsified** their results—this is the scientific equivalent of fake news. When false reports are discovered then studies are often rescinded or retracted from the scientific journals. Unfortunately, fake scientific news can sometimes have very disturbing and widespread consequences. One example of this comes from false data about side effects of vaccinations in the field of medicine. In 1998, a team of doctors in the UK led by Andrew Wakefield published a study on 12 children in a highly influential journal called *The Lancet*, claiming that autism was linked to the measles, mumps, and rubella (MMR) vaccine. Subsequently, the article has been retracted, and the main author, Wakefield, has been discredited by indisputable evidence that he falsified the data. He also failed to get proper consent from the participants and their parents. Since then, many, many other studies have overwhelmingly shown that vaccination is NOT linked to autism. However, Wakefield's article got a lot of media attention, and parents became reluctant to vaccinate their children, despite the fact that measles can result in blindness, encephalitis (swelling of the brain), pneumonia, and even death. This legacy continues today; increasing numbers of children are unvaccinated and the prevalence of measles is rising dramatically in Europe and the USA (Godlee et al. 2011; Offit & Coffin, 2003; Taylor et al., 2014).

2.2. Qualitative Research

There are pros and cons to all of the various types of research design in Psychology. We have already seen that experiments allow us to study cause and effect. However, correlational and quasi-experimental research also have value, because they allow us to study topics we cannot experimentally manipulate (these might include whether people have a large or small income; whether they identify as male, female, non-binary, trans, etc.). However, quantitative methods often reduce complex psychological phenomena to simple processes studied in artificial settings and so it is sometimes difficult to reconcile this with how people behave, think, and feel in real world settings. Qualitative research offers the potential for researchers to contextualize their findings within a broader social, cultural, and historical context, and is gaining more popularity as an alternative methodology in the field of psychology. While quantitative research focuses on generating specific answers to research questions using numbers and statistics, qualitative research aims to provide detailed descriptions of human behavior that help to understand the personal experiences of the research participants.

Figure 2.9. Qualitative research seeks to understand the lived experiences of

people, such as successful Black women in corporate America.

Qualitative research is not hypothesis driven, and questions tend to be more general, with the goal of understanding what people feel or understand about certain situations in their lives. For example, counseling psychologist, Dr. Aisha Holder and colleagues wanted to understand how successful Black women in senior management positions in corporate America experienced and coped with racism in the work place. They wanted to provide rich, detailed, and contextualized descriptions and so they conducted a qualitative research study where they asked ten participants about their lived experiences (Holder et al. 2015).

METHODS OF DATA COLLECTION

The ways that psychological data is collected varies a lot across studies, and methods used in qualitative studies sometimes overlap with those used in quantitative studies (interviews, archival data, and surveys). However, the data generated in qualitative studies are more complex, because they do not need to be represented by numerical values. Thus, questions are typically more open-ended. For example, in her qualitative research study, psychologist, Luo Lu from Taiwan asked Chinese students to write an essay in response to the prompt: *What is happiness?* (Lu, 2008). As you can imagine, this generates a lot more data than simply asking people to rate how happy they feel on a numerical scale (Dunn et al., 2008). Qualitative studies are also inherently flexible in their design, researchers frequently adapt their methods and questions in response to new findings as they emerge in the study.

INTERVIEWS

One of the most common ways to gather data in qualitative

psychological research is to conduct **interviews**. Interviews vary in their format; they can be **unstructured, structured,** or **semi-structured**. Unstructured interviews consist of asking a few, very broad questions and letting the participants talk as much as possible. In structured interviews, the researchers have a list of questions that they ask in a systematic fashion. Most interviews in qualitative research are semi-structured, which is somewhere in between the two other types of interview. The researcher has a list of scripted questions, but they can ask other follow-up questions about the topics that participants bring up. Dr. Holder and her colleagues used semi-structured interviews in their study of successful business women and their experiences of racism (Holder et al., 2015).

Interviews in qualitative psychology studies are often lengthy and generate a lot of information, researchers frequently record the interviews (with the participants' permission) and/or make detailed notes on what everyone said. Recordings are later transcribed into text. The researcher might also observe non-verbal cues and behaviors during the interviews, such as changes in tone of voice or body language. As you can imagine, it is very time consuming to collect and analyze interview data and so the total number of participants in a qualitative study using interviews is typically small. Dr. Holder's research study had 10 participants, but in some studies, there are even fewer. Interviews can be conducted in private with a single individual; but focus groups (where small groups of people are interviewed together) are also popular.

Figure 2.10. A focus group.

In focus groups, participants interact with each other and share ideas, which can often make them feel more at ease and open. This can help to generate richer, more comprehensive responses. **Consumer psychologists** often use focus groups to understand people's reactions to new products and how they might influence their buying behaviors. However, focus groups may be less appropriate for sensitive topics. In Dr. Holder's study about racism in the workplace, she conducted individual interviews to ensure that participants had an increased feeling of privacy and anonymity. Sometimes, researchers conduct both individual interviews and focus groups and compare people's responses in the two settings. As you know (and will learn more about in the modules on Social Psychology), people often behave differently in groups compared to when they are alone. For example, in their ethnographic study on the views of poor and working class Gen-

Xers from various racial backgrounds, Fine and Weis found that White women only reported being victims of domestic violence in individual interviews but not in focus groups (Fine & Weis, 1998).

OBSERVATIONAL DATA COLLECTION

Data about behavior can also be collected by observation—watching and recording how people (or animals) behave in natural settings in their everyday lives—this type of research is often referred to as **field research**. This differentiates between studies where researchers collect data in a research lab. Observations in field settings can occur in one of two ways. During **naturalistic observation**, the researchers try to be as unobtrusive as possible, and collect data without influencing the situation. For example, researchers might be interested in learning more about the dynamics of bullying and peer relationships in young children. They want to observe how young children play together in a pre-school setting, but they don't want to change the dynamics of the group by sitting in the same room as the children. The researchers might set up an adjoining room with a two-way mirror, so that they can see and record the children's behavior, but the children cannot see or hear them. Alternatively, a researcher may use **participant observation**, where they become a participant in the culture or context they are observing. For example, in 1973, psychologist, David Rosenhan conducted a qualitative research study where he and a few others went to psychiatric institutions and pretended to be mentally ill. They told the doctors that they had been hearing voices. After they were admitted as patients, they then recorded their experiences of the ward, patients, and staff in their notebooks (Rosenhan, 1973). Once admitted they told staff that they no longer heard voices, and behaved normally. However, their behavior was often interpreted by staff in the context of their diagnosis of schizophrenia.

OTHER FORMS OF QUALITATIVE DATA COLLECTION

Psychological data can also be collected using **surveys**. In qualitative research, the questions are open-ended, for example, "*What makes you happy?* " allowing for extensive, individualized responses. **Surveys** are also used in quantitative research, but they consist of closed-ended questions that can be coded numerically (like the rating scale that Dr. Dunn's team used to ask people "How happy do you feel?"). Psychologists may also use **archival data** in both qualitative or quantitative studies. Archival data are records of people's lives that were collected by other entities, for other purposes. They can include public health records, census and other population data, records of births, deaths, and marriages, medical and police records, real estate transactions, newspaper stories, films, videos, art work, music, and personal items, such as letters, diaries, emails, and social media. For example, Sanger and Veach (2008) conducted an archival study where they analyzed suicide notes to try and understand more about social interactions among people who commit suicide.

DATA ANALYSIS IN QUALITATIVE STUDIES

As you have already seen, the ways that data in qualitative and quantitative studies are collected is often somewhat similar (surveys, interviews, archives, observation). However, it is *the way that data are analyzed and reported t*hat really sets the two approaches apart. For example, researchers using qualitative techniques are acutely aware of their influence on the research process and often disclose positionality and identity as part of their research article. There are many different ways to analyze qualitative data. Researchers frequently use a coding system to look for recurring patterns and often provide rich and detailed descriptions of their participants' behavior in real-world contexts.

This might include quotes from participants, as well as descriptions of their feelings and actions. This is often referred to as "**thick description**" (Geertz, 1973). Some qualitative studies use the **phenomenological approach,** which aims to richly portray their participants lived experiences using thick descriptions, moreover, it does not try to explain why people behave the way they do. Dr. Holder's study of Black women managers employed a phenomenological approach where they used thick descriptions to convey their participants' experiences of racism in the workplace and the many different ways that they coped with it.

GROUNDED THEORY

Another approach in qualitative research analysis is called **grounded theory**; researchers collect data and then develop a theory or an interpretation that is based on, or "grounded in" those data (Glaser & Strauss, 1967). First, the researchers identify ideas that are repeated throughout the data. They then organize these ideas into a smaller number of broader themes. Finally, they interpret the data by writing a theoretical narrative of these themes. This narrative focuses on the subjective experience of the participants and is usually supported by quotes. For example, Yosso and colleagues interviewed Latinx students attending three elite predominantly White academic higher education institutions about their academic aspirations, perceptions of campus climate, experiences of racism and coping mechanisms. They found that Latinx students experienced a good deal of racism on campus, which organized into three main types: racial jokes, interpersonal microaggressions (being excluded or insulted by White students), and institutional microaggressions (lack of representation among faculty and staff and no institutional support for Latinx students). Another theme was that Latinx students felt empowered to deal with the negative environment by banding together and supporting each other. The researchers interpreted their data within the

context of critical race theory, which posits that racism is embedded systemically within society and institutions, like universities (Yosso et al., 2009).

2.3. Mixed Methods and Triangulation

Given their differences, it may come as no surprise that quantitative and qualitative research in psychology and related fields do not coexist in complete harmony. Some quantitative researchers criticize qualitative methods on the grounds that they lack objectivity, are difficult to evaluate in terms of reliability and validity, and do not allow generalization to people or situations other than those actually studied. At the same time, some qualitative researchers criticize quantitative methods on the grounds that they overlook the richness of human behavior and experiences and instead answer simple questions about easily quantifiable variables.

Some researchers agree that quantitative and qualitative approaches can and should be combined into what has come to be called **mixed-methods** research (Todd et al., 2004). One approach is to conduct a qualitative study first and then use the results for hypothesis generation and testing in a follow-up quantitative study. A second approach for combining quantitative and qualitative research is called **triangulation**, where researchers use both quantitative and qualitative methods simultaneously to study the same general questions and to compare the results. If the results of the quantitative and qualitative methods come to the same general conclusion, they reinforce and enrich each other. Using qualitative

research can often help clarify quantitative results. For example, Trenor and colleagues investigated the university experiences of female engineering students. In the first phase, female engineering students completed a survey, where they assessed their perceptions of the University, including their sense of belonging. Statistically, the various ethnic groups showed no differences in their ratings of sense of belonging. One might conclude from these data that ethnicity does not have anything to do with sense of belonging to a college or university. However, in the second phase of their study, the authors also conducted interviews with the students, in which many minority students reported how diversity at the university enhanced their sense of belonging (Trenor et al., 2008). Without the qualitative component, they might have drawn the wrong conclusion about the quantitative results.

There are two other important types of research study that we want to discuss in this section, they are Case Studies and Participatory Action Research. Both often use qualitative and quantitative data.

CASE STUDIES

Case studies are detailed, often lengthy studies of a very small number of individuals (sometimes as few as only one). Case studies have been used throughout the history of psychology to better understand the mind and behavior. For example, Sigmund Freud developed psychoanalytical theory based on case studies of his patients. Case studies are frequently used in clinical psychology to better understand psychological and neurological disorders. One downside of case studies is that we do not know how generalizable the findings are to the general population, because only small numbers of participants are studied.

PARTICIPATORY ACTION RESEARCH

Participatory Action Research (PAR) is a research framework that aims to bring about social change. The research is guided by the needs and desires of community members who are grappling with specific social justice issues in their community. Community members work in collaboration with academic researchers as partners in a research team. Unlike most other kinds of research study in psychology, community members are active participants in all stages of PAR projects. PAR projects not only provide information about a community issue, but they also result in some kind of action designed to bring about improvements (Carleton College, n.d.; Fine & Torre, 2019). An example of PAR includes the Morris Justice Project. Maria Elena Torre and Brett Stoudt, two scholar-activists from City University of New York (CUNY) collaborated with community members in the Morris Avenue area of the Bronx to document the experiences and impact of aggressive policing on the community. This area had particularly high rates of "stop and frisk" of young Black men. Together they formulated a research plan, and community members surveyed over 1000 people in the neighborhood. They disseminated this information in multiple formats, including a report, buttons, and t-shirts, and at various community events. The project was successful in raising awareness of the problem among multiple groups, including the NYPD and people who lived in other neighborhoods. It was also used to support legal action against police officers, and recent legislation (Safer Communities Act, 2022) to protect people of color from discriminatory police practices.

2.4. Ethics

INSTITUTIONAL REVIEW BOARD (IRB)

Today, scientists agree that good research should be ethical in nature and guided by a basic respect for human dignity and safety, and the humane treatment of animals. However historically, there have been multiple studies that were highly unethical, both in medicine and psychology. In later chapters, you will read about the Milgram Study and the Stanford Prison study, two so-called "classical" studies in social psychology that caused considerable distress to the research participants. Modern-day protections for human participants grew from the actions of Peter Buxton, a journalist who exposed an exceptionally long-lasting, highly unethical medical study, called the Tuskegee study in the national press (see below for more details). As a consequence, the National Research Act was passed in 1974, which imposed strict ethical guidelines for research on humans. Modern-day researchers must demonstrate that the research they propose is ethically sound before it can begin. At every research institution, each study that involves human participants is reviewed by a committee of people. This committee is called the **Institutional Review Board** (IRB). The IRB typically consists of members of the institution's administration and scientists, in addition to community members (Figure 2.11). The IRB reviews study proposals to ensure that they do not cause

harm or undue distress to participants. Research studies must have IRB approval before they can begin.

Figure 2.11. An institution's IRB meets regularly to review experimental proposals that involve human participants.

The IRB has four main requirements for a research study to be considered ethical:

- *Informed consent*. A participant must provide informed consent before a study begins. This means that participants are *informed* about the purpose of the research, and what will happen to them during the study. They then are free to choose whether they *consent* to participate. They will often sign a consent form to indicate their agreement and to confirm that they have been informed about the study.
- *Privacy*. Researchers should not make observations of people in private places, such as restrooms or their homes, without their knowledge and consent.

Researchers should not seek confidential information from others, such as school authorities, without consent of the participant or his or her guardian. Information that researchers learn about individual participants must not be made public without the consent of the individual.

- *Risks and Benefits:* People who participate in psychological studies should be exposed to risk only if they fully understand the risks and only if the potential benefits clearly outweigh those risks.
- *Deception*. Some researchers need to deceive participants in order to hide the true nature of the study. Deception is typically only permitted (a) when the benefits of the study outweigh the risks, (b) participants are not reasonably expected to be harmed, (c) the research question cannot be answered without the use of deception, and (d) participants are informed about the deception as soon as possible, usually through debriefing.

ETHICS AND THE TUSKEGEE SYPHILIS STUDY

Unfortunately, the strict ethical guidelines that exist for research today were not always applied in the past. The 40-year-long Tuskegee Study is a particularly egregious example. In 1932, extremely poor Black men from Tuskegee, a rural area in Alabama, were recruited to participate in a study looking at how untreated syphilis affected Black men. The study was conducted by the U.S. Public Health Service (Figure 2.12). 600 men agreed to participate in the study, which they were told was to designed to treat "bad blood". Bad blood was a colloquial name for any number of mostly minor ailments. More than half of the men tested positive for syphilis. However, none of the participants were informed whether that they had the disease, and were never informed about the true nature of the study. Moreover, the incentives they were offered

(free medical care, meals, and burial insurance), can be considered coercive; the participants were on the poverty line and so it was probably hard for them to turn these benefits down.

Figure 2.12. Tuskegee study participant being examined by a study doctor.

Although there was no effective treatment for syphilis when the study began, by 1947 penicillin was being widely prescribed to cure the disease. Despite this, none of the participants were ever given penicillin. The study continued for 40 years and many of the participants unknowingly spread syphilis to their partners (and subsequently their children). Most eventually died as a consequence of having the untreated disease. This study was discontinued in 1972 when the study was discovered by the national press. In 1974, the National Research Act was passed and provided strict guidelines for research involving human participants. In 1973, survivors of the study sued the US government, resulting in a $10 million, out-of-court settlement in

1974. The settlement was paid to the families of the study participants, and any surviving participants. It was not until 1997 that President Bill Clinton issued a formal Presidential Apology for the study (Center for Disease Control, 2022).

RESEARCH INVOLVING ANIMAL SUBJECTS

Some psychologists conduct research involving animals (Figure 2.13). Because many basic processes in many animals are sufficiently similar to those in humans, these animals are acceptable substitutes for research that would be considered unethical in human participants. However, the humane and ethical treatment of animal research subjects is a critical aspect of this type of research. Researchers must design their studies to minimize any pain or distress experienced by animal subjects.

Figure 2.13. Rats often serve as the subjects of animal research in psychology studies.

Whereas IRBs review research proposals that involve human participants, animal experimental proposals are reviewed by an Institutional Animal Care and Use Committee (IACUC). An IACUC

consists of institutional administrators, scientists, veterinarians, and community members. This committee is charged with ensuring the humane treatment of animal research subjects. It also conducts regular inspections of all animal facilities to ensure that the research protocols are being followed. No animal research project can proceed without the committee's approval.

Chapter 2. Conclusions

In conclusion, traditional research methods have provided valuable insights into human behavior, however, it is important to recognize the limitations and biases inherent in these studies. The overrepresentation of college students and individuals from Western, educated, industrialized, rich, and democratic cultures, calls into question the generalizability of the research findings to the broader population. The exploration of non-traditional research methods, such as qualitative and mixed method approaches, participatory action research, and the use of diverse samples, opens up new avenues for understanding the complexities of human behavior across different cultural contexts.

Chapter 2. Key Terms

archival research
method of research using past records or data sets to answer various research questions, or to search for interesting patterns or relationships

cause and effect
changes in one variable cause the changes in the other variable; can be determined only through an experimental research design

case study
in depth research study focusing on one or a few people

consumer psychologist
studies how consumers behave, think, and feel, and how they make decisions about buying and using products and services.

control group
serves as a basis for comparison and controls for chance factors that might influence the results of the study—by holding such factors constant across groups so that the experimental manipulation is the only difference between groups

correlation
relationship between two or more variables; when two variables are correlated, one variable changes as the other does

correlation coefficient
number from -1 to +1, indicating the strength and direction of the relationship between variables, and usually represented by *r*

debriefing

when an experiment involved deception, participants are told complete and truthful information about the experiment at its conclusion

deception
purposely misleading experiment participants in order to maintain the integrity of the experiment

dependent variable
variable that the researcher measures to see how much effect the independent variable had

double-blind study
experiment in which both the researchers and the participants are unaware of which participants are assigned to which groups

empirical
grounded in objective, tangible evidence that can be observed time and time again, regardless of who is observing

experimental group
group designed to answer the research question; experimental manipulation is the only difference between the experimental and control groups, so any differences between the two are due to experimental manipulation rather than chance

experimenter bias
researcher expectations skew the results of the study

falsified
made up false information

field research
data collected in a natural setting outside of a laboratory or controlled environment

generalizable
inferring that the results for a sample apply to the larger population

grounded theory
the systematic analysis of data to develop a theory about a particular phenomenon or social process.

hypothesis

(plural: hypotheses) tentative and testable prediction about the relationship between two or more variables

independent variable

variable that is influenced or controlled by the experimenter

informed consent

process of informing a research participant about what to expect during an experiment, any risks involved, and the implications of the research, and then obtaining the person's agreement to participate

Institutional Animal Care and Use Committee (IACUC)

committee of administrators, scientists, veterinarians, and community members that reviews proposals for research involving non-human animals

Institutional Review Board (IRB)

committee of administrators, scientists, and community members that reviews proposals for research involving human participants

interviews

a method of gathering information about an individual's thoughts, feelings, or behaviors through a structured or unstructured conversation with a trained interviewer

naturalistic observation

observation of behavior in its natural setting, without any intervention or manipulation by the researcher.

negative correlation

two variables change in different directions, with one becoming larger as the other becomes smaller

operational definition

precise description of what actions and operations will be used to measure the dependent variables and manipulate the independent variables

participants

people who take part in psychological research studies and from whom data is collected

participant observation

the researcher immerses themselves in a particular social setting and observes and records the behavior of the individuals within that setting.

peer-reviewed
article read by several other scientists (usually anonymously) with expertise in the subject matter, who provide feedback regarding the quality of the manuscript and determine whether it is accepted for publication

phenomenological method
qualitative research method that involves describing an individual's subjective experiences and perceptions of a particular phenomenon or event

placebo effect
people's expectations or beliefs influencing or determining their experience in a given situation

population
overall group of individuals that the researchers are interested in

positive correlation
two variables change in the same direction, both becoming either larger or smaller

qualitative research
seeks to understand a specific phenomenon through in-depth exploration and analysis of non-numerical data, often using rich descriptions

quantitative research
uses numerical data and statistical analysis to study a specific phenomenon. This type of research is often used to test hypotheses

random assignment
method of experimental group assignment in which all participants have an equal chance of being assigned to either group

replication
repeating an experiment using different samples to determine the research's reliability

sample
subset of individuals selected from the larger population

semi-structured interview
interviewer has a set of pre-determined questions, but is also able to follow up with additional questions based on the participant's responses

single-blind study
experiment in which the researcher knows which participants are in the experimental group and which are in the control group

statistics
mathematical analysis determines how likely any difference between experimental groups is reliable/meaningful or due to chance

structured interview
interviewer has a set of pre-determined questions that are asked in a specific order

survey
list of questions to be answered by research participants—allowing researchers to collect data from a large number of people

theory
well-developed set of ideas that propose an explanation for observed phenomena

thick description
provides a detailed and comprehensive description of the social and cultural context in which a particular behavior or phenomenon occurs.

triangulation
a research method using multiple methods or sources of data to confirm or support a particular finding or theory.

unstructured interview
interviewer has a general topic or set of questions, but allows the conversation to flow freely, without a set format or specific questions

Chapter 2. Personal Application Questions

- Consider a situation where you are trying to make a personal decision based on information from a research study, such as choosing a particular therapy or adopting a specific study habit. What are some potential pitfalls or biases that you should be aware of when interpreting the results of a single study? How might you seek additional information or perspectives to make a more informed decision?
- As a college student, you are no doubt concerned about the grades that you earn while completing your coursework. If you wanted to know how overall grade point average (GPA) is related to success in life after college, how would you choose to approach this question and what kind of resources would you need to conduct this research?
- Take a few minutes to think about all of the advancements that our society has achieved as a function of research involving animal subjects. How have you, a friend, or a family member benefited directly from this kind of research?
- Imagine you are conducting a Participatory Action

Research (PAR) project in your community to address a social justice issue. Describe the issue and the steps you would take to ensure effective collaboration with community members and academic researchers, and how you would implement the findings to bring about positive change. Refer to the Morris Justice Project as an example.

- Reflect on your own experiences with surveys or questionnaires in your life. Based on these experiences, identify potential sources of bias or limitations in self-report measures and discuss strategies you would employ to mitigate these issues if you were to design your own research study.
- Consider the influence of personal biases and expectations on the interpretation of research findings that you might read about in the news or on social media. How might your own beliefs, values, or preconceived notions about a topic influence how you interpret research studies? What strategies can you employ to minimize the impact of bias and approach research findings with a more objective mindset?
- Reflect on the ethical considerations associated with research studies involving human participants. Why is it important to be aware of the ethical guidelines and practices in research, even as a consumer of research findings? How might your awareness of ethical issues influence your support for or skepticism towards certain research studies or practices?

Chapter 2. Critical Thinking Questions

- Which of the research methods discussed in this chapter would be best suited to study the impact of diet and exercise on the prevalence of a disease such as diabetes? Why?
- Aside from biomedical research, what other areas of research could greatly benefit from archival research?
- Recently a study was published in the journal, *Nutrition and Cancer*, which established a negative correlation between coffee consumption and breast cancer. Specifically, it was found that women consuming more than 5 cups of coffee a day were less likely to develop breast cancer than women who never consumed coffee (Lowcock et al., 2013). Imagine you see a newspaper story about this research that says, "Coffee Protects Against Cancer." Why is this headline misleading and why would a more accurate headline draw less interest?
- Sometimes, true random sampling can be very difficult to obtain. Many researchers make use of convenience samples as an alternative. For example, one popular convenience sample would involve students enrolled in Introduction to Psychology courses. What are the pros

and cons of using this sampling technique?

- Peer review is an important part of publishing research findings in many scientific disciplines. This process is normally conducted anonymously; in other words, the author of the article being reviewed does not know who is reviewing the article, and the reviewers are unaware of the author's identity. Why would this be an important part of this process?
- Some argue that animal research is inherently flawed in terms of being ethical because unlike human participants, animals do not consent to be involved in research. Do you agree with this perspective? Given that animals do not consent to be involved in research projects, what sorts of extra precautions should be taken to ensure that they receive the most humane treatment possible?
- Suppose you are interested in exploring the relationship between sleep quality and academic performance among college students. How would you design a correlational study to examine this relationship? Identify the variables involved, explain the appropriate statistical analysis, and discuss the potential limitations of correlational research in establishing causality.
- Considering the advantages and disadvantages of naturalistic observation, how would you approach conducting an observational study on children's social interactions in a school setting? Discuss the ethical considerations involved and propose strategies to minimize observer bias and maintain the ecological validity of your study.

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CHAPTER 3. BIOPSYCHOLOGY

This text has been modified by Jill Grose-Fifer from the original text from *Psychology 2e*. OpenStax.org

Have you ever taken a device apart to find out how it works? Many of us have done so, often to attempt a repair or simply to satisfy our curiosity. In this chapter, we will look inside the nervous system to try and explain the biological mechanisms that underlie behavior, thoughts and feelings. In this chapter, you will read about how genetics influence both physiological and psychological traits. You will become familiar with the structure and function of the nervous system. And, finally, you will learn how the nervous system interacts with the endocrine system.

LEARNING OBJECTIVES

- Describe basic characteristics of genes
- Discuss how gene-environment interactions are critical for expression of physical and psychological characteristics
- Identify the basic parts of a neuron
- Describe how neurons communicate with each other
- Explain how drugs act as agonists or antagonists for a given neurotransmitter system
- Describe the difference between the central and peripheral nervous systems
- Explain the difference between the somatic and autonomic nervous systems
- Differentiate between the sympathetic and parasympathetic divisions of the autonomic nervous system
- Explain the functions of the spinal cord
- Describe the hemispheres and lobes of the brain and their basic functions
- Describe the types of techniques available to clinicians and researchers to image the brain or record its activity
- Identify the major glands of the endocrine system
- Identify the hormones secreted by each gland
- Describe each hormone's role in regulating bodily functions

3.1. Human Genetics

Psychologists study genetics in order to better understand the biological factors that contribute to certain behaviors, thoughts, and feelings. Our genes provide a "blue-print" for making the proteins that are the building blocks of our bodies, including our nervous system. Let's take a look at how we inherit our genes and how they can affect our "psychology".

GENETIC VARIATION

After a sperm fertilizes an egg, it forms a zygote. The egg and the sperm each contain 23 chromosomes, and so the zygote has 23 pairs of chromosomes. Therefore, each child inherits half of their genetic information from each biological parent. Chromosomes are long strings of genetic material known as deoxyribonucleic acid (DNA). Each chromosome contains multiple genes, which are formed by specific sequences of DNA (Figure 3.1a). A single gene often has multiple variations, which we refer to as alleles. So, if there was a specific gene that coded for hair color, different alleles of that gene will determine the hair color of an individual. Thus, scientists investigate how our genotypes (or genetic makeup) determine our phenotypes, such as hair color, eye color, skin color, whether we have a cleft chin, or other psychological traits (see Figure 3.1b).

Most traits are controlled by multiple genes (i.e., they are polygenic), but there are a few traits that are controlled by one single gene, such as having wet or dry earwax in humans or the length of fur in a cat (McDonald, 2013). We inherit one "earwax" allele from each parent and there two possible variants of the allele – the dominant form (B) and the recessive form (b). When someone has two copies of the same allele, they are said to be homozygous for that allele (BB or bb). When someone has a mixture of alleles for a given gene, they are said to be heterozygous (Bb). Having wet earwax is a dominant trait (B), so if we have that dominant allele in our genotype we will have wet earwax, regardless of the other allele we inherit (BB or Bb). Having dry earwax is a recessive trait, which means that we will only have dry earwax if we are homozygous for that recessive allele (bb).

(a)

(b)

Figure 3.1. (a) Genotype refers to the genetic makeup of an individual based on the genetic material (DNA) inherited from one's biological parents. (b) Phenotype describes an individual's observable characteristics, such as hair color, skin color, height, and build.

If you AND your partner both have wet earwax, you might be thinking that it is inevitable that your offspring will have wet earwax too. But, actually it's not that simple. It depends on the specific alleles that you and your partner carry. If one of you is homozygous (BB), then all your offspring *will* have wet earwax. It gets a little more complicated, however, if you and your partner are both heterozygous for this gene (Bb). If we use a Punnett square to

look at the four possible combinations of genes (Figure 3.2), we see there is a 75% chance your offspring will have wet earwax and a 25% chance they will have dry ear wax (Figure 3.2). We recognize that earwax is rather a strange trait to talk about, but we did not want to perpetuate the common myth that traits such as eye color, cleft chins, presence of earlobes, and tongue rolling are determined by a single gene. These traits are polygenic and occur on a continuum.

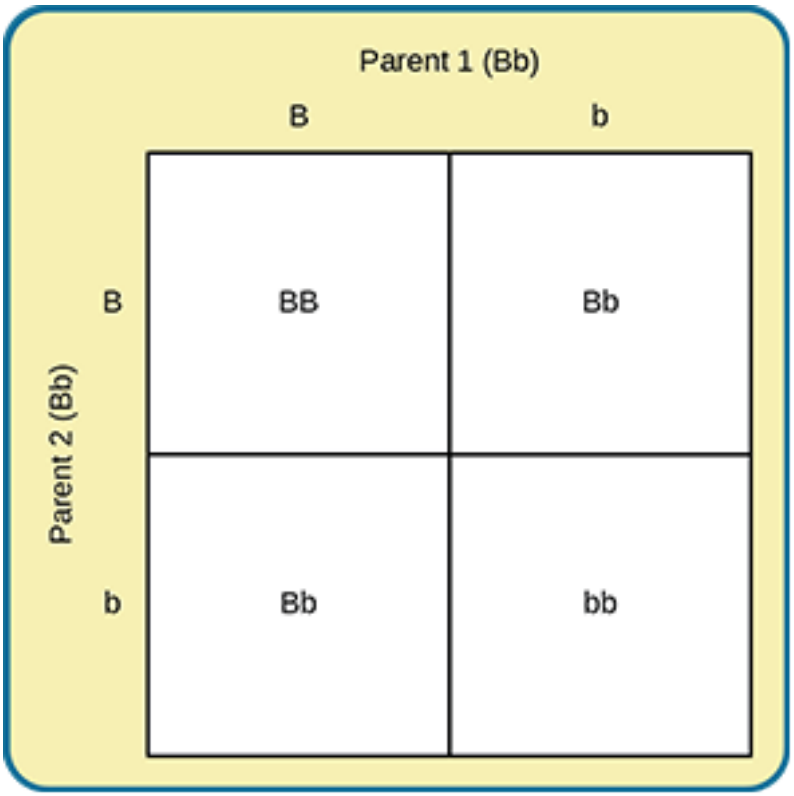


Figure 3.2. A Punnett square is a tool used to predict how genes will interact in the production of offspring. The capital B represents the dominant allele, and the lowercase b represents the recessive allele. In the example of the type of earwax, where B is wet (dominant allele), wherever a pair contains the dominant allele, B, there is a wet ear wax phenotype. You will have a dry earwax phenotype only when there are two copies of the recessive allele, bb.

Some genetic disorders come from inheriting two recessive alleles. For example, phenylketonuria (PKU) is a metabolic condition where individuals lack an enzyme that normally converts harmful amino acids in foods into harmless byproducts. If this condition goes untreated, people with PKU can experience significant deficits in

cognitive function, seizures, and an increased risk of various psychiatric disorders. Because PKU is a recessive trait, each parent must have at least one copy of the recessive allele in order to produce a child with the condition.

Where do genes that contribute to diseases like PKU come from? It has been shown that PKU is due to a gene mutation. This is where the DNA sequencing in a specific gene is slightly altered. Gene mutations can arise spontaneously but once present, it can be passed down to future generations. While many mutations can be harmful or lethal, once in a while, a mutation benefits an individual by giving that person an advantage over those who do not have the mutation. Recent research shows that people who have long lives despite an unhealthy lifestyle, may be benefiting from gene mutations that protect from heart disease and high cholesterol (Williams, 2016).

THEORIES OF EVOLUTION

Theories of evolution aim to explain how genetic variation among a population *changes* over time. Charles Darwin, an English biologist, proposed an influential evolutionary theory called the theory of natural selection of evolution. He suggested that the individuals within a species that were best suited to their particular environments were more likely to survive and reproduce and pass on their genes to future generations. This is often referred to as the survival of the fittest. Let's consider the story of the peppered moth in the UK. In the early 19^{th} century, these moths were predominantly light in color, which provided them with good camouflage from predatory birds. However, during the industrial revolution, the trees in and around large cities became covered in black soot and the light colored moths were virtually eradicated because they were eaten by birds. However, there were also some rarer, dark-colored moths that were now much harder to see when

they landed on the trees. Thus, they were more likely to thrive and mate, and pass their dark coloring genes to their offspring. So, over time, there were more and more dark colored peppered moths. Darwin assumed that each successive generation would evolve to be better adapted for the environment. We certainly see evidence for evolution through natural selection among multiple species. However, scientists now acknowledge that evolution simply means *changes* in genetic variations, and not necessarily improvements. Darwin's studies focused on plants and animals, but modern-day scientists have found genetic linkages to a number of behavioral characteristics, ranging from basic personality traits to sexual orientation to spirituality (for examples, see Mustanski et al., 2005; Comings et al., 2000). Genes are also associated with temperament and a number of psychological disorders, such as depression and schizophrenia.

Darwin's theories on evolution took the world by storm, because it was in direct opposition to the popular religious doctrine of the time. Most believed that people were created by God. However, Darwin believed that humans evolved from other species, Unfortunately, like many other White men of his time, Darwin mistakenly believed that race was genetically determined and Anglo-Europeans were "genetically superior" to people of color. Darwin's theory of natural selection was abused by eugenicists to provide a "scientific rationale" for racism, sexism, and xenophobia. We now know that race is a social, not a biological construct. There is actually more genetic variation within a given racial category than there is between racial categories, so two white people typically share more genetic overlap with a person of color, than they do with each other.

Modern biology acknowledges that most human characteristics are controlled by multiple genes. Each of us represents a unique interaction between our genetic makeup *and* our environment. For example, intelligence is a complex construct that has a polygenetic component to it, but it is also highly dependent on education,

nutrition, and other environmental factors, such as stress. People have questioned whether evolution through natural selection still applies to humans in our modern world given the advances in modern medicine. However, humans have not stopped evolving. Evolution does not necessarily result in improvements; evolution simply means that there are changes in genetic variation over time.

GENE-ENVIRONMENT INTERACTIONS

Environmental factors can turn some genes on and others off, we call these **epigenetic** changes. Epigenetics helps to explain why two identical twins (who have identical genes) show an incredible amount of variability in their phenotypes. For example, one twin could develop a mental illness such as schizophrenia, while the other never does. Their unique environmental interactions determine how their genetic information is expressed, which in turn gives rise to different phenotypes for various traits. Tienari and colleagues found that among adoptees whose biological mothers had schizophrenia, those who were raised in a disturbed family environment were about 6 times more likely to develop schizophrenia (or related disorder) than those raised in a healthy environment. Adoptees with a biological mother who had schizophrenia and who raised in a disturbed environment were also much more likely to develop schizophrenia than participants with no biological family history of schizophrenia, regardless of type of childhood environment (Tienari et al., 2004). This research highlights that both genetic vulnerability and environmental stress are necessary for schizophrenia to develop, and that genes alone do not tell the full tale (Figure 3.3).

Link to Learning

Watch this video about the epigenetics of twin studies to learn more.

We also know that the genes can influence the environment, which in turn can amplify the effects of any genetic predispositions we have, this is often referred to as the **multiplier effect.** For example, imagine a child who, due to their genes, is naturally agile and athletic. Their parents notice this at an early age and they set up a mini-basketball court and teach their child how to dribble and shoot. The child is encouraged by their enthusiasm and likes to practice. The parents sign their child up for basketball camp and a local team. Thus, the child's genes have influenced their environment. As a consequence, the child is likely to be a better player than other children.

Figure 3.3. Nature and nurture work together like complex pieces of a human puzzle. The interaction of our environment and genes makes us the individuals we are.

3.2 Cells of the Nervous System

Biological psychologists striving to understand the mind and behavior, study the nervous system in various ways. The nervous system is an electro-chemical communication system, which receives and processes information from the outside world, and allows us to react to it. The nervous system is composed of two basic cell types: neurons and glial cells. The neurons (nerve cells) are the central building blocks of the nervous system, whereas various types of glial cells provide support services.

NEURON STRUCTURE

Like all cells, neurons consist of several different parts, each serving a specialized function (Figure 3.4). The cell body or soma contains the nucleus (which contains the genetic material of the cell), and other structures that are important for cell maintenance. The soma is surrounded by a semipermeable membrane, which regulates what can pass in and out of the neuron. Information enters the neuron through the dendrites, which are branching tree-like projections on the soma. Information is then transmitted to the other end of the neuron down a tail-like process called the axon. Information travels through the neuron in the form of electrical energy. At the end of the axon, there are small structures called terminal buttons which contain chemical messengers called

neurotransmitters; these allow the neuron to communicate with other cells.

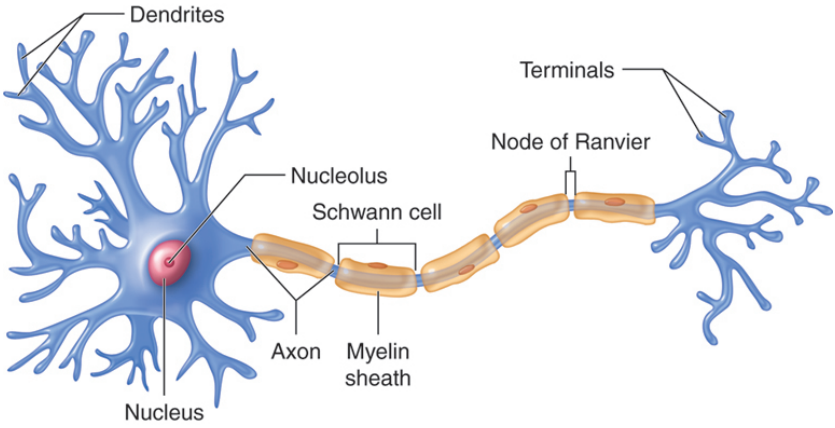


Figure 3.4. Structures within a neuron

The length of the axon determines how far one neuron carries information through our bodies, some axons are microscopic while others, like the ones running from our toes to the base of our spine, are several feet long. Many axons are coated with a fatty substance known as the **myelin sheath**, which acts as an insulator and speeds up the transmission of the electrical signals down the axon. The myelin has small gaps along the axon called the nodes of Ranvier. In myelinated axons, the electrical signals are only generated at the nodes, like a train making express stops. In unmyelinated neurons, information travels more slowly because electrical signals need to be generated along the entire length of the axon—acting more like a local train. Multiple sclerosis (MS) is an autoimmune disorder that causes degeneration of the myelin sheath on axons throughout the nervous system. This can result in inefficient transmission, and sometimes total loss, of neural information. Some of the first symptoms of MS include blurry

vision, sensations of numbness (like pins and needles) and difficulty walking.

Link to Learning

Watch this video about neuronal communication to learn more.

NEURONAL COMMUNICATION

Now that we have learned about the basic structures of the neuron, let's take a closer look at the signals that the neurons transmit. We will look first at how the electrical signals are generated and move through the neuron. Then we will look at how the chemical messengers are released and communicate with other cells, including other neurons.

The neuron exists in a fluid environment—it is surrounded by extracellular fluid and contains intracellular fluid. The neuronal membrane keeps these two salty fluids separate. Both fluids contain sodium (Na+) and potassium (K+) ions, and other charged molecules. Differences in the overall electrical charge of molecules in the two fluids contribute to a difference in voltage across the membrane, called the membrane potential. When the neuron is resting, the membrane resting potential is about 70 mV and the inside of the neuron is more negatively charged than the outside. In general, ions move quickly across concentration gradients from areas of high concentration to areas of low concentration through the process of osmosis. Na^+ is at higher concentrations outside the neuron, so it strongly wants to move into the neuron. K^+ on the

other hand, is more concentrated inside the neuron, and so wants to move out of the neuron (Figure 3.5). Because the inside of the neuron is slightly negatively charged compared to the outside, this provides an additional drive for Na+ to be drawn into the neuron (opposite electrical charges attract). We call this an electrical gradient. However, there is relatively little ionic movement across the membrane when the neuron is resting. This is because most of the Na+ and K+ gates in the semi-permeable membrane, which regulate the flow of ions, are closed when the neuron is at rest. This results in a high state of tension across the membrane; the Na+ and K+ ions are waiting for the gates to open so they can move through them, and so the neuron is primed and ready for action. For the most part, the Na+ and K+ gates do a good job of staying closed, but a few open and there is a little bit of leakage of Na+ and K+ in the direction described above. To keep the resting potential stable, there is an active sodium-potassium pump, which pumps three sodium ions out of the neuron for every two potassium ions in.

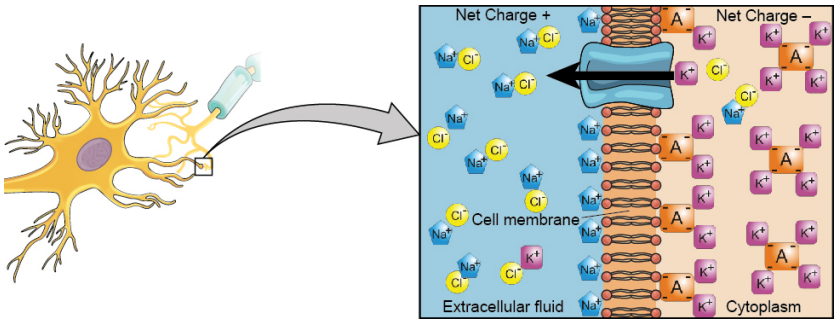


Figure 3.5. At resting potential, Na+ (blue pentagons) is more highly concentrated outside the neuron in the extracellular fluid (shown in blue), whereas K+ (purple squares) is more highly concentrated near the membrane in the cytoplasm or intracellular fluid. Other molecules, such as chloride ions (yellow circles) and negatively charged proteins (brown squares), help contribute to a positive net charge in the extracellular fluid and a negative net charge in the intracellular fluid.

From this resting potential state, if the neuron receives a signal at the dendrites (Figure 3.5), the Na+ gates open on the neuronal membrane, and Na+ ions rush into the cell. These ions make the inside of the cell become more positively charged. If that charge reaches a certain level, called the threshold of excitation, the neuron becomes active and the action potential begins. The action potential, sometimes called a nerve impulse, is a large, rapid change in voltage across the membrane. (Figure 3.6).

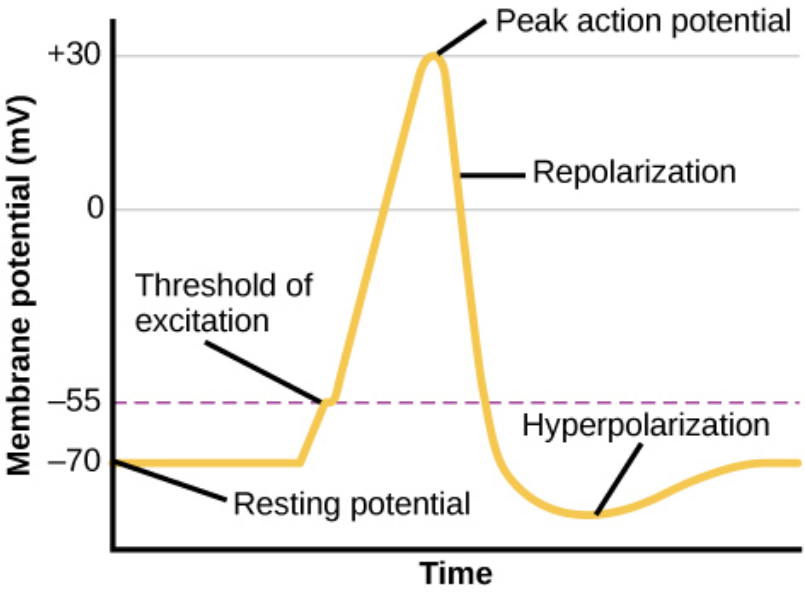


Figure 3.6. During the action potential, the electrical charge across the membrane changes dramatically.

Whether or not the threshold of excitation is reached depends on the strength of the signal being received. When the signal is strong, more Na+ gates open and more Na+ ions enter and the inside of the cell becomes more positive. When the threshold of excitation is reached, all the Na+ channels open rapidly, causing a massive influx of Na^+ ions and a huge positive spike in the membrane potential. At the peak of the spike, the Na+ gates close and the K+ gates open. As positively charged potassium ions rapidly leave the neuron, the inside of the cell quickly starts to become less positive (or more negative). This process is called repolarization. After an action potential is generated there is a resting period, called the refractory period, where the inside of the cell is in a state of hyperpolarization. This means that it is slightly more negative than the resting potential for a few milliseconds; it is not possible to generate another action potential during the refractory period.

As the K+ gates close, the neuron returns to the resting potential (Figure 3.6).

The action potentials are generated in the cell body first and then are produced along the axon until they reach its end. At each point in the journey, some of the sodium ions that enter the neuron diffuse to the next section of the axon, raising the charge past the threshold of excitation and triggering a new influx of sodium ions. So, multiple action potentials are generated sequentially all the way down the axon until they reach the terminal buttons.

The action potential is an all-or-none phenomenon. In simple terms, this means that an incoming signal from another neuron must be strong enough to reach the threshold of excitation, otherwise, there is no action potential (none). The action potential is always generated at its full strength at every point along the axon. It does not fade away as it travels down the axon. This ensures that there is no loss of information along the way. The size of the action potential does not depend on the strength of the stimulus, it is always the same. Strong stimuli simply produce more action potentials.

In many ways, the way that a neuron produces action potentials is similar to the way that an old fashioned toilet produces a flush. Both work on the all or none principle. A toilet also has a threshold of excitation – if you do not push the handle hard enough, it will not flush. However, as long as you press the handle hard enough, the toilet always produces a flush of the same size. So, you have all or none. If you push very hard, you don't get any more water than if you push more gently, provided you meet the threshold of excitation. Like a neuron, a toilet also has a refractory period (when the tank is refilling); the toilet cannot flush during this time.

COMMUNICATION ACROSS THE SYNAPSE

At the end of every axon, there is a tiny gap called a synapse

between the terminal buttons and the next cell, which is often another neuron. This gap is filled with extracellular fluid and is too big for an action potential to jump across. Instead, when the action potential arrives at the terminal button, small vesicles release their neurotransmitters into the synapse. The neurotransmitters travel across the synapse and bind to receptors on the dendrites of the adjacent neuron. This action then affects the ionic gates in that neuron. Some neurotransmitters are excitatory (they open Na+) gates, whereas others are inhibitory (they open K+) gates. So, neurotransmitters give more flexibility in terms of how neurons can respond to a stimulus.

NEUROTRANSMITTERS AND DRUGS

There are several different types of neurotransmitters in our nervous system, each one acts on specific types of receptors and has a particular action (see Table 3.1).

Table 3.1 *Major Neurotransmitters and How They Affect Behavior*

Neurotransmitter	Involved in	Potential Effect on Behavior
Acetylcholine	Muscle action, memory	Increased arousal, enhanced cognition
Beta-endorphin	Pain, pleasure	Decreased anxiety, decreased tension
Dopamine	Mood, sleep, learning	Increased pleasure, suppressed appetite
Gamma-aminobutyric acid (GABA)	Brain function, sleep	Decreased anxiety, decreased tension
Glutamate	Memory, learning	Increased learning, enhanced memory
Norepinephrine	Heart, intestines, alertness	Increased arousal, suppressed appetite
Serotonin	Mood, sleep	Modulated mood, suppressed appetite

Much of what psychologists know about the functions of neurotransmitters comes from research on the effects of drugs in psychological disorders. There is some biological evidence that psychological disorders like depression and schizophrenia are associated with imbalances in one or more neurotransmitter systems. Psychoactive medications affect neurotransmitter balance and can help improve the symptoms associated with these disorders.

Psychoactive drugs generally act as agonists or antagonists for a given neurotransmitter system, which is why they can affect our behaviors, thoughts, and feelings. Agonists are chemicals that mimic the action of a neurotransmitter at the receptor site. An antagonist, on the other hand, blocks or impedes the normal activity of a neurotransmitter at the receptor. GABA-agonists such as benzodiazepines mimic the effects of GABA and so produce calming effects for people who have anxiety. Conversely, psychotic symptoms in schizophrenia are associated with high levels of dopamine and so are treated with dopamine antagonists. These drugs stop dopamine's effects by blocking its receptors. Thus, they can prevent one dopamine-producing neuron from sending information to adjacent neurons.

In contrast to agonists and antagonists, which both operate by binding to receptor sites, some psychoactive drugs affect the deactivation of the neurotransmitter. Once the neurotransmitter has done its job it is typically removed in some way so that it does not continue to act on the post-synaptic neuron. In some cases, the neurotransmitters in the synapse diffuse away, or they are broken down by an enzyme or pumped back into the pre-synaptic neuron in a process known as reuptake (Figure 3.7).

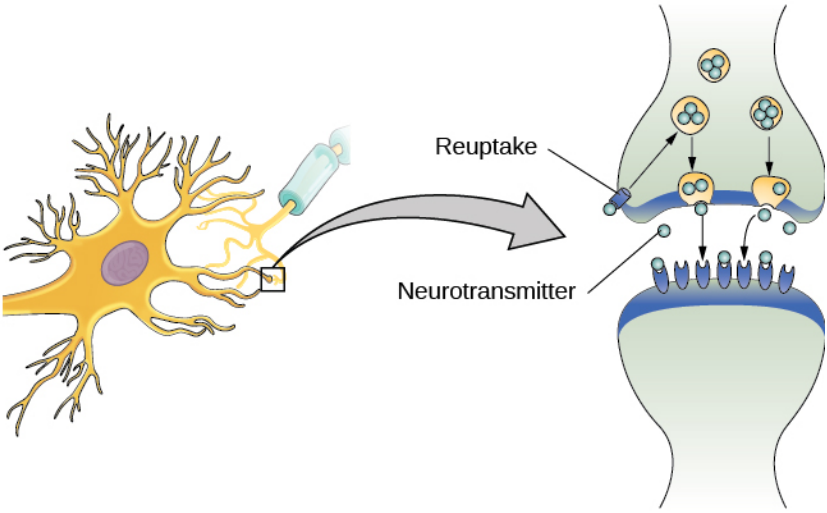


Figure 3.7. Reuptake involves moving a neurotransmitter from the synapse back into the axon terminal from which it was released.

Some psychoactive drugs can prevent the removal of the neurotransmitters. This allows neurotransmitters to remain active in the synapse for longer durations, increasing their effectiveness. Depression, which has been linked with reduced serotonin levels, is frequently treated with selective serotonin reuptake inhibitors (SSRIs). By preventing reuptake, SSRIs prolong the action of serotonin, giving it more time to interact with serotonin receptors on dendrites. So, SSRIs act as serotonin agonists, even though they do not work directly on the receptors. Common SSRIs include Prozac, Paxil, and Zoloft. Psychotropic drugs are not instant solutions for people suffering from psychological disorders. Often, an individual must take a drug for several weeks before seeing improvement, and many psychoactive drugs have significant negative side effects. Furthermore, individuals vary dramatically in how they respond to the drugs. To improve chances for success, it is common for people receiving medications to undergo psychological and/or behavioral therapies as well. Research

suggests that combining drug therapy with other forms of therapy tends to be more effective than any one treatment alone for the treatment of depression (Cuijpers, Noma, et al., 2020) and some anxiety disorders like panic disorder and obsessive-compulsive disorder (Cuijpers, Sijbrandij, et al., 2020).

3.3. Parts of the Nervous System

Now you have some idea about how the nervous system works on the microscopic level, let us zoom out and look at the big picture in terms of how it is organized. The nervous system can be divided into two major subdivisions: the central nervous system (CNS) and the peripheral nervous system (PNS), shown in Figure 3.8. The CNS is comprised of the brain and spinal cord, which are the control centers of the nervous system. The PNS connects the CNS to the rest of the body.

PERIPHERAL NERVOUS SYSTEM

The peripheral nervous system is made up of thick bundles of axons, commonly called nerves, which carry messages back and forth between the CNS and the muscles, glands, and organs in the periphery of the body (i.e., everything outside the CNS). The PNS has two major subdivisions: the somatic nervous system and the autonomic nervous system (Figure 3.8).

The **somatic nervous system** is associated with activities traditionally thought of as conscious or voluntary, like picking up or putting down a cup of coffee, or writing a paper. The somatic nervous system relays information about our senses (via sensory neurons) to the CNS; and brings information from the CNS to muscles, glands, and organs via motor neurons. Sensory neurons

carrying sensory information into the CNS are afferent fibers (afferent means "moving toward"), whereas motor neurons carrying information from the CNS are efferent fibers. A helpful way to remember this is that **e**fferent = **e**xit and **a**fferent = **a**rrive.

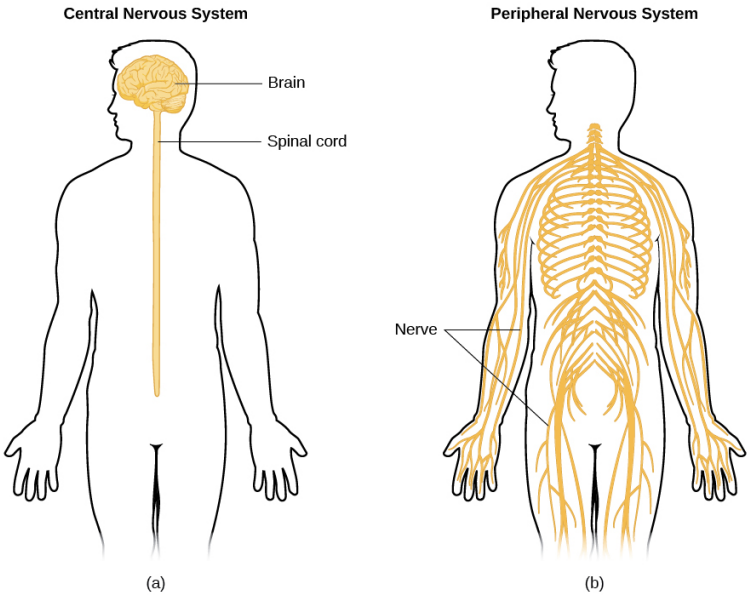


Figure 3.8. The nervous system consists of (a) the central nervous system and (b) the peripheral nervous system

The **autonomic nervous system** controls our internal organs and glands and is generally considered to be outside of our voluntary control. It can be further subdivided into the **sympathetic** and **parasympathetic** divisions (Figures 3.9 and 3.10.). The sympathetic nervous system is often referred to as the fight or flight system, because it is involved in preparing the body for stress-related activities. The parasympathetic nervous system is associated with resting and digesting, and returning the body to routine, day-to-day operations. The two systems have complementary functions, operating in tandem to maintain the

body's homeostasis. Homeostasis is a state of equilibrium, or balance, in which bodily functions (such as body temperature) are maintained at optimal levels.

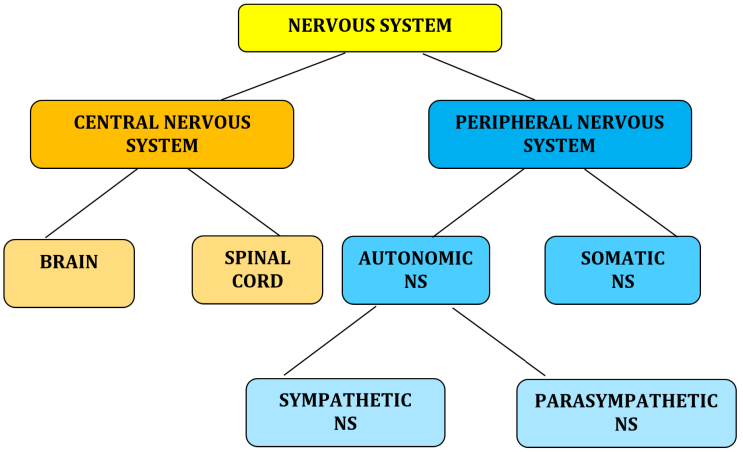


Figure 3.9. The subdivisions of the nervous system

The sympathetic nervous system is activated when we are faced with stressful or very arousing situations. This system was highly adaptive for our ancestors, as it increased their chances of survival in dangerous situations. Imagine, for example, if one of our early ancestors, suddenly disturbed a large bear. At that moment, the sympathetic nervous system would immediately activate and their body would undergo a rapid series of changes, which would prepare them to face the threat. Their pupils would dilate, their heart rate and blood pressure would increase, their bladder would relax, and their liver would releases glucose; adrenaline would surge into the bloodstream, and blood would flow to large muscles. This constellation of physiological changes allows the body access to energy reserves and provides heightened sensory capacity so that they would be able to fight off the threat or run away to safety. Once the threat has been resolved, the parasympathetic

nervous system would take over and return bodily functions to a relaxed state. Their heart rate and blood pressure would return to normal, their pupils would constrict, their bladder control would be restored, and their liver would begin to store glucose in the form of glycogen for future use.

Link to Learning

Watch this video about the Freeze, Fight and Flight response to learn more. Sometimes we initially freeze when we experience a threatening situation. Freezing helps us to "hide" from potential predators, because moving objects (including people and animals) are easier to spot than stationary ones. Freezing is initiated by the parasympathetic nervous system and may be important for getting the sympathetic nervous system ready to respond (Roelofs, 2017).

While it is clear that the "fight or flight" response was critical for survival for our ancestors, who lived in a world full of physical threats, many of the highly arousing situations we face in the modern world are more psychological in nature. For example, think about how you might feel when you have to stand up and give a presentation in front of a roomful of people, or when you are about to take a big test. You are in no real physical danger in those situations, and yet you have evolved to respond with the fight or flight response. This kind of response is not nearly as adaptive in the modern world; in fact, we can suffer negative health consequences when faced constantly with psychological threats that we can neither fight nor flee. Recent research suggests that

an increase in susceptibility to heart disease (Chandola et al., 2006) and impaired function of the immune system (Glaser & Kiecolt-Glaser, 2005) are among the many negative consequences of persistent and repeated exposure to stressful situations. Some of this tendency for stress reactivity can be wired by early experiences of trauma.

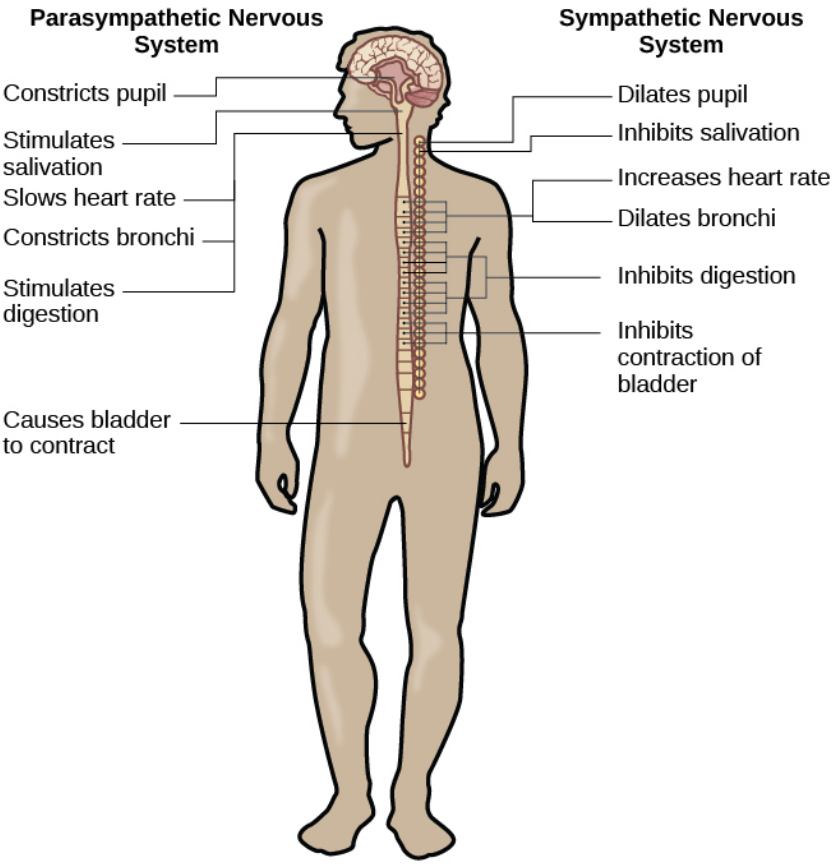


Figure 3.10. The sympathetic and parasympathetic divisions of the autonomic nervous system have the opposite effects on various systems

3.4. The Brain and Spinal Cord

The central nervous system is the control center for our thoughts, feelings, and behaviors. The brain is the "boss" of the whole system, it is a complex organ comprised of billions of interconnected neurons and glia, and is capable of very sophisticated processing. The spinal cord, on the other hand, is a simpler structure. It relays messages back and forth to the brain, but it also controls basic, automatic behaviors known as reflexes.

SPINAL CORD

The spinal cord is functionally organized into 31 segments; each segment is encased within one of the vertebrae (bones) of the spine for protection (Figure 3.11a).

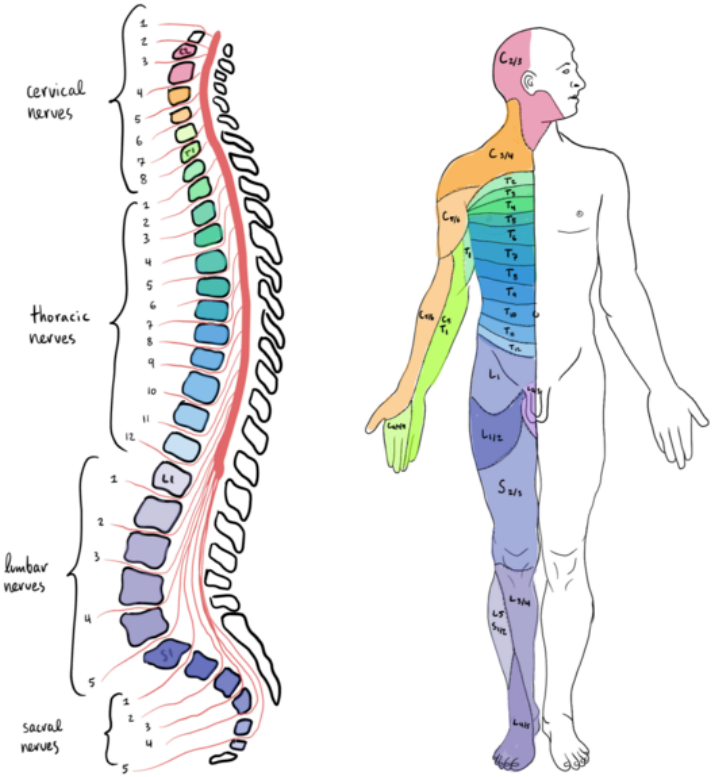


Figure 3.11a. Each of the segments of the spinal cord is named for the type of bone and its position in the spinal column. C1 is at the top of the cervical segments. The coccygeal section (not shown here) is at the base of the spinal cord.

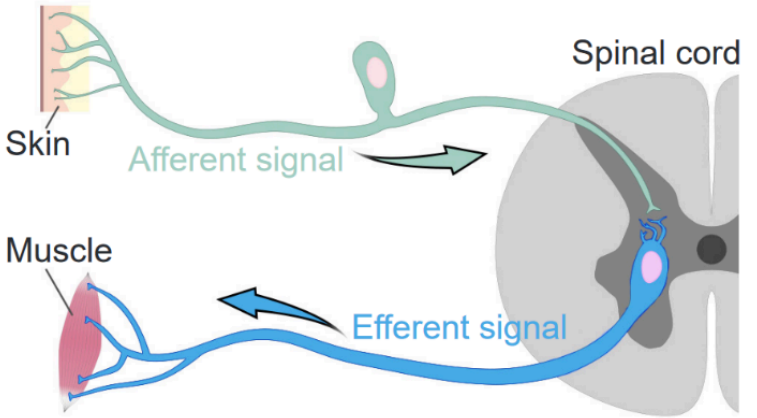


Figure 3.11b. Each segment of the spinal cord has an afferent and an efferent nerve that serves specific parts of the body.

The spinal cord (like the brain) is also cushioned in cerebrospinal fluid. There are 8 cervical segments, 12 thoracic segments, 5 lumbar, 5 sacral, and one coccygeal (not shown in Figure 3.11a, but it is located at the base of the spine). The spinal cord connects to rest of the body via the peripheral nervous system, in the following ways. Each spinal cord segment connects to a sensory nerve and a motor nerve that serve a specific part of the body; thus, there are 31 pairs of spinal nerves. Sensory nerves bring messages into the spinal cord, which are then sent to the brain. The brain also sends messages to the spinal cord, which are conveyed via motor nerves to the muscles, glands, and organs (Figure 3.11b) Figure 3.11a shows the connections between each spinal cord segment and the specific body areas it serves. The cervical segments connect to muscles in the face, head, neck, shoulders, arms and hands, as well as the diaphragm that controls our breathing. The thoracic segments connect to the middle trunk area, the intercostal muscles between the ribs, and the abdominal muscles. The lumbar segments connect to the hips and legs, and the sacral regions to

the toes, backs of the thighs and pelvis. The coccygeal segment sends information to and from the tailbone area.

REFLEXES

Clearly, the spinal cord is an important relay station between the brain and the rest of the body. However, the spinal cord is also able to initiate some basic behaviors in direct response to sensory messages, without any input from the brain. These very rapid automatic responses are called reflexes and are mostly protective. They include the actions of moving quickly away from a hot object and blinking your eye(s) when something comes too close to them.

Reflexes are controlled by very simple neural circuits called reflex arcs. In a reflex arc, information (e.g., sensing when something is hot) passes from the outside world via the sensory nerve to an interneuron in the spinal cord, this then sends a command via a motor neuron to a muscle (Figure 3.12). There are only three neurons are involved in the circuit—a sensory neuron, interneuron and a motor neuron. The distance between the sensory and motor neurons is very small because the interneuron is tiny. Consequently, the time between receiving the information and responding to it is extremely fast. Of course, the spinal cord does ultimately send the information to the brain as well, so we are aware of what has happened. However, if we had to wait for the brain to respond, it would take a lot longer for the action to occur. The information would have to travel up the spinal cord to the brain where it would be processed, and then the brain would send a command down the spinal cord to the motor neuron and then to the muscle.

Because the brain is not involved in simple reflexes, it is possible to have brain damage where the reflexes are unaffected. The parts of the body that are affected by spinal cord injuries depend on the location of the damage. Damage in the upper parts of the spinal cord will have more serious consequences than damage

lower down. This is because the damage will not only affect the part of the body that segment serves (see Figure 3.11a), but all the axons from all the lower segments have to pass through this area too, so they are cut off from the brain, causing widespread paralysis and lack of sensory input.

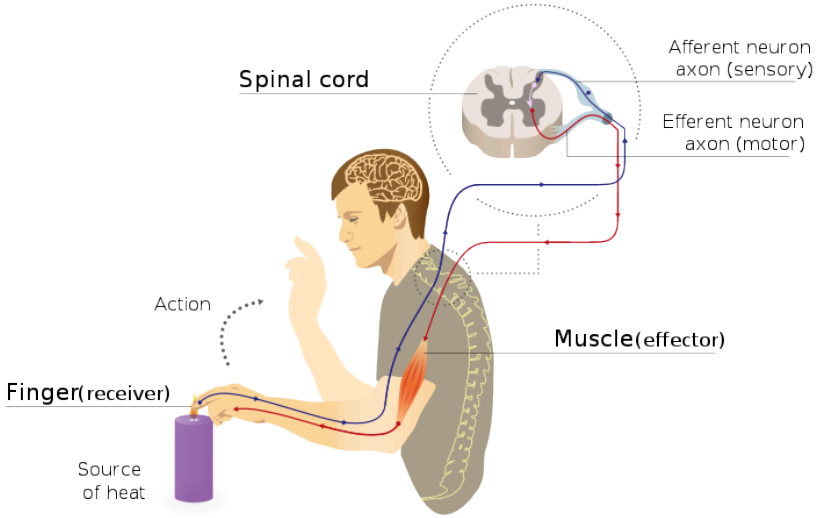


Figure 3.12. We pull our hand away from a hot candle flame automatically. This rapid action is mediated by a simple neural circuit called a reflex arc.

BRAINSTEM

The brainstem sits on top of the spinal cord and contains the medulla and the pons. The medulla controls primitive but vital life-support functions, such as breathing, blood pressure, and heart rate. The word pons literally means "bridge," and as the name suggests, the pons is a crossing point for many axons traveling back and forth to the brain. It also is involved in regulating brain activity during sleep (see Figure 3.13).

CEREBELLUM

The cerebellum (Latin for "little brain") is a structure that is located close to the brainstem because it developed from the same general area in the fetal brain (see Figure 3.13). It receives messages from muscles, tendons, joints, and structures in our ear to control balance, coordination, movement, and motor skills. The cerebellum is also thought to be an important area for procedural memories, which are actions that we do not consciously think too much about, like walking, brushing our teeth, and riding a bike. The basal ganglia also contribute to procedural memories.

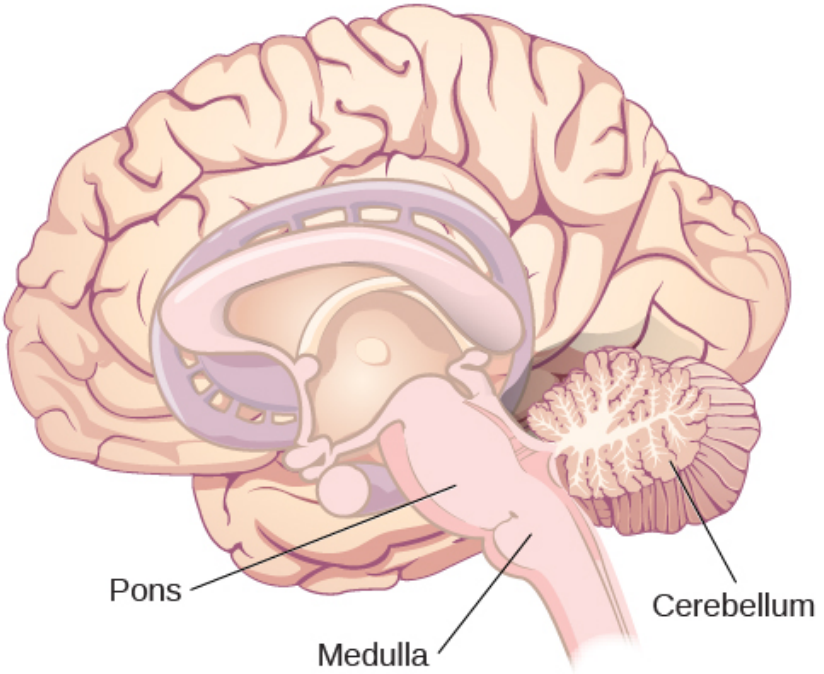


Figure 3.13. The pons, medulla, and cerebellum

MIDBRAIN

The midbrain extends up from the brain stem. Midbrain structures are located deep within the brain and are not visible from the surface, again they have fairly primitive functions. The reticular formation, which is important in regulating sleep/wake cycles, arousal, alertness, and motor activity, passes through the midbrain as it sends messages back and forth between the brainstem and higher brain areas, like the cerebral cortex.

The substantia nigra (Latin for "black substance") and the ventral tegmental area (VTA) are also located in the midbrain, Figure 3.14). Both regions contain cell bodies that produce the neurotransmitter, dopamine. The substantia nigra and VTA along with other interconnected brain structures are collectively known as the basal ganglia. The basal ganglia are critical for movement, and are also involved in mood, reward, and addiction behaviors (Berridge & Robinson, 1998; Gardner, 2011; George et al., 2012). Degeneration of the substantia nigra and VTA is involved in Parkinson's disease.

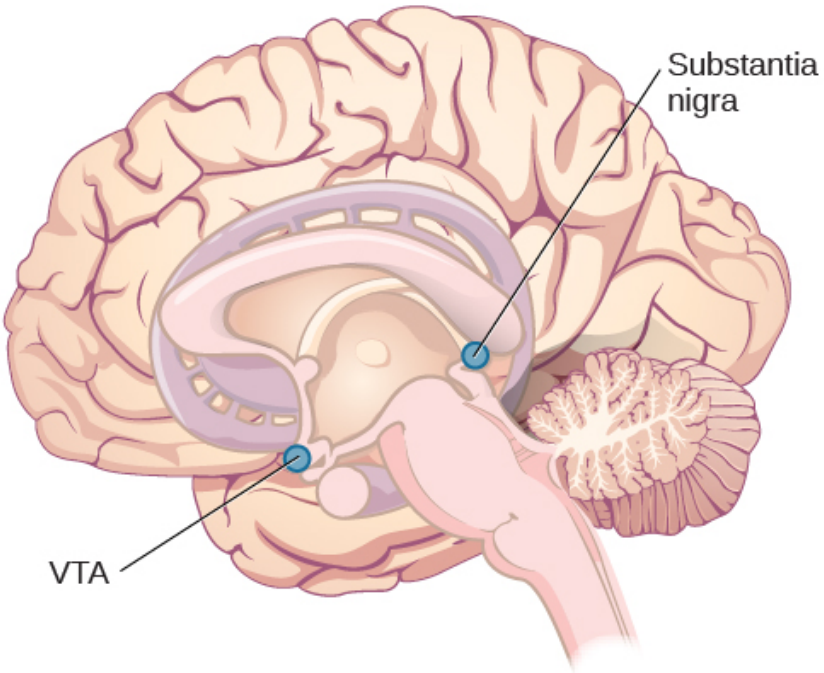


Figure 3.14. The substantia nigra and ventral tegmental area (VTA) are located in the midbrain.

OTHER IMPORTANT SUBCORTICAL AREAS

Other important structures that lie beneath the surface of the brain include the **thalamus** and the limbic system. The thalamus sits on top of the midbrain structures and relays information from all of our senses, with the exception of smell, to other areas of the brain for processing (Figure 3.15). It is often referred to as a sensory switchboard.

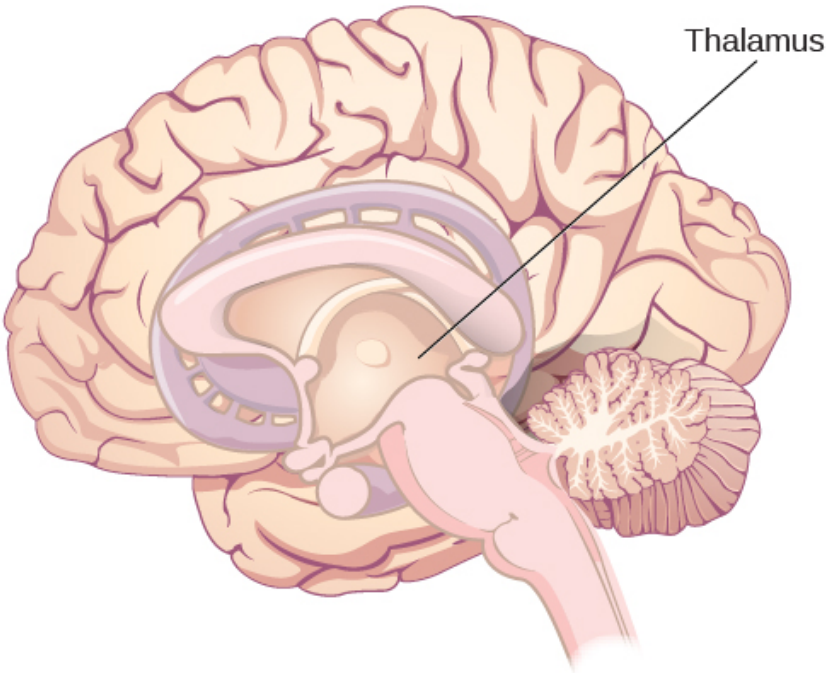


Figure 3.15. The thalamus serves as the relay center of the brain where most sensory information is routed for more sophisticated levels of processing.

The **limbic system** is involved in processing information relating to both emotion and memory. Interestingly, the sense of smell projects to the limbic system; therefore, not surprisingly, smell can evoke emotional responses in ways that other sensory modalities do not. The limbic system is made up of a number of different structures, including the hippocampal structures, the amygdala, and the hypothalamus (Figure 3.16). The hippocampus is an essential structure for learning and memory. The amygdala is active when experiencing emotion and is important for tying emotional meaning to our memories. The hypothalamus regulates a number of homeostatic processes, including the regulation of body temperature, appetite, and blood pressure. The

hypothalamus also serves as an interface between the nervous system and the endocrine system and in the regulation of sexual motivation and behavior.

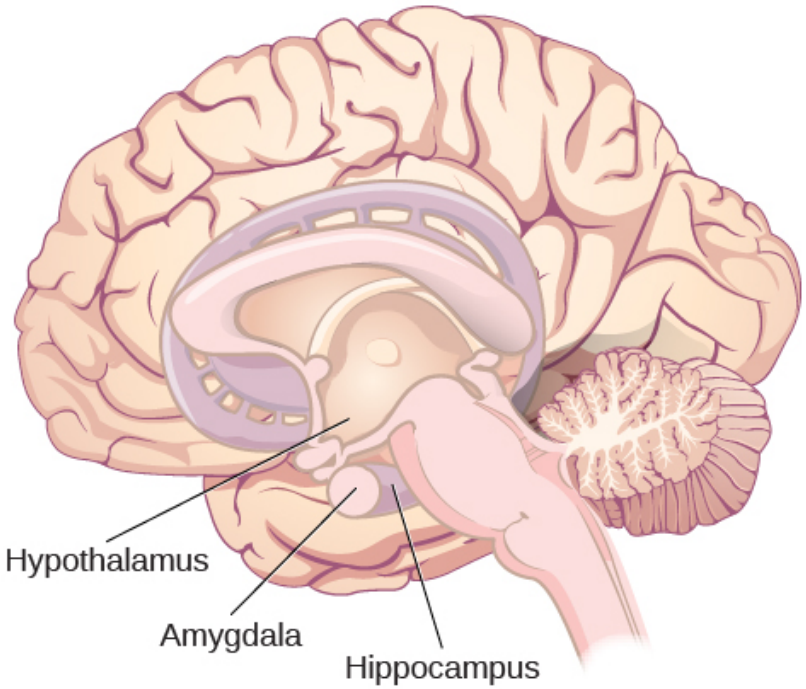


Figure 3.16. The limbic system is involved in mediating emotional response and memory.

THE CASE OF HENRY MOLAISON (H.M.)

In 1953, surgeons performed experimental brain surgery on Henry Gustav Molaison (H. M.). He was a 27-year-old man who been experiencing severe epileptic seizures since he fell off his bicycle as a child. Surgeons removed the hippocampal structures and much of the amygdala from both sides of his brain. After the surgery, H. M's seizures became much less severe, but he also suffered some

unexpected and devastating consequences of the surgery. He was no longer able to form most types of new memories. For example, he was unable to learn new facts. He could not remember new faces, and he was unable to remember anything that someone had said to him for more than about 30 seconds. He was able to learn new motor skills, such as drawing while looking in a mirror, but afterward he had no recollection of learning how to do it. Researchers were fascinated by his experience, and he is considered one of the most famous case studies in medical and psychological history (Hardt, Einarsson, & Nader, 2010; Squire, 2009). His case provided vital insights as to the importance of the hippocampal structures for explicit memories.

Link to Learning

Clive Wearing, an accomplished musician, lost the ability to form new memories when his hippocampus was damaged when a virus affected his brain. Check out the first few minutes of this documentary video about this man and his condition to learn more.

THE TWO HEMISPHERES OF THE BRAIN

Probably the most noticeable feature of the brain is its wrinkly surface, this is a thin layer of cells called the cerebral cortex. The cerebral cortex has a distinctive pattern of bumps, known as gyri (singular: gyrus), and grooves, known as sulci (singular: sulcus), shown in Figure 3.17. The cerebral cortex is associated with higher level processes such as consciousness, thought, emotion,

reasoning, language, and memory. Its wrinkled structure allows more cells to be packed into a smaller area; if it were flattened out—it would be about the size of a pillow case and we would need to have much larger skulls to accommodate this! The structures that we have already discussed in this chapter are housed below the cerebral cortex which is why we refer to them as subcortical structures. They perform more basic functions than the cortex.

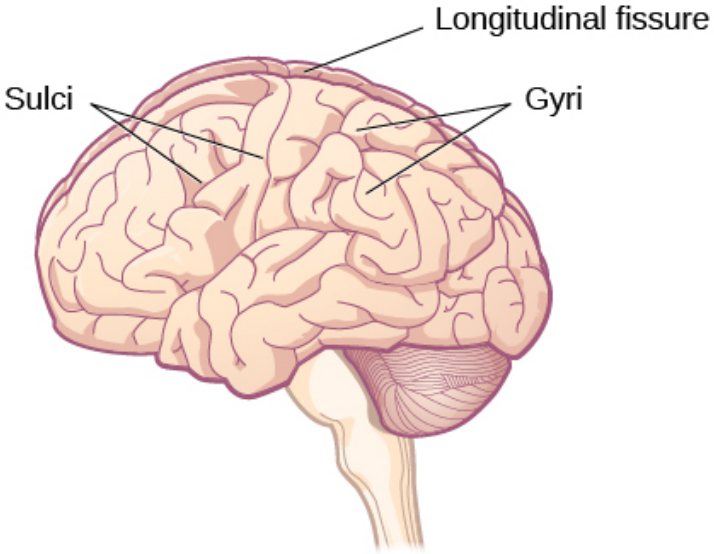


Figure 3.17. The surface of the brain is covered with gyri and sulci. A deep sulcus is called a fissure, such as the longitudinal fissure that divides the brain into left and right hemispheres.

BRAIN LATERALIZATION

The brain has two halves called hemispheres that are divided by the longitudinal fissure. The two sides of the brain work in concert together, however, each side of the brain is somewhat specialized for some functions. For example, we know that each hemisphere controls the opposite (contralateral) side of the body. So, people

with damage to their left motor cortex will have paralysis in the right side of their body. Dr. Michael Gazzaniga and his colleagues have performed decades of research on lateralization of brain function, largely based on studies performed with split-brain patients (Gazzaniga, 2005).

SPLIT BRAIN PATIENTS

The right and left hemispheres are interconnected by the corpus callosum, a thick band of about 200 million axons. The corpus callosum allows information being processed on one side of the brain to be shared with the other side. In some cases of severe epilepsy, doctors elect to sever the corpus callosum as a means of controlling the spread of seizures (Figure 3.18). While this is an effective treatment option, it results in "split brain" behaviors after surgery. In a "split-brain", the two sides of the brain do not communicate with each other. Therefore, split-brain patients are good at using their hands independently, for example, they are able to simultaneously draw a circle with one hand and a square with the other. Give it a try—it's not as easy as it sounds!

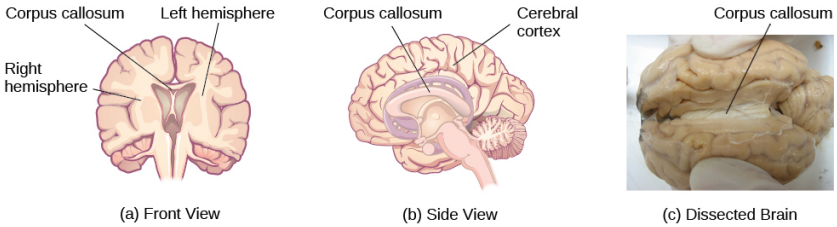


Figure 3.18. (a) and (b) The corpus callosum connects the left and right hemispheres of the brain. (c) A scientist spreads this dissected sheep brain apart to show the corpus callosum between the hemispheres.

In a lab setting, we can also use a technique known as hemi-field presentation that capitalizes on the anatomical arrangement of the visual pathways to present information to each hemisphere of the

brain separately (Figure 3.19). If we stare at a center spot on a computer screen – information to the right of the spot (i.e., the right visual field of each eye) goes to the left side of the brain and information on the left of the spot (left visual field) goes to the right side of the brain Figure 3.19). In typical healthy brains, this information is quickly shared across the corpus callosum but in a split-brain patient, the information stays within each hemisphere. Thus, split brain patients provide scientists with a way to study the function of each hemisphere separately. Split-brain studies have shown that for most people, the left hemisphere is responsible for speech production. This finding comes from hemi-field studies where split-brain people are shown different pictures on each side of the screen. If a split brain person looks at a picture with a key in the left visual field (sent to right hemisphere) and a ring in the right visual field (sent to left hemisphere) and are asked what they see, they would **say** that they only saw a ring. However, if they were asked to **draw** what they saw with their left hand, which is also controlled by the right hemisphere, they would draw the key (Figure 3.19).

Other studies with split brain patients show that the right hemisphere is better at recognizing faces than the left hemisphere, and is more likely to process information holistically rather than in terms of its individual parts. For example, showing a picture like the one by the artist, Arcimboldo, depicted in Figure 3.20, gives rise to different perceptions depending on which visual field it is shown in. If shown in the left visual field (right hemisphere) split brain patients are likely to perceive a face, but if it is shown in the opposite visual field, they are more likely to perceive the individual elements that the face is made from, such as the fruits and the flowers.

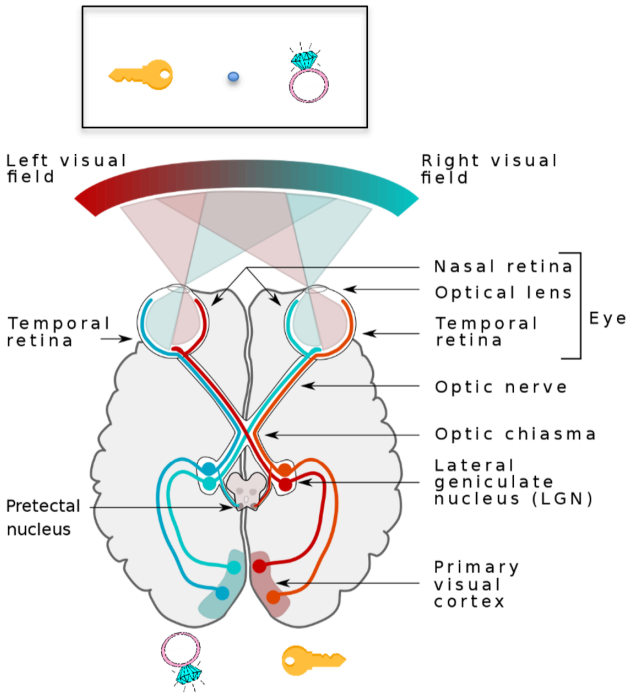


Figure 3.19. Visual pathways. Information from the right visual field (ring) goes to the left hemisphere. Information from the left visual field (key) goes to the right hemisphere.

Figure 3.20. In studies with split brain patients; their perception of this picture depends on which visual field it appears in.

LOBES OF THE BRAIN

The brain can be divided into four lobes called the frontal, parietal, temporal, and occipital lobes (Figure 3.21). Each of the four lobes is somewhat specialized in terms of its function. However, many thoughts, feelings, and behaviors activate networks of neurons across the four lobes.

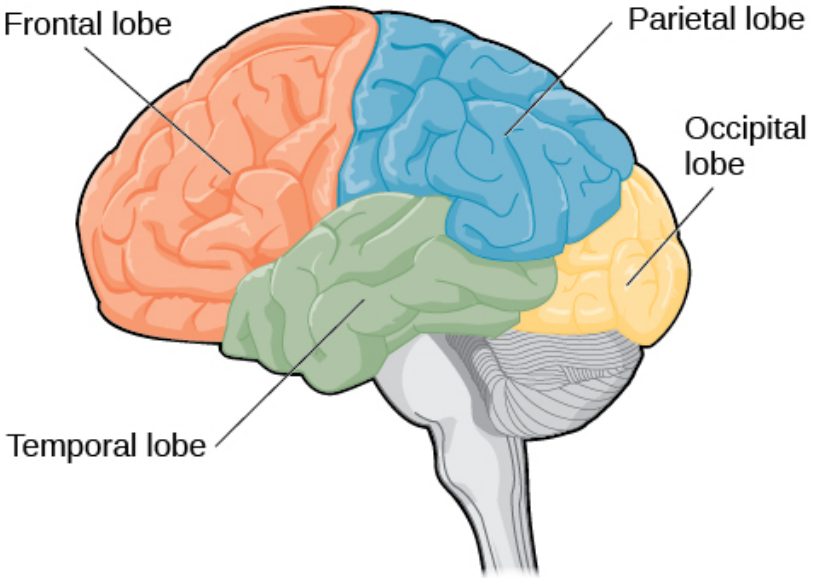


Figure 3.21. The lobes of the brain.

FRONTAL LOBE

The frontal lobe is located in the most anterior part of the brain at the front of the head, and extends back to the central sulcus. The frontal lobe is involved in reasoning, emotion, and language. It is subdivided into many different regions, including the motor cortex (Figure 3.22a), which is involved in planning and coordinating

movement; the prefrontal cortex, which is responsible for higher-level cognitive functioning; and Broca's area (Figure 3.22b), which is essential for language production. People who suffer damage to Broca's area have great difficulty producing language of any kind (Broca's aphasia). For example, Padma was in a car accident and sustained damage to her Broca's area. She completely lost the ability to speak and form any kind of meaningful language. There is nothing wrong with her mouth or her vocal cords, but she is unable to produce words. She can follow directions but can't respond verbally, and she can read but no longer write. She can do routine tasks like running to the market to buy milk, but she cannot ask for assistance in the store.

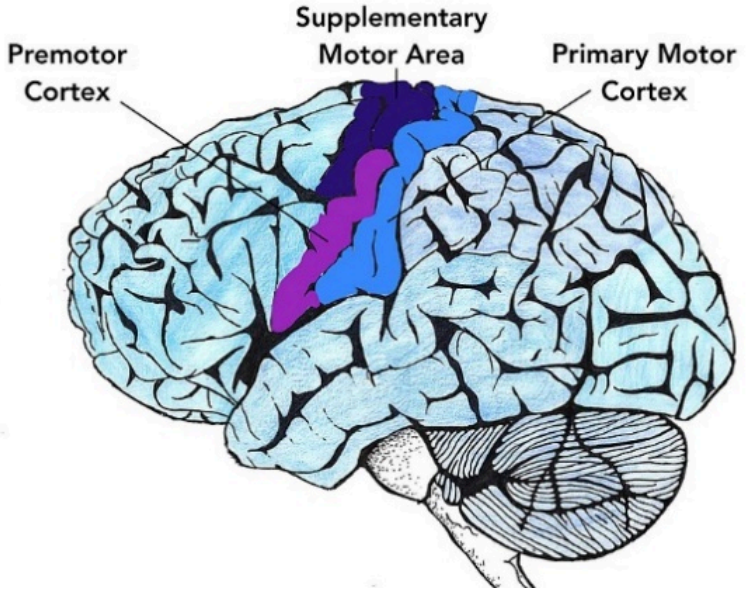


Figure 3.22a. The motor cortex.

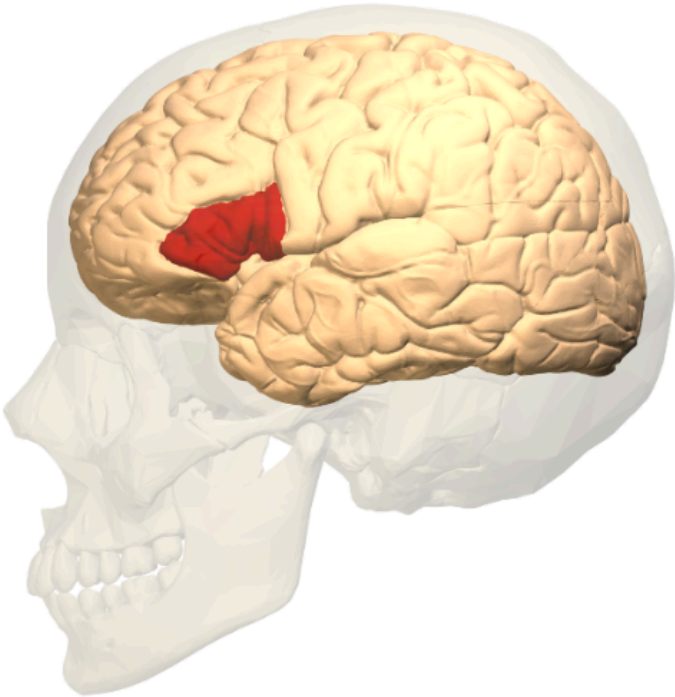
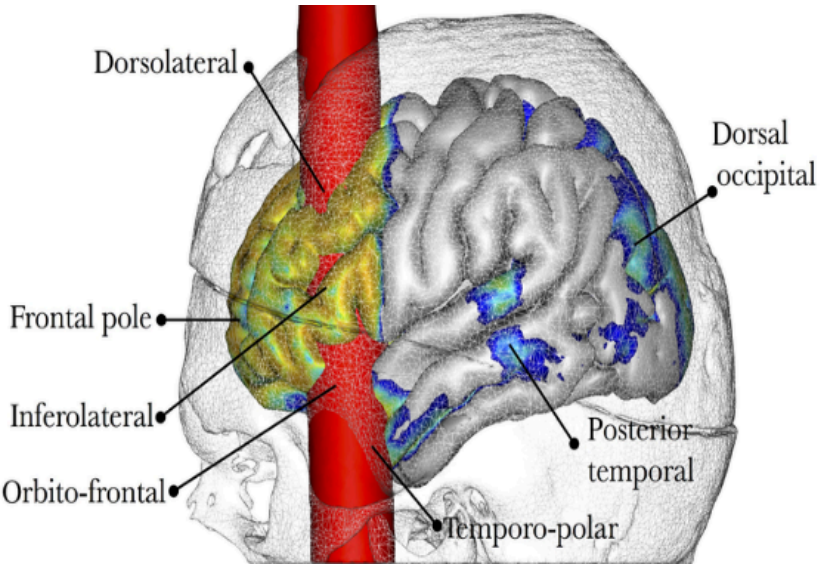


Figure 3.22b. Broca's area in frontal lobe

Probably the most famous case study of prefrontal lobe damage is the story of Phineas Gage. In1848, Gage (age 25) was working as a railroad foreman in Vermont. He and his crew were using an iron rod to tamp explosives down into a hole to remove rock along the railway's path. Unfortunately, the iron rod created a spark and caused the rod to explode out of the hole, through Gage's left cheek and out of the top of his skull (Figure 3.23). Gage quickly regained consciousness and was able to get up, walk, and speak. But in the months following his accident, people noticed that his personality had changed. Before the accident, Gage was a well-mannered, soft-spoken man, but afterwards, he became uncaring

and unreliable, and frequently used inappropriate language. These changes in personality are consistent with the loss of impulse control—a prefrontal cortex function (Macmillan, 1999). Figure 3.23.c depicts a reconstruction of the brain damage that the iron rod caused in Phineas Gage.

Figure 3.23. (a) Phineas Gage holds the iron rod that penetrated his skull in a railroad construction accident. (b) Gage's prefrontal cortex was severely damaged in the left hemisphere. The rod entered Gage's face on the left side, passed behind his eye, and exited through the top of his skull, before landing about 80 feet away.



(c). Reconstruction of areas in Phineas Gage's brain that were affected by the tamping pole

PARIETAL LOBE

The brain's parietal lobe is located immediately behind the frontal lobe, and is involved in attentional processes and processing somatosensory information such as touch, temperature, and pain. Each part of your body sends information to a particular area of the primary somatosensory cortex, so it is organized like a map. Body parts that are very sensitive (like our lips) have many sensory neurons and so send a lot of information to the brain. Consequently, the lips take up more space in the primary somatosensory cortex compared to a less sensitive, but larger body part, like a leg (Figure 3.24).

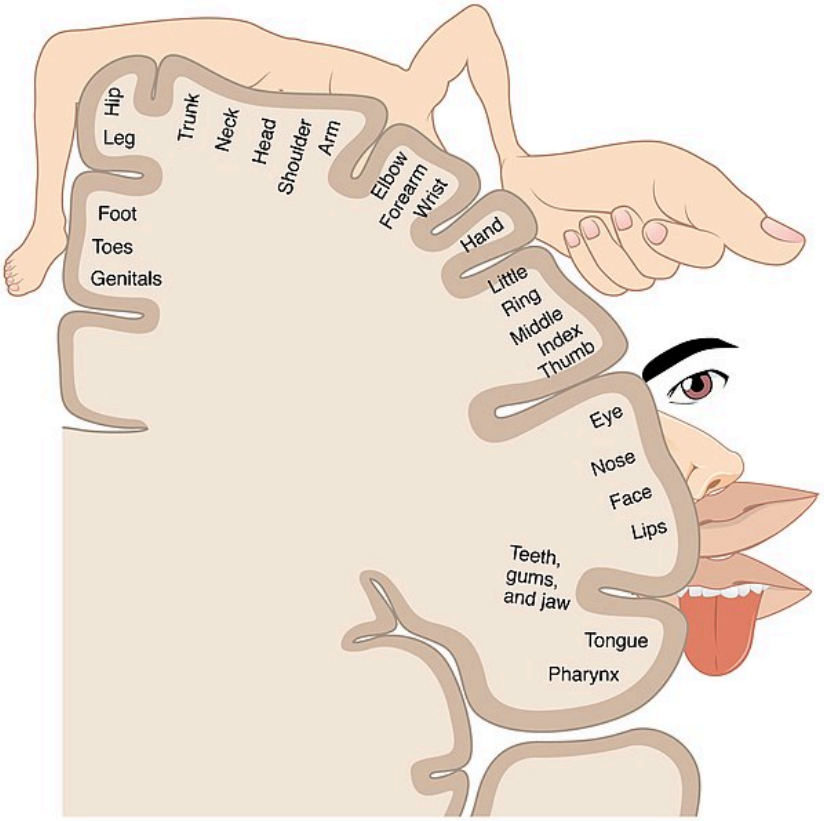


Figure 3.24. Organization of the primary somatosensory cortex. The word primary means that this is the first area of the cortex that receives somatosensory information. The size of the body part in the picture represents the cortical area it takes up in the primary somatosensory cortex

TEMPORAL LOBE

The temporal lobes are located on each side of the head and are associated with hearing, memory, emotion, and language comprehension. The primary auditory cortex, the first cortical area that processes auditory information, is located within the temporal

lobe. Wernicke's area, which is important for speech comprehension, is located nearby. Whereas individuals with damage to Broca's area have difficulty producing language, those with damage to Wernicke's area have difficulty understanding language (Figure 3.25). People with Wernicke's aphasia can pronounce words. However, what they say does not make sense and is often referred to as word salad.

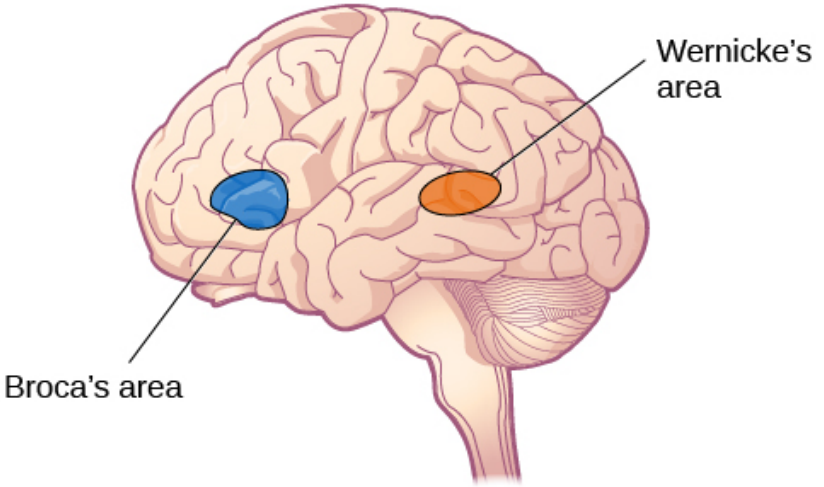


Figure 3.25. Damage to either Broca's area or Wernicke's area result in different kinds of language deficits.

OCCIPITAL LOBE

The occipital lobe is located at the very back of the brain, and contains the primary visual cortex, which is the first cortical area to receive and process visual information. As you saw earlier when reading about hemi-field presentations of visual information, the occipital cortex is organized such that each hemisphere receives information from each visual field of each eye (Figure 3.19). You will learn more about how visual information is processed in the

occipital lobe when you read about sensation and perception in an upcoming chapter.

BRAIN IMAGING

Although early studies of the brain focused on case studies of patients with brain damage, there are now multiple different ways to image brains. This allows scientists to study healthy brains too. Some techniques provide information about the layout of the physical structures of the brain, these are called structural imaging techniques. They are useful for understanding the anatomy of the brain, locating areas of damage or measuring the size of tumors etc. Other techniques measure which part of the brain is active or how the brain processes different kinds of information, these are called functional imaging techniques. Researchers use functional brain imaging to learn about how the brain works.

CT SCANS

A computerized tomography (CT or CAT) scan involves taking multiple x-rays of a particular section of a person's body or brain from different angles (Figure 3.26). The x-rays pass through tissues of different densities at different rates, allowing a computer to construct a detailed structural image (Figure 3.27). A CT scan is useful for identifying the location and size of damage or tumors.

Figure 3.26. CT scanner

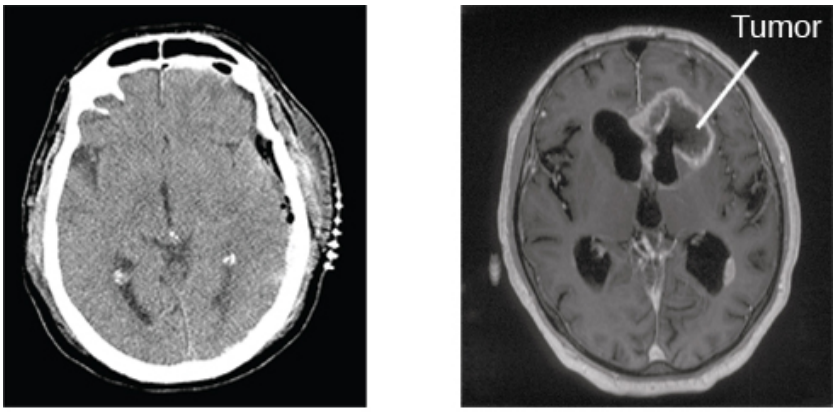
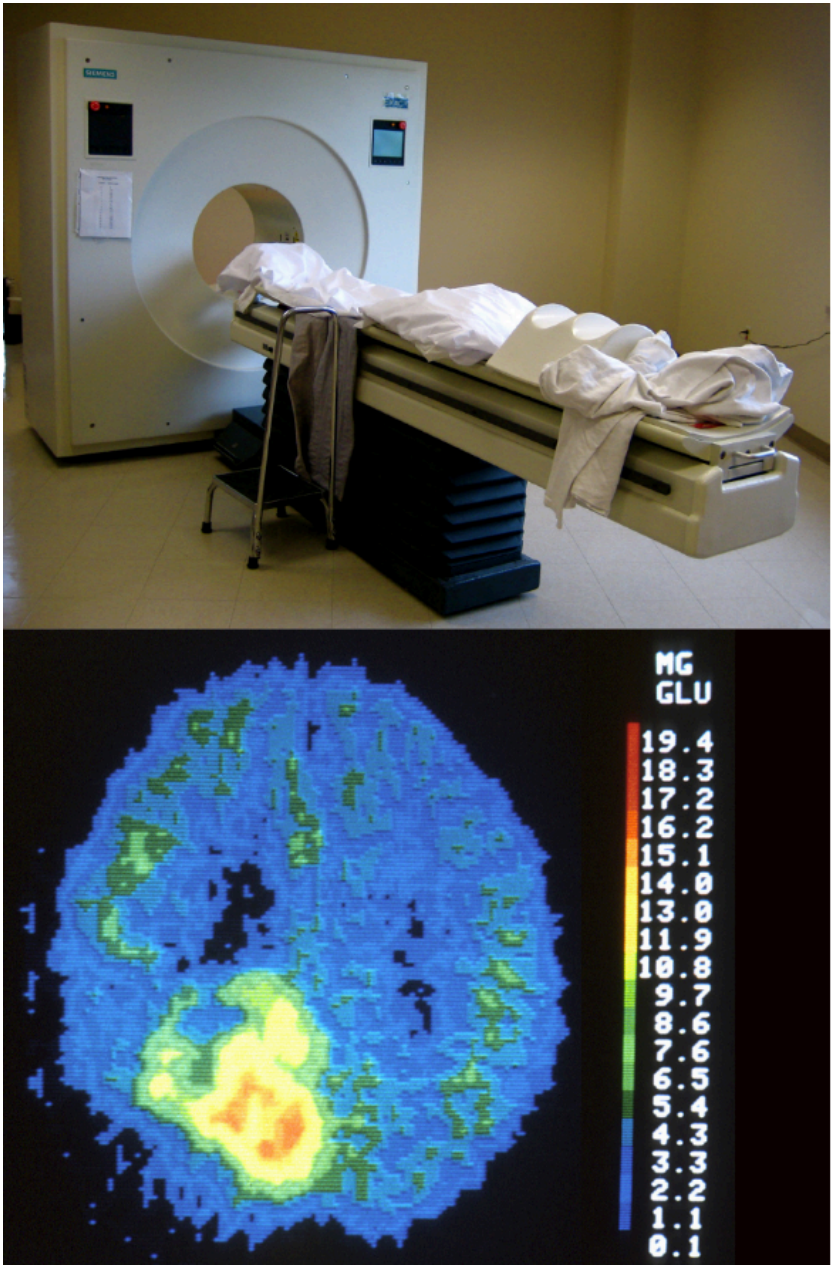


Figure 3.27. A CT scan can be used to show brain tumors. (a) The image on the left shows a healthy brain, whereas (b) the image on the right indicates a brain tumor in the left frontal lobe.

PET SCANS

Positron emission tomography (PET) scans are functional brain scans because they create pictures of the living, active brain (Figure 3.28a and b). An individual receiving a PET scan drinks, inhales or is injected with a mildly radioactive substance, called a tracer. Common tracers include glucose and oxygen. Once in the bloodstream, the amount of tracer in any given region of the brain can be monitored. As a brain area becomes more active, more blood flows to that area. This technique was frequently used to assess whether tumors are metabolically active or benign. PET scans are sometimes used in conjunction with CT scans to provide better information about the location of the tumor. CT scans contribute clear images of brain structures, while PET scans show the brain's activity. This combined technique is also useful for studies looking at how psychological disorders affect neurotransmitter function. PET scans can create a rough map of

active and inactive areas of the brain during a given behavior, e.g., listening to someone speaking compared to listening to white noise. However, PET scans lack detail, are unable to pinpoint events precisely in time, and require that the brain be exposed to radiation. Therefore, these kinds of research studies now use fMRI instead.


MG
GLU
19.4
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17.2
16.2
15.1
14.0
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11.9
10.8
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7.6
6.5
5.4
4.3
3.3
2.2
1.1
0.1

Figure 3.28. (a) PET scanner. (b) A PET scan is helpful for showing activity in different parts of the brain. In this figure, we can see a yellow area, which corresponds to a metabolically active tumor.

MAGNETIC RESONANCE IMAGING (MRI) AND FMRI

In magnetic resonance imaging (MRI), a person lies inside a scanner that generates a strong magnetic field (Figure 3.29a). The hydrogen atoms in the body's cells align with the magnetic field. Then a radio wave is turned on, which knocks the hydrogen atoms off their alignment, when the radio wave is turned off, the hydrogen atoms emit electromagnetic signals as they return to their original positions. Tissues of different densities give off different signals, which a computer interprets and displays on a monitor giving highly detailed anatomical images of the structures of the brain (Figure 3.29b). MRI provides more detailed images of the brain than CT scans, and does not expose participants to radiation, but it is more expensive.

Figure 3.29a. MRI scanner.

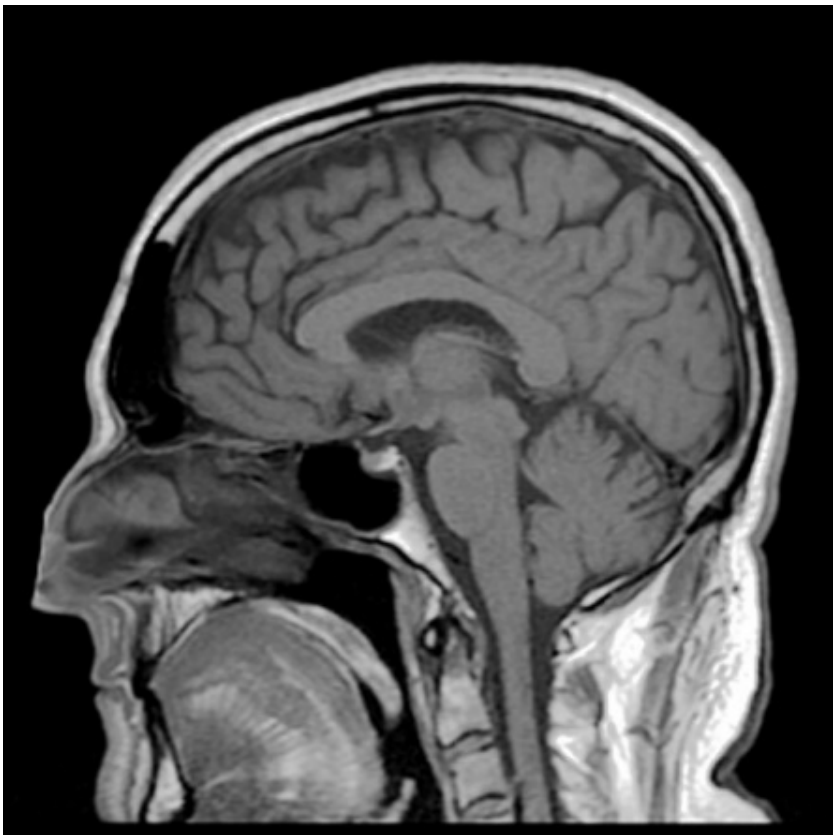

Figure 3.29(b). MRI scan of the brain.

FMRI

Functional magnetic resonance imaging (fMRI) operates on the same principles as MRI, but it tracks blood flow and oxygen levels as a measure of brain activity. So, while MRI tells us about structure, fMRI tells us about function. fMRI scans (Figure 3.30) are safer and provide more detailed information than PET scans. Brain activity on an fMRI is usually shown in different colors, which are then superimposed on an MRI scan. In Figure 3.30, the scale shows us

that the yellow regions are more active than the orange and red ones for this particular task comparison.

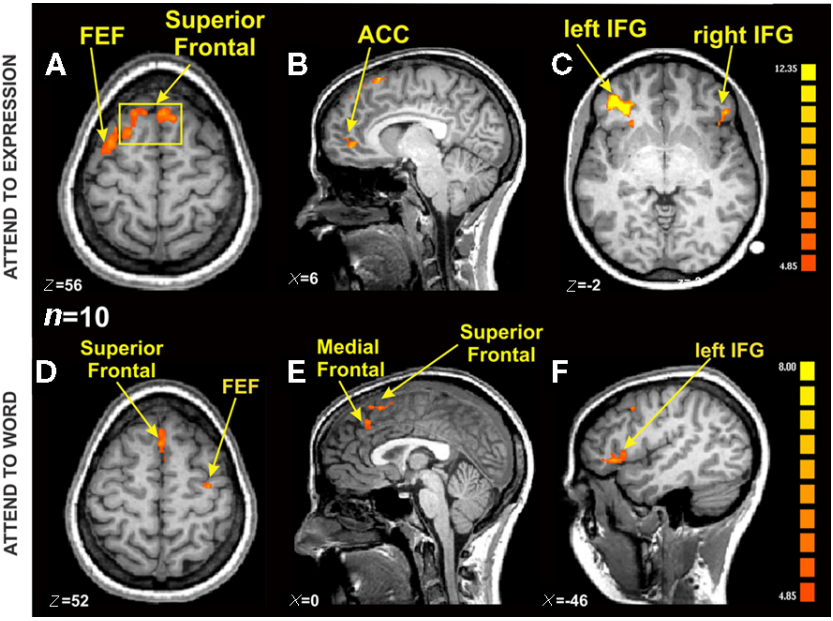

Figure 3.30. An fMRI scan during an emotional Stroop task.

The data in Figure 3.30 come from an emotional Stroop study where participants were asked to look at emotional faces with a word superimposed on them (Figure 3.31). In half of the stimuli, the words matched the facial expression, these are called congruent stimuli, and for the half, the words and expressions did not match, these are called incongruent stimuli. In one part of the study, participants were asked to identify the facial expression and in the other part of the study, they identified the word. Participants were faster and more accurate when the label matched the facial expression. Figure 3.30 shows the increased activity that occurs for incongruent (non-matching) compared to congruent (matching) stimuli, which we can see is mostly in the frontal lobe. We can see that one of active areas is the left inferior frontal gyrus (IFG), also

known as Broca's area, which activates when people speak or think about speaking.

Figure 3.31. Example of an incongruent emotional Stroop stimulus. Here, the word "sad" does not match the happy facial expression.

ELECTROENCEPHALOGRAPHY (EEG) AND EVENT-RELATED POTENTIALS (ERPS)

In some situations, it is helpful to understand how the brain is processing information without needing information on the actual location of the activity. Electroencephalography (EEG) provides this information by measuring the electrical activity produced by the brain. In EEG recording, electrodes are placed on a person's head using conductive gel (Figure 3.32a). The signals received by the electrodes are displayed on a computer screen and show how the brain voltage changes over time, with an accuracy within milliseconds (Figure 3.32b). EEG recordings are especially helpful to

researchers studying sleep patterns among individuals with sleep disorders and for studies of epilepsy.

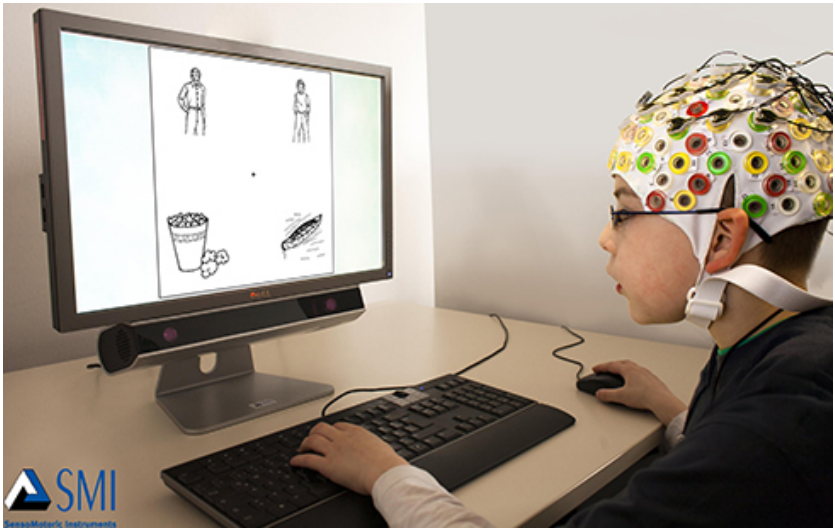

Figure 3.32a. A boy wearing an EEG cap, which has electrodes sewn into it. He is performing a task and so the researchers can measure changes in brain activity (ERPs) in responses to the different stimuli

Figure 3.32b. EEG recording from 16 electrodes

Researchers also use EEG technology to study changes in brain activity in response to different kinds of stimuli. These averaged electrical brain responses are often referred to as brain waves or event-related potentials (ERPs). For example, the emotional Stroop stimuli described in Figure 3.31 were also used in an ERP study by a different group of researchers (Zhu et al., 2018). The ERP study provides information about the timing of the increased effort needed to process conflicting information—when the label and

facial expression do not match. Sad faces with incongruent labels produced larger P3 brain waves (positive brain waves starting about 300 ms after seeing the face) compared to sad faces with the correct label. This difference in neural effort was apparent starting around 300 milliseconds after seeing the face/label combination (see Figure 3.33).

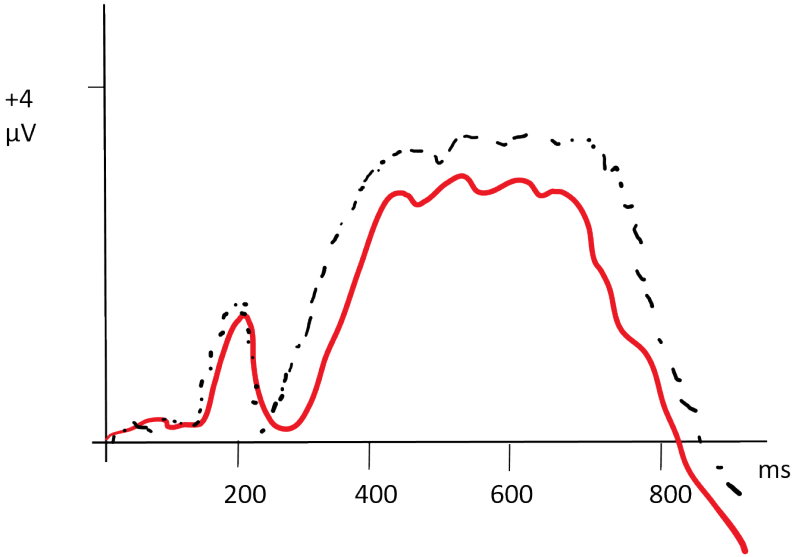

Figure 3.33. ERPs to emotional Stroop face/word stimuli. The P3 wave to the incongruent (black dotted) condition is larger (more positive) than the congruent condition (red)

NEUROPLASTICITY

Although new neurons are only found in a few locations (hippocampus and olfactory bulb) in the central nervous system after we are born, the nervous system is plastic – meaning that it can change and adapt. In fact, it is constantly changing and adapting. Neuroplasticity occurs in response to personal

experiences as we learn new things. It often involves the creation of new synapses, pruning of synapses that are no longer used and changes in glial cells. However, neuroplasticity can also occur in response to damage or injury. Our brains are most plastic when we are young, but even older people, who suffer brain damage, such as from a stroke often recover at least partially, especially with the help of therapy.

One remarkable story relating to neuroplasticity is that of Bob Woodruff, a reporter for ABC, who suffered a traumatic brain injury after a bomb exploded next to the vehicle he was in while covering a news story in Iraq. Woodruff was put in a medical coma for 36 days to reduce his brain swelling. However, as a consequence of his injuries, Woodruff experienced many cognitive problems, including difficulties with memory and language. However, over time and with the aid of intensive amounts of cognitive and speech therapy, Woodruff has shown an incredible recovery of function (Fernandez, 2008). Mostly neuroplasticity is helpful. But, in some cases, the changes that occur in the brain in response to injury can cause further problems, such as phantom limb syndrome, where people experience pain in a limb that has been amputated.

Link to Learning
Learn more about Bob Woodruff's story here

Link to Learning

Check out this NPR segment about amputees' experiences of "phantom limbs" following amputation to learn more.

Brain Dead and on Life Support

What would you do if a loved one was declared brain dead but their body was being kept alive by medical equipment? Whose decision should it be to remove a feeding tube? Should medical care costs be a factor?

On February 25, 1990, a Florida woman named Terri Schiavo went into cardiac arrest, apparently triggered by a bulimic episode. She was eventually revived, but her brain had been deprived of oxygen for a long time. Brain scans indicated that there was no activity in her cerebral cortex, and she suffered from severe and permanent cerebral atrophy. Schiavo was diagnosed as being in a vegetative state. Medical professionals determined that she would never again be able to move, talk, or respond in any way. To

remain alive, she required a feeding tube, and there was no chance that her situation would ever improve.

On occasion, Schiavo's eyes would move, and sometimes she would groan. Despite the doctors' insistence to the contrary, her parents believed that these were signs that she was trying to communicate with them.

After 12 years, Schiavo's husband argued that his wife would not have wanted to be kept alive with no feelings, sensations, or brain activity. Her parents, however, were very much against removing her feeding tube. Eventually, the case made its way to the courts, both in the state of Florida and at the federal level. By 2005, the courts found in favor of Schiavo's husband, and the feeding tube was removed on March 18, 2005. Schiavo died 13 days later.

Why did Schiavo's eyes sometimes move, and why did she groan? Although the parts of her brain that control thought, voluntary movement, and feeling were completely damaged, her brainstem was still intact. Her medulla and pons maintained her breathing and caused involuntary movements of her eyes and the occasional groans. Over the 15-year period that she was on a feeding tube, Schiavo's medical costs may have topped $7 million (Arnst, 2003).

Decisions surrounding the medical response to

patients declared brain dead are complex. What do you think about these issues?

3.5. The Endocrine System

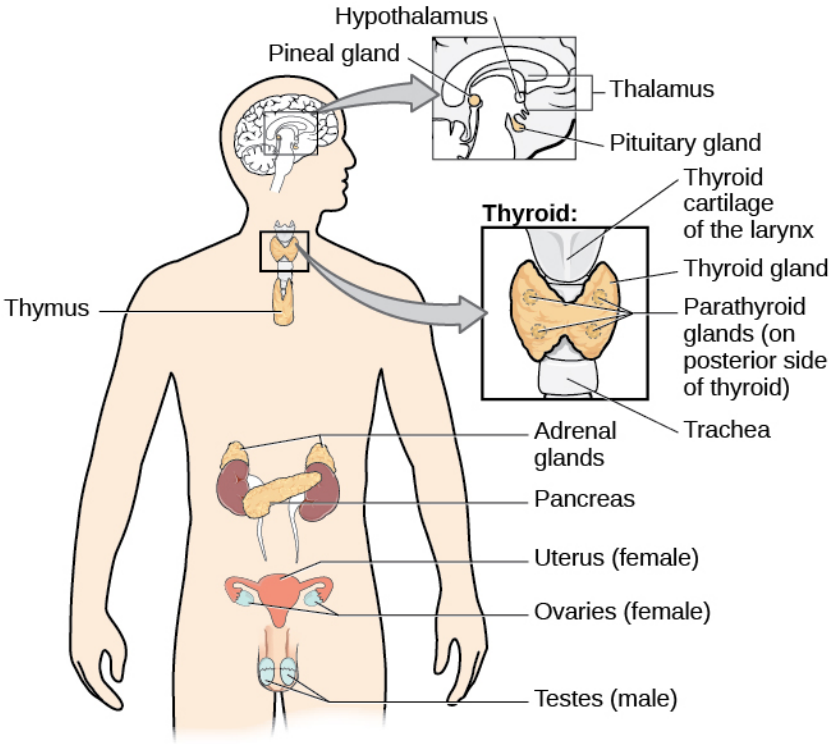

Figure 3.34. The major glands of the endocrine system

The endocrine system consists of a network of glands that produce chemical substances known as hormones (Figure 3.34). Like

neurotransmitters, hormones are chemical messengers that must bind to a receptor in order to have an effect. However, unlike neurotransmitters, which are released in close proximity to the receptors they affect, hormones are secreted into the bloodstream and travel widely throughout the body. Thus, whereas neurotransmitters' effects are localized, the effects of hormones are widespread. Also, hormones are slower to take effect, and tend to be longer lasting.

Hormones are involved in regulating all sorts of bodily functions, and they are controlled by the hypothalamus (in the central nervous system) through its action on the pituitary gland (in the endocrine system). Imbalances in hormones are related to a number of disorders, including diabetes, obesity, infertility, thyroid disease, insomnia, and even acne! This section explores some of the major glands that make up the endocrine system and the hormones secreted by these glands (Table 3.2).

Table 3.2. *Major Endocrine Glands and Associated Hormone Functions*

Endocrine Gland	Associated Hormones	Function
Pituitary	Growth hormone, releasing and inhibiting hormones (such as thyroid stimulating hormone)	Regulate growth, regulate hormone release
Thyroid	Thyroxine, triiodothyronine	Regulate metabolism and appetite
Pineal	Melatonin	Regulate some biological rhythms such as sleep cycles
Adrenal	Epinephrine, norepinephrine	Stress response, increase metabolic activities
Pancreas	Insulin, glucagon	Regulate blood sugar levels
Ovaries	Estrogen, progesterone	Mediate sexual motivation and behavior, reproduction
Testes	Androgens, such as testosterone	Mediate sexual motivation and behavior, reproduction

MAJOR GLANDS

The pituitary gland is located at the base of the brain, and receives messages from the hypothalamus. The pituitary is often referred to as the "master gland" because its messenger hormones control all the other glands in the endocrine system. In addition to messenger hormones, the pituitary also secretes growth hormone, endorphins for pain relief, and a number of key hormones that regulate fluid levels in the body.

The thyroid gland is located in the neck and releases hormones that regulate growth, metabolism, and appetite. In hyperthyroidism, or Grave's disease, the thyroid secretes too much of the hormone thyroxine, causing feelings of agitation, bulging eyes, and weight loss. In hypothyroidism, reduced hormone levels cause people with this condition to experience tiredness, weight gain, and they often complain of feeling cold. Fortunately, thyroid disorders are often treatable with medications that help reestablish a balance in the thyroid hormones.

The adrenal glands sit on top of our kidneys and secrete hormones involved in the stress response, such as epinephrine (adrenaline) and norepinephrine (noradrenaline). The pancreas is an internal organ that secretes two hormones, insulin and glucagon, which regulate blood sugar levels. These pancreatic hormones are essential for maintaining stable levels of blood sugar throughout the day by lowering blood glucose levels (insulin) when levels are too high or raising them when they are too low (glucagon). People who suffer from diabetes do not produce enough insulin; therefore, they must take medications that stimulate or replace insulin production, and they must closely control the amount of sugars and carbohydrates they consume.

The gonads secrete sexual hormones, which are important in reproduction, and mediate both sexual motivation and behavior. The female gonads are the ovaries; the male gonads are the testes.

Ovaries secrete estrogens and progesterone, and the testes secrete androgens, such as testosterone.

Dig Deeper

Athletes and Anabolic Steroids

Although it is against Federal laws and many professional athletic associations (The National Football League, for example) have banned their use, anabolic steroid drugs continue to be used by amateur and professional athletes. Anabolic steroid drugs mimic the effects of the body's own steroid hormones, like testosterone and its derivatives. These drugs have the potential to provide a competitive edge by increasing muscle mass, strength, and endurance, although not all users experience these results. Moreover, use of performance-enhancing drugs (PEDs) does not come without risks. Anabolic steroid use has been linked with a wide variety of potentially negative outcomes, ranging in severity from largely cosmetic (acne) to life threatening (heart attack). Furthermore, use of these substances can result in profound changes in mood and can increase aggressive behavior (National Institute on Drug Abuse, 2001).

Baseball player Alex Rodriguez (A-Rod) spent the latter part of his playing career at the center of a media storm regarding his use of illegal PEDs. Rodriguez excelled while using the drugs; his success

played a large role in negotiating a contract that made him the highest paid player in professional baseball. A subsequent scandal and suspension tarnished his reputation and, according to a statement he made once retired, cost him over $40 million (Gaines, 2013). Even lower-profile athletes, particularly in cycling (e.g., US Lance Armstrong who had several Tour de France trophies revoked) and Olympic sports, have been revealed as steroid users. What are your thoughts on athletes and doping? Why or why not should the use of PEDs be banned? What advice would you give an athlete who was considering using PEDs?

Chapter 3. Conclusions

In conclusion, biopsychology plays a critical role in the field of psychology by exploring the intricate connections between biology and behavior. The study of biopsychology offers a comprehensive understanding of how the brain, nervous system, and genetics contribute to various psychological processes and behaviors. Biopsychology provides valuable insights into the biological foundations of cognition, emotion, perception, social interaction, and mental health.

Chapter 3. Key Terms

action potential
electrical signal that moves down the neuron's axon

adrenal gland
sits atop our kidneys and secretes hormones involved in the stress response

agonist
drug that mimics or strengthens the effects of a neurotransmitter

all-or-none
phenomenon that incoming signal from another neuron is either sufficient or insufficient to reach the threshold of excitation

allele
specific version of a gene

amygdala
structure in the limbic system involved in our experience of emotion and tying emotional meaning to our memories

antagonist
drug that blocks or impedes the normal activity of a given neurotransmitter

auditory cortex
strip of cortex in the temporal lobe that is responsible for processing auditory information

autonomic nervous system
controls our internal organs and glands

axon
major extension of the soma

biological perspective
view that psychological disorders like depression and schizophrenia are associated with imbalances in one or more neurotransmitter systems

Broca's area
region in the left hemisphere that is essential for language production

central nervous system (CNS)
brain and spinal cord

cerebellum
hindbrain structure that controls our balance, coordination, movement, and motor skills, and it is thought to be important in processing some types of memory

cerebral cortex
surface of the brain that is associated with our highest mental capabilities

chromosome
long strand of genetic information

computerized tomography (CT) scan
imaging technique in which a computer coordinates and integrates multiple x-rays of a given area

corpus callosum
thick band of neural fibers connecting the brain's two hemispheres

dendrite
branch-like extension of the soma that receives incoming signals from other neurons

deoxyribonucleic acid (DNA)
helix-shaped molecule made of nucleotide base pairs

diabetes
disease related to insufficient insulin production

dominant allele

allele whose phenotype will be expressed in an individual that possesses that allele

electroencephalography (EEG)
recording the electrical activity of the brain via electrodes on the scalp

endocrine system
series of glands that produce chemical substances known as hormones

epigenetics
study of gene-environment interactions, such as how the same genotype leads to different phenotypes

fight or flight response
activation of the sympathetic division of the autonomic nervous system, allowing access to energy reserves and heightened sensory capacity so that we might fight off a given threat or run away to safety

forebrain
largest part of the brain, containing the cerebral cortex, the thalamus, and the limbic system, among other structures

fraternal twins
twins who develop from two different eggs fertilized by different sperm, so their genetic material varies the same as in non-twin siblings

frontal lobe
lobe involved in reasoning, motor control, emotion, and language; contains motor cortex

functional magnetic resonance imaging (fMRI)
MRI that shows changes in metabolic activity over time

gene
sequence of DNA that controls or partially controls physical characteristics

genetic environmental correlation
view of gene-environment interaction that asserts our genes affect

our environment, and our environment influences the expression of our genes

genotype
genetic makeup of an individual

glial cell
nervous system cell that provides physical and metabolic support to neurons, including neuronal insulation and communication, and nutrient and waste transport

gonad
secretes sexual hormones, which are important for successful reproduction, and mediate both sexual motivation and behavior

gyrus
(plural: gyri) bump or ridge on the cerebral cortex

hemisphere
left or right half of the brain

heterozygous
consisting of two different alleles

hindbrain
division of the brain containing the medulla, pons, and cerebellum

hippocampus
structure in the temporal lobe associated with learning and memory

homeostasis
state of equilibrium—biological conditions, such as body temperature, are maintained at optimal levels

homozygous
consisting of two identical alleles

hormone
chemical messenger released by endocrine glands

hypothalamus
forebrain structure that regulates sexual motivation and behavior and a number of homeostatic processes; serves as an interface between the nervous system and the endocrine system

identical twins
twins that develop from the same sperm and egg

lateralization
concept that each hemisphere of the brain is associated with specialized functions

limbic system
collection of structures involved in processing emotion and memory

longitudinal fissure
deep groove in the brain's cortex

magnetic resonance imaging (MRI)
magnetic fields used to produce a picture of the tissue being imaged

medulla
hindbrain structure that controls automated processes like breathing, blood pressure, and heart rate

membrane potential
difference in charge across the neuronal membrane

midbrain
division of the brain located between the forebrain and the hindbrain; contains the reticular formation

motor cortex
strip of cortex involved in planning and coordinating movement

mutation
sudden, permanent change in a gene

myelin sheath
fatty substance that insulates axons

neuron
cells in the nervous system that act as interconnected information processors, which are essential for all of the tasks of the nervous system

neuroplasticity
nervous system's ability to change

neurotransmitter
chemical messenger of the nervous system

Nodes of Ranvier
open spaces that are found in the myelin sheath that encases the axon

occipital lobe
lobe associated with visual processing; contains the primary visual cortex

pancreas
secretes hormones that regulate blood sugar

parasympathetic nervous system
associated with routine, day-to-day operations of the body

parietal lobe
brain lobe involved in processing various sensory and perceptual information; contains the primary somatosensory cortex

peripheral nervous system (PNS)
connects the brain and spinal cord to the muscles, organs and senses in the periphery of the body

phenotype
individual's inheritable physical characteristics

pituitary gland
secretes a number of key hormones, which regulate fluid levels in the body, and a number of messenger hormones, which direct the activity of other glands in the endocrine system

polygenic
multiple genes affecting a given trait

pons
hindbrain structure that connects the brain and spinal cord; involved in regulating brain activity during sleep

positron emission tomography (PET) scan
involves injecting individuals with a mildly radioactive substance and monitoring changes in blood flow to different regions of the brain

prefrontal cortex

area in the frontal lobe responsible for higher-level cognitive functioning

psychotropic medication
drugs that treat psychiatric symptoms by restoring neurotransmitter balance

receptor
protein on the cell surface where neurotransmitters attach

recessive allele
allele whose phenotype will be expressed only if an individual is homozygous for that allele

resting potential
the state of readiness of a neuron membrane's potential between signals

reticular formation
midbrain structure important in regulating the sleep/wake cycle, arousal, alertness, and motor activity

reuptake
neurotransmitter is pumped back into the neuron that released it

semipermeable membrane
cell membrane that allows smaller molecules or molecules without an electrical charge to pass through it, while stopping larger or highly charged molecules

soma
cell body

somatic nervous system
relays sensory and motor information to and from the CNS

somatosensory cortex
essential for processing sensory information from across the body, such as touch, temperature, and pain

substantia nigra
midbrain structure where dopamine is produced; involved in control of movement

sulcus
(plural: sulci) depressions or grooves in the cerebral cortex

sympathetic nervous system
involved in stress-related activities and functions

synaptic cleft
small gap between two neurons where communication occurs

synaptic vesicle
storage site for neurotransmitters

temporal lobe
brain lobe associated with hearing, memory, emotion, and some aspects of language; contains primary auditory cortex

terminal button
axon terminal containing synaptic vesicles

thalamus
sensory relay for the brain

theory of evolution by natural selection
states that organisms that are better suited for their environments will survive and reproduce compared to those that are poorly suited for their environments

threshold of excitation
level of charge in the membrane that causes the neuron to become active

thyroid
secretes hormones that regulate growth, metabolism, and appetite

ventral tegmental area (VTA)
midbrain structure where dopamine is produced: associated with mood, reward, and addiction

Wernicke's area
important for speech comprehension

Chapter 3. Personal Reflection Questions

- You share half of your genetic makeup with each of your parents, but you are no doubt very different from both of them. Spend a few minutes jotting down the similarities and differences between you and your parents. How do you think your unique environment and experiences have contributed to some of the differences that you see?
- You probably have your fair share of stress. What situations are your most common sources of stress? What can you do to try to minimize the negative consequences of these particular stressors in your life? From a biological standpoint, why is this important to do?
- Consider the impact of hormones on your own physiological and psychological states. Reflect on how fluctuations in hormone levels, such as during puberty or stress, have influenced your mood, energy levels, or behavior.
- Reflect on the concept of neuroplasticity and its implications for personal growth and learning. How might an understanding of neuroplasticity influence your approach to personal development and lifelong learning?

Chapter 3. Critical Thinking Questions

- Before the advent of modern imaging techniques, scientists and clinicians relied on autopsies of people who suffered brain injury with resultant change in behavior to determine how different areas of the brain were affected. What are some of the limitations associated with this traditional kind of approach?
- Chemical messengers are used in both the nervous system and the endocrine system. What properties do these two systems share? What properties are different? Which one would be faster? Which one would result in long-lasting changes?
- Hormone secretion is often regulated through a negative feedback mechanism, which means that once a hormone is secreted it will cause the hypothalamus and pituitary to shut down the production of signals necessary to secrete the hormone in the first place. Most oral contraceptives are made of small doses of estrogen and/or progesterone. Why would this be an effective means of contraception?
- Biopsychology often relies on studying animal models to gain insights into human behavior. What are the ethical

considerations involved in conducting research on animals, and how do you personally navigate the balance between scientific advancement and animal welfare?

- The study of biopsychology often involves complex research methods, such as brain imaging techniques and genetic studies. What are the potential limitations and biases that can arise in biopsychological research, and how can you be a discerning consumer of scientific information?

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CHAPTER 4. CONSCIOUSNESS AND SLEEP

Figure 4.1. Two sleeping children are depicted in this 1895 oil painting titled Zwei schlafende Mädchen auf der Ofenbank, which translates as "two sleeping girls on the stove," by Swiss painter Albert Anker.)

Our lives involve regular, dramatic changes in the degree to which we are aware of our surroundings and our internal states. While

awake, we often feel alert and aware of the many important things going on around us. Our experiences change dramatically while we are in deep sleep and again when we are dreaming. Some people also experience altered states of consciousness through meditation, hypnosis, trances, or by using alcohol and other drugs.

This chapter will discuss states of consciousness with a particular emphasis on sleep. The different stages of sleep will be identified, and sleep disorders will be described. The chapter will close with discussions of altered states of consciousness produced by hypnosis, meditation, drugs, and trance-like states.

LEARNING OBJECTIVES

- Describe what is meant by consciousness
- Explain how circadian rhythms are involved in regulating the sleep-wake cycle, and how circadian cycles can be disrupted
- Discuss the consequences of having a sleep debt and who is most likely to have them
- Describe areas of the brain and the hormones involved in sleep
- Describe theories aimed at explaining the function of sleep, and why we dream
- Describe the different stages of sleep
- Explain the role that REM and non-REM sleep play in learning and memory
- Describe the symptoms and treatments of insomnia
- Differentiate between the parasomnias
- Describe treatments for different types of parasomnias
- Describe how psychoactive substances affect

consciousness

- Differentiate between hypnosis, meditation, and trance
- Discuss the use of altered states of consciousness in different world cultures

4.1. What Is Consciousness?

Consciousness describes our awareness of our internal states and external stimuli. Internal states include feeling pain, hunger, thirst, sleepiness, thoughts, and emotions. External stimuli include the light from the sun, the warmth of a room, and the voice of a friend.

We experience different states of consciousness and different levels of awareness on a regular basis. We can describe consciousness as a continuum that ranges from full awareness to a deep sleep or coma. Sleep is a state marked by relatively low levels of physical activity and reduced sensory awareness that is distinct from periods of rest that occur during wakefulness. Wakefulness is characterized by high levels of sensory awareness, thought, and behavior. Beyond being awake or asleep, there are many other states of consciousness people experience. These include daydreaming, intoxication, and unconsciousness due to anesthesia. Often, we are not completely aware of our surroundings, even when we are fully awake. For instance, have you ever daydreamed while walking home from work or school without really thinking about the walk itself? You were capable of finding your way even though you were not aware of doing so. Many of these processes, like much of psychological behavior, are rooted in our biology.

BIOLOGICAL RHYTHMS

Biological rhythms are internal rhythms of biological activity. A woman's menstrual cycle is one example of a biological rhythm—a recurring, cyclical pattern of bodily changes. One complete menstrual cycle takes about 28 days—a lunar month—but many biological cycles are much shorter. For example, body temperature fluctuates cyclically over a 24-hour period (Figure 4.2). Alertness is associated with higher body temperatures, and sleepiness with lower body temperatures.

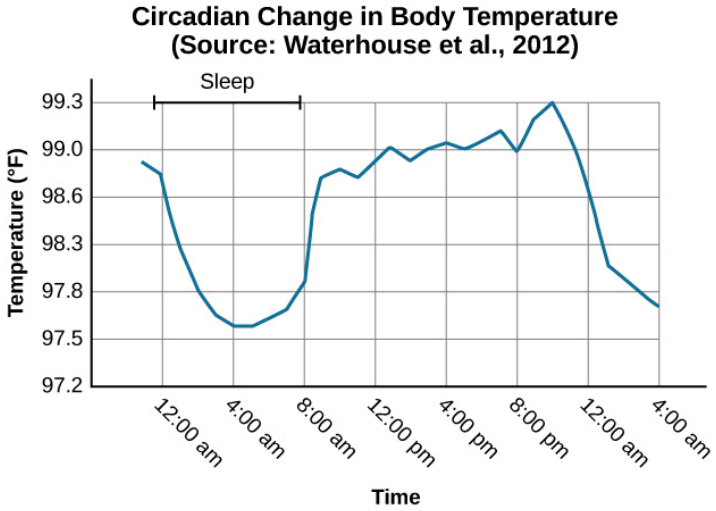


Figure 4.2. This chart illustrates the circadian change in body temperature over 28 hours in a group of eight young men. Body temperature rises throughout the waking day, peaking in the afternoon, and falls during sleep with the lowest point occurring during the very early morning hours.

This pattern of temperature fluctuation, which repeats every day, is one example of a circadian rhythm. A circadian rhythm is a biological rhythm that takes place over a period of about 24 hours. Our sleep-wake cycle, which is linked to our environment's natural light-dark cycle, is perhaps the most obvious example of a circadian

rhythm, but we also have daily fluctuations in heart rate, blood pressure, blood sugar, and body temperature. Some circadian rhythms play a role in changes in our state of consciousness.

Biological rhythms are controlled by an internal clock in the brain. As shown in Figure 4.3. the brain's clock mechanism is located in an area of the hypothalamus known as the suprachiasmatic nucleus (SCN). The axons of light-sensitive neurons in the retina provide information to the SCN based on the amount of light present, allowing this internal clock to be synchronized with the outside world (Klein et al., 1991; Welsh et al., 2010).

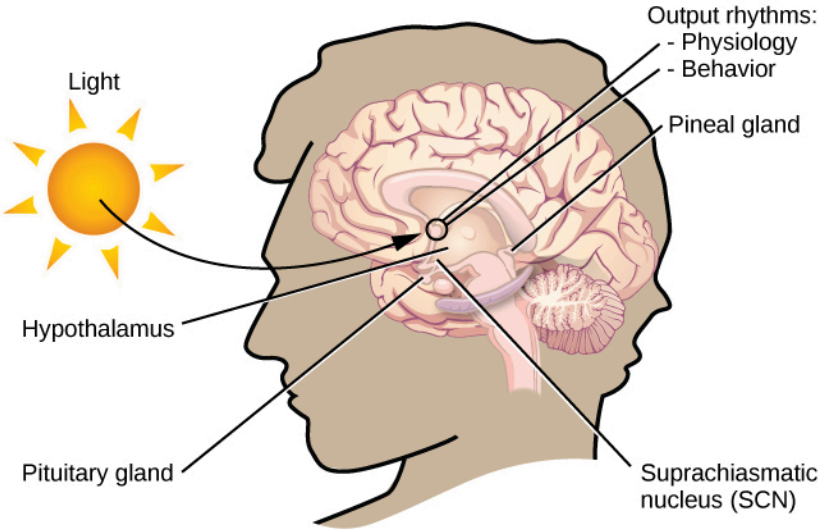


Figure 4.3. The suprachiasmatic nucleus (SCN) serves as the brain's clock mechanism. The clock can reset itself with light information received through projections from the retina.

PROBLEMS WITH CIRCADIAN RHYTHMS

Generally, and for most people, our circadian cycles are aligned

with the outside world. For example, most people sleep during the night and are awake during the day. One important regulator of sleep-wake cycles is the hormone melatonin, which is released by the pineal gland. Melatonin is thought to be involved in the regulation of various other biological rhythms and has been linked to effective immune function (Hardeland et al., 2006). Melatonin release is stimulated by darkness and inhibited by light.

There are considerable individual differences in our sleep-wake cycles. For instance, some people would say they are morning people, while others would consider themselves to be night owls. These individual differences in circadian patterns of activity are known as a person's chronotype, and research demonstrates that morning larks and night owls differ with regard to sleep regulation (Taillard et al., 2003). Sleep regulation refers to the brain's control of switching between sleep and wakefulness, as well as coordinating this cycle with the outside world.

Link to Learning

Watch this brief video about circadian rhythms and how they affect sleep to learn more.

Link to learning

Spend 5-10 minutes taking this online assessment to find out whether you are an Owl or a Lark

DISRUPTIONS OF NORMAL SLEEP

Whether lark, owl, or somewhere in between, there are situations in which a person's circadian clock gets out of synchrony with the external environment. One way that this happens involves traveling across multiple time zones. When we do this, we often experience jet lag. Jet lag is a collection of symptoms that results from the mismatch between our internal circadian cycles and our environment. These symptoms include fatigue, sluggishness, irritability, and insomnia, i.e., a consistent difficulty in falling or staying asleep for at least three nights a week over the period of a month (Roth, 2007).

Individuals who do rotating shift work are also likely to experience disruptions in circadian cycles. Rotating shift work refers to a work schedule that changes from early to late on a daily or weekly basis. For example, a person may work from 7:00 a.m. to 3:00 p.m. on Monday, 3:00 a.m. to 11:00 a.m. on Tuesday, and 11:00 a.m. to 7:00 p.m. on Wednesday. In such instances, the individual's schedule changes so frequently that it becomes difficult for a normal circadian rhythm to be maintained. This often results in sleeping problems, and it can lead to signs of depression and anxiety. These kinds of schedules are common for individuals working in health care professions and service industries, and they are often associated with persistent feelings of exhaustion and

agitation that can make someone more prone to making mistakes on the job (Gold et al., 1992; Presser, 1995).

Rotating shift work has pervasive effects on the lives and experiences of individuals engaged in that kind of work, which is clearly illustrated in stories reported in a qualitative study that researched the experiences of middle-aged nurses who worked rotating shifts (West et al., 2009). Several of the nurses interviewed commented that their work schedules affected their relationships with their family. One of the nurses said,

> *"If you've had a partner who does work regular job 9 to 5 office hours . . . the ability to spend time, good time with them when you're not feeling absolutely exhausted . . . that would be one of the problems that I've encountered*." (West et al., 2009, p. 114)

While disruptions in circadian rhythms can have negative consequences, there are things we can do to help us realign our biological clocks with the external environment. Some of these approaches, such as using a bright light as shown in Figure 4.4, have been shown to alleviate some of the problems experienced by individuals suffering from jet lag or from the consequences of rotating shift work. Because the biological clock is driven by light, exposure to bright light during working shifts and dark exposure when not working, can help combat insomnia and symptoms of anxiety and depression (Huang et al., 2013).

Figure 4.4. Devices like this are designed to provide exposure to bright light to help people maintain a regular circadian cycle. They can be helpful for people working night shifts or for people affected by seasonal variations in light.

Link to Learning

Watch this video about overcoming jet lag to learn some tips.

INSUFFICIENT SLEEP

When people have difficulty getting enough sleep due to their work or the demands of day-to-day life, they accumulate a sleep debt.

A person with a sleep debt does not get sufficient sleep on a chronic basis. The consequences of sleep debt include decreased levels of alertness and mental efficiency. Interestingly, since the advent of electric light, the amount of sleep that people get has declined. While we certainly welcome the convenience of having artificial lighting in our lives, we also suffer the consequences of reduced amounts of sleep because we are more active during the nighttime hours than our ancestors were. As a result, many of us sleep less than 7–8 hours a night and accrue a sleep debt. There is tremendous variation in any given individual's sleep needs. The amount of sleep we need varies across the lifespan. The National Sleep Foundation (n.d.) cites research estimating that newborns require the most sleep (between 12 and 18 hours a day) but this amount declines to just 7–9 hours by the time we are adults. A meta-analysis, which is a study that combines the results of many related studies, indicates that by the time we are 65 years old, we average fewer than 7 hours of sleep per day (Ohayon et al., 2004).

If you lie down to take a nap and fall asleep very easily, you may have sleep debt. Given that college students are notorious for suffering from significant sleep debt (Hicks et al., 1992; 2001; Miller et al., 2010), it is likely that you and your classmates deal with sleep debt-related issues on a regular basis. In 2015, the National Sleep Foundation updated their sleep duration hours, to better accommodate individual differences. Table 4.1 shows the new recommendations, which describes sleep durations which vary from "recommended" and "may be appropriate", to "not recommended".

Table 4.1. *Sleep Needs at Different Ages*

AGE	RECOMMENDED	MAY BE APPROPRIATE	NOT RECOMMENDED
0–3 months	14–17 hours	11–13 hours 18–19 hours	Fewer than 11 hours More than 19 hours
4–11 months	12–15 hours	10–11 hours 16–18 hours	Fewer than 10 hours More than 18 hours
1–2 years	11–14 hours	9–10 hours 15–16 hours	Fewer than 9 hours More than 16 hours
3–5 years	10–13 hours	8–9 hours 14 hours	Fewer than 8 hours More than 14 hours
6–13 years	9–11 hours	7–8 hours 12 hours	Fewer than 7 hours More than 12 hours
14–17 years	8–10 hours	7 hours 11 hours	Fewer than 7 hours More than 11 hours
18–25 years	7–9 hours	6 hours 10–11 hours	Fewer than 6 hours More than 11 hours
26–64 years	7–9 hours	6 hours 10 hours	Fewer than 6 hours More than 10 hours
≥65 years	7–8 hours	5–6 hours 9 hours	Fewer than 5 hours More than 9 hours

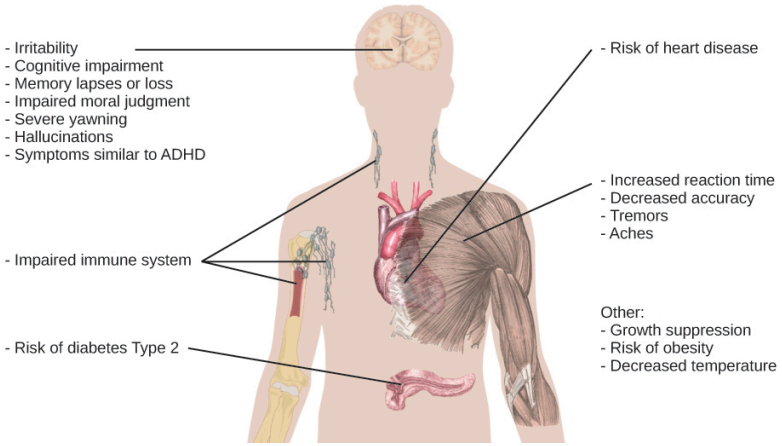


Figure 4.5. Some of the negative consequences of sleep deprivation..

Sleep deprivation can have significant effects on the body, cognition, and health (see Figure 4.5). As mentioned earlier, lack

of sleep can result in decreased mental alertness and impaired cognitive function. In addition, sleep deprivation often results in depression-like symptoms. These effects can occur as a function of accumulated sleep debt or in response to more acute periods of sleep deprivation. It may surprise you to know that sleep deprivation is associated with chronic disease including diabetes, obesity, increased blood pressure, heart disease, increased levels of stress hormones, and reduced immune functioning (Banks & Dinges, 2007; Center for Disease Control, 2022a). Studies have shown that having less than the optimal amount of sleep reduces antibody production after a vaccination. So, being sleep-deprived can reduce the efficacy of vaccinations (Stickgold, 2015).

The Center for Disease Control estimates that over one-third of adults (35%) in the United States do not get enough sleep (defined as less than seven hours per day). Short sleep duration is slightly more common among males (33.4%) than females (32.2%) and is most common among people in the 25 to 44-year-old age group (36.4%). There is considerable racial variation among who is most likely to be sleep deprived. Hawaiian and other Pacific Islanders (47%) and African-American and non-Hispanic Blacks (43.5%) are the two groups with the highest proportion of people with short sleep duration. In contrast, non-Hispanic Whites (30.2%)and Asians (30.5%) have the lowest percentage of people who sleep less than seven hours per day (Center for Disease Control, 2022b).

A sleep deprived individual generally will fall asleep more quickly than if they were not sleep deprived. Some sleep-deprived individuals have difficulty staying awake when they stop moving (for example, when sitting and watching television or driving a car). That is why individuals suffering from sleep deprivation can put themselves and others at risk when they drive a car or work with dangerous machinery. Some research suggests that sleep deprivation affects cognitive and motor function as much as alcohol intoxication (Williamson & Feyer, 2000). Research shows

that if a person stays awake for more than 24 hours (Killgore & Weber, 2014; Killgore et al., 2007), or has repeated nights with fewer than four hours in bed (Wickens et al., 2015), they are likely to experience symptoms of extreme sleep deprivation, including irritability, distractibility, and impairments in cognitive and moral judgment. If someone stays awake for 48 hours in a row, they could start to hallucinate.

Links to Learning

Read this article about sleep needs to assess your own sleeping habits.

Use the Epworth Sleepiness Scale to assess whether you are sleep deprived

Visit this CDC website to learn more about national and state trends in sleep debt and its health consequences in adults.

Watch this video interview of Tricia Hersey (the Nap Bishop) who advocates for rest as a form of resistance against capitalism and white supremacy, which perpetuates the culture of working all the time.

4.2. What is Sleep?

We spend approximately one-third of our lives sleeping. Given the average life expectancy for U.S. citizens falls between 73 and 79 years old (Singh & Siahpush, 2006), we can expect to spend approximately 25 years of our lives sleeping. Why do we devote so much time to sleeping? Is it absolutely essential that we sleep? The two following sections will consider these questions and explore various explanations for why we sleep.

WHAT IS SLEEP?

Sleep is characterized by low levels of physical activity and reduced sensory awareness. Sleep is regulated by circadian rhythms and homeostatic mechanisms (Siegel, 2008). Homeostatic regulation of sleep is evidenced by sleep rebound following sleep deprivation. Sleep rebound refers to the fact that a sleep-deprived individual will fall asleep more quickly during subsequent opportunities for sleep. Sleep is characterized by distinct stages of brain activity of the brain that can be visualized using electroencephalography (EEG). Sleep-wake cycles are controlled by multiple brain areas acting in conjunction with one another. Some of these areas include the thalamus, the hypothalamus, and the pons (Figure 4.6).

Sleep is also associated with the secretion and regulation of a number of hormones that are important for growth, immune

system function, reproduction, and other metabolic functions. As previously mentioned, the pineal gland releases melatonin during sleep (Figure 4.6), which is involved in the regulation of various biological rhythms and immune system function (Hardeland et al., 2006). During sleep, the pituitary gland secretes both FSH and LH which are important in regulating the reproductive system (Christensen et al., 2012; Sofikitis et al., 2008). The pituitary gland also secretes growth hormone, during sleep, which plays a role in physical growth and maturation, as well as other metabolic processes (Bartke et al., 2013).

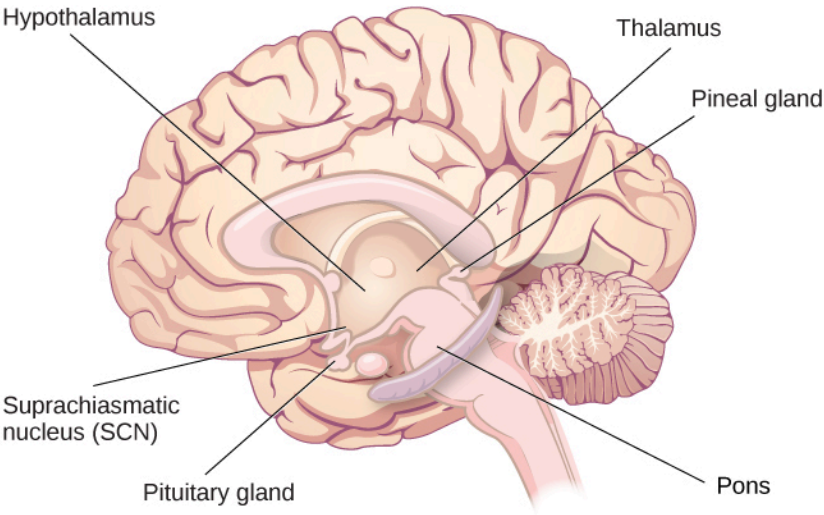


Figure 4.6. The pineal and pituitary glands secrete a number of hormones during sleep.

Sleep is not a uniform state of being. Instead, sleep is composed of several different stages that can be differentiated from one another by the patterns of brain wave activity that occur during each stage. EEG recordings allow us to measure changes in the frequency and amplitude of brain waves in different sleep stages. The frequency of a brain wave is how many peaks(or oscillations) occur in a second and it is measured in Hertz (Hz). The amplitude

is the height (size) of the brain wave. While awake, our EEG is dominated by high frequency brain waves called beta waves (13-30 Hz). Each EEG electrode measures average brain activity from the overlapping peaks and troughs produced by thousands of neurons. When we are awake, the brain is very active and neurons are more variable in the frequencies they produce and so the amplitude of the EEG is low. As we begin to fall asleep, our brain wave activity changes. Sleep can be divided into two different general phases: rapid eye movement (REM) sleep and non-REM (NREM) sleep.

REM sleep is characterized by darting movements of the eyes under closed eyelids. Brain waves during REM sleep appear very similar to brain waves during wakefulness. The pons is important for regulating REM sleep (National Institutes of Health, n.d.). NREM sleep is subdivided into three different stages. When we first fall asleep, we typically cycle through the three stages of NREM sleep, followed by a period of REM sleep. In this section, we will discuss each of these sleep stages and their associated patterns of brain wave activity (Figure 4.7).

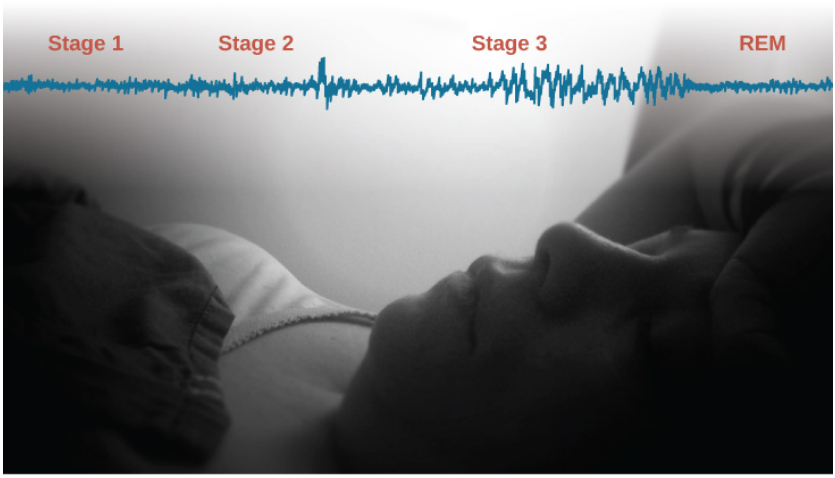


Figure 4.7. Brainwave activity changes dramatically across the different stages of sleep.

NREM STAGES OF SLEEP

As we begin to fall asleep, we enter NREM sleep, and brain wave patterns decrease in frequency and increase in amplitude, indicating that the brain activity is becoming more synchronized and less variable. This increased EEG amplitude indicates that the brain is less busy now. The first stage of NREM sleep is known as stage 1 sleep. Stage 1 sleep is a transitional phase that occurs between wakefulness and sleep, and occurs when we first drift off to sleep. During this time, both our respiration and heartbeat slow down, and we experience a marked decrease in overall muscle tension as well as a decrease in core body temperature.

The early portion of stage 1 sleep produces alpha waves (8–12 Hz), which can also be produced when someone is very relaxed and awake (Figure 4.8). In the later parts of stage 1 sleep, there is an increase in theta wave (4–7 Hz) activity. It is relatively easy to wake someone from stage 1 sleep; in fact, people often report that they have not been asleep if they are awoken during this sleep stage.

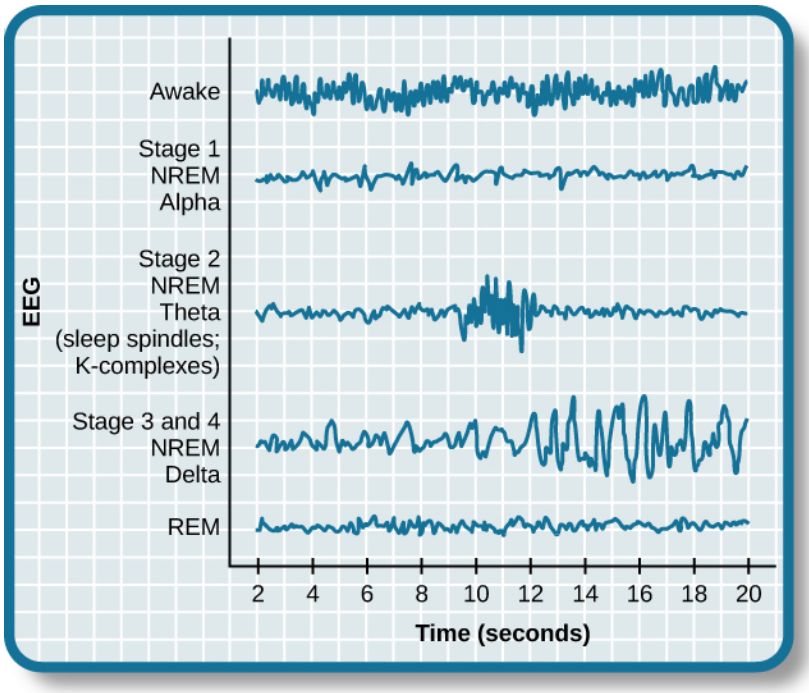


Figure 4.8. Brain activity varies considerably across different sleep stages

As we move into stage 2 sleep, the body becomes deeply relaxed. Theta waves still dominate the activity of the brain, but they are interrupted by brief bursts of high frequency activity known as sleep spindles (Figures 4.8 and 4.9). Sleep spindles may be important for learning and memory (Fogel & Smith, 2011; Poe, et al., 2010). In addition, stage 2 sleep is often associated with the appearance of K-complexes. A K-complex is a very high amplitude pattern of brain activity that may, in some cases, occur in response to environmental stimuli. Thus, K-complexes might help us to know what is going on in our environment even when we are asleep (Halász, 1993; Steriade & Amzica, 1998).

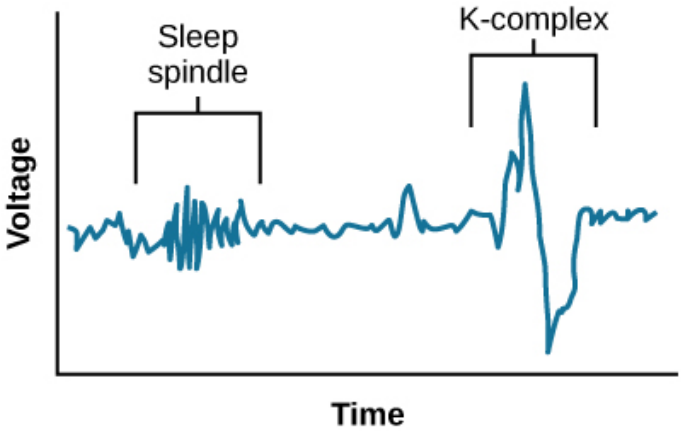


Figure 4.9. Stage 2 sleep is characterized by the appearance of both sleep spindles and K-complexes.

NREM stage 3 sleep is often referred to as deep sleep or slow-wave sleep because this stage is characterized by low frequency (less than 3 Hz), very high amplitude delta waves (Figure 4.10). Slow wave sleep is regulated by the hypothalamus and the thalamus (National Sleep Foundation, n.d.). During this time, an individual's heart rate and respiration slow dramatically, and it is much more difficult to awaken someone from sleep during stage 3 than during earlier stages. Individuals who have increased levels of alpha brain wave activity (more often associated with wakefulness and transition into stage 1 sleep) during stage 3 sleep often report that they do not feel refreshed upon waking, regardless of how long they slept (Stone et al., 2008).

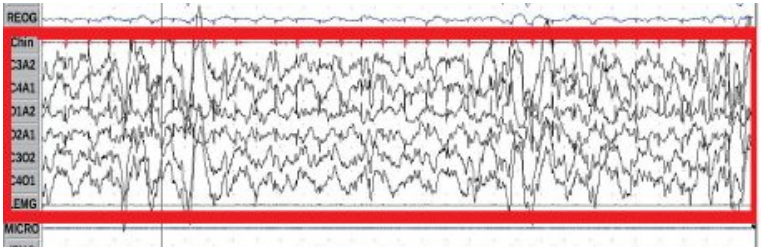


Figure 4.10. Slow wave sleep is characterized by high frequency delta waves.

REM SLEEP

As mentioned earlier, REM sleep is marked by rapid movements of the eyes. The brain waves associated with this stage of sleep are very similar to those observed when a person is awake, as shown in Figure 4.11, this is the period of sleep in which most dreaming occurs. REM sleep is also associated with paralysis of large muscle systems in the body, with the exception of those concerned with circulation and respiration. Therefore, no movement of large voluntary muscles occurs during REM sleep in a normal individual; REM sleep is often referred to as paradoxical sleep because of the high level of brain activity but lack of muscle tone. Like NREM sleep, REM has been implicated in various aspects of learning and memory (Wagner et al., 2001; Siegel, 2001).

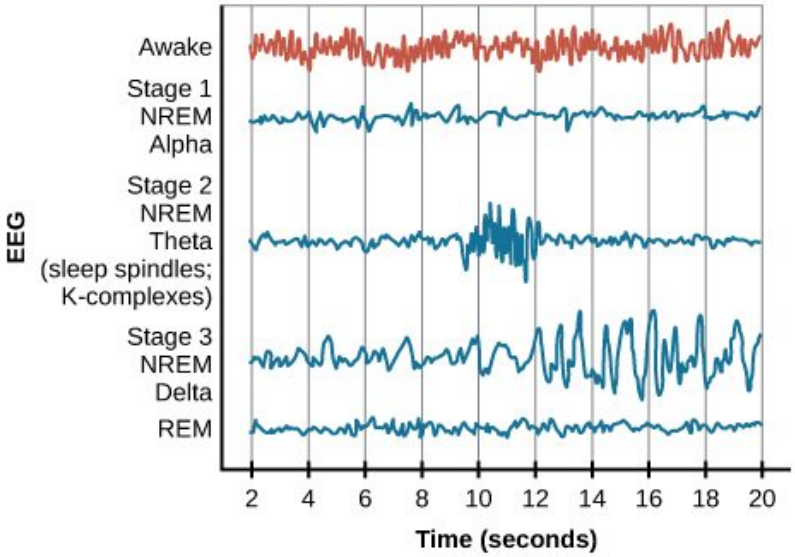


Figure 4.11. Brain activity during REM sleep (blue line at bottom) is similar to that while we are awake (red line at the top).

If people are deprived of REM sleep and then allowed to sleep without disturbance, they will spend more time in REM sleep in an effort to recoup the lost time in REM. This is known as REM rebound, and it suggests that REM sleep is also homeostatically regulated. In addition to the role that REM sleep may play in processes related to learning and memory, REM sleep may also be involved in emotional processing and regulation. In such instances, REM rebound may actually be an adaptive response to stress in nondepressed individuals, because it reduces the emotional salience of aversive events that occurred during the day (Suchecki et al., 2012). The hypnogram below (Figure 4.12) shows a person's passage through the stages of sleep. We cycle through the different stages of sleep through the night, but we have more slow wave sleep in the early parts of the night and more REM sleep later on.

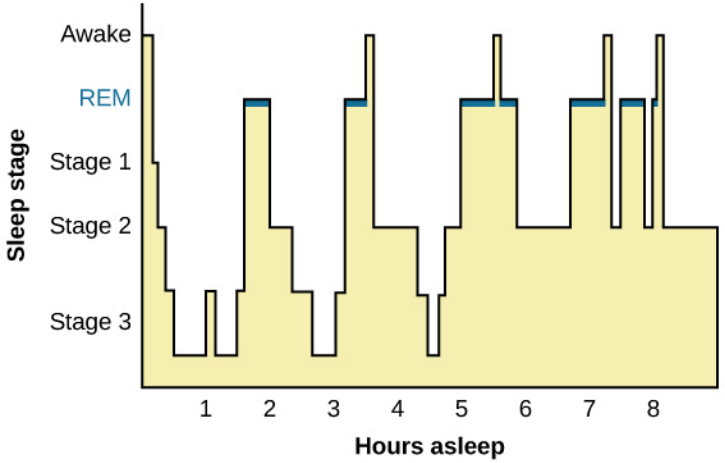


Figure 4.12. A hypnogram is a diagram of the stages of sleep as they occur during a period of sleep. This hypnogram illustrates how an individual moves through the various stages of sleep.

Link to Learning
View this video about sleep stages to learn more.

4.3. Why do we sleep?

Given the central role that sleep plays in our lives and the number of adverse consequences that have been associated with sleep deprivation, one would think that we would have a clear understanding of why it is that we sleep. Unfortunately, this is not the case; however, researchers have proposed various different hypotheses to explain the function of sleep. Let's take a look at some of them.

ADAPTIVE FUNCTION OF SLEEP

One popular hypothesis of sleep incorporates the perspective of evolutionary psychology. Evolutionary psychology is a discipline that studies how universal patterns of behavior and cognitive processes have evolved over time as a result of natural selection. One hypothesis from this perspective argues that sleep is essential to restore resources that are expended during the day. Just as bears hibernate in the winter when resources are scarce, perhaps people sleep at night to reduce their energy expenditures. While this is an intuitive explanation of sleep, there is little research to support this explanation. In fact, it has been suggested that there is no reason to think that energetic demands could not be addressed with periods of rest and inactivity while we are awake (Frank, 2006; Rial et al., 2007), and some research has actually found a negative

correlation between energetic demands and the amount of time spent sleeping (Capellini et al., 2008).

Another evolutionary hypothesis of sleep suggests that our sleep patterns evolved as an adaptive response to predatory risks, which increase in darkness. Thus we sleep in safe areas to reduce the chance of harm. Again, this is an intuitive and appealing explanation for why we sleep. Perhaps our ancestors spent extended periods of time asleep to reduce attention to themselves from potential predators. Comparative research indicates, however, that the relationship between predatory risk and sleep is very complex and equivocal. Some research suggests that species that face higher predatory risks sleep fewer hours than other species (Capellini et al., 2008), while other researchers suggest there is no relationship between the amount of time a given species spends in deep sleep and its predation risk (Lesku et al., 2006).

It is quite possible that sleep serves no single universally adaptive function, and different species have evolved different patterns of sleep in response to their unique evolutionary pressures. While we have discussed the negative outcomes associated with sleep deprivation, it should be pointed out that there are many benefits that are associated with having adequate amounts of sleep. A few such benefits listed by the National Sleep Foundation (n.d.) include maintaining healthy weight, lowering stress levels, improving mood, and increasing motor coordination, as well as better cognitive function and memory.

COGNITIVE FUNCTION OF SLEEP

Another theory regarding why we sleep involves the importance of sleep for cognitive function and memory formation (Rattenborg et al., 2007). Indeed, we know sleep deprivation results in disruptions in cognition (Brown, 2012), leading to difficulties in maintaining

attention, making decisions, and recalling long-term memories. Moreover, these impairments become more severe as the amount of sleep deprivation increases (Alhola & Polo-Kantola, 2007). Furthermore, slow-wave sleep seems essential for effective memory formation (Stickgold, 2005). Having periods of slow wave sleep after learning a new task improves subsequent performance on that task (Huberet al., 2004). Understanding the impact of sleep on cognitive function should help you understand that cramming all night for a test is often not effective and can even prove counterproductive.

Link to Learning

Watch this brief video that gives sleep tips for college students to learn more.

Getting the optimal amount of sleep has also been associated with other cognitive benefits, including better creative thinking (Cai et al., 2009; Wagner et al., 2004) and language learning (Fenn et al., 2003; Gómez et al., 2006), and improved ability in making inferential judgments (Ellenbogen et al., 2007). Effective emotion processing is also influenced by certain aspects of sleep (Walker, 2009).

Link to Learning

Watch this brief video about the relationship between sleep and memory to learn more.

DREAMS

How people think about dreams varies across different cultures and the psychology of dreaming in the Western world has also changed over time. By the late 19th century, Austrian psychiatrist Sigmund Freud had become convinced that dreams represented an opportunity to gain access to the unconscious. By analyzing dreams, Freud thought people could increase self-awareness and gain valuable insight to help them deal with the problems they faced in their lives. Freud made distinctions between the manifest and the latent content of dreams. Manifest content is the actual content, or storyline, of a dream. Latent content, on the other hand, refers to the hidden meaning of a dream. For instance, if a woman dreams about being chased by a snake, Freud might have argued that this represents the woman's fear of sexual intimacy, with the snake serving as a symbol of a man's penis.

Freud was not the only Western theorist to focus on the content of dreams. The 20th century Swiss psychiatrist, Carl Jung, believed that dreams allowed us to tap into the collective unconscious. The collective unconscious, as described by Jung, is a theoretical repository of information which he believed was shared by everyone. According to Jung, certain symbols in dreams reflect

universal archetypes, and their meanings are similar for all people regardless of culture or location.

The sleep and dream researcher, Rosalind Cartwright, however, believes that dreams simply reflect life events that are important to the dreamer. Cartwright's ideas about dreaming have found empirical support. For example, she and her colleagues published a study in which women going through divorce were asked several times over a five-month period to report the degree to which their former spouses were on their minds. These same women were awakened during REM sleep in order to provide a detailed account of their dream content. There was a significant positive correlation between the degree to which women thought about their former spouses while they were awake and the number of times their former spouses appeared in their dreams (Cartwright et al. 2006).

Allan Hobson, a neuroscientist, is credited for developing the activation-synthesis theory of dreaming. Early versions of this theory proposed that dreams were not the meaning-filled representations of angst proposed by Freud and others, but were rather the result of our brain attempting to make sense of ("synthesize") the neural activity ("activation") that was happening during REM sleep. Recent adaptations (e.g., Hobson, 2002) continue to update the theory based on accumulating evidence. For example, Hobson (2009) suggests that dreaming may represent a state of protoconsciousness. In other words, dreaming involves constructing a virtual reality in our heads based on the activation that occurs in different parts of our brain during REM sleep. For example, when we experience random activation of the motor cortex we dream that we are moving and activation of the vestibular system might lead us to dream that we are falling or flying. Hobson suggests that learning to interpret our brain activity while we are asleep, might serve as a model so that we can use similar interpretive processes when we are conscious. Among a variety of neurobiological evidence, Hobson cites research on lucid dreams as an opportunity to better understand dreaming in

general. In a lucid dream, a person becomes aware of the fact that they are dreaming, and as such, they can control the dream's content (LaBerge, 1990). Other research (Horikawa et al., 2013) has uncovered new techniques by which researchers may effectively detect and classify the visual images that occur during dreaming by using fMRI to measure brain activity. This may open the way for additional research in this area.

There are considerable cultural variations in how people view their dreams. Whites are more likely to dismiss them as unimportant compared to African Americans, people from Africa and Latinx people. In many non-White cultures, dreams are considered an important part of consciousness and are central to peoples' lives. In collectivistic cultures that embrace spirituality, religion, and magical realism, dreams are often a way to connect with the dead. Dreams often provide messages either for the dreamer or for someone else, which is consistent with the fact that people in collectivistic cultures typically have strong ties with their community. Messages in dreams vary in content, for example, they might give reassurance that the deceased is at peace, or they might provide guidance, approval, or inspiration for their loved ones for their life choices. Alternatively, messages may be prophetic or premonitory in nature. It is also usual for people who value dreams to share them with other members of their community (Comas-Diaz, 2006; Nwoye, 2017; Shafton, 2002).

4.4 Sleep Problems and Disorders

Sleep disorders are very common; between 30% and 50% of the United States population suffers from a sleep disorder at some point in their lives (Bixler et al., 1979; Hossain & Shapiro, 2002; Ohayon, 1997, 2002; Ohayon & Roth, 2002). This section will describe several sleep disorders as well as some of their treatment options.

INSOMNIA

One of the most common sleep disorders is insomnia, which is characterized by difficulty falling or staying asleep. One of the criteria for insomnia involves experiencing these symptoms for at least three nights a week for at least one month (Roth, 2007). People suffering from insomnia often experience increased levels of anxiety about their inability to fall asleep. This can lead to a self-perpetuating cycle where increased anxiety leads to increased arousal, making it even more difficult for them to fall asleep. Chronic insomnia is almost always associated with feeling overtired and may be associated with symptoms of depression.

Multiple factors can contribute to insomnia, including being elderly, drug use, lack of exercise or exercising too close to bedtime, mental health issues, and lack of healthy bedtime routines. As such, insomnia treatment often takes different

approaches depending on the cause. Individuals may need to limit their use of stimulant drugs like caffeine or increase their amount of physical exercise during the day. However, relying on over-the-counter (OTC) or prescribed sleep medications should be done with caution as many sleep medications can result in dependence, alter the nature of the sleep cycle, and can even worsen insomnia over time. People with persistent insomnia, particularly if it impacts their quality of life, should seek professional advice.

Cognitive-behavioral therapy (CBT) has been shown to be highly effective in treating insomnia (Savard et al., 2005; Williams et al., 2013). CBT is a type of psychotherapy that focuses on cognitive processes and problem behaviors. CBT treatment for insomnia may include learning stress management techniques and changing problematic behaviors that could contribute to insomnia, such as spending too much time in bed during the day.

Everyday Connection

Solutions to Support Healthy Sleep

Has something like this ever happened to you? My sophomore college housemate got so stressed out during finals sophomore year he drank almost a whole bottle of Nyquil to try to fall asleep. When he told me, I made him go and see the college therapist. Many college students struggle getting the recommended 7–9 hours of sleep each night. However, for some, it's not because of all-night partying or late-night study sessions. It's simply that they feel so overwhelmed and stressed that they

cannot fall asleep or stay asleep. One or two nights of sleep difficulty is not unusual, but if you experience anything more than that, you should seek professional advice. **Your College Counseling Center can help to advise you.**

TIPS TO MAINTAIN HEALTHY SLEEP:

1. Stick to the same sleep schedule, even on the weekends. Try going to bed and waking up at the same time every day to keep your biological clock in sync so your body gets in the habit of sleeping every night.
2. Avoid stimulating activities for an hour before bed. That includes exercise and bright light from devices, like phones and computers.
3. Exercise daily, but not right before bedtime.
4. Avoid naps.
5. Keep your bedroom temperature between 60 and 67 degrees. People sleep better in cooler temperatures.
6. Avoid alcohol, cigarettes, caffeine, and heavy meals before bed. It may feel like alcohol helps you sleep, but it actually disrupts REM sleep and leads to frequent awakenings. Heavy meals may make you sleepy, but they can also lead to frequent awakenings due to gastric distress.

7. If you cannot fall asleep, leave your bed and do something else until you feel tired again. Train your body to associate your bed with sleeping rather than other activities like studying, eating, or watching television shows.

PARASOMNIAS

A parasomnia is a type of sleep disorder characterized by unwanted and disruptive motor activity and/or experiences during sleep. These events can occur during either rapid eye movement (REM) or non-rapid eye movement (NREM) phases of sleep. Examples of parasomnias include sleepwalking, restless leg syndrome, and night terrors (Mahowald & Schenck, 2000).

SLEEPWALKING

In sleepwalking, or somnambulism, the sleeper engages in relatively complex behaviors ranging from wandering about to driving an automobile. During periods of sleepwalking, sleepers often have their eyes open, but they are not responsive to attempts to communicate with them. Sleepwalking typically occurs during slow-wave sleep, but it can occur at any time during a sleep period (Mahowald & Schenck, 2000).

Historically, somnambulism has been treated with a variety of drugs ranging from benzodiazepines to antidepressants. However, the success rate of such treatments is questionable. Guilleminault et al. (2005) found that sleepwalking was not alleviated with the use of benzodiazepines. However, patients who also suffered from sleep-related breathing problems, like sleep apnea, showed a marked decrease in sleep walking when their breathing problems were effectively treated.

Dig Deeper

A Sleepwalking Defense?

On January 16, 1997, Scott Falater sat down to dinner with his wife and children and told them about the difficulties he was experiencing on a project at work. After dinner, he prepared some materials for a church youth group that was scheduled for the following morning, and attempted to repair the family's swimming pool pump before retiring to bed. The following morning, he awoke to barking dogs and unfamiliar voices from downstairs. As he went to investigate what was going on, he was met by a group of police officers who arrested him for the murder of his wife (Cartwright, 2004; CNN, 1999).

Yarmila Falater's body was found in the family's pool with 44 stab wounds. A neighbor called the police after witnessing Falater standing over his wife's body before dragging her into the pool, and telling his dog to be quiet. He then returned to the house and changed his clothes. Upon a search of the premises, police found blood-stained clothes and a bloody knife in a bag in a Tupperware container hidden in the wheel well of Falater's car. Falater also had blood stains on his neck.

Remarkably, Falater insisted that he had no recollection of hurting his wife in any way. His children and his wife's parents all agreed that Falater had an excellent relationship with his wife and they couldn't

think of a reason that would provide any sort of motive to murder her (Cartwright, 2004).

The defense claimed that Scott Falater had killed his wife in his sleep (Cartwright, 2004; CNN, 1999). They argued that he had a history of regular episodes of sleepwalking as a child, and he had even behaved violently toward his sister once when she tried to prevent him from leaving their home in his pajamas during a sleepwalking episode. However, childhood sleep walking is fairly common. Falater had no history of sleep walking as an adult, nor did he have any apparent anatomical brain anomalies or psychological disorders. In Falater's case, a jury found him guilty of first degree murder in June of 1999 (CNN, 1999). The expert witness for the prosecution pointed out that sleep walking rarely lasts more than 10-20 minutes and the whole series of events took 45 minutes, also they said it would be unusual for the sleep walker not to be awakened by the screams of his victim. There are a number of murder cases where the sleepwalking defense has been offered, but acquittals since 1990 are very rare. In such cases, sleep scientists are now often called upon to produce physiological evidence of a parasomnia (Broughton et al., 1994; Cartwright, 2004; Mahowald, et al., 2005; Pressman, 2007).

REM SLEEP BEHAVIOR DISORDER (RBD)

People often have vivid dreams in REM sleep but our brains send messages to our muscles that prevent us acting out our dreams. In REM behavior disorder (RBD) there is a muscle paralysis failure. Consequently, individuals with RBD are often physically very active during REM sleep, especially during disturbing dreams. These behaviors vary widely, but they can include kicking, punching, scratching, yelling, and behaving like an animal that has been frightened or attacked. People with RBD can injure themselves or their sleeping partners when engaging in these behaviors. Furthermore, these types of behaviors ultimately disrupt sleep; affected individuals have no memories that these behaviors have occurred but they are likely to remember the dream associated with the activity (Arnulf, 2012).

RBD is associated with a number of neurodegenerative diseases such as Parkinson's disease. In fact, this relationship is so robust that some view the presence of RBD as a potential aid in the diagnosis of a number of neurodegenerative diseases (Ferini-Strambi, 2011). Clonazepam, an anti-anxiety medication with sedative properties, is most often used to treat RBD. It can be administered alone or in conjunction with melatonin (sleep-inducing hormone secreted by the pineal gland). As part of treatment, the sleeping environment is often modified to make it a safer place for those suffering from RBD (Zangini et al, 2011).

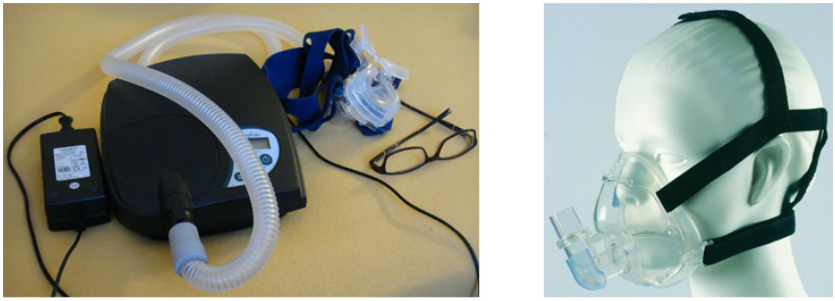
SLEEP APNEA

Sleep apnea occurs when a sleeper periodically stops breathing for 10–20 seconds (or longer) during which time they may briefly waken. People with sleep apnea often feel tired, have frequent headaches, and may experience difficulties staying mentally alert while driving or working (Henry & Rosenthal, 2013). Many individuals first seek treatment because their sleeping partners

notice that they snore loudly and/or stop breathing for extended periods of time while sleeping (Henry & Rosenthal, 2013). Sleep apnea is more common in overweight people, and is also linked with smoking and alcohol consumption. Sleep apnea may exacerbate cardiovascular disease (Sánchez-de-la-Torre et al., 2012). While sleep apnea is less common in people who are not overweight, anyone, regardless of their weight, who snores loudly or gasps for air while sleeping, should be checked for sleep apnea.

There are two types of sleep apnea: obstructive sleep apnea and central sleep apnea. Obstructive sleep apnea occurs when an individual's airway becomes blocked during sleep, and air is prevented from entering the lungs. In central sleep apnea, periods of interrupted breathing are caused by a lack of appropriate signals from the brain (White, 2005).

One of the most effective treatments for sleep apnea involves the use of a special device worn during sleep that pumps air into a person's nose and mouth to keep their airways open (Figure 4.13; McDaid et al., 2009). The continuous positive airway pressure (CPAP) devices usually have a mask that fits over the sleeper's nose and mouth, however new smaller masks cover only the nose. Many people with sleep apnea dislike wearing the CPAP devices, prompting exploration of alternative treatment options. A newer smaller device, the EPAP (expiratory positive air pressure) device, fits in the nostrils, and has shown promise in double-blind trials as one such alternative (Berry et al., 2011).



(a) (b)

Figure 4.13 (a) A typical CPAP device used in the treatment of sleep apnea is (b) affixed to the head with straps, and a mask that covers the nose and mouth.

SUDDEN INFANT DEATH SYNDROME (SIDS)

In sudden infant death syndrome (SIDS), an infant stops breathing during sleep and dies. Infants younger than 12 months are at the highest risk for SIDS, and boys are at greater risk than girls. A number of risk factors have been associated with SIDS, including premature birth, maternal smoking during pregnancy, smoking within the home, and hyperthermia (overheating). There may also be differences in both brain structure and function in infants that die from SIDS (Berkowitz, 2012; Mage & Donner, 2006; Thach, 2005).

Decades of research on SIDS has led to a number of recommendations for parents to protect their children (Figure 4.14). Infants should be placed on their backs when put down to sleep, and their cribs should not contain any items which pose suffocation or overheating threats, such as blankets, pillows, or padded crib bumpers (cushions that cover the bars of a crib). Similarly, infants should not wear hats or be overly dressed when sleeping to prevent overheating, and people in the child's household should abstain from smoking in the home. Recommendations like these have helped to dramatically decrease

the number of infant deaths from SIDS (DeLuca et al., 2016; Mitchell, 2009; Task Force on Sudden Infant Death Syndrome, 2011).

Figure 4.14 The Safe to Sleep campaign educates the public about how to minimize risk factors associated with SIDS. This campaign is sponsored in part by the National Institute of Child Health and Human Development.

NARCOLEPSY

Unlike the other sleep disorders described thus far in this section, a person with narcolepsy cannot stop falling asleep, often at inopportune moments. These sleep episodes are often associated with cataplexy, which is a lack of muscle tone or muscle weakness, and in some cases involves complete paralysis of the voluntary muscles, which may cause the person to collapse. This is similar to the kind of paralysis experienced by healthy individuals during REM sleep (Burgess & Scammell, 2012; Hishikawa & Shimizu, 1995; Luppi et al., 2011). Narcoleptic episodes take on other features of REM sleep. For example, around one third of individuals diagnosed with narcolepsy experience vivid, dream-like hallucinations during narcoleptic attacks (Chokroverty, 2010).

Surprisingly, narcoleptic episodes are often triggered by states of

heightened arousal or stress. The typical episode can last from a minute or two to half an hour. Once awakened from a narcoleptic attack, people report that they feel refreshed (Chokroverty, 2010). Frequent narcoleptic episodes often interfere with a person's ability to perform their job or complete schoolwork, and in some situations can result in significant harm and injury (e.g., driving a car or operating machinery). There is a genetic component to developing narcolepsy, which may be an auto-immune disease. Narcolepsy with cataplexy is associated with reduced secretion of a chemical called orexin from the hypothalamus. Orexin helps to keep us awake (De la Herrán-Arita & Drucker-Colín, 2012; Han, 2012; National Institute of Neurological Diseases, n.d.). Animal models suggest that orexin agonists might be effective treatment for narcolepsy, but research in humans is still lacking (Pizza et al., 2022).Generally, narcolepsy is treated using non-amphetamine psychomotor stimulant drugs, or if they fail, amphetamines are used. More recently, histamine agonists have also shown promise as non-addictive alternatives (National Institute of Neurological Diseases, n.d.).

OTHER PARASOMNIAS

Restless leg syndrome is another parasomnia that is quite common. People with restless leg syndrome experience discomfort in their legs when trying to fall asleep or if they are inactive. Deliberately moving their legs relieves the discomfort, but makes it difficult to fall or stay asleep. Restless leg syndrome has been associated with a number of other medical diagnoses, such as iron deficiency, pregnancy, chronic kidney disease and diabetes (Mahowald & Schenck, 2000). Treatments for this condition include benzodiazepines, opiates, and anticonvulsants (Restless Legs Syndrome Foundation, n.d.).

Night terrors are episodes of screaming and feelings of intense panic while asleep (Mahowald & Schenck, 2000). Individuals

suffering from night terrors often appear to be awake—they might sit up and open their eyes and seem very frightened, however, attempts to console them are ineffective. Night terrors occur during the NREM phase of sleep and people typically have no memory of them on waking (Provini et al., 2011). Night terrors are common among children (40%), but most grow out of them. Generally, night terrors are not treated unless there is some contributing medical or psychological condition (Mayo Clinic, n.d.).

4.5. Other States of Consciousness

Our states of consciousness change as we move from wakefulness to sleep. We also can alter our consciousness through other means. This final section will consider how we can alter our states of consciousness through the use of psychoactive drugs, hypnosis, trances, and meditation. In addition, we will briefly introduce some cross-cultural perspectives on states of altered consciousness.

PSYCHOACTIVE DRUGS

On April 16, 1943, Albert Hoffman—a Swiss chemist working in a pharmaceutical company— accidentally ingested a newly synthesized drug. The drug—lysergic acid diethylimide (LSD)—turned out to be a powerful hallucinogen. Hoffman went home and later reported the effects of the drug, describing them as seeing the world through a "warped mirror" and experiencing visions of "extraordinary shapes with intense, kaleidoscopic play of colors." Hoffman had discovered what members of many traditional cultures around the world already knew, there are substances that, when ingested, can have a powerful effect on perception and consciousness. Drugs operate on human physiology in a variety of ways and researchers and medical doctors tend to classify drugs

according to their effects (Biswas-Diener & Teeny, 2023). Here we will briefly cover three categories of drugs: *hallucinogens, depressants, and stimulants*.

HALLUCINOGENS

It is possible that hallucinogens are the chemicals that have, historically, been used the most widely to alter consciousness. Traditional societies have used plant-based hallucinogens such as peyote, ebene, and psilocybin mushrooms in a wide range of religious ceremonies. Hallucinogens are substances that alter a person's perceptions, often by creating visions or hallucinations (sensory experiences that are not based on physical realities in the environment). There are a wide range of hallucinogens and many are used as recreational substances in industrialized societies. Common examples include marijuana, LSD, and MDMA (also known as "ecstasy"). Marijuana is made from the dried flowers of the hemp plant and is often smoked to produce euphoria (a feeling of intense excitement and happiness). The active ingredient in marijuana is called THC and it can also produce distortions in the perception of time, and rambling, unrelated thoughts. THC is sometimes associated with increased hunger or excessive laughter. The use of marijuana has been legalized in many states and is increasingly used for medical purposes, such as for the management of nausea or treating glaucoma (Biswas-Diener & Teeny, 2023). Marijuana is also used throughout the word as an entheogen—a substance that alters consciousness for religious purposes (Ferrara, 2021).

DEPRESSANTS

Depressants are substances that, as their name suggests, slow down the body's physiology and mental processes. Alcohol is the most widely used depressant. Alcohol reduces inhibition, meaning

that intoxicated people are more likely to act in ways they normally would not. Alcohol works by increasing the action of a common inhibitory neurotransmitter, GABA. Increased GABA also affects the visual and motor systems of the brain and so produces physical effects, such as loss of balance and coordination (Biswas-Diener & Teeny, 2023).

Other common depressants include opiates/opioids (also called "narcotics"), which are substances synthesized from the poppy flower. Opiates stimulate endorphin production in the brain and because of this they are often prescribed as pain killers by medical professionals. Unfortunately, because opiates such as Oxycontin, reliably produce euphoria, their use is often abused, which has contributed to huge numbers of opioid-related deaths in the United States every year (Biswas-Diener & Teeny, 2023).

Link to Learning

Visit this webpage to learn more about the National Opioid Crisis

STIMULANTS

Stimulants are substances that "speed up" the body's physiological and mental processes. Two commonly used stimulants are caffeine —the chemical found in coffee and tea—and nicotine, the active chemical in cigarettes and other tobacco products, including e-cigarettes. These substances are both legal and relatively inexpensive, leading to their widespread use. Many people are attracted to stimulants because they feel more alert when under

the influence of these drugs. As with any drug there are health risks associated with consumption. For example, excessive consumption of caffeine can result in anxiety, headaches, and insomnia. Also, smoking cigarettes— the most common means of ingesting nicotine—is associated with higher risks of cancer (Biswas-Diener & Teeny, 2023). 90% of lung cancer among heavy smokers is directly attributable to smoking (Stewart & Kleihues, 2003).

Illegal stimulants. such as cocaine and methamphetamine (also known as "crystal meth" or "ice"), are also commonly used. These substances act by blocking the "re-uptake" of dopamine in the brain. This means that the brain does not naturally clear out the dopamine and that it builds up in the synapses, creating euphoria and alertness. As the effects wear off, it stimulates strong cravings for more of the drug. Because of this these powerful effects, stimulants are highly addictive (Biswas-Diener & Teeny, 2023).

HYPNOSIS

Hypnosis is a state of extreme focus where people pay minimal attention to external stimuli. You may have experienced a similar sensation called "flow", when you were fully immersed in a task, such as playing a video game. When you experience flow, you are concentrating entirely on what you were doing such that you lose track of time and became less aware of what is around you (Csikszentmihalyi & Nakamura, 2018). Flow is an altered state of consciousness, which most of us experience from time to time in our normal lives. Although both flow and hypnosis feel relaxing and pleasant, one important difference is that during hypnosis, people become very open to suggestion. However, this does not mean that you would do things that you wouldn't normally do (Lynn et al., 1990). Also, contrary to many depictions of hypnosis in books and movies, you can't be hypnotized against your will (Lynn & Kirsh, 2006). In therapeutic settings, clinicians may use hypnosis

to reduce anxiety and help patients relax. Some psychologists use hypnosis to draw out forgotten (repressed) memories, but as you will read in the upcoming chapter on Memory, this technique is highly controversial. Hypnosis can also provide pain relief and reduce the side effects of medications. In one study, burn patients were treated with either hypnotherapy, pseudo-hypnosis (i.e., a placebo condition), or no treatment at all. Afterward, even though people in the placebo condition experienced a 16% decrease in pain, those in the true hypnosis condition experienced a pain reduction of nearly 50% (Patterson et al., 1996). Hypnosis has also been used in the treatment of depression and anxiety, smoking cessation, and weight loss (Alladin, 2012; Elkins et al., 2012; Golden, 2012; Montgomery et al., 2012; Williamson, 2019).

Brain imaging studies have demonstrated that hypnotic states are associated with global changes in brain functioning (Del Casale et al., 2012; Guldenmund et al., 2012). Hypnosis generally causes reduced activation in the medial prefrontal cortex, an area associated with day dreaming and sense of self. Hypnosis frequently uses visual imagery and so not surprisingly, hypnosis reliably activates visual areas in the occipital cortex. Similarly, imagining pain relief during hypnosis, reduces activity in pain-related areas of the brain, such as the somatosensory cortex, and the anterior cingulate cortex in the limbic system (Landry et al., 2017; Williamson, 2019).

Historically, hypnosis has been viewed with some suspicion because of its portrayal in popular media and entertainment. Therefore, it is important to make a distinction between hypnosis as an empirically based therapeutic approach versus as a form of entertainment. Contrary to popular belief, individuals undergoing hypnosis usually have clear memories of the hypnotic experience and are in control of their own behaviors. While hypnosis may be effective in enhancing memory or a skill, such enhancements are very modest in nature (Raz, 2011).

How exactly does a hypnotist bring a participant to a state of

hypnosis? Although techniques vary, there are four consistent components associated with hypnosis (National Research Council, 1994). These include:

1. The participant is guided to focus on one thing, such as the hypnotist's words or a ticking watch.
2. The participant is made comfortable and is directed to be relaxed and sleepy.
3. The participant is told to be open to the process of hypnosis, to trust the hypnotist and let go.
4. The participant is encouraged to use their imagination.

People vary in terms of their ability to be hypnotized, but research suggests that most people are at least moderately hypnotizable (Kihlstrom, 2013). People can also be trained to self-hypnotize, which is an effective way to alleviate pain and psychological distress (Eason & Parris, 2019).

MEDITATION

There are a number of different meditation techniques but a central feature of all of them is to regulate physical and emotional states to bring about feelings of inner peace (Chen et al., 2013; Lang et al., 2012). Some practices focus the person's attention to help them quiet their mind. The focus might be a single sound, image, or sensation, such as in transcendental meditation, where a person silently repeats a sound called a mantra. Other focused attention practices include using breathing techniques, body scanning, or silently repeating phrases relating to love and kindness. Mind-body practices like yoga, qigong, and tai chi also serve to focus the mind. In contrast, other meditation practices involve open monitoring, where instead of blocking out thoughts and feelings, people are encouraged to be aware of their thoughts, feelings, and sensations,

in a non-judgmental way (Lutz et al., 2008). Mindfulness meditation is a popular example of open monitoring (Zeidan et al., 2012).

Meditative techniques have their roots in religious practices (Figure 4.15), but their use has grown in popularity among practitioners of alternative medicine in the West. Research indicates that meditation may help reduce blood pressure, and the American Heart Association suggests that meditation might be used in conjunction with more traditional treatments as a way to manage hypertension, although there is not sufficient data for a recommendation to be made (Brook et al., 2013). Like hypnosis, meditation also shows promise in stress management, improving sleep quality (Caldwell et al., 2010), treating mood and anxiety disorders (Chen et al., 2013; Freeman et al., 2010; Vøllestad et al., 2012), and for pain management (Reiner et al., 2013).

CULTURE AND ALTERED STATES OF CONSCIOUSNESS

In many parts of the world, different cultures embrace experiencing altered states of consciousness as part of religious, wellness and/or healing practices, and other rituals related to problem solving. In tribal societies and Eastern cultures, practices in which people experience altered states of consciousness are important, mystical, and sacred. They may allow people to feel connected with God(s), ancestors, and/or themselves. These altered states include samadhi in yoga, moksha in Hinduism, satori in Zen Buddhism, fana in Sufism and Ruach Hakodesh in Kabbalah. Even in the West, people have reported experiencing altered states of consciousness during mystical or spiritual practices, such as religious ecstasy or speaking in tongues, which is common within the Pentecostal religion. Multiple different cultures in Asia, the Middle East, Africa, and the Americas have used psychoactive substances as entheogens (Ferrara, 2021).

Many world cultures use shamanic healers (magico-religious leaders) who typically enter into trance states during various rituals. During the trances, the shamans often "travel" to a supernatural location where they experience vivid imagery filled with symbolic meaning. Trance states increase the kinds of slow waves that we see in deep sleep (Winkelman, 2012) and can be induced in a number of ways, including rhythmic drumming, chanting, dancing, or through sensory deprivation, celibacy, fasting, and/or the use of plants that produce psychedelic effects. Many shamans used a combination of techniques to induce a trance state. Shamanistic rituals also bring communities together and help to promote bonding, which may also be an important part of the healing process (Winkelman, 2012).

(a) (b)

Figure 4.15 (a) This is a statue of a meditating Buddha, representing one of the many religious traditions in which meditation plays a part. (b) People practicing meditation may experience an alternate state of consciousness.

Link to Learning

Feeling stressed? Think meditation might help? Watch this instructional video about using Buddhist meditation techniques to alleviate stress to learn more.

Link to Learning

Watch this video about the results of a brain imaging study in individuals who underwent specific mindfulness meditative techniques to learn more.

Chapter 4. Conclusions

In conclusion, the study of sleep and states of consciousness provides valuable insights into the functions of sleep, the nature of dreams, and the impact of altered states of consciousness. Exploring the influence of technology, cultural norms, and sleep disorders highlights the importance of maintaining healthy sleep habits and understanding the diverse states of consciousness that shape our lives. By exploring these topics, we gain a deeper understanding of ourselves and the complexities of the human experience.

Chapter 4. Key Terms

alpha wave
type of brain wave characteristic during the early part of NREM stage 1 sleep, which has fairly low amplitude and a frequency of 8–12 Hz

beta wave
type of brain wave characteristic during wakefulness, which has a very low amplitude and a frequency of 13–30 Hz

biological rhythm
internal cycle of biological activity

cataplexy
lack of muscle tone or muscle weakness, and in some cases complete paralysis of the voluntary muscles

central sleep apnea
sleep disorder with periods of interrupted breathing due to a disruption in signals sent from the brain that regulate breathing

circadian rhythm
biological rhythm that occurs over approximately 24 hours

cognitive-behavioral therapy (CBT)
psychotherapy that focuses on cognitive processes and problem behaviors that is sometimes used to treat sleep disorders such as insomnia

collective unconscious
theoretical repository of information shared by all people across cultures, as described by Carl Jung

consciousness
awareness of internal and external stimuli

continuous positive airway pressure (CPAP)
device used to treat sleep apnea; includes a mask that fits over the sleeper's nose and mouth, which is connected to a pump that pumps air into the person's airways, forcing them to remain open

delta wave
type of brain wave characteristic during stage 3 NREM sleep, which has a high amplitude and low frequency of less than 3 Hz

depressants
a class of drugs that slow down the body's physiological and mental processes

entheogen
psychoactive substance used to produce an altered sense of consciousness for religious purposes

euphoria
an intense feeling of pleasure, excitement or happiness.

evolutionary psychology
discipline that studies how universal patterns of behavior and cognitive processes have evolved over time as a result of natural selection

hallucinogens
substances that, when ingested, alter a person's perceptions, often by creating hallucinations (perceptions that are not based on environmental stimuli) or distorting their perceptions of time.

homeostasis
tendency to maintain a balance, or optimal level, within a biological system

hypnosis
state of extreme self-focus and attention in which minimal attention is given to external stimuli

hypnotherapy
use of hypnotic techniques such as relaxation and suggestion to

help engineer desirable change such as lower pain or quitting smoking

insomnia
consistent difficulty in falling or staying asleep for at least three nights a week over a month's time

jet lag
collection of symptoms brought on by travel from one time zone to another that results from the mismatch between our internal circadian cycles and our environment

K-complex
very high amplitude pattern of brain activity associated with stage 2 sleep that may occur in response to environmental stimuli

latent content
hidden meaning of a dream, per Sigmund Freud's view of the function of dreams

lucid dream
people become aware that they are dreaming and can control the dream's content

manifest content
storyline of events that occur during a dream, per Sigmund Freud's view of the function of dreams

meditation
learning to regulate attentional processes to achieve a sense of wellbeing and peace

melatonin
hormone secreted by the pineal gland that serves as an important regulator of the sleep-wake cycle

meta-analysis
study that combines the results of several related studies

narcolepsy
sleep disorder in which the sufferer cannot resist falling to sleep at inopportune times

night terror
sleep disorder in which the sleeper experiences a sense of panic

and may scream or attempt to escape from the immediate environment

non-REM (NREM)
period of sleep outside periods of rapid eye movement (REM) sleep

obstructive sleep apnea
sleep disorder defined by episodes when breathing stops during sleep as a result of blockage of the airway

parasomnia
one of a group of sleep disorders characterized by unwanted, disruptive motor activity and/or experiences during sleep

pineal gland
endocrine structure located inside the brain that releases melatonin

psychedelics
also known as hallucinogens, are a class of psychoactive substances that induce profound alterations in perception, thoughts, and emotions

rapid eye movement (REM) sleep
period of sleep characterized by brain waves very similar to those during wakefulness and by darting movements of the eyes under closed eyelids

REM sleep behavior disorder (RBD)
sleep disorder in which the muscle paralysis associated with the REM sleep phase does not occur; sleepers have high levels of physical activity during REM sleep, especially during disturbing dreams

restless leg syndrome
sleep disorder in which the sufferer has uncomfortable sensations in the legs when trying to fall asleep that are relieved by moving the legs

rotating shift work
work schedule that changes from early to late on a daily or weekly basis

sleep

state marked by relatively low levels of physical activity and reduced sensory awareness that is distinct from periods of rest that occur during wakefulness

sleep apnea
sleep disorder defined by episodes during which breathing stops during sleep

sleep debt
result of insufficient sleep on a chronic basis

sleep rebound
sleep-deprived individuals will experience shorter sleep latencies during subsequent opportunities for sleep

sleep regulation
brain's control of switching between sleep and wakefulness as well as coordinating this cycle with the outside world

sleep spindle
rapid burst of high frequency brain waves during stage 2 sleep that may be important for learning and memory

sleepwalking
(also, somnambulism) sleep disorder in which the sleeper engages in relatively complex behaviors

stage 1 sleep
first stage of sleep; transitional phase that occurs between wakefulness and sleep; the period during which a person drifts off to sleep

stage 2 sleep
second stage of sleep; the body goes into deep relaxation; characterized by the appearance of sleep spindles

stage 3 sleep
third stage of sleep; deep sleep characterized by low frequency, high amplitude delta waves

stimulants
a class of drugs that speed up the body's physiological and mental processes

sudden infant death syndrome (SIDS)

infant (one year old or younger) with no apparent medical condition suddenly dies during sleep

suprachiasmatic nucleus (SCN)
area of the hypothalamus in which the body's biological clock is located

theta wave
type of brain wave characteristic of the end of stage 1 NREM sleep, which has a moderately low amplitude and a frequency of 4–7 Hz

tolerance
state of requiring increasing quantities of the drug to gain the desired effect

wakefulness
characterized by high levels of sensory awareness, thought, and behavior

Chapter 4. Personal Reflection Questions

- Reflect on your own sleep patterns and routines. How would you describe the quality and duration of your sleep? Are there any factors that may be affecting your sleep, such as stress, technology use, or environmental factors? How do you think improving your sleep habits could impact your overall well-being and cognitive functioning?
- Consider your own experiences with altered states of consciousness, such as daydreaming, meditation, or even moments of flow during engaging activities. How do these states of consciousness differ from your everyday waking state? What are the potential benefits and challenges of accessing these altered states intentionally?
- Consider your own experiences with altered states of consciousness induced by substances such as caffeine, alcohol, or recreational drugs. Reflect on the potential benefits and risks associated with these substances and how they can impact your cognitive functioning, mood, and overall well-being. How do you navigate the choices you make regarding substance use and its influence on your states of consciousness?

- Reflect on the impact of technology on your own states of consciousness, particularly in relation to sleep. How often do you use electronic devices before bed, and how does this affect your ability to fall asleep and the quality of your sleep? How might you modify your technology use to promote better sleep and healthier states of consciousness?
- Think about the role of dreams in your life. How often do you remember your dreams, and do you notice any patterns or themes? How do you personally interpret your dreams, and do you believe they hold any significance or meaning?
- Reflect on the concept of mindfulness and its connection to states of consciousness. Have you ever practiced mindfulness or engaged in activities that promote present-moment awareness? How might incorporating mindfulness practices into your daily routine enhance your ability to be fully present and engaged in various aspects of your life?

Chapter 4. Critical Thinking Questions

- Healthcare professionals often work rotating shifts. Why is this problematic? What can be done to deal with potential problems?
- Do you think attitudes toward drug use are changing over time? If so, how? Why do you think these changes occur?
- Students in high school and college are increasingly using stimulants such as Adderol as study aids and "performance enhancers." What is your opinion of this trend?
- Consider the cultural and societal influences on sleep patterns and states of consciousness. How do cultural norms, work schedules, and societal expectations impact our sleep habits and the value we place on states of consciousness other than wakefulness? How might cultural variations in sleep practices and beliefs provide insights into the universality or subjectivity of sleep experiences?
- Reflect on the potential implications of sleep disorders and disturbances on individuals and society as a whole. How do conditions such as insomnia, sleep apnea, or narcolepsy affect individuals' daily functioning and quality

of life? What are the social and economic consequences of insufficient sleep and sleep-related disorders? How can increased awareness and education about sleep disorders contribute to better support and treatment for affected individuals?

- Tricia Hersey advocates for rest as a form of resistance against the oppressive forces of capitalism and white supremacy that have shaped many of us to believe that we should be working hard all the time. How can using rest as resistance improve the quality of your life?

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OERS

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A smaller part of the text comes from the following: Biswas-Diener, R. & Teeny, J. (2023). In: R. Biswas-Diener & E. Diener (Eds .) *Textbook Series: Psychology*. Champ DEF Publishers. https://nobaproject.com

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CHAPTER 5. SENSATION AND PERCEPTION

Figure 5.1. If you were standing in the midst of this street scene, you would be absorbing and processing numerous pieces of sensory input.

The next few chapters in this book focus on the field of cognitive psychology. Cognitive psychology is the study of the mental processes and activities involved in acquiring, processing, storing, and using information. It encompasses a wide range of functions such as sensation and perception, attention, memory, language,

problem-solving, decision-making, reasoning and intelligence. This chapter focuses on one subfield of cognitive psychology—sensation and perception. But, what does that mean exactly? Imagine standing on a city street corner (like the one in Figure 5.1). You might notice the rapid movement of the cars and people as they go about their business, the sound of a street musician's melody or a horn honking in the distance, the smell of exhaust fumes or of food being sold by a nearby vendor, and the sensation of hard pavement under your feet. You are experiencing sensation and perception!

We rely on our sensory systems to provide important information about our surroundings. We use this information to successfully navigate and interact with our environment so that we can find nourishment, seek shelter, maintain social relationships, and avoid potentially dangerous situations.

This chapter will provide an overview of how sensory information is received and processed by the nervous system and how that affects our conscious experience of the world. We begin with the distinction between sensation and perception. Then we give an overview of the basic structure and function of the major sensory systems. Along the way, we will consider how culture and other experiences, influence our perceptions of the world around us.

LEARNING OBJECTIVES

- Distinguish between sensation and perception
- Discuss the roles that attention, motivation, experience and culture play in perception
- Describe the basic anatomy of the visual system
- Discuss how rods and cones contribute to different aspects of vision
- Describe the trichromatic and opponent process theories

of color vision

- Describe how monocular and binocular cues are used in the perception of depth
- Explain how Gestalt psychology influenced the field of sensation and perception
- Describe basic Gestalt principles of grouping
- Explain the concept of perceptual set and its influence on perception
- Explain the concept of implicit bias and its effects on perception
- Describe the basic anatomy and function of the auditory system
- Explain how we encode and perceive pitch
- Discuss how we localize sound
- Describe the basic functions of the chemical senses
- Explain the basic functions of the somatosensory, nociceptive, and thermoceptive sensory systems
- Describe the basic functions of the vestibular, proprioceptive, and kinesthetic sensory systems
- Describe the importance of multisensory neurons in perception

5.1 Sensation versus Perception

SENSATION

What does it mean to sense something? Our sense organs contain sensory receptors, which are specialized neurons that respond to specific types of information in our physical world. They convert the external information into neural energy and then send this to the brain. These biological processes collectively are known as sensation. For example, light causes chemical changes in cells that line the back of the eye. These cells relay messages (For review see Chapter 3 on Biopsychology), to the central nervous system. Because information from the world around us stimulates our senses, we refer to various types of sensory information as stimuli (singular: stimulus).

You probably know about our five major senses: vision, hearing (audition), smell (olfaction), taste (gustation), and touch (somatosensation). However, we also have sensory systems that provide information about balance (vestibular sense), body position (proprioception), body movement (kinesthesia), pain (nociception), and temperature (thermoception).

The sensitivity of a given sensory system to a particular stimulus can be expressed as an absolute threshold. Absolute threshold refers to the minimum amount of stimulus energy that must be present for the stimulus to be reliably detected. Another way to

think about this is by asking how dim can a light be, or how soft can a sound be for us to reliably notice it. The sensitivity of our sensory receptors can be quite amazing. For example, on a clear night, the receptors in the back of the eye can detect a candle flame 1.6 miles away (Silver, 2015). Also, under quiet conditions, the receptor cells of the ear can detect the tick of a clock 20 feet away (Galanter, 1962).

It is also possible for us to get messages that are presented below the threshold for conscious awareness—these are called subliminal messages. A stimulus reaches the absolute threshold when it is strong enough to excite sensory receptors and send nerve impulses to the brain. However, in some cases our brains may receive a message, but we are not consciously aware of it. Over the years there has been a great deal of speculation about the use of subliminal messages in advertising, rock music, and self-help audio programs. Research shows that in laboratory settings, people can process and respond to information outside of awareness. But this does not mean that we obey these messages like zombies; in fact, subliminal messages have little effect on behavior outside the laboratory (Kunst-Wilson & Zajonc, 1980; Rensink, 2004; Nelson, 2008; Radel et al., 2009; Loersch et al., 2013).

Sometimes, we are interested in how much we need to change a stimulus to detect a difference. This is known as the just noticeable difference (jnd) or difference threshold. The difference threshold changes depending on the stimulus intensity. For example, imagine you are in a very dark movie theater. If you receive a text message, it is likely that many people would notice your cell phone light up. However, if you were in a brightly lit arena during a basketball game, very few people would notice. The cell phone brightness does not change, but its noticeability changes dramatically between the two contexts. Ernst Weber described the importance of context on difference threshold in the 1830s, and it has become known as Weber's law: The difference threshold is a constant fraction of the original stimulus. Thresholds are generally measured under

incredibly controlled conditions in situations that are optimal for sensitivity.

PERCEPTION

Our sensory receptors are constantly collecting information from the environment. However, our interactions with the world are affected by *how* we interpret that information. Perception refers to the way sensory information is interpreted and consciously experienced. As shown in Figure 5.2, perception involves both bottom-up and top-down processing. Bottom-up processing refers to the information we receive from our sensory systems (sensation), and top-down processing refers to the ways we use knowledge and expectancy to process this information (Egeth & Yantis, 1997; Fine & Minnery, 2009; Yantis & Egeth, 1999). Bottom-up processing is automatic, but top-down processing is often under our conscious control.

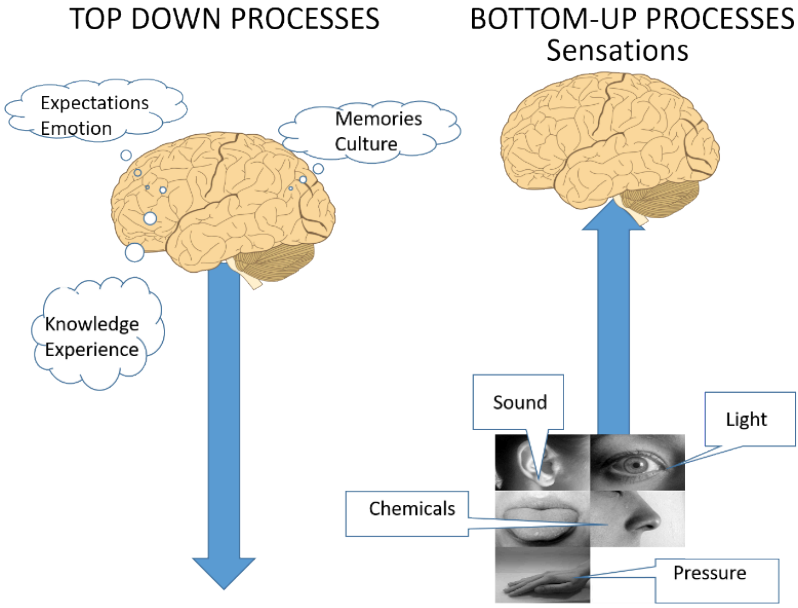

Figure 5.2. Top-down and bottom-up processes affect our perceptions.

Imagine that you and some friends are sitting in a crowded restaurant eating lunch and talking. It is very noisy, and you are concentrating on your friend's face to hear what they are saying. Suddenly, you are startled by the sound of breaking glass and clang of metal pans hitting the floor. One of the servers dropped a large tray of food. Although you were attending to your meal and conversation, that crashing sound would likely grab your attention. You would have no choice but to notice it. Because this attentional capture was caused by the sound from the environment: it would be considered a bottom-up process.

Alternatively, top-down processes are generally goal directed, slow, deliberate, effortful, and under your control (Fine & Minnery, 2009; Miller & Cohen, 2001; Miller & D'Esposito, 2005). For instance, if you misplaced your keys, how would you look for them? If you had a yellow key chain, you would probably look for a yellow object

of a certain size in specific locations where you might leave your keys. You would not look for yellow on your ceiling fan, because you know keys are unlikely to be there. That act of searching for a certain size of yellowness in some locations and not others, would be top-down—under your control and based on your experience. Even very simple perceptions, such as recognizing an object, are influenced by your expectations, prior experiences, and culture.

One way to think of the difference between sensation and perception, is to remember that sensation is a physical process, whereas perception is psychological. For example, imagine walking into a kitchen and smelling the scent of baking cinnamon rolls, the *sensation* is the smell receptors reacting to the odor of cinnamon and sending information to the brain, but the *perception* may be "*Mmm, this smells like the holiday bread that Grandma used to bake when I was a kid*".

Our sensations (and perceptions) can change over time. In fact, when sensory stimuli remain relatively constant over prolonged periods of time, our perception of them often changes. This is known as sensory adaptation. Imagine that you have been baking a cake but forget to take it out of the oven and it burns. It fills the kitchen with a strong smell, but after a while you don't notice it anymore and feel pretty convinced that no one will ever know about your cooking disaster. Then your roommate comes back and says – Wow! It really smells in here– what did you burn? If you go outside of your home for a short period of time and then return, you too will notice the smell. The chemicals were still present in the air, but your olfactory receptors had stopped responding to them.

ATTENTION

Attention is another factor that affects sensation and perception. Attention plays a significant role in determining whether we perceive what is sensed. Imagine you are at a party full of music,

chatter, and laughter. You get involved in an interesting conversation with a friend, and you tune out all the background noise. This is a top-down process. Your ears are still being stimulated by the music, but, if someone interrupted you to ask what song had just finished playing, you would probably be unable to answer that question. We need to pay attention to stimuli (either voluntarily or automatically) in order for us to be aware of them.

The importance of attention for perception of the environment is demonstrated in a famous study conducted by Daniel Simons and Christopher Chabris (1999). In this study, participants watched a video of two teams of people (dressed in black or white) passing two basketballs among them. Participants were asked to count the number of times the team dressed in white passed the ball. During the video, a person dressed in a black gorilla costume walks among the two teams. You would think that everyone would notice the gorilla, right? Nearly half of the people who watched the video didn't notice the gorilla at all, despite the fact that it was in plain view for nine seconds. Because participants were so focused on counting the passes, they completely tuned out other visual information. Inattentional blindness is the failure to notice something that is completely visible because the person was actively attending to something else and did not pay attention to other things (Mack & Rock, 1998; Simons & Chabris, 1999).

Link to Learning

See for yourself how inattentional blindness works by checking out this selective attention test from Simons and Chabris (1999).

A similar attentional phenomenon is change blindness. Change blindness is when something changes in the environment (or a picture) and you fail to notice. Like inattentional blindness, change blindness is also very common. It explains why we often fail to notice when someone we see every day gets new glasses or a haircut, or shaves off their facial hair. Psychology experiments often investigate change blindness by alternating two slightly different pictures within a video and measuring how quickly participants spot the difference. Can you see the differences between the two pictures in Figure 5.3? – look at the end of this chapter (above the References) to find the answer. Even when there is a large difference between two pictures, about 40% of people fail to find the difference. Simons and Levi (1988) showed that change blindness affects a similar number of people in real world settings. Fifty percent of participants failed to notice that the person to whom they were giving directions changed to a different person! It sounds like a magic trick but check out the Door Study video to see how the researchers managed to make the switch without people noticing.

Figure 5.3. Can you spot the differences between the two pictures? In change blindness studies, experimenters measure how long participants take to find the difference(s) between the pictures. We can see four differences – but let us know if you see more.

CULTURE AND EXPERIENCE

Cultural psychologists, like Shinoba Kitayama and Richard Nisbett and their colleagues, have conducted multiple studies on the effects of culture on visual perception (Masuda, 2017). In general, White people from the USA tend to be more individualistic in their outlook, whereas people from East Asia are more collectivistic and place a greater emphasis on the importance of community. Multiple studies have shown that US participants tend to focus on central elements in a picture or scene—like the globe in Figure 5.3, whereas East Asian participants attend more to context. Therefore, East Asian participants are more likely to notice the background objects as well as the larger objects in the foreground. So, in a change blindness study, participants from the US and East Asia take a similar amount of time to see changes in the large, central objects but US participants are slower to notice changes in the background (Masuda & Nisbett, 2006).

Culture can also affect our susceptibility to visual illusions. Marshall and colleagues conducted a multi-national study in which they found that individuals from industrialized cultures were more prone to experience certain types of visual illusions than individuals from non-industrialized cultures (Marshall et al., 1963). Westerners are more likely to experience the Müller-Lyer illusion than people living in non-industrialized countries (Figure 5.4). In the illusion, the two black lines in Figure 5.4a appear to be different lengths, but they are actually the same. This illusion also applies to the lines in Figure 5.4b.

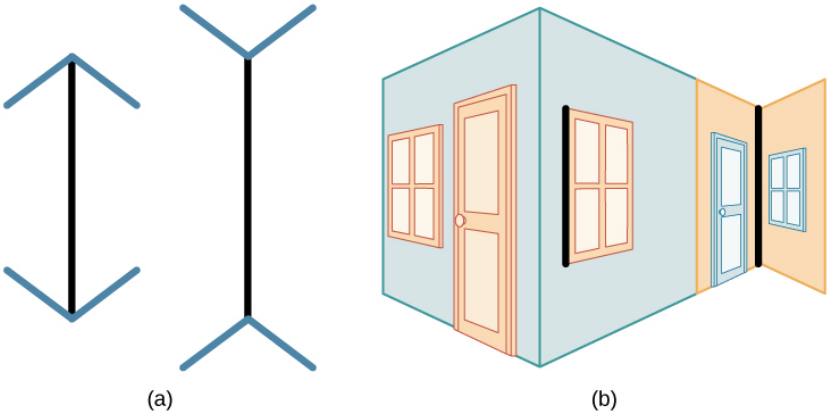

Figure 5.4. In the Müller-Lyer illusion, lines appear to be different lengths although they are identical. (a) Arrows at the ends of lines may make the line on the right appear longer, although the lines are the same length. (b) When applied to a three-dimensional image, the line on the right again may appear longer although both black lines are the same length.

Cultural differences in line length perception are consistent with differences in the types of environment that people experience. People who are more susceptible to the illusion are used to seeing buildings with straight lines, often referred to as a carpentered world (Segall et al., 1966). In contrast, the Zulu people of South Africa, whose villages are made up of round huts arranged in circles, are less susceptible to the illusion (Segall et al., 1999).

CROSS-RACE EFFECT

The cross-race effect is another example of how visual perception is affected by culture and experience. In general, we are better at recognizing faces of people of our own race, better than those from other races (Young et al., 2012). Most studies looking at the cross-race effect have focused on White participants, however, in one study Lee and Penrod found that participant race determines the extent of the cross-race effect. White participants showed larger

cross-race effects than Asian participants. However, Asian, White, and Latinx participants all showed larger cross-race effects for Black faces compared to faces of other races (Lee & Penrod, 2022). In other words, people who are not Black, have more difficulty recognizing Black faces than White, Asian or Latinx faces. Poorer recognition of Black faces has widespread societal implications, ranging from hurtful classroom interactions where Black students are mistaken for other students (Griffith et al., 2019), to the increased rates of misidentification of Black people by people witnessing crimes and misdemeanors (Lee & Penrod, 2022). However, research by McKone et al. (2019) offers hope for addressing these issues. They found that exposure to faces of different races before the age of 12 years can help mitigate the cross-race effect. This suggests that by promoting diversity and inclusivity from a young age, we can work towards a future where BIPOC individuals experience less discrimination.

FOOD PERCEPTION

It is not just visual perception that is affected by cultural factors. Culture affects our perception across all sensory modalities. For example, Japanese participants rated Japanese foods (e.g., soy sauce, fermented soybeans etc.) as more pleasant smelling than German participants who were unfamiliar with them (Ayabe-Kanamura et al., 1998).

MOTIVATION

As we have already see, motivation can also affect perception. Have you ever been expecting a really important phone call and you keep thinking that you hear the phone ringing, only to discover that it is not? If so, then you have experienced how motivation to detect a meaningful stimulus can shift our ability to discriminate between a true sensory stimulus and background noise. The ability to identify

a stimulus when it is embedded in a distracting background is explained by signal detection theory—the idea that our ability to detect a stimulus depends not only on its physical properties, but also on the psychological state of the observer. This might also explain why a mother is awakened by a quiet murmur from her baby but not by other sounds that occur while she is asleep. Signal detection theory has practical applications, such as increasing air traffic controller accuracy. Controllers need to be able to detect planes among many signals that appear on the radar screen and follow those planes as they move through the sky (Swets, 1964).

OTHER FACTORS THAT AFFECT PERCEPTION

Many other factors can influence perception. Basic aspects of personality are associated with differences in taste perception. Children described as thrill seekers are more likely to show taste preferences for intense sour flavors than children who are not (Liem et al., 2004). Our values and expectations also affect our taste ratings of food. Individuals with positive attitudes toward reduced-fat foods are more likely to rate foods labeled as reduced fat as better tasting than people with less negative attitudes about these products (Aaron et al., 1994). Our expectations about taste are also strongly influenced by brands and packaging. Some studies have used deceptive packaging to demonstrate its power on consumer preferences. For example, in a taste test, participants who identified as Coke drinkers preferred Pepsi when it was served from a Coke bottle more than Coke served from a Pepsi bottle, and vice versa for Pepsi drinkers (Woolcott et al., 1983). An fMRI study showed that participants rated the same wine as tasting better if it had an expensive price tag compared to a cheap one (Plassman et al., 2008). Also, popcorn tastes sweeter if served in a red bowl and saltier if served in a blue one (Wang & Chang, 2022).

CRITICAL PERIODS OF SENSORY DEVELOPMENT

Sensory experiences during early development are critical for perception. All sensory systems need to be stimulated early in life in order for them to develop normally (Cisneros-Franco et al., 2020). All of our senses are stimulated to some extent when we are still in the womb, with the exception of vision. Hence, when we are born, our visual system is less developed than our other sensory systems. The exact timing of critical periods varies across our senses. However, permanent deficits can arise if normal stimulation does not occur during a critical period. For example, if a newborn has visual problems that prevent them from seeing normally in the first year of their life they have permanent difficulties with face perception for the rest of their lives (Pascalis et al., 2020). Similarly, young children with eye problems often develop a condition called amblyopia (lazy eye), where they are unable to see fine details, such as small letters. Amblyopia is permanent if the eye problem is not corrected before the age of 8 years old. Therefore, it is important to detect and resolve any sensory issues, e.g., eye and ear problems as early as possible (Cisneros-Franco et al., 2020; Pascalis et al., 2020).

5.2. Vision

The visual system constructs a mental representation of the world around us (Figure 5.5). This helps us to successfully navigate through physical space and interact with other individuals and objects in our environments. This section will provide an overview of the basic anatomy and function of the visual system. In addition, we will explore our ability to perceive color and depth.

Figure 5.5. Our eyes take in sensory information that helps us understand the world around us.

ANATOMY OF THE VISUAL SYSTEM

The eyes work a little like a camera (Figure 5.6). Light passes first through the cornea, then the pupil and lens to reach the receptors at the back of the eye. The cornea is the transparent covering over the iris. It serves as a barrier between the eye and the outside world, and helps to focus the light as it enters the eye. The pupil is the small opening in the iris (colored part of the eye). When it is dark, the pupil dilates (expands) to allow more light to enter the eye. When it is very bright, the pupil will constrict to reduce the amount of light that enters the eye. Pupil size is controlled by the muscles in the iris. The size of the pupil is affected by the autonomic nervous system, it becomes large when we are scared or emotionally aroused (sympathetic NS) and smaller when we are resting and digesting (parasympathetic NS).

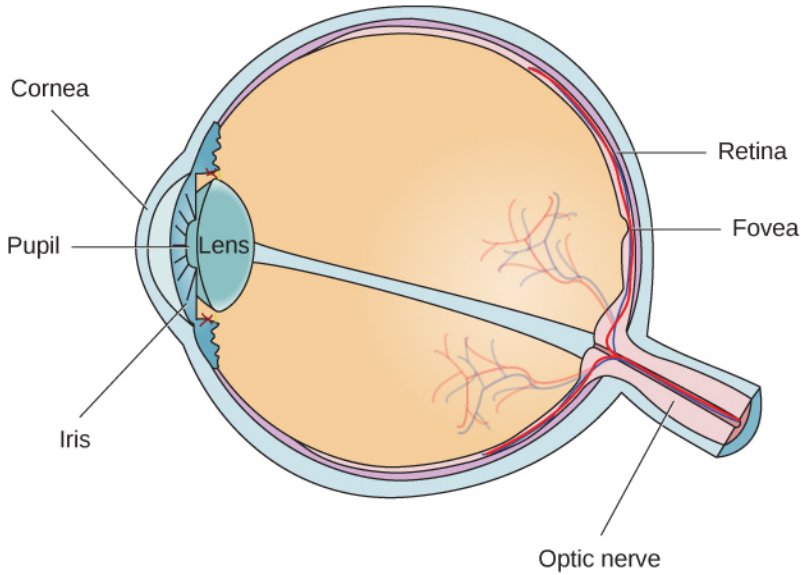

Figure 5.6. The anatomy of the eye is illustrated in this diagram.

The lens is attached to muscles that can change its shape so that we can see objects that are close to us, as well as those that are far away. Ideally, the optics of the eye (cornea and lens) produce a picture of the outside world that is perfectly focused on the retina at the back of the eye. The retina contains specialized photoreceptor cells called rods and cones (Figure 5.7). Cones work best in bright light conditions, they are sensitive to detail (like small letters), and help us to perceive color. At the very center of the retina, there is a tiny area, called the fovea, which has only cones. When we are looking directly at an object, e.g., when reading, we move our eyes so the words fall on the fovea, this allows us to see them better. The rest of the retina is important for our peripheral vision and contains both rods and cones. Rods work well in low light conditions, and while they are unable to detect color and fine detail, they help us to see general shapes in dimly lit environments and to detect movement in our peripheral vision.

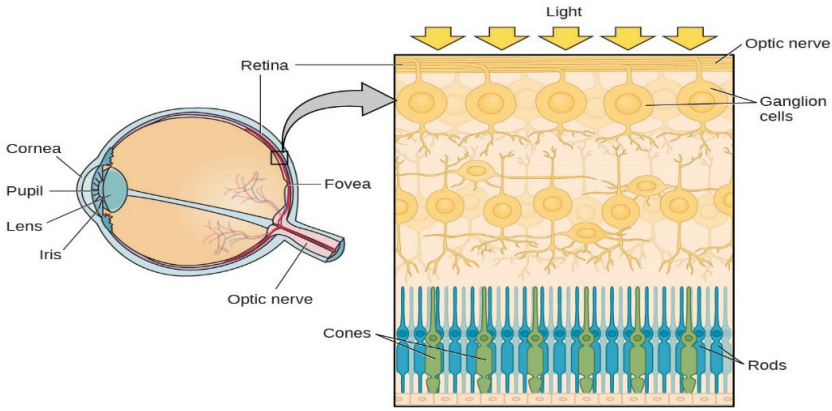

Figure 5.7. The two types of photoreceptors are shown in this image. Cones are shown in green and rods in blue.

The different sensitivities of rods and cones are apparent when we compare our vision in a brightly lit environment to a dark one. Imagine you are at home in the evening. The lights are on and you can see the small print in your textbook without much effort and

you can also easily avoid eating the orange candies you dislike from the packet of candy that you are eating. But, suddenly, there is a power cut and the lights go out. It's impossible to continue reading. You decide to console yourself by eating the rest of your candy, but, you can no longer see which ones are orange. You can see the basic outlines of the furniture and so you make your way to bed instead.

Rods and cones are connected (via interneurons) to retinal ganglion cells. Axons from the retinal ganglion cells converge to form the optic nerves, which exits at the back of each eye. The optic nerves carry visual information from the retina to the brain. There are no receptors where the optic nerve leaves the eye, which results in a blind spot in each eye. We are not consciously aware of our blind spots for two reasons: First, each eye gets a slightly different view of the visual field; therefore, the blind spots do not overlap. Second, even if we are looking with just one eye, our brain fills in what it thinks is in the blind spot so we unaware that information is missing.

The optic nerve from each eye merges at a point called the optic chiasm before it reaches the thalamus. As you can see in Figure 5.8, the optic chiasm is an X-shaped structure. Information from the right visual field (from both eyes) is sent to the left side of the brain, and information from the left visual field is sent to the right side of the brain.

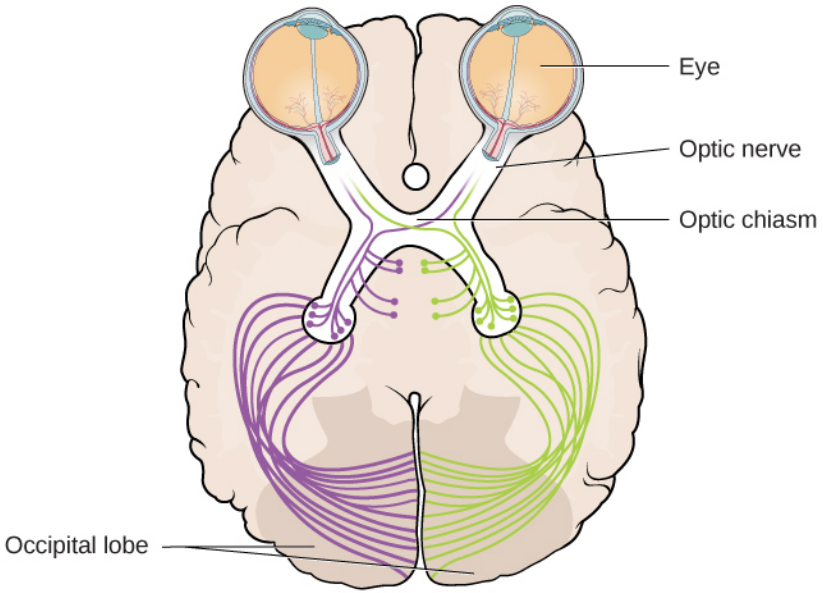

Figure 5.8.This illustration shows the optic chiasm of the brain and the pathways to the occipital lobe at the back of the brain.

Link to Learning

We can "see" our blind spot under certain conditions. This video will show you how. Tip: you probably need to be about 4 inches from your computer screen to experience this.

Visual information is sent via the thalamus to the primary visual cortex in the occipital lobe at the back of the brain for initial processing. Visual information is then sent to other parts of the

brain (association areas) via two major pathways (see Figure 5. 9). The "what" pathway, which projects to the temporal lobe, is involved in object recognition and identification. While the "where/how" pathway projects to the parietal lobe and is involved with processing the location and movement of an object (Milner & Goodale, 2008; Ungerleider & Haxby, 1994). For example, when you see a red ball rolling down the street, the "what" pathway helps you to know what the object is, and the "where/how" pathway helps you to intercept it and pick it up.

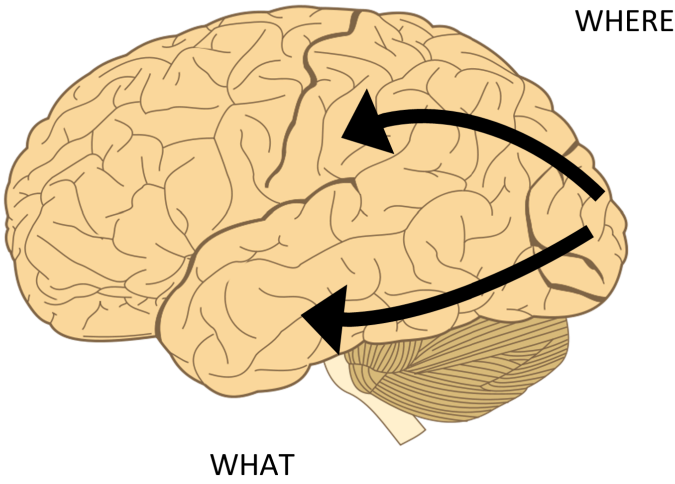

Figure 5.9. What and Where Visual Pathways

What Do You Think?

Ethics of Research Using Animals

David Hubel and Torsten Wiesel were awarded the Nobel Prize in Medicine in 1981 for their research on the visual system. They collaborated for more than twenty years and made significant discoveries about the neurology of visual perception (Hubel & Wiesel, 1959, 1962, 1963, 1970; Wiesel & Hubel, 1963). They studied mostly cats and monkeys. Although they used several techniques, they did a considerable amount of single unit recordings, where they inserted tiny electrodes in the animal's brain to determine when a single cell was activated. Among their many discoveries, they found that the retina mapped onto the visual cortex in alternating columns called ocular dominance columns. Each column was dominated by input from each eye separately.

In some of their research, they sewed one eye of newborn kittens, closed to measure its impact on the development of vision. They discovered there was a critical period of development for vision. If kittens were deprived of input from one eye, the ocular dominance columns in the brain were taken over by the other eye. When the eye was opened after a few weeks, the kitten was blind in that eye. In other words, they discovered that sensory input was needed for normal visual development.

What do you think about sewing a kitten's eye closed for research? Bear in mind, this research was helpful in

preventing blindness and permanent vision damage in children born with eye problems. Would you conduct that research, even if it meant causing some harm to cats? Would you think the same way if you were the parent of a child with vision problems? What if you worked at an animal shelter?

COLOR VISION

Light is not made of different colors—we perceive color by detecting light of different wavelengths. Short wavelength light is perceived as blue, medium wavelengths as green and long wavelengths as red. By mixing these three primary wavelengths in different combinations we are able to match any color. The trichromatic theory of color vision states that we have three different types of cone, each of which is maximally sensitive to one of the three primary wavelengths. Any wavelength that we can see is coded by the relative activity of the three different cone types, which are commonly referred to as blue (short wavelength), green (medium wavelength), and red (long wavelength) cones (Figure 5.10).

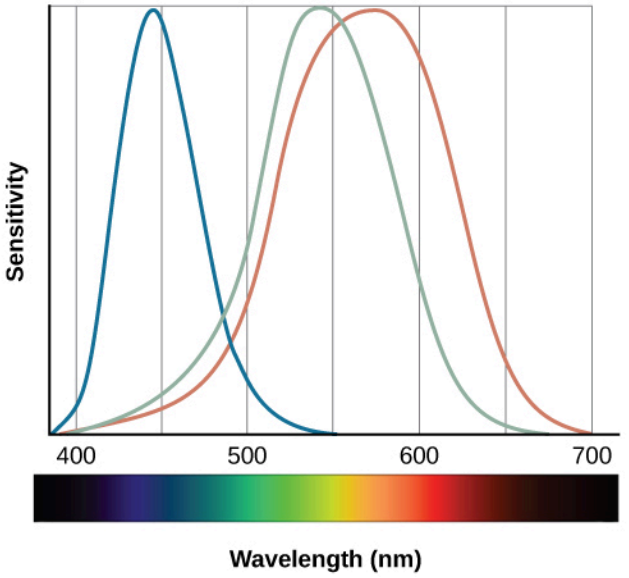

Figure 5.10. This figure illustrates the different sensitivities for the three cone types

You have probably heard of the term “color-blind”, most people who are “color-blind” can see colors, but they confuse certain ones because they look similar to them. The psychological term for this condition is color vision deficiency. The circles in Figure 5.11 are part of a common test used to see whether people are able to tell red from green.

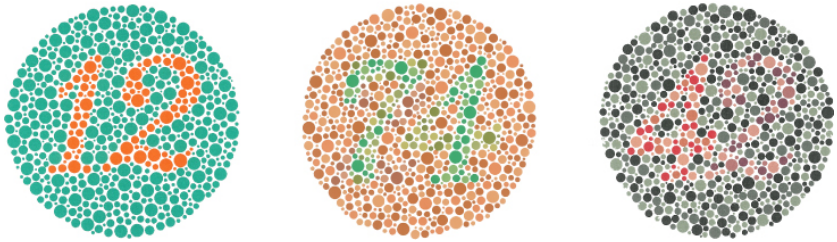

Figure 5.11. The Ishihara test evaluates color perception by assessing whether individuals can discern numbers that appear in a circle of dots of varying colors and sizes. Can you see a number in each of these circles?

The most common color vision deficiency is confusion between reds and greens due to the absence or malfunction of the green or red cone types. This red-green color deficiency affects males more than females (Birch, 2012). Among males, it affects approximately 8% of European Whites, 5% of Asians, 4% of Africans, and less than 2% of indigenous Americans, Australians, and Polynesians (Birch, 2012). Only about 0.4% in females of European Caucasian descent have red-green color deficiency (Birch, 2012). Males are affected more than females because this is a sex-linked recessive genetic trait. Males inherit this gene from their mothers via the X chromosome. The Y chromosome in males is small and so lacks many of the genes that are present on the X chromosome (like the ones associated with color vision). So, if the recessive gene is present on the X chromosome in a male, it will be expressed—giving rise to red-green color deficiency. Typically, females are not affected because they are likely to also have a dominant normal color vision gene on their other X chromosome.

The trichromatic theory explains color vision at the level of the cones. The opponent process theory is a complementary theory, which explains how color is coded at the level of the ganglion cells and thalamus (Land, 1959; Kaiser, 1997). It states that color is coded in opponent pairs: black-white, yellow-blue, and green-red. The basic idea is that these cells in the visual system are excited by one of the opponent colors and inhibited by the other. So, a cell

excited by wavelengths associated with green would be inhibited by wavelengths associated with red, and vice versa. Opponent processing helps to explain why we cannot experience greenish-reds or yellowish-blues as colors. The opponent process theory also explains our experience of negative afterimages. You have probably experienced an afterimage after looking directly at a lightbulb. You still see an image of the lightbulb (albeit of a different color) after you look away. The colors of the afterimage that we see can be predicted by the color pairings identified in the opponent-process theory. If we stare at a red color for a minute and then look at a white background, we will see green. You can test this concept using the flag in Figure 5.12.

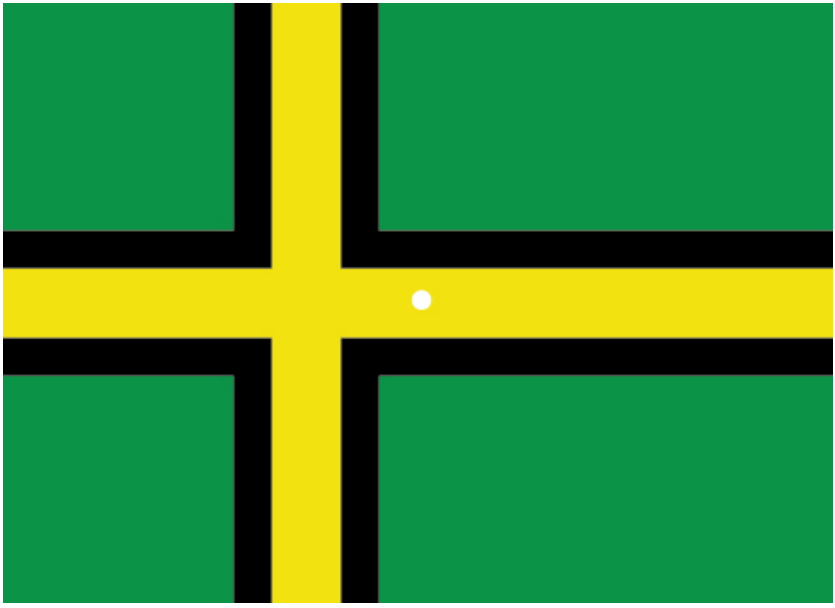

Figure 5.12. Stare at the white dot inside the yellow cross for 30–60 seconds and then move your eyes to a blank piece of white paper. What do you see? This is known as a negative afterimage, and it provides empirical support for the opponent-process theory of color vision.

Link to Learning

Watch this video about color perception to learn more.

DEPTH PERCEPTION

Our ability to perceive three-dimensional (3-D) space is known as depth perception. Our depth perception allows us to describe things as being in front, behind, above, below, or to the side of other things. We can also see real objects as three-dimensional (they have height, breadth and depth) rather than being flat, like a photo.

We use a variety of cues in a visual scene to perceive depth. Some of these are binocular cues, which means that they rely on the use of both eyes working together. Binocular depth perception relies on binocular disparity, in other words, each of our eyes receives a slightly different view of the world. You can experience binocular disparity by doing this simple exercise: Extend your arm fully and hold up one of your fingers. Now, close your left eye and line your finger up with an object in the distance – like a clock or a picture. Then open your left eye and close your right eye without moving your head. You will notice that your finger seems to shift position as you alternate between the two eyes. Our brain has cells that detect this disparity, so when our brain fuses the two different images from the eyes into a single object, we are able to see in 3-D.

A 3-D movie uses the same principle. It is created using specialized cameras equipped with two lenses that are separated

by the same distance that we have between our two eyes. This allows for the production of two separate movies—one provides a view for the left eye and one for the right eye. The special glasses you wear allow the two slightly different images projected onto the screen to be seen separately by your left and your right eye, your brain then has to put them together. This can give the illusion that an animal, person, or object is coming right toward you.

Although we use binocular cues to experience depth in our 3-D world, we also use monocular (or 2-D) cues. These are also used by artists to give the perception of distance in pictures. We rely on monocular cues to make judgements about the relative position of objects in pictures and photographs. For example, in Figure 5.13, a monocular cue called linear perspective allows us to perceive a road stretching out into the distance. Linear perspective is where two parallel lines (like those on the sides of the road in Figure 5.13) converge to give the perception of depth. Some other monocular depth cues include the partial overlap of objects (the telegraph pole is covering the clouds – so we know the clouds are further away), relative size of objects (smaller clouds seem further away than larger ones), and relative height of the base of the objects (the bottom of the small white poles are closer than the bottom of the telegraph pole).

Figure 5.13. We perceive depth in a two-dimensional figure like this one through the use of monocular cues like linear perspective, the parallel lines converging as the road narrows in the distance.

5.3. Gestalt Principles of Perception

In the early part of the 20th century, inspired by a children's toy, Max Wertheimer a German psychologist published a paper demonstrating that individuals perceived motion when watching rapidly presented still images. This is similar to how a Disney cartoon is made —photos are taken of sets of drawings, each one with a very slight difference. Then when they are presented in quick succession, it looks like the objects and people in the pictures are moving. A flip book works on the same principle. Wertheimer (along with Kohler and Koffka) used this example to demonstrate that perception involved more than combining simple sensations. We are not aware of each individual picture – we perceive something new—apparent motion. This discovery led to a new movement within the field of psychology known as Gestalt psychology. The word *gestalt* is used to mean the "whole structure", but its use reflects the idea that the whole perception is different from the sum of its parts. In other words, the brain creates a perception (such as motion) that is more than simply the sum of available sensory inputs (lots of static images). Gestalt psychologists discovered that visual perception depends on a number of predictable principles that the brain uses to organize sensory information. As a result, Gestalt psychology has been

extremely influential in the area of sensation and perception (Rock & Palmer, 1990).

Gestalt psychologists often used optical illusions to demonstrate the principles. For example, if we look at a picture (or a scene in the real world) we make assumptions about figure-ground relationships. Figure refers to central object(s) in the scene, while the ground refers to the background. In general, the figure is more meaningful (i.e., we can recognize it), and tends to be smaller and lower down in the scene than the ground. *What do you see in Figure 5.14?* It is likely that your perception will switch between seeing faces or a vase. So, one minute, you perceive the vases as the figure, and the next, you perceive the faces. If we think about Gestalt figure-ground principles, neither black or white object(s) are clearly the figure– they are both in a similar position, meaningful, and about the same size (Peterson & Gibson, 1994; Vecera & O'Reilly, 1998).

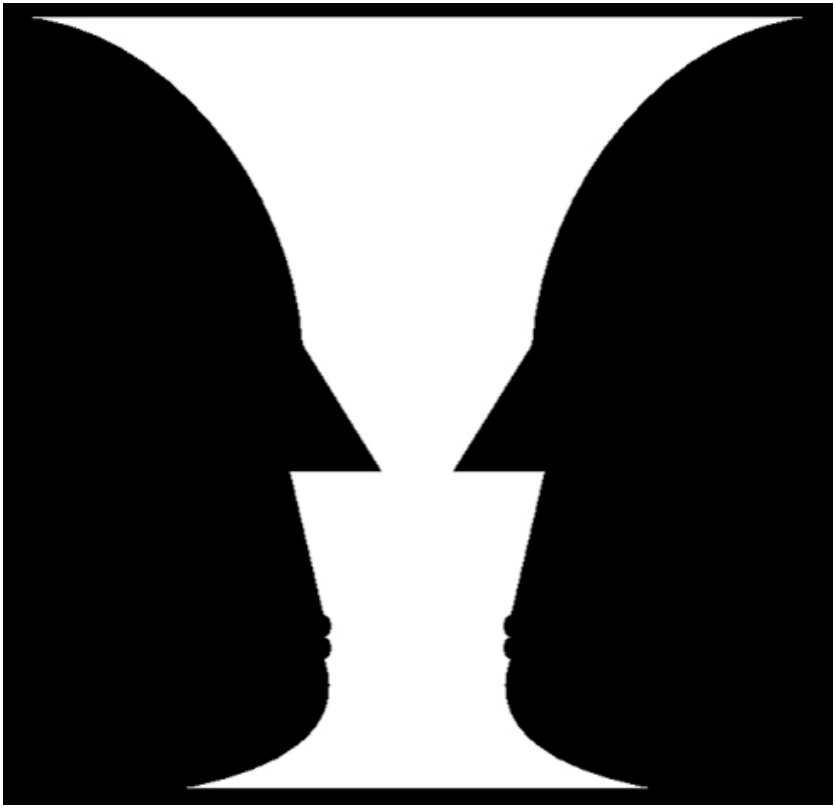

Figure 5.14. The concept of figure-ground relationship explains why this image can be perceived either as a vase or as a pair of faces.

Another Gestalt principle for organizing sensory stimuli into meaningful perceptions is proximity. When things are close to one another, we tend to group them together to form objects. In Figure 5.15a we see a square, and in Figure 5.15b we perceive three columns. The organization of the dots changes our perception.

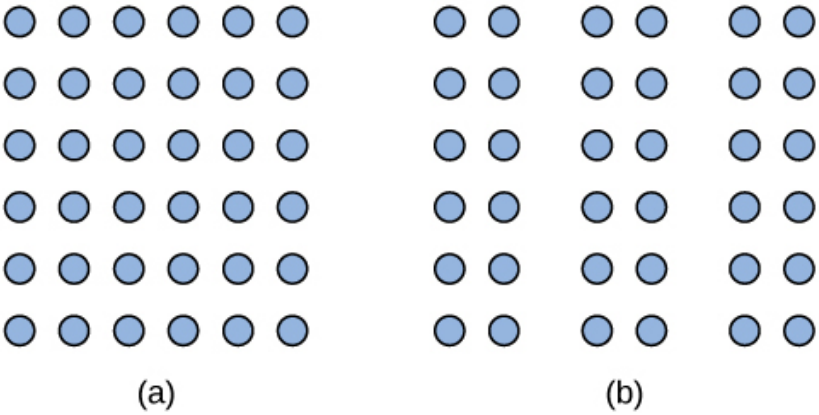

Figure 5.15. The Gestalt principle of proximity suggests that you see (a) one square of dots on the left side and (b) three columns on the right side.

We also use the principle of similarity to group elements into "whole" objects. Things that are alike tend to be grouped together. In Figure 5.16 you probably perceive the figure as being organized in alternating rows (same color) rather than columns.

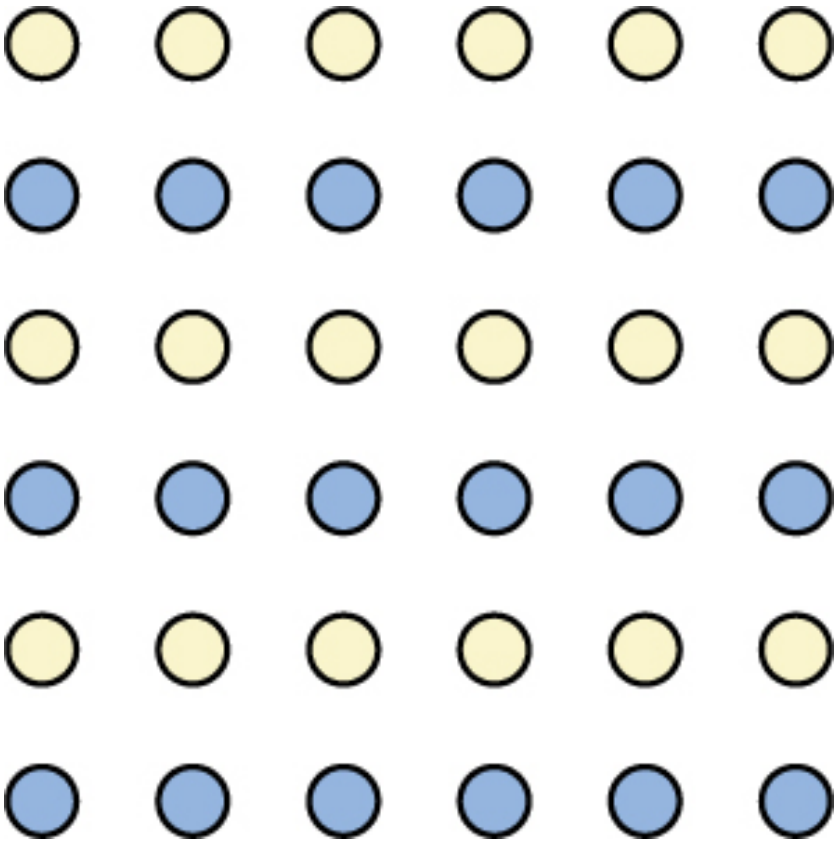

Figure 5.16. When looking at this array of dots, we likely perceive alternating rows of colors. We are grouping these dots according to the principle of similarity.

Two additional Gestalt principles are continuity (or good continuation) and closure. The principle of continuity suggests that we are more likely to perceive lines as continuous and smooth rather than broken or going in strange angles (Figure 5.17).

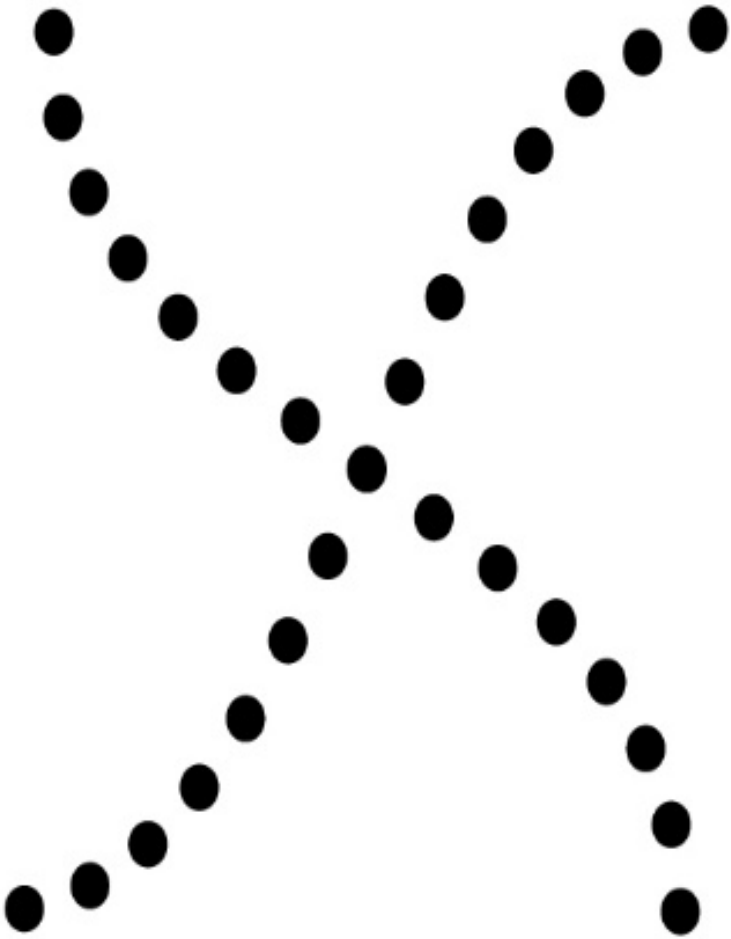

Figure 5.17. Good continuation suggest that we are more likely to perceive this as two overlapping lines, rather than four lines meeting in the center.

The principle of closure states that we organize our perceptions into complete objects rather than as a series of parts (Figure 5.18). Closure allows our brain to fill in the parts we cannot see.

Figure 5.18. Closure suggests that we will perceive a complete circle and rectangle rather than a series of segments.

Link to Learning

Watch this video showing real world examples of Gestalt principles to learn more.

According to Gestalt theorists, our ability to discriminate among shapes and objects occurs because our brains gain experience of the world and then automatically use the Gestalt principles described above to interpret what we see. You probably feel fairly certain that your perception accurately matches the real world, but this is not always the case. Our perceptions are based on educated guesses. These guesses are influenced by a number of factors, including our experiences, expectations, personality, and even what we have just been doing, thinking, and feeling. So, each time we make an interpretation of the world around us, we do so with a particular perceptual set, i.e., some expectation of what we are going to perceive. Priming studies in psychology show us that

it is easy to change someone's perceptual set, simply by presenting some other stimulus prior to a perceptual task. Priming explains why it is easier to startle someone after watching a scary movie. The content of the movie changes our perceptual set.

Stereotypes and bias can also affect perceptual set and perceptual task performance. For example, the ability of non-Black participants to quickly decide whether they are looking at a picture of a gun or a tool (weapon identification task) is influenced by whether they see a picture of a Black or White male just before it (Figure 5.19). Non-Black participants identify guns faster and are more likely to identify tools as guns when the image is paired with a picture of a Black man (Payne, 2001; Payne et al., 2005; Payne & Correll, 2020). Research shows that this type of bias is largely unconscious or implicit. Non-Black participants do not necessarily endorse explicitly prejudiced views toward Black and Brown people. Nevertheless, disproportionately large numbers of unarmed Black and Brown people have been killed by police officers and others claiming that they believed that the victim had a gun and/or represented some threat to their personal safety. Psychologists like Stanford Professor and recipient of the MacArthur Genius Award, Jennifer Eberhardt, and Director of the Center for Policing Equity, Phillip Goff, are fierce advocates for social justice. Both have worked with police departments to help officers recognize implicit bias and to learn strategies to reduce its impact on policing.

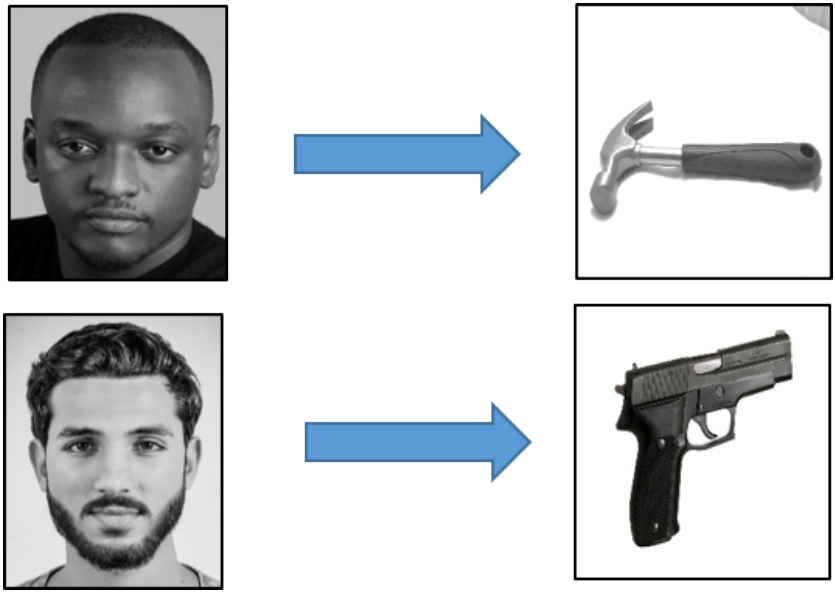

Figure 5.19. Weapon identification task. A Black or a White face is shown first quickly followed by a picture of a gun or a tool. Participants have to rapidly press a button to indicate whether they see a gun or a tool.

Link to Learning

Watch this interview with Dr. Goff and read this article about Dr. Eberhardt's implicit bias training with police officers, to learn more about the ways psychologists can influence police behaviors and training.

5.4. Hearing

Our auditory system converts sound waves, which are changes in air pressure, into meaningful sounds. This translates into our ability to hear the sounds of nature, to appreciate the beauty of music, and to communicate with one another through spoken language. This section will provide an overview of the basic anatomy and function of the auditory system. It will include a discussion of how sensory stimuli are translated into neural impulses, the parts of the brain that process that information, how we perceive pitch, and how we know where sound is coming from.

ANATOMY OF THE AUDITORY SYSTEM

The ear is designed to receive and transmit sound to the receptors deep inside it. It can be separated into three main sections. The outer ear structures include the pinna, which is the visible part of the ear, the auditory canal, and the tympanic membrane or eardrum. The middle ear structures consist of three tiny bones known as the ossicles. The inner ear contains the cochlea, a fluid-filled, snail-shaped structure that contains the sensory receptor cells (hair cells) of the auditory system (Figure 5.20). The vestibular systems is also located in the inner ear, it is involved in balance and movement (the vestibular sense) but not hearing.

Sound waves cause vibration of the air in the auditory canal,

when these vibrations reach the tympanic membrane, it too begins to move. It is more difficult for sound to travel through the fluid-filled middle and inner ear. However, the movement of the ossicles amplifies the vibration, which is then passed onto the inner ear through the flexing of the oval window. This then causes the fluid inside the cochlea to move, which in turn moves the hair cells.

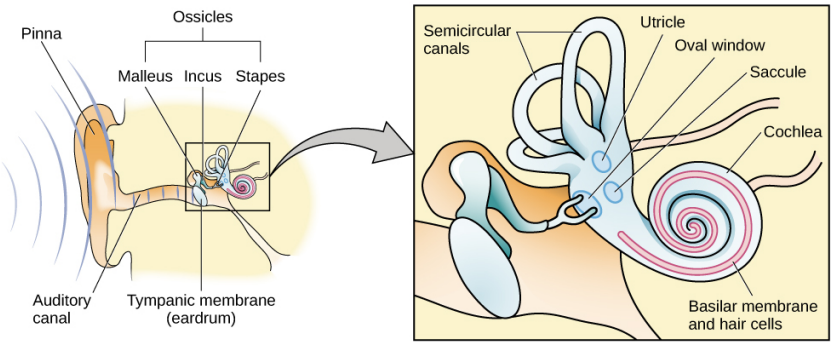

Figure 5.20. The ear is divided into outer (pinna and tympanic membrane), middle (the three ossicles: malleus, incus, and stapes), and inner (cochlea) divisions.

The activation of hair cells is a mechanical process: the physical movement of the hairs on the cell leads to the generation of action potentials (neural impulses) that travel along the auditory nerve to the brain. Auditory information is shuttled to the pons, then the midbrain (inferior colliculus), to the medial geniculate nucleus of the thalamus, and finally to the primary auditory cortex in the temporal lobe of the brain for processing. This information is then passed to other association brain areas (Figure 5.21).

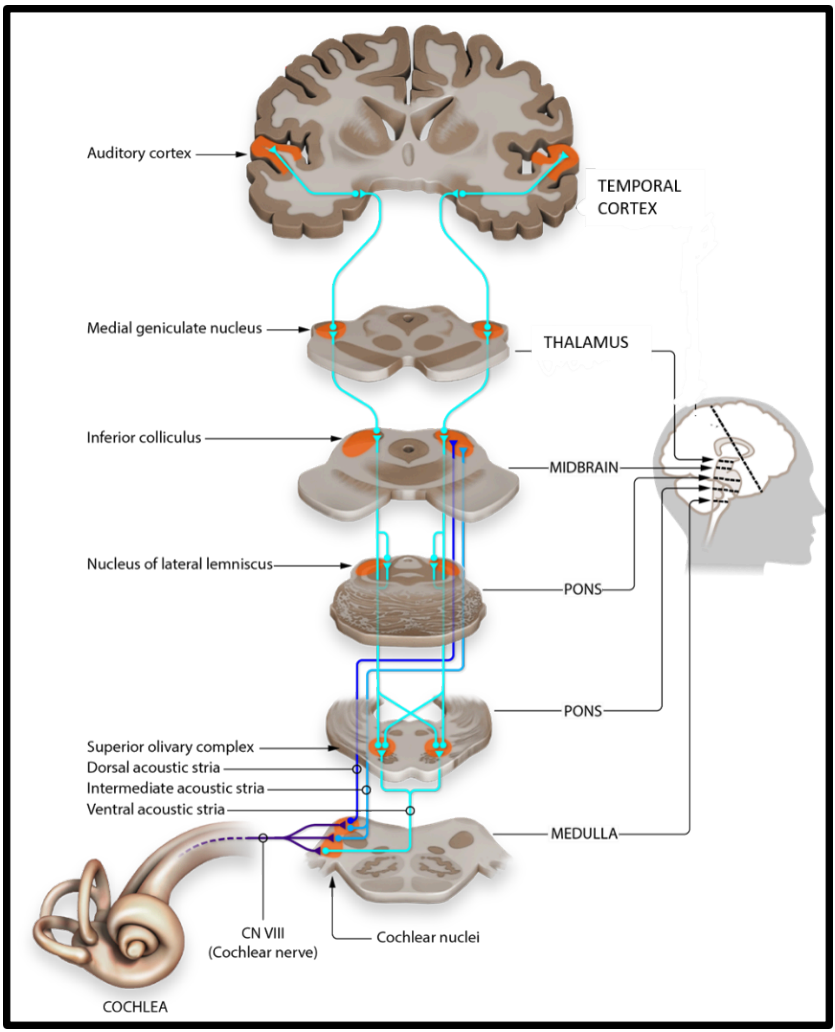

Figure 5.21. Auditory pathways from the ear to the brain.

Like the visual system, information about what a sound is and where it is coming from is processed simultaneously in two parallel neural pathways (Rauschecker & Tian, 2000; Renier et al., 2009). Information that helps us to recognize the sound is processed in the "what" pathway, which extends to the cortex via other parts

of the temporal lobe to the inferior frontal lobe. Information that helps us to understand where the sound is coming from is processed in the "where" pathway, which extends via the parietal lobe to the superior frontal lobe (see Figure 5.22).

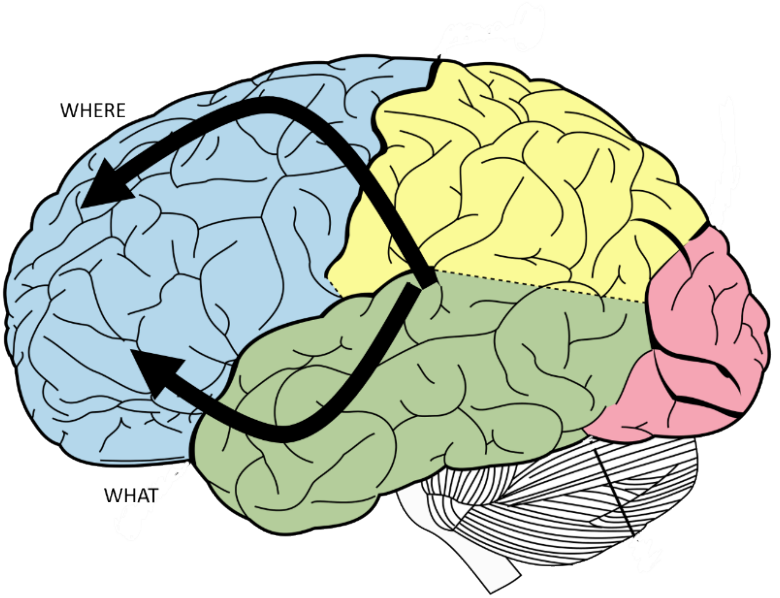

Figure 5.22. Auditory information passes from the primary auditory cortex to association brain areas via the "what" and the "where" pathways.

PITCH PERCEPTION

We perceive sound waves of different frequencies as tones of different pitches. Low-frequency sounds are lower pitched, and high-frequency sounds are higher pitched (see Figure 5.23). In other words, if a person has a deep voice, it is because they are producing a lot of low frequency sound waves, but high pitched voices contain sound waves of higher frequency. How does the auditory system differentiate among various pitches?

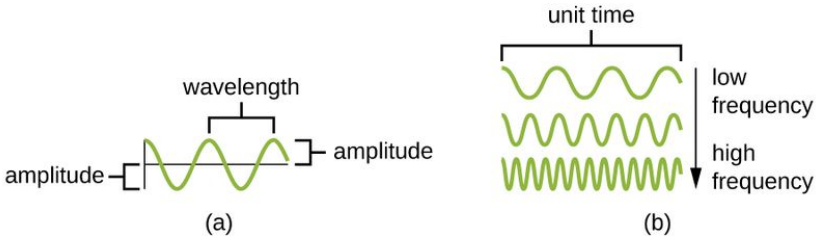

Figure 5.23. Sound waves vary in frequency and amplitude. Higher frequency sounds consist of more vibrations per second than low frequency sounds.

Several theories have been proposed to account for our ability to translate frequency into pitch. We'll discuss two of them here: temporal theory and place theory. The temporal theory of pitch perception suggests that a given hair cell produces action potentials at the same frequency as the sound wave. So, a 200 Hz sound will produce 200 action potentials per second. This seems to work for lower frequencies (up to 4,000 vibrations per second). However, we know that neurons need to have a short rest between producing action potentials so this theory cannot explain how we can detect frequencies of up to 20,000 vibrations per second (Shamma, 2001).

The place theory of pitch perception complements the temporal theory and suggests that auditory receptors in different portions of the cochlea are sensitive to sounds of different frequencies (Figure 5.24). More specifically, hair cells in the base of the cochlea respond best to high frequencies and hair cells in the tip of the cochlea respond best to low frequencies (Shamma, 2001). We can say that the cochlea is arranged tonotopically (according to tone). The primary auditory cortex is also arranged tonotopically.

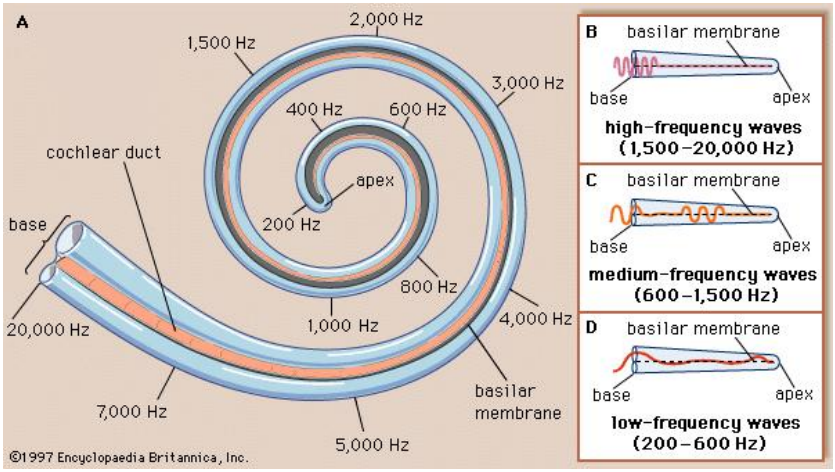

Figure 5.24. Map of how different parts of the cochlear respond to different frequencies.

SOUND LOCALIZATION

The ability to locate sound in our environment is an important part of hearing. Like the monocular and binocular cues that provided information about depth in the visual system, the auditory system uses both monaural (one-eared) and binaural (two-eared) cues to help us to know where sound is coming from.

Sound coming into the ear hits the nooks and crannies in the pinna at different angles depending on its location (above, below in front or behind us). Thus, each sound has a slightly different mixture of frequencies and amplitudes depending on its location (Grothe et al., 2010).

If we have good hearing in both ears, we can also use binaural cues to help localize a sound. Binaural cues are particularly helpful for letting us know whether something is on our right, left, straight ahead, or directly behind us. When sound comes from the side, it reaches the nearest ear first – thus, we have an interaural timing

difference. It is also louder in the nearest ear, which we refer to as an interaural level difference (Figure 5.25). Structures in the brainstem and midbrain (Figure 5.21) are important for determining where a sound originates (Grothe et al., 2010).

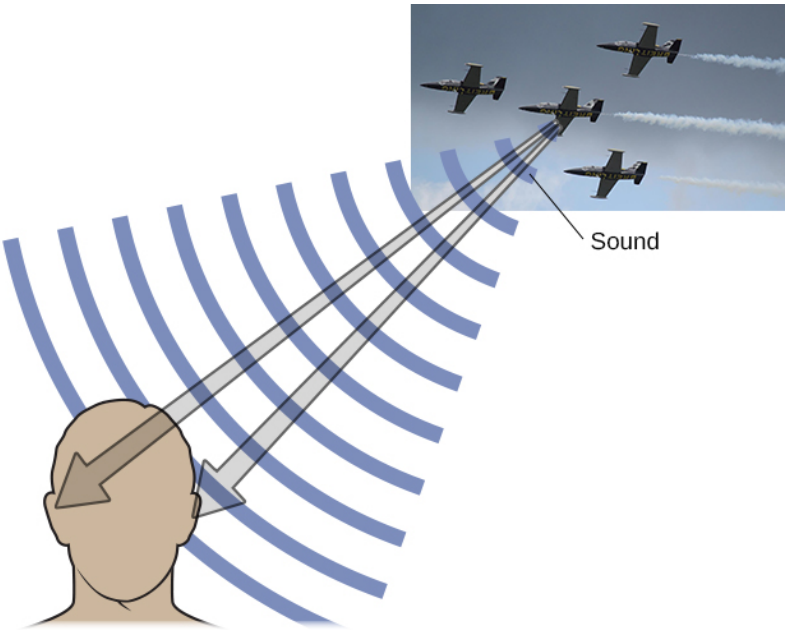

Figure 5.25. Localizing sound involves the use of both monaural and binaural cues.

HEARING LOSS

Deafness is the partial or complete inability to hear. There are two main types of deafness. Conductive hearing loss is when sound fails to reach the cochlea. Causes for conductive hearing loss include blockage of the ear canal, a hole in the tympanic membrane, or problems with the ossicles. However, sensorineural hearing loss is the most common form of hearing loss.

Sensorineural hearing loss occurs when neural signals from the cochlea fail to be transmitted to the brain, and can be caused by many factors, such as aging, head or acoustic trauma, infections and diseases (such as measles or mumps), medications, environmental effects such as noise exposure (Figure 5.26), and toxins (such as those found in certain solvents and metals). Some people are born without hearing, which is known as congenital deafness. This could be inherited or due to birth complications, prematurity, or infections during pregnancy. Most states mandate that babies have their hearing tested shortly after birth (Penn State Health, 2023). This increases the likelihood that problems can be corrected within the critical period for the development of hearing, which is important for language development.

(a) (b)

Figure 5.26. Sensorineural hearing loss can result from excessive exposure to loud noises, like (a) loud music or (b) construction noises.

Conductive hearing loss and sensorineural hearing loss are often helped with hearing aids that amplify the vibrations of incoming sound waves. Some individuals might also be candidates for a cochlear implant as a treatment option. Cochlear implants are electronic devices that consist of a microphone, a speech

processor, and an electrode array. The device receives sound waves and directly stimulates the auditory nerve to transmit information to the brain. However, because hearing and language develop relatively early in life, cochlear implants work best for very young children or for people who had normal hearing and language before becoming deaf.

Link to Learning

Watch this video about cochlear implant surgeries to learn more.

DEAF CULTURE

In the United States and other places around the world, deaf people have their own culture, defined by specific language(s), schools, and customs. In the United States, deaf and hard of hearing individuals often communicate using American Sign Language (ASL); ASL has no verbal component and is based entirely on visual signs, facial expressions, and gestures. Deaf culture particularly values using sign language and creating community among the deaf and hard of hearing. Deaf culture highlights that children who are hard of hearing or deaf often feel isolated, stigmatized and lonely in mainstream education. In contrast, being educated within a school for the deaf fosters a greater sense of belonging (Cripps, 2023). Some members of the deaf community strongly object to the use of cochlear implants and learning how to lip read, arguing that this approach treats deafness as a medical problem that needs to be cured rather than a treasured cultural

identity (Cooper, 2019). When a child is diagnosed as deaf or hard of hearing, parents have difficult decisions to make. They have to decide whether to enroll their child in mainstream schools and teach them to verbalize and read lips or send them to a school for deaf children to learn ASL and have significant exposure to deaf culture? *How would you feel if you were to face this situation?*

5.5. The Other Senses

Vision (our most highly developed sense) and hearing have received an incredible amount of attention from researchers over the years. While there is still more to be learned about how these sensory systems work, we have a much better understanding of them than of our other sensory modalities. In this section, we will explore our chemical senses (taste and smell) and our body senses (touch, temperature, pain, balance, and body position).

CHEMICAL SENSES

Taste (gustation) and smell (olfaction) are called chemical senses because both have sensory receptors that respond to chemicals in the food we eat or in the air we breathe. These senses help promote appetite for edible foods and warn us when something is potentially harmful. There is a prominent interaction between our chemical senses. When we describe the flavor of a given food, we are really referring to the interaction between our gustatory and olfactory processes. This helps to explain why food loses its flavor when we have a blocked nose and cannot smell it.

TASTE (GUSTATION)

You may have learned in grade school that there are four basic

categories of taste: sweet, salty, sour, and bitter. However, we probably have at least five categories of taste receptors in the mouth and throat. Umami was the fifth taste category to be discovered. Umami is a Japanese word that roughly translates to "yummy", and it is associated with the savory taste found in foods like meat, cheese, and mushrooms (Kinnamon & Vandenbeuch, 2009).

Taste receptors cells are grouped inside of taste buds; taste receptor cells have hair-like extensions called cilia that poke through the central pore of the taste bud into the saliva. Chemicals in the saliva bind to the receptors on the cilia (Figure 5.27). Taste receptors are found on the tongue, mouth and throat, and have a life cycle of ten days to two weeks, so burning your tongue won't have any long-term effect; they just grow right back. We have different taste receptors for each of the five basic tastes. When a chemical binds to a receptor, it produces action potentials, which are transmitted to the brain via one of three different nerves, depending on where the receptor is located. Taste information is transmitted to the brainstem, thalamus, and limbic system, and then to the primary gustatory cortex, which is tucked underneath the overlap between the frontal and temporal lobes (Maffei et al., 2012; Roper, 2013). Information is then sent to a multi-sensory area in the brain called the orbitofrontal cortex (OFC), which is responsible for the perception of flavor. The OFC also receives visual, somatosensory, and olfactory information, all of which influence our perception of flavor and whether we are enjoying what we are eating.

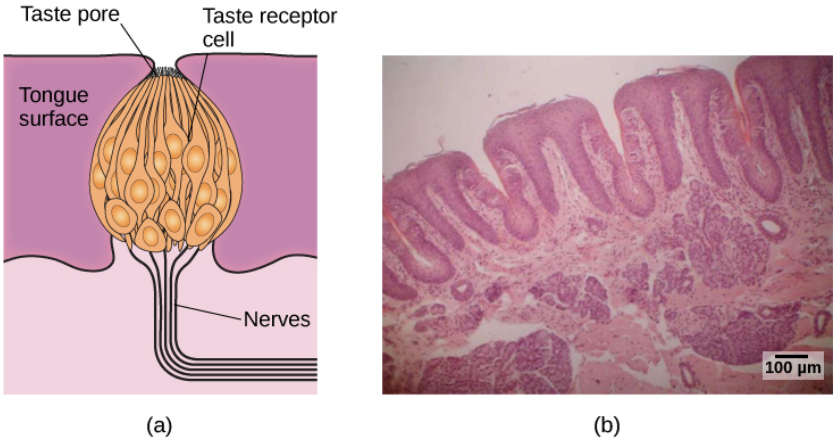

Figure 5.27. (a) Taste buds are composed of a number of individual taste receptors cells that transmit information to nerves. (b) This micrograph shows a close-up view of the tongue's surface.

SMELL (OLFACTION)

Olfactory receptor cells are located in the mucous membrane inside the top of the nose. Chemicals from odorants dissolve in the nasal mucous and stimulate the olfactory receptors that sit on cilia of the olfactory receptor cells (Figure 5.28). Once an odor molecule binds to a given receptor, action potentials are sent to the olfactory bulb: a subcortical structure which sits under the frontal lobe. From there, information is sent to regions of the limbic system and to the primary olfactory cortex, which is located very near to the primary gustatory cortex (Lodovichi & Belluscio, 2012; Spors et al., 2013). Information is then also sent to the OFC.

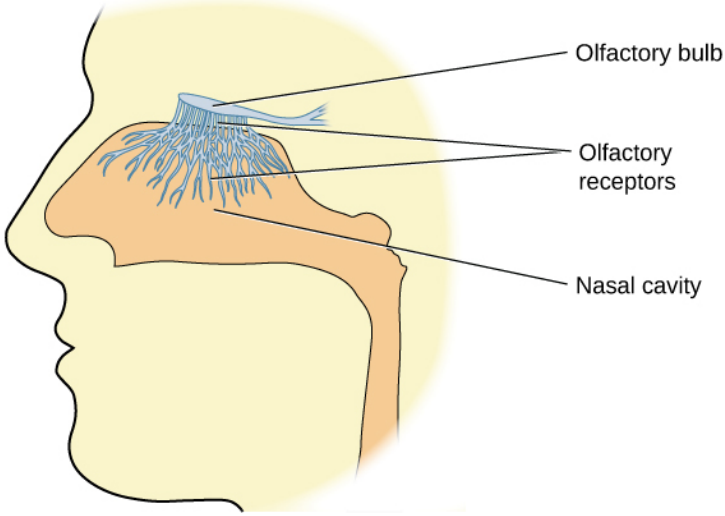

Figure 5.28. Olfactory receptors are located on the hair-like parts of the olfactory cells in the nose. These send messages to the olfactory bulb.

There is tremendous variation in the sensitivity of the olfactory systems across different species. In general, dogs have far superior olfactory systems than our own and are better at detecting most smells than humans. However, there are actually a few odors (like bananas) that we can detect better than dogs (Handwerk, 2017). Some dogs can do some remarkable things with their noses. There is some evidence to suggest that dogs can "smell" dangerous changes in blood glucose levels in people with diabetes, and can detect the presence of cancerous tumors (Wells, 2010). Dogs' extraordinary olfactory abilities are due to having greater numbers and more types of olfactory receptors compared to humans. It is estimated that dogs have about 220 million olfactory receptors consisting of 800 –1200 different types, whereas humans have about 5 million receptors with fewer than 400 different types (Lippi & Cervellin, 2012; Niimura & Nei, 2007).

Many species respond both behaviorally and physiologically to chemical messages, known as pheromones, that are secreted by

other individuals (Wysocki & Preti, 2004). Pheromonal communication often provides information about the reproductive status of a potential mate. For example, when a female rat is fertile, she secretes pheromonal signals that elicit sexual behavior in the male rat (Furlow, 1996, 2012; Purvis & Haynes, 1972; Sachs, 1997). Studies show that humans can also detect odors related to reproduction, which helps to explain why women living together tend to have synchronized menstrual cycles (Cherry & Baum, 2020; Miller & Maner, 2010; Stern & McClintock, 1998). The vomeronasal organ in the nose, which detects pheromones, is very small in humans, which has cast some doubt as to whether these phenomena are actually mediated by pheromones. However, alternative neural pathways for processing pheromones have recently been suggested (Cherry & Baum, 2020).

TOUCH, THERMOCEPTION, AND NOCICEPTION

In Chapter 3, you read about the primary somatosensory cortex in the parietal lobe, which receives information about touch from different parts of the body. However, there are many different aspects of body sensation, including heat and pain detection. We have different somatosensory receptors at different levels of the skin, which respond to specific aspects of stimuli in different ways (Figure 5.29). For example, both Merkel's disks and Meissner's corpuscles are present in the surface of the skin and respond to light touch, but in different ways. Meissner's disks respond to changes in touch, whereas Merkel's disks respond to sustained touch. So, if a small insect hops onto our hand, the Meissner's corpuscles in that part of the skin will respond, but they stop responding if the insect stays still. However, Merkel's disks let you know that the insect is still there. Pacinian corpuscles are found deeper in the skin, so respond to more intense pressure, like a small poke. However, they too only respond to changes in touch,

i.e., when the poking begins or ends. Ruffini corpuscles are present in the collagen, deep in the skin and respond in a sustained way to stretch (Abraira & Ginty, 2013). All of these receptors are called mechanoreceptors because they undergo physical movement when they are stimulated, which is converted into neural energy and sent to the brain to be processed.

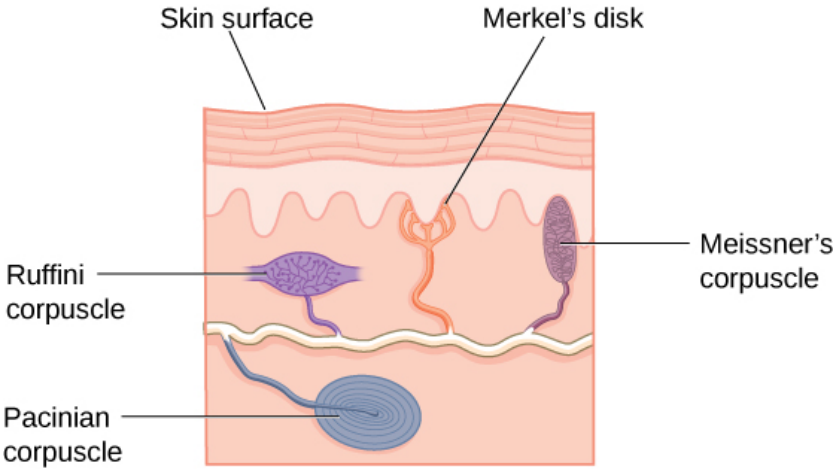

Figure 5.29. There are many types of sensory receptors located in the skin, each attuned to specific touch-related stimuli.

We also have proprioceptors in our joints, muscles, and tendons that send information about the position and movement of our body to our brain (Lackner & DiZio, 2005; Proske, 2006; Proske & Gandevia, 2012). This information travels via the spinal column, to the cerebellum and other parts of the brain. These receptors allow us to perceive the relative position of our body (proprioception), which is important for our sense of balance. They also contribute to our awareness of how our bodies are moving (kinesthesia).

There are also a number of free nerve endings in the skin that respond to thermoception (temperature perception) and nociception, which are the unpleasant sensations that can be elicited by stimuli that are too hot or too cold, or by excessive

pressure (Garland, 2012; Petho & Reeh, 2012; Spray, 1986). All signals from the body's sense receptors are sent to the primary somatosensory cortex via the pons and thalamus.

PAIN PERCEPTION

Pain is an unpleasant experience that involves both physical and psychological components. Feeling pain is adaptive because it makes us aware of an injury, and it motivates us to move away from the cause of that injury. In addition, pain also makes us less likely to suffer additional injury because we are typically more careful with our injured body parts.

Nociceptors provide information about transient threats, such as briefly touching a hot stove. But, they also respond to more prolonged pain that can arise from inflammation (tissue damage) or neuropathy (peripheral or central nervous system damage). In neuropathy, pain signals that are sent to the brain are amplified resulting in intense and often chronic pain. The unpleasantness of pain is an emotional response, which activates the limbic system, and in particular, the anterior cingulate cortex (ACC). Multiple treatment options for pain relief range from relaxation therapy to the use of analgesic medications to deep brain stimulation. The most effective treatment option for a given individual will depend on a number of considerations, including the severity and persistence of the pain and any medical/psychological conditions. Pain relief reduces activation in the ACC. In the previous chapter on consciousness, we talked about how hypnosis can reduce ACC activity and alleviate pain.

Some individuals are born without the ability to feel pain, a condition known as congenital analgesia. This is a very rare genetic disorder, where people can detect differences in temperature and pressure, but they cannot experience pain. As a result, they often suffer significant injuries. Young children suffer serious mouth and tongue injuries because they have bitten themselves repeatedly.

Not surprisingly, individuals with this disorder have much shorter life expectancies due to more injuries and related secondary infections (U.S. National Library of Medicine, 2013).

Link to Learning

Watch this video about congenital insensitivity to pain to learn more.

THE VESTIBULAR SENSE, PROPRIOCEPTION, AND KINESTHESIA

The vestibular sense contributes to our ability to maintain balance and body posture. As seen in Figure 5.30, the major vestibular organs (utricle, saccule, and the three semicircular canals) are located in the inner ear. The vestibular organs are fluid-filled and have hair cells, similar to the ones found in the auditory system. These respond to movement of the head and gravitational forces. When these hair cells are stimulated, they send signals to the brain via the vestibular nerve. Although we may not be consciously aware of our vestibular system sending information under normal circumstances, we can experience unpleasant symptoms such as motion sickness and/or dizziness when the vestibular system is stimulated by movement or when we have infections of the inner ear (Khan & Chang, 2013). In the brain, information from the vestibular system also interacts with that coming from proprioceptors, which is used to help control body position.

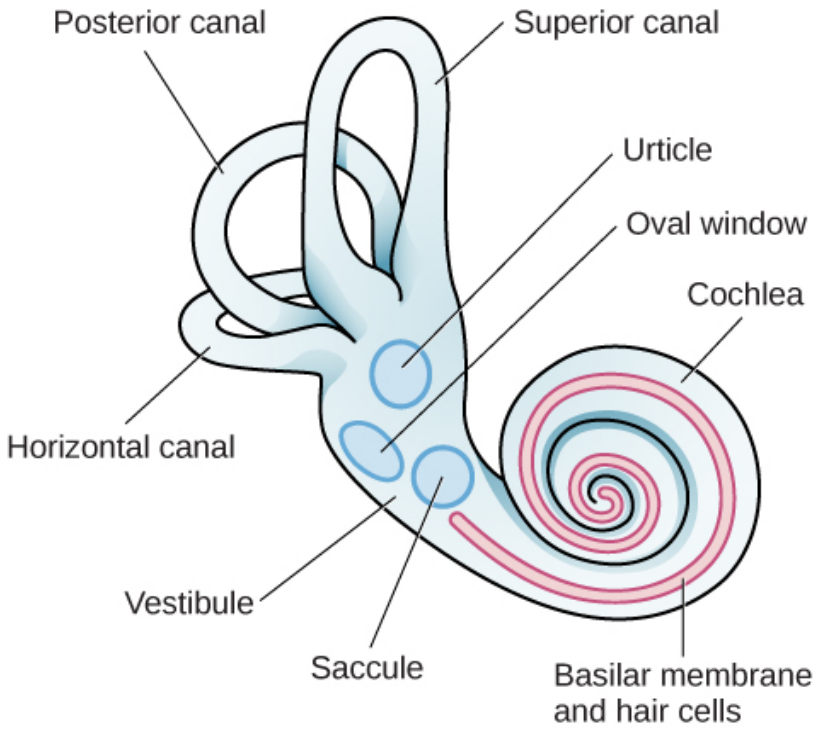

Figure 5.30. The major sensory organs of the vestibular system are located next to the cochlea in the inner ear. These include the utricle, saccule, and the three semicircular canals

MULTIMODAL PERCEPTION

In this chapter we have largely considered our senses as separate from each other, because this is mostly how they have been studied. However, real-life perception mostly depends on information from multiple senses interacting at the same time. Imagine a having special meal at a restaurant – you experience the smell, taste, texture, temperature and flavor of the food, but also the ambience in terms of the lighting, décor, music, and the

loudness of the diners around you. Your dining experience – like most real-life experiences—is multi-modal, it clearly depends on all of your senses. The brain usually processes multi-modal information more efficiently, than information from just one sensory system. We have specialized neurons in multiple parts of our brain that allow us to integrate multimodal sensory information. We have already mentioned the OFC – a multisensory brain area that is important for the perception of flavor. Another multisensory cortical area is the superior temporal sulcus. The superior temporal sulcus has neurons that respond to both visual and auditory stimuli, which are particularly helpful in understanding speech. Watching the movements of people's mouths helps us to understand what they are saying – especially in a noisy environment. Even subcortical areas like the superior colliculus in the midbrain have multisensory neurons. These allow us to rapidly respond to sudden loud sounds in our environment. Imagine if someone behind you suddenly drops a glass and it crashes to the floor. The sound rapidly activates the neurons in the superior colliculi, which also enable us to move our heads quickly and use our eyes in combination with our hearing to figure out whether we are in danger. This is considered to be an automatic orientation response, that helps to protect us from potential threats.

Chapter 5. Conclusions

In conclusion, our sensory systems play a crucial role in providing us with information about the world around us. Sensation involves the process of converting external stimuli into neural energy and transmitting it to the brain, while perception refers to the interpretation and conscious experience of sensory information. Our perception is influenced by factors such as attention, motivation, experience, biases, and culture. Studying sensation and perception provides insights into the mechanisms of sensory processing and how they shape our experiences. This knowledge is crucial for fields such as psychology, neuroscience, medicine, and design, enabling us to improve human well-being, create effective interventions, and design environments that optimize sensory experiences.

Chapter 5. Key Terms

absolute threshold
minimum amount of stimulus energy that must be present for the stimulus to be reliably detected

afterimage
continuation of a visual sensation after removal of the stimulus

amplitude
height of a wave

basilar membrane
floor of the cochlea, which contains the hair cells (sensory receptors for the auditory system)

binaural cue
two-eared cue to localize sound

binocular cue
cue that relies on the use of both eyes

binocular disparity
each eye receives a slightly different view of the world

blind spot
point where optic nerve leaves eye – has no receptors therefore we are unable to respond to visual information falling on this part of the eye

bottom-up processing
system in which perceptions are built from sensory input

closure

organizing elements into complete objects rather than as a series of parts

cochlea

fluid-filled, snail-shaped structure that contains the sensory receptor cells of the auditory system

cochlear implant

electronic device that consists of a microphone, a speech processor, and an electrode array to directly stimulate the auditory nerve to transmit information to the brain

conductive hearing loss

failure transmit sound to the auditory receptors (hair cells)

cone

specialized photoreceptor that works best in bright light conditions and detects color

congenital deafness

deafness from birth

congenital insensitivity to pain (congenital analgesia)

genetic disorder that results in the inability to experience pain from birth

cornea

curved transparent covering over the iris

deafness

partial or complete inability to hear

decibel (dB)

logarithmic unit of sound intensity

depth perception

ability to perceive distance and see in 3-D

figure-ground relationship

how we decide what is in front (figure) and background (ground) in our world

fovea

central part of the retina that we use when we look directly at a visual object, this area contains only cones

frequency

number of waves in a time period

Gestalt psychology

field of psychology based on the idea that the whole is different from the sum of its parts

good continuation

(also, continuity) we are more likely to perceive lines as continuous and smoothly flowing lines rather than jagged and broken

hair cell

auditory receptor cell of the inner ear

hertz (Hz)

cycles per second; measure of frequency of sound waves

inattentional blindness

failure to notice something that is completely visible because of a lack of attention

incus

middle ear ossicle; also known as the anvil

inflammatory pain

signal that some type of tissue damage has occurred

interaural level difference

sound coming from one side of the body is more intense at the closest ear because of the attenuation of the sound wave as it passes through the head

interaural timing difference

small difference in the time at which a given sound wave arrives at each ear

iris

colored portion of the eye

just noticeable difference

smallest difference between two stimuli required to detect that they are different

kinesthesia

perception of the body's movement through space

lens

curved, transparent structure that provides ability to change focus from near back to distance

linear perspective

two parallel lines converge to give perception of depth – monocular cue

malleus

middle ear ossicle; also known as the hammer

Meissner's corpuscle

touch receptor in skin that responds to light changes in pressure (but not sustained pressure)

Merkel's disk

touch receptor that responds to sustained light touch

monaural cue

one-eared cue to localize sound

monocular cue

cue that requires only one eye – often in context of depth perception

neuropathic pain

pain from damage to neurons of either the peripheral or central nervous system

nociception

sensory signal indicating potential harm and maybe pain

olfactory bulb

bulb-like structure under the frontal lobe processes information about smell from nose

olfactory receptor

sensory cell for the olfactory system

opponent-process theory of color perception

color is coded in opponent pairs: black-white, yellow-blue, and red-green

optic chiasm

X-shaped structure that sits just below the brain's ventral surface; represents the merging of the optic nerves from the two eyes and

the separation of information from the two sides of the visual field to the opposite side of the brain

optic nerve

axons that carry visual information from the retina to the brain

Pacinian corpuscle

touch receptor that detects changes in deep pressure

pattern perception

ability to discriminate among different figures and shapes

perception

way that sensory information is interpreted and consciously experienced

perceptual hypothesis

prediction used to explain how we interpret sensory information

pheromone

chemical message secreted by one individual that affects behavior and physiology of another

photoreceptor

cell that detects light and converts it into neural energy

pinna

visible part of the ear that protrudes from the head

pitch

perception of the frequency of a sound

place theory of pitch perception

different portions of the cochlea are sensitive to different sounds of specific frequencies

principle of closure

organize elements into complete objects, rather than small broken lines

proprioception

perception of body position

proximity

elements that are close to one another tend to be grouped together

pupil

small opening in the iris through which light passes, can change in size

retina

light-sensitive lining at the back of the eye

rod

specialized photoreceptor that works well in low light conditions

Ruffini corpuscle

touch receptor deep in the skin that detects stretch

sensation

what happens when sensory information is detected by a sensory receptor and sent to the brain

sensorineural hearing loss

failure to transmit neural signals from the cochlea to the brain

sensory adaptation

failure to perceive stimuli that remain relatively constant over prolonged periods of time

signal detection theory

change in stimulus detection as a function of current mental state

similarity

elements that are alike tend to be grouped together

stapes

middle ear ossicle; also known as the stirrup

subliminal message

message presented below the threshold of conscious awareness

taste bud

structure that contains a group of taste receptor cells

temporal theory of pitch perception

auditory neurons codes for frequency by producing action potentials at the same frequency of the sound

thermoception

temperature perception

top-down processing

interpretation of sensations is influenced by available knowledge, experiences, and thoughts

trichromatic theory of color perception

color vision is mediated by the relative activity across the three groups of cones

tympanic membrane

eardrum

umami

fifth basis taste category – detects savory

vertigo

spinning sensation

vestibular sense

contributes to our ability to maintain balance and body posture

wavelength

length of a wave from one peak to the next

Chapter 5. Personal Application Questions

- How has your cultural background shaped your perception of sensory stimuli (e.g., taste/smell perception, ability to discern specific language sounds, etc.) How is your experience(s) different to other cultures that you know?
- Think about a time when you were in a crowded and noisy environment. How did your attentional processes influence your ability to focus on specific sensory information and filter out distractions?
- Think about a personal experience where you had to rely on both bottom-up and top-down processing to understand a sensory event or situation. How did these two processes interact, and how did they contribute to your overall perception and interpretation?
- Describe a situation where you encountered physical pain and consider how your perception of pain was influenced by factors such as your emotional state, expectations, and the context in which the pain occurred. What helped to get you through the pain?
- Think about a time when you use one sense by itself and other times where you needed to use more than one

sense at a time to do something successfully.

Chapter 5. Critical Thinking Questions

- Imagine you are designing a product packaging for a food item. How can you utilize principles of sensory perception, such as color, texture, and visual cues, to create packaging that not only captures attention but also enhances consumers' perception of the product?
- How does our perception of the world differ from objective reality? Discuss the role of sensation and perception in shaping our subjective experience.
- How do cultural factors influence our perception of sensory stimuli? Give examples of how cultural norms and values can shape our interpretation of sensory information.
- Consider the concept of selective attention. How does our ability to selectively attend to certain stimuli impact our perception of the environment? Can selective attention lead to biases or distortions in our perceptions?
- Discuss the concept of perceptual organization. How do we organize and interpret sensory information to create meaningful perceptions? Provide examples of how principles such as figure-ground relationship and Gestalt principles influence our perception.

- Reflect on the role of illusions in studying sensation and perception. Why do illusions occur, and what do they reveal about the workings of our sensory and perceptual systems? How can illusions help us understand the limitations and biases of our perceptions?
- Reflect on the ways in which our perception of sensory stimuli can be influenced by psychological factors such as emotions, motivations, and past experiences. How might our subjective states shape our interpretation and response to sensory information?

Chapter 5. References

Here are the differences that we spotted in Figure 5.3. 1) People on the path (10 o'clock from globe); 2) Tower on distant shoreline (11 o' clock from the globe); 3) Tree branch (12 o'clock from globe); and 4) Vertical bars in railing around the globe (4 o'clock from globe).

Jill Grose-Fifer edited this chapter, by modifying and adding to the original which can be found at Psychology 2e Openstax.org

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CHAPTER 6. LEARNING

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I*ntroduction to Psychology* (2010). University of Minnesota Libraries Publishing. http://open.lib.umn.edu/intropsyc/

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Figure 6.1. Childhood is a time of incredible learning, such as learning to walk and talk and make friends.

Childhood and adolescence are significant periods for learning and development. However, we continue to learn throughout our lives. We absorb new facts about the world by reading, watching the news, or listening to a podcast. Our friends and family often teach us new skills when we play games or sports, or when we dance or cook together. We frequently learn about new foods and customs when we travel or meet someone from a different part of the world or a different culture to our own. We discover new routes as we navigate our neighborhoods. We learn to avoid foods that make us sick, and to steer clear of dangerous situations. Importantly, we learn to make changes to our lives in response to the demands we face. Without learning, our lives would be dangerous and inefficient. But, how exactly do we learn? This chapter will introduce you to some basic concepts about learning, with an emphasis on behavioral studies by some of the first Western psychologists.

LEARNING OBJECTIVES:

- Define learning
- Explain how learned behaviors are different from reflexes
- Compare and contrast across three basic forms of learning—classical conditioning, operant conditioning, and observational learning
- Explain how classical conditioning occurs
- Summarize the processes of acquisition, extinction, spontaneous recovery, generalization, and discrimination
- Explain the importance of classical conditioning in learning to avoid dangerous situations or foods that make us sick
- Define operant conditioning
- Explain the difference between reinforcement and punishment
- Distinguish between the different types of reinforcement schedules and their efficacy
- Describe latent learning and how it is different from operant conditioning
- Define observational learning
- Discuss the steps in the modeling process
- Explain the prosocial and antisocial effects of observational learning
- Reflect on cultural differences across basic types of learning

6.1. What Is Learning?

Human babies have a lifetime of learning ahead of them. However, they are born with a small number of reflexes that increase their chances for survival. These innate behaviors are unlearned. For example, healthy newborns are born knowing how to suck—this reflex increases the chances that they will get adequate nourishment to grow and thrive. However, a baby quickly learns other new behaviors that help them to adapt to their environment. Unlike innate behaviors, learning involves a relatively permanent change in behavior, knowledge, or skills, and depends on experience. For example, babies would have a hard time picking up small objects (like raisins or small toys) if everything in their environment was larger than their fists. They need experience touching small objects for their fine motor skills to develop.

A lot of learning involves the integration of multiple complex processes. However, traditionally, psychologists studied learning in the simplest way— by investigating the mechanisms underlying associative learning. Associative learning refers to our natural tendency to make connections between events that occur closely together or in a sequence. Although each event is represented by different patterns of neural activity—a process we refer to as encoding—our brains organize the information so that the patterns of activity become linked together. You will see that associative learning is central to all three basic learning processes discussed in this chapter: classical conditioning, operant conditioning and

observational learning. However, they differ in important ways. In classical conditioning; we make associations without any conscious intention. In operant conditioning, our ability to learn involves consciously associating a behavior with a consequence, while observational learning adds social and cognitive layers to both conscious and unconscious associative processes.

In classical conditioning, animals and humans learn to associate a new stimulus with an existing stimulus that always produces a reflex (unlearned behavior). You probably remember that humans have a number of reflexes, which often serve as protective mechanisms. For example, we pull our hand away from a stove if it is very hot, and we blink our eyes when something comes close to our face, or we startle when we hear a loud noise, like a balloon popping or the sound of thunder. However, many people also startle when they see a flash of lightning that comes before the thunder. This is because they have learned through experience that a crash of thunder is on its way. So, they have been classically conditioned to jump when they see the lightning—this is an unconscious process—it happens automatically.

In operant conditioning, animals and humans learn to associate a behavior with its consequence. A pleasant consequence encourages more of that behavior in the future, whereas a punishment deters the behavior. Imagine you are teaching your dog, Hodor, to sit. You tell Hodor to sit, and you give him a treat when he does. After repeated experiences, Hodor begins to associate the act of sitting with receiving a treat (Figure 6.2). Conversely, if you have an invisible electric fence that gives Hodor a small electric shock if he leaves your property, he will soon learn to stay in the yard.

Figure 6.2. In operant conditioning, a response is associated with a consequence. This dog has learned that certain behaviors result in receiving a treat.

Observational learning extends the range of learning beyond classical and operant conditioning. In classical and operant conditioning, learning occurs only through direct experience. However, observational learning occurs by watching others and then imitating what they do. A lot of learning among humans and other animals comes from observational learning. For example, young babies learn to imitate facial expressions at an early age, and wave "Bye" after watching their caregivers doing the same thing. Can you think of something you have learned how to do after watching someone else?

The learning processes covered in this chapter are part of a particular tradition in psychology, called behaviorism. Behaviorists measure learning in terms of changes in observable behavior. However, these approaches do not represent the entire study of

learning. We will cover other types of learning in the chapters on Memory and Cognition.

6.3. Operant Conditioning

The previous section of this chapter focused on the type of associative learning known as classical conditioning. Remember that in classical conditioning, an organism learns to associate new stimuli with other stimuli that produce natural, biological responses such as salivation, fear, or vomiting. The organism does not learn a new behavior, instead, a new signal produces the behavior. Now we turn to a second type of associative learning, operant conditioning. In operant conditioning, organisms learn to associate a behavior with its consequence (Table 6.1). A pleasant consequence makes that behavior more likely to be repeated in the future. For example, if Spirit, a dolphin at the aquarium, does a flip in the air when her trainer blows a whistle, they get a fish. This makes them eager to do it again. The fish reinforces the behavior.

Behaviorist psychologists in the United States, such as Edward Thorndike and B. F. Skinner realized that classical conditioning cannot account for new behaviors, such as riding a bike. So, they went to work on studies centered on operant conditioning. In some of Thorndike's studies (1911), he would put a cat inside a "puzzle box" to see if they could figure out how to escape. Through trial and error, they eventually pressed the lever that opened the door, allowing them to leave the box to eat a scrap of fish placed just outside. The cat got quicker at opening the door, each time it was put in the box. Thorndike took this as an indication of learning. Observing these changes in the cats' behavior led Thorndike to

develop his Law of Effect—the principle that behaviors that lead to a pleasant outcome in a particular situation are more likely to be repeated in a similar situation, whereas behaviors that produce unpleasant outcomes are less likely to occur again. The Law of Effect works for human behavior too—if you pay someone to work for you, it makes it more likely they will continue to want to work for you.

Working with Thorndike's Law of Effect as his foundation, Skinner also began conducting scientific experiments on animals (mainly rats and pigeons) to determine how they learned through operant conditioning (Skinner, 1938). He placed an animal inside an operant conditioning chamber, which has come to be known as a "Skinner box" (Figure 6.6). A Skinner box contains a lever (for rats) or disk (for pigeons) that the animal can press or peck for a food reward via the dispenser. A person counts the number of responses made by the animal. For example, a rat can be trained to press a bar in response to a flashing light, by rewarding the rat with food.

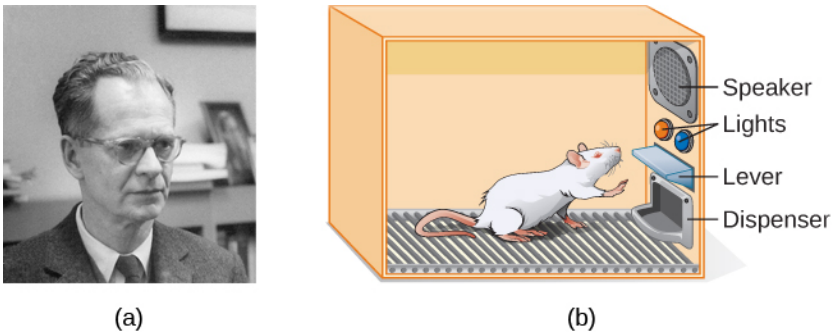

Figure 6.6. In a Skinner box, a rat learns to press a lever in an operant conditioning chamber to receive a food reward.

Link to Learning

Watch this brief video to see Skinner's interview and a demonstration of operant conditioning of pigeons to learn more.

Psychologists use several everyday words—positive, negative, reinforcement, and punishment—in a specialized manner when describing operant conditioning. This can be confusing at first. In operant conditioning, positive and negative do not mean good and bad. Instead, *positive (+)* means you are adding something, and *negative (-)* means you are taking something away. *Reinforcement* means you are increasing a behavior, and *punishment* means you are decreasing a behavior. Reinforcement can be positive or negative, and punishment can also be positive or negative. All reinforcers *increase* the likelihood of a behavioral response. All punishments *decrease* the likelihood of a behavioral response. Now let's combine these four terms: positive reinforcement, negative reinforcement, positive punishment, and negative punishment (Table 6.1).

Table 6.1. *Positive and Negative Reinforcement and Punishment*

	REINFORCEMENT	PUNISHMENT
POSITIVE	Something is *added* to *increase* the likelihood of a behavior.	Something is *added* to *decrease* the likelihood of a behavior.
NEGATIVE	Something is *removed* to *increase* the likelihood of a behavior.	Something is *removed* to *decrease* the likelihood of a behavior.

REINFORCEMENT

Thorndike and Skinner both found that the most effective way to teach a person or animal a new behavior, is with positive reinforcement. In positive reinforcement, a desirable stimulus is given to increase a behavior. For example, you might tell your five-year-old son, Jerome, that if he cleans up his room, he will get some money. Jerome quickly cleans up his room because he wants the $2 you promised him. The money is a positive reinforcement—something that was given to increase the likelihood of a desired behavior (cleaning up). Rewards for a child might include being read a story, receiving money, toys, a favorite food, or a sticker. You can probably think of many more incentives. Fryer (2010) found that paying children in school districts with below-average reading scores to read books increased their reading comprehension. Specifically, second-grade students in Dallas were paid $2 each time they passed a short quiz about a book they had just read (Fryer, 2010). Fryer found that incentives benefited boys (especially Latinx) more than girls. However, Fryer also found no increase in reading and math test scores among 4th graders in New York when they were paid $5 for completing the test and $25 for a perfect score. What do you think about

paying children to do school work? Why do you think the incentives worked for 2nd graders, but not 4th graders?

Early psychologists, such as Pressey (in the 1920s) and Skinner (in the 1950s), created teaching machines that used operant conditioning principles to improve school performance (Ludy, 1988; Skinner, 1961). These machines were early forerunners of computer-assisted learning. In one version of the machine, students were shown a series of multiple choice questions. If they answered a question correctly, a light came on providing immediate positive reinforcement and allowing them to continue to the next question. If they answered incorrectly, nothing happened and they were able to try a different answer. The idea was that students would spend additional time studying the material to increase their chance of being reinforced the next time (Skinner, 1961). This was a way to allow children to learn at their own pace.

Negative reinforcement is another way to increase a behavior, this works by removing an undesirable stimulus. For example, car manufacturers use the principles of negative reinforcement in their seatbelt systems, which beep or ding until you fasten your seatbelt. The annoying sound stops as soon as you exhibit the desired behavior, increasing the likelihood that you will buckle up in the future. The principles of negative reinforcement also determine how likely we are to take an over the counter drug (like aspirin). If it was successful in getting rid of a headache previously, we are more willing to take it again if we have a headache in the future.

PUNISHMENT

Many people confuse negative reinforcement with punishment in operant conditioning, but they are two very different mechanisms. Remember that reinforcement, even when it is negative, always increases a behavior. In contrast, punishment always decreases a

behavior. In positive punishment, you add an undesirable stimulus to decrease a behavior. An example of positive punishment is scolding a student when they are texting in class. In this case, a stimulus (the reprimand) is added in order to decrease the behavior (texting in class). With negative punishment, you take away a pleasant stimulus to decrease behavior. For example, if a child misbehaves, a caregiver might confiscate a favorite toy for a period of time. In this case, a stimulus (the toy) is removed in order to decrease the behavior.

Punishment, especially when it is immediate, can decrease undesirable behavior, especially in the short term. Positive punishments include getting a speeding ticket for driving too fast, or getting blasted by a car horn when stepping into the road without looking. Spanking is another example of positive punishment. However, multiple studies have shown that when children are spanked they are more likely to act out or have mental health issues in the future (Gershoff, 2002; Gershoff et al., 2010).

It is important to note that reinforcement and punishment are not simply opposites. The use of positive reinforcement in changing behavior is almost always more effective than using punishment. This is because positive reinforcement makes the person or animal feel better, helping create a positive relationship with the person providing the reinforcement. Types of positive reinforcement that are effective in everyday life include verbal praise or approval, the awarding of status or prestige, and direct financial payment. Punishment, on the other hand, is more likely to create only temporary changes in behavior because it is based on coercion and may create a negative and adversarial relationship with the person providing the punishment. When the person who provides the punishment leaves the situation, the unwanted behavior is likely to return. Today's Western psychologists and parenting experts favor reinforcement over punishment—they recommend that you catch your child doing something good and reward them for it.

SHAPING

In his operant conditioning experiments, Skinner often used an approach called shaping. Shaping is a technique that guides an animal or person toward a complex behavior, by rewarding successive approximations of the behavior. Shaping is often needed to encourage the development of complex behaviors, because it is extremely unlikely that any organism will display anything but the simplest of behaviors spontaneously. In shaping, behaviors are broken down into many small, achievable steps. For each step, the animal or person is reinforced if they exhibit the behavior. Skinner used shaping to teach pigeons to enact many unusual and entertaining behaviors, such as turning in circles, walking in figure eights, and even playing ping pong; shaping is commonly used by animal trainers today.

Link to Learning

Watch this brief video of Skinner's pigeons playing ping pong to learn more.

Watch this video clip of veterinarian Dr. Sophia Yin shaping a dog's behavior using the steps outlined above to learn more.

It's easy to see how shaping is effective in teaching behaviors to animals, but how does shaping work with humans? Let's consider a common childhood milestone—learning to use the toilet. Few toddlers can get the hang of using a toilet all by themselves. Instead, most caregivers break the learning down into simple steps.

For example, you might first reward your child for sitting on the potty with their clothes on. Then after a while you only reward your child if they sit on the potty without a diaper. The next step might be to reward your child only if they actually use the potty. Before long (hopefully) your child can remove their own clothing and use the potty or toilet when they need to go. Once a behavior like this is reliably established—we rarely have to provide a reinforcement to maintain it.

PRIMARY AND SECONDARY REINFORCERS

For animals, food is typically used as a reinforcer in operant conditioning studies; and to incentivize them further, most researchers stop feeding them prior to a study. What would be a good reinforcer for humans? For Jerome, it was the promise of some money if they cleaned their room. But what if you gave ice cream or candy instead? In this case, you would be using food, which is known as a primary reinforcer. Primary reinforcers relate to innate behaviors, like eating, drinking, sexual activity, temperature regulation, feeling pleasure, or relief from pain. For example, on a hot day, you might be motivated to clean your room, if you know you can have an ice-cold drink when you finish.

On the other hand, secondary reinforcers have no innate value and only have reinforcing qualities when linked with a primary reinforcer. For example, money is only worth something when you can use it to buy other things—either things that satisfy basic needs (food, water, shelter—all primary reinforcers) or other secondary reinforcers. If you were on a remote island in the middle of the Pacific Ocean and you had stacks of money, the money would not be useful if you could not spend it. Some settings use tokens as secondary reinforcers, which can then be traded in for rewards and prizes. Token economies have been found to be very effective at modifying behavior in a variety of settings such as schools, prisons,

and hospitals for the mentally ill. For example, Adibsereshki and Abkenar (2015) found that using tokens to reward appropriate classroom behaviors improved the science test scores in a group of eighth grade girls with intellectual disabilities.

Everyday Connection

Behavior Modification in Children

Parents and teachers often use behavior modification to change a child's behavior. Behavior modification uses the principles of operant conditioning to accomplish behavior change so that undesirable behaviors are switched for more socially acceptable ones. Some teachers and parents create a sticker chart, in which several behaviors are listed (Figure 6.7). Sticker charts are a form of token economy. Each time children perform the behavior, they get a sticker, and after a certain number of stickers, they get a prize, or reinforcer. The goal is to increase acceptable behaviors and decrease misbehavior. In the classroom, the teacher often reinforces a wide range of behaviors, from students raising their hands, to walking quietly in the hall, to turning in their homework. At home, parents might create a behavior chart that rewards children for things such as putting away toys, brushing their teeth, and helping with dinner. In order for behavior modification to be effective, the reinforcement should be connected with

the behavior and the reinforcement should matter to the child and be done consistently.

Figure 6.7. Sticker charts are a form of positive reinforcement and a tool for behavior modification. Once this child earns a certain number of stickers for demonstrating a desired behavior, she will be rewarded with a trip to the ice cream store.

Time-out is another common technique used in behavior modification with children. It operates on the principle of negative punishment. When a child demonstrates an undesirable behavior, they are removed from the desirable activity at hand (Figure 6.8). For example, say that Sophia and her brother Mario are playing with building blocks. Sophia throws some blocks at her brother, so you give her a warning that she will have a time-out if she does it again. A few minutes later, she throws more blocks at Mario. You

remove Sophia from the room for a few minutes. When she comes back, she plays with the blocks without throwing them.

(a) (b)

Figure 6.8. Time-out is a popular form of negative punishment used by caregivers. When a child misbehaves, he or she is removed from a desirable activity in an effort to decrease the unwanted behavior. For example, (a) a child might be playing on the playground with friends and push another child; (b) the child who misbehaved would then be removed from the activity for a short period of time.

REINFORCEMENT SCHEDULES

Western psychology suggests that the best way for a person or animal to learn a behavior through operant conditioning is by using positive reinforcement. For example, you might use positive reinforcement to teach your dog to roll over. At first, the dog might randomly happen to rollover after you tell it to, and so you give it a biscuit. If it does it again, it gets another biscuit. If you give your dog a reinforcer each time it displays the desired behavior (rolling over on command), it is called *continuous reinforcement*. This reinforcement schedule is often the quickest way to teach a new behavior. Providing a reinforcement *immediately* after the behavior is important for learning the association between the behavior and the reward. However, if you stop reinforcing the behavior—it is possible that it will disappear (extinction).

Once a behavior is trained, researchers and trainers often turn to a different reinforcement schedule—partial reinforcement. In partial reinforcement, also referred to as intermittent reinforcement, the person or animal does not get reinforced every time they perform the desired behavior. This ensures that the behavior is maintained but you don't need to have a constant supply of dog biscuits or other rewards. There are several different types of partial reinforcement schedules (Table 6.2). These schedules are described as either fixed or variable, and as either interval or ratio. *Interval* means the schedule is based on the **time** between reinforcements, and *ratio* means the schedule is based on the **number of responses** between reinforcements; either one can be fixed (predictable) or variable. For example, a slot machine pays out on a variable schedule (Figure 6.9).

As we can see from Table 6.2 and Figure 6.10 – different reinforcement schedules affect the behavior of the individual quite dramatically. You will probably notice that ratio schedules are more effective at increasing behaviors than interval schedules. For example, a sales person will tend to make more sales if they are paid a small commission each time they make a sale (fixed ratio) rather than being paid weekly or monthly (fixed interval). Variable schedules also tend to increase behaviors and make them more consistent than fixed schedules. Thus, adding some uncertainty to the schedule, keeps the person or animal more on their toes. Notice that the best (and quickest) way to increase behavior with partial reinforcement is to use a variable ratio schedule. Variable ratio schedules require a person or animal to keep producing the behavior to receive a reward – because they cannot predict when it will come. Slot machines pay out using a variable ratio, which is why it is often difficult to stop playing once we start. Understanding reinforcement schedules can also be helpful when caring for children. Variable ratios of reinforcement are especially effective for increasing desirable behaviors such as doing homework, etc. However, they can also serve to increase undesirable behaviors,

like temper tantrums. If a child is having a tantrum because they want something you said they can't have, giving in every now and then (variable ratio), is likely to encourage future tantrums. So, consistency is best.

Table 6.2. *Reinforcement Schedules*

REINFORCEMENT SCHEDULE	DESCRIPTION	RESULT	EXAMPLE
Fixed interval	Reinforcement is delivered at predictable time intervals (e.g., every week).	Moderate response rate with significant pauses after reinforcement	Shoe salesperson gets paid every week
Variable interval	Reinforcement is delivered at unpredictable time intervals (e.g., after 5, 7, 10, and 20 minutes).	Moderate yet steady response rate	Checking social media
Fixed ratio	Reinforcement is delivered after a predictable number of responses (e.g., after 2, 4, 6, & 8 responses).	High response rate with pauses after reinforcement	Piecework—factory worker getting paid for every x number of items manufactured
Variable ratio	Reinforcement is delivered after an unpredictable number of responses (e.g., after 1, 4, & 9 responses).	High and steady response rate	Gambling—slot machines

Figure 6.9. Slot machines pay out on a variable ratio schedule

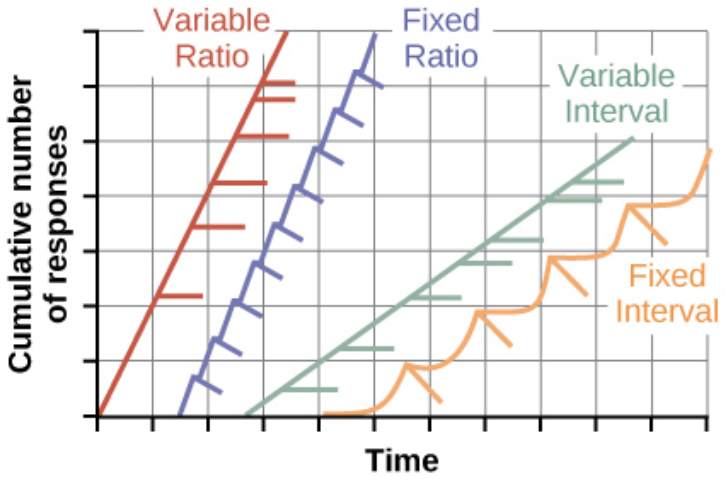

Figure 6.10. The four reinforcement schedules yield different response patterns. The variable ratio schedule is unpredictable and yields high and steady response rates, with little if any pause after reinforcement. A fixed ratio schedule is predictable and produces a high response rate, with a short pause after reinforcement. The variable interval schedule is unpredictable and produces a moderate, steady response rate. The fixed interval schedule yields a scallop-shaped response pattern, reflecting a significant pause after reinforcement.

Table 6.3. *Comparisons of Classical and Operant Conditioning*

	CLASSICAL CONDITIONING	OPERANT CONDITIONING
Conditioning approach	Neutral stimulus paired with an unconditioned stimulus that elicits a reflex behavior	The target behavior is followed by a consequence that influences future behavior
Stimulus timing	The neutral stimulus typically occurs immediately before the response.	The consequence (reinforcement or punishment) occurs soon after the response.

What do you think?

Operant conditioning is often used in child-raising, however, there is considerable cultural variations in how reinforcement and punishment are used. Some cultures (like White middle-class Americans) value praise as a way to reinforce behaviors. However, other cultures (such as in South Asia and China) fear that too much praise can spoil a child (Raj & Raval, 2013). In some cultures, spanking children is relatively common, and in other countries it has been banned (Pace et al., 2019). What are your childhood experiences of operant conditioning and how have they influenced your views on using operant conditioning to shape children's behavior?

COGNITION AND LATENT LEARNING

Watson and Skinner were behaviorists who believed that *all* learning could be explained by conditioning (see Table 6.3 for a reminder about the differences between the two types of conditioning). But some kinds of learning are very difficult to explain using only conditioning. One example of this is when we suddenly find the solution to a problem, as if the idea just popped into our head. This cognitive process is known as insight, the sudden understanding of a solution to a problem. The German psychologist, Wolfgang Köhler (1925), found that chimpanzees who were given a problem that was not easy for them to solve also often

had flashes of insight. When he placed some food in a part of their cage that was too high for them to reach, he found that the chimps first engaged in trial-and-error attempts at solving the problem. However, when these failed they seemed to stop and contemplate for a while. Then, all of a sudden, they would seem to know how to solve the problem, for instance by using a stick to knock the food down or by standing on a chair to reach it.

Edward C. Tolman also believed that organisms can learn even if they do not receive immediate reinforcement (Tolman & Honzik, 1930; Tolman et al., 1946). In one of his studies, Tolman studied the behavior of two groups of hungry rats that were placed in a maze (Figure 6.11). The first group always received a reward of food at the end of the maze. The second group only received food on the 11th day of the experimental period. As you might expect, the rats in the first group quickly learned to navigate the maze, but the rats in the second group seemed to wander aimlessly through it. However, on day 11, they too quickly found their way to the end of the maze to eat the food that was placed there. It was clear to Tolman that the rats had created a cognitive (mental) map of the maze in their earlier explorations but had no motivation to go through it quickly, he called this latent learning. Latent learning refers to learning that is not observable until there is a reason to demonstrate it.

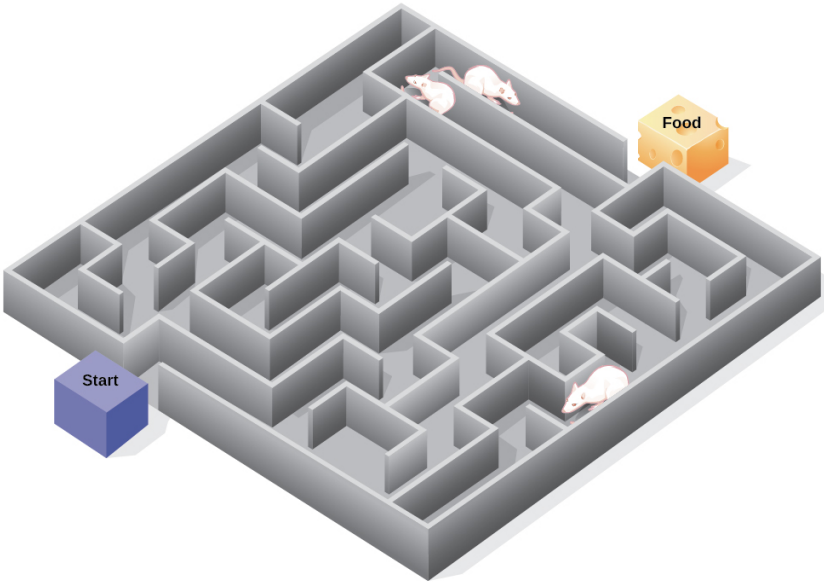

Figure 6.11. Psychologist Edward Tolman found that rats create cognitive maps to navigate through a maze.

Latent learning also occurs in humans. Children may learn by watching the actions of their parents but only demonstrate it at a later date, when the knowledge is needed. For example, suppose that Ravi's dad drives him to school every day. Ravi has learned the route from his house to his school during the rides with his dad, but he's never gone there by himself, so he has not had a chance to demonstrate that he's learned the way. One morning Ravi's dad has to leave early for a meeting, so he can't drive Ravi to school. Instead, Ravi rides his bike and follows the same route that his dad would have taken in the car. This demonstrates latent learning. Ravi had learned the route to school, but had no need to demonstrate this knowledge earlier.

Everyday Connection

This Place Is Like a Maze

Have you ever gotten lost in a building and couldn't find your way back out? While that can be frustrating, you're not alone. At one time or another we've all gotten lost in places like a museum, hospital, or university library. Whenever we go someplace new, we build a mental representation—or cognitive map of the location, as Tolman's rats built a cognitive map of their maze. However, some buildings are confusing because they include many areas that look alike or have short lines of sight. Because of this, it's often difficult to predict what's around a corner or decide whether to turn left or right to get out of a building. Psychologist Laura Carlson (2010) suggests that what we place in our cognitive map can impact our success in navigating through the environment. She suggests that paying attention to specific features upon entering a building, such as a picture on the wall, a fountain, a statue, or an escalator, adds information to our cognitive map that can be used later to help find our way out of the building.

Link to Learning

Watch this video about Carlson's studies on cognitive maps and navigation in buildings to learn more.

6.2. Classical Conditioning

Pavlov (1849–1936) was one of the first scientists to demonstrate how animals learn through classical conditioning. Pavlov was a Russian physiologist, not a psychologist, who was studying the digestive system in dogs (Hunt, 2007). He accidentally discovered how animals learn . Pavlov was measuring the amount of saliva that dogs produced in response to various foods. However, over time, he also noticed that the dogs began to salivate not only at the taste of the food, but also at the sight of food or the food bowl, and even at the sound of the laboratory assistants' footsteps (Pavlov, 1927). Salivating to food in the mouth is a reflex, so no learning is involved. However, dogs do not naturally salivate at the sight of an empty bowl or the sound of footsteps—they had learned to make the associations.

These unusual responses intrigued Pavlov, and he wondered what accounted for the behavior (Pavlov, 1927). He designed a series of carefully controlled experiments to see what other stimuli could cause the dogs to salivate. He was able to train the dogs to associate food with other unrelated (neutral) stimuli, such as the sound of a bell, a light, and a touch on the leg. Pavlov proposed that the dogs had two types of responses to their environment: (1) unconditioned (unlearned) responses, or reflexes, and (2) conditioned (learned) responses, which they learned through experience.

In Pavlov's experiments, the dogs always salivated (response)

when given some meat (stimulus). Thus, the meat is an example of an unconditioned stimulus (UCS) and the dogs' salivation is an unconditioned response (UCR). In his classical conditioning experiments, to teach the dog a new association, Pavlov presented a neutral stimulus (NS) immediately before an unconditioned stimulus (UCS), and repeated the pairings many times. For example, Pavlov would ring a bell and then give the dogs the meat (Figure 6.3). The bell is a neutral stimulus (NS), because it does not naturally elicit a salivation response. Prior to conditioning, the dogs did not salivate when they just heard the bell.

By pairing the bell sound with giving the dog some meat many times, eventually, the previously neutral stimulus (the bell ringing) also began to elicit salivation. Thus, the neutral stimulus had become the conditioned stimulus (CS). The dogs began to salivate to the bell alone. The behavior caused by the conditioned stimulus is called the conditioned response (CR). Pavlov's dogs had learned to associate the bell (CS) with being fed, and they began to salivate (CR) in anticipation of food.

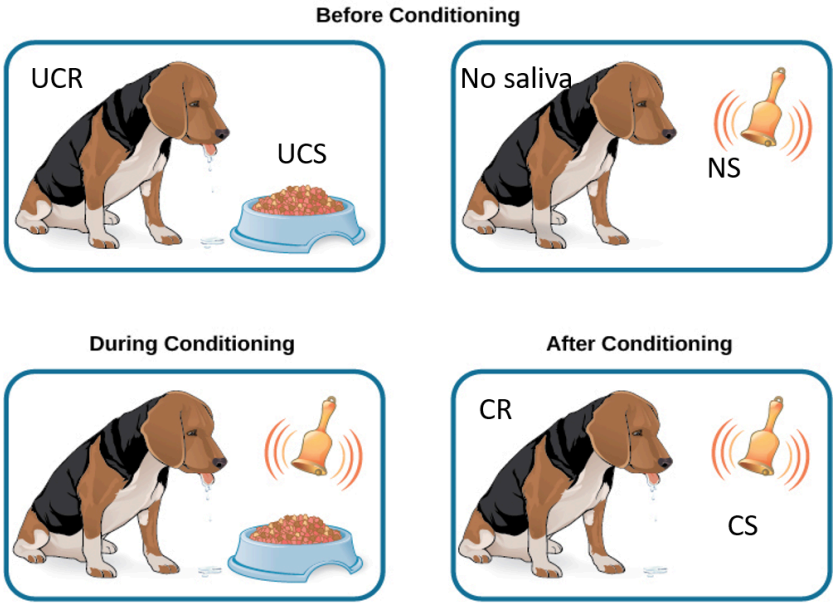

Figure 6.3. Before conditioning, an unconditioned stimulus (food) automatically produces an unconditioned response (salivation), and a neutral stimulus (bell) does not produce a response. During conditioning, the unconditioned stimulus (food) is presented repeatedly just after the presentation of the neutral stimulus (bell). After conditioning, the neutral stimulus alone produces a conditioned response (salivation), thus becoming a conditioned stimulus.

ACQUISITION, EXTINCTION, AND SPONTANEOUS RECOVERY

Pavlov conducted many experiments to try better understand the constraints under which the dogs could be conditioned to associate a neutral stimulus with food. For example, how long did the learning last for? He found that during the acquisition (learning) phase of the studies, when he always paired the neutral stimulus

with food—the dogs always salivated. But, what if he played the bell many times but never followed up with any food? As you might have guessed, over time, the dogs began to produce less and less saliva, until the conditioned response disappeared—this is called extinction. On the following day, Pavlov was quite surprised to find that when he rang the bell, but did not give any meat—the dog started to salivate again. The CR had returned—this is called spontaneous recovery. Acquisition and extinction involve the strengthening and weakening, respectively, of a learned association. Pavlov found that the CR seems to persist for a long time. He also discovered that the CR could be quickly returned to previous levels simply by reminding the dog of the association by presenting a few bell—meat pairings.

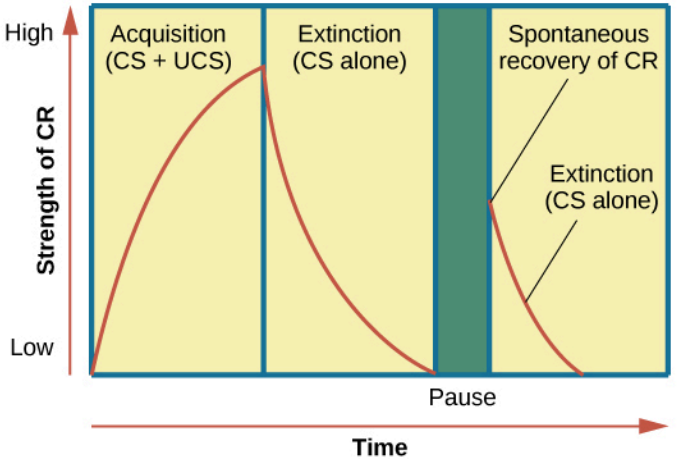

Figure 6.4. This figure shows how the conditioned response changes across acquisition, extinction, and spontaneous recovery. The rising curve shows the conditioned response quickly getting stronger through the repeated pairing of the conditioned stimulus and the unconditioned stimulus (acquisition). Then the curve decreases, which shows how the conditioned response weakens when only the conditioned stimulus is presented (extinction). After a break or pause from conditioning, the conditioned response reappears (spontaneous recovery).

Link to Learning

View this video about Pavlov and his dogs to learn more.

LITTLE ALBERT, FEAR CONDITIONING, AND

PHOBIAS

You might be wondering whether people also learn through classical conditioning? The answer is yes, they do. Before we go into several real-life applications of classical conditioning in people, we will describe a historical study with a human infant. The study was conducted by John B. Watson and his research assistant, Rosalie Rayner, at Johns Hopkins University in Baltimore at the beginning of the 20th century. Watson was one of a group of psychologists in the United States who felt that the methods of introspection and psychoanalysis that were popular in Europe, were too subjective to be considered scientific. Watson believed that psychology should focus on behaviorism—that is, studies that depended on measuring outward observable behaviors (Hunt, 2007). Building on the work of Pavlov, Watson and Rayner conducted a series of studies designed to try and evoke learned fears in an 9-month-old boy, whom they referred to as Albert B. This study has been cited many times—especially as an example of an unethical study in Psychology. There has also been much speculation about the fate of "Little Albert" as the study participant came to be known.

Initially, Little Albert was shown and touched various stimuli, including a rabbit, a dog, a monkey, some masks, some cotton, and a white rat. He was not afraid of any of them, and so they can be considered neutral stimuli. In one study, Watson and Rayner conditioned Little Albert to associate a white rat with feeling scared. Every time they handed Albert the rat, Watson would make a loud clanging sound by hitting a hammer on a metal bar behind the little boy. The sound was very loud and scary and made Little Albert cry. Watson repeated the rat-clanging sound pairings seven times until Albert became frightened of the sight of the rat, even before the noise began. Can you figure out the UCS, CS, UCR, and CR in this study? This highly distressing study (which would not be permitted today) showed that emotions could become conditioned responses. According to Watson, Albert exhibited another

phenomenon of classical conditioning—stimulus generalization. He was also scared by a somewhat similar stimulus—a Santa Claus mask with a white beard, even though he was not directly conditioned to fear it. Stimulus generalization is quite common in classical conditioning. For example, Pavlov found that once trained with the bell—meat pairings, dogs would also salivate to other similar sounds, such as a musical note on a piano. However, Pavlov then taught the dogs to recognize the difference between the two stimuli, by playing the piano note—without any meat repeatedly, and consistently ringing the bell before the meat all the other times. Thus, over time the dogs learned that the piano note was not associated with meat, but the bell was. This is called stimulus discrimination.

Figure 6.5. Little Albert was conditioned to fear rabbits and rats.

Link to Learning

View scenes from this video on John Watson's experiment in which Little Albert was conditioned to respond in fear to various furry objects. At first, you will see Albert's initial reactions to a rat, dog and a monkey. The acquisition phase is not shown in the video. But, you will then see Albert's reactions to the animals after conditioning, in addition to his reaction to a Santa Claus mask and a fur coat.

As you watch the video, look closely at Little Albert's reactions and the manner in which Watson and Rayner present the stimuli before and after conditioning. Based on what you see, would you come to the same conclusions as the researchers? Why is this study considered to be unethical? What safeguards are there today to protect research participants who are babies, like Little Albert, and who are not able to give their consent to participate in a study?

According to Watson, Little Albert was also conditioned to fear a rabbit (Figure 6.5) and a dog. We do not know whether these fears persisted through his life or how strongly he feared the animals. Albert B apparently left the hospital soon after the study and was not traceable. However, there have been many inaccurate portrayals of these studies in the literature, including embellishments by Watson himself (Harris, 1979).

Some clinical psychologists have suggested that phobias—*a*

strong and irrational fear of a specific object, activity, or situation can arise though classical conditioning. For example, a child who is bitten or extremely frightened by a barking dog may learn to fear other dogs too (Oar et al., 2019). Somewhat relatedly, classical conditioning has also been used to help explain the experiences of people with posttraumatic stress disorder (PTSD). PTSD is a severe trauma and stressor-related disorder that can develop after exposure to a serious, often life-threatening traumatic event, (American Psychiatric Association, 2013). PTSD occurs when the individual develops a strong association between the situational factors surrounding the traumatic event (e.g., sounds or smells associated with guns, bombs, etc.) and the trauma itself. The trauma is a UCS because it automatically produces a fear response (UCR). Loud noises, flashing lights, or thinking about the situation in which the trauma occurred can become conditioned stimuli (CS) and so will produce a conditioned response (CR) of feeling extremely scared (Keane, Zimering, & Caddell, 1985). Thus some of the symptoms of PTSD may be explained by classical conditioning.

TASTE AVERSION

In general, learning often takes a lot of practice. For example, we saw that Pavlov's dogs needed many repetitions of the CS-UCS pairings to learn the association. However, conditioned fears can develop after a single CS-UCS pairing, if the response that is evoked is especially intense. Taste-aversion, is another example of classical conditioning that can occur after one single CS-UCS pairing. You probably have had the experience of being ill after eating something and never wanting to eat it again. Even looking at the food or thinking about it makes you feel nauseous. Typically, we only learn to associate events that are close together in time. However, with taste aversion, several hours can pass between the CS (something eaten) and the UCR (nausea, vomiting etc.) caused

by something like bacteria in the food (UCS). Garcia and Koelling (1966) showed that there are some biological constraints when learning to avoid specific foods. The conditioned stimulus has to be food-related, e.g., taste or smell; other non-food stimuli (like flashing lights) will not work. They discovered this by conducting a study investigating how rats learn to avoid poisons. They put a bottle of water containing a chemical that made rats sick, in each test cage. For one group of rats, they had sweetened the water so it had a distinctive taste, and for the other group, they flashed lights and played sounds whenever the rats drank from the water bottle. They found that rats that drank the sweet water learned to avoid the flavor, but none of the rats exposed to lights and sounds learned to avoid the bottle when the lights or sounds were presented.

Taste aversion and fear conditioning are two ways in which classical conditioning can contribute to a species' survival by helping organisms learn to avoid stimuli that pose real dangers to health and welfare (Garcia & Rusiniak, 1980; Garcia & Koelling, 1966). However, cancer patients who are treated with chemotherapy also often develop taste aversions, but these are not helpful. These aversions can occur when a healthy food is eaten just prior to a chemotherapy session, and then the patient gets sick to their stomach after the session (Holmes, 1993; Jacobsen et al., 1993; Hutton et al., 2007; Skolin et al., 2006). Chemotherapy drugs often make people sick, but one way to avoid developing taste aversions to healthy foods during treatment is to ensure that the patient eats something with relatively low nutritional value just before a session. Broberg and Bernstein (1987) used this approach by having children undergoing chemotherapy eat a strongly flavored candy just before a session. The children developed an aversion to the candy flavor, but not to the nutrition-rich food they ate before the candy.

Everyday Connection

Advertising and Associative Learning

Advertising executives often apply the principles of classical conditioning in commercials and advertisements. Many of these revolve around the use of images that are overtly sexual (especially images of women). The idea is that the sexual images will act as unconditioned stimuli in that they often elicit feelings of sexual arousal. Therefore, if a product like a cologne, beer, or car is shown at the same time (or just afterwards), then the product will also become associated with feeling aroused. Of course, advertising executives understand that learning this association depends on repetition, which is why marketing campaigns often saturate us with their advertisements on television, magazines, Internet webpages, billboards, and public transport (Reichert, 2002).

What examples of these types of advertisements can you find on television, in magazines, or on the Internet? How do these advertisements make you feel about the product(s)?

6.4. Observational Learning (Modeling)

Previous sections of this chapter focused on classical and operant conditioning, which are forms of associative learning. In contrast, in observational learning, we learn by watching others and then imitating what they do or say. For instance, have you ever gone to YouTube to find a video when you need to figure out how to do something? The individuals performing the action in the video are called models. We often use instructors in exercise or yoga classes as models too (see Figure 6.12). Research suggests that this imitative learning involves a specific type of neuron in our brains, called a mirror neuron (Hickock, 2010; Rizzolatti et al., 2002; Rizzolatti et al., 2006). Mirror neurons help us to know if we are performing an action correctly. For example, we may have some mirror neurons that fire when we watch someone jumping rope—these same neurons will fire when we jump rope ourselves. In other words, they respond both to specific actions and observation of that action, and can help us to learn if we are correctly copying the actions we observe.

Humans and other animals frequently engage in observational learning. For example, Yamamoto and colleagues in a study of social learning in chimpanzees, gave juice boxes with straws to two groups of chimpanzees. Some of the chimps dipped the straw into the juice box, and then sucked on the small amount of juice at the

end of the straw. Others sucked through the straw directly, getting much more juice. When the "dippers," observed the "suckers," they quickly switched to sucking through the straws directly. Simply observing the other chimps and modeling their behavior, helped them to learn that this was a more efficient method of getting juice (Yamamoto et al., 2013).

Link to Learning

Are chimps smarter than children? Watch this video showing chimps and children performing tasks and contemplate who performed the task better. How about quicker? Why do you think that they behaved in such different ways with the transparent box?

Psychologist Albert Bandura and other researchers proposed a brand of behaviorism called social learning theory, which acknowledged the importance of cognitive processes for learning. Like Tolman, Bandura felt that pure behaviorism could not explain why learning can take place in the absence of external reinforcement. He felt that internal mental states must also have a role in learning.

Figure 6.12. Yoga students learn by observation as their yoga instructor demonstrates the correct stance and movement for the students.

Of course, we don't learn a behavior simply by observing a model. Bandura described specific steps in the process of modeling that must occur for learning to be successful: *attention, retention, reproduction, and motivation*. First, you must pay sufficient attention to notice all the different aspects of the behavior that you are observing. Next, you must be able to retain, or remember, what you observed. Third, you must be able to physically perform the behavior that you observed and committed to memory. Finally, you must have motivation for modeling the behavior. Bandura showed that if a model is reinforced for their behavior, this increases the motivation for others to copy them. This is known as vicarious reinforcement. On the other hand, if you vicariously observe a model being "punished—or having an unpleasant consequence to their behavior, you would be less motivated to copy them. This is called vicarious punishment. Vicarious punishment can help us from making the same mistakes as other people, for example, we

can learn to avoid touching objects (like a stove) that could burn us if we see someone else do it first.

Some of Bandura's research focused on how children modeled aggressive and violent behaviors they saw in adults (Bandura et al., 1961). Bandura conducted an experiment with a five-foot inflatable doll that he called a Bobo doll. Children watched an adult referred to in the study as a "teacher" interacting with the doll. In one scenario, the teacher acted aggressively with the doll, hitting, throwing, and even punching the doll, while a child watched. When the teacher was punished for their bad behavior, the children decreased their tendency to act as the teacher had. When the teacher was praised (or not punished) for his behavior, the children imitated what they did, and even what they said. The children punched, kicked, and yelled at the doll. In a later similar study, Bandura showed that the children's behavior could be modified by offering them incentives to copy what they had seen. When given juice and candy, even children who had watched the adult being punished (with a scolding and a spanking) imitated the behavior just as much as the other groups of children (Bandura, 1965).

Link to Learning

Watch this video clip about the famous Bobo doll experiment to see a portion of the experiment and an interview with Albert Bandura.

What are the implications of Bandura's Bobo doll study? Bandura concluded that we watch and learn, and that this learning can have both prosocial and antisocial effects. Prosocial models can be used to encourage socially acceptable behavior. For example, children

can be encouraged to read, if others read to them. And if they see other people reading (and enjoying it). Children who live in collectivistic societies see how people in their community help each other and follow suit. Who have been the most effective prosocial models in your life?

The antisocial effects of observational learning are also worth mentioning. Children who are victims of violence are more likely to be violent as adults (Widom, 1989; Murrell et al., 2007). In fact, about 30% of child abuse victims become abusive parents (U.S. Department of Health & Human Services, 2013). Children who grow up witnessing their parents deal with anger and frustration through violent and aggressive acts sometimes learn to behave in that manner themselves. It is important to note however, that the *majority* of people who experience abuse and neglect as children, do not show the same behaviors toward their own children.

Multiple studies suggest that violent television shows, movies, and video games may also have antisocial effects on children (Anderson & Gentile, 2008; Kirsch, 2010; Miller et al., 2012). These findings may not be surprising, given that a child graduating from high school has been exposed to around 200,000 depictions of violent acts including murder, robbery, torture, bombings, beatings, and rape through various forms of media (Huston et al., 1992). Viewing media violence can model aggressive behavior, thus encouraging people to act that way in real life situations, but it has also been suggested that repeated exposure to violent acts can desensitize people to its impact. Psychologists are working to understand this dynamic (Figure 6.13).

Figure 6.13. Can viewing violent media make us violent? Psychological researchers study this topic.

Link to Learning

View this video about the connection between violent video games and violent behavior to learn more.

What Do You Think?

Violent Media and Aggression

Does watching violent media or playing violent video games cause aggression? Albert Bandura's early studies suggested television violence increased aggression in children, and more recent studies support these findings. For example, research by

Craig Anderson and colleagues (Anderson et al., 2015; Anderson et al., 2010; Bushman et al., 2016) found extensive evidence to suggest a causal link between hours of exposure to violent media and aggressive thoughts and behaviors. However, studies by Christopher Ferguson and others suggests that while there may be a link between violent media exposure and aggression, research to date has not accounted for other risk factors for aggression including mental health and family life (Ferguson, 2011; Gentile, 2016). *What do think? What are some of the ethical issues around conducting experiments on exposure to violence in children?*

The extent to which children rely on observational learning varies widely across cultures, which in turn contributes to cultural differences in the efficacy of this type of learning (Rogoff et al., 2011). Children from indigenous cultures (e.g., Mayans from Mexico and Guatemala) are frequently present as the adults around them work and engage in social activities. They have multiple opportunities to learn by observing interactions between others in their daily lives (see Figure 6.14). Indigenous children frequently perform simple chores and participate in other community activities from an early age. In contrast, most young children from middle-class Euro-American families spend most of their day apart from adults engaged in child-centered activities (e.g., in preschool settings). Children exposed to indigenous practices are more likely to pay greater attention to a number

of activities at the same time, and so are able to reproduce the behaviors that they observe more accurately than their non-indigenous counterparts. Rogoff and colleagues conducted studies with Mayan children exposed to indigenous practices and middle-class European American children. Each child was told that they were going to be shown how to make an origami figure and were given a toy to distract them. At the same time, their sibling was being shown how to make a different toy at another table. Video recording showed that the Mayan children (but not the Euro-American children) paid close attention to what their sibling was doing. A week later, only the Mayan children were able to create the toy they had seen their sibling make, with relatively little assistance (Rogoff et al., 2011).

Figure 6.14. Mayan families at the market together. Children observe the mothers' interactions with the vendors at the market.

Chapter 6. Conclusions

Learning is a fundamental process that encompasses acquiring various types of knowledge and skills. Classical conditioning involves the association of a neutral stimulus with a natural reflex response. This form of learning typically occurs through repeated pairings, leading to the conditioned stimulus triggering the conditioned response. Though in the case of fear learning and taste aversion it can occur after a single pairing. Operant conditioning focuses on the consequences of behavior. Individuals learn to associate their actions with reinforcements (which increase behavior) or punishments (that reduce behavior), shaping future actions. Not all learning can be explained by conditioning. Latent learning suggests that learning can occur without immediate reinforcement and is not always immediately expressed. Lastly, observation learning highlights the importance of observing and imitating others' behaviors, fostering the acquisition of new skills through modeling. Together, these different types of learning contribute to our understanding of how individuals acquire knowledge and adapt their behavior to meet life's challenges.

Chapter 6. Key Terms

acquisition
period of initial learning in classical conditioning in which a human or an animal begins to connect a neutral stimulus and an unconditioned stimulus so that the neutral stimulus will begin to elicit the conditioned response

associative learning
form of learning that involves connecting certain stimuli or events that occur together in the environment (classical and operant conditioning)

classical conditioning
learning in which the stimulus or experience occurs before the behavior and then gets paired or associated with the behavior

cognitive map
mental picture of the layout of the environment

conditioned response (CR)
response caused by the conditioned stimulus

conditioned stimulus (CS)
stimulus that elicits a response due to its being paired with an unconditioned stimulus

continuous reinforcement
rewarding a behavior each time it occurs

extinction
decrease in the conditioned response when the unconditioned stimulus is no longer paired with the conditioned stimulus

fixed interval reinforcement schedule
behavior is rewarded after a set amount of time

fixed ratio reinforcement schedule
set number of responses must occur before a behavior is rewarded

instinct
unlearned knowledge, involving complex patterns of behavior; instincts are thought to be more prevalent in lower animals than in humans

latent learning
learning that occurs, but it may not be evident until there is a reason to demonstrate it

law of effect
behavior that is followed by consequences satisfying to the organism will be repeated and behaviors that are followed by unpleasant consequences will be discouraged

learning
change in behavior or knowledge that is the result of experience

model
person who performs a behavior that serves as an example (in observational learning)

negative punishment
taking away a pleasant stimulus to decrease or stop a behavior

negative reinforcement
taking away an undesirable stimulus to increase a behavior

neutral stimulus (NS)
stimulus that does not initially elicit a response

observational learning
type of learning that occurs by watching others

operant conditioning
form of learning in which the stimulus/experience happens after the behavior is demonstrated

partial reinforcement
rewarding behavior only some of the time

positive punishment
adding an undesirable stimulus to stop or decrease a behavior
positive reinforcement
adding a desirable stimulus to increase a behavior
primary reinforcer
has innate reinforcing qualities (e.g., food, water, shelter, sex)
punishment
implementation of a consequence in order to decrease a behavior
reflex
unlearned, automatic response by an organism to a stimulus in the environment
reinforcement
implementation of a consequence in order to increase a behavior
secondary reinforcer
has no inherent value unto itself and only has reinforcing qualities when linked with something else (e.g., money, gold stars, poker chips)
shaping
rewarding successive approximations toward a target behavior
spontaneous recovery
return of a previously extinguished conditioned response
stimulus discrimination
ability to respond differently to similar stimuli
stimulus generalization
demonstrating the conditioned response to stimuli that are similar to the conditioned stimulus
unconditioned response (UCR)
natural (unlearned) behavior to a given stimulus
unconditioned stimulus (UCS)
stimulus that elicits a reflexive response
variable interval reinforcement schedule
behavior is rewarded after unpredictable amounts of time have passed

variable ratio reinforcement schedule
number of responses differ before a behavior is rewarded

vicarious punishment
process where the observer sees the model punished, making the observer less likely to imitate the model's behavior

vicarious reinforcement
process where the observer sees the model rewarded, making the observer more likely to imitate the model's behavior

Chapter 6. Personal Application Questions

- What is your personal definition of learning? How do your ideas about learning compare with the definition of learning presented in this text?
- What kinds of things have you learned through the process of classical conditioning? Operant conditioning? Observational learning? How did you learn them?
- Can you think of an example in your life of how classical conditioning has produced a positive emotional response, such as happiness or excitement? How about a negative emotional response, such as fear, anxiety, or anger?
- Think of a behavior that you have that you would like to change. How could you use behavior modification, specifically positive reinforcement, to change your behavior? What is your positive reinforcer?
- What is something you have learned how to do after watching someone else?

Chapter 6. Critical Thinking Questions

- Compare and contrast classical and operant conditioning. How are they alike? How do they differ?
- What is the difference between a reflex and a learned behavior?
- If the sound of your toaster popping up your toast causes your mouth to water, what are the UCS, CS, and CR?
- Explain how the processes of stimulus generalization and stimulus discrimination are considered opposites.
- How does a neutral stimulus become a conditioned stimulus?
- What is the difference between negative reinforcement and punishment?
- What is shaping and how would you use shaping to teach a dog to roll over?
- What is the effect of prosocial modeling and antisocial modeling?
- Cara is 17 years old. Cara's mother and father both drink alcohol every night. They tell Cara that drinking is bad and

she shouldn't do it. Cara goes to a party where beer is being served. What do you think Cara will do? Why?

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CHAPTER 7. MEMORY

Figure 7.1 We have many memorable events in our lives

Western psychologists have often conceptualized memory as a computer-like information processing system—information is encoded, stored, and retrieved. However, until recently there has been little attention to how environmental influences shape these processes. Although, there is still much to be learned, this chapter tries to integrate current knowledge about culture and memory with traditional Western memory models. In this chapter, we will explore the fundamental role that memory plays in human cognition. We will delve into different models of memory to provide a comprehensive understanding of how information is processed and retained, and we will look at how some of the ways that culture shapes our memory. We will discuss the fallibility of memory by exploring common memory errors and their implications in our daily lives. Furthermore, we will explore the intricate relationship between different parts of the brain and various aspects of memory, including amnesia. Lastly, we will equip you with memory-enhancing strategies and study techniques that you can apply to optimize your learning as a college student.

LEARNING OBJECTIVES

- Explain the importance of memory in human cognition.
- Explain what happens in the three stages of memory: encoding, storage, and retrieval.
- Differentiate between sensory memory, short-term memory/working memory, and long-term memory.
- Discuss the role of rehearsal, deep processing and elaboration in memory storage.
- Discuss the factors that influence the retrieval of information from long-term memory.
- Describe the processes and types of explicit and implicit

memory

- Describe common memory errors and their implications in people's lives
- Describe how different parts of the brain play a role in different aspects of memory
- Explain how cultural norms, values, and practices shape memory processes
- Recognize and apply memory-enhancing strategies and study techniques

Materials in this chapter were edited by Jill Grose-Fifer (2023) and are based on original OER materials from the following sources:

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7.1 How Memory Functions

Western cognitive psychologists have historically viewed memory as a computer-like information processing system (Figure 7.2). We get information "into" our memories via a process called encoding—energy from the outside world is converted by our sense organs into neural energy and is sent to the brain. Thus, our experiences of the world are represented or coded in terms of patterns of neural activity. The information needs to be stored in such a way that we can retrieve it when needed. However, memory does not work like a video recorder; information is modified during each stage of processing—encoding, storage, and retrieval. Therefore, our memories often do not provide a wholly accurate representation of our external world. We will return to this theme when we discuss memory errors later in the chapter.

Figure 7.2. Encoding involves inputting information into the memory system. Storage is the retention of the encoded information. Retrieval, or getting the information out of memory and back into awareness, is the third function.

Information-processing models of memory suggest that we have three different types of memory stores, each of which retains information for a different length of time. When we are studying for a test, our goal is to store information permanently, which requires that it reaches long-term memory (LTM). According to Atkinson and Shiffrin's popular model of memory (Figure 7.3), if we want information to be retained permanently, it has to pass through three distinct stages: sensory memory, short-term memory, and finally long-term memory (Atkinson & Shiffrin, 1968). Information gets re-encoded at each stage of the model.

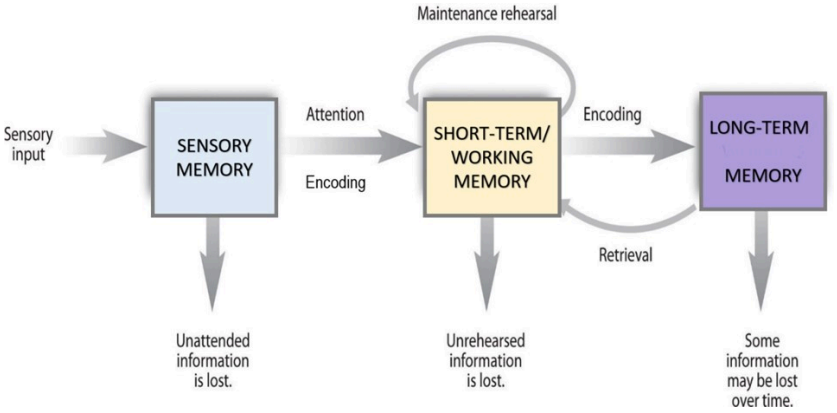


Figure 7.3. According to the Atkinson-Shiffrin model of memory, information passes through three distinct stages in order for it to be stored in long-term memory. Credit: Modified from http://open.lib.umn.edu/intropsyc/

SENSORY MEMORY

In the Atkinson-Shiffrin model, information from the environment is processed first by our various sensory systems (vision, hearing, taste, etc.) and this information is held very briefly in sensory memory. Visual sensory memories (commonly referred to as iconic

memories) last about 0.5 seconds longer than the actual stimulus. In contrast, auditory sensory memories (echoic memories) last about 3 seconds longer than the actual stimulus. This brief persistence of sensory information in our memory allows us to perceive the world in a continuous and coherent fashion. For example, as we move our eyes from one location to another, we create overlapping sensory memories. The persistence of visual memory can be demonstrated by moving a flashlight or sparkler very quickly to form a shape. For example, if we move our sparkler very quickly in a circular motion, we can often see a circle (see Figure 7.4). This is because the sensory memory of the starting position and all the other intermediate positions of the sparkler are still present by the time we finish drawing the shape. Similarly, sounds stay in our echoic memory for a few seconds allowing us to integrate them with other sounds, such as other syllables in a word, or words in a sentence. Researchers believe that although the duration of sensory memory is brief, it can hold a massive amount of information at one time—in other words, it has a very large capacity.

Figure 7.4. Visual sensory (iconic) memory—if you move your sparkler quickly enough—you might see a circle.

SHORT-TERM MEMORY/WORKING MEMORY

As you know from Chapter 5 when you read about inattentional blindness, we need to pay attention (at least at some level) for information to enter into our conscious thoughts. Otherwise the information is lost. This ensures that we do not waste time

processing irrelevant stimuli in our environment. Provided we pay sufficient attention to it, information will make its way into our intermediate memory store. Psychologists use different terms to describe this kind of memory, including short-term memory (STM) and more recently, working memory (WM). Baddeley and Hitch (1974, 2000) proposed that WM is a more accurate description of what happens between sensory and LTM. Working memory describes whatever we are thinking about in a given moment—in other words, what our minds are working on. As you can see in the model in Figure 7.3, information can enter working memory either from LTM (what we already know) or from sensory memory (what is happening in our environment at a given moment).

Because our cultural values influence how we prioritize different kinds of information, they also determine how we direct our attention. Our attentional focus in turn determines how much (and what kinds of information) we remember. For example, people from East Asian countries are more likely to pay equal attention to foreground and background objects in a scene, and so not surprisingly, they have better memory for contextual details and non-focal objects compared to Euro-Americans who attend more to central objects (Wang, 2021).

WM/STM has limited processing abilities. This is why it is so difficult for us to multi-task. For example, taking notes in class can be very demanding, especially if your instructor is talking very quickly about a topic that you have never heard about before. Usually psychologists test WM/STM with simple tasks like asking people to repeat back a sequence of numbers in the same order. There is a limit to the number of digits that we can remember in this kind of task. Test out your working memory using the demonstration below:

Exercise: LET'S TEST YOUR WORKING MEMORY

Have a partner read the strings of random numbers in Figure 7.5 out loud to you, beginning each string by saying, "Ready?" and ending each by saying, "Recall," at which point you should try to write down the string of numbers from memory—they have to be in the correct order.

9754 68259 913825 5316842 86951372 719384273
6419 67148 648327 5963827 51739826 163875942

Figure 7.5. Work through this series of numbers using the recall exercise explained above to determine the longest string of digits that you can store.

How did you do? In his much cited article, Miller (1956) estimated the capacity of working memory to be 7 (plus or minus 2), and referred to this as the "magic number". In other words, most of us are limited to holding between 5 and 9 digits in our conscious thoughts at any one time. However, we can potentially remember more than 7 numbers, if we divide the number string into parts—a strategy known as chunking. For example, a telephone number is typically chunked into three parts, e.g., 212-237-8000. More recent research suggests that we can remember 4 (plus or minus 1) chunks of information (Bodie et al., 2006; Cowan, 2010; Mathy & Feldman, 2012). In the STM/WM test you just did, chunking the numbers into groups of three can be helpful for remembering more of the number string, e.g., 719-384-273. Did you use that strategy? Apart from telephone numbers, can you think of other numbers in our daily lives that are also chunked to help us to remember them? Chunking might help you remember other important numbers that are in a continuous string– like your student ID number.

In addition to the limited capacity of working memory, we can only hold incoming information in our minds for a limited amount of time (about 15 to 30 seconds). For example, if someone tells you a string of numbers (like a phone number) you will probably

be able to repeat them back to them in the same order straight away, especially if you use chunking. However, if they give you a distracting task to do first—like counting backwards in 3s from 99, by the time you get to zero, it is unlikely that you will still be able to recite the numbers correctly. The information has been lost. If we want to keep the numbers in STM/WM we need to repeat them – either out loud or silently to ourselves—this is called maintenance rehearsal. However, the distraction of having to count backwards prevents us from rehearsing.

Peterson and Peterson (1959) used a similar technique to investigate how long we can hold new information in working memory, without rehearsing or using other memory strategies. They asked participants to try and remember a list of trigrams. Trigrams are 3-letter nonsense words (e.g., CLS, TPQ etc.) that are used a lot in memory research, because they are unfamiliar to participants. However, they also used distracting tasks of various lengths of time to prevent rehearsal. Participants remembered about 80% of the trigrams after a 3-second delay, but only 10% after a delay of 18 seconds. Therefore, they concluded that a lot of the content of working memory decayed in about 18 seconds (see Figure 7.6. It is also likely that trying to remember the earlier trigrams interfered with the ability to remember later ones—interference is a common issue in memory (Keppel & Underwood, 1962).

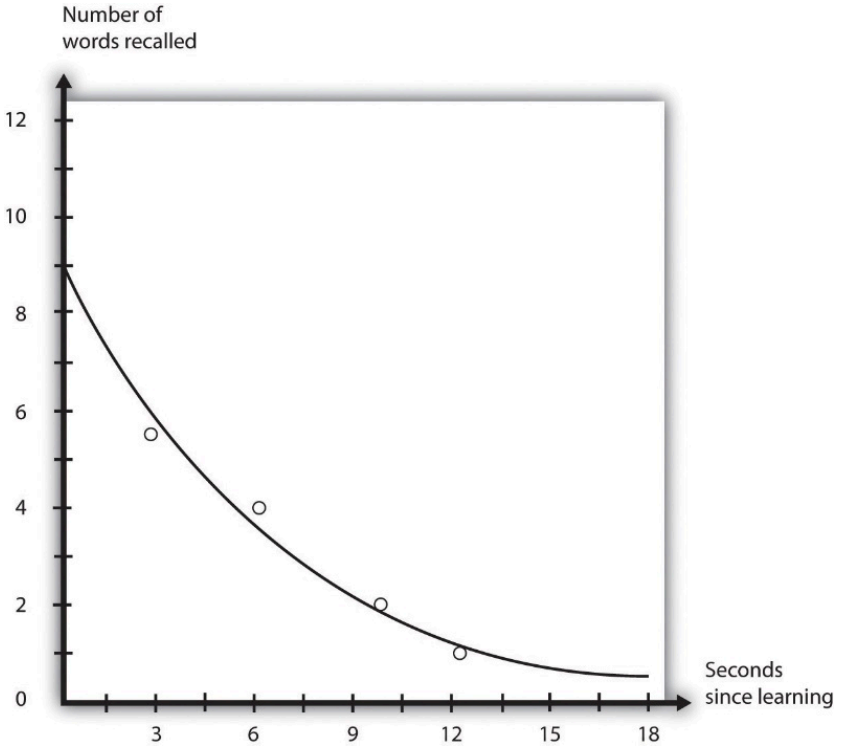


Figure 7.6. This graph shows that we quickly forget items held in STM after about 18 seconds (after Peterson & Peterson, 1959)

According to Baddeley (2000), we process information in working memory within separate systems (Figure 7.7). There is a *visuospatial sketchpad*, which processes visual information from the world around us– such as shapes and colors of objects, as well as their location and movement. In contrast, the *phonological loop* processes sound-related information, such as words that we hear or read. Therefore, our working memory's "contents" often consists of pictures and internal dialog. Another system, the *episodic buffer* keeps track of the timing and order of events, and helps to integrate the information from all the different systems. These three systems are controlled by the *central executive* (like the main

boss). The central executive directs our attention towards different kinds of information in our environments and helps us to choose appropriate encoding strategies to move new information into more permanent LTM storage. As you can see in Figure 7.7, information flows between the systems, thus, at any given moment we can pay attention to what is around us, but we can also retrieve existing knowledge from LTM to help us to organize and encode information.

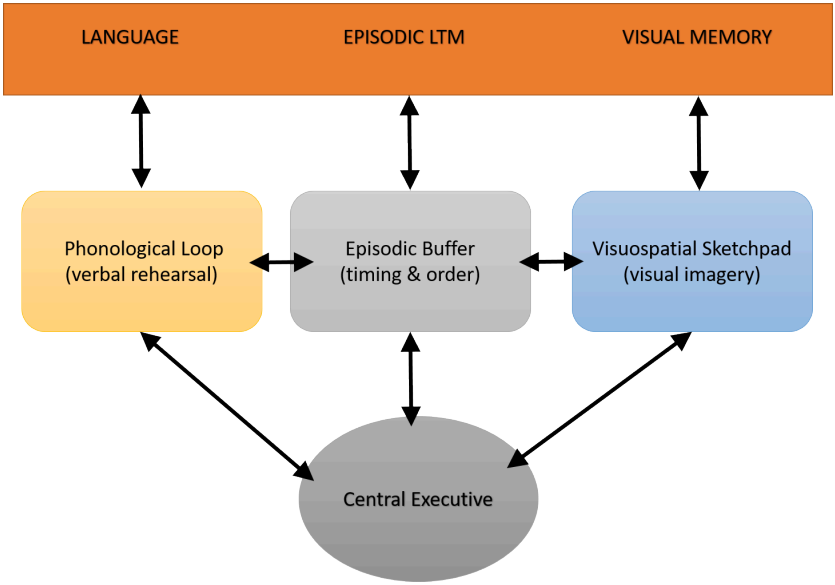


Figure 7.7. Baddeley's model of working memory. Adapted from Baddeley (2003)

ENCODING INFORMATION FOR LONG-TERM MEMORY

We often want to move information into LTM, so that we can remember it for long periods of time. One strategy is to use simple

maintenance rehearsal by repeating the information over and over. This is also known as rote learning. You might have learned your alphabet like this, or possibly your times table. However, it is not the most effective way to learn most information. Let's take a look at other encoding strategies that have been shown to be more effective—deep processing and semantic encoding. Semantic encoding refers to processing the meaning of the word and making associations with other concepts we already know. These techniques are sometimes referred to as elaborative techniques. Although these strategies are more effortful than rote learning they are more efficient and effective.

Deep processing requires that we pay close attention to information and try to encode it in meaningful ways. Research by Paivio (1986) shows that information that is encoded using both auditory and visual systems of working memory (Figure 7.7) are more easily remembered than information processed from only one system alone. This explains why it is easier to remember concrete words like *car, dog,* and *book*, than abstract words such as *level, truth,* and *value*. We can easily form mental images of concrete words, but not abstract ones. When we can think about the picture, as well as the sound of the word, it helps us to build a stronger memory (Paivio, 1986).

Craik and Tulving used a different approach to show that deep processing resulted in better memory for words, than shallow processing (Craik & Lockhart, 1972; Craik & Tulving, 1975). Craik and Tulving showed participants a list of words. For each word, the participants were asked one of three questions about them. 1) Was the word written in upper or lower case (visual encoding)? 2) Did the word rhyme with a second word on the screen (auditory encoding)? or 3) Did the word fit in a specific sentence (semantic encoding)? After seeing all the words, participants were given a surprise memory test. Participants remembered the words that they encoded semantically, better than those they encoded visually or acoustically. Semantic encoding involves a deeper level of

processing than the visual or acoustic encoding, because participants have to elaborate further to think about the meaning of the word.

You probably already know about the benefits of using semantic encoding when trying to remember new information. It is easiest for us to learn concepts if they make sense to us. You have probably had the experience of memorizing a passage of text—such as a speech, poem, or the lyrics of a song at some point. Was it difficult to do? Some kinds of text are easier to remember than others. Which of the two quotes below do you think would be easier for you to remember and why?

"O, reason not the need!
Our basest beggars
Are in the poorest thing superfluous.
Allow not nature more than nature needs,
Man's life's as cheap as beast's."

From Shakespeare's play "*King Lear*" (Act 2, Scene 4)

"You may shoot me with your words,
You may cut me with your eyes,
You may kill me with your hatefulness,
But still, like air, I'll rise."

From Still I Rise by Maya Angelou

Multiple studies have shown that sentences are easier to recall if they are meaningful to us (Anderson, 1984). So, whichever of the two quotes made more sense to you, is the one that you are most likely to remember. For most of us, it's probably the quote by Maya Angelou. Try out this adaptation of a study by Bransford and McCarrell (1974), which provided scientific evidence showing that meaning is important when trying to learn sentences.

Exercise: READ THESE SENTENCES

- *The child was hungry until they broke it.*
- *The street was full of holes because the turning stopped.*
- *The woman saw her face in the body.*

Now look away and count backwards from 30 in threes (30, 27, 24, etc.) all the way to zero.

Then try to write down the sentences, word for word (no peeking back at this page!).

How well did you do? The statements above do not make a lot of sense and so they were probably difficult for you to remember accurately. Here are some prompts for each sentence that should help them make more sense to you: #1—an egg; #2 —a cement mixer; and #3— a new car. Read the sentences out loud again, keeping the prompts in mind. Next, count backwards from 40 by fours, then check yourself to see how well you recall the sentences this time. You probably did better than before. Of course, you now have had the benefits of already studying the sentences once before, and we know that practice helps information to be retained. But, the sentences are also probably much more memorable because each one was placed in context. This is another example that shows that material is more easily encoded when you make it meaningful.

Neuroscientists believe that concepts in memory are represented within nodes, which are networks of interconnected neurons (Collins & Loftus, 1975). We form semantic networks by building connections between nodes for concepts that are related (Figure 7.8). Activating any part of a semantic network also activates other related concepts, which makes them easier to access and remember. These semantic networks differ from person to person depending on personal experiences and so they are shaped by cultural values.

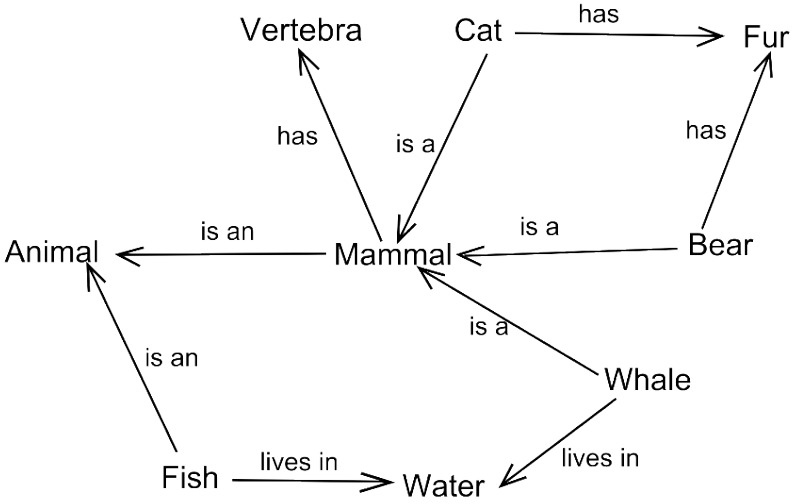


Figure 7.8. Example of a semantic network for the word "Mammal". Notice how the word "Mammal" will automatically activate other words in the network like Whale, Bear, Animal, Cat, and Vertebra. These words are also connected to other related concepts, e.g., Whale activates Water, which activates Fish. The word Animal also activates Fish.

Semantic encoding strategies vary across cultures. When trying to remember lists of words, American participants often automatically organize them in terms of semantic categories. For example, Bousfield (1953) asked American participants to memorize a list of 60 words, which belonged to four different categories (animals, names, jobs, and vegetables). Although the words were presented in random order and the participants were not told about the categories, participants tended to recall them in category clusters, such as a group of animals first and then a group of names. In contrast, when East Asian participants were asked to recall words in a similar kind of task, they used a different semantic encoding strategy. They were more likely to cluster objects thematically in terms of their perceptual and functional similarities, or by associations. For example, they might remember dog, bone, and

kennel together in one cluster, and long objects, such as river, serpent, and broom, together in another cluster (Wang, 2021).

Multiple studies with Western participants have shown that people are also more likely to remember information that they feel a personal connection to—such as adjectives that describe themselves—this is known as the self-reference effect (Rogers et al.,1977). However, recent studies suggest that this effect is specific to people from cultures that endorse individualism. In contrast, people from more collectivistic cultures are more likely to remember words relating to relationships, such as roommate or brother (Wang, 2021).

We will return to the importance of semantic encoding at the end of this chapter when we discuss effective learning strategies.

SUBDIVISIONS OF LONG-TERM MEMORY

Although our goal is often to get information into long-term memory, there are several types of biologically distinct LTM (Figure 7.9). All LTMs are divided into explicit (requiring conscious effort for recall) or implicit (recalled automatically) memories. For example, if you are studying for your psychology exam, the material you are learning will (hopefully) become an explicit memory. Explicit memory is sometimes referred to as declarative memory, because it can be "declared" or put into words. Explicit memory is further subdivided into semantic and episodic memory.

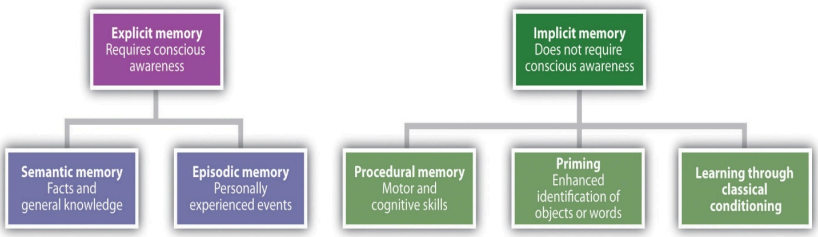


Figure 7.9. Subtypes of long-term memory.

Link to Learning

View this video that explains short-term and long-term memory to learn more about how memories are stored and retrieved.

EXPLICIT MEMORY

Semantic memory encompasses knowledge about words, concepts, and language-based knowledge and facts. Semantic memory is typically reported as "facts". In contrast, episodic memory, which is also known as autobiographical memory, includes details about events we have personally experienced. We can think of episodic memory as recalling the episodes of our life—a bit like a TV show. Episodic memories help us to remember the "what, where, and when" of an event (Tulving, 1972; 2002). Episodic memories often involve visual imagery and are typically in story form (Hassabis & Maguire, 2007). The content of episodic memories is also driven by cultural values. For example, Chinese people often value low-arousal harmonic relationships among

members of their community, and are more likely to recall episodic memories focusing on loved ones and social interactions. In contrast, people from individualistic cultures typically remember more unique personal details in their stories. The kinds of events most likely to be remembered also vary across cultures, Chinese people are more likely to focus on peaceful occasions and achievements, whereas people from Brazil and Portugal often recall rites of passage more easily (Wang, 2021).

IMPLICIT MEMORIES

Implicit memories are long-term memories that we recall without conscious effort—in other words, we do not summon them—they happen automatically. They include memories learned through classical conditioning and procedural memories, which are actions that once learned become automatic behaviors—such as riding a bike or brushing our teeth. Priming is another example of implicit memory.

Implicit procedural memory stores information about physical actions needed for complex behaviors, such as brushing your teeth, riding a bicycle, typing on a computer keyboard, or driving a car (Adams, 1957; Lacey & Smith, 1954; Lazarus & McCleary, 1951). Even walking and talking are procedural memories. Procedural memories are sometimes referred to as "muscle memories"—we can only learn them through physically doing something with our bodies. Procedural memories are formed through repetition and practice, and once learned they allow behaviors to occur automatically without effort or conscious thought. We often see toddlers struggling to balance and having a hard time coordinating their body movements when they take their first steps, but before long, walking becomes automatic and easy. Similarly, when people first learn to drive, they typically think a lot about how to operate the car – where to put their hands on the steering wheel, which foot to use for the brake and the accelerator, and the meanings

of the traffic signs and signals. Learner drivers gradually shift from explicit, conscious thinking to implicit, automatic actions. The actions become streamlined, requiring less cognitive effort and allowing the driver to focus on higher-level tasks such as navigation and other kinds of decision-making.

Priming is another type of implicit memory (Schacter, 1992). If you remember from chapter 5, we say that priming has occurred if exposure to a stimulus influences subsequent thoughts, perceptions, or behaviors, without conscious awareness. Priming stimuli may include single words, sentences or short stories, pictures, videos, and sounds, etc. Priming stimuli can affect our perceptual set and bias the focus of our attention or expectations, without us realizing. Let's see whether this example of priming affects you.

Priming Exercise

First, read this text:

Some people really enjoy picnics. They love going to new places, spreading a blanket on the ground, and eating a delicious meal.

Now, unscramble the following letters to make a word: AETPL

What word did you come up with? Was it "plate?" If you had read a different priming story – such as "*Some people really enjoy growing flowers. They love going outside to their garden, fertilizing their plants, and watering their flowers,*" you might have come up with the word "petal" instead of "plate".

The reason that people are more likely to come up with the word "plate" after reading about a picnic is that "plate" and "picnic" are semantically related. Similarly, reading about flowers activates other related concepts within that semantic network – such as "petal". Therefore, "picnic" primes "plate" and "flower" primes "petal".

Memories that are formed through classical conditions are a third type of implicit memory. These memories are typically formed and recalled without conscious awareness. For example, Pavlov's dogs starting to salivate when they heard a bell that was rung prior to them being fed is an example of an implicit memory. Pavlov's dogs did not consciously have to learn the association between the bell and the food – it was effortless – similarly, their salivation to the bell was also automatic. Humans also have classically conditioned memories, as we discussed in Chapter 6. For example, fear conditioning helps us to avoid places and activities that we have learned were harmful to us, and taste aversion helps us to avoid foods that make us sick.

Everyday Connection

Can You Remember Everything You Ever Did or Said?

What were you wearing exactly five years ago today? What did you eat for lunch on April 10, 2018? You probably find it difficult, if not impossible, to answer these questions about your episodic memory. However, there are a few individuals who have amazing and highly superior autobiographical memory – or hyperthymesia. They are often able to remember every event they have experienced over the course of their life—meals, conversations, clothing choices, weather conditions, and so on. As of 2022, around 50 people have been identified as having this hyperthymesia, but not all have been studied (Parker et al., 2006; University of California Irvine, 2022).

Link to Learning

Watch this video about superior autobiographical memory from the television news show *60 Minutes* to learn more.

7.2. Parts of the Brain Involved with Memory

The storage of long-term memories (regardless of their type) involves creating and strengthening connections between neurons in the brain. Each thought, feeling, or behavior is associated with a circuit or network of neurons that are activated at the same time. If neurons within the circuit are frequently and repeatedly stimulated, then it becomes easier to reactivate the circuit in the future. This process is known as long-term potentiation (LTP; Lynch, 2004). Long-term potentiation happens gradually. For example, as soon as you start learning a new skill – such as learning to play the guitar— you slowly modify existing neural circuits and create new ones. Each time you practice, you strengthen the connections between the neurons that you are using. The more you practice the guitar, the stronger (more potentiated) the connections become, and the easier it gets. If you practice a lot, the memory can become consolidated, making it relatively permanent, allowing you to play the guitar for the rest of your life. Memory consolidation involves further synaptic changes in multiple parts of the brain (Alberini et al., 2013). Repetition and attention are essential for LTP to occur, otherwise the connections between neurons can weaken and may even disappear. Under some circumstances, this weakening process is quite useful because it allows our brains to discard information that is likely not very important to us because we

have not practiced or tried to retrieve it. When we are learning something new, paying close attention and practicing, signals to our brains that something is important.

Different parts of your brain are important for different kinds of memory. In this section, we will focus on the role of the hippocampus, the amygdala, the cerebellum and basal ganglia, and the prefrontal cortex (Figure 7.10).

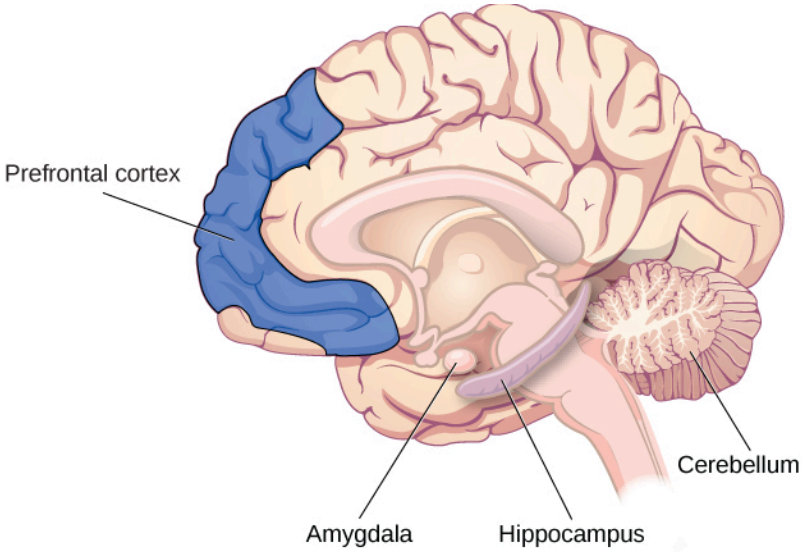


Figure 7.10. The amygdala, hippocampus, cerebellum, and prefrontal cortex are all important for different aspects of memory.

HIPPOCAMPUS

The hippocampus (and nearby hippocampal-related structures) play an essential role in the consolidation of explicit memories. Hippocampal structure damage leaves us unable to store new episodic memories. One famous patient, known for years as H. M. to protect his identity, had hippocampal structures removed

from both hemispheres of his brain when he was 27 years old, in an attempt to help control the epileptic seizures he had been suffering since childhood (Corkin et al., 1997). The surgery stopped the seizures, but he was not able to form new explicit memories for the rest of his life. However, he was able to remember everything from his childhood and up till 2-3 years before the surgery. It is thought that the hippocampal structures play a role in encoding, storing, and retrieving memories. The hippocampal structures are thought to keep a record of the neural circuits that are active in many different parts of the brain during encoding. During retrieval, the hippocampal structures reactivate the same circuitry. Over time, as memories become consolidated, the hippocampal structures are no longer needed to activate the circuits, they become active automatically. This is why H.M. did not lose most of his memories from before the surgery, only the more recent ones. H.M. was able to learn new implicit memories, such as new procedural memories—which shows that these types of memories are governed by other brain areas. For example, he learned to accurately trace the outline of a shape while looking in a mirror (mirror-tracing). He did the activity multiple days in a row and improved over time, however, on each occasion he never remembered doing it previously (Corkin, 1968; Squire, 2009).

The posterior part of the hippocampus is also important for spatial memory, e.g., remembering your way to school from home and other routes (Clark et al., 2000). London taxi drivers who rely heavily on spatial memory in their daily lives have larger than average posterior hippocampal structures. Children from hunter-gatherer societies (e.g., Australian aborigines, and the Tsimané people who live in the Amazon) who grow up learning to track animals and forage, also develop superior visuospatial skills (Wang, 2021). It is highly likely that they too, have larger than average hippocampi.

AMYGDALA

The main job of the amygdala is to process and regulate emotions. The amygdala also plays a part in how emotional memories are formed and stored. Emotional memories are often easier to encode than other memories, this is because the amygdala is activated when we experience strong emotions. This is why emotional memories are often so memorable. Similarly, when we remember an emotional situation, the amygdala is often reactivated and we may re-experience the emotions (McGaugh, 2003). The amygdala is essential for fear conditioning to develop. Damage to the amygdala prevents people and animals from learning to avoid dangerous situations and stimuli (Josselyn, 2010).

CEREBELLUM AND BASAL GANGLIA

Although the hippocampus is important for explicit memories, implicit memories (procedural memory, motor learning, and classical conditioning) are mediated by other parts of the brain. The amygdala is important for fear-conditioning, but other classically conditioned behaviors, such as learning to blink to a stimulus that comes before a puff of air to the eye, is dependent on the cerebellum (Steinmetz, 1999; Green & Woodruff-Pak, 2000). Procedural memories are highly dependent on activation of the basal ganglia (these are a group of subcortical structures deep within the brain) and to a lesser extent, the cerebellum also plays a role in procedural memories (Ullman, 2004).

PREFRONTAL CORTEX

The prefrontal cortex (PFC) is important for many aspects of memory. For example, during most WM tasks, there is strong

activation of the PFC. The PFC is also likely to play a role in encoding information, by directing our attention to different stimuli in the environment. The PFC is also activated when we retrieve semantic and episodic memories, which may be explained in part by the presence of strong PFC-hippocampal connections (Nyberg et al., 2003).

Link to Learning

In this TED Talk called "A Mouse. A Laser Beam. A Manipulated Memory," Steve Ramirez and Xu Liu from MIT talk about using laser beams to manipulate fear memory in rats. Find out why their work caused a media frenzy when it was published in a leading scientific journal called *Science*.

FLASHBULB MEMORIES

Our ability to relatively easily encode events that arouse very strong emotions is linked to a particular type of memory, called a flashbulb memory. The word was used to indicate the memory is like a picture (taken with a flash). Flashbulb memories are unusually detailed and vivid episodic memories of unexpected, highly emotionally arousing events, that are frequently of historical significance (Figure 7.11). In recent United States history, flashbulb memories might include the January 6th 2021 attack on the United States Capitol, the mass shooting of many LGBTQ individuals at the Pulse nightclub in Orlando on June 12, 2016, and the 9/11 terrorist

attacks in 2001. Many people continue to have very strong, often emotional memories, about these events.

In addition to the biological mechanisms that facilitate the encoding of unexpected emotional events, there are many other factors that likely contribute to the memorability of flashbulb events. Firstly, people engage in frequent discussion about flashbulb events and are more likely to spend a lot of time thinking about them, compared to more everyday events. People are also likely to be inundated by coverage of the events on news and social media. These factors contribute to the perceived memorability of a flashbulb memory. Flashbulb memories seem to have a particularly long duration. A Pew Research Center survey (2011) found that for Americans who were age 8 or older at the time of 9/11 terrorist attacks, 97% can recall the moment they learned of this event, even a decade after it happened.

Although most flashbulb memory research has focused on historically significant, negative events, strictly speaking, any surprising highly emotional event (positive or negative, personal or global) could become a flashbulb memory (Brown & Kulik, 1977; Hirst & Phelps, 2016). Although flashbulb memories are typically detailed and vivid, they are not always accurate.

Figure 7.11. Most people can remember where they were when they first heard about the 9/11 terrorist attacks. This is an example of a flashbulb memory: a record of an unexpected event that has very strong emotional associations.

Dig Deeper

Inaccurate and False Memories

Even flashbulb memories for important events can become less accurate with the passage of time. For example, four months after the terrorist attacks of 9/11, President George W. Bush misremembered how he had first heard about the attacks. He said:

"*I was sitting there, and my Chief of Staff—well, first of all, when we walked into the classroom, I had seen this plane fly into the first building. There was a TV set on. And you know, I thought it was pilot error and I was amazed that anybody could make such a terrible mistake*." (Greenberg, 2004, p. 2)

However, there was no live footage of the plane hitting the building. This was the event that alerted people, including the TV stations, who then quickly went to the scene.

George W. Bush is not the only person who has inaccurate flashbulb memories, they are quite common (Talarico & Rubin, 2007). As we pointed out at the beginning of the chapter, memory is not like a video recording. Human memory, even flashbulb memories, can be vulnerable to change and misinformation. Every time we retrieve a memory, we reconstruct it. *Do you have a flashbulb memory? Who were you with and what were you doing? What did you talk about? Can you contact those people you were with? Do they have the same memories as you or do they have different memories?*

7.3. Problems with Memory

As we have seen, our cultural values and encoding strategies can affect the memories that we store. Similarly, a memory pulled from long-term storage into short-term memory is flexible and subject to change. New events can be added and we can unintentionally alter what we think we remember about past events, resulting in inaccuracies and distortions (Roediger & DeSoto, 2015). Therefore, memories are constructed, rather than accurate action replays of what actually happened.

MEMORY ERRORS

Daniel Schacter (2001, 2003; 2020), a well-known memory researcher, offers seven different ways that our memories can (and often do) fail us. He calls them the seven sins of memory and categorizes them into three groups: forgetting, distortion, and intrusion (Table 7.1). Let us focus first on why he suggests that we sometimes forget. One caveat about these "sins" is that they are based on studies with WEIRD, mostly undergraduate samples, we have provided information from other studies when available.

Table 7.1. Schacter's Seven Sins of Memory

Sin	Type	Description	Example
Transience	Forgetting	Memory fades over time	Forget events that occurred long ago
Absentmindedness	Forgetting	Forgetting caused by lapses in attention- encoding failure	Forget where your phone is
Blocking	Forgetting	Accessibility of information is temporarily blocked	Tip of the tongue
Misattribution	Distortion	Source of memory is confused	Recalling a dream memory as a waking memory
Suggestibility	Distortion	False memories	Result from leading questions
Bias	Distortion	Memories distorted by current belief system	Align memories to current beliefs
Persistence	Intrusion	Inability to forget undesirable memories	Traumatic events

WHY DO WE FORGET?

TRANSIENCE

Most memories are somewhat transient, meaning that they tend to fade over time, especially if they are not activated regularly. In 1885, German psychologist, Hermann Ebbinghaus conducted a study to measure the transience of memory. He was the sole participant in this famous study! First, he memorized long lists of meaningless trigrams (TQM, LMX, etc.) Then he measured how many of them he remembered at various intervals of time for a month (Figure 7.12). The curve in Figure 7.12 shows his rate of forgetting and is called the Ebbinghaus forgetting curve (Ebbinghaus, 1885/1964). Its general shape has been replicated many times in studies of LTM. We can see that initially, there is a rapid, steep loss of information—he forgot more than half the trigrams in the first hour! Then the curve begins to level off, so that after a month, a little bit of information is still retained. You may have had similar experiences if you cram for an exam. You remember very little information a month after the exam.

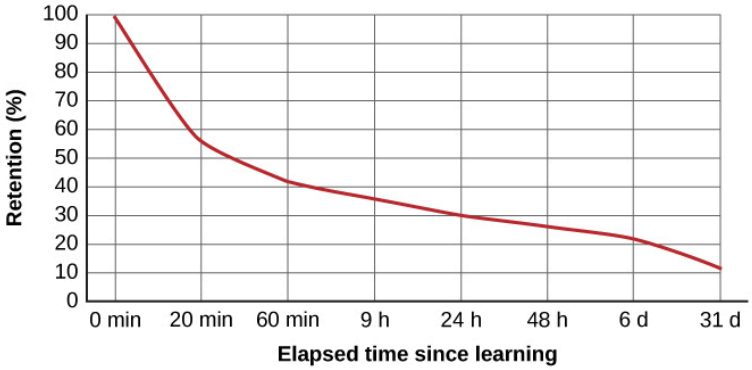


Figure 7.12. The Ebbinghaus forgetting curve shows how quickly memory for new information decays.

ABSENTMINDEDNESS AND ENCODING FAILURE

Are you constantly losing your cell phone or your ID? Have you ever gone back home to make sure you turned off the stove or locked the door? Have you ever walked into a room for something, but forgotten what it was? You probably answered yes to at least one, if not all, of these examples—but don't worry, you are not alone. We are all prone to the memory error that Schacter refers to as absentmindedness, these are lapses in memory when we fail to pay adequate attention to what we were doing. This then makes it difficult to remember what we actually did. We can consider absentmindedness to be a failure of encoding. We know that information only can move into LTM if we pay attention to it and process it deeply. For example, people from individualistic societies are less likely to encode background information in a scene than people from collectivistic cultures, and therefore are less likely to remember that information in future (Wang, 2021).

When we are trying to multi-task, we cannot pay attention to everything and so something that we are doing – like putting our phone down—does not get encoded properly. Other times, we

think we have paid attention to something, but we have not processed it deeply enough to remember it. There are many instances of this in our daily lives. For example, you might spend many hours re-reading your notes to study for a test, but when you get to the test, you realize that you only skimmed your notes because you were tired or distracted by other tasks and people. The final section of this chapter gives suggestions for better ways to study. Failing to recognize coins that we use in our daily lives is another example where frequent shallow processing makes us believe that we know something better than we actually do. For instance, think of how many times in your life you've seen a U.S. penny. Do you think you can accurately remember what the front of it actually looks like? Nickerson and Adams (1979) found that most Americans are not able to pick out a picture of a real penny from similar pictures of "fake" pennies. Most of us never look closely enough to encode the details of the penny. We only encode enough information to be able to distinguish it from other coins. Try it for yourself with another US coin. See if you can figure out which of the nickels in Figure 7.13 is the real one. Look for the answer at the end of the chapter (at the top of the Reference section).

(a) (b) (c) (d)

Figure 7.13. Can you tell which coin, (a), (b), (c), or (d) is the accurate depiction of a US nickel? You will found the answer above the Reference section at the end of this chapter.

BLOCKING

Blocking refers to a failure in the retrieval of information that has been successfully encoded and stored in LTM. For example, if someone asks you a question such as – What's the name of the Black actor who starred in *The Shawshank Redemption*? You might feel very confident that you know the answer, but you cannot remember it in the moment. This is a very frustrating feeling for most of us, and is often referred to as the "tip of the tongue" (ToT) phenomenon. You feel that the information is right there – just waiting to come out of your mouth. Often you can recall other details about the information that you are searching for. For example, you might be able to clearly picture the actor's face, or name other movies he starred in, such as *Oblivion* and the *Dark Knight* trilogy. You might even remember that his first name begins with an M and that his last name has two syllables. As you activate all of this other information in your memory, more and more parts of the semantic network that includes the actor's name become active, and then suddenly it pops into your head (Morgan Freeman!) Research suggests that when we experience a strong ToT sensation, it is likely that we are correct in thinking that we know the information, we are just slow in retrieving it. Older adults experience ToT more than younger ones, and it is common across many languages and cultures, including people who use American Sign Language (Schwartz & Metcalfe, 2011). *What do you do when you experience ToT?* D'Angelo and Humphreys (2015) found that encouraging people to say what they are thinking out loud (e.g., He was in *Oblivion* and the *Dark Knight* trilogy – I think his name begins with M) often helped them to remember the ToT information. Or do you (as I often do) Google it, rather than waiting for your brain to provide the answer? It certainly resolves feelings of frustration and allows me to move on with other things, but how does it affect my memory? Psychologists have begun to explore how the Internet affects our memory. Sparrow and colleagues found that

people who rely on the Internet are less likely to put effort into remembering information (Sparrow & Chatman, 2013). Also, D'Angelo and Humphreys (2015) showed that when people experienced ToT for newly learned words, they were more likely to experience ToT in the future if they were given the answer, than if they remembered the word by themselves. Therefore, putting effort into trying to retrieve ToT information yourself, probably makes it more memorable in the future.

Interference

Another reason why we fail to successfully retrieve familiar information is because other information that we know is interfering with our remembering. There are two types of interference: proactive and retroactive interference (Figure 7.14). In proactive interference, older information that you learned previously prevents you remembering newer information. For example, you might remember your friend's old phone number, where they used to live, or their previous partner's name, but you cannot remember their current information. Retroactive interference happens in the opposite direction, newly learned information interferes with the recall of older information. For example, you learn your college email address, but then have difficulty remembering your email address from high school.

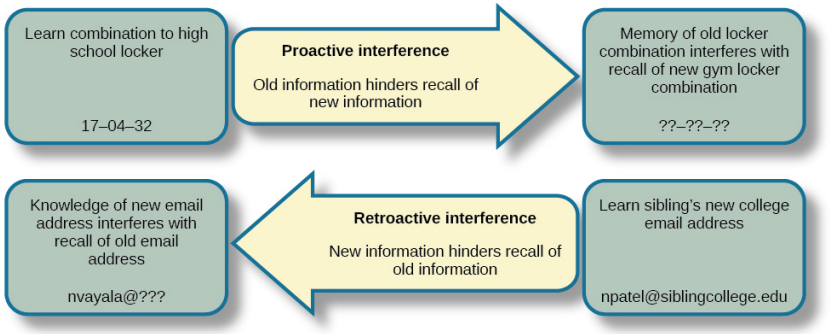


Figure 7.14. Sometimes forgetting is caused by a failure to retrieve information. This can be due to interference, either retroactive or proactive.

MEMORY DISTORTIONS

It is often hard for us to believe that our memory is not as accurate as we think it is, especially in the case of vivid and detailed flashbulb memories. However, it is important to remember that memory is constructed and so can be influenced by multiple different factors, including our beliefs, expectations, emotions, prior experiences, and interference from external sources. Memory distortions are common in our daily lives, but can have devastating effects when people testify about the details of a crime that they witnessed. People's freedom and lives often rest on the accuracy of these memories. Unfortunately, eyewitness misidentification is the leading cause for wrongful conviction in the United States (Innocence Project, 2023). Among the 375 people exonerated on the basis of DNA evidence between 1989 and 2020, 69% were convicted on the basis of eye witness testimony (Innocence Project, 2023). Let's take a look at three common errors of memory distortion: misattribution, suggestibility, and bias, and consider how they affect eyewitness memory and other kinds of memories in our daily lives.

MISATTRIBUTION/SOURCE CONFUSION

Misattribution happens when you make a mistake about where you got your information—this is sometimes called source confusion or a source-monitoring error. You might remember something—but where did that knowledge come from? Did you read it online? Did someone tell you? Or did you experience it yourself? Sometimes people believe that they witnessed a detail personally, but they actually learned about it by reading, or watching something on TV or on the Internet. Source confusion led to the wrongful arrest of Australian eyewitness expert, Donald Thomson, in 1975.

> *Australian eyewitness expert, Donald Thomson, appeared on a live TV discussion about the unreliability of eyewitness memory. He was later arrested, placed in a lineup and identified by a victim as the man who had raped her. The police charged Thomson although the rape had occurred at the time he was on TV. Initially, they dismissed his alibi that he was in plain view of a TV audience and in the company of the other discussants, including an assistant commissioner of police. Eventually, the investigators discovered that the rapist had attacked the woman as she was watching TV—the very program on which Thomson had appeared. Authorities eventually cleared Thomson. The woman had confused the rapist's face with the face that she had seen on TV* (Baddeley, 2004, p. 133)

Source confusion can also influence jurors' decision making

abilities. Ruva and Guenther (2015) found that mock jurors often confused negative publicity about a case with actual evidence they had heard during a trial, which affected their verdicts. It is often difficult to remember where exactly we learned something, because we tend to remember general information (the gist), rather than specific details (Schacter, 2003; 2021).

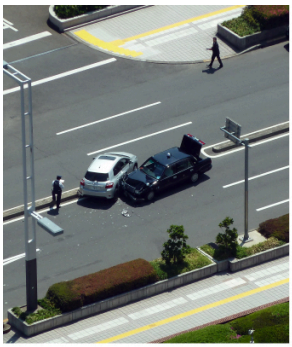
Psychologists have studied the phenomena of misattribution in the lab, using a technique referred to as the Deese-Roediger-McDermott (DRM) paradigm (Roediger & McDermott, 2000). Participants are shown (or read) a list of words—many of which are semantically related to a common theme, e.g., FURNITURE, and so could include words like chair, stool, desk, sofa, cabinet, etc., in addition to unrelated words (e.g., elephant, banana, ocean). Participants are more likely to remember the related words than the unrelated ones, because they are inter-connected within a semantic network. However, they will also often falsely remember a common item of furniture that was not among the words presented. The DRM effect also helps us to understand how people organize information in semantic memory. We frequently build schemas (cognitive structures) for situations that we experience frequently, which help us to quickly recognize places and situations. For example, you likely have a schema for a classroom (students, teacher, desks, board etc.), which is very different from that for a restaurant or a dentist's office. False memories can be elicited with the DRM paradigm when we activate existing schema— showing it is easy to misremember something if we would normally expect it to be present in a commonly encountered setting.

SUGGESTIBILITY/MISINFORMATION EFFECT

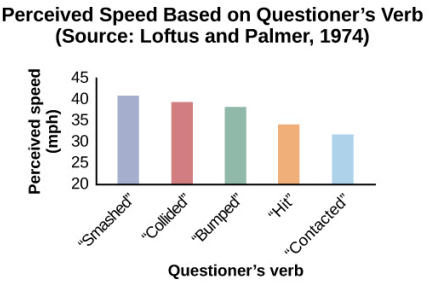
Our memories are also vulnerable to suggestion. People around us can influence our memories by asking leading questions or giving us incorrect information. Elizabeth Loftus, a much cited memory

researcher, refers to this as the misinformation effect and has conducted multiple studies with WEIRD participants showing how misinformation can generate false memories. Loftus has testified in court many times about the unreliability of eyewitness testimony.

In one well-known study on the misinformation effect, Loftus and Palmer (1974) asked 45 U.S. college students to watch a video of two cars colliding, and then asked them to estimate the speed of the cars involved (Figure 7.15).



(a)



(b)

Figure 7.15. Misinformation Effect. When people are asked leading questions about an event, their memory of the event may be altered.

The participants' answers depended on the format of the question. Participants who were asked*: How fast were the cars going when they "smashed" into each other?* estimated that the cars were traveling at a much higher speed than participants who heard the verb "*contacted*" instead of *"smashed".* In a follow-up study one week later, they asked the participants whether they saw any broken glass at the scene of the accident (none was actually present in the video). Participants who had been in the "smashed" group were more than twice as likely to indicate that they did remember seeing glass. Loftus and Palmer demonstrated that a leading question can generate false memories. Misinformation can corrupt memory quite easily when it is encountered in social situations (Gabbert et al., 2004). This is a problem particularly in cases where more than

one person witnesses a crime, because witnesses often talk to one another as they wait for the police to arrive. However, different witnesses are likely to see or notice different things, so when they talk about the crime, they not only reinforce common memories for the event, they also contaminate each other's memories (Gabbert et al., 2003; Paterson & Kemp, 2006; Takarangi et al., 2006).

Most studies of the misinformation effect have been conducted with WEIRD participants and point to the dangers of sharing information with others. However, Vredeveldt and de Bruïne (2022) have emphasized the benefits and cultural importance of memory sharing within some non-WEIRD communities. Some communities, particularly those in Rwanda, East Timor, and Sierra Leone particularly value the importance of collective memory, they embrace the fact that multiple versions of an event are pooled together because they believe that they enrich the memory. In such cultures, memory is considered social rather than individual, and contributes to feelings of community.

Link to Learning

In 1984, Jennifer Thompson, a 22-year-old college student in North Carolina, was brutally raped at knife-point. She tried to memorize every detail of her rapist's face and physical characteristics, vowing that if she survived, she would help get him convicted. However, in two line-ups (photo and in-person) she misidentified an innocent man, Ronald Cotton, as her assailant. Ronald Cotton spent 11 years in jail for a crime he did not commit, until he was finally

exonerated on the basis of DNA evidence. There were many factors that contributed to Jennifer Thompson's misidentification of her rapist.

Watch this first video about Ronald Cotton who was falsely convicted and then watch this second video about the task of his accuser to learn more about why Jennifer Thompson identified the wrong person as her attacker.

Can you think of any other factors that might have influenced Jennifer Thompson's memory about her attacker?

Ever since Loftus published her first studies on the suggestibility of eyewitness testimony in the 1970s, social scientists, police officers, therapists, and legal practitioners have been aware of the flaws in interview practices. Consequently, steps have been taken to decrease suggestibility of witnesses. One way is to modify how witnesses are questioned. When interviewers use neutral and less leading language, children more accurately recall what happened and who was involved (Goodman, 2006; Pipe, 1996; Pipe et al., 2004). Another change is in how police lineups are conducted. It's recommended that a blind photo lineup be used. This way the person administering the lineup doesn't know which photo belongs to the suspect, minimizing the possibility of giving leading cues. Additionally, judges in some states now inform jurors about the possibility of mis-identification and can suppress eyewitness testimony if they deem it unreliable. *Imagine that you were a juror*

listening to an eyewitness identify someone in court as the perpetrator of a crime. How persuasive do you think you would find the evidence?

Controversies over Repressed and Recovered Memories

Loftus and colleagues have conducted several studies where they have used the power of suggestion to implant false autobiographical memories. The researchers used fake materials, which they told the participants had come from their family members. These included fake stories and digitally altered photographs showing their participants in situations that had never occurred, such as meeting Bugs Bunny at Disney Land (Bugs Bunny isn't a Disney character) or riding in a hot air balloon (Braun et al., 2002; Wade et al., 2002). Loftus has used this research when testifying in court cases to demonstrate the fallibility of memory and its susceptibility to suggestion. Some of these court cases have involved individuals with recovered repressed memories. Repressed memories are a controversial area in psychology. Some psychologists believe that it is possible to bury (repress) deeply traumatic memories, which can be recovered many years later during therapy sessions. They argue that some children's experiences were so traumatizing and distressing that they had to lock those memories away in order to lead some semblance of a normal life. They believe that repressed memories can be buried for decades, but can be accurately recalled through hypnosis and guided imagery techniques (Devilly, 2007). Cheit (2007) has suggested that repressing these memories creates psychological distress in adulthood. The Recovered Memory Project was created so that victims of childhood sexual abuse can recall these memories and allow the healing process to begin (Cheit, 2007; Devilly, 2007).

Repressed memories have received a lot of publicity, particularly as they frequently relate to memories of events (such as sexual abuse) that do not have independent witnesses. Loftus and other

memory researchers have challenged the idea that individuals can repress memories of traumatic events from childhood, including sexual abuse, and then recover those memories years later during therapy. Loftus is not denying the occurrence of childhood sexual abuse, but she questions the accuracy of "recovered memories". She is particularly skeptical of the questioning process used to access these memories, given her research on misinformation effects. Of relevance to her arguments are studies that have demonstrated that young children do not always have reliable memories about where people have touched them. For example, Ceci and Brucks (1993, 1995) asked three-year-old children to use an anatomically correct doll to show where their pediatricians had touched them during an exam. Fifty-five percent of the children pointed to the genital/anal area on the dolls, even though they had not received any form of genital examination.

BIAS

Schacter (2001) points out that our feelings and view of the world can distort our memory of past events. Here are some examples of different types of bias that are known to affect memory.

Stereotype bias

Our stereotypes about people can affect what we remember about them. For example, racial stereotypes were shown to affect the memory of Asian American and European American research participants when they were asked to remember a name presented with a job (basketball player or politician). Half of the names in the study were commonly associated with African Americans, such as Jamal and Tyrone, and half were typically White names such as Greg and Howard. They found that participants made more mistakes when "African American" names were paired with "politician" and when "White names were paired with "basketball

player" (Payne et al., 2004). The study shows that people have better memory for information when it is consistent with their stereotypes about particular groups of people.

Egocentric bias

Egocentric bias is a phenomenon found with WEIRD participants, where they tend remember past events that boost self-esteem, by focusing on more positive events than negative ones. This bias is likely exclusive to individualistic societies, as it is not found among East Asian participants (Wang, 2021).

Hindsight bias

Hindsight bias describes the "I knew it all along" phenomenon. In other words, we often misremember what we knew or believed before an event. For example, you might be watching a murder-mystery and at some point in the movie, you announce that you know who committed the crime. At the end of the movie, however, it turns out that someone else was the murderer. However, you are pretty convinced that despite your earlier incorrect announcement, you really knew who it was – you just didn't say it out loud. In an Internet study, Pohl and colleagues found that hindsight bias was similar among participants from Asia, Australia and North America (Pohl et al., 2002).

PERSISTENCE

Persistence of memory occurs when highly unpleasant memories of traumatic events intrude into our thoughts in such a way that we are unable to function effectively. An example of this includes frequent involuntary flashbacks that are a common symptom among people with PTSD. Arousal theory explains why emotional memories are more easily encoded than non-emotional ones. In

the case of traumatic memories, stress hormones may speed the consolidation of memories, and may contribute to strong re-experiencing of emotions when memories of the trauma are recalled.

AMNESIA

People's memory can also be affected by medical conditions (e.g., stroke, viral infections, Alzheimer's disease and other forms of dementia, substance abuse, and traumatic head injury)—resulting in a condition known as amnesia. There are two common types of amnesia: anterograde amnesia and retrograde amnesia (Figure 7.16).

Figure 7.16. Timeline of retrograde and anterograde amnesia. Memory problems that extend back in time before the injury and prevent retrieval of information previously stored in long-term memory is known as retrograde amnesia. Conversely, memory problems that extend forward in time from the point of injury and prevent the formation of new memories is called anterograde amnesia.

ANTEROGRADE AMNESIA

People with anterograde amnesia cannot make new explicit (semantic or autobiographical) memories. The hippocampal structures are usually affected, which prevents the successful transfer of information from short-term to long-term memory.

Early Alzheimer's disease is the most common cause of anterograde amnesia (Markowitsch, 2011).

Patient H.M. who had his hippocampal structures removed to treat his epilepsy, had severe anterograde amnesia. He could read the same magazine over and over again but would have no memory of ever reading it before—it was always new to him. H.M. was unable to remember meeting new people after his surgery. If they were introduced to H. M. and then left the room, he would not know them when they returned a few minutes later. However, he, like other people with hippocampal damage, was still able to form new procedural memories, even though he had no explicit memory of doing so (Bayley & Squire, 2002; Corkin, 1968; Squire, 2009).

RETROGRADE AMNESIA

Retrograde amnesia is the loss of explicit memories that occurred prior to damage to the brain. People with retrograde amnesia cannot remember some, or even all of their past episodic memories. People with damage to their hippocampus often also have some degree of retrograde amnesia for memories that have not yet been consolidated. HM had retrograde amnesia that affected memories from the 2-3 years before his surgery.

Link to learning

Watch this video about EP who had a virus that destroyed the hippocampal structures in his brain.

Watch this video about the neuroscience of

memory for a brief summary about different aspects of memory and how memories are biologically stored.

7.4. Ways to Enhance Memory

Most of us suffer from memory failures of one kind or another on a daily basis, and would like to improve our memories so that we don't forget where we put our keys, wallet, or other important items in our daily lives. For college students, knowing how to effectively retain information can improve test performance and can facilitate the transfer of knowledge between settings, such as from one course to another or from school to the workplace. In this next section, we'll look at some ways to help you improve your memory, including strategies for effective studying. It might be helpful to know that psychologists have debunked the popular and persistent myth that different people have different learning styles (visual or auditory learners, etc.). In reality, we all benefit from encoding information in a variety of ways.

Link to learning

Watch this video to find out more about why the myth about learning styles has been so pervasive in education

MEMORY-ENHANCING STRATEGIES

Practice strengthens the connections between neurons, and so is essential for effective remembering. Moreover, elaborative rehearsal, where we process information at a deep level, is much more effective for remembering, than simple rote repetition. In a nutshell, we have to pay careful attention to whatever it is we are trying to learn and make connections between new information and what we already know. Then we need to practice remembering this information regularly. This tells our brains that what we are trying to learn is important and needs to be consolidated. These are the techniques that memory champions use to improve their memory, which allows them to learn extensive lists of information. Importantly, these are skills that can be learned and improved with practice.

MAKE A SEMANTIC CONNECTION

Whenever you need to learn something new it is very helpful to associate it with another concept you already know. For example, you might remember the difference between positive and negative reinforcement by thinking about positive (+) and negative (-) math signs and concepts. The plus sign (+) is connected with adding and the minus sign (–) with subtracting. So, it can help us to remember that a positive (+) reinforcer is something we add and a negative (-) one is something we take away. Alternatively, you might connect a new vocabulary word (reprehensible) by thinking about how it sounds and connecting it to other words you know, e.g., **repr**imand and incom**prehensible**. This might help you remember that reprehensible describes a behavior that people would find bad and unacceptable. Creating a sentence using the word also helps to remember, e.g., Donald stole from the charity he was working for and was sacked for his reprehensible behavior. Or, you might

create a mental image in your mind of some **rep**tiles (snakes and lizards) robbing a bank—possibly with one lizard swinging through the window on a **prehens**ile tail. Often the sillier the connection, the more memorable it is. For example, think of an enormous hippopotamus sitting on the top of your college campus – when trying to remember the name of the hippocampus—a structure that is important for memory consolidation. These strategies work for just about any information, including people's names, e.g., when introduced to Lily imagine her holding a large bouquet of lilies. We call strategies like this that make information more memorable, mnemonic devices. Let us take a look at some popular mnemonic devices that people find effective.

SILLY PHRASES

We often need to remember steps, stages, phases, and parts of a system in a particular order, like the names of the 12 different cranial nerves in biological psychology or the order of the planets in the solar system in earth sciences (Bellezza, 1981). You can create a silly phrase (often referred to as a mnemonic) using the first letter of each of the words in the correct order. For example to remember the planets in the solar system: **M**ercury, **V**enus, **E**arth, **M**ars, **J**upiter, **S**aturn, **U**ranus, and **N**eptune), you could use the phrase: *my very educated mother just served us noodles.* This method is also helpful if you have to memorize a string of random letters and numbers for a password.

RHYMES

Many people use rhymes to help them remember information (like the alphabet song). These might include rhymes for general spelling rules (I before E except after C), or for remembering the number of days in each month of the year (30 days have September, April, June, and November, all the rest have 31, except

for February alone, which has 28 days clear, and 29 in each leap year). These are very old rhymes, but, maybe you know or can create other ones that are even more personally memorable to you. The key to success is finding connections that work for you.

METHOD OF LOCI OR A MEMORY PALACE

A lot of memory champions use a "memory palace" to learn incredibly long lists of words or other kinds of information. This might be helpful to you for remembering lists of information—including your shopping lists. For most people, our "palace" is simply a place that we know really well, like the place where we live. The idea is that you make mental pictures of things you need to remember and then you mentally place them in different locations in your home. This is the encoding part of the device (you might imagine **eggs** in your bed, your kitchen sink being full of **milk**, **lettuce** on the toilet, etc.). Then when you go to the store, you mentally walk through your "palace" and think about the different things you have stashed there in your mind. This method is also known as the method of loci.

Link to Learning

Watch this interview of memory champion, Dominic O'Brien. O'Brien struggled at school as a child, but he won the World Memory Championships eight times. He explains how at the age of 30, he learned how to dramatically improve his memory.

HOW TO STUDY EFFECTIVELY

Decades of research show that there are several strategies that you can use to "study smarter" (Dunlosky, 2013; Friedman, 2014). It would be awesome if we could just listen to someone talk, or read a book and then remember everything – but as you know, memory takes effort. Here are some of the things that you can do to help:

BEFORE CLASS

Read (or at least skim) the assigned material and do any pre-class assignments before coming to class. This will help to familiarize you with the materials—which will start to activate the semantic networks in your brain that you will need to reactivate to remember the information. It also will help you to identify any areas that you do not understand. Make note of any questions to ask in class.

DURING CLASS

Mind-wandering is very common in our lives, especially in lecture-based classes. To help you stay focused, unless you are having a situation that requires you to monitor your phone, switch it to "do not disturb" and put it away. Taking notes will help you to stay engaged and will encourage the process of elaborate encoding. Take notes even if your instructor provides handouts or access to slides, and even when you think everything feels easy to understand and remember. It won't take long for information to disappear from your memory. However, don't try to write down everything you hear and see, instead, focus on understanding and writing down the most important pieces of information. Write short bullet points in your own words, and use abbreviations, diagrams, and symbols. Underlining headings will help keep you organized. Ask questions to help clarify concepts and deepen your

understanding. Consider using the Cornell system for taking notes, where you draw a line down your notebook and only write in the right-hand column in class. This then allows you to go back and use the left-hand column to add questions, headings, and summaries when you review your notes. Note taking is a skill that relies on working memory and can take practice. If you are struggling to take notes in class, ask your professor for help.

AFTER CLASS

Review your notes as soon after class as possible (preferably the same day), this helps to strengthen the information in memory, but it also gives you an opportunity to make sure that you understand what you have written and to add missing information or make other edits as needed. Use other resources to help clarify any muddy points (such as instructor-provided materials, textbooks, friends, and of course, your instructor). If time allows, test yourself on the main points. Ask a question and then answer it out loud, or write down your answer. This helps you to organize your thoughts and further strengthens the neural pathways. It also helps you to know what you do not know (meta-cognition) and where to focus your efforts.

BEFORE A TEST

John Dunlosky and colleagues have conducted many studies on how students study and which methods are most effective. The most common study techniques (re-reading notes and highlighting) are largely ineffective for most students because they rely on shallow processing. These techniques can also fool students into thinking that they have put a lot of effort into studying.

1. The number one best method for studying is to **test yourself** on the information that you are trying to learn **over several spaced out study sessions.**

2. Use elaborative rehearsal: Link new information with other memorable concepts (as discussed above). Make sure the information is personally meaningful to you and that you understand it. Write definitions in your own words that make sense to you. When you do this, you are building a semantic network of retrieval cues that will help you access the material when you want to remember it.

3. Use distributed practice: Study for short durations across several sessions rather than trying to cram it all in at once. Memory consolidation takes time. Spreading out your studying signals to your brain that this information is important. After you have organized your notes and read them to ensure you understand them, test yourself as much as possible, across multiple study periods. You can test yourself with practice quizzes, sample tests, flash cards or just make up questions to ask yourself. Some students like using Quizlet for this.

4. Study efficiently: Learning is effortful. When quizzing ourselves, it is tempting to read a question and then quickly look at the answer and think that we know it. The reason that we *think* we know it, is because it is familiar – but we are not really encoding it deeply (just like the nickel demonstration you tried out earlier in this chapter). However, if we make the effort to search for the answer in our memory by saying the answer out loud or writing it down, it increases the likelihood we will remember it in future. It also helps us meta-cognitively, by showing us how well we know the material. If you use flash cards—put a question on one side and an elaborative answer on the other. This might include a definition in your own words and a real-life example. When testing yourself, separate your cards into those you got right and those you got wrong. Go over the ones you got wrong first (maybe think of a more memorable association to help you remember) and keep sorting. Eventually, all your cards will be in the pile you answered correctly. But, don't stop, reviewing the cards again will encourage consolidation.

5. Be aware of interference: To reduce the likelihood of interference, study during a quiet time without interruptions or distractions (like television or music) – or use headphones playing white noise.

6. Get enough sleep: While you are sleeping, your brain is still at work. During sleep the brain organizes and consolidates information to be stored in long-term memory (Abel & Bäuml, 2013).

7. Keep moving: Regular aerobic exercise (anything that gets your heart rate elevated) is beneficial for memory (van Praag, 2008). Aerobic exercise promotes neurogenesis: the growth of new brain cells in the hippocampus, an area of the brain known to play a role in memory and learning.

Link to learning

Read this article to see what psychology research can tell you about the efficacy of your current study habits.

Chapter 7. Conclusions

Memory is a fundamental cognitive process that involves the encoding, storage, and retrieval of information. It plays a crucial role in everyday life, allowing us to retain and recall knowledge, experiences, and skills. The process begins with encoding, where information is transformed into a format that can be stored in the brain. Cultural differences among people determine what we pay attention to and encode in our memories. Storage refers to the retention of encoded information over time, and it occurs in different memory systems. These systems include sensory memory, short-term memory, and long-term memory. Retrieval involves accessing stored information from memory when needed. While memory is generally reliable, it can be influenced by various factors, leading to errors and distortions.

Chapter 7. Key Terms

absentmindedness
lapses in memory that are caused by breaks in attention or our focus being somewhere else, during encoding

acoustic encoding
input of sounds, words, and music

amnesia
loss of long-term memory that occurs as the result of disease, physical trauma, or psychological trauma

anterograde amnesia
loss of memory for events that occur after the brain damage

arousal theory
strong emotions trigger the formation of strong memories and weaker emotional experiences form weaker memories

Atkinson-Shiffrin model
memory model that states we process information through three systems: sensory memory, short-term memory, and long-term memory

automatic processing
encoding of informational details like time, space, frequency, and the meaning of words

bias
how feelings and view of the world distort memory of past events

blocking
memory error in which you cannot access stored information

chunking
organizing information into manageable bits or chunks
collective memory
memories, knowledge, and cultural information often of important historical events that is passed down and maintained within a group or society over time.,
construction
formulation of memories, memories are "built" and subject to multiple influences
declarative memory
type of long-term memory of facts and events we personally experience
deep processing
encoding of information that takes effort and attention
effortful processing
encoding of information that takes effort and attention
elaborative rehearsal
thinking about the meaning of new information and its relation to knowledge already stored in your memory
encoding
input of information into the memory system
episodic memory
type of declarative memory that contains information about events we have personally experienced, also known as autobiographical memory
explicit memory
memories we consciously try to remember and recall
flashbulb memory
exceptionally clear recollection of an important event
forgetting
loss of information from long-term memory
implicit memory
memories that are not part of our consciousness
levels of processing

information that is thought of more deeply becomes more meaningful and thus better committed to memory

long-term memory (LTM)

continuous storage of information

memory

set of processes used to encode, store, and retrieve information over different periods of time

misattribution

memory error in which you confuse the source of your information

misinformation effect

after exposure to additional and possibly inaccurate information, a person may misremember the original event

mnemonic device

memory aids that help organize information for more effective encoding and retrieval

persistence

failure of the memory system that involves the involuntary recall of unwanted memories, particularly unpleasant ones

proactive interference

old information hinders the recall of newly learned information

procedural memory

type of long-term memory for making skilled actions, such as how to brush your teeth, how to drive a car, and how to swim

recall

accessing information without cues

recognition

identifying previously learned information after encountering it again, usually in response to a cue

rehearsal

repetition of information to be remembered

relearning

learning information that was previously learned

retrieval

act of getting information out of long-term memory storage and back into conscious awareness

retroactive interference

information learned more recently hinders the recall of older information

retrograde amnesia

loss of memory for events that occurred prior to brain trauma

self-reference effect

tendency for WEIRD individuals to have better memory for information that relates to oneself in comparison to material that has less personal relevance

semantic encoding

input of words and their meaning

semantic memory

type of declarative memory about words, concepts, and language-based knowledge and facts

sensory memory

storage of brief sensory events, such as sights, sounds, and tastes

shallow processing

limited attention to information results in inefficient processing

short-term memory (STM)

holds about seven bits of information before it is forgotten or stored, as well as information that has been retrieved and is being used

storage

creation of a permanent record of information

suggestibility

effects of misinformation from external sources that leads to the creation of false memories

transience

memory error in which unused memories fade with the passage of time

visual encoding

input of images

Chapter 7. Personal Application Questions

- Reflect on a time when you struggled to remember something important. What strategies did you employ to enhance your memory? How effective were those strategies? Would you approach it differently now based on what you've learned about memory enhancement techniques?
- Describe something you have learned as an adult that is now in your procedural memory. Discuss how you learned this information.
- Describe something you learned in high school that is now in your semantic memory. Why do you think you still remember this information?
- Describe a flashbulb memory of a significant event in your life. Do you think this memory is accurate?
- Which of the seven memory errors presented by Schacter is most troublesome for you? Provide an example of each kind of error in your life.
- Create a mnemonic device to help you remember a term or concept from this chapter.
- Think about your current study habits. Are you using any

memory-enhancing strategies while studying? If yes, which ones? If no, which strategies mentioned in the course material do you think would be beneficial for improving your memory while studying?

- Reflect on your sleep habits and their potential impact on memory. How many hours of sleep do you typically get each night? Do you notice any differences in your memory and cognitive abilities on nights when you have sufficient sleep compared to nights when you have inadequate sleep?
- Consider the role of exercise in memory improvement. Do you engage in regular aerobic exercise? If yes, have you noticed any changes in your memory or cognitive function as a result? If you don't exercise regularly, do you think incorporating exercise into your routine could potentially benefit your memory?
- Think about a challenging course or subject you have encountered in the past. How did you approach studying for that subject? Were you aware of the most effective study techniques at that time? Looking back, what study strategies could you have employed to enhance your memory and improve your understanding of the material?
- Reflect on your note-taking habits during lectures. How do you currently take notes? Are there any strategies mentioned in this chapter that you could incorporate into your note-taking process to enhance memory encoding and organization?

Chapter 7. Critical Thinking Questions

- Compare and contrast implicit and explicit memory.
- According to the Atkinson-Shiffrin model, name and describe the three stages of memory.
- Compare and contrast some of the ways in which we encode information.
- What might happen to your memory system if you sustained damage to your hippocampus?
- Compare and contrast the two types of interference.
- Compare and contrast the two types of amnesia.
- You and your roommate spent all of last night studying for your psychology test. You think you know the material; however, you suggest that you study again the next morning an hour prior to the test. Your roommate asks you to explain why you think this is a good idea. What do you tell them?
- Jurors place a lot of weight on eyewitness testimony. Imagine you are an attorney representing a defendant who is accused of robbing a convenience store. Several eyewitnesses have been called to testify against your

client. What would you tell the jurors about the reliability of eyewitness testimony?

Chapter 7. References

The answer to the question about the nickel in Figure 7.13 is (c).

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CHAPTER 8. HIGHER ORDER COGNITION: LANGUAGE AND INTELLIGENCE

Hello
Guten Tag
Привет
สวัสดี
こんにちは
你好
Sawubona
Bonjour

Cognitive psychologists study a wide range of mental processes. We have already delved into the subfields of sensation and perception, attention, and learning and memory. In this chapter, we will focus on other higher aspects of cognition: language and intelligence, and we will briefly discuss how language can affect thinking. Language is a remarkable human ability that lies at the heart of our communication and cognitive processes. Moreover, language and culture are inextricably entwined. In this chapter, we will explore the use and development of language, and how language shapes our identities and how we think. We will also look at the various ways that psychologists have conceptualized and measured intelligence. We will discuss psychology's abhorrent role in the eugenics movement, where misguided theories of intelligence and biased testing were used to justify discriminatory practices that unjustly marginalized women, BIPOC individuals, sexual minorities, and those with mental illness.

LEARNING OBJECTIVES

- Define language and demonstrate familiarity with the components of language
- Explain the development of language
- Discuss the cultural and social aspects of language including how language shapes cultural identity
- Compare and contrast some pragmatic language differences, such as differences in expressing gratitude, making requests, and refusing invitations.
- Describe the benefits of bilingualism and multilingualism
- Describe the ways that language can influence metacognition, self-evaluation, perceptions, and attributions.
- Define linguistic racism and describe its impact
- Describe the influence of cultural values on the conceptualization of intelligence.
- Compare and contrast theories of intelligence
- Describe how the eugenics movement used pseudoscientific information about intelligence to oppress large groups of marginalized people
- Explain why race is not a biological construct and describe its social construction.
- Analyze the impact of genes and environment on IQ scores and understand the complex interplay between genetic and environmental factors.
- Describe the roles of academic social activists in challenging the discriminatory practices associated with intelligence testing.

- Critically analyze the limitations and cultural biases present in intelligence tests, highlighting the need for more diverse and culturally sensitive assessments.
- Recognize the phenomenon of stereotype threat and its influence on test performance, particularly for marginalized groups, and explore strategies to mitigate its effects.
- Describe alternative theories of intelligence, including Howard Gardner's multiple intelligences theory and Robert Sternberg's triarchic theory, and understand how these theories broaden the understanding of intelligence beyond traditional IQ tests.
- Describe the concept of emotional intelligence and its relevance in social relationships and job performance, while acknowledging the challenges in measuring and studying emotional intelligence.

8.1. Language

Language is defined as a communication system that involves the systematic use of words, signs, or symbols, to transmit information from one individual to another. We engage in a vast array of language-based activities every day, from speaking and listening (or signing and interpreting signs), to reading and writing. However, not all communication is language. Many species communicate with one another through their postures, movements, odors, or vocalizations. This communication is crucial for their survival and social interactions, but, unlike language, the complexity of ideas that can be communicated is limited. Language is thought to be uniquely human because it allows us to communicate abstract, complex ideas, about the past, present and future. Language is uniquely generative – this means that we can create infinite numbers of sentences with different meanings by combining words and phrases in different ways (Corballis & Suddendorf, 2007; Tomasello & Rakoczy, 2003).

COMPONENTS OF LANGUAGE

Language, be it spoken, signed, or written, has a lexicon and grammar (Fernández & Cairns, 2011). The lexicon refers to all the words in a given language, in other words, the vocabulary. Grammar refers to the set of rules that are used to convey

meaning, for instance, in English, we typically indicate past tense by adding –ed to the end of a verb (e.g., walk becomes walk**ed**). Syntax (word order) and semantics (word meaning) are part of a language's grammar.

In most languages, each word is made up of phonemes and morphemes. A phoneme is the basic sound unit of a given language. If you learned to read by sounding out your letters – you were pronouncing the phonemes. In English, there are 26 letters in the alphabet but about 40 phonemes (depending on your accent). This is because some letters have more than one sound; for example, the letter "i" has two different sounds, think about how "i" sounds in the word "***ice***" versus "***in***". Other letters combine to make a single phoneme, such as "th". Some languages have very few phonemes (e.g., Hawaiian has 13) and other languages have a lot of phonemes, for example, Khoisan click languages have more than 100. Phonemes are combined to form morphemes, which are the smallest units of language that convey some type of meaning. For example, the word "unhappiness" has three morphemes. "Un" means not; "happy" means glad; and "ness" means a state of being. In addition to using phonemic systems of language, some cultures also use little pictures (characters) that depict morphemes or words. For example, Hanzi in China, Kanji in Japan, and Hanja in Korea all use the same ancient Chinese characters in one of their writing systems, see Figure 8.2 for an example.

Figure 8.2. The words "Chinese characters" written in traditional Chinese characters used in Hanzi, Kanji, and Hanja.

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Figure 8.3. American Sign Language uses hand gestures and facial expressions.

Sign language is a nonverbal form of communication, which involves signaling using the hands and facial expressions (see Figure 8.3), but words can still be broken down into smaller phoneme-like visual units, which in turn can be combined into morphemes.

Intonation in language is also very important in conveying meaning. In many languages (e.g., Mandarin, Vietnamese, Thai, Punjabi, Yoruba, Cherokee) are tonal – the change in the pitch of a single syllable can change the meaning of the word dramatically.

Link to Learning

Watch this video . How easy is it for you to make the correct tonal variations in the Mandarin word – Ma?

Emphasis and tone of voice (prosody) are also vitally important for conveying meaning in both tonal and atonal languages. Consider for example, how your voice tends to go up at the end when asking a question, and how your voice changes when you feel happy or sad. In addition to these basic elements of language, pragmatic aspects of language also determine how we communicate in social and cultural contexts.

8.2 Culture and Language

Culture and language are inextricably entwined. For example, the metaphorical meanings of simple words, such as colors, vary across cultures. In English and Serbian, if we are feeling "blue", then it means we feel sad, and a "blue day" in Arabic is a bad one. However, if someone describes themselves as "blue" in German, they are drunk. A gay man in Russian might describe himself as a "blue man", and a blue prince in Spanish is a Prince Charming. If you describe yourself as" turning blue" in Japanese, then you are shocked or scared. Alternatively, if someone tells you that "the fire in the stove is pure blue" in Mandarin, they are paying you a compliment by saying that you are very accomplished. Clearly, colors have very different metaphorical meanings in different languages.

There is also considerable cross-cultural divergence in language practices in social situations, we refer to these as the pragmatic aspects of language. Let's take a look at a few of these practices, such as saying thank you, and making and refusing requests. It is common in Anglo-American and Australian (and Italian) cultures to frequently thank people, even when they are simply doing their job or if they are doing something that does not require much effort. For example, they thank family members for passing the salt and for cooking dinner, as well as the bus driver when they get off the bus. This is done out of politeness, rather than expressing heart-felt emotion. In other cultures, expressions of gratitude are used much

more sparingly, and are reserved for occasions when someone feels genuine gratitude. For example, Liddicoat (2014) describes how a teenager from Chile contrasts all the thanking she observes when staying with an Australian family, with the experiences in her own home. She remarks that her mother would be upset if she thanked her for making dinner—because it is something that she does daily. It would imply that something was wrong with all the other dinners she had made or that something was wrong with that particular meal. In other cultures, such as in South and South East Asia, saying thank you to a family member or close friend for a trivial act, is perceived as overly formal and a way to distance yourself from them. Instead gratitude is assumed in everyday life (Engels & Hsieh, 2021). Thus, in some cultures, saying thank you can be perceived as rude behavior in some situations, whereas in others, not saying it is considered impolite.

Cultural variations in how people make requests are common. Some cultures use direct speech (e.g., Open the door) more frequently, whereas others favor indirect speech (e.g., Would you mind opening the door, please?). People from Persia and Israeli Jews are more likely to make direct requests than people from North America, who tend to use indirect requests most of the time. People from Greece and China also favor indirect requests, whereas people from Spain make more direct requests than people from the Netherlands (Eslami et al., 2014; Le Pair, 1996; Wierzbicka, 2003). Differences in directness of speech also extend to other areas, such as refusing requests or invitations. Again, Israeli Jews tend to be more direct than Americans and Arabs, who are more likely to provide a reason for their refusal. Native speakers of Arabic provide more extensive explanations for the refusal than Americans (Stavans & Shafran, 2017). In some cultures, refusals are so indirect they might sound like permission has been granted. For example, imagine that as a teenager, you asked a parent or older relative if it was OK with them for you to go to a party. If they said "*You can go, if you want.*" What would that

mean to you? Are they giving you permission? The way that the words are said to you, in addition to your language and culture will probably influence how you interpret their response. This example comes from an article by Tannen, who describes a Greek woman whose father and husband never directly refused a request from her. However, she had to interpret what they were really saying from the level of enthusiasm they expressed (Tannen, 1981).

BENEFITS OF BI/MULTILINGUALISM

Although historically, Western psychology has studied language from a monolingual perspective, the field today more accurately reflects the fact that the majority of people in the world are bilingual or multilingual. Being fluent in two or more languages has been shown to provide both a cognitive advantage and greater open-mindedness (Anoniou, 2019; Cárdenas & Verkuyten, 2021; van den Noort, 2019; Ware et al., 2020). The cognitive benefits of speaking more than one language have often been attributed to the constant practice that bi/multilinguals have in selecting one language and inhibiting the other when reading, listening, speaking or writing. The majority of studies have demonstrated that bilinguals have better **executive function** than monolinguals, especially on tasks that require ignoring distracting information (Anoniou, 2019; van den Noort, 2019; Ware et al., 2020). Studies have also shown other learning benefits for bilingual and multilingual toddlers (Brito et al., 2014). Bi/multilingual children who watched a researcher interact with a puppet were able to apply what they had learned by interacting in similar ways with different toys. Monolingual children were less able to generalize their learning. Being able to generalize learning is very helpful for figuring out how to interact with new objects and in new situations. Bilingualism has also been associated with greater creativity, better intercultural sensitivity, and improved chances of employment. In

addition, bilingualism also seems to be protective against some of the cognitive decline that often happens as we age (Fox et al., 2019).

LANGUAGE AND IDENTITY

Our use of language is often closely linked to our national and cultural identities (Cárdenas & Verkuyten, 2021; Garcia, 2020). Some bilinguals embrace both of their languages and cultures, but others feel that there is too much difference between them, often leading to abandonment of heritage languages and cultural practices, because of their lack of socioeconomic power.

Multiple studies have suggested that bilinguals feel different when they switch languages, some people feel more authentic and better able to express emotion in their native language (Dewaele & Nakano, 2013). Some bilinguals exhibit cultural shifts in attitudes and behaviors depending on the context, which is often (but not always) determined by their language use (Ramírez-Esparza & García-Sierra, 2014). For example, Mexican-American bilinguals were rated as more agreeable when speaking Spanish in an interview, than when speaking English (Ramírez-Esparza & García-Sierra, 2014). However, Chinese-English bilinguals were rated as more open and extroverted when speaking to a Caucasian interviewer than a Chinese one, regardless of the language spoken (Ramírez-Esparza & García-Sierra, 2014). Thus, bilinguals are able to shift between languages easily, but they also show cross-cultural changes in their body language and pragmatics of their speech.

8.3. Language and Thinking

We have already discussed how bilingualism provides multiple cognitive advantages, however, studies show that the language that we use can also affect *how* we think. Bicultural individuals sometimes switch how they think, depending on the language that they are using. For example, metacognition – the ability to accurately self-evaluate how well one has done on a task—has been shown to vary across cultures. A meta-analysis by Zell and colleagues on the Overconfidence Bias (also known as the Better than Average effect) shows that people around the world have a tendency to think they have done better on a task than they actually did. However, European-Americans are more likely to over-estimate their performance than people from East Asia (Zell et al., 2020). Stankov and Lee (2014) also found that people in South Asia, the sub-Sahara, and Latin American countries were more likely to over-estimate how well they had done on a test compared to people from Western Europe, East Asian, and English-speaking countries (US, UK, Canada, Australia, etc.). But, what about bicultural individuals? Lechuga and Wiebe (2011) found that overconfidence in bilinguals depends on language context. When Mexican-Americans assessed their performance on a general knowledge test they expressed greater over-confidence when tested in Spanish, than when tested in English (Lechuga & Wiebe, 2011).

Other studies have also shown how language context can

influence our thinking. In many languages, nouns are masculine or feminine. A study by Boroditsky and colleagues showed that we tend to attribute more masculine qualities to masculine nouns and more feminine qualities to feminine nouns (Boroditsky et al., 2003). Spanish-English and German-English bilinguals were asked to describe the attributes of some common objects. Half of the objects were masculine nouns in German and feminine in Spanish, and vice versa. Participants used words that were consistent with the gender of the noun in their descriptions. For example, "key" is masculine in German and feminine in Spanish. It was described as hard, metal, jagged, and heavy by German speakers, but shiny, intricate, and tiny by Spanish speakers (Boroditsky et al., 2003). However, language does not always influence how we think. Berlin and Kay (1969) compared how English speakers and the Dani people of Papua, New Guinea, think about colors. The Dani have only two words for color: one word for *light* and one word for *dark*. In contrast, the English language has 11 basic color words. The researchers hypothesized that the lack of color terms would limit the ways that the Dani people conceptualized color. However, there was no evidence for this. The Dani were able to distinguish between colors just as well as English speakers, despite having fewer color words at their disposal.

LINGUISTIC RACISM

It is important to acknowledge that enthusiasm for bilingualism (including bilingual educational programs) is relatively recent in the history of the United States. Flawed psychological studies conducted prior to the 1960s were widely used to advocate that children only use one language—Standard American English. Parents were told that they were harming their children's cognitive development if they were raised bilingual and so many children were stopped from using their heritage languages. This practice

often severed communication with older members of a child's community, who did not speak English. These ideas were so pervasive they still persist among many teachers in early education today in the United States (De Houwer, 2020).

Some children are reluctant to use their heritage language(s) because they perceive them to have less status than the dominant societal language (De Houwer, 2020). These views are often a consequence of linguistic racism, where people are attacked, demeaned, and marginalized for speaking in a non-dominant language (Ek et al., 2013; Wang & Dovchin, 2022). Relatedly, from 1869 until 1969, tens of thousands of native American children were actively prevented from speaking their native language(s), when the United States Government mandated they were taken from their parents and forced into residential boarding schools. The children were punished for speaking native languages or engaging in the spiritual rituals that they had practiced at home. The Government claimed that this forced acculturation would benefit the children from a socioeconomic standpoint. Instead, this violent action has left lasting scars on the native American population, including high rates of substance abuse and mental illness (American Psychological Association, 2003). The American Psychological Association acknowledges that it failed to intervene against these harmful practices, and they not issue a protest until it was much too late (American Psychological Association, 2003). Recent studies have shown that revitalization and maintenance of indigenous ancestral languages is associated with increased wellbeing among native Americans and other indigenous people (Whalen et al., 2016).

8.4. Language Development

Given the complexity of language, it seems remarkable that young children master it very quickly and with relative ease. We are apparently born with a biological predisposition to acquire language (Chomsky, 1965; Fernández & Cairns, 2011). Language develops in the absence of formal instruction and appears to follow a very similar pattern in children across the world. However, early language exposure is important for proficient language development (Johnson & Newport, 1989; Lenneberg, 1967; Singleton, 1995). It becomes more difficult to learn new languages (especially grammar and pronunciation) as we get older, however, people often have to do so, especially when immigrating to a new country (Hakuta et al., 2003).

Language development begins very early on in life (see Table 8.1). The fetus can hear muffled voices from the outside world, and so learns about language even before birth. Their mother's voice is the loudest and most consistent voice they hear during the fetal period, which explains why newborn babies can identify their mother's voice and her native language (DeCasper & Fifer, 1980; Moon et al., 1993).

Table 8.1. *Stages of Language and Communication Development*

Stage	Age	Developmental Language and Communication
1	0–3 months	Reflexive communication
2	3–8 months	Reflexive communication; interest in others
3	8–13 months	Intentional communication; sociability
4	12–18 months	First words
5	18–24 months	Simple sentences of two words
6	2–3 years	Sentences of three or more words
7	3–5 years	Complex sentences; has conversations

You may recall that each language has its own set of phonemes (or speech sounds). At birth, all babies can differentiate between phonemes from all human languages, even if they are not being used in their environments. At about 7 months, babies start to produce speech sounds – a process known as babbling. Babies exposed to sign language also babble with hand gestures (Petitto et al., 2004). Babies are just as likely to babble when they are alone as when they are with their caregivers (Fernández & Cairns, 2011), but they also have periods of social babbling, where they look to others to respond to their babble (Laing & Bergelson, 2020). By about 11-12 months, babbling becomes more speech-like, and contains more phonemes from the language(s) that babies are hearing around them (Laing & Bergelson, 2020). Around the same time, babies lose their ability to hear the difference between sounds that do not occur in their speech environments (Jensen, 2011; Werker & Lalonde, 1988; Werker & Tees, 1984). For example, in Hindi, there are two "*t*" sounds that are easy for people who speak Hindi to differentiate between, but often sound the same to English speakers. Similarly, the phonemes associated with the letters *r* and *l* in English, often sound the same to Japanese speakers.

Link to Learning

Watch this video about how babies lose the ability to discriminate among all possible human phonemes as they age to learn more.

Generally, a child utters their first word between the ages of 1 year to 18 months. For the next few months, a child will learn a lot of new words, but will only produce one-word utterances. A child's early vocabulary is limited to familiar objects or events, often nouns. These words often carry larger meaning (Fernández & Cairns, 2011), for example, a child saying "cookie" could be identifying a cookie or asking for a cookie.

As a child's lexicon grows, they begin to use simple sentences and acquire new vocabulary at a very rapid pace. In addition, children begin to demonstrate a clear understanding of the specific rules that apply to their language(s). During this time, children frequently overgeneralize grammatical rules. For example, in English, the letter "s" is often added to the end of a noun to make it plural, e.g., one dog versus two dogs. Young children might also say "two gooses" or "three mouses." Clearly, they understand the rules of the language but have not yet learned all the irregular exceptions (Moskowitz, 1978).

Dig Deeper

The Case of Genie

In the fall of 1970, a social worker in the Los Angeles area found a 13-year-old girl who had been extremely abused and neglected. "Genie" had lived most of her life tied to a potty chair or confined to a crib in a small locked room with the curtains drawn. Genie had virtually no social interaction and no access to the outside world. As a result of these conditions, Genie was unable to stand up, chew solid food, or speak (Fromkin et al., 1974; Rymer, 1993). Genie's abilities improved dramatically after she was rescued from her abusive environment (Fromkin et al., 1974). She managed to amass an impressive vocabulary in a relatively short amount of time. However, she never mastered the grammatical aspects of language (Curtiss, 1981).

8.5. Intelligence

This module is heavily influenced by Dr. Robert Guthrie's work:
Guthrie, R.V. (1998). *Even the rat was white*, 2nd edition. Allyn & Bacon.

Learn more about Dr. Guthrie here https://www.apa.org/monitor/nov01/american

Some of this content comes from the following open source materials:

Biswas-Diener, R. (2018). Intelligence. In: R. Biswas-Diener & E. Diener (Eds.), *Noba textbook series: Psychology.* Champaign, IL: DEF Publishers. DOI:nobaproject.com

Stangor, C., & Walinga, J. (2014). Intelligence. *Introduction to Psychology* (1st Canadian Edition). https://opentextbc.ca/introductiontopsychology/

Psychology 2e. Openstax.org

WHAT IS INTELLIGENCE?

Defining intelligence is highly controversial. The conceptualization of intelligence varies widely across the world and is heavily influenced by cultural values (Sternberg & Grigorenko, 2004). Some cultures emphasize practical skills and the ability to adapt to the challenges of everyday life, and so place particular value on the ability to communicate and empathize with others. Collectivistic cultures value collective intelligence: the ability to contribute to group cohesion by work collaboratively and communicating effectively. Other cultures (e.g., Native Americans and people from Africa) particularly respect the elders in their community because they believe that wisdom comes from life experiences. Anglo-European/American cultures commonly emphasize cognitive abilities that are related to academic achievement, such as rapid information processing, memory, language skills, mathematical aptitude, and logical reasoning (Sternberg & Grigorenko, 2004). However, even Western psychologists disagree on a single definition of intelligence. In this chapter we will discuss how different psychologists conceptualize intelligence, but we will particularly emphasize how psychology has contributed to the oppression of large groups of people (especially people of color), through the misuse and misinterpretation of intelligence tests.

There is a long history of unfounded prejudicial attitudes about the intelligence of people of color. Africans were engaging in sophisticated practices, such as bronze casting, iron making, and pyramid building, long before the Europeans did. However, Europeans arriving in the continent of Africa assumed that Black and Brown people were uncivilized and had low levels of intelligence (Guthrie, 1998). These dehumanizing views were used to justify a long history of colonization and the enslavement of African people in Europe and the United States. Even after slavery was abolished, psychologists in the United States used questionable "scientific evidence" to perpetuate racist beliefs

(Guthrie, 1998). As you will read in this chapter, their "science" was based on poorly designed studies and biased tests (Guthrie, 1998). Even today, modern intelligence tests and other cognitive aptitude tests (including the SAT) are frequently criticized as being culturally unfair, and their continued use excludes many people from marginalized groups from multiple educational and career opportunities (Helms, 2012; Washington et al., 2016).

8.6. Development of Intelligence Testing in the United States

Many psychology textbooks suggest that the first intelligence tests were developed in France during the early 20th century. However, the earliest known intelligence tests were actually conducted 3,000 years ago in China, where people wrote essays to determine whether they would be hired as a civil servant. The tests were later expanded to include problem solving questions that bear some similarities to some modern-day intelligence tests (Saklofske et al., 2015). It was not until the 20th century, that intelligence testing came to the Western world. In France in1905, self-trained psychologist, Alfred Binet, and a doctor, Theophile Simon, developed one of the first Western intelligence tests—the Binet-Simon Scale. They hoped that it could be used to identify students who might struggle in school with the aim of providing additional support (Gredler, 2020). Binet conceptualized intelligence as the ability to engage in focused, but flexible thinking, and so the test questions consisted of basic everyday problems designed to evaluate attention, memory, and verbal skills (Guthrie, 1998; National Institutes of Health, 2014). However, Binet warned that this measure could *only* be considered a test of intelligence if comparisons were made among children from similar backgrounds, because disparities in education or socioeconomic status (SES) would affect the scores (Guthrie, 1998).

INTELLIGENCE TESTING AND THE EUGENICS MOVEMENT

Binet's warnings were largely ignored. Instead, in the first half of the 20th century, many psychologists, along with huge numbers of other influential people in society, including President Roosevelt, embraced the horrific field of eugenics. The eugenics movement was founded by an English scientist called Francis Galton, who misappropriated Charles Darwin's work on evolution and Gregor Mendel's work on heritable traits in plants and animals, by arguing that these ideas could be extended to human traits. Eugenicists believed that genes, not environment, determined intelligence (among other traits), and that people with low intelligence should be prevented from "spreading their genes". These beliefs were extremely widespread across Europe and the United States (US). Many of the Western psychologists involved in developing intelligence tests were eugenicists, including Spearman in the UK, and Terman in the US. As a result, intelligence testing was weaponized as evidence of low intelligence among people with little formal education. This included women, people of color, and people living in poverty (Guthrie, 1998). Eugenics was used to promote and justify social and economic inequalities in the US, United Kingdom, and the countries that they colonized. The long-term consequences of eugenics-inspired oppressive socioeconomic policies still persist today.

In addition to his eugenicist beliefs, Spearman (1904) also popularized the idea that intelligence could be adequately measured by a single score, which represented a person's general intelligence (Boake, 2002; Guthrie, 1998). Spearman justified this approach because he had found that people who were proficient in one intellectual domain, e.g., verbal skills, were often equally adept in other domains, such as quantitative reasoning (Biswas-Diener, 2008; Guthrie, 1998). In the 1940s, Raymond Cattell proposed that

intelligence consists of two major factors: crystallized intelligence and fluid intelligence (Cattell, 1963). Crystallized intelligence is characterized as the ability to retrieve knowledge related to your educational and life experiences. When you learn, remember, and recall information in your classes, you are using crystallized intelligence. Fluid intelligence on the other hand, encompasses the ability to cope with new situations, see complex relationships, and solve abstract problems (Cattell, 1963). More recent theories suggest that intelligence consists of multiple different domains that go beyond cognitive skills; a person may be strong in one, but not others (Gardner, 1983; Sternberg 1999). The first widely used intelligence test in the US was designed to measure general intelligence, and so minimized the strengths of people who were not good "all-rounders". The test was developed in 1916 by Lewis Terman, a eugenicist and a professor at Stanford University. Terman adapted the Binet-Simon Scale for use in the US and renamed it the Stanford-Binet scale. Scores on the Stanford-Binet scale were referred to as the intelligence quotient or IQ (Guthrie, 1998). Terman standardized the Stanford-Binet scale using data from 1000 children and 400 adults in the US (Guthrie, 1998), all of whom were White.

Standardization is a commonly used procedure when developing psychological tests. Raw scores are converted so that they fall in a normal distribution, which is shaped like a "bell curve" (see Figure 8.5), allowing for easy comparisons between individuals (Biswas-Diener, 2018). It is therefore important that the scores are standardized using a large sample of people who are representative of the population. In the Stanford-Binet (and other subsequent IQ tests) the mean IQ score is set at 100. As you can see in Figure 8.6, in the standardized sample, 68% of people have an IQ between 85 and 115; and 96% of people have an IQ of between 70 and 130. There are very few people at the extremes of the scale. You probably have taken at least one standardized aptitude test (like the SAT) in your life. The scores on these tests are standardized

in a similar way, but the average score is set around 500 on each SAT subtest.

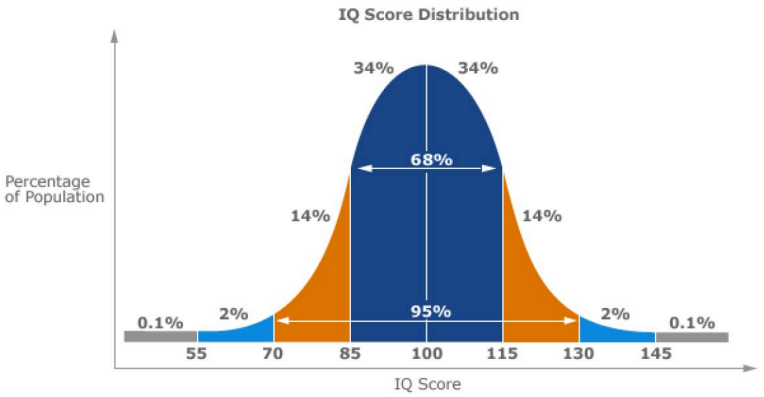

Figure 8.5. Standardized IQ scores. The average score is 100.

Excluding people of color from the standardization process was a serious flaw in Terman's Stanford-Binet intelligence test, as the standardized scores were not representative of the population (Guthrie, 1998). The test relied heavily on verbal skills and asked questions that reflected middle-class values, and so children with little schooling were often misclassified as having low intelligence. The same issues that made the Stanford-Binet test culturally unfair for less educated children also plagued the intelligence tests for adults that were developed for the army during the First World War (Guthrie, 1998).

In 1917, Yerkes (another eugenicist), Terman, and other psychologists from the American Psychological Association helped to design intelligence tests for the US army. These were used to determine work assignments for army recruits (Guthrie, 1998). Not only were the tests culturally biased, but the military psychologists who administered them were all White men, many of whom were blatantly racist, which probably influenced their scoring (Guthrie, 1998). The army then released reports based on the test scores

stating that non-Whites, Jews, and Eastern and Southern European immigrants were intellectually inferior to Whites. Moreover, they reported that people with darker skins were less intelligent than light-skinned people (Guthrie, 1998). This misguided report was widely disseminated and was used to further justify and promote many harmful socioeconomic policies. In the US, interracial marriages were banned and racial segregation was strongly enforced to decrease the likelihood of inter-racial relationships developing. Policies such as redlining and racially segregating schools, increased wealth and resources for Whites, and reduced economic opportunities and educational resources for people of color. The 1924 Immigration law severely restricted the admittance of Jews, Eastern and Southern Europeans, and people who were mentally ill, to the United States (Guthrie, 1998).

In keeping with eugenicist ideals, by 1944, 40,000 people had been subjected to forced sterilization in the United States, and 30 states had passed laws legalizing this practice. Forced sterilization targeted mostly poor Black women and immigrants, and included people living in institutions for the mentally ill, people accused of crimes, people with disabilities or chronic illnesses. Scandinavia and Canada also embraced forced sterilization during the same general time period (Kevles, 1999). Adolf Hitler and other German eugenicists were inspired by the widespread ways that the US embraced White supremacy through its sterilization and immigration laws. However, the policies that Germany subsequently created were even more extreme; they mandated killing young children with physical or mental disabilities. Eugenics was also used to justify the persecution and mass murders of six million Jews during the holocaust (Guthrie, 1998). Sterilizations in the United States declined dramatically in the aftermath of the atrocities of the holocaust and in the light of new studies showing that intelligence and mental illness are not solely determined by genes. However, even as laws began to change, the US still carried out about 22,000 more forced sterilizations across 27 states

between 1943 and 1963. Some of these sterilizations occurred in some southern and mid-Western states shortly after World War II, when reproductive clinics began "offering" sterilization as a method of birth control to young, mostly Black, poor women in rural areas, and to young native American women. Many of these sterilizations were coerced or done without the woman's consent (Reilly, 2015). It was not until the 60s and 70s, that US sterilization laws were largely repealed (Sofair & Kaldjian, 2000; Suarez-Balcazar, 2022). However, reproductive injustice continues even today. In 2020, a nurse working for Immigration and Customs Enforcement (ICE) made it publicly known that several Latina women were coerced into having unnecessary hysterectomies during their detention by ICE (Suarez-Balcazar, 2022).

8.7. Genes, Environment, and IQ scores

Let us briefly review some of the more recent scientific studies about genes and IQ scores. Genetic factors do play some role in the determination of IQ scores, but these are hard to separate from environmental ones. For example, biological sex is genetically determined, and for centuries, patriarchal societies, like the United States, have regarded females as intellectually inferior to males. Despite the fact that many eugenicists were married men, their wives, who were White middle-class women, were also harmed by eugenicist claims that they lacked intelligence and were only fit for child-rearing, and home-making (Yakushko, 2019). However, there is no scientific evidence to support that males are intellectually superior to females. In fact, there are very few innate sex differences in cognitive aptitudes. Historically, females have been shown to do better on verbal tasks than males, whereas males do better on math tests, but these differences are largely environmental (Grose-Fifer & diFilipo, 2017). Boys and girls are frequently treated differently at home, in school, and by society in general. For example, children are often given gender-specific toys to play with and are encouraged to engage in behaviors that are perceived to be gender-specific. They are also discouraged from behaving in ways that are not considered gender-consistent. Can you think of how the toys that you were given and the games that

you were encouraged to play as a child might have shaped your cognitive abilities as a young adult?

RACE IS NOT BIOLOGICAL

It is also important for us to highlight again in this chapter, that in contrast to sex, race is not a biological construct. In other words, you cannot tell a person's race from looking at their genes (Berry et al., 2014; Guthrie, 1998; Nisbett et al, 2012). This is a long-perpetuated myth. Race is NOT based on genes, it is socially constructed—people decide who belongs to one race or another, based on skin color, hair texture, and the shape of facial features. People within a single family share a lot of common genes, but often vary widely in their physical characteristics. People show more genetic variation within a race than across races. Therefore, racial differences in IQ are NOT genetically determined. However, a person's race affects how they are treated in society and often determines the socioeconomic opportunities they receive (Weiss & Saklofske, 2020).

Environmental influences including SES, neighborhood, family structure, schooling, parents' level of education, nutrition, language, and culture, all affect IQ scores (Berry et al., 2014; Guthrie, 1998; Helms, 2012; Nisbett et al., 2012; Weiss & Saklofske, 2020). SES is a particularly important predictor of IQ (Weiss & Saklofske, 2020). Families with higher incomes generally have higher IQs (Berry et al., 2014). Importantly, individual IQ scores (and other measures of intelligence) can increase with environmental improvements. Infants and young children in poverty who attend Head Start programs do much better on IQ tests (and in school) than those who do not attend (Bauer, 2019). Furthermore, students, regardless of race, in well-resourced schools do better on IQ tests than those in poorly funded schools (Guthrie, 1998).

ACADEMIC ACTIVISTS FIGHT AGAINST INTELLIGENCE TESTING.

For many years, eugenicists used IQ testing to justify the oppression of women, immigrants, and people of color. However, during the late 1920s, voices of dissent began to emerge (Guthrie, 1998). Academic activists began to conduct scientific programs of research that revealed the prejudicial nature of previous reports on IQ. Many of the scientists that contributed to this research were people of color and/or women.

Some of the first women to earn PhDs in psychology spent their academic careers conducting research showing that gender did not influence intelligence. In 1900, Helen Thompson Woolley was awarded a PhD from the University of Chicago. Her doctoral research showed that men and women performed similarly on IQ tests (Rodkey, 2010). Thompson Woolley spent her academic career showing that education affected IQ scores. Leta S. Hollingworth was awarded a PhD from Columbia University in 1916. Her doctoral research refuted the popular belief that women's decision making abilities were affected by menstruation. Hollingworth's research was even more remarkable, given that her mentor, Edward Thorndike, held the opposite view. However, like many psychologists of her time, Hollingworth believed that intelligence was inherited and she also endorsed eugenicist views (Benjamin 1990; Leta Stetter Hollingworth, 2023).

In 1927, Horace Mann Bond (Figure 8.6), a young Black sociology professor at Langston University, Oklahoma, reported that IQ testing conditions disadvantaged Black children. He showed that White psychologists took time to build rapport with White, but not Black children, which artificially depressed the IQ scores of Black children (Guthrie, 1998). Bond also demonstrated that low IQ scores among Black children were linked to environmental factors, like education and socioeconomic status (SES). He showed that

many Black children from middle and upper-middle class families had IQ scores in the exceptional range (Guthrie, 1998). Bond spent his career working against segregation and later became the president of Lincoln University.

Figure 8.6. Horace Mann Bond

George Sanchez was an American educational psychologist of Mexican descent who demonstrated that Indigenous children and Chicano (Mexican-American) children were disadvantaged by IQ tests in similar ways to Black children. Low IQ scores in BIPOC children were used as justification for them to be educated in segregated, poorly resourced schools (Guthrie, 1998). In 1932, George Sanchez, took advantage of his position as Director of the Information and Statistics Division of the New Mexico State Department of Education to speak out against the inadequacies of the tests that were being used to assess the intelligence of Chicano children, many of whom spoke Spanish and lived in poverty (Tevis, 2020; Guthrie, 1998). Although IQ tests were translated into Spanish, the translations were poor and reflected the cultural

values of White middle-class Americans (Guthrie, 1998). Sanchez later became a professor at the University of Texas, and remained an academic activist throughout his life. He continued to protest and testify against racist policies, and advocated for better education for children from marginalized groups, especially Chicano children (Tevis, 2020).

In 1935, Otto Klineberg, a Canadian psychologist, drew attention to the cultural inappropriateness of intelligence testing of indigenous children, pointing out that children of the Dakota nation were used to cooperating rather than competing. Often they did not answer the psychologists' questions unless they felt sure that their peers also knew the answer (Guthrie, 1998). This was an important insight, because when children refused to answer the questions on the intelligence tests, White psychologists interpreted their silence as a lack of intelligence.

In the 1930s, as more Black students were admitted to graduate schools, many focused their graduate school research on demonstrating that Black people were in no way less intelligent than Whites (Guthrie, 1998). For example, in 1935, Herman Canady's thesis for his Masters' degree followed Horace Mann Bond's lead by showing that Black children scored higher when the Stanford-Binet scale was administered by Black psychologists compared to White psychologists (Guthrie, 1998). However, it was not until 1979, that a team of Black psychologists and civil rights lawyers brought the issue to federal court in California. The state of California then ruled that intelligence tests – like the Stanford Binet, were racially and culturally biased, and banned their use for Black children. This helped to reduce the high number of Black students who were being misdirected into special education classes on the basis of their scores (Guthrie, 1998). Improved versions of the Stanford-Binet and other intelligence tests are still widely used today in many other US States (Gibbons & Warne, 2019) and Black children are still being inappropriately placed in special education at unjustifiably high rates (US Department of Education, 2016).

However, IQ testing is not the only reason for this discriminatory practice. Although 50% of students in the United States are BIPOC, most teachers are White. White teachers disproportionately classify Black children (especially boys) as disruptive and aggressive, even when engaging in the same behaviors as their White peers (Association of Psychological Science, 2014).

In response to this problem, there are increasing numbers of initiatives designed to improve teacher education and the outcomes of their BIPOC students. For example, in 2012, New York City launched a program called the Expanded Success Initiative in 40 schools, which impacted 20,000 students. Teachers were trained to provide culturally relevant instruction that was more inclusive of the experiences of BIPOC students, especially males, and which encouraged social activism. Relationships between teachers and students were improved through extra-curricular experiences, and a sense of belonging to the school was also fostered through peer-tutoring. Students were given more academically rigorous instruction, and were helped with college preparation, including visits to colleges and help with applications. Students who participated in the programs felt that they were treated more fairly and felt more connected to their schools than students in schools who did not participate in the initiative (Villavicencio, 2023).

MODERN-DAY INTELLIGENCE TESTS

Modern-day IQ tests are heavily influenced by the work of David Wechsler. In 1939, Wechsler created a new kind of intelligence test designed to tap a wider range of abilities than the original Stanford-Binet scale. After he graduated with a Bachelor's degree from City College, CUNY, Wechsler became a private in the army where he administered some of the intelligence tests developed by Terman and his team (Boake, 2002; Saxon, 1981). Wechsler was struck by the fact that many army recruits, who functioned perfectly well in

their skilled civilian jobs, scored way below average on the army intelligence tests (Boake, 2002). After the army, Wechsler earned a doctoral degree from Columbia University and became the chief psychologist at Bellevue Hospital in New York City. He became increasingly frustrated with the limitations of the Stanford-Binet scale and so decided to create a new measure of intelligence (Boake, 2002). Wechsler was particularly critical of the Stanford-Binet scale's heavy reliance on verbal ability and the use of a single IQ score to measure general intelligence (Biswas-Diener, 2018). To try and address these issues, Wechsler created the Wechsler-Bellevue test, which was designed to test a wider range of cognitive abilities (Saxon, 1981). In 1955, Wechsler's initial test was revised and renamed as the Wechsler Adult Intelligence Scale (WAIS; Saxon, 1981). The WAIS assessed people's ability to remember, compute, understand language, reason well, and process information quickly (Biswas-Diener, 2018). Wechsler developed similar tests for children: The Wechsler Intelligence Scale for Children (WISC) in 1949, and the Wechsler Pre-School and Primary Scale of Intelligence (WPPSI) in 1967 (Saxon, 1981). Wechsler scales became more popular measures of intelligence than the Stanford Binet scale, in part, because they provided separate assessments of verbal and non-verbal skills. These scores were then summed together to provide a measure of IQ. However, just like Terman, Wechsler standardized all of these tests, using only White participants, which was a major shortcoming (Boake, 2002).

Since the 1930s, the various Wechsler and Stanford-Binet scales have been revised multiple times and newer versions are widely used today (Gibbons & Warne, 2019). The Stanford-Binet and Wechsler scales have also now been standardized using more diverse samples (Boake, 2002). The current versions of the Wechsler and Stanford-Binet scales (and serval other modern cognitive aptitude tests) have been greatly influenced by the Cattell-Horn-Carroll (CHC) theory of cognitive abilities (Schneider & McGrew, 2018). This comprehensive theory rests on the idea

that general intelligence can be subdivided into at least 9-10 broad categories of cognitive abilities, some of which come under the general heading of fluid intelligence while others rely on crystallized intelligence (Cattell, 1963). These domains include quantitative knowledge; reading and writing; comprehension and general knowledge; fluid reasoning – the ability to solve novel problems; short-term memory, long-term memory, visual processing, auditory processing and processing speed. Each of these domains can be further subdivided into other abilities. Both the current version of the WAIS (WAIS IV) and the Stanford Binet (Stanford-Binet 5) aim to tap many of these abilities. The WAIS IV consists of 10 subtests, which yields four separate factor scores: perceptual reasoning, verbal comprehension, working memory and processing speed. The Stanford-Binet 5 yields five different factor measures: fluid reasoning, general knowledge, quantitative reasoning, visual-spatial reasoning, and working memory. However, despite these refinements, all of these tests were developed by White psychologists and reflect their cultural values (Washington et al., 2016). To underscore this point, in 1972, Black psychologist and professor, Dr. Robert Williams, designed the Black Intelligence Test of Cultural Homogeneity (B.I.T.C.H.), which focused on testing knowledge about the lives and culture of Black people in the US. Williams coined the term “ebonics” to describe the use of African American Vernacular English (AAVE) that he used in his test questions. As predicted, Black children and college students did much better on this test than their White counterparts (Williams, 1975).

Links to Learning

Learn more about the different subtests of the WAIS IV here

Watch this interview of Dr. Williams explaining why he developed the B.I.T.C.H. to show that culturally-specific language affects performance on tests designed to measure intelligence.

Researchers have underscored that many people misunderstand AAVE to be a form of broken English and a sign of low intelligence (Edwards, 2008). However, AAVE is a dialect of English, with roots in Africa. Just like any other language, it has a lexicon and a grammar, but these are different to those in standard English. Using AAVE does not indicate a lack of intelligence. Indeed, some researchers have argued that AAVE speakers who are also fluent in standard English are likely to have the same cognitive advantages as other bilinguals who switch between two languages (Edwards, 2008; Washington et al., 2023). However, most AAVE-speaking children, particularly those from working-class backgrounds, are not given the same educational support as other bilinguals when learning standard American English. This has been shown to impact reading and writing development (Washington et al., 2023). Julie Washington argues that many AAVE-speaking children are discouraged in school settings by teachers who lack cultural sensitivity and knowledge of AAVE. She has shown that culturally appropriate reading instruction improves the abilities of working class African American children who use AAVE (Washington et al., 2023).

STEREOTYPE THREAT

We have already discussed some of the reasons why people from marginalized backgrounds may do poorly on IQ tests or other tests of cognitive aptitudes. In addition, people are vulnerable to a phenomenon known as stereotype threat. This is when people internalize negative stereotypes about themselves, which then negatively affects their performance (Stangor & Walinga, 2014). In 1995, two psychologists, Claude Steele and Joshua Aronson tested the hypothesis that the differences in performance on IQ tests between Blacks and Whites might be due to the activation of negative stereotypes about intelligence in Blacks (Steele & Aronson, 1995). They found that Black university students performed worse on standardized test questions when they were told that their verbal ability was being tested (thus the stereotype was activated). In contrast, their performance was not affected when the test was described simply as problem solving. They also found that when students were asked to write down their race before they took a math test, Black students performed more poorly than they had on prior exams, whereas White students were unaffected (Stangor & Walinga, 2014). Stereotype threat can also negatively impact women's performance on math tests and exams (Marx & Roman, 2002; McIntyre et al., 2003). Importantly, stereotype threat is not absolute; we can reduce the feelings of self-doubt, by thinking about the positive characteristics that we (or a group with which we identify) possess (Johns et al., 2005; Marx & Roman, 2002; McIntyre et al., 2003). Just knowing that stereotype threat exists can help to alleviate its negative impact (Johns et al., 2005).

MEASURING INTELLIGENCE BEYOND IQ TESTS

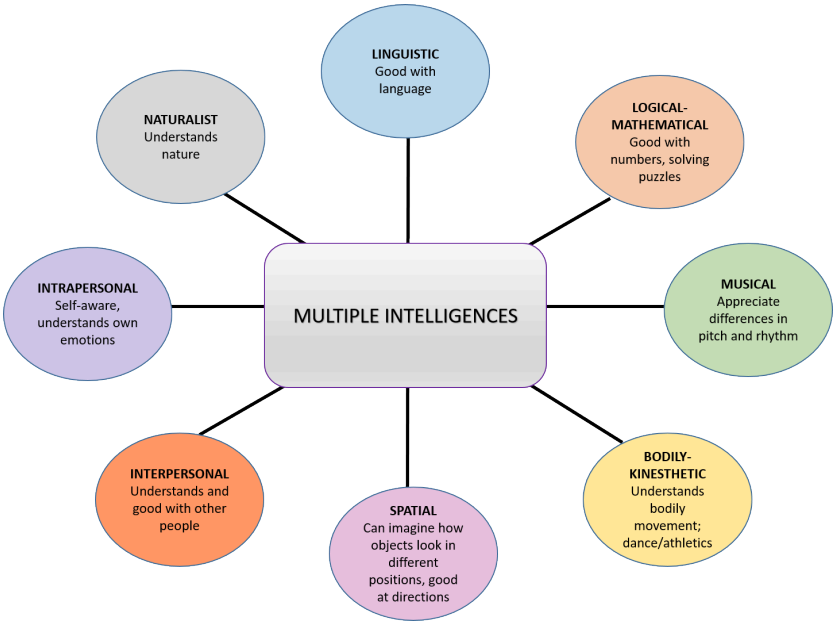

Figure 8.7. Gardner's Theory of Multiple Intelligences

Many psychologists have argued that intelligence is not adequately measured by IQ tests. For example, Howard Gardner, a Harvard psychologist, believes that intelligence is extremely multi-faceted and should be broadly defined across multiple domains. In 1983, Gardner published a book explaining his Multiple Intelligences Theory (see Figure 8.7). In Gardner's theory, each person possesses at least eight different kinds of intelligence. These include linguistic, logical-mathematical, musical, bodily kinesthetic, spatial, interpersonal, intrapersonal intelligence, and naturalistic intelligences (Gardner, 1983). Among cognitive psychologists, Gardner's theory has been criticized for lacking empirical evidence. However, educators continue to study and use Gardner's theory to improve learning experiences for students.

Robert Sternberg agreed with Gardener that there are multiple different types of intelligence. Sternberg developed a theory of intelligence, which he called the Triarchic theory of Intelligence because it conceives of intelligence as being comprised of three kinds (Sternberg, 1988): practical, creative, and analytical intelligence (Figure 8.8).

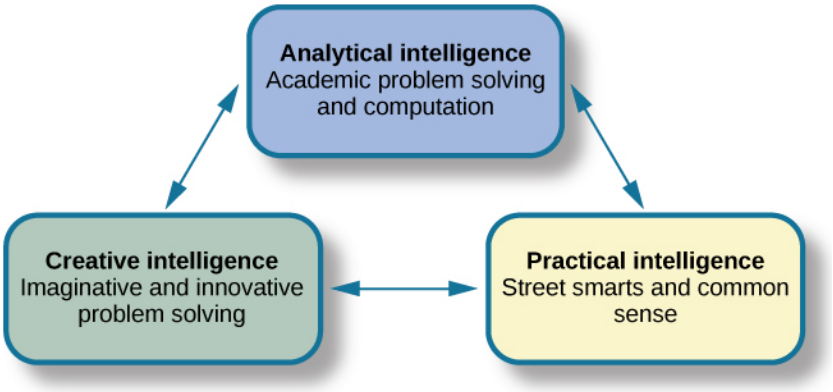

Figure 8.8. Sternberg's theory identifies three types of intelligence: practical, creative, and analytical.

Practical intelligence is sometimes compared to "street smarts." Being practical means you find solutions that work in your everyday life by applying knowledge based on your experiences. Analytical intelligence is closely aligned with academic problem solving and computational abilities. Creative intelligence is marked by inventing or imagining a solution to a problem or situation. Sternberg expected that someone's background would have different effects on the three types of intelligence. Sternberg grouped student applicants to a summer program at Yale University according to their scores on the Triarchic Intelligence test. He found that more White middle-class students from well-resourced schools scored high on analytical intelligence, but there was much more diversity among people with high scores on practical intelligence or creative intelligence, in terms of their race/ethnicity and SES (Sternberg,

1999). Sternberg has worked with colleges to make admissions criteria more fair and found that creative intelligence was a much better predictor of first-year grade-point average than SAT scores (Sternberg & The Rainbow Project Collaborators, 2006). Importantly, Sternberg (1995) has also shown that practical intelligence may be a better predictor of career success than analytical intelligence. Sternberg has also pointed out that intelligence is subjective and culturally specific (Sternberg, 1999). Some cultures value cognitive intelligence over social intelligence, but the opposite is true in other cultures.

Gardner's inter- and intrapersonal intelligences have often been combined by other psychologists into a single type: emotional intelligence. Emotional intelligence encompasses the ability to identify and understand the emotions of yourself and others, show empathy, understand social relationships and cues, and regulate your own emotions and respond in culturally appropriate ways (Mayer & Salovey, 1997; Parker et al., 2009). People with high emotional intelligence typically have well-developed social skills. Studies have shown a link between emotional intelligence and success, including job performance (Goleman, 1995; Lopes et al., 2006). However, emotional intelligence has been widely debated, with researchers pointing out that it is a concept that is difficult to measure and study empirically (Locke, 2005; Mayer et al., 2004).

WHY ARE IQ TESTS STILL USED?

As discussed in this chapter, there are many reasons as to why, on average, BIPOC people, especially African Americans and Latinx people, do less well than Whites on IQ tests and other standardized tests of cognitive ability, such as the SAT. In particular, IQ and SAT tests continue to use content that predominantly reflects White middle class values, not surprisingly, these tests lack predictive power about job and academic success for people of color (Berry et

al., 2014; Helms et al., 2014). In particular, Au (2020) has argued that high-stakes standardized tests promote whiteness and undermine multicultural education. So, why and how are IQ and other cognitive skills test still used? One reason for a lack of change is that alternative conceptualizations of intelligence (other than IQ) are relatively hard to measure. Robert Sternberg has worked with various colleges and Universities to try and improve admissions criteria, but the research program is small and tests are time consuming to score (Sternberg et al., 2020; Sternberg and the Rainbow Project Collaborators, 2006). However, one of the very few benefits of the recent Covid-19 pandemic was that SAT testing was suspended and colleges and universities were unable to use the scores in their admissions decisions. SAT testing has now resumed, but a large number of institutions, including many Ivy leagues, are now questioning its value. Many colleges either no longer require SAT/ACT test results, or have decided to make testing optional. This change increased the diversity of admitted students at both Cornell and the University of Chicago (Einhorn, 2022). *What do you think about this decision? What metrics do you think that colleges should use for admissions decisions?*

The most common use of IQ tests is in educational settings to identify children who may need extra assistance in schools because of a learning disability. IQ tests, like the Stanford-Binet scale, are also used to identify gifted children for selective academic programs. Neuropsychologists also use IQ tests to identify cognitive issues following a head injury, stroke, or the onset of illnesses, such as dementia. In recognition of racial and socioeconomic differences in scores, many IQ tests now have different norms for different groups of people. However, some neuropsychologists have acknowledged that making adjustments to scores is a poor substitute for culture-fair testing (Byrd & Rivera-Mindt, 2022). Given the long entrenched history of many of these tests it is unlikely that change will occur rapidly, especially given that the majority of psychologists are White, and may lack the

cultural awareness needed to spur transformative change (Hiermeier & Verity, 2022). However, psychologists of color like Dr. Desiree Byrd of Queens College, CUNY, are leading the way to make change.

PRACTICE CAN INCREASE INTELLIGENCE (AND IQ SCORES)

Given that IQ and other cognitive tests are likely to be around for a long time, it is helpful to know that cognitive abilities, just like other abilities, can improve with practice. When we learn and practice skills that help us to do better in school – we also do better on IQ tests (Nisbett et al., 2012). Performing any task (physical or mental) causes new connections between neurons to grow in the brain, these connections grow stronger with subsequent practice, making it easier and easier each time we repeat a task.

Link to Learning

Check out this video to remind yourself about what is happening in your brain when you learn something new.

MINDSET CAN INCREASE INTELLIGENCE

Making mistakes and figuring out what to do differently next time is an important part of learning, though admittedly, sometimes it can be frustrating. The way a person thinks about making mistakes

and their own intelligence is very important because it predicts how they perform in school and on IQ tests (Biswas-Diener, 2018; Dweck, 1986). Dr. Carol Dweck has focused her life's work on looking at the differences between children who perform well in school and those who do not. Dweck found that children who believe that their abilities (including intelligence) are fixed and cannot change, tend to underperform. In contrast, kids who have a "growth mindset" and believe that intelligence is changeable and evolving, are able to handle failure and perform better in school (Dwek, 1986; Biswas-Diener, 2018).

Dig Deeper

Assess your mindset by taking this assessment based on Dweck's work. What did you learn about yourself? Can you use the information in this book about neuroplasticity to shift toward a stronger growth mindset?

Link to Learning Watch this video that compares different theories of intelligence to learn more.

Chapter 8. Summary

Language is a communication system that has both a lexicon and a system of grammar. Language acquisition occurs naturally and effortlessly during the early stages of life, and this acquisition occurs in a predictable sequence for individuals around the world. Language has a strong influence on identity and thought, and the concept of how language may influence cognition remains an area of study and debate in psychology.

Intelligence is a controversial topic in Psychology. The field of eugenics spread misinformation about intelligence and race/ethnicity and gender, which contributed to the long-lasting oppression of BIPOC people. Definitions of intelligence vary across cultures. In the West, intelligence is conceptualized as a complex characteristic of cognition. Various theories have been developed to explain what intelligence is and how it works. Sternberg generated his Triarchic theory of intelligence, whereas Gardner posits that intelligence is comprised of many factors. Still others focus on the importance of emotional intelligence.

Chapter 8. Key Terms

analytical intelligence
aligned with academic problem solving and computations
ancestral language
language spoken by our ancestors
bicultural
belonging to two cultures
bilingualism
ability to speak two languages fluently
cognition
thinking, including perception, learning, problem solving, judgment, and memory
cognitive psychology
field of psychology dedicated to studying every aspect of how people think
creative intelligence
ability to produce new products, ideas, or inventing a new, novel solution to a problem
crystallized intelligence
characterized by acquired knowledge and the ability to retrieve it
cultural identity
shared beliefs, norms, behaviors, and values with people of same culture
cultural intelligence

ability with which people can understand and relate to those in another culture

emotional intelligence

ability to understand emotions and motivations in yourself and others

eugenics

scientifically erroneous and immoral *theory* designed to manipulate gene pool, centered on incorrect belief that racial differences in intelligence were genetic

executive function

mental processes that enable us to plan, focus attention, remember, and juggle multiple tasks

fluid intelligence

ability to see complex relationships and solve problems

grammar

set of rules that are used to convey meaning through the use of a lexicon

heritage language

minority language (either immigrant or indigenous) learned by its speakers at home as children

intelligence quotient

(also, IQ) score on a test designed to measure intelligence

intercultural sensitivity

understanding and appreciation of cross-cultural differences

language

communication system that involves using words to transmit information from one individual to another

lexicon

the words of a given language

linguistic racism

acts of racism are perpetuated against individuals on the basis of their language use.

metaphorical meaning

something used symbolically to represent something else, e.g., a color may mean something other than an actual color

monolingual

person who speaks only one language

morpheme

smallest unit of language that conveys some type of meaning

Multiple Intelligences Theory

Gardner's theory that each person possesses at least eight types of intelligence

multilingualism

ability to speak two or more languages fluently

norming

administering a test to a large population so data can be collected to reference the normal scores for a population and its groups

phoneme

basic sound unit of a given language

practical intelligence

aka "street smarts"

pragmatic language

use of appropriate communication in social situations

problem-solving strategy

method for solving problems

redlining

discriminatory practice that designates certain geographic areas as high risk (based on race/ethnicity); residents are denied services such as mortgages, insurance loans, and other financial services

representative sample

subset of the population that accurately represents the general population

semantics

process by which we derive meaning from morphemes and words

standard deviation

measure of variability that describes the difference between a set of scores and their mean

standardization

method of testing in which administration, scoring, and interpretation of results are consistent

stereotype threat

when people internalize negative stereotypes about themselves, which then negatively affects their performance

syntax

manner by which words are organized into sentences

Triarchic theory of intelligence

Sternberg's theory of intelligence; three facets of intelligence: practical, creative, and analytical

White supremacy

belief that white people are superior to other races and should therefore dominate society, justifying the oppression of other racial and ethnic groups.

Chapter 8. Personal Application Questions

- Consider the components of language mentioned in the text, such as lexicon, grammar, phonemes, and morphemes. Can you identify some examples from your native language that demonstrate these components? How do these components contribute to effective communication?
- Think about the role of intonation and prosody in language. How does the use of emphasis, tone of voice, and pitch variations influence the meaning of a spoken sentence? Can you recall any personal experiences where the tone of voice or intonation had a significant impact on communication?
- Explore the cultural variations in language practices mentioned in the text, such as expressing gratitude and making requests. Reflect on your own cultural background or experiences with different cultures. How do cultural norms and language influence the ways in which gratitude is expressed or requests are made or refused in your culture?
- Consider the benefits of bilingualism and multilingualism discussed in the text. Are you fluent in multiple

languages? If yes, have you noticed any cognitive or other advantages associated with speaking multiple languages in your own life?

- If you only speak one language, what might be some of the best ways to learn about how to use another language effectively.
- Reflect on the connection between language and identity. How does your use of language relate to your national or cultural identity? Have you ever experienced shifts in attitudes or behaviors when switching between languages? Can you provide personal examples that demonstrate the influence of language on your sense of self?
- Explore the influence of language on thinking. Have you ever noticed differences in your thought processes or self-evaluation when using different languages? Can you recall situations where language context affected your perception of performance or influenced your thinking patterns?
- How do your cultural values and beliefs influence your conceptualization of intelligence? In what ways do you emphasize practical skills, adaptability, or cognitive abilities in your understanding of intelligence?
- In your own educational experiences, how have cognitive abilities related to academic achievement, such as rapid information processing, memory, language skills, mathematical aptitude, and logical reasoning, been emphasized?
- Have you ever encountered situations where intelligence tests or cognitive aptitude tests felt culturally unfair? How do you think such tests can exclude individuals from marginalized groups from educational and career

opportunities?

- What influence do you think emotional intelligence plays in your personal life?
- Do you believe your level of intelligence was improved because of the experiences in your childhood environment? Why or why not?
- Reflecting on your own intellectual journey, can you describe a specific instance where you encountered a challenge or setback that tested your intellectual abilities? How did you approach the situation, and what mindset did you adopt to overcome it? How did this experience shape your understanding of intelligence and the importance of mindset in achieving personal growth and success?

Chapter 8. Critical Thinking Questions

- How does language shape our perception of the world and influence our thoughts and behaviors?
- What impact does bilingualism or multilingualism have on cognitive processes and brain development?
- Can language limit our ability to think or express certain concepts? Provide examples to support your answer.
- How could grammatical errors actually be indicative that children are learning language well?
- Describe a situation in which you would need to use practical intelligence.
- Describe a situation in which emotional intelligence would help you communicate better.
- How do different theories of intelligence shape our understanding of human capabilities and potential?
- What are the potential biases and limitations of intelligence tests in accurately assessing an individual's intellectual abilities?
- How does the concept of intelligence vary across different cultures and societies? How does cultural context

influence our understanding of intelligence?

- In what ways can stereotypes and societal expectations about intelligence impact individuals' self-perception and performance on cognitive tasks?
- How do environmental factors, such as access to quality education and socio-economic status, influence the development and expression of intelligence?
- What role does genetics play in determining intelligence? How does the interaction between genetic and environmental factors affect intelligence?
- Can intelligence be accurately measured by a single IQ score? What other factors should be considered in assessing an individual's intellectual abilities?
- What are the potential consequences, both positive and negative, of emphasizing intelligence as a measure of human worth and potential?
- How do different domains of intelligence, such as emotional intelligence or creative intelligence, contribute to overall human intelligence?

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This module is heavily influenced by Dr. Robert Guthrie's work:

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Learn more about Dr. Guthrie here https://www.apa.org/monitor/nov01/american

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LIFESPAN DEVELOPMENT

How have you changed since childhood? How are you the same? What will your life be like 25 years from now? Fifty years from now? Lifespan development studies how you change as well as how you remain the same over the course of your life. (credit: modification of work by Giles Cook)

Welcome to the story of your life. In this chapter we explore the fascinating tale of how you have grown and developed into the person you are today. We also look at some ideas about who you

will grow into tomorrow. Yours is a story of lifespan development, from the start of life to the end.

The process of human growth and development is more obvious in infancy and childhood, yet your development is happening this moment and will continue, minute by minute, for the rest of your life. Who you are today and who you will be in the future depends on a blend of genetics, environment, culture, relationships, and more, as you continue through each phase of life. You have experienced firsthand much of what is discussed in this chapter. Now consider what psychological science has to say about your physical, cognitive, and psychosocial development, from the womb to the tomb.

Learning Objectives

By the end of this section, you will be able to:

- Define and distinguish between the three domains of development: physical, cognitive and psychosocial
- Discuss the normative approach to development
- Understand the three major issues in development: continuity and discontinuity, one common course of development or many unique courses of development, and nature versus nurture

> My heart leaps up when I behold
> A rainbow in the sky:
> So was it when my life began;
> So is it now I am a man;
> So be it when I shall grow old,
> Or let me die!
> The Child is father of the Man;
> I could wish my days to be
> Bound each to each by natural piety. (Wordsworth, 1802)

In this poem, William Wordsworth writes, "the child is father of the man." What does this seemingly incongruous statement mean, and what does it have to do with lifespan development? Wordsworth might be suggesting that the person he is as an adult depends largely on the experiences he had in childhood. Consider the following questions: To what extent is the adult you are today influenced by the child you once were? To what extent is a child fundamentally different from the adult he grows up to be?

These are the types of questions developmental psychologists try to answer, by studying how humans change and grow from conception through childhood, adolescence, adulthood, and death. They view development as a lifelong process that can be studied scientifically across three developmental domains—physical, cognitive, and psychosocial development. Physical development involves growth and changes in the body and brain, the senses, motor skills, and health and wellness. Cognitive development involves learning, attention, memory, language, thinking, reasoning, and creativity. Psychosocial development involves emotions, personality, and social relationships. We refer to these domains throughout the chapter.

Research Methods in Developmental Psychology

You've learned about a variety of research methods used by psychologists. Developmental psychologists use many of these approaches in order to better understand how individuals change mentally and physically over time. These methods include naturalistic observations, case studies, surveys, and experiments, among others.

Naturalistic observations involve observing behavior in its natural context. A developmental psychologist might observe how

children behave on a playground, at a daycare center, or in the child's own home. While this research approach provides a glimpse into how children behave in their natural settings, researchers have very little control over the types and/or frequencies of displayed behavior.

In a case study, developmental psychologists collect a great deal of information from one individual in order to better understand physical and psychological changes over the lifespan. This particular approach is an excellent way to better understand individuals, who are exceptional in some way, but it is especially prone to researcher bias in interpretation, and it is difficult to generalize conclusions to the larger population.

In one classic example of this research method being applied to a study of lifespan development Sigmund Freud analyzed the development of a child known as "Little Hans" (Freud, 1909/1949). Freud's findings helped inform his theories of psychosexual development in children, which you will learn about later in this chapter. Little Genie, the subject of a case study discussed in the chapter on thinking and intelligence, provides another example of how psychologists examine developmental milestones through detailed research on a single individual. In Genie's case, her neglectful and abusive upbringing led to her being unable to speak until, at age 13, she was removed from that harmful environment. As she learned to use language, psychologists were able to compare how her language acquisition abilities differed when occurring in her late-stage development compared to the typical acquisition of those skills during the ages of infancy through early childhood (Fromkin, Krashen, Curtiss, Rigler, & Rigler, 1974; Curtiss, 1981).

The survey method asks individuals to self-report important information about their thoughts, experiences, and beliefs. This particular method can provide large amounts of information in relatively short amounts of time; however, validity of data collected in this way relies on honest self-reporting, and the data is relatively

shallow when compared to the depth of information collected in a case study.

Experiments involve significant control over extraneous variables and manipulation of the independent variable. As such, experimental research allows developmental psychologists to make causal statements about certain variables that are important for the developmental process. Because experimental research must occur in a controlled environment, researchers must be cautious about whether behaviors observed in the laboratory translate to an individual's natural environment.

Later in this chapter, you will learn about several experiments in which toddlers and young children observe scenes or actions so that researchers can determine at what age specific cognitive abilities develop. For example, children may observe a quantity of liquid poured from a short, fat glass into a tall, skinny glass. As the experimenters question the children about what occurred, the subjects' answers help psychologists understand at what age a child begins to comprehend that the volume of liquid remained the same although the shapes of the containers differs.

Across these three domains—physical, cognitive, and psychosocial—the normative approach to development is also discussed. This approach asks, "What is normal development?" In the early decades of the 20th century, normative psychologists studied large numbers of children at various ages to determine norms (i.e., average ages) of when most children reach specific developmental milestones in each of the three domains (Gesell, 1933, 1939, 1940; Gesell & Ilg, 1946; Hall, 1904). Although children develop at slightly different rates, we can use these age-related averages as general guidelines to compare children with same-age peers to determine the approximate ages they should reach specific normative events called developmental milestones (e.g., crawling, walking, writing, dressing, naming colors, speaking in sentences, and starting puberty).

Not all normative events are universal, meaning they are not

experienced by all individuals across all cultures. Biological milestones, such as puberty, tend to be universal, but social milestones, such as the age when children begin formal schooling, are not necessarily universal; instead, they affect most individuals in a particular culture (Gesell & Ilg, 1946). For example, in developed countries children begin school around 5 or 6 years old, but in developing countries, like Nigeria, children often enter school at an advanced age, if at all (Huebler, 2005; United Nations Educational, Scientific, and Cultural Organization [UNESCO], 2013).

To better understand the normative approach, imagine two new mothers, Louisa and Kimberly, who are close friends and have children around the same age. Louisa's daughter is 14 months old, and Kimberly's son is 12 months old. According to the normative approach, the average age a child starts to walk is 12 months. However, at 14 months Louisa's daughter still isn't walking. She tells Kimberly she is worried that something might be wrong with her baby. Kimberly is surprised because her son started walking when he was only 10 months old. Should Louisa be worried? Should she be concerned if her daughter is not walking by 15 months or 18 months?

The Centers for Disease Control and Prevention (CDC) describes the developmental milestones for children from 2 months through 5 years old. After reviewing the information, take this quiz to see how well you recall what you've learned. If you are a parent with concerns about your child's development, contact your pediatrician: CDC's Developmental

Milestones.

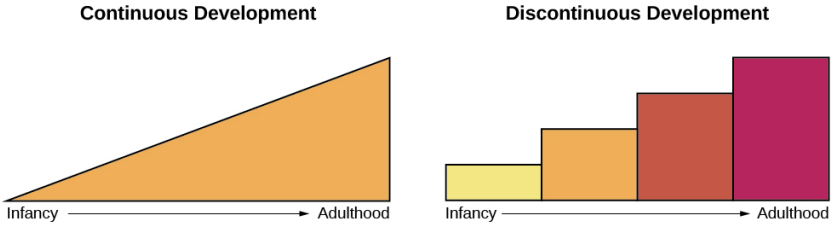
ISSUES IN DEVELOPMENTAL PSYCHOLOGY

There are many different theoretical approaches regarding human development. As we evaluate them in this chapter, recall that developmental psychology focuses on how people change, and keep in mind that all the approaches that we present in this chapter address questions of change: Is the change smooth or uneven (continuous versus discontinuous)? Is this pattern of change the same for everyone, or are there many different patterns of change (one course of development versus many courses)? How do genetics and environment interact to influence development (nature versus nurture)?

IS DEVELOPMENT CONTINUOUS OR DISCONTINUOUS?

Continuous development views development as a cumulative process, gradually improving on existing skills. With this type of development, there is gradual change. Consider, for example, a child's physical growth: adding inches to her height year by year. In contrast, theorists who view development as discontinuous believe that development takes place in unique stages: It occurs at specific times or ages. With this type of development, the change is more sudden, such as an infant's ability to conceive object permanence. The concept of continuous development can be visualized as a smooth slope of progression, whereas discontinuous development

sees growth in more discrete stages.



IS THERE ONE COURSE OF DEVELOPMENT OR MANY?

Is development essentially the same, or universal, for all children (i.e., there is one course of development) or does development follow a different course for each child, depending on the child's specific genetics and environment (i.e., there are many courses of development)? Do people across the world share more similarities or more differences in their development? How much do culture and genetics influence a child's behavior?

Stage theories hold that the sequence of development is universal. For example, in cross-cultural studies of language development, children from around the world reach language milestones in a similar sequence (Gleitman & Newport, 1995). Infants in all cultures coo before they babble. They begin babbling at about the same age and utter their first word around 12 months old. Yet we live in diverse contexts that have a unique effect on each of us. For example, researchers once believed that motor development follows one course for all children regardless of culture. However, child care practices vary by culture, and different practices have been found to accelerate or inhibit achievement of developmental milestones such as sitting, crawling, and walking (Karasik, Adolph, Tamis-LeMonda, & Bornstein, 2010).

For instance, let's look at the Aché society in Paraguay. They spend a significant amount of time foraging in forests. While foraging, Aché mothers carry their young children, rarely putting them down in order to protect them from getting hurt in the forest.

Consequently, their children walk much later: They walk around 23–25 months old, in comparison to infants in Western cultures who begin to walk around 12 months old. However, as Aché children become older, they are allowed more freedom to move about, and by about age 9, their motor skills surpass those of U.S. children of the same age: Aché children are able to climb trees up to 25 feet tall and use machetes to chop their way through the forest (Kaplan & Dove, 1987). As you can see, our development is influenced by multiple contexts, so the timing of basic motor functions may vary across cultures. However, the functions themselves are present in all societies.

(a)

(b)

All children across the world love to play. Whether in (a) Florida or (b) South Africa, children enjoy exploring sand, sunshine, and the sea. (credit a: modification of work by "Visit St. Pete/Clearwater"/Flickr; credit b: modification of work by "stringer_bel"/Flickr)

HOW DO NATURE AND NURTURE INFLUENCE DEVELOPMENT?

Are we who we are because of nature (biology and genetics), or are we who we are because of nurture (our environment and culture)? This longstanding question is known in psychology as the nature versus nurture debate. It seeks to understand how our personalities and traits are the product of our genetic makeup and biological factors, and how they are shaped by our environment,

including our parents, peers, and culture. For instance, why do biological children sometimes act like their parents—is it because of genetics or because of early childhood environment and what the child has learned from the parents? What about children who are adopted—are they more like their biological families or more like their adoptive families? And how can siblings from the same family be so different?

We are all born with specific genetic traits inherited from our parents, such as eye color, height, and certain personality traits. Beyond our basic genotype, however, there is a deep interaction between our genes and our environment: Our unique experiences in our environment influence whether and how particular traits are expressed, and at the same time, our genes influence how we interact with our environment (Diamond, 2009; Lobo, 2008). This chapter will show that there is a reciprocal interaction between nature and nurture as they both shape who we become, but the debate continues as to the relative contributions of each.

The Achievement Gap: How Does Socioeconomic Status Affect

Development?

The achievement gap refers to the persistent difference in grades, test scores, and graduation rates that exist among students of different ethnicities, races, and—in certain subjects—sexes (Winerman, 2011). Research suggests that these achievement gaps are strongly influenced by differences in socioeconomic factors that exist among the families of these children. While the researchers acknowledge that programs aimed at reducing such socioeconomic discrepancies would likely aid in equalizing the aptitude and performance of children from different backgrounds, they recognize that such large-scale interventions would be difficult to achieve. Therefore, it is recommended that programs aimed at fostering aptitude and achievement among disadvantaged children may be the best option for dealing with issues related to academic achievement gaps (Duncan & Magnuson, 2005).

Low-income children perform significantly more poorly than their middle- and high-income peers on a number of educational variables: They have significantly lower standardized test scores, graduation rates, and college entrance rates, and they have much higher school dropout rates. There have been attempts to correct the achievement gap through state and federal legislation, but what if the problems start before the children even enter school?

Psychologists Betty Hart and Todd Risley (2006) spent their careers looking at early language ability and progression of children in various income levels. In one longitudinal study, they found that although all the parents in the study engaged and interacted with their children, middle- and high-income parents interacted with their children differently than low-income parents. After analyzing 1,300 hours of parent-child interactions, the researchers found that middle- and high-income parents talk to their children significantly more, starting when the children are

infants. By 3 years old, high-income children knew almost double the number of words known by their low-income counterparts, and they had heard an estimated total of 30 million more words than the low-income counterparts (Hart & Risley, 2003). And the gaps only become more pronounced. Before entering kindergarten, high-income children score 60% higher on achievement tests than their low-income peers (Lee & Burkam, 2002).

There are solutions to this problem. At the University of Chicago, experts are working with low-income families, visiting them at their homes, and encouraging them to speak more to their children on a daily and hourly basis. Other experts are designing preschools in which students from diverse economic backgrounds are placed in the same classroom. In this research, low-income children made significant gains in their language development, likely as a result of attending the specialized preschool (Schechter & Byeb, 2007). What other methods or interventions could be used to decrease the achievement gap? What types of activities could be implemented to help the children of your community or a neighboring community?

SUMMARY

Lifespan development explores how we change and grow from conception to death. This field of psychology is studied by developmental psychologists. They view development as a lifelong process that can be studied scientifically across three developmental domains: physical, cognitive development, and psychosocial. There are several theories of development that focus on the following issues: whether development is continuous or discontinuous, whether development follows one course or many, and the relative influence of nature versus nurture on development.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=124#h5p-125

CRITICAL THINKING QUESTIONS

Describe the nature versus nurture controversy, and give an example of a trait and how it might be

influenced by each?

The nature versus nurture controversy seeks to understand whether our personalities and traits are the product of our genetic makeup and biological factors, or whether they are shaped by our environment, which includes such things as our parents, peers, and culture. Today, psychologists agree that both nature and nurture interact to shape who we become, but the debate over the relative contributions of each continues. An example would be a child learning to walk: Nature influences when the physical ability occurs, but culture can influence when a child masters this skill, as in Aché culture.

Compare and contrast continuous and discontinuous development.

Continuous development sees our development as a cumulative process: Changes are gradual. On the other hand, discontinuous development sees our development as taking place in specific steps or stages: Changes are sudden.

Why should developmental milestones only be used as a general guideline for normal child development?

Children develop at different rates. For example, some children may walk and talk as early as 8 months old, while others may not do so until well after their

first birthday. Each child's unique contexts will influence when he reaches these milestones.

PERSONAL APPLICATION QUESTIONS

How are you different today from the person you were at 6 years old? What about at 16 years old? How are you the same as the person you were at those ages?

Your 3-year-old daughter is not yet potty trained. Based on what you know about the normative approach, should you be concerned? Why or why not?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

https://pressbooks.cuny.edu/jjcpsy101/?p=124#h5p-126

Learning Objectives

By the end of this section, you will be able to:

- Discuss Freud's theory of psychosexual development
- Describe the major tasks of child and adult psychosocial development according to Erikson
- Discuss Piaget's view of cognitive development and apply the stages to understanding childhood cognition
- Describe Kohlberg's theory of moral development

There are many theories regarding how babies and children grow and develop into happy, healthy adults. We explore several of these theories in this section.

PSYCHOSEXUAL THEORY OF DEVELOPMENT

Sigmund Freud (1856–1939) believed that personality develops during early childhood. For Freud, childhood experiences shape our personalities and behavior as adults. Freud viewed development as discontinuous; he believed that each of us must pass through a series of stages during childhood, and that if we lack proper nurturance and parenting during a stage, we may become stuck, or fixated, in that stage. Freud's stages are called the stages of psychosexual development. According to Freud, children's pleasure-seeking urges are focused on a different area of the body,

called an erogenous zone, at each of the five stages of development: oral, anal, phallic, latency, and genital.

While most of Freud's ideas have not found support in modern research, we cannot discount the contributions that Freud has made to the field of psychology. Psychologists today dispute Freud's psychosexual stages as a legitimate explanation for how one's personality develops, but what we can take away from Freud's theory is that personality is shaped, in some part, by experiences we have in childhood. These stages are discussed in detail in the chapter on personality.

PSYCHOSOCIAL THEORY OF DEVELOPMENT

Erik Erikson (1902–1994), another stage theorist, took Freud's theory and modified it as psychosocial theory. Erikson's psychosocial development theory emphasizes the social nature of our development rather than its sexual nature. While Freud believed that personality is shaped only in childhood, Erikson proposed that personality development takes place all through the lifespan. Erikson suggested that how we interact with others is what affects our sense of self, or what he called the ego identity.

Erik Erikson proposed the psychosocial theory of development. In each stage of Erikson's theory, there is a psychosocial task that we must master in order to feel a sense of competence.

Erikson proposed that we are motivated by a need to achieve competence in certain areas of our lives. According to psychosocial theory, we experience eight stages of development over our lifespan, from infancy through late adulthood. At each stage there is a conflict, or task, that we need to resolve. Successful completion of each developmental task results in a sense of competence and a healthy personality. Failure to master these tasks leads to feelings of inadequacy.

According to Erikson (1963), trust is the basis of our development during infancy (birth to 12 months). Therefore, the primary task of this stage is trust versus mistrust. Infants are dependent upon their caregivers, so caregivers who are responsive and sensitive to

their infant's needs help their baby to develop a sense of trust; their baby will see the world as a safe, predictable place. Unresponsive caregivers who do not meet their baby's needs can engender feelings of anxiety, fear, and mistrust; their baby may see the world as unpredictable.

As toddlers (ages 1–3 years) begin to explore their world, they learn that they can control their actions and act on the environment to get results. They begin to show clear preferences for certain elements of the environment, such as food, toys, and clothing. A toddler's main task is to resolve the issue of autonomy versus shame and doubt, by working to establish independence. This is the "me do it" stage. For example, we might observe a budding sense of autonomy in a 2-year-old child who wants to choose her clothes and dress herself. Although her outfits might not be appropriate for the situation, her input in such basic decisions has an effect on her sense of independence. If denied the opportunity to act on her environment, she may begin to doubt her abilities, which could lead to low self-esteem and feelings of shame.

Once children reach the preschool stage (ages 3–6 years), they are capable of initiating activities and asserting control over their world through social interactions and play. According to Erikson, preschool children must resolve the task of initiative versus guilt. By learning to plan and achieve goals while interacting with others, preschool children can master this task. Those who do will develop self-confidence and feel a sense of purpose. Those who are unsuccessful at this stage—with their initiative misfiring or stifled—may develop feelings of guilt. How might over-controlling parents stifle a child's initiative?

During the elementary school stage (ages 6–12), children face the task of industry versus inferiority. Children begin to compare themselves to their peers to see how they measure up. They either develop a sense of pride and accomplishment in their schoolwork, sports, social activities, and family life, or they feel inferior and inadequate when they don't measure up. What are some things

parents and teachers can do to help children develop a sense of competence and a belief in themselves and their abilities?

In adolescence (ages 12–18), children face the task of identity versus role confusion. According to Erikson, an adolescent's main task is developing a sense of self. Adolescents struggle with questions such as "Who am I?" and "What do I want to do with my life?" Along the way, most adolescents try on many different selves to see which ones fit. Adolescents who are successful at this stage have a strong sense of identity and are able to remain true to their beliefs and values in the face of problems and other people's perspectives. What happens to apathetic adolescents, who do not make a conscious search for identity, or those who are pressured to conform to their parents' ideas for the future? These teens will have a weak sense of self and experience role confusion. They are unsure of their identity and confused about the future.

People in early adulthood (i.e., 20s through early 40s) are concerned with intimacy versus isolation. After we have developed a sense of self in adolescence, we are ready to share our life with others. Erikson said that we must have a strong sense of self before developing intimate relationships with others. Adults who do not develop a positive self-concept in adolescence may experience feelings of loneliness and emotional isolation.

When people reach their 40s, they enter the time known as middle adulthood, which extends to the mid-60s. The social task of middle adulthood is generativity versus stagnation. Generativity involves finding your life's work and contributing to the development of others, through activities such as volunteering, mentoring, and raising children. Those who do not master this task may experience stagnation, having little connection with others and little interest in productivity and self-improvement.

From the mid-60s to the end of life, we are in the period of development known as late adulthood. Erikson's task at this stage is called integrity versus despair. He said that people in late adulthood reflect on their lives and feel either a sense of

satisfaction or a sense of failure. People who feel proud of their accomplishments feel a sense of integrity, and they can look back on their lives with few regrets. However, people who are not successful at this stage may feel as if their life has been wasted. They focus on what "would have," "should have," and "could have" been. They face the end of their lives with feelings of bitterness, depression, and despair.

Erikson's Psychosocial Stages of Development

Stage	Age (years)	Developmental Task	Description
1	0–1	Trust vs. mistrust	Trust (or mistrust) that basic needs, such as nourishment and affection, will be met
2	1–3	Autonomy vs. shame/doubt	Develop a sense of independence in many tasks
3	3–6	Initiative vs. guilt	Take initiative on some activities—may develop guilt when unsuccessful or boundaries overstepped
4	7–11	Industry vs. inferiority	Develop self-confidence in abilities when competent or sense of inferiority when not
5	12–18	Identity vs. confusion	Experiment with and develop identity and roles
6	19–29	Intimacy vs. isolation	Establish intimacy and relationships with others
7	30–64	Generativity vs. stagnation	Contribute to society and be part of a family
8	65–	Integrity vs. despair	Assess and make sense of life and meaning of contributions

COGNITIVE THEORY OF DEVELOPMENT

Jean Piaget (1896–1980) is another stage theorist who studied childhood development. Instead of approaching development from a psychoanalytical or psychosocial perspective, Piaget focused on children's cognitive growth. He believed that thinking is a central aspect of development and that children are naturally inquisitive. However, he said that children do not think and reason like adults (Piaget, 1930, 1932). His theory of cognitive development holds

that our cognitive abilities develop through specific stages, which exemplifies the discontinuity approach to development. As we progress to a new stage, there is a distinct shift in how we think and reason.

Jean Piaget spent over 50 years studying children and how their minds develop.

Piaget said that children develop schemata to help them understand the world. Schemata are concepts (mental models) that are used to help us categorize and interpret information. By the time children have reached adulthood, they have created schemata for almost everything. When children learn new information, they

adjust their schemata through two processes: assimilation and accommodation. First, they assimilate new information or experiences in terms of their current schemata: assimilation is when they take in information that is comparable to what they already know. Accommodation describes when they change their schemata based on new information. This process continues as children interact with their environment.

For example, 2-year-old Blake learned the schema for dogs because his family has a Labrador retriever. When Blake sees other dogs in his picture books, he says, "Look mommy, dog!" Thus, he has assimilated them into his schema for dogs. One day, Blake sees a sheep for the first time and says, "Look mommy, dog!" Having a basic schema that a dog is an animal with four legs and fur, Blake thinks all furry, four-legged creatures are dogs. When Blake's mom tells him that the animal he sees is a sheep, not a dog, Blake must accommodate his schema for dogs to include more information based on his new experiences. Blake's schema for dog was too broad, since not all furry, four-legged creatures are dogs. He now modifies his schema for dogs and forms a new one for sheep.

Like Freud and Erikson, Piaget thought development unfolds in a series of stages approximately associated with age ranges. He proposed a theory of cognitive development that unfolds in four stages: sensorimotor, preoperational, concrete operational, and formal operational.

Piaget's Stages of Cognitive Development

Age (years)	Stage	Description	Developmental issues
0–2	Sensorimotor	World experienced through senses and actions	Object permanence Stranger anxiety
2–6	Preoperational	Use words and images to represent things, but lack logical reasoning	Pretend play Egocentrism Language development
7–11	Concrete operational	Understand concrete events and analogies logically; perform arithmetical operations	Conservation Mathematical transformations
12–	Formal operational	Formal operations Utilize abstract reasoning	Abstract logic Moral reasoning

The first stage is the sensorimotor stage, which lasts from birth to about 2 years old. During this stage, children learn about the world through their senses and motor behavior. Young children put objects in their mouths to see if the items are edible, and once they can grasp objects, they may shake or bang them to see if they make sounds. Between 5 and 8 months old, the child develops object permanence, which is the understanding that even if something is out of sight, it still exists (Bogartz, Shinskey, & Schilling, 2000). According to Piaget, young infants do not remember an object after it has been removed from sight. Piaget studied infants' reactions when a toy was first shown to an infant and then hidden under a blanket. Infants who had already developed object permanence would reach for the hidden toy, indicating that they knew it still existed, whereas infants who had not developed object permanence would appear confused.

Please take a few minutes to view this brief video demonstrating different children's ability to understand object permanence: Piaget – Stage 1 – Sensorimotor stage : Object Permanence.

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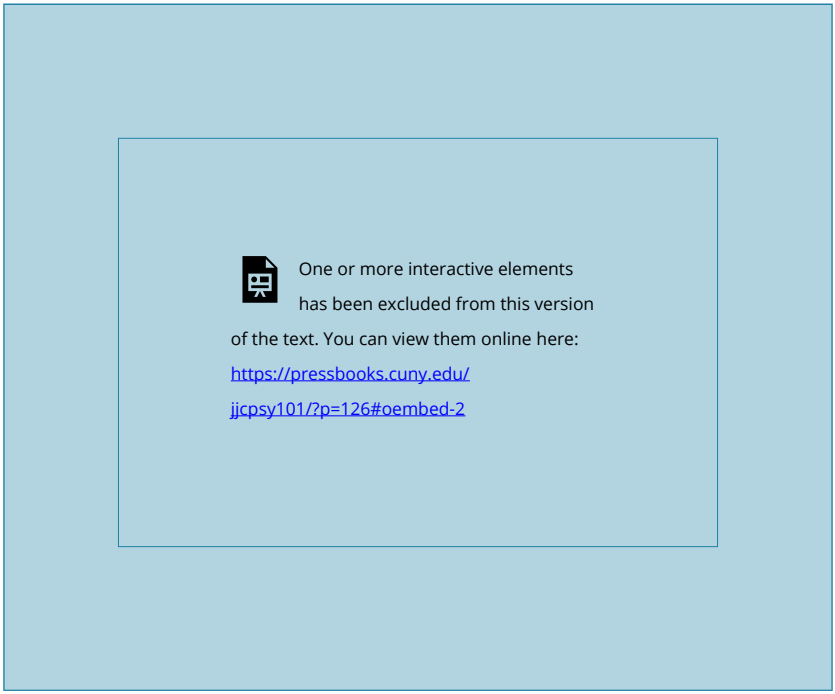
In Piaget's view, around the same time children develop object permanence, they also begin to exhibit stranger anxiety, which is a fear of unfamiliar people. Babies may demonstrate this by crying and turning away from a stranger, by clinging to a caregiver, or by attempting to reach their arms toward familiar faces such as parents. Stranger anxiety results when a child is unable to assimilate the stranger into an existing schema; therefore, she can't predict what her experience with that stranger will be like, which results in a fear response.

Piaget's second stage is the preoperational stage, which is from approximately 2 to 7 years old. In this stage, children can use

symbols to represent words, images, and ideas, which is why children in this stage engage in pretend play. A child's arms might become airplane wings as he zooms around the room, or a child with a stick might become a brave knight with a sword. Children also begin to use language in the preoperational stage, but they cannot understand adult logic or mentally manipulate information (the term *operational* refers to logical manipulation of information, so children at this stage are considered to be *pre*-operational). Children's logic is based on their own personal knowledge of the world so far, rather than on conventional knowledge. For example, dad gave a slice of pizza to 10-year-old Keiko and another slice to her 3-year-old brother, Kenny. Kenny's pizza slice was cut into five pieces, so Kenny told his sister that he got more pizza than she did. Children in this stage cannot perform mental operations because they have not developed an understanding of conservation, which is the idea that even if you change the appearance of something, it is still equal in size as long as nothing has been removed or added.

This video shows a 4.5-year-old boy in the preoperational stage as he responds to Piaget's conservation tasks: A typical child on Piaget's conservation tasks.



During this stage, we also expect children to display egocentrism, which means that the child is not able to take the perspective of others. A child at this stage thinks that everyone sees, thinks, and feels just as they do. Let's look at Kenny and Keiko again. Keiko's birthday is coming up, so their mom takes Kenny to the toy store to choose a present for his sister. He selects an Iron Man action figure for her, thinking that if he likes the toy, his sister will too. An egocentric child is not able to infer the perspective of other people and instead attributes his own perspective.

Piaget developed the Three-Mountain Task to determine the level of egocentrism displayed by children. Children view a 3-dimensional mountain scene from one viewpoint, and are asked what another person at a different viewpoint would see in the same scene. Watch the Three-Mountain Task in action in this short video from the University of Minnesota and the Science Museum of Minnesota: Piaget's Mountains Task.

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Piaget's third stage is the concrete operational stage, which occurs from about 7 to 11 years old. In this stage, children can think logically about real (concrete) events; they have a firm grasp on the use of numbers and start to employ memory strategies. They can perform mathematical operations and understand transformations, such as addition is the opposite of subtraction,

and multiplication is the opposite of division. In this stage, children also master the concept of conservation: Even if something changes shape, its mass, volume, and number stay the same. For example, if you pour water from a tall, thin glass to a short, fat glass, you still have the same amount of water. Remember Keiko and Kenny and the pizza? How did Keiko know that Kenny was wrong when he said that he had more pizza?

Children in the concrete operational stage also understand the principle of reversibility, which means that objects can be changed and then returned back to their original form or condition. Take, for example, water that you poured into the short, fat glass: You can pour water from the fat glass back to the thin glass and still have the same amount (minus a couple of drops).

The fourth, and last, stage in Piaget's theory is the formal operational stage, which is from about age 11 to adulthood. Whereas children in the concrete operational stage are able to think logically only about concrete events, children in the formal operational stage can also deal with abstract ideas and hypothetical situations. Children in this stage can use abstract thinking to problem solve, look at alternative solutions, and test these solutions. In adolescence, a renewed egocentrism occurs. For example, a 15-year-old with a very small pimple on her face might think it is huge and incredibly visible, under the mistaken impression that others must share her perceptions.

BEYOND FORMAL OPERATIONAL THOUGHT

As with other major contributors of theories of development, several of Piaget's ideas have come under criticism based on the results of further research. For example, several contemporary studies support a model of development that is more continuous than Piaget's discrete stages (Courage & Howe, 2002; Siegler, 2005, 2006). Many others suggest that children reach cognitive

milestones earlier than Piaget describes (Baillargeon, 2004; de Hevia & Spelke, 2010).

According to Piaget, the highest level of cognitive development is formal operational thought, which develops between 11 and 20 years old. However, many developmental psychologists disagree with Piaget, suggesting a fifth stage of cognitive development, known as the postformal stage (Basseches, 1984; Commons & Bresette, 2006; Sinnott, 1998). In postformal thinking, decisions are made based on situations and circumstances, and logic is integrated with emotion as adults develop principles that depend on contexts. One way that we can see the difference between an adult in postformal thought and an adolescent in formal operations is in terms of how they handle emotionally charged issues.

It seems that once we reach adulthood our problem solving abilities change: As we attempt to solve problems, we tend to think more deeply about many areas of our lives, such as relationships, work, and politics (Labouvie-Vief & Diehl, 1999). Because of this, postformal thinkers are able to draw on past experiences to help them solve new problems. Problem-solving strategies using postformal thought vary, depending on the situation. What does this mean? Adults can recognize, for example, that what seems to be an ideal solution to a problem at work involving a disagreement with a colleague may not be the best solution to a disagreement with a significant other.

TEST YOUR UNDERSTANDING

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THEORY OF MORAL DEVELOPMENT

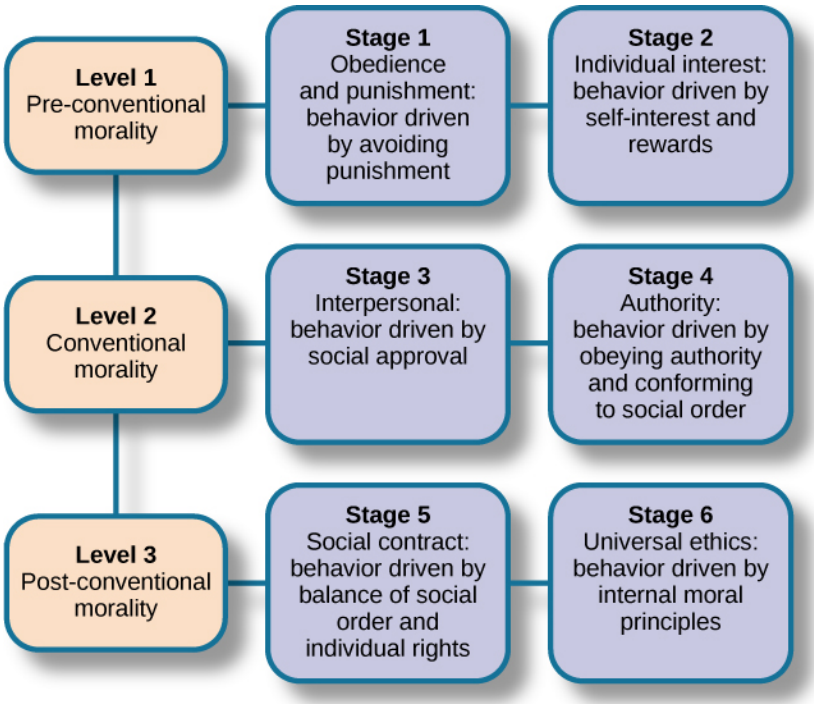
A major task beginning in childhood and continuing into adolescence is discerning right from wrong. Psychologist Lawrence Kohlberg (1927–1987) extended upon the foundation that Piaget built regarding cognitive development. Kohlberg believed that moral development, like cognitive development, follows a series of stages. To develop this theory, Kohlberg posed moral dilemmas to people of all ages, and then he analyzed their answers to find evidence of their particular stage of moral development. Before reading about the stages, take a minute to consider how you would

answer one of Kohlberg's best-known moral dilemmas, commonly known as the Heinz dilemma:

> In Europe, a woman was near death from a special kind of cancer. There was one drug that the doctors thought might save her. It was a form of radium that a druggist in the same town had recently discovered. The drug was expensive to make, but the druggist was charging ten times what the drug cost him to make. He paid $200 for the radium and charged $2,000 for a small dose of the drug. The sick woman's husband, Heinz, went to everyone he knew to borrow the money, but he could only get together about $1,000, which is half of what it cost. He told the druggist that his wife was dying and asked him to sell it cheaper or let him pay later. But the druggist said: "No, I discovered the drug and I'm going to make money from it." So Heinz got desperate and broke into the man's store to steal the drug for his wife. Should the husband have done that? (Kohlberg, 1969, p. 379)

How would you answer this dilemma? Kohlberg was not interested in whether you answer yes or no to the dilemma: Instead, he was interested in the reasoning behind your answer.

After presenting people with this and various other moral dilemmas, Kohlberg reviewed people's responses and placed them in different stages of moral reasoning. According to Kohlberg, an individual progresses from the capacity for pre-conventional morality (before age 9) to the capacity for conventional morality (early adolescence), and toward attaining post-conventional morality (once formal operational thought is attained), which only a few fully achieve. Kohlberg placed in the highest stage responses that reflected the reasoning that Heinz should steal the drug because his wife's life is more important than the pharmacist making money. The value of a human life overrides the pharmacist's greed.



Kohlberg identified three levels of moral reasoning: pre-conventional, conventional, and post-conventional: Each level is associated with increasingly complex stages of moral development.

It is important to realize that even those people who have the most sophisticated, post-conventional reasons for some choices may make other choices for the simplest of pre-conventional reasons. Many psychologists agree with Kohlberg's theory of moral development but point out that moral reasoning is very different from moral behavior. Sometimes what we say we would do in a situation is not what we actually do in that situation. In other words, we might "talk the talk," but not "walk the walk."

How does this theory apply to males and females? Kohlberg (1969) felt that more males than females move past stage four in their moral development. He went on to note that women seem

to be deficient in their moral reasoning abilities. These ideas were not well received by Carol Gilligan, a research assistant of Kohlberg, who consequently developed her own ideas of moral development. In her groundbreaking book, *In a Different Voice: Psychological Theory and Women's Development*, Gilligan (1982) criticized her former mentor's theory because it was based only on upper class White men and boys. She argued that women are not deficient in their moral reasoning—she proposed that males and females reason differently. Girls and women focus more on staying connected and the importance of interpersonal relationships. Therefore, in the Heinz dilemma, many girls and women respond that Heinz should not steal the medicine. Their reasoning is that if he steals the medicine, is arrested, and is put in jail, then he and his wife will be separated, and she could die while he is still in prison.

TEST YOUR UNDERSTANDING

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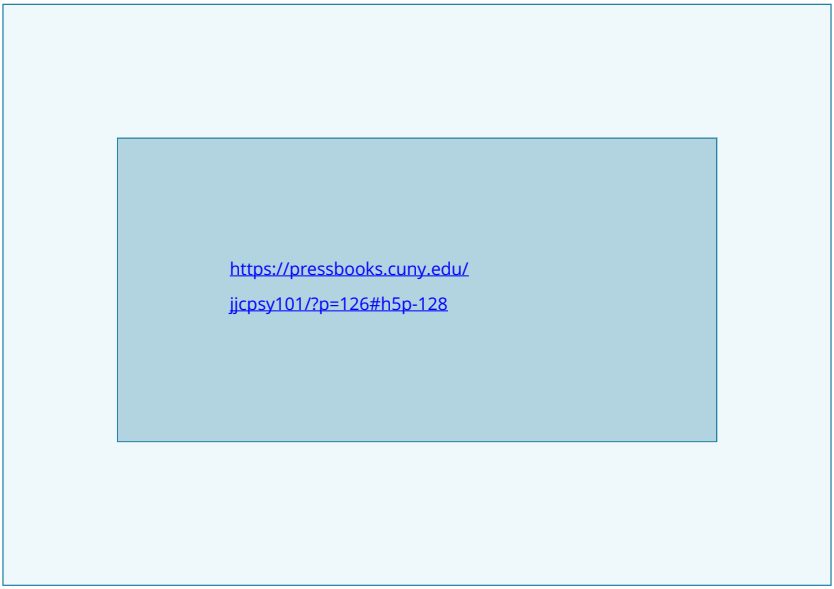

SUMMARY

There are many theories regarding how babies and children grow and develop into happy, healthy adults. Sigmund Freud suggested that we pass through a series of psychosexual stages in which our energy is focused on certain erogenous zones on the body. Eric Erikson modified Freud's ideas and suggested a theory of psychosocial development. Erikson said that our social interactions and successful completion of social tasks shape our sense of self. Jean Piaget proposed a theory of cognitive development that explains how children think and reason as they move through various stages. Finally, Lawrence Kohlberg turned his attention to moral development. He said that we pass through three levels of moral thinking that build on our cognitive development.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

What is the difference between assimilation and accommodation? Provide examples of each.

Assimilation is when we take in information that

is comparable to what we already know. Accommodation is when we change our schemata based on new information. An example of assimilation is a child's schema of "dog" based on the family's golden retriever being expanded to include two newly adopted golden retrievers. An example of accommodation is that same child's schema of "dog" being adjusted to exclude other four-legged furry animals such as sheep and foxes.

Why was Carol Gilligan critical of Kohlberg's theory of moral development?

Gilligan criticized Kohlberg because his theory was based on the responses of upper class White men and boys, arguing that it was biased against women. While Kohlberg concluded that women must be deficient in their moral reasoning abilities, Gilligan disagreed, suggesting that female moral reasoning is not deficient, just different.

What is egocentrism? Provide an original example.

Egocentrism is the inability to take the perspective of another person. This type of thinking is common in young children in the preoperational stage of cognitive development. An example might be that upon seeing his mother crying, a young child gives her his favorite stuffed animal to make her feel better.

PERSONAL APPLICATION QUESTIONS

Explain how you would use your understanding of one of the major developmental theories to deal with each of the difficulties listed below:

1. Your infant daughter puts everything in her mouth, including the dog's food.
2. Your eight-year-old son is failing math; all he cares about is baseball.
3. Your two-year-old daughter refuses to wear the clothes you pick for her every morning, which makes getting dressed a twenty-minute battle.
4. Your sixty-eight-year-old neighbor is chronically depressed and feels she has wasted her life.
5. Your 18-year-old daughter has decided not to go to college. Instead she's moving to Colorado to become a ski instructor.
6. Your 11-year-old son is the class bully.

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Describe the stages of prenatal development and recognize the importance of prenatal care
- Discuss physical, cognitive, and emotional development that occurs from infancy through childhood
- Discuss physical, cognitive, and emotional development that occurs during adolescence
- Discuss physical, cognitive, and emotional development that occurs in adulthood

From the moment we are born until the moment we die, we continue to develop.

As discussed at the beginning of this chapter, developmental psychologists often divide our development into three areas: physical development, cognitive development, and psychosocial development. Mirroring Erikson's stages, lifespan development is divided into different stages that are based on age. We will discuss prenatal, infant, child, adolescent, and adult development.

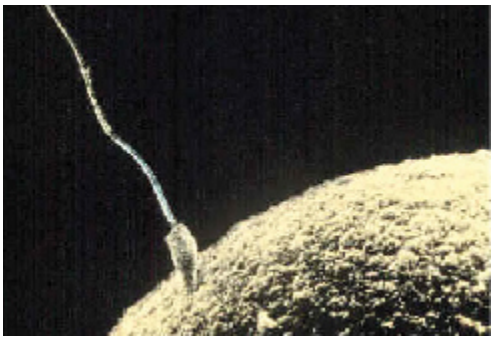
PRENATAL DEVELOPMENT

How did you come to be who you are? From beginning as a one-cell structure to your birth, your prenatal development occurred in an orderly and delicate sequence.

There are three stages of prenatal development: germinal, embryonic, and fetal. Let's take a look at what happens to the developing baby in each of these stages.

GERMINAL STAGE (WEEKS 1–2)

In the discussion of biopsychology earlier in the book, you learned about genetics and DNA. A mother and father's DNA is passed on to the child at the moment of conception. Conception occurs when sperm fertilizes an egg and forms a zygote. A zygote begins as a one-cell structure that is created when a sperm and egg merge. The genetic makeup and sex of the baby are set at this point. During the first week after conception, the zygote divides and multiplies, going from a one-cell structure to two cells, then four cells, then eight cells, and so on. This process of cell division is called mitosis. Mitosis is a fragile process, and fewer than one-half of all zygotes survive beyond the first two weeks (Hall, 2004). After 5 days of mitosis there are 100 cells, and after 9 months there are billions of cells. As the cells divide, they become more specialized, forming different organs and body parts. In the germinal stage, the mass of cells has yet to attach itself to the lining of the mother's uterus. Once it does, the next stage begins.



Sperm and ovum fuse at the point of conception.

EMBRYONIC STAGE (WEEKS 3–8)

After the zygote divides for about 7–10 days and has 150 cells, it travels down the fallopian tubes and implants itself in the lining of the uterus. Upon implantation, this multi-cellular organism is called an embryo. Now blood vessels grow, forming the placenta. The placenta is a structure connected to the uterus that provides nourishment and oxygen from the mother to the developing embryo via the umbilical cord. Basic structures of the embryo start to develop into areas that will become the head, chest, and abdomen. During the embryonic stage, the heart begins to beat and organs form and begin to function. The neural tube forms along the back of the embryo, developing into the spinal cord and brain.

FETAL STAGE (WEEKS 9–40)

When the organism is about nine weeks old, the embryo is called a fetus. At this stage, the fetus is about the size of a kidney bean and begins to take on the recognizable form of a human being as the "tail" begins to disappear.

From 9–12 weeks, the sex organs begin to differentiate. At about 16 weeks, the fetus is approximately 4.5 inches long. Fingers and toes are fully developed, and fingerprints are visible. By the time the fetus reaches the sixth month of development (24 weeks), it weighs up to 1.4 pounds. Hearing has developed, so the fetus can respond to sounds. The internal organs, such as the lungs, heart, stomach, and intestines, have formed enough that a fetus born prematurely at this point has a chance to survive outside of the mother's womb. Throughout the fetal stage the brain continues to grow and develop, nearly doubling in size from weeks 16 to 28. Around 36 weeks, the fetus is almost ready for birth. It weighs about 6 pounds and is about 18.5 inches long, and by week 37 all of the fetus's organ systems are developed enough that it could

survive outside the mother's uterus without many of the risks associated with premature birth. The fetus continues to gain weight and grow in length until approximately 40 weeks. By then, the fetus has very little room to move around and birth becomes imminent.

During the fetal stage, the baby's brain develops and the body adds size and weight, until the fetus reaches full-term development.

For an amazing look at prenatal development and the process of birth, view the video *Life's Greatest Miracle* from Nova and PBS: Life's Greatest Miracle.

PRENATAL INFLUENCES

During each prenatal stage, genetic and environmental factors can affect development. The developing fetus is completely dependent on the mother for life. It is important that the mother takes good care of herself and receives prenatal care, which is medical care during pregnancy that monitors the health of both the mother and the fetus. According to the National Institutes of Health ([NIH], 2013), routine prenatal care is important because it can reduce the risk of complications to the mother and fetus during pregnancy. In fact, women who are trying to become pregnant or who may become pregnant should discuss pregnancy planning with their doctor. They may be advised, for example, to take a vitamin containing folic acid, which helps prevent certain birth defects, or to monitor aspects of their diet or exercise routines.

A pregnant woman receives an ultrasound as part of her prenatal care. (credit: United States Agency for International Development)

Recall that when the zygote attaches to the wall of the mother's uterus, the placenta is formed. The placenta provides nourishment and oxygen to the fetus. Most everything the mother ingests, including food, liquid, and even medication, travels through the placenta to the fetus, hence the common phrase "eating for two." Anything the mother is exposed to in the environment affects the fetus; if the mother is exposed to something harmful, the child can show life-long effects.

A teratogen is any environmental agent—biological, chemical, or physical—that causes damage to the developing embryo or fetus. There are different types of teratogens. Alcohol and most drugs cross the placenta and affect the fetus. Alcohol is not safe to drink in any amount during pregnancy. Alcohol use during pregnancy has been found to be the leading preventable cause of intellectual disabilities in children in the United States (Maier & West, 2001). Excessive maternal drinking while pregnant can cause fetal alcohol spectrum disorders with life-long consequences for the child ranging in severity from minor to major. Fetal alcohol spectrum disorders (FASD) are a collection of birth defects associated with heavy consumption of alcohol during pregnancy. Physically,

children with FASD may have a small head size and abnormal facial features. Cognitively, these children may have poor judgment, poor impulse control, higher rates of ADHD, learning issues, and lower IQ scores. These developmental problems and delays persist into adulthood (Streissguth et al., 2004). Based on studies conducted on animals, it also has been suggested that a mother's alcohol consumption during pregnancy may predispose her child to like alcohol (Youngentob et al., 2007).

Fetal Alcohol Syndrome Facial Features

Facial Feature	Potential Effect of Fetal Alcohol Syndrome
Head size	Below-average head circumference
Eyes	Smaller than average eye opening, skin folds at corners of eyes
Nose	Low nasal bridge, short nose
Midface	Smaller than average midface size
Lip and philtrum	Thin upper lip, indistinct philtrum

Smoking is also considered a teratogen because nicotine travels through the placenta to the fetus. When the mother smokes, the developing baby experiences a reduction in blood oxygen levels. According to the Centers for Disease Control and Prevention (2013), smoking while pregnant can result in premature birth, low-birth-weight infants, stillbirth, and sudden infant death syndrome (SIDS).

Heroin, cocaine, methamphetamine, almost all prescription medicines, and most over-the counter medications are also considered teratogens. Babies born with a heroin addiction need heroin just like an adult addict. The child will need to be gradually weaned from the heroin under medical supervision; otherwise, the child could have seizures and die. Other teratogens include radiation, viruses such as HIV and herpes, and rubella (German measles). Women in the United States are much less likely to be afflicted with rubella because most women received childhood immunizations or vaccinations that protect the body from disease.

Each organ of the fetus develops during a specific period in

the pregnancy, called the critical or sensitive period. For example, research with primate models of FASD has demonstrated that the time during which a developing fetus is exposed to alcohol can dramatically affect the appearance of facial characteristics associated with fetal alcohol syndrome. Specifically, this research suggests that alcohol exposure that is limited to day 19 or 20 of gestation can lead to significant facial abnormalities in the offspring (Ashley, Magnuson, Omnell, & Clarren, 1999). Given regions of the brain also show sensitive periods during which they are most susceptible to the teratogenic effects of alcohol (Tran & Kelly, 2003).

SHOULD WOMEN WHO USE DRUGS DURING PREGNANCY BE ARRESTED AND JAILED?

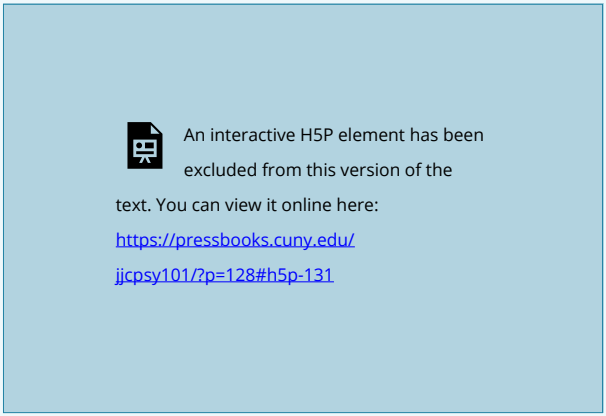
As you now know, women who use drugs or alcohol during pregnancy can cause serious lifelong harm to their child. Some people have advocated mandatory screenings for women who are pregnant and have a history of drug abuse, and if the women continue using, to arrest, prosecute, and incarcerate them (Figdor & Kaeser, 1998). This policy was tried in Charleston, South Carolina, as recently as 20 years ago. The policy was called the Interagency Policy on Management of Substance Abuse During Pregnancy, and had disastrous results.

> The Interagency Policy applied to patients attending the obstetrics clinic at MUSC, which primarily serves patients who are indigent or on Medicaid. It did not apply to private obstetrical patients. The policy required patient education about the harmful effects of substance abuse during pregnancy. . . . [A] statement also warned patients that protection of unborn and newborn children from the harms of illegal drug abuse could involve the Charleston police, the Solicitor of the Ninth Judicial Court, and the Protective Services Division of the Department of Social Services (DSS). (Jos, Marshall, & Perlmutter, 1995, pp. 120–121)

This policy seemed to deter women from seeking prenatal care,

deterred them from seeking other social services, and was applied solely to low-income women, resulting in lawsuits. The program was canceled after 5 years, during which 42 women were arrested. A federal agency later determined that the program involved human experimentation without the approval and oversight of an institutional review board (IRB). What were the flaws in the program and how would you correct them? What are the ethical implications of charging pregnant women with child abuse?

TEST YOUR UNDERSTANDING

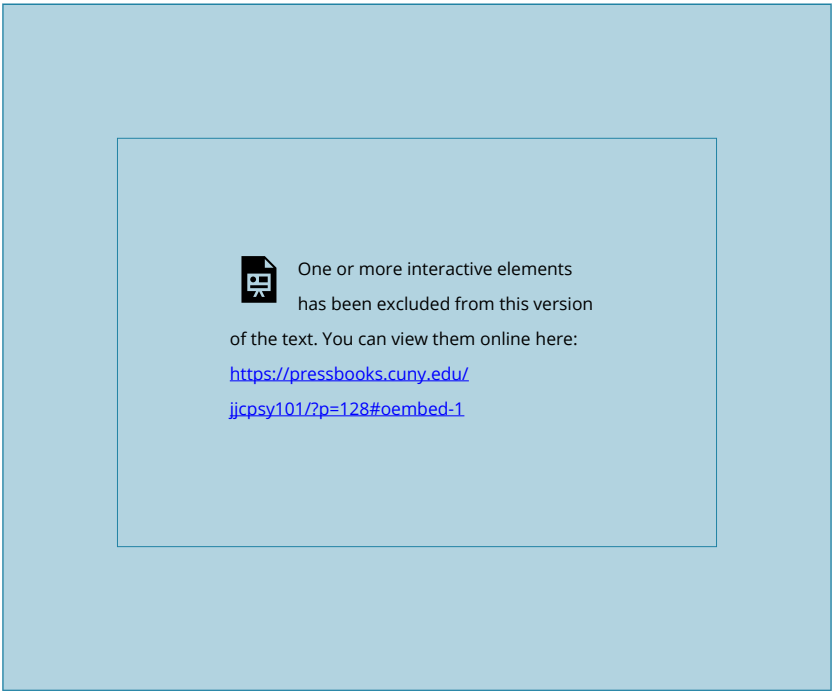


INFANCY THROUGH CHILDHOOD

The average newborn weighs approximately 7.5 pounds. Although

small, a newborn is not completely helpless because his reflexes and sensory capacities help him interact with the environment from the moment of birth. All healthy babies are born with newborn reflexes: inborn automatic responses to particular forms of stimulation. Reflexes help the newborn survive until it is capable of more complex behaviors—these reflexes are crucial to survival. They are present in babies whose brains are developing normally and usually disappear around 4–5 months old. Let's take a look at some of these newborn reflexes. The rooting reflex is the newborn's response to anything that touches her cheek: When you stroke a baby's cheek, she naturally turns her head in that direction and begins to suck. The sucking reflex is the automatic, unlearned, sucking motions that infants do with their mouths. Several other interesting newborn reflexes can be observed. For instance, if you put your finger into a newborn's hand, you will witness the grasping reflex, in which a baby automatically grasps anything that touches his palms. The Moro reflex is the newborn's response when she feels like she is falling. The baby spreads her arms, pulls them back in, and then (usually) cries. How do you think these reflexes promote survival in the first months of life?

Take a few minutes to view this brief video clip illustrating several newborn reflexes: Reflexes in newborn babies.



What can young infants see, hear, and smell? Newborn infants' sensory abilities are significant, but their senses are not yet fully developed. Many of a newborn's innate preferences facilitate interaction with caregivers and other humans. Although vision is their least developed sense, newborns already show a preference for faces. Babies who are just a few days old also prefer human voices, they will listen to voices longer than sounds that do not involve speech (Vouloumanos & Werker, 2004), and they seem to prefer their mother's voice over a stranger's voice (Mills & Melhuish, 1974). In an interesting experiment, 3-week-old babies were given pacifiers that played a recording of the infant's mother's voice and of a stranger's voice. When the infants heard their mother's voice, they sucked more strongly at the pacifier (Mills & Melhuish, 1974). Newborns also have a strong sense of smell. For instance, newborn babies can distinguish the smell of their own mother from that of

others. In a study by MacFarlane (1978), 1-week-old babies who were being breastfed were placed between two gauze pads. One gauze pad was from the bra of a nursing mother who was a stranger, and the other gauze pad was from the bra of the infant's own mother. More than two-thirds of the week-old babies turned toward the gauze pad with their mother's scent.

PHYSICAL DEVELOPMENT

In infancy, toddlerhood, and early childhood, the body's physical development is rapid. On average, newborns weigh between 5 and 10 pounds, and a newborn's weight typically doubles in six months and triples in one year. By 2 years old the weight will have quadrupled, so we can expect that a 2 year old should weigh between 20 and 40 pounds. The average length of a newborn is 19.5 inches, increasing to 29.5 inches by 12 months and 34.4 inches by 2 years old (WHO Multicentre Growth Reference Study Group, 2006).

Children experience rapid physical changes through infancy and early childhood. (credit "left": modification of work by Kerry Ceszyk; credit "middle-left": modification of work by Kristi Fausel; credit "middle-right": modification of work by "devinf"/Flickr; credit "right": modification of work by Rose Spielman)

During infancy and childhood, growth does not occur at a steady rate (Carel, Lahlou, Roger, & Chaussain, 2004). Growth slows between 4 and 6 years old: During this time children gain 5–7 pounds and grow about 2–3 inches per year. Once girls reach 8–9

years old, their growth rate outpaces that of boys due to a pubertal growth spurt. This growth spurt continues until around 12 years old, coinciding with the start of the menstrual cycle. By 10 years old, the average girl weighs 88 pounds, and the average boy weighs 85 pounds.

We are born with all of the brain cells that we will ever have—about 100–200 billion neurons (nerve cells) whose function is to store and transmit information (Huttenlocher & Dabholkar, 1997). However, the nervous system continues to grow and develop. Each neural pathway forms thousands of new connections during infancy and toddlerhood. This period of rapid neural growth is called blooming. Neural pathways continue to develop through puberty. The blooming period of neural growth is then followed by a period of pruning, where neural connections are reduced. It is thought that pruning causes the brain to function more efficiently, allowing for mastery of more complex skills (Hutchinson, 2011). Blooming occurs during the first few years of life, and pruning continues through childhood and into adolescence in various areas of the brain.

The size of our brains increases rapidly. For example, the brain of a 2-year-old is 55% of its adult size, and by 6 years old the brain is about 90% of its adult size (Tanner, 1978). During early childhood (ages 3–6), the frontal lobes grow rapidly. Recalling our discussion of the 4 lobes of the brain earlier in this book, the frontal lobes are associated with planning, reasoning, memory, and impulse control. Therefore, by the time children reach school age, they are developmentally capable of controlling their attention and behavior. Through the elementary school years, the frontal, temporal, occipital, and parietal lobes all grow in size. The brain growth spurts experienced in childhood tend to follow Piaget's sequence of cognitive development, so that significant changes in neural functioning account for cognitive advances (Kolb & Whishaw, 2009; Overman, Bachevalier, Turner, & Peuster, 1992).

Motor development occurs in an orderly sequence as infants

move from reflexive reactions (e.g., sucking and rooting) to more advanced motor functioning. For instance, babies first learn to hold their heads up, then to sit with assistance, and then to sit unassisted, followed later by crawling and then walking.

Motor skills refer to our ability to move our bodies and manipulate objects. Fine motor skills focus on the muscles in our fingers, toes, and eyes, and enable coordination of small actions (e.g., grasping a toy, writing with a pencil, and using a spoon). Gross motor skills focus on large muscle groups that control our arms and legs and involve larger movements (e.g., balancing, running, and jumping).

As motor skills develop, there are certain developmental milestones that young children should achieve. For each milestone there is an average age, as well as a range of ages in which the milestone should be reached. An example of a developmental milestone is sitting. On average, most babies sit alone at 7 months old. Sitting involves both coordination and muscle strength, and 90% of babies achieve this milestone between 5 and 9 months old. In another example, babies on average are able to hold up their head at 6 weeks old, and 90% of babies achieve this between 3 weeks and 4 months old. If a baby is not holding up his head by 4 months old, he is showing a delay. If the child is displaying delays on several milestones, that is reason for concern, and the parent or caregiver should discuss this with the child's pediatrician. Some developmental delays can be identified and addressed through early intervention.

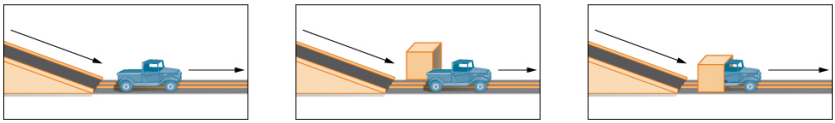
Developmental Milestones, Ages 2–5 Years

Age (years)	Physical	Personal/Social	Language	Cognitive
2	Kicks a ball; walks up and down stairs	Plays alongside other children; copies adults	Points to objects when named; puts 2–4 words together in a sentence	Sorts shapes and colors; follows 2-step instructions
3	Climbs and runs; pedals tricycle	Takes turns; expresses many emotions; dresses self	Names familiar things; uses pronouns	Plays make believe; works toys with parts (levers, handles)
4	Catches balls; uses scissors	Prefers social play to solo play; knows likes and interests	Knows songs and rhymes by memory	Names colors and numbers; begins writing letters
5	Hops and swings; uses fork and spoon	Distinguishes real from pretend; likes to please friends	Speaks clearly; uses full sentences	Counts to 10 or higher; prints some letters and copies basic shapes

COGNITIVE DEVELOPMENT

In addition to rapid physical growth, young children also exhibit significant development of their cognitive abilities. Piaget thought that children's ability to understand objects—such as learning that a rattle makes a noise when shaken—was a cognitive skill that develops slowly as a child matures and interacts with the environment. Today, developmental psychologists think Piaget was incorrect. Researchers have found that even very young children understand objects and how they work long before they have experience with those objects (Baillargeon, 1987; Baillargeon, Li, Gertner, & Wu, 2011). For example, children as young as 3 months old demonstrated knowledge of the properties of objects that they had only viewed and did not have prior experience with them. In one study, 3-month-old infants were shown a truck rolling down a track and behind a screen. The box, which appeared solid but was actually hollow, was placed next to the track. The truck rolled past the box as would be expected. Then the box was placed on the track to block the path of the truck. When the truck was rolled down

the track this time, it continued unimpeded. The infants spent significantly more time looking at this impossible event. Baillargeon (1987) concluded that they knew solid objects cannot pass through each other. Baillargeon's findings suggest that very young children have an understanding of objects and how they work, which Piaget (1954) would have said is beyond their cognitive abilities due to their limited experiences in the world.



In Baillargeon's study, infants observed a truck (a) roll down an unobstructed track, (b) roll down an unobstructed track with an obstruction (box) beside it, and (c) roll down and pass through what appeared to be an obstruction.

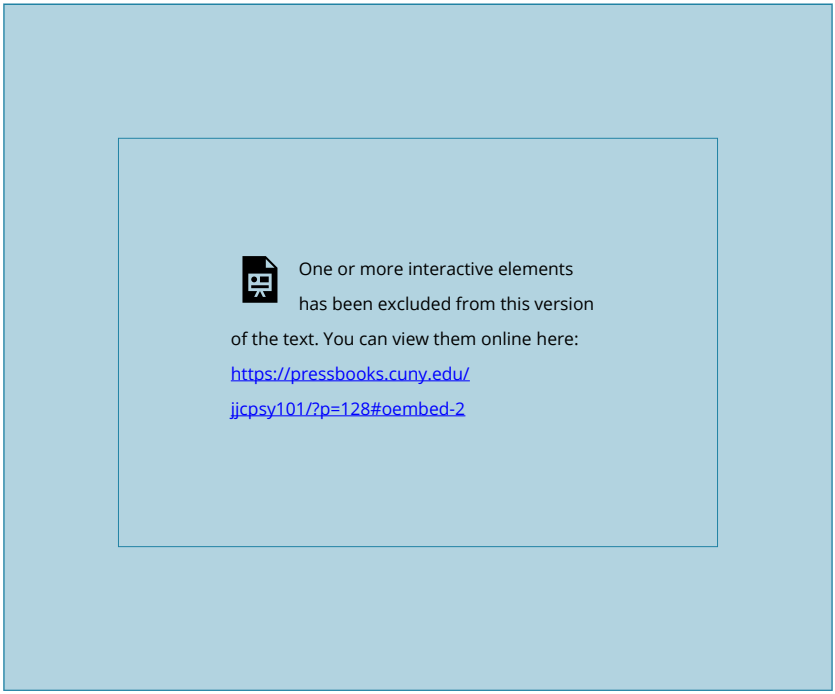
Just as there are physical milestones that we expect children to reach, there are also cognitive milestones. It is helpful to be aware of these milestones as children gain new abilities to think, problem solve, and communicate. For example, infants shake their head "no" around 6–9 months, and they respond to verbal requests to do things like "wave bye-bye" or "blow a kiss" around 9–12 months. Remember Piaget's ideas about object permanence? We can expect children to grasp the concept that objects continue to exist even when they are not in sight by around 8 months old. Because toddlers (i.e., 12–24 months old) have mastered object permanence, they enjoy games like hide and seek, and they realize that when someone leaves the room they will come back (Loop, 2013). Toddlers also point to pictures in books and look in appropriate places when you ask them to find objects.

Preschool-age children (i.e., 3–5 years old) also make steady progress in cognitive development. Not only can they count, name colors, and tell you their name and age, but they can also make some decisions on their own, such as choosing an outfit to wear.

Preschool-age children understand basic time concepts and sequencing (e.g., before and after), and they can predict what will happen next in a story. They also begin to enjoy the use of humor in stories. Because they can think symbolically, they enjoy pretend play and inventing elaborate characters and scenarios. One of the most common examples of their cognitive growth is their blossoming curiosity. Preschool-age children love to ask "Why?"

An important cognitive change occurs in children this age. Recall that Piaget described 2–3 year olds as egocentric, meaning that they do not have an awareness of others' points of view. Between 3 and 5 years old, children come to understand that people have thoughts, feelings, and beliefs that are different from their own. This is known as theory-of-mind (TOM). Children can use this skill to tease others, persuade their parents to purchase a candy bar, or understand why a sibling might be angry. When children develop TOM, they can recognize that others have false beliefs (Dennett, 1987; Callaghan et al., 2005).

False-belief tasks are useful in determining a child's acquisition of theory-of-mind (TOM). Take a look at this video clip showing a false-belief task involving a box of crayons: The "False Belief" Test: Theory of Mind.



Cognitive skills continue to expand in middle and late childhood (6–11 years old). Thought processes become more logical and organized when dealing with concrete information. Children at this age understand concepts such as the past, present, and future, giving them the ability to plan and work toward goals. Additionally, they can process complex ideas such as addition and subtraction and cause-and-effect relationships. However, children's attention spans tend to be very limited until they are around 11 years old. After that point, it begins to improve through adulthood.

Because they understand luck and fairness, children in middle and late childhood (6–11 years old) are able to follow rules for games. (credit: Edwin Martinez)

One well-researched aspect of cognitive development is language acquisition. As mentioned earlier, the order in which children learn language structures is consistent across children and cultures (Hatch, 1983). You've also learned that some psychological researchers have proposed that children possess a biological predisposition for language acquisition.

Starting before birth, babies begin to develop language and communication skills. At birth, babies apparently recognize their mother's voice and can discriminate between the language(s) spoken by their mothers and foreign languages, and they show preferences for faces that are moving in synchrony with audible language (Blossom & Morgan, 2006; Pickens, 1994; Spelke & Cortelyou, 1981).

Children communicate information through gesturing long before they speak, and there is some evidence that gesture usage predicts subsequent language development (Iverson & Goldin-Meadow, 2005). In terms of producing spoken language, babies begin to coo almost immediately. Cooing is a one-syllable

combination of a consonant and a vowel sound (e.g., coo or ba). Interestingly, babies replicate sounds from their own languages. A baby whose parents speak French will coo in a different tone than a baby whose parents speak Spanish or Urdu. After cooing, the baby starts to babble. Babbling begins with repeating a syllable, such as ma-ma, da-da, or ba-ba. When a baby is about 12 months old, we expect her to say her first word for meaning, and to start combining words for meaning at about 18 months.

At about 2 years old, a toddler uses between 50 and 200 words; by 3 years old they have a vocabulary of up to 1,000 words and can speak in sentences. During the early childhood years, children's vocabulary increases at a rapid pace. This is sometimes referred to as the "vocabulary spurt" and has been claimed to involve an expansion in vocabulary at a rate of 10–20 new words per week. Recent research may indicate that while some children experience these spurts, it is far from universal (as discussed in Ganger & Brent, 2004). It has been estimated that, 5 year olds understand about 6,000 words, speak 2,000 words, and can define words and question their meanings. They can rhyme and name the days of the week. Seven year olds speak fluently and use slang and clichés (Stork & Widdowson, 1974).

What accounts for such dramatic language learning by children? Behaviorist B. F. Skinner thought that we learn language in response to reinforcement or feedback, such as through parental approval or through being understood. For example, when a two-year-old child asks for juice, he might say, "me juice," to which his mother might respond by giving him a cup of apple juice. Noam Chomsky (1957) criticized Skinner's theory and proposed that we are all born with an innate capacity to learn language. Chomsky called this mechanism a language acquisition device (LAD). Who is correct? Both Chomsky and Skinner are right. Remember that we are a product of both nature and nurture. Researchers now believe that language acquisition is partially inborn and partially learned

through our interactions with our linguistic environment (Gleitman & Newport, 1995; Stork & Widdowson, 1974).

ATTACHMENT

Psychosocial development occurs as children form relationships, interact with others, and understand and manage their feelings. In social and emotional development, forming healthy attachments is very important and is the major social milestone of infancy. Attachment is a long-standing connection or bond with others. Developmental psychologists are interested in how infants reach this milestone. They ask such questions as: How do parent and infant attachment bonds form? How does neglect affect these bonds? What accounts for children's attachment differences?

Researchers Harry Harlow, John Bowlby, and Mary Ainsworth conducted studies designed to answer these questions. In the 1950s, Harlow conducted a series of experiments on monkeys. He separated newborn monkeys from their mothers. Each monkey was presented with two surrogate mothers. One surrogate monkey was made out of wire mesh, and she could dispense milk. The other monkey was softer and made from cloth: This monkey did not dispense milk. Research shows that the monkeys preferred the soft, cuddly cloth monkey, even though she did not provide any nourishment. The baby monkeys spent their time clinging to the cloth monkey and only went to the wire monkey when they needed to be fed. Prior to this study, the medical and scientific communities generally thought that babies become attached to the people who provide their nourishment. However, Harlow (1958) concluded that there was more to the mother-child bond than nourishment. Feelings of comfort and security are the critical components to maternal-infant bonding, which leads to healthy psychosocial development.

Harlow's studies of monkeys were performed before modern ethics guidelines were in place, and today his experiments are widely considered to be unethical and even cruel. Watch this video to see actual footage of Harlow's monkey studies: Harlow's Studies on Dependency in Monkeys.

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Building on the work of Harlow and others, John Bowlby developed the concept of attachment theory. He defined attachment as the affectional bond or tie that an infant forms with the mother (Bowlby, 1969). An infant must form this bond with a primary caregiver in order to have normal social and emotional development. In addition, Bowlby proposed that this attachment bond is very powerful and continues throughout life. He used the concept of secure base to define a healthy attachment between parent and child (1988). A secure base is a parental presence that

gives the child a sense of safety as he explores his surroundings. Bowlby said that two things are needed for a healthy attachment: The caregiver must be responsive to the child's physical, social, and emotional needs; and the caregiver and child must engage in mutually enjoyable interactions (Bowlby, 1969).

Mutually enjoyable interactions promote the mother-infant bond. (credit: Peter Shanks)

While Bowlby thought attachment was an all-or-nothing process, Mary Ainsworth's (1970) research showed otherwise. Ainsworth wanted to know if children differ in the ways they bond, and if so, why. To find the answers, she used the Strange Situation procedure to study attachment between mothers and their infants (1970). In the Strange Situation, the mother (or primary caregiver) and the infant (age 12-18 months) are placed in a room together. There are toys in the room, and the caregiver and child spend some time alone in the room. After the child has had time to explore her surroundings, a stranger enters the room. The mother then leaves her baby with the stranger. After a few minutes, she returns to comfort her child.

Based on how the infants/toddlers responded to the separation

and reunion, Ainsworth identified three types of parent-child attachments: secure, avoidant, and resistant (Ainsworth & Bell, 1970). A fourth style, known as disorganized attachment, was later described (Main & Solomon, 1990). The most common type of attachment—also considered the healthiest—is called secure attachment. In this type of attachment, the toddler prefers his parent over a stranger. The attachment figure is used as a secure base to explore the environment and is sought out in times of stress. Securely attached children were distressed when their caregivers left the room in the Strange Situation experiment, but when their caregivers returned, the securely attached children were happy to see them. Securely attached children have caregivers who are sensitive and responsive to their needs.

In secure attachment, the parent provides a secure base for the toddler, allowing him to securely explore his environment. (credit: Kerry Ceszyk)

With avoidant attachment, the child is unresponsive to the parent,

does not use the parent as a secure base, and does not care if the parent leaves. The toddler reacts to the parent the same way she reacts to a stranger. When the parent does return, the child is slow to show a positive reaction. Ainsworth theorized that these children were most likely to have a caregiver who was insensitive and inattentive to their needs (Ainsworth, Blehar, Waters, & Wall, 1978).

In cases of resistant attachment, children tend to show clingy behavior, but then they reject the attachment figure's attempts to interact with them (Ainsworth & Bell, 1970). These children do not explore the toys in the room, as they are too fearful. During separation in the Strange Situation, they became extremely disturbed and angry with the parent. When the parent returns, the children are difficult to comfort. Resistant attachment is the result of the caregivers' inconsistent level of response to their child.

Finally, children with disorganized attachment behaved oddly in the Strange Situation. They freeze, run around the room in an erratic manner, or try to run away when the caregiver returns (Main & Solomon, 1990). This type of attachment is seen most often in kids who have been abused. Research has shown that abuse disrupts a child's ability to regulate their emotions.

While Ainsworth's research has found support in subsequent studies, it has also met criticism. Some researchers have pointed out that a child's temperament may have a strong influence on attachment (Gervai, 2009; Harris, 2009), and others have noted that attachment varies from culture to culture, a factor not accounted for in Ainsworth's research (Rothbaum, Weisz, Pott, Miyake, & Morelli, 2000; van Ijzendoorn & Sagi-Schwartz, 2008).

Watch this video to view a clip of the Strange Situation. Try to identify which type of attachment baby Lisa exhibits: The Strange Situation – Mary Ainsworth.

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SELF-CONCEPT

Just as attachment is the main psychosocial milestone of infancy, the primary psychosocial milestone of childhood is the development of a positive sense of self. How does self-awareness develop? Infants don't have a self-concept, which is an understanding of who they are. If you place a baby in front of a mirror, she will reach out to touch her image, thinking it is another baby. However, by about 18 months a toddler will recognize that the person in the mirror is herself. How do we know this? In a well-known experiment, a researcher placed a red dot of paint

on children's noses before putting them in front of a mirror (Amsterdam, 1972). Commonly known as the mirror test, this behavior is demonstrated by humans and a few other species and is considered evidence of self-recognition (Archer, 1992). At 18 months old they would touch their own noses when they saw the paint, surprised to see a spot on their faces. By 24–36 months old children can name and/or point to themselves in pictures, clearly indicating self-recognition.

Children from 2–4 years old display a great increase in social behavior once they have established a self-concept. They enjoy playing with other children, but they have difficulty sharing their possessions. Also, through play children explore and come to understand their gender roles and can label themselves as a girl or boy (Chick, Heilman-Houser, & Hunter, 2002). By 4 years old, children can cooperate with other children, share when asked, and separate from parents with little anxiety. Children at this age also exhibit autonomy, initiate tasks, and carry out plans. Success in these areas contributes to a positive sense of self. Once children reach 6 years old, they can identify themselves in terms of group memberships: "I'm a first grader!" School-age children compare themselves to their peers and discover that they are competent in some areas and less so in others (recall Erikson's task of industry versus inferiority). At this age, children recognize their own personality traits as well as some other traits they would like to have. For example, 10-year-old Layla says, "I'm kind of shy. I wish I could be more talkative like my friend Alexa."

Development of a positive self-concept is important to healthy development. Children with a positive self-concept tend to be more confident, do better in school, act more independently, and are more willing to try new activities (Maccoby, 1980; Ferrer & Fugate, 2003). Formation of a positive self-concept begins in Erikson's toddlerhood stage, when children establish autonomy and become confident in their abilities. Development of self-concept continues in elementary school, when children compare themselves to

others. When the comparison is favorable, children feel a sense of competence and are motivated to work harder and accomplish more. Self-concept is re-evaluated in Erikson's adolescence stage, as teens form an identity. They internalize the messages they have received regarding their strengths and weaknesses, keeping some messages and rejecting others. Adolescents who have achieved identity formation are capable of contributing positively to society (Erikson, 1968).

What can parents do to nurture a healthy self-concept? Diana Baumrind (1971, 1991) thinks parenting style may be a factor. The way we parent is an important factor in a child's socioemotional growth. Baumrind developed and refined a theory describing four parenting styles: authoritative, authoritarian, permissive, and uninvolved. With the authoritative style, the parent gives reasonable demands and consistent limits, expresses warmth and affection, and listens to the child's point of view. Parents set rules and explain the reasons behind them. They are also flexible and willing to make exceptions to the rules in certain cases—for example, temporarily relaxing bedtime rules to allow for a nighttime swim during a family vacation. Of the four parenting styles, the authoritative style is the one that is most encouraged in modern American society. American children raised by authoritative parents tend to have high self-esteem and social skills. However, effective parenting styles vary as a function of culture and, as Small (1999) points out, the authoritative style is not necessarily preferred or appropriate in all cultures.

In authoritarian style, the parent places high value on conformity and obedience. The parents are often strict, tightly monitor their children, and express little warmth. In contrast to the authoritative style, authoritarian parents probably would not relax bedtime rules during a vacation because they consider the rules to be set, and they expect obedience. This style can create anxious, withdrawn, and unhappy kids. However, it is important to point out that authoritarian parenting is as beneficial as the authoritative style in

some ethnic groups (Russell, Crockett, & Chao, 2010). For instance, first-generation Chinese American children raised by authoritarian parents did just as well in school as their peers who were raised by authoritative parents (Russell et al., 2010).

For parents who employ the permissive style of parenting, the kids run the show and anything goes. Permissive parents make few demands and rarely use punishment. They tend to be very nurturing and loving, and may play the role of friend rather than parent. In terms of our example of vacation bedtimes, permissive parents might not have bedtime rules at all—instead they allow the child to choose his bedtime whether on vacation or not. Not surprisingly, children raised by permissive parents tend to lack self-discipline, and the permissive parenting style is negatively associated with grades (Dornbusch, Ritter, Leiderman, Roberts, & Fraleigh, 1987). The permissive style may also contribute to other risky behaviors such as alcohol abuse (Bahr & Hoffman, 2010), risky sexual behavior especially among female children (Donenberg, Wilson, Emerson, & Bryant, 2002), and increased display of disruptive behaviors by male children (Parent et al., 2011). However, there are some positive outcomes associated with children raised by permissive parents. They tend to have higher self-esteem, better social skills, and report lower levels of depression (Darling, 1999).

With the uninvolved style of parenting, the parents are indifferent, uninvolved, and sometimes referred to as neglectful. They don't respond to the child's needs and make relatively few demands. This could be because of severe depression or substance abuse, or other factors such as the parents' extreme focus on work. These parents may provide for the child's basic needs, but little else. The children raised in this parenting style are usually emotionally withdrawn, fearful, anxious, perform poorly in school, and are at an increased risk of substance abuse (Darling, 1999).

As you can see, parenting styles influence childhood adjustment, but could a child's temperament likewise influence parenting?

Temperament refers to innate traits that influence how one thinks, behaves, and reacts with the environment. Children with easy temperaments demonstrate positive emotions, adapt well to change, and are capable of regulating their emotions. Conversely, children with difficult temperaments demonstrate negative emotions and have difficulty adapting to change and regulating their emotions. Difficult children are much more likely to challenge parents, teachers, and other caregivers (Thomas, 1984). Therefore, it's possible that easy children (i.e., social, adaptable, and easy to soothe) tend to elicit warm and responsive parenting, while demanding, irritable, withdrawn children evoke irritation in their parents or cause their parents to withdraw (Sanson & Rothbart, 1995).

The Importance of Play and Recess

According to the American Academy of Pediatrics (2007), unstructured play is an integral part of a child's development. It builds creativity, problem solving skills, and social relationships. Play also allows children to develop a theory-of-mind as they imaginatively take on the perspective of others.

Outdoor play allows children the opportunity to directly experience and sense the world around them. While doing so, they may collect objects that they come across and develop lifelong interests and hobbies. They also benefit from increased exercise, and engaging in outdoor play can actually increase how much they

enjoy physical activity. This helps support the development of a healthy heart and brain. Unfortunately, research suggests that today's children are engaging in less and less outdoor play (Clements, 2004). Perhaps, it is no surprise to learn that lowered levels of physical activity in conjunction with easy access to calorie-dense foods with little nutritional value are contributing to alarming levels of childhood obesity (Karnik & Kanekar, 2012).

Despite the adverse consequences associated with reduced play, some children are over scheduled and have little free time to engage in unstructured play. In addition, some schools have taken away recess time for children in a push for students to do better on standardized tests, and many schools commonly use loss of recess as a form of punishment. Do you agree with these practices? Why or why not?

TEST YOUR UNDERSTANDING

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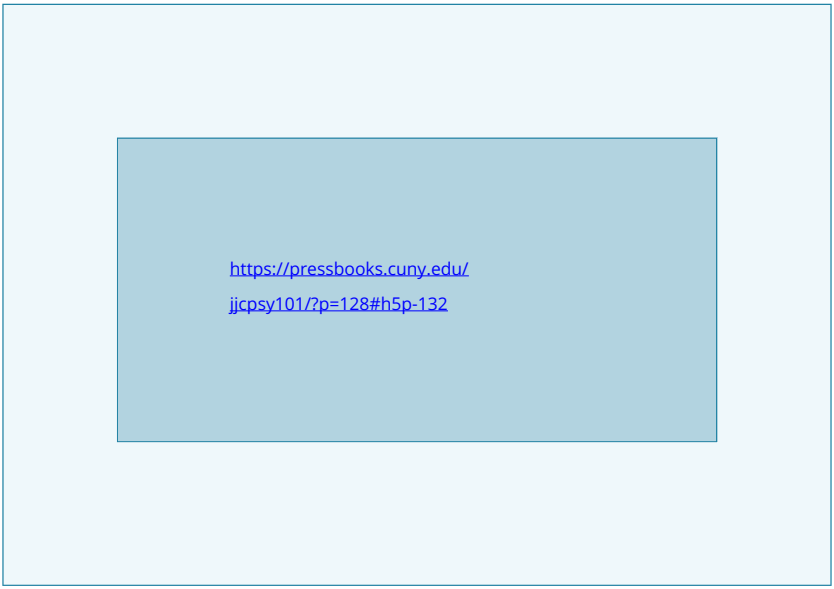


ADOLESCENCE

Adolescence is a socially constructed concept. In pre-industrial society, children were considered adults when they reached physical maturity, but today we have an extended time between childhood and adulthood called adolescence. Adolescence is the period of development that begins at puberty and ends at emerging adulthood, which is discussed later. In the United States, adolescence is seen as a time to develop independence from parents while remaining connected to them. The typical age range of adolescence is from 12 to 18 years, and this stage of development also has some predictable physical, cognitive, and psychosocial milestones.

Peers are a primary influence on our development in adolescence. (credit: Sheila Tostes)

PHYSICAL DEVELOPMENT

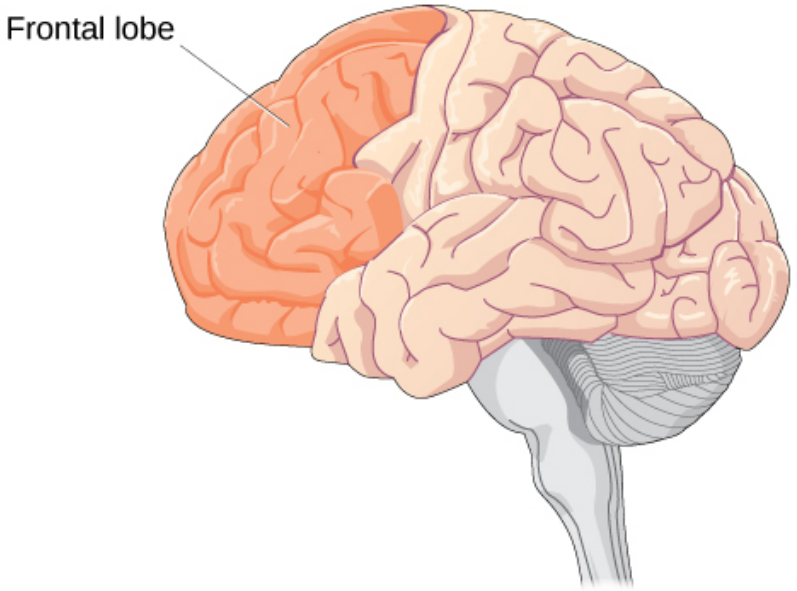
As noted above, adolescence begins with puberty. While the sequence of physical changes in puberty is predictable, the onset and pace of puberty vary widely. Several physical changes occur during puberty, such as adrenarche and gonadarche, the maturing of the adrenal glands and sex glands, respectively. Also during this time, primary and secondary sexual characteristics develop and mature. Primary sexual characteristics are organs specifically needed for reproduction, like the uterus and ovaries in females and testes in males. Secondary sexual characteristics are physical signs of sexual maturation that do not directly involve sex organs, such as development of breasts and hips in girls, and development of facial hair and a deepened voice in boys. Girls experience menarche, the beginning of menstrual periods, usually around 12–13 years old, and boys experience spermarche, the first ejaculation, around 13–14 years old.

During puberty, both sexes experience a rapid increase in height (i.e., growth spurt). For girls this begins between 8 and 13 years

old, with adult height reached between 10 and 16 years old. Boys begin their growth spurt slightly later, usually between 10 and 16 years old, and reach their adult height between 13 and 17 years old. Both nature (i.e., genes) and nurture (e.g., nutrition, medications, and medical conditions) can influence height.

Because rates of physical development vary so widely among teenagers, puberty can be a source of pride or embarrassment. Early maturing boys tend to be stronger, taller, and more athletic than their later maturing peers. They are usually more popular, confident, and independent, but they are also at a greater risk for substance abuse and early sexual activity (Flannery, Rowe, & Gulley, 1993; Kaltiala-Heino, Rimpela, Rissanen, & Rantanen, 2001). Early maturing girls may be teased or overtly admired, which can cause them to feel self-conscious about their developing bodies. These girls are at a higher risk for depression, substance abuse, and eating disorders (Ge, Conger, & Elder, 2001; Graber, Lewinsohn, Seeley, & Brooks-Gunn, 1997; Striegel-Moore & Cachelin, 1999). Late blooming boys and girls (i.e., they develop more slowly than their peers) may feel self-conscious about their lack of physical development. Negative feelings are particularly a problem for late maturing boys, who are at a higher risk for depression and conflict with parents (Graber et al., 1997) and more likely to be bullied (Pollack & Shuster, 2000).

The adolescent brain also remains under development. Up until puberty, brain cells continue to bloom in the frontal region. Adolescents engage in increased risk-taking behaviors and emotional outbursts possibly because the frontal lobes of their brains are still developing. Recall that this area is responsible for judgment, impulse control, and planning, and it is still maturing into early adulthood (Casey, Tottenham, Liston, & Durston, 2005).



Brain growth continues into the early 20s. The development of the frontal lobe, in particular, is important during this stage.

According to neuroscientist Jay Giedd in the Frontline video "Inside the Teenage Brain" (2013), "It's sort of unfair to expect [teens] to have adult levels of organizational skills or decision-making before their brains are finished being built." Watch this segment on "The Wiring of the Adolescent Brain" to find out more about the developing brain during adolescence: The Teenage Brain.

COGNITIVE DEVELOPMENT

More complex thinking abilities emerge during adolescence. Some researchers suggest this is due to increases in processing speed and efficiency rather than as the result of an increase in mental capacity—in other words, due to improvements in existing skills rather than development of new ones (Bjorkland, 1987; Case, 1985). During adolescence, teenagers move beyond concrete thinking and become capable of abstract thought. Recall that Piaget refers to this stage as formal operational thought. Teen thinking is also characterized by the ability to consider multiple points of view, imagine hypothetical situations, debate ideas and opinions (e.g., politics, religion, and justice), and form new ideas. In addition, it's not uncommon for adolescents to question authority or challenge established societal norms.

Cognitive empathy, also known as theory-of-mind (which we discussed earlier with regard to egocentrism), relates to the ability to take the perspective of others and feel concern for others (Shamay-Tsoory, Tomer, & Aharon-Peretz, 2005). Cognitive empathy begins to increase in adolescence and is an important component of social problem solving and conflict avoidance. According to one longitudinal study, levels of cognitive empathy begin rising in girls around 13 years old, and around 15 years old in boys (Van der Graaff et al., 2013). Teens who reported having supportive fathers with whom they could discuss their worries were found to be better able to take the perspective of others (Miklikowska, Duriez, & Soenens, 2011).

Teenage thinking is characterized by the ability to reason logically and solve hypothetical problems such as how to design, plan, and build a structure. (credit: U.S. Army RDECOM)

PSYCHOSOCIAL DEVELOPMENT

Adolescents continue to refine their sense of self as they relate to others. Erikson referred to the task of the adolescent as one of identity versus role confusion. Thus, in Erikson's view, an adolescent's main questions are "Who am I?" and "Who do I want to be?" Some adolescents adopt the values and roles that their parents expect for them. Other teens develop identities that are in opposition to their parents but align with a peer group. This is common as peer relationships become a central focus in adolescents' lives.

As adolescents work to form their identities, they pull away from their parents, and the peer group becomes very important (Shanahan, McHale, Osgood, & Crouter, 2007). Despite spending less time with their parents, most teens report positive feelings toward them (Moore, Guzman, Hair, Lippman, & Garrett, 2004). Warm and healthy parent-child relationships have been associated

with positive child outcomes, such as better grades and fewer school behavior problems, in the United States as well as in other countries (Hair et al., 2005).

It appears that most teens don't experience adolescent storm and stress to the degree once famously suggested by G. Stanley Hall, a pioneer in the study of adolescent development. Only small numbers of teens have major conflicts with their parents (Steinberg & Morris, 2001), and most disagreements are minor. For example, in a study of over 1,800 parents of adolescents from various cultural and ethnic groups, Barber (1994) found that conflicts occurred over day-to-day issues such as homework, money, curfews, clothing, chores, and friends. These types of arguments tend to decrease as teens develop (Galambos & Almeida, 1992).

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EMERGING ADULTHOOD

The next stage of development is emerging adulthood. This is a relatively newly defined period of lifespan development spanning from 18 years old to the mid-20s, characterized as an in-between time where identity exploration is focused on work and love.

When does a person become an adult? There are many ways to answer this question. In the United States, you are legally considered an adult at 18 years old. But other definitions of adulthood vary widely; in sociology, for example, a person may be considered an adult when she becomes self-supporting, chooses a career, gets married, or starts a family. The ages at which we achieve these milestones vary from person to person as well as from culture to culture. For example, in the African country of Malawi, 15-year-old Njemile was married at 14 years old and had her first child at 15 years old. In her culture she is considered an adult. Children in Malawi take on adult responsibilities such as marriage and work (e.g., carrying water, tending babies, and working fields) as early as 10 years old. In stark contrast, independence in Western cultures is taking longer and longer, effectively delaying the onset of adult life.

Why is it taking twentysomethings so long to grow up? It seems that emerging adulthood is a product of both Western culture and our current times (Arnett, 2000). People in developed countries are living longer, allowing the freedom to take an extra decade to start a career and family. Changes in the workforce also play a role. For example, 50 years ago, a young adult with a high school diploma could immediately enter the work force and climb the corporate ladder. That is no longer the case. Bachelor's and even graduate degrees are required more and more often—even for entry-level jobs (Arnett, 2000). In addition, many students are taking longer (five or six years) to complete a college degree as a result of working and going to school at the same time. After graduation, many young adults return to the family home because they have

difficulty finding a job. Changing cultural expectations may be the most important reason for the delay in entering adult roles. Young people are spending more time exploring their options, so they are delaying marriage and work as they change majors and jobs multiple times, putting them on a much later timetable than their parents (Arnett, 2000).

ADULTHOOD

Adulthood begins around 20 years old and has three distinct stages: early, middle, and late. Each stage brings its own set of rewards and challenges.

PHYSICAL DEVELOPMENT

By the time we reach early adulthood (20 to early 40s), our physical maturation is complete, although our height and weight may increase slightly. In young adulthood, our physical abilities are at their peak, including muscle strength, reaction time, sensory abilities, and cardiac functioning. Most professional athletes are at the top of their game during this stage. Many women have children in the young adulthood years, so they may see additional weight gain and breast changes.

Middle adulthood extends from the 40s to the 60s. Physical decline is gradual. The skin loses some elasticity, and wrinkles are among the first signs of aging. Visual acuity decreases during this time. Women experience a gradual decline in fertility as they approach the onset of menopause, the end of the menstrual cycle, around 50 years old. Both men and women tend to gain weight: in the abdominal area for men and in the hips and thighs for women. Hair begins to thin and turn gray.

Physical declines of middle and late adulthood can be minimized with proper exercise, nutrition, and an active lifestyle. (credit: modification of work by Peter Stevens)

Late adulthood is considered to extend from the 60s on. This is the last stage of physical change. The skin continues to lose elasticity, reaction time slows further, and muscle strength diminishes. Smell, taste, hearing, and vision, so sharp in our twenties, decline significantly. The brain may also no longer function at optimal levels, leading to problems like memory loss, dementia, and Alzheimer's disease in later years.

Aging doesn't mean a person can't explore new pursuits, learn new skills, and continue to grow. Watch this inspiring story about Neil Unger who is a newbie

to the world of skateboarding at 60 years old: NEAL UNGER – 60 YEAR OLD SKATEBOARDER.

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COGNITIVE DEVELOPMENT

Because we spend so many years in adulthood (more than any other stage), cognitive changes are numerous. In fact, research suggests that adult cognitive development is a complex, ever changing process that may be even more active than cognitive development in infancy and early childhood (Fischer, Yan, & Stewart, 2003).

There is good news for the middle age brain. View this brief video to find out what it is: Middle Age Brains – NJN News Healthwatch Report.

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Unlike our physical abilities, which peak in our mid-20s and then begin a slow decline, our cognitive abilities remain steady throughout early and middle adulthood. Our crystalized intelligence (information, skills, and strategies we have gathered through a lifetime of experience) tends to hold steady as we age—it may even improve. For example, adults show relatively stable to increasing scores on intelligence tests until their mid-30s to mid-50s (Bayley & Oden, 1955). However, in late adulthood we begin to experience a decline in another area of our cognitive abilities—fluid intelligence (information processing abilities, reasoning, and memory). These processes become slower. How can we delay the onset of cognitive decline? Mental and physical

activity seems to play a part. Research has found adults who engage in mentally and physically stimulating activities experience less cognitive decline and have a reduced incidence of mild cognitive impairment and dementia (Hertzog, Kramer, Wilson, & Lindenberger, 2009; Larson et al., 2006; Podewils et al., 2005).

Cognitive activities such as playing mahjong, chess, or other games, can keep you mentally fit. The same is true for solo pastimes like reading and completing crossword puzzles. (credit: Philippe Put)

PSYCHOSOCIAL DEVELOPMENT

There are many theories about the social and emotional aspects of aging. Some aspects of healthy aging include activities, social connectedness, and the role of a person's culture. According to many theorists, including George Vaillant (2002), who studied and analyzed over 50 years of data, we need to have and continue to find meaning throughout our lives. For those in early and middle adulthood, meaning is found through work (Sterns & Huyck, 2001) and family life (Markus, Ryff, Curan, & Palmersheim, 2004). These areas relate to the tasks that Erikson referred to as generativity

and intimacy. As mentioned previously, adults tend to define themselves by what they do—their careers. Earnings peak during this time, yet job satisfaction is more closely tied to work that involves contact with other people, is interesting, provides opportunities for advancement, and allows some independence (Mohr & Zoghi, 2006) than it is to salary (Iyengar, Wells, & Schwartz, 2006). How might being unemployed or being in a dead-end job challenge adult well-being?

Positive relationships with significant others in our adult years have been found to contribute to a state of well-being (Ryff & Singer, 2009). Most adults in the United States identify themselves through their relationships with family—particularly with spouses, children, and parents (Markus et al., 2004). While raising children can be stressful, especially when they are young, research suggests that parents reap the rewards down the road, as adult children tend to have a positive effect on parental well-being (Umberson, Pudrovska, & Reczek, 2010). Having a stable marriage has also been found to contribute to well-being throughout adulthood (Vaillant, 2002).

Another aspect of positive aging is believed to be social connectedness and social support. As we get older, socioemotional selectivity theory suggests that our social support and friendships dwindle in number, but remain as close, if not more close than in our earlier years (Carstensen, 1992).

Social support is important as we age. (credit: Gabriel Rocha)

To learn more, view this video on aging in America: The Many Faces of Aging In America.

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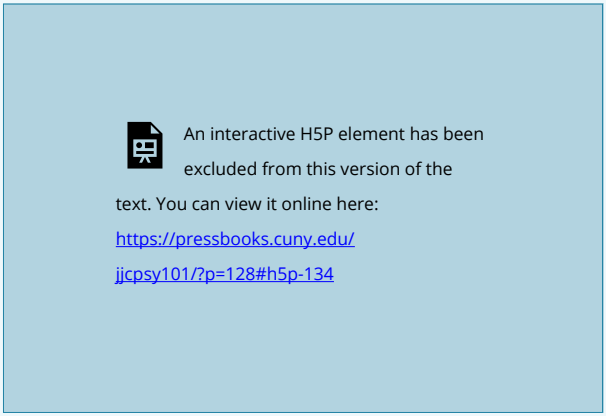
SUMMARY

At conception the egg and sperm cell are united to form a zygote, which will begin to divide rapidly. This marks the beginning of the first stage of prenatal development (germinal stage), which lasts about two weeks. Then the zygote implants itself into the lining of the woman's uterus, marking the beginning of the second stage of prenatal development (embryonic stage), which lasts about six weeks. The embryo begins to develop body and organ structures, and the neural tube forms, which will later become the brain and spinal cord. The third phase of prenatal development (fetal stage) begins at 9 weeks and lasts until birth. The body, brain, and organs grow rapidly during this stage. During all stages of pregnancy it is important that the mother receive prenatal care to reduce health risks to herself and to her developing baby.

Newborn infants weigh about 7.5 pounds. Doctors assess a newborn's reflexes, such as the sucking, rooting, and Moro reflexes. Our physical, cognitive, and psychosocial skills grow and change as we move through developmental stages from infancy through late adulthood. Attachment in infancy is a critical component of healthy development. Parenting styles have been found to have an effect on childhood outcomes of well-being. The transition from adolescence to adulthood can be challenging due to the timing of puberty, and due to the extended amount of time spent in emerging adulthood. Although physical decline begins in middle adulthood, cognitive decline does not begin until later. Activities that keep the body and mind active can help maintain good physical and cognitive health as we age. Social supports through family and friends remain important as we age.

REVIEW QUESTIONS



CRITICAL THINKING QUESTIONS

What are some known teratogens, and what kind of damage can they do to the developing fetus?

Alcohol is a teratogen. Excessive drinking can cause

intellectual disabilities in children. The child can also have a small head and abnormal facial features, which are characteristic of fetal alcohol syndrome (FAS). Another teratogen is nicotine. Smoking while pregnant can lead to low-birth weight, premature birth, stillbirth, and SIDS.

What is prenatal care and why is it important?

Prenatal care is medical care during pregnancy that monitors the health of both the mother and fetus. It's important to receive prenatal care because it can reduce complications to the mother and fetus during pregnancy.

Describe what happens in the embryonic stage of development. Describe what happens in the fetal stage of development.

In the embryonic stage, basic structures of the embryo start to develop into areas that will become the head, chest, and abdomen. The heart begins to beat and organs form and begin to function. The neural tube forms along the back of the embryo, developing into the spinal cord and brain. In the fetal stage, the brain and body continue to develop. Fingers and toes develop along with hearing, and internal organs form.

What makes a personal quality part of someone's

personality?

The particular quality or trait must be part of an enduring behavior pattern, so that it is a consistent or predictable quality.

Describe some of the newborn reflexes. How might they promote survival?

The sucking reflex is the automatic, unlearned sucking motions that infants do with their mouths. It may help promote survival because this action helps the baby take in nourishment. The rooting reflex is the newborn's response to anything that touches her cheek. When you stroke a baby's cheek she will naturally turn her head that way and begin to suck. This may aid survival because it helps the newborn locate a source of food.

Compare and contrast the four parenting styles and describe the kinds of childhood outcomes we can expect with each.

With the authoritative style, children are given reasonable demands and consistent limits, warmth and affection are expressed, the parent listens to the child's point of view, and the child initiates positive standards. Children raised by authoritative parents tend to have high self-esteem and social skills. Another parenting style is authoritarian: The parent places a high value on conformity and obedience. The

parents are often strict, tightly monitor their children, and express little warmth. This style can create anxious, withdrawn, and unhappy kids. The third parenting style is permissive: Parents make few demands, rarely use punishment, and give their children free rein. Children raised by permissive parents tend to lack self-discipline, which contributes to poor grades and alcohol abuse. However, they have higher self-esteem, better social skills, and lower levels of depression. The fourth style is the uninvolved parent: They are indifferent, uninvolved, and sometimes called neglectful. The children raised in this parenting style are usually emotionally withdrawn, fearful, anxious, perform poorly in school, and are at an increased risk of substance abuse.

What is emerging adulthood and what are some factors that have contributed to this new stage of development?

Emerging adulthood is a relatively new period of lifespan development from 18 years old to the mid-20s, characterized as a transitional time in which identity exploration focuses on work and love. According to Arnett, changing cultural expectations facilitate the delay to full adulthood. People are spending more time exploring their options, so they are delaying marriage and work as they change

majors and jobs multiple times, putting them on a much later timetable than their parents.

PERSONAL APPLICATION QUESTIONS

1. Which parenting style describes how you were raised? Provide an example or two to support your answer.
2. Would you describe your experience of puberty as one of pride or embarrassment? Why?
3. Your best friend is a smoker who just found out she is pregnant. What would you tell her about smoking and pregnancy?
4. Imagine you are a nurse working at a clinic that provides prenatal care for pregnant women. Your patient, Anna, has heard that it's a good idea to play music for her unborn baby, and she wants to know when her baby's hearing will develop. What will you tell her?

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Discuss hospice care
- Describe the five stages of grief
- Define living will and DNR

Every story has an ending. Death marks the end of your life story. Our culture and individual backgrounds influence how we view death. In some cultures, death is accepted as a natural part of life and is embraced. In contrast, until about 50 years ago in the United States, a doctor might not inform someone that they were dying, and the majority of deaths occurred in hospitals. In 1967 that reality began to change with Cicely Saunders, who created the first modern hospice in England. The aim of hospice is to help provide a death with dignity and pain management in a humane and comfortable environment, which is usually outside of a hospital setting. In 1974, Florence Wald founded the first hospice in the United States. Today, hospice provides care for 1.65 million Americans and their families. Because of hospice care, many terminally ill people are able to spend their last days at home.

In some cultures, people's bodies may be buried in a cemetery after death. (credit: Christina Rutz)

Research has indicated that hospice care is beneficial for the patient (Brumley, Enquidanos, & Cherin, 2003; Brumley et al., 2007; Godkin, Krant, & Doster, 1984) and for the patient's family (Rhodes, Mitchell, Miller, Connor, & Teno, 2008; Godkin et al., 1984). Hospice patients report high levels of satisfaction with hospice care because they are able to remain at home and are not completely dependent on strangers for care (Brumley et al., 2007). In addition, hospice patients tend to live longer than non-hospice patients (Connor, Pyenson, Fitch, Spence, & Iwasaki, 2007; Temel et al., 2010). Family members receive emotional support and are regularly informed of their loved one's treatment and condition. The family member's burden of care is also reduced (McMillan et al., 2006). Both the patient and the patient's family members report increased family support, increased social support, and improved coping while receiving hospice services (Godkin et al., 1984).

How do you think you might react if you were diagnosed with a terminal illness like cancer? Elizabeth Kübler-Ross (1969), who worked with the founders of hospice care, described the process of

an individual accepting his own death. She proposed five stages of grief: denial, anger, bargaining, depression, and acceptance. Most individuals experience these stages, but the stages may occur in different orders, depending on the individual. In addition, not all people experience all of the stages. It is also important to note that some psychologists believe that the more a dying person fights death, the more likely he is to remain stuck in the denial phase. This could make it difficult for the dying person to face death with dignity. However, other psychologists believe that not facing death until the very end is an adaptive coping mechanism for some people.

Whether due to illness or old age, not everyone facing death or the loss of a loved one experiences the negative emotions outlined in the Kübler-Ross model (Nolen-Hoeksema & Larson, 1999). For example, research suggests that people with religious or spiritual beliefs are better able to cope with death because of their hope in an afterlife and because of social support from religious or spiritual associations (Hood, Spilka, Hunsberger, & Corsuch, 1996; McIntosh, Silver, & Wortman, 1993; Paloutzian, 1996; Samarel, 1991; Wortman & Park, 2008).

A prominent example of a person creating meaning through death is Randy Pausch, who was a well-loved and respected professor at Carnegie Mellon University. Diagnosed with terminal pancreatic cancer in his mid-40s and given only 3–6 months to live, Pausch focused on living in a fulfilling way in the time he had left. Instead of becoming angry and depressed, he presented his now famous last lecture called “Really Achieving Your Childhood Dreams.” In his moving, yet humorous talk, he shares his insights on seeing the good in others, overcoming obstacles, and experiencing zero gravity, among many other things. Despite his terminal diagnosis, Pausch lived the final year of his life with joy and hope, showing us that our plans for the future still matter, even if we know that we are dying.

Really Achieving Your Childhood Dreams is Randy Pausch's last lecture. Listen to his inspiring talk: Randy Pausch Last Lecture: Achieving Your Childhood Dreams.

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SUMMARY

Death marks the endpoint of our lifespan. There are many ways that we might react when facing death. Kübler-Ross developed a five-stage model of grief as a way to explain this process. Many people facing death choose hospice care, which allows their last days to be spent at home in a comfortable, supportive environment.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Describe the five stages of grief and provide examples of how a person might react in each stage.

The first stage is denial. The person receives news

that he is dying, and either does not take it seriously or tries to escape from the reality of the situation. He might say something like, "Cancer could never happen to me. I take good care of myself. This has to be a mistake." The next stage is anger. He realizes time is short, and he may not have a chance to accomplish what he wanted in life. "It's not fair. I promised my grandchildren that we would go to Disney World, and now I'll never have the chance to take them." The third stage is bargaining. In this stage, he tries to delay the inevitable by bargaining or pleading for extra time, usually with God, family members, or medical care providers. "God, just give me one more year so I can take that trip with my grandchildren. They're too young to understand what's happening and why I can't take them." The fourth stage is depression. He becomes sad about his impending death. "I can't believe this is how I'm going to die. I'm in so much pain. What's going to become of my family when I'm gone?" The final stage is acceptance. This stage is usually reached in the last few days or weeks before death. He recognizes that death is inevitable. "I need to get everything in order and say all of my good-byes to the people I love."

What is the purpose of hospice care?

Hospice is a program of services that provide

medical, social, and spiritual support for dying people and their families.

PERSONAL APPLICATION QUESTIONS

1. Have you ever had to cope with the loss of a loved one? If so, what concepts described in this section provide context that may help you understand your experience and process of grieving?
2. If you were diagnosed with a terminal illness would you choose hospice care or a traditional death in a hospital? Why?

GLOSSARY

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EMOTION AND MOTIVATION

Emotions can change in an instant, especially in response to an unexpected event. Surprise, fear, anger, and sadness are some immediate emotions that people experienced in the aftermath of the April 15, 2013 Boston Marathon bombing. What are emotions? What causes them? What motivated some bystanders to immediately help others, while other people ran for safety? (credit: modification of work by Aaron "tango" Tang)

What makes us behave as we do? What drives us to eat? What drives us toward sex? Is there a biological basis to explain the feelings we experience? How universal are emotions?

In this chapter, we will explore issues relating to both motivation and emotion. We will begin with a discussion of several theories

that have been proposed to explain motivation and why we engage in a given behavior. You will learn about the physiological needs that drive some human behaviors, as well as the importance of our social experiences in influencing our actions.

Next, we will consider both eating and having sex as examples of motivated behaviors. What are the physiological mechanisms of hunger and satiety? What understanding do scientists have of why obesity occurs, and what treatments exist for obesity and eating disorders? How has research into human sex and sexuality evolved over the past century? How do psychologists understand and study the human experience of sexual orientation and gender identity? These questions—and more—will be explored.

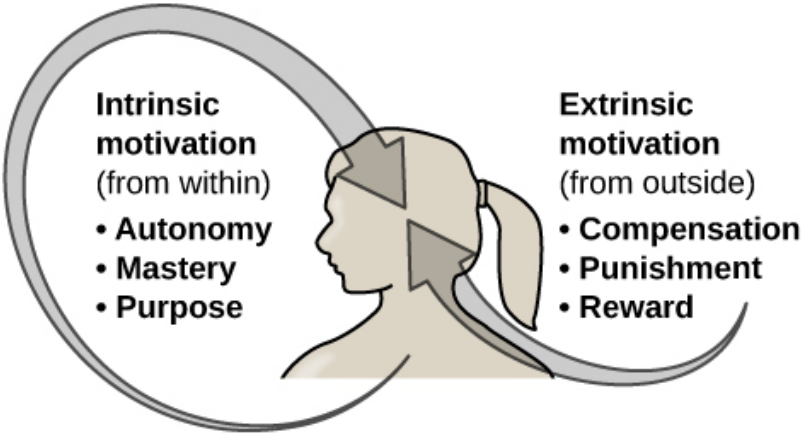
This chapter will close with a discussion of emotion. You will learn about several theories that have been proposed to explain how emotion occurs, the biological underpinnings of emotion, and the universality of emotions.

Learning Objectives

By the end of this section, you will be able to:

- Define intrinsic and extrinsic motivation
- Understand that instincts, drive reduction, self-efficacy, and social motives have all been proposed as theories of motivation
- Explain the basic concepts associated with Maslow's hierarchy of needs

Why do we do the things we do? What motivations underlie our behaviors? Motivation describes the wants or needs that direct behavior toward a goal. In addition to biological motives, motivations can be intrinsic (arising from internal factors) or extrinsic (arising from external factors). Intrinsically motivated behaviors are performed because of the sense of personal satisfaction that they bring, while extrinsically motivated behaviors are performed in order to receive something from others.



Intrinsic motivation comes from within the individual, while extrinsic motivation comes from outside the individual.

Think about why you are currently in college. Are you here because you enjoy learning and want to pursue an education to make yourself a more well-rounded individual? If so, then you are intrinsically motivated. However, if you are here because you want to get a college degree to make yourself more marketable for a high-paying career or to satisfy the demands of your parents, then your motivation is more extrinsic in nature.

In reality, our motivations are often a mix of both intrinsic and extrinsic factors, but the nature of the mix of these factors might change over time (often in ways that seem counter-intuitive). There is an old adage: "Choose a job that you love, and you will never have to work a day in your life," meaning that if you enjoy your occupation, work doesn't seem like . . . well, work. Some research suggests that this isn't necessarily the case (Daniel & Esser, 1980; Deci, 1972; Deci, Koestner, & Ryan, 1999). According to this research, receiving some sort of extrinsic reinforcement (i.e., getting paid) for engaging in behaviors that we enjoy leads to those behaviors being thought of as work no longer providing that same enjoyment. As a result, we might spend less time engaging in these

reclassified behaviors in the absence of any extrinsic reinforcement. For example, Odessa loves baking, so in her free time, she bakes for fun. Oftentimes, after stocking shelves at her grocery store job, she often whips up pastries in the evenings because she enjoys baking. When a coworker in the store's bakery department leaves his job, Odessa applies for his position and gets transferred to the bakery department. Although she enjoys what she does in her new job, after a few months, she no longer has much desire to concoct tasty treats in her free time. Baking has become work in a way that changes her motivation to do it. What Odessa has experienced is called the overjustification effect—intrinsic motivation is diminished when extrinsic motivation is given. This can lead to extinguishing the intrinsic motivation and creating a dependence on extrinsic rewards for continued performance (Deci et al., 1999).

Research suggests that when something we love to do, like icing cakes, becomes our job, our intrinsic and extrinsic motivations to do it may change. (credit: Agustín Ruiz)

Other studies suggest that intrinsic motivation may not be so vulnerable to the effects of extrinsic reinforcements, and in fact, reinforcements such as verbal praise might actually increase

intrinsic motivation (Arnold, 1976; Cameron & Pierce, 1994). In that case, Odessa's motivation to bake in her free time might remain high if, for example, customers regularly compliment her baking or cake decorating skills.

These apparent discrepancies in the researchers' findings may be understood by considering several factors. For one, physical reinforcement (such as money) and verbal reinforcement (such as praise) may affect an individual in very different ways. In fact, tangible rewards (i.e., money) tend to have more negative effects on intrinsic motivation than do intangible rewards (i.e., praise). Furthermore, the expectation of the extrinsic motivator by an individual is crucial: If the person expects to receive an extrinsic reward, then intrinsic motivation for the task tends to be reduced. If, however, there is no such expectation, and the extrinsic motivation is presented as a surprise, then intrinsic motivation for the task tends to persist (Deci et al., 1999).

In educational settings, students are more likely to experience intrinsic motivation to learn when they feel a sense of belonging and respect in the classroom. This internalization can be enhanced if the evaluative aspects of the classroom are de-emphasized and if students feel that they exercise some control over the learning environment. Furthermore, providing students with activities that are challenging, yet doable, along with a rationale for engaging in various learning activities can enhance intrinsic motivation for those tasks (Niemiec & Ryan, 2009). Consider Hakim, a first-year law student with two courses this semester: Family Law and Criminal Law. The Family Law professor has a rather intimidating classroom: He likes to put students on the spot with tough questions, which often leaves students feeling belittled or embarrassed. Grades are based exclusively on quizzes and exams, and the instructor posts results of each test on the classroom door. In contrast, the Criminal Law professor facilitates classroom discussions and respectful debates in small groups. The majority of the course grade is not exam-based, but centers on a student-designed research project

on a crime issue of the student's choice. Research suggests that Hakim will be less intrinsically motivated in his Family Law course, where students are intimidated in the classroom setting, and there is an emphasis on teacher-driven evaluations. Hakim is likely to experience a higher level of intrinsic motivation in his Criminal Law course, where the class setting encourages inclusive collaboration and a respect for ideas, and where students have more influence over their learning activities.

THEORIES ABOUT MOTIVATION

William James (1842–1910) was an important contributor to early research into motivation, and he is often referred to as the father of psychology in the United States. James theorized that behavior was driven by a number of instincts, which aid survival. From a biological perspective, an instinct is a species-specific pattern of behavior that is not learned. There was, however, considerable controversy among James and his contemporaries over the exact definition of instinct. James proposed several dozen special human instincts, but many of his contemporaries had their own lists that differed. A mother's protection of her baby, the urge to lick sugar, and hunting prey were among the human behaviors proposed as true instincts during James's era. This view—that human behavior is driven by instincts—received a fair amount of criticism because of the undeniable role of learning in shaping all sorts of human behavior. In fact, as early as the 1900s, some instinctive behaviors were experimentally demonstrated to result from associative learning (recall when you learned about Watson's conditioning of fear response in "Little Albert") (Faris, 1921).

(a) (b)

(a) William James proposed the instinct theory of motivation, asserting that behavior is driven by instincts. (b) In humans, instincts may include behaviors such as an infant's rooting for a nipple and sucking. (credit b: modification of work by "Mothering Touch"/Flickr)

Another early theory of motivation proposed that the maintenance of homeostasis is particularly important in directing behavior. You may recall from your earlier reading that homeostasis is the tendency to maintain a balance, or optimal level, within a biological system. In a body system, a control center (which is often part of the brain) receives input from receptors (which are often complexes of neurons). The control center directs effectors (which may be other neurons) to correct any imbalance detected by the control center.

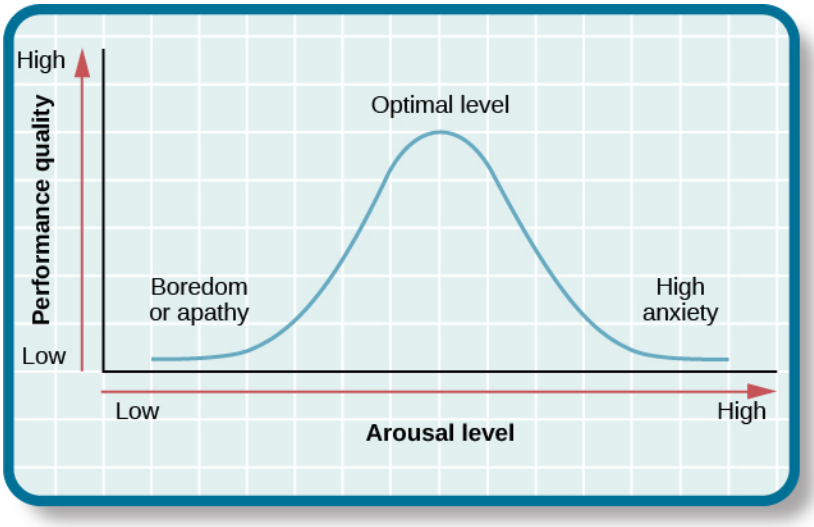
According to the drive theory of motivation, deviations from homeostasis create physiological needs. These needs result in psychological drive states that direct behavior to meet the need and, ultimately, bring the system back to homeostasis. For example, if it's been a while since you ate, your blood sugar levels will drop below normal. This low blood sugar will induce a physiological need and a corresponding drive state (i.e., hunger)

that will direct you to seek out and consume food. Eating will eliminate the hunger, and, ultimately, your blood sugar levels will return to normal. Interestingly, drive theory also emphasizes the role that habits play in the type of behavioral response in which we engage. A habit is a pattern of behavior in which we regularly engage. Once we have engaged in a behavior that successfully reduces a drive, we are more likely to engage in that behavior whenever faced with that drive in the future (Graham & Weiner, 1996).

Hunger and subsequent eating are the result of complex physiological processes that maintain homeostasis. (credit "left": modification of work by "Gracie and Viv"/Flickr; credit "center": modification of work by Steven Depolo; credit "right": modification of work by Monica Renata)

Extensions of drive theory take into account levels of arousal as potential motivators. As you recall from your study of learning, these theories assert that there is an optimal level of arousal that we all try to maintain. If we are underaroused, we become bored and will seek out some sort of stimulation. On the other hand, if we are overaroused, we will engage in behaviors to reduce our arousal (Berlyne, 1960). Most students have experienced this need to maintain optimal levels of arousal over the course of their academic career. Think about how much stress students experience toward the end of spring semester. They feel overwhelmed with seemingly endless exams, papers, and major assignments that must be completed on time. They probably yearn for the rest and relaxation that awaits them over the extended summer break. However, once they finish the semester, it doesn't

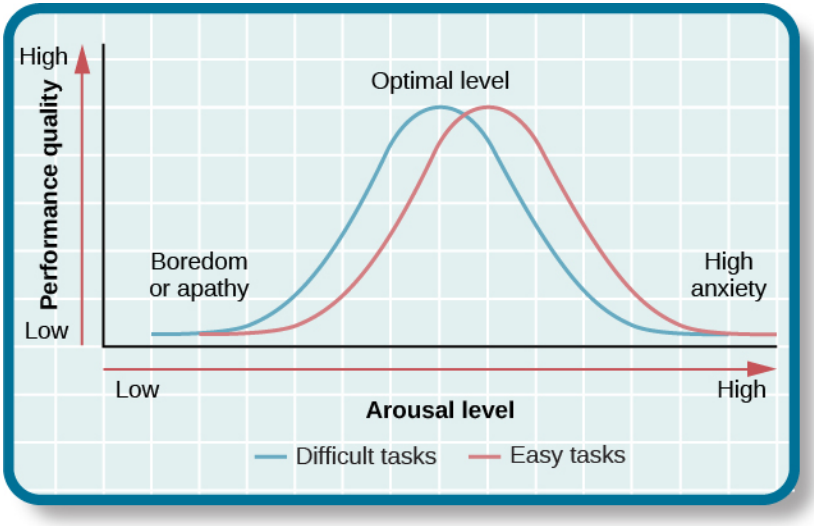
take too long before they begin to feel bored. Generally, by the time the next semester is beginning in the fall, many students are quite happy to return to school. This is an example of how arousal theory works.



The concept of optimal arousal in relation to performance on a task is depicted here. Performance is maximized at the optimal level of arousal, and it tapers off during under- and overarousal.

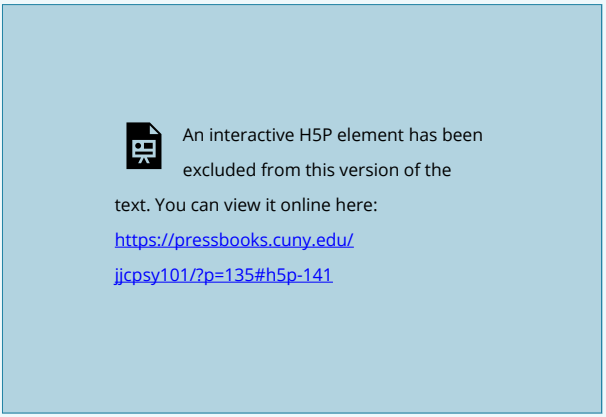
So what is the optimal level of arousal? What level leads to the best performance? Research shows that moderate arousal is generally best; when arousal is very high or very low, performance tends to suffer (Yerkes & Dodson, 1908). Think of your arousal level regarding taking an exam for this class. If your level is very low, such as boredom and apathy, your performance will likely suffer. Similarly, a very high level, such as extreme anxiety, can be paralyzing and hinder performance. Consider the example of a softball team facing a tournament. They are favored to win their first game by a large margin, so they go into the game with a lower level of arousal and get beat by a less skilled team.

But optimal arousal level is more complex than a simple answer that the middle level is always best. Researchers Robert Yerkes (pronounced "Yerk-EES") and John Dodson discovered that the optimal arousal level depends on the complexity and difficulty of the task to be performed. This relationship is known as Yerkes-Dodson law, which holds that a simple task is performed best when arousal levels are relatively high and complex tasks are best performed when arousal levels are lower.



Task performance is best when arousal levels are in a middle range, with difficult tasks best performed under lower levels of arousal and simple tasks best performed under higher levels of arousal.

TEST YOUR UNDERSTANDING



SELF-EFFICACY AND SOCIAL MOTIVES

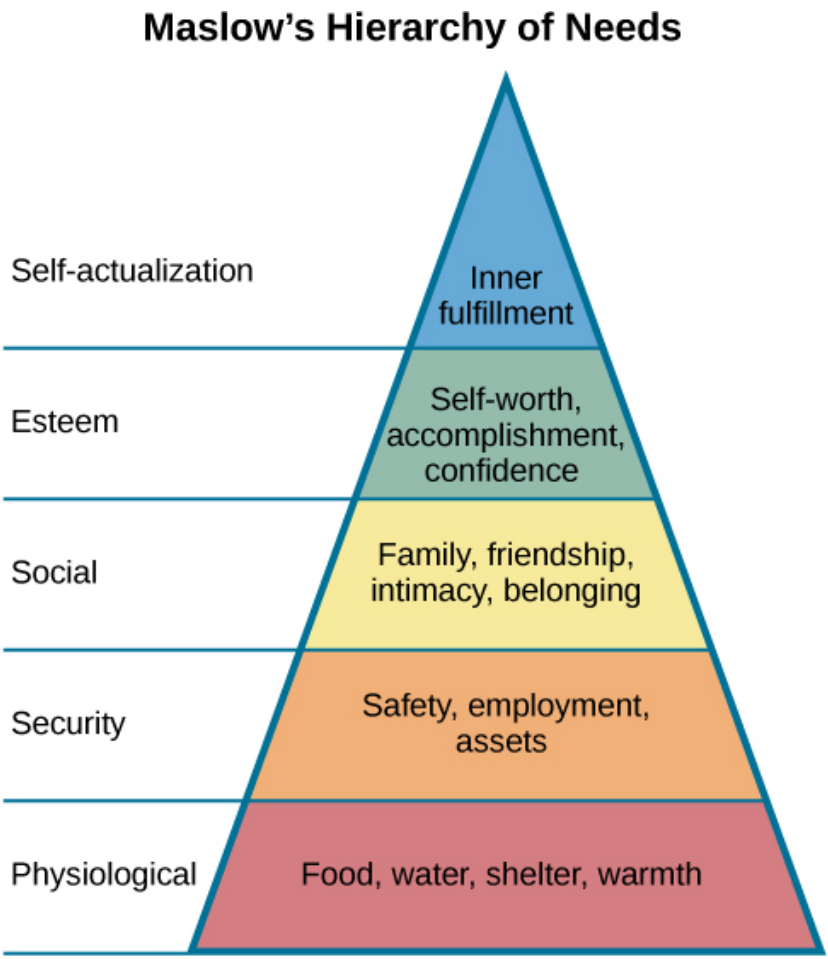
Self-efficacy is an individual's belief in her own capability to complete a task, which may include a previous successful completion of the exact task or a similar task. Albert Bandura (1994) theorized that an individual's sense of self-efficacy plays a pivotal role in motivating behavior. Bandura argues that motivation derives from expectations that we have about the consequences of our behaviors, and ultimately, it is the appreciation of our capacity to engage in a given behavior that will determine what we do and the future goals that we set for ourselves. For example, if you have

a sincere belief in your ability to achieve at the highest level, you are more likely to take on challenging tasks and to not let setbacks dissuade you from seeing the task through to the end.

A number of theorists have focused their research on understanding social motives (McAdams & Constantian, 1983; McClelland & Liberman, 1949; Murray et al., 1938). Among the motives they describe are needs for achievement, affiliation, and intimacy. It is the need for achievement that drives accomplishment and performance. The need for affiliation encourages positive interactions with others, and the need for intimacy causes us to seek deep, meaningful relationships. Henry Murray et al. (1938) categorized these needs into domains. For example, the need for achievement and recognition falls under the domain of ambition. Dominance and aggression were recognized as needs under the domain of human power, and play was a recognized need in the domain of interpersonal affection.

MASLOW'S HIERARCHY OF NEEDS

While the theories of motivation described earlier relate to basic biological drives, individual characteristics, or social contexts, Abraham Maslow (1943) proposed a hierarchy of needs that spans the spectrum of motives ranging from the biological to the individual to the social. These needs are often depicted as a pyramid.



Maslow's hierarchy of needs is illustrated here. In some versions of the pyramid, cognitive and aesthetic needs are also included between esteem and self-actualization. Others include another tier at the top of the pyramid for self-transcendence.

At the base of the pyramid are all of the physiological needs that are necessary for survival. These are followed by basic needs for security and safety, the need to be loved and to have a sense of belonging, and the need to have self-worth and confidence. The

top tier of the pyramid is self-actualization, which is a need that essentially equates to achieving one's full potential, and it can only be realized when needs lower on the pyramid have been met. To Maslow and humanistic theorists, self-actualization reflects the humanistic emphasis on positive aspects of human nature. Maslow suggested that this is an ongoing, life-long process and that only a small percentage of people actually achieve a self-actualized state (Francis & Kritsonis, 2006; Maslow, 1943).

According to Maslow (1943), one must satisfy lower-level needs before addressing those needs that occur higher in the pyramid. So, for example, if someone is struggling to find enough food to meet his nutritional requirements, it is quite unlikely that he would spend an inordinate amount of time thinking about whether others viewed him as a good person or not. Instead, all of his energies would be geared toward finding something to eat. However, it should be pointed out that Maslow's theory has been criticized for its subjective nature and its inability to account for phenomena that occur in the real world (Leonard, 1982). Other research has more recently addressed that late in life, Maslow proposed a self-transcendence level above self-actualization—to represent striving for meaning and purpose beyond the concerns of oneself (Koltko-Rivera, 2006). For example, people sometimes make self-sacrifices in order to make a political statement or in an attempt to improve the conditions of others. Mohandas K. Gandhi, a world-renowned advocate for independence through nonviolent protest, on several occasions went on hunger strikes to protest a particular situation. People may starve themselves or otherwise put themselves in danger displaying higher-level motives beyond their own needs.

Check out this interactive exercise that illustrates some of the important concepts in Maslow's hierarchy of needs: Maslow's Hierarchy of Needs Exercise.

SUMMARY

Motivation to engage in a given behavior can come from internal and/or external factors. Multiple theories have been put forward regarding motivation. More biologically oriented theories deal with the ways that instincts and the need to maintain bodily homeostasis motivate behavior. Bandura postulated that our sense of self-efficacy motivates behaviors, and there are a number of theories that focus on a variety of social motives. Abraham Maslow's hierarchy of needs is a model that shows the relationship among multiple motives that range from lower-level physiological needs to the very high level of self-actualization.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=135#h5p-142

CRITICAL THINKING QUESTIONS

How might someone espousing an arousal theory of motivation explain visiting an amusement park?

The idea of optimal levels of arousal is similar to a drive theory of motivation. Presumably, we all seek to maintain some intermediate level of arousal. If we are underaroused, we are bored. If we are overaroused, we experience stress. The rides at an amusement park would provide higher arousal (however, we would hope that these rides don't actually pose significant threats to personal safety that would lead to a state of panic) to push us toward our own optimal level of arousal. Individuals at the park would choose different rides based on their specific arousal thresholds; for example, one person might find a simple water ride optimally arousing and an extreme roller coaster overarousing, while others would find the extreme roller coaster optimally arousing.

Schools often use concrete rewards to increase adaptive behaviors. How might this be a disadvantage for students intrinsically motivated to learn? What are educational implications of the potential for concrete rewards to diminish intrinsic motivation for a given task?

We would expect to see a shift from learning for the sake of learning to learning to earn some reward. This would undermine the foundation upon which traditional institutions of higher education are built. For a student motivated by extrinsic rewards, dependence on those may pose issues later in life

(post-school) when there are not typically extrinsic rewards for learning.

PERSONAL APPLICATION QUESTION

Can you think of recent examples of how Maslow's hierarchy of needs might have affected your behavior in some way?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

https://pressbooks.cuny.edu/jjcpsy101/?p=135#h5p-143

Learning Objectives

By the end of this section, you will be able to:

- Describe how hunger and eating are regulated
- Differentiate between levels of overweight and obesity and the associated health consequences
- Explain the health consequences resulting from anorexia and bulimia nervosa

Eating is essential for survival, and it is no surprise that a drive like hunger exists to ensure that we seek out sustenance. While this chapter will focus primarily on the physiological mechanisms that regulate hunger and eating, powerful social, cultural, and economic influences also play important roles. This section will explain the regulation of hunger, eating, and body weight, and we will discuss the adverse consequences of disordered eating.

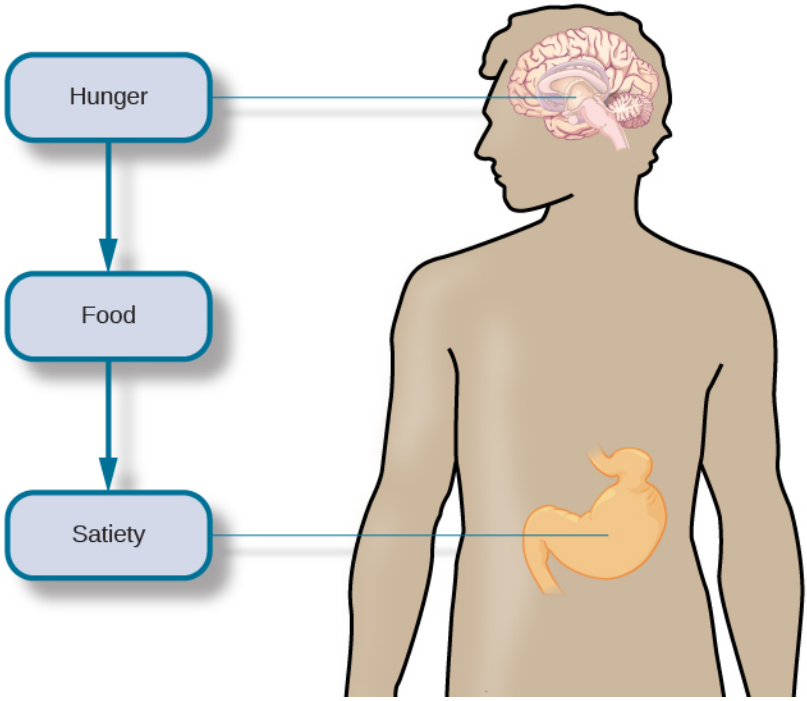
PHYSIOLOGICAL MECHANISMS

There are a number of physiological mechanisms that serve as the basis for hunger. When our stomachs are empty, they contract, causing both hunger pangs and the secretion of chemical messages that travel to the brain to serve as a signal to initiate feeding behavior. When our blood glucose levels drop, the pancreas and liver generate a number of chemical signals that

induce hunger (Konturek et al., 2003; Novin, Robinson, Culbreth, & Tordoff, 1985) and thus initiate feeding behavior.

For most people, once they have eaten, they feel satiation, or fullness and satisfaction, and their eating behavior stops. Like the initiation of eating, satiation is also regulated by several physiological mechanisms. As blood glucose levels increase, the pancreas and liver send signals to shut off hunger and eating (Drazen & Woods, 2003; Druce, Small, & Bloom, 2004; Greary, 1990). The food's passage through the gastrointestinal tract also provides important satiety signals to the brain (Woods, 2004), and fat cells release leptin, a satiety hormone.

The various hunger and satiety signals that are involved in the regulation of eating are integrated in the brain. Research suggests that several areas of the hypothalamus and hindbrain are especially important sites where this integration occurs (Ahima & Antwi, 2008; Woods & D'Alessio, 2008). Ultimately, activity in the brain determines whether or not we engage in feeding behavior.



Hunger and eating are regulated by a complex interplay of hunger and satiety signals that are integrated in the brain.

METABOLISM AND BODY WEIGHT

Our body weight is affected by a number of factors, including gene-environment interactions, and the number of calories we consume versus the number of calories we burn in daily activity. If our caloric intake exceeds our caloric use, our bodies store excess energy in the form of fat. If we consume fewer calories than we burn off, then stored fat will be converted to energy. Our energy expenditure is obviously affected by our levels of activity, but our body's metabolic rate also comes into play. A person's metabolic rate is the amount of energy that is expended in a given period of time, and there

is tremendous individual variability in our metabolic rates. People with high rates of metabolism are able to burn off calories more easily than those with lower rates of metabolism.

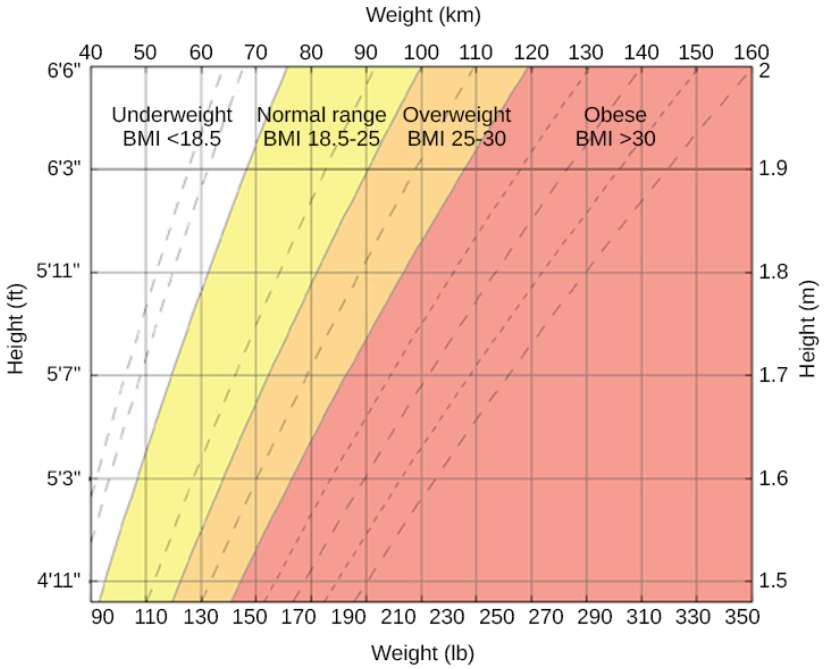
We all experience fluctuations in our weight from time to time, but generally, most people's weights fluctuate within a narrow margin, in the absence of extreme changes in diet and/or physical activity. This observation led some to propose a set-point theory of body weight regulation. The set-point theory asserts that each individual has an ideal body weight, or set point, which is resistant to change. This set-point is genetically predetermined and efforts to move our weight significantly from the set-point are resisted by compensatory changes in energy intake and/or expenditure (Speakman et al., 2011).

Some of the predictions generated from this particular theory have not received empirical support. For example, there are no changes in metabolic rate between individuals who had recently lost significant amounts of weight and a control group (Weinsier et al., 2000). In addition, the set-point theory fails to account for the influence of social and environmental factors in the regulation of body weight (Martin-Gronert & Ozanne, 2013; Speakman et al., 2011). Despite these limitations, set-point theory is still often used as a simple, intuitive explanation of how body weight is regulated.

OBESITY

When someone weighs more than what is generally accepted as healthy for a given height, they are considered overweight or obese. According to the Centers for Disease Control and Prevention (CDC), an adult with a body mass index (BMI) between 25 and 29.9 is considered overweight. An adult with a BMI of 30 or higher is considered obese (Centers for Disease Control and Prevention [CDC], 2012). People who are so overweight that they are at risk for death are classified as morbidly obese. Morbid obesity is defined

as having a BMI over 40. Note that although BMI has been used as a healthy weight indicator by the World Health Organization (WHO), the CDC, and other groups, its value as an assessment tool has been questioned. The BMI is most useful for studying populations, which is the work of these organizations. It is less useful in assessing an individual since height and weight measurements fail to account for important factors like fitness level. An athlete, for example, may have a high BMI because the tool doesn't distinguish between the body's percentage of fat and muscle in a person's weight.

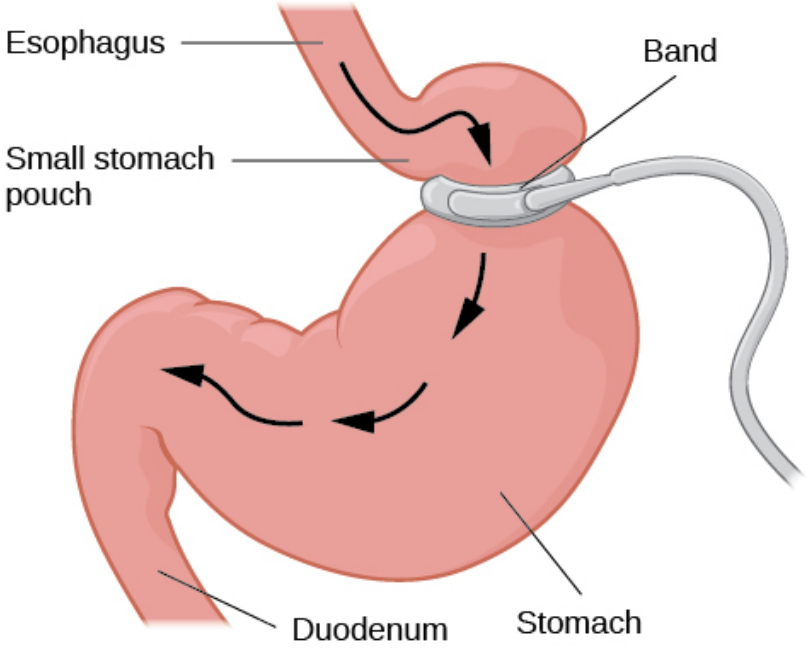


Being extremely overweight or obese is a risk factor for several negative health consequences. These include, but are not limited to, an increased risk for cardiovascular disease, stroke, Type 2 diabetes, liver disease, sleep apnea, colon cancer, breast cancer, infertility, and arthritis. Given that it is estimated that in the United

States around one-third of the adult population is obese and that nearly two-thirds of adults and one in six children qualify as overweight (CDC, 2012), there is substantial interest in trying to understand how to combat this important public health concern.

What causes someone to be overweight or obese? You have already read that both genes and environment are important factors for determining body weight, and if more calories are consumed than expended, excess energy is stored as fat. However, socioeconomic status and the physical environment must also be considered as contributing factors (CDC, 2012). For example, an individual who lives in an impoverished neighborhood that is overrun with crime may never feel comfortable walking or biking to work or to the local market. This might limit the amount of physical activity in which he engages and result in an increased body weight. Similarly, some people may not be able to afford healthy food options from their market, or these options may be unavailable (especially in urban areas or poorer neighborhoods); therefore, some people rely primarily on available, inexpensive, high fat, and high calorie fast food as their primary source of nutrition.

Generally, overweight and obese individuals are encouraged to try to reduce their weights through a combination of both diet and exercise. While some people are very successful with these approaches, many struggle to lose excess weight. In cases in which a person has had no success with repeated attempts to reduce weight or is at risk for death because of obesity, bariatric surgery may be recommended. Bariatric surgery is a type of surgery specifically aimed at weight reduction, and it involves modifying the gastrointestinal system to reduce the amount of food that can be eaten and/or limiting how much of the digested food can be absorbed (Mayo Clinic, 2013). A recent meta-analysis suggests that bariatric surgery is more effective than non-surgical treatment for obesity in the two-years immediately following the procedure, but to date, no long-term studies yet exist (Gloy et al., 2013).



Gastric banding surgery creates a small pouch of stomach, reducing the size of the stomach that can be used for digestion.

Watch this video that describes two different types of bariatric surgeries: Bariatric Surgery.
https://youtube.com/watch?v=I8D8zpLqpw8

PRADER-WILLI SYNDROME

Prader-Willi Syndrome (PWS) is a genetic disorder that results in persistent feelings of intense hunger and reduced rates of metabolism. Typically, affected children have to be supervised around the clock to ensure that they do not engage in excessive eating. Currently, PWS is the leading genetic cause of morbid obesity in children, and it is associated with a number of cognitive deficits and emotional problems.

Eugenia Martínez Vallejo, depicted in this 1680 painting, may have had Prader-Willi syndrome. At just eight years old, she weighed approximately 120 pounds, and she was nicknamed "La Monstrua" (the monster).

While genetic testing can be used to make a diagnosis, there are a number of behavioral diagnostic criteria associated with PWS.

From birth to 2 years of age, lack of muscle tone and poor sucking behavior may serve as early signs of PWS. Developmental delays are seen between the ages of 6 and 12, and excessive eating and cognitive deficits associated with PWS usually onset a little later.

While the exact mechanisms of PWS are not fully understood, there is evidence that affected individuals have hypothalamic abnormalities. This is not surprising, given the hypothalamus's role in regulating hunger and eating. However, as you will learn in the next section of this chapter, the hypothalamus is also involved in the regulation of sexual behavior. Consequently, many individuals suffering from PWS fail to reach sexual maturity during adolescence.

There is no current treatment or cure for PWS. However, if weight can be controlled in these individuals, then their life expectancies are significantly increased (historically, sufferers of PWS often died in adolescence or early adulthood). Advances in the use of various psychoactive medications and growth hormones continue to enhance the quality of life for individuals with PWS (Cassidy & Driscoll, 2009; Prader-Willi Syndrome Association, 2012).

EATING DISORDERS

While nearly two out of three US adults struggle with issues related to being overweight, a smaller, but significant, portion of the population has eating disorders that typically result in being normal weight or underweight. Often, these individuals are fearful of gaining weight. Individuals who suffer from bulimia nervosa and anorexia nervosa face many adverse health consequences (Mayo Clinic, 2012a, 2012b).

People suffering from bulimia nervosa engage in binge eating behavior that is followed by an attempt to compensate for the large amount of food consumed. Purging the food by inducing vomiting or through the use of laxatives are two common compensatory

behaviors. Some affected individuals engage in excessive amounts of exercise to compensate for their binges. Bulimia is associated with many adverse health consequences that can include kidney failure, heart failure, and tooth decay. In addition, these individuals often suffer from anxiety and depression, and they are at an increased risk for substance abuse (Mayo Clinic, 2012b). The lifetime prevalence rate for bulimia nervosa is estimated at around 1% for women and less than 0.5% for men (Smink, van Hoeken, & Hoek, 2012).

As of the 2013 release of the *Diagnostic and Statistical Manual, fifth edition*, Binge eating disorder is a disorder recognized by the American Psychiatric Association (APA). Unlike with bulimia, eating binges are not followed by inappropriate behavior, such as purging, but they are followed by distress, including feelings of guilt and embarrassment. The resulting psychological distress distinguishes binge eating disorder from overeating (American Psychiatric Association [APA], 2013).

Anorexia nervosa is an eating disorder characterized by the maintenance of a body weight well below average through starvation and/or excessive exercise. Individuals suffering from anorexia nervosa often have a distorted body image, referenced in literature as a type of body dysmorphia, meaning that they view themselves as overweight even though they are not. Like bulimia nervosa, anorexia nervosa is associated with a number of significant negative health outcomes: bone loss, heart failure, kidney failure, amenorrhea (cessation of the menstrual period), reduced function of the gonads, and in extreme cases, death. Furthermore, there is an increased risk for a number of psychological problems, which include anxiety disorders, mood disorders, and substance abuse (Mayo Clinic, 2012a). Estimates of the prevalence of anorexia nervosa vary from study to study but generally range from just under one percent to just over four percent in women. Generally, prevalence rates are considerably lower for men (Smink et al., 2012).

While both anorexia and bulimia nervosa occur in men and women of many different cultures, Caucasian females from Western societies tend to be the most at-risk population. Recent research indicates that females between the ages of 15 and 19 are most at risk, and it has long been suspected that these eating disorders are culturally-bound phenomena that are related to messages of a thin ideal often portrayed in popular media and the fashion world (Smink et al., 2012). While social factors play an important role in the development of eating disorders, there is also evidence that genetic factors may predispose people to these disorders (Collier & Treasure, 2004).

Young women in our society are inundated with images of extremely thin models (sometimes accurately depicted and sometimes digitally altered to make them look even thinner). These images may contribute to eating disorders. (credit: Peter Duhon)

TEST YOUR UNDERSTANDING

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=137#h5p-144

SUMMARY

Hunger and satiety are highly regulated processes that result in a person maintaining a fairly stable weight that is resistant to change. When more calories are consumed than expended, a person will store excess energy as fat. Being significantly overweight adds substantially to a person's health risks and problems, including cardiovascular disease, type 2 diabetes, certain cancers, and other medical issues. Sociocultural factors that emphasize thinness as a beauty ideal and a genetic predisposition contribute to the

development of eating disorders in many young females, though eating disorders span ages and genders.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=137#h5p-145

CRITICAL THINKING QUESTIONS

The index that is often used to classify people as being underweight, normal weight, overweight, obese, or morbidly obese is called BMI. Given that BMI is calculated solely on weight and height, how could it be misleading?

Using BMI as a sole metric can actually be misleading because people who have large amounts of lean muscle mass can actually be characterized as being overweight or obese based on their height and weight. Weight versus height is a somewhat crude measurement as it doesn't distinguish the amount of body weight that comes from lean versus fatty tissue.

As indicated in this section, Caucasian women from industrialized, Western cultures tend to be at the highest risk for eating disorders like anorexia and bulimia nervosa. Why might this be?

These disorders are closely associated with sociocultural emphasis on a thin-ideal that is often portrayed in media. Given that non-Caucasians are under-represented in popular media in the West and that the thin-ideal is more heavily emphasized for women, this particular group is most vulnerable.

PERSONAL APPLICATION QUESTION

Think about popular television programs on the air right now. What do the women in these programs look like? What do the men look like? What kinds of messages do you think the media is sending about men and women in our society?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

https://pressbooks.cuny.edu/jjcpsy101/?p=137#h5p-146

Media Attributions

- "Bariatric Surgery" by Nucleus Medical Media. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

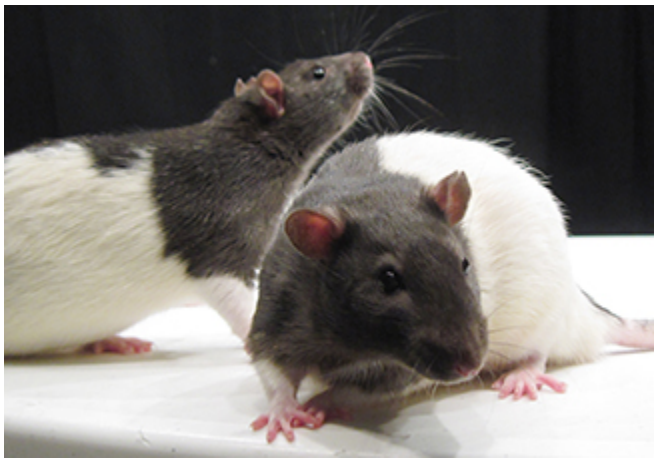
- Understand basic biological mechanisms regulating sexual behavior and motivation
- Appreciate the importance of Alfred Kinsey's research on human sexuality
- Recognize the contributions that William Masters and Virginia Johnson's research made to our understanding of the sexual response cycle
- Define sexual orientation and gender identity

Like food, sex is an important part of our lives. From an evolutionary perspective, the reason is obvious—perpetuation of the species. Sexual behavior in humans, however, involves much more than reproduction. This section provides an overview of research that has been conducted on human sexual behavior and motivation. This section will close with a discussion of issues related to gender and sexual orientation.

PHYSIOLOGICAL MECHANISMS OF SEXUAL BEHAVIOR AND MOTIVATION

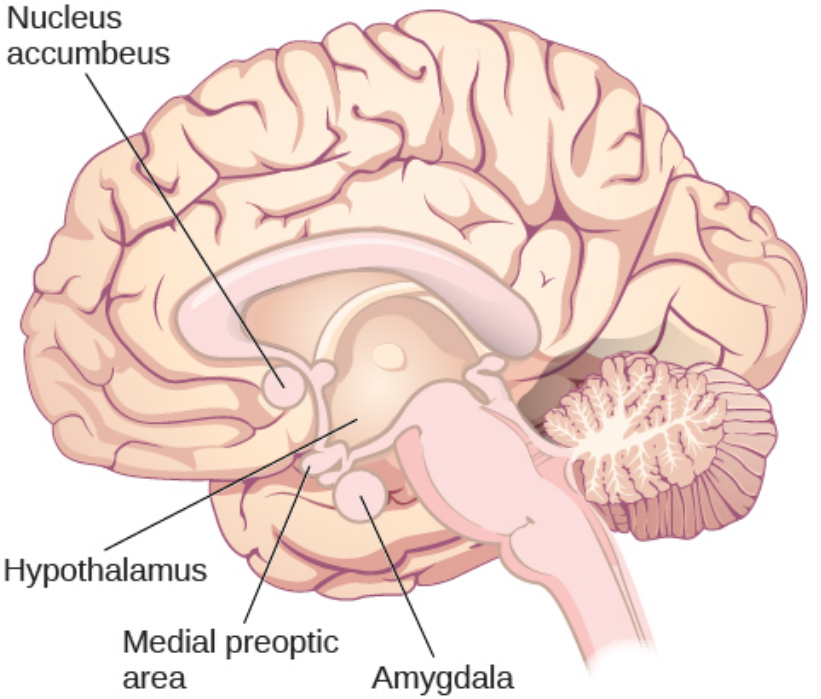
Much of what we know about the physiological mechanisms that underlie sexual behavior and motivation comes from animal research. As you've learned, the hypothalamus plays an important

role in motivated behaviors, and sex is no exception. In fact, lesions to an area of the hypothalamus called the medial preoptic area completely disrupt a male rat's ability to engage in sexual behavior. Surprisingly, medial preoptic lesions do not change how hard a male rat is willing to work to gain access to a sexually receptive female. This suggests that the ability to engage in sexual behavior and the motivation to do so may be mediated by neural systems distinct from one another.



A male rat that cannot engage in sexual behavior still seeks receptive females, suggesting that the ability to engage in sexual behavior and the motivation to do so are mediated by different systems in the brain. (credit: Jason Snyder)

Animal research suggests that limbic system structures such as the amygdala and nucleus accumbens are especially important for sexual motivation. Damage to these areas results in a decreased motivation to engage in sexual behavior, while leaving the ability to do so intact (Everett, 1990). Similar dissociations of sexual motivation and sexual ability have also been observed in the female rat (Becker, Rudick, & Jenkins, 2001; Jenkins & Becker, 2001).



The medial preoptic area, an area of the hypothalamus, is involved in the ability to engage in sexual behavior, but it does not affect sexual motivation. In contrast, the amygdala and nucleus accumbens are involved in motivation for sexual behavior, but they do not affect the ability to engage in it.

Although human sexual behavior is much more complex than that seen in rats, some parallels between animals and humans can be drawn from this research. The worldwide popularity of drugs used to treat erectile dysfunction (Conrad, 2005) speaks to the fact that sexual motivation and the ability to engage in sexual behavior can also be dissociated in humans. Moreover, disorders that involve abnormal hypothalamic function are often associated with hypogonadism (reduced function of the gonads) and reduced sexual function (e.g., Prader-Willi syndrome). Given the hypothalamus's role in endocrine function, it is not surprising that

hormones secreted by the endocrine system also play important roles in sexual motivation and behavior. For example, many animals show no sign of sexual motivation in the absence of the appropriate combination of sex hormones from their gonads. While this is not the case for humans, there is considerable evidence that sexual motivation for both men and women varies as a function of circulating testosterone levels (Bhasin, Enzlin, Coviello, & Basson, 2007; Carter, 1992; Sherwin, 1988).

TEST YOUR UNDERSTANDING

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=139#h5p-147

KINSEY'S RESEARCH

Before the late 1940s, access to reliable, empirically-based information on sex was limited. Physicians were considered authorities on all issues related to sex, despite the fact that they had little to no training in these issues, and it is likely that most of what people knew about sex had been learned either through their own experiences or by talking with their peers. Convinced that people would benefit from a more open dialogue on issues related to human sexuality, Dr. Alfred Kinsey of Indiana University initiated large-scale survey research on the topic. The results of some of these efforts were published in two books—*Sexual Behavior in the Human Male* and *Sexual Behavior in the Human Female*—which were published in 1948 and 1953, respectively (Bullough, 1998).

In 1947, Alfred Kinsey established The Kinsey Institute for Research, Sex, Gender and Reproduction at Indiana University, shown here in 2011. The Kinsey Institute has continued as a research site of important psychological studies for decades.

At the time, the Kinsey reports were quite sensational. Never before had the American public seen its private sexual behavior

become the focus of scientific scrutiny on such a large scale. The books, which were filled with statistics and scientific lingo, sold remarkably well to the general public, and people began to engage in open conversations about human sexuality. As you might imagine, not everyone was happy that this information was being published. In fact, these books were banned in some countries. Ultimately, the controversy resulted in Kinsey losing funding that he had secured from the Rockefeller Foundation to continue his research efforts (Bancroft, 2004).

Although Kinsey's research has been widely criticized as being riddled with sampling and statistical errors (Jenkins, 2010), there is little doubt that this research was very influential in shaping future research on human sexual behavior and motivation. Kinsey described a remarkably diverse range of sexual behaviors and experiences reported by the volunteers participating in his research. Behaviors that had once been considered exceedingly rare or problematic were demonstrated to be much more common and innocuous than previously imagined (Bancroft, 2004; Bullough, 1998).

Watch this trailer from the 2004 film *Kinsey* that depicts Alfred Kinsey's life and research: Kinsey (2004) – Movie Trailer.

Among the results of Kinsey's research were the findings that women are as interested and experienced in sex as their male counterparts, that both males and females masturbate without adverse health consequences, and that homosexual acts are fairly common (Bancroft, 2004). Kinsey also developed a continuum

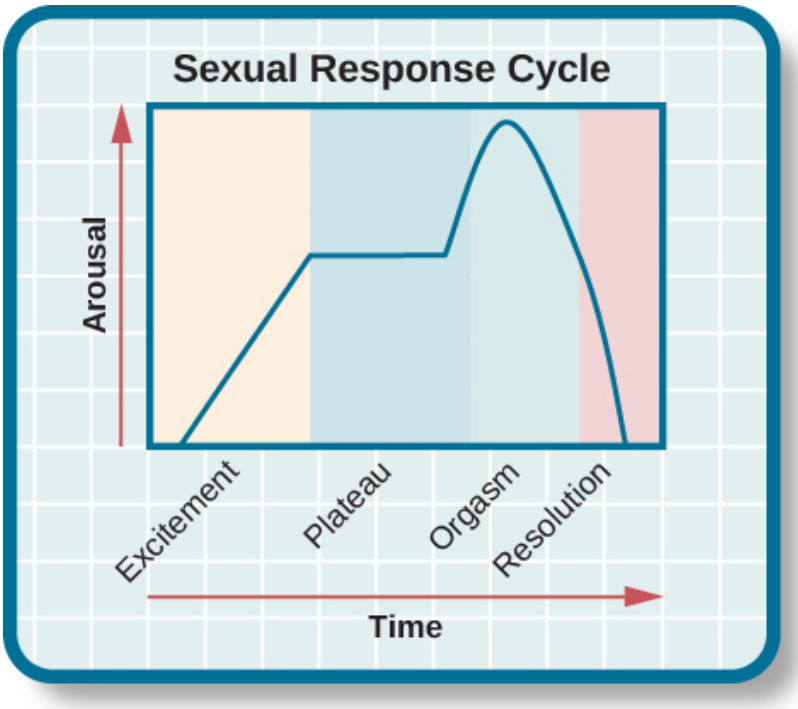
known as the Kinsey scale that is still commonly used today to categorize an individual's sexual orientation (Jenkins, 2010). Sexual orientation is an individual's emotional and erotic attractions to same-sexed individuals (homosexual), opposite-sexed individuals (heterosexual), or both (bisexual).

MASTERS AND JOHNSON'S RESEARCH

In 1966, William Masters and Virginia Johnson published a book detailing the results of their observations of nearly 700 people who agreed to participate in their study of physiological responses during sexual behavior. Unlike Kinsey, who used personal interviews and surveys to collect data, Masters and Johnson observed people having intercourse in a variety of positions, and they observed people masturbating, manually or with the aid of a device. While this was occurring, researchers recorded measurements of physiological variables, such as blood pressure and respiration rate, as well as measurements of sexual arousal, such as vaginal lubrication and penile tumescence (swelling associated with an erection). In total, Masters and Johnson observed nearly 10,000 sexual acts as a part of their research (Hock, 2008).

Based on these observations, Masters and Johnson divided the sexual response cycle into four phases that are fairly similar in men and women: excitement, plateau, orgasm, and resolution. The excitement phase is the arousal phase of the sexual response cycle, and it is marked by erection of the penis or clitoris and lubrication and expansion of the vaginal canal. During plateau, women experience further swelling of the vagina and increased blood flow to the labia minora, and men experience full erection and often exhibit pre-ejaculatory fluid. Both men and women experience increases in muscle tone during this time. Orgasm is marked in women by rhythmic contractions of the pelvis and uterus along

with increased muscle tension. In men, pelvic contractions are accompanied by a buildup of seminal fluid near the urethra that is ultimately forced out by contractions of genital muscles, (i.e., ejaculation). Resolution is the relatively rapid return to an unaroused state accompanied by a decrease in blood pressure and muscular relaxation. While many women can quickly repeat the sexual response cycle, men must pass through a longer refractory period as part of resolution. The refractory period is a period of time that follows an orgasm during which an individual is incapable of experiencing another orgasm. In men, the duration of the refractory period can vary dramatically from individual to individual with some refractory periods as short as several minutes and others as long as a day. As men age, their refractory periods tend to span longer periods of time.



This graph illustrates the different phases of the sexual response cycle as described by Masters and Johnson.

In addition to the insights that their research provided with regards to the sexual response cycle and the multi-orgasmic potential of women, Masters and Johnson also collected important information about reproductive anatomy. Their research demonstrated the oft-cited statistic of the average size of a flaccid and an erect penis (3 and 6 inches, respectively) as well as dispelling long-held beliefs about relationships between the size of a man's erect penis and his ability to provide sexual pleasure to his female partner. Furthermore, they determined that the vagina is a very elastic structure that can conform to penises of various sizes (Hock, 2008).

SEXUAL ORIENTATION

As mentioned earlier, a person's sexual orientation is their emotional and erotic attraction toward another individual. While the majority of people identify as heterosexual, there is a sizable population of people within the United States who identify as either homosexual or bisexual. Research suggests that somewhere between 3% and 10% of the population identifies as homosexual (Kinsey, Pomeroy, & Martin, 1948; LeVay, 1996; Pillard & Bailey, 1995).

Between 3% and 10% of the adult population identifies as homosexual. (credit: Till Krech)

Issues of sexual orientation have long fascinated scientists interested in determining what causes one individual to be heterosexual while another is homosexual. For many years, people believed that these differences arose because of different socialization and familial experiences. However, research has

consistently demonstrated that the family backgrounds and experiences are very similar among heterosexuals and homosexuals (Bell, Weinberg, & Hammersmith, 1981; Ross & Arrindell, 1988).

Genetic and biological mechanisms have also been proposed, and the balance of research evidence suggests that sexual orientation has an underlying biological component. For instance, over the past 25 years, research has demonstrated gene-level contributions to sexual orientation (Bailey & Pillard, 1991; Hamer, Hu, Magnuson, Hu, & Pattatucci, 1993; Rodriguez-Larralde & Paradisi, 2009), with some researchers estimating that genes account for at least half of the variability seen in human sexual orientation (Pillard & Bailey, 1998). Other studies report differences in brain structure and function between heterosexuals and homosexuals (Allen & Gorski, 1992; Byne et al., 2001; Hu et al., 2008; LeVay, 1991; Ponseti et al., 2006; Rahman & Wilson, 2003a; Swaab & Hofman, 1990), and even differences in basic body structure and function have been observed (Hall & Kimura, 1994; Lippa, 2003; Loehlin & McFadden, 2003; McFadden & Champlin, 2000; McFadden & Pasanen, 1998; Rahman & Wilson, 2003b). In aggregate, the data suggest that to a significant extent, sexual orientations are something with which we are born.

MISUNDERSTANDINGS ABOUT SEXUAL ORIENTATION

Regardless of how sexual orientation is determined, research has made clear that sexual orientation is not a choice, but rather it is a relatively stable characteristic of a person that cannot be changed. Claims of successful gay conversion therapy have received wide criticism from the research community due to significant concerns with research design, recruitment of experimental participants, and interpretation of data. As such, there is no credible scientific evidence to suggest that individuals can change their sexual orientation (Jenkins, 2010).

Dr. Robert Spitzer, the author of one of the most widely-cited examples of successful conversion therapy, apologized to both the scientific community and the gay community for his mistakes, and he publically recanted his own paper in a public letter addressed to the editor of *Archives of Sexual Behavior* in the spring of 2012 (Carey, 2012). In this letter, Spitzer wrote,

> I was considering writing something that would acknowledge that I now judge the major critiques of the study as largely correct. . . . I believe I owe the gay community an apology for my study making unproven claims of the efficacy of reparative therapy. I also apologize to any gay person who wasted time or energy undergoing some form of reparative therapy because they believed that I had proven that reparative therapy works with some "highly motivated" individuals. (Becker, 2012, pars. 2, 5)

Citing research that suggests not only that gay conversion therapy is ineffective, but also potentially harmful, legislative efforts to make such therapy illegal have either been enacted (e.g., it is now illegal in California) or are underway across the United States, and many professional organizations have issued statements against this practice (Human Rights Campaign, n.d.)

Read this draft of Dr. Spitzer's letter:EXCLUSIVE: Dr. Robert Spitzer Apologizes to Gay Community for Infamous 'Ex-Gay' Study.

GENDER IDENTITY

Many people conflate sexual orientation with gender identity because of stereotypical attitudes that exist about homosexuality. In reality, these are two related, but different, issues. Gender identity refers to one's sense of being male or female. Generally, our gender identities correspond to our chromosomal and phenotypic sex, but this is not always the case. When individuals do not feel comfortable identifying with the gender associated with their biological sex, then they experience gender dysphoria. Gender dysphoria is a diagnostic category in the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) that describes individuals who do not identify as the gender that most people would assume they are. This dysphoria must persist for at least six months and result in significant distress or dysfunction to meet DSM-5 diagnostic criteria. In order for children to be assigned this diagnostic category, they must verbalize their desire to become the other gender.

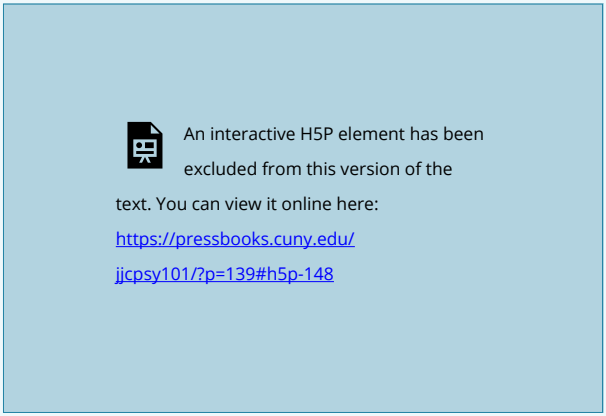
Many people who are classified as gender dysphoric seek to live their lives in ways that are consistent with their own gender identity. This involves dressing in opposite-sex clothing and assuming an opposite-sex identity. These individuals may also undertake transgender hormone therapy in an attempt to make their bodies look more like the opposite sex, and in some cases, they elect to have surgeries to alter the appearance of their external genitalia to resemble that of their gender identity. While these may sound like drastic changes, gender dysphoric individuals take these steps because their bodies seem to them to be a mistake of nature, and they seek to correct this mistake.

(a)

(b)

Chaz Bono, a transgender male, is a well-known person who transitioned from female to male. (a) In the 1970s, the world knew Chaz as Chastity Bono, the daughter of the famous entertaining duo Sonny and Cher; here young Chastity is pictured with Sonny. (b) Later in life, Chaz transitioned to align his physical body with his gender identity. (credit b: modification of work by "dvsross"/Flickr)

CHECK YOUR UNDERSTANDING



CULTURAL FACTORS IN SEXUAL ORIENTATION AND GENDER IDENTITY

Issues related to sexual orientation and gender identity are very much influenced by sociocultural factors. Even the ways in which we define sexual orientation and gender vary from one culture to the next. While in the United States exclusive heterosexuality is viewed as the norm, there are societies that have different attitudes regarding homosexual behavior. In fact, in some instances, periods of exclusively homosexual behavior are socially prescribed as a part of normal development and maturation. For

example, in parts of New Guinea, young boys are expected to engage in sexual behavior with other boys for a given period of time because it is believed that doing so is necessary for these boys to become men (Baldwin & Baldwin, 1989).

There is a two-gendered culture in the United States. We tend to classify an individual as either male or female. However, in some cultures there are additional gender variants resulting in more than two gender categories. For example, in Thailand, you can be male, female, or kathoey. A kathoey is an individual who would be described as intersexed or transgendered in the United States (Tangmunkongvorakul, Banwell, Carmichael, Utomo, & Sleigh, 2010).

The Case of David Reimer

In August of 1965, Janet and Ronald Reimer of Winnipeg, Canada, welcomed the birth of their twin sons, Bruce and Brian. Within a few months, the twins were experiencing urinary problems; doctors recommended the problems could be alleviated by having the boys circumcised. A malfunction of the medical equipment used to perform the circumcision resulted in Bruce's penis being irreparably damaged. Distraught, Janet and Ronald looked to expert advice on what to do with their baby boy. By happenstance, the couple became aware of Dr. John Money at Johns Hopkins University and his theory of psychosexual neutrality (Colapinto, 2000).

Dr. Money had spent a considerable amount of time researching transgendered individuals and individuals born with ambiguous genitalia. As a result of this work, he developed a theory of psychosexual neutrality. His theory asserted that we are essentially neutral at birth with regard to our gender identity and that we don't assume a concrete gender identity until we begin to master language. Furthermore, Dr. Money believed that the way in which we are socialized in early life is ultimately much more important than our biology in determining our gender identity (Money, 1962).

Dr. Money encouraged Janet and Ronald to bring the twins to Johns Hopkins University, and he convinced them that they should raise Bruce as a girl. Left with few other options at the time, Janet and Ronald agreed to have Bruce's testicles removed and to raise him as a girl. When they returned home to Canada, they brought with them Brian and his "sister," Brenda, along with specific instructions to never reveal to Brenda that she had been born a boy (Colapinto, 2000).

Early on, Dr. Money shared with the scientific community the great success of this natural experiment that seemed to fully support his theory of psychosexual neutrality (Money, 1975). Indeed, in early interviews with the children it appeared that Brenda was a typical little girl who liked to play with "girly" toys and do "girly" things.

However, Dr. Money was less than forthcoming with information that seemed to argue against the success of the case. In reality, Brenda's parents were constantly concerned that their little girl wasn't really behaving as most girls did, and by the time Brenda was nearing adolescence, it was painfully obvious to the family that she was really having a hard time identifying as a female. In addition, Brenda was becoming increasingly reluctant to continue her visits with Dr. Money to the point that she threatened suicide if her parents made her go back to see him again.

At that point, Janet and Ronald disclosed the true nature of Brenda's early childhood to their daughter. While initially shocked,

Brenda reported that things made sense to her now, and ultimately, by the time she was an adolescent, Brenda had decided to identify as a male. Thus, she became David Reimer.

David was quite comfortable in his masculine role. He made new friends and began to think about his future. Although his castration had left him infertile, he still wanted to be a father. In 1990, David married a single mother and loved his new role as a husband and father. In 1997, David was made aware that Dr. Money was continuing to publicize his case as a success supporting his theory of psychosexual neutrality. This prompted David and his brother to go public with their experiences in attempt to discredit the doctor's publications. While this revelation created a firestorm in the scientific community for Dr. Money, it also triggered a series of unfortunate events that ultimately led to David committing suicide in 2004 (O'Connell, 2004).

This sad story speaks to the complexities involved in gender identity. While the Reimer case had earlier been paraded as a hallmark of how socialization trumped biology in terms of gender identity, the truth of the story made the scientific and medical communities more cautious in dealing with cases that involve intersex children and how to deal with their unique circumstances. In fact, stories like this one have prompted measures to prevent unnecessary harm and suffering to children who might have issues with gender identity. For example, in 2013, a law took effect in Germany allowing parents of intersex children to classify their children as indeterminate so that children can self-assign the appropriate gender once they have fully developed their own gender identities (Paramaguru, 2013).

Watch this news story about the experiences of David Reimer and his family: Circumcision Damage Leads to More Tragedy.

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SUMMARY

The hypothalamus and structures of the limbic system are important in sexual behavior and motivation. There is evidence to suggest that our motivation to engage in sexual behavior and our ability to do so are related, but separate, processes. Alfred Kinsey conducted large-scale survey research that demonstrated the incredible diversity of human sexuality. William Masters and Virginia Johnson observed individuals engaging in sexual behavior in developing their concept of the sexual response cycle. While

often confused, sexual orientation and gender identity are related, but distinct, concepts.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

While much research has been conducted on how an individual develops a given sexual orientation, many people question the validity of this research citing that the participants used may not be representative. Why do you think this might be a legitimate concern?

Given the stigma associated with being non-heterosexual, participants who openly identify as homosexual or bisexual in research projects may not be entirely representative of the non-heterosexual population as a whole.

There is no reliable scientific evidence that gay conversion therapy actually works. What kinds of evidence would you need to see in order to be convinced by someone arguing that she had successfully converted her sexual orientation?

Answers will vary, but it should be indicated that something more than self-reports of successful conversion would be necessary to support such a claim. Longitudinal, objective demonstrations of a real switch in both erotic attraction and the actual behavior in which the individual engaged would need to be presented in addition to assurances that this type of therapy was safe.

PERSONAL APPLICATION QUESTION

Issues related to sexual orientation have been at the forefront of the current political landscape. What do you think about current debates on legalizing same-sex marriage?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=139#h5p-150

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Learning Objectives

By the end of this section, you will be able to:

- Explain the major theories of emotion
- Describe the role that limbic structures play in emotional processing
- Understand the ubiquitous nature of producing and recognizing emotional expression

As we move through our daily lives, we experience a variety of emotions. An emotion is a subjective state of being that we often describe as our feelings. The words emotion and mood are sometimes used interchangeably, but psychologists use these words to refer to two different things. Typically, the word emotion indicates a subjective, affective state that is relatively intense and that occurs in response to something we experience. Emotions are often thought to be consciously experienced and intentional. Mood, on the other hand, refers to a prolonged, less intense, affective state that does not occur in response to something we experience. Mood states may not be consciously recognized and do not carry the intentionality that is associated with emotion (Beedie, Terry, Lane, & Devonport, 2011). Here we will focus on emotion, and you will learn more about mood in the chapter that covers psychological disorders.

Toddlers can cycle through emotions quickly, being (a) extremely happy one moment and (b) extremely sad the next. (credit a: modification of work by Kerry Ceszyk; credit b: modification of work by Kerry Ceszyk)

We can be at the heights of joy or in the depths of despair or. We might feel angry when we are betrayed, fear when we are threatened, and surprised when something unexpected happens. This section will outline some of the most well-known theories explaining our emotional experience and provide insight into the biological bases of emotion. This section closes with a discussion of the ubiquitous nature of facial expressions of emotion and our abilities to recognize those expressions in others.

THEORIES OF EMOTION

Our emotional states are combinations of physiological arousal, psychological appraisal, and subjective experiences. Together, these are known as the components of emotion. These appraisals

are informed by our experiences, backgrounds, and cultures. Therefore, different people may have different emotional experiences even when faced with similar circumstances. Over time, several different theories of emotion, have been proposed to explain how the various components of emotion interact with one another.

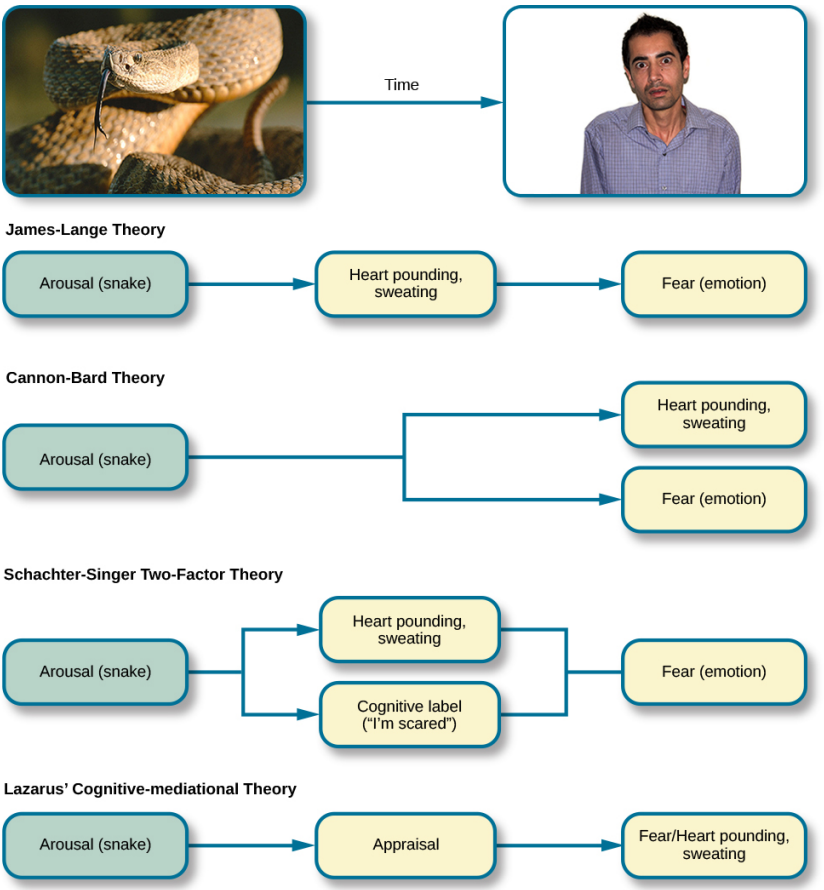
The James-Lange theory of emotion asserts that emotions arise from physiological arousal. Recall what you have learned about the sympathetic nervous system and our fight or flight response when threatened. If you were to encounter some threat in your environment, like a venomous snake in your backyard, your sympathetic nervous system would initiate significant physiological arousal, which would make your heart race and increase your respiration rate. According to the James-Lange theory of emotion, you would only experience a feeling of fear after this physiological arousal had taken place. Furthermore, different arousal patterns would be associated with different feelings.

Other theorists, however, doubted that the physiological arousal that occurs with different types of emotions is distinct enough to result in the wide variety of emotions that we experience. Thus, the Cannon-Bard theory of emotion was developed. According to this view, physiological arousal and emotional experience occur simultaneously, yet independently (Lang, 1994). So, when you see the venomous snake, you feel fear at exactly the same time that your body mounts its fight or flight response. This emotional reaction would be separate and independent of the physiological arousal, even though they co-occur.

The James-Lange and Cannon-Bard theories have each garnered some empirical support in various research paradigms. For instance, Chwalisz, Diener, and Gallagher (1988) conducted a study of the emotional experiences of people who had spinal cord injuries. They reported that individuals who were incapable of receiving autonomic feedback because of their injuries still experienced emotion; however, there was a tendency for people

with less awareness of autonomic arousal to experience less intense emotions. More recently, research investigating the facial feedback hypothesis suggested that suppression of facial expression of emotion lowered the intensity of some emotions experienced by participants (Davis, Senghas, & Ochsner, 2009). In both of these examples, neither theory is fully supported because physiological arousal does not seem to be necessary for the emotional experience, but this arousal does appear to be involved in enhancing the intensity of the emotional experience.

The Schachter-Singer two-factor theory of emotion is another variation on theories of emotions that takes into account both physiological arousal and the emotional experience. According to this theory, emotions are composed of two factors: physiological and cognitive. In other words, physiological arousal is interpreted in context to produce the emotional experience. In revisiting our example involving the venomous snake in your backyard, the two-factor theory maintains that the snake elicits sympathetic nervous system activation that is labeled as fear given the context, and our experience is that of fear.



This figure illustrates the major assertions of the James-Lange, Cannon-Bard, and Schachter-Singer two-factor theories of emotion. (credit "snake": modification of work by "tableatny"/Flickr; credit "face": modification of work by Cory Zanker)

It is important to point out that Schachter and Singer believed that physiological arousal is very similar across the different types of emotions that we experience, and therefore, the cognitive appraisal of the situation is critical to the actual emotion experienced. In fact, it might be possible to misattribute arousal to

an emotional experience if the circumstances were right (Schachter & Singer, 1962).

To test their idea, Schachter and Singer performed a clever experiment. Male participants were randomly assigned to one of several groups. Some of the participants received injections of epinephrine that caused bodily changes that mimicked the fight-or-flight response of the sympathetic nervous system; however, only some of these men were told to expect these reactions as side effects of the injection. The other men that received injections of epinephrine were told either that the injection would have no side effects or that it would result in a side effect unrelated to a sympathetic response, such as itching feet or headache. After receiving these injections, participants waited in a room with someone else they thought was another subject in the research project. In reality, the other person was a confederate of the researcher. The confederate engaged in scripted displays of euphoric or angry behavior (Schachter & Singer, 1962).

When those subjects who were told that they should expect to feel symptoms of physiological arousal were asked about any emotional changes that they had experienced related to either euphoria or anger (depending on how their confederate behaved), they reported none. However, the men who weren't expecting physiological arousal as a function of the injection were more likely to report that they experienced euphoria or anger as a function of their assigned confederate's behavior. While everyone that received an injection of epinephrine experienced the same physiological arousal, only those who were not expecting the arousal used context to interpret the arousal as a change in emotional state (Schachter & Singer, 1962).

Strong emotional responses are associated with strong physiological arousal. This has led some to suggest that the signs of physiological arousal, which include increased heart rate, respiration rate, and sweating, might serve as a tool to determine whether someone is telling the truth or not. The assumption is that

most of us would show signs of physiological arousal if we were being dishonest with someone. A polygraph, or lie detector test, measures the physiological arousal of an individual responding to a series of questions. Someone trained in reading these tests would look for answers to questions that are associated with increased levels of arousal as potential signs that the respondent may have been dishonest on those answers. While polygraphs are still commonly used, their validity and accuracy are highly questionable because there is no evidence that lying is associated with any particular pattern of physiological arousal (Saxe & Ben-Shakhar, 1999).

The relationship between our experiencing of emotions and our cognitive processing of them, and the order in which these occur, remains a topic of research and debate. Lazarus (1991) developed the cognitive-mediational theory that asserts our emotions are determined by our appraisal of the stimulus. This appraisal mediates between the stimulus and the emotional response, and it is immediate and often unconscious. In contrast to the Schachter-Singer model, the appraisal precedes a cognitive label. You will learn more about Lazarus's appraisal concept when you study stress, health, and lifestyle.

Two other prominent views arise from the work of Robert Zajonc and Joseph LeDoux. Zajonc asserted that some emotions occur separately from or prior to our cognitive interpretation of them, such as feeling fear in response to an unexpected loud sound (Zajonc, 1998). He also believed in what we might casually refer to as a gut feeling—that we can experience an instantaneous and unexplainable like or dislike for someone or something (Zajonc, 1980). LeDoux also views some emotions as requiring no cognition: some emotions completely bypass contextual interpretation. His research into the neuroscience of emotion has demonstrated the amygdala's primary role in fear (Cunha, Monfils, & LeDoux, 2010; LeDoux 1996, 2002). A fear stimulus is processed by the brain through one of two paths: from the thalamus (where it is perceived)

directly to the amygdala or from the thalamus through the cortex and then to the amygdala. The first path is quick, while the second enables more processing about details of the stimulus. In the following section, we will look more closely at the neuroscience of emotional response.

TEST YOUR UNDERSTANDING

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SUPPLEMENTAL VIDEO

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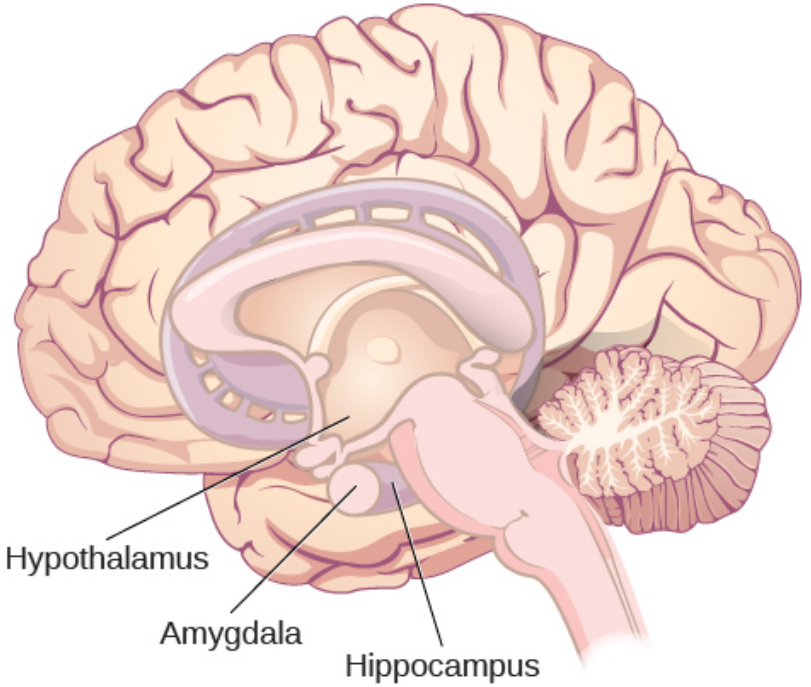
THE BIOLOGY OF EMOTIONS

Earlier, you learned about the limbic system, which is the area of the brain involved in emotion and memory. The limbic system includes the hypothalamus, thalamus, amygdala, and the hippocampus. The hypothalamus plays a role in the activation of the sympathetic nervous system that is a part of any given emotional reaction. The thalamus serves as a sensory relay center whose neurons project to both the amygdala and the higher cortical regions for further processing. The amygdala plays a role in processing emotional information and sending that information on

to cortical structures (Fossati, 2012).The hippocampus integrates emotional experience with cognition (Femenía, Gómez-Galán, Lindskog, & Magara, 2012).



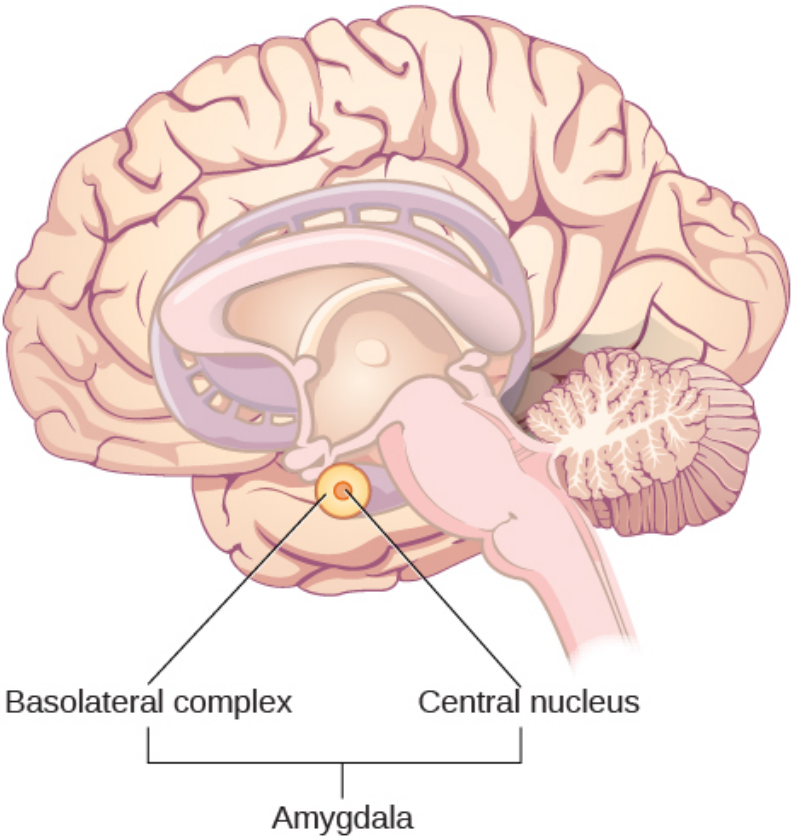
The limbic system, which includes the hypothalamus, thalamus, amygdala, and the hippocampus, is involved in mediating emotional response and memory.

Work through this Open Colleges interactive 3D brain simulator for a refresher on the brain's parts

and their functions. To begin, click the "Start Exploring" button. To access the limbic system, click the plus sign in the right-hand menu (set of three tabs): Open Colleges Interactive Brain.

AMYGDALA

The amygdala has received a great deal of attention from researchers interested in understanding the biological basis for emotions, especially fear and anxiety (Blackford & Pine, 2012; Goosens & Maren, 2002; Maren, Phan, & Liberzon, 2013). The amygdala is composed of various subnuclei, including the basolateral complex and the central nucleus. The basolateral complex has dense connections with a variety of sensory areas of the brain. It is critical for classical conditioning and for attaching emotional value to learning processes and memory. The central nucleus plays a role in attention, and it has connections with the hypothalamus and various brainstem areas to regulate the autonomic nervous and endocrine systems' activity (Pessoa, 2010).



The anatomy of the basolateral complex and central nucleus of the amygdala are illustrated in this diagram.

Animal research has demonstrated that there is increased activation of the amygdala in rat pups that have odor cues paired with electrical shock when their mother is absent. This leads to an aversion to the odor cue that suggests the rats learned to fear the odor cue. Interestingly, when the mother was present, the rats actually showed a preference for the odor cue despite its association with an electrical shock. This preference was associated with no increases in amygdala activation. This suggests a

differential effect on the amygdala by the *context* (the presence or absence of the mother) determined whether the pups learned to fear the odor or to be attracted to it (Moriceau & Sullivan, 2006).

Raineki, Cortés, Belnoue, and Sullivan (2012) demonstrated that, in rats, negative early life experiences could alter the function of the amygdala and result in adolescent patterns of behavior that mimic human mood disorders. In this study, rat pups received either abusive or normal treatment during postnatal days 8–12. There were two forms of abusive treatment. The first form of abusive treatment had an insufficient bedding condition. The mother rat had insufficient bedding material in her cage to build a proper nest that resulted in her spending more time away from her pups trying to construct a nest and less times nursing her pups. The second form of abusive treatment had an associative learning task that involved pairing odors and an electrical stimulus in the absence of the mother, as described above. The control group was in a cage with sufficient bedding and was left undisturbed with their mothers during the same time period. The rat pups that experienced abuse were much more likely to exhibit depressive-like symptoms during adolescence when compared to controls. These depressive-like behaviors were associated with increased activation of the amygdala.

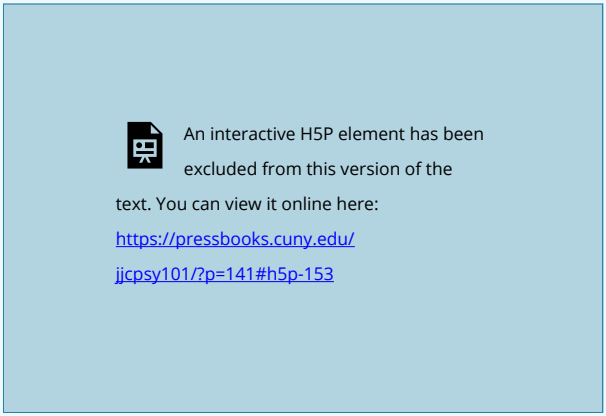
Human research also suggests a relationship between the amygdala and psychological disorders of mood or anxiety. Changes in amygdala structure and function have been demonstrated in adolescents who are either at-risk or have been diagnosed with various mood and/or anxiety disorders (Miguel-Hidalgo, 2013; Qin et al., 2013). It has also been suggested that functional differences in the amygdala could serve as a biomarker to differentiate individuals suffering from bipolar disorder from those suffering from major depressive disorder (Fournier, Keener, Almeida, Kronhaus, & Phillips, 2013).

HIPPOCAMPUS

As mentioned earlier, the hippocampus is also involved in emotional processing. Like the amygdala, research has demonstrated that hippocampal structure and function are linked to a variety of mood and anxiety disorders. Individuals suffering from posttraumatic stress disorder (PTSD) show marked reductions in the volume of several parts of the hippocampus, which may result from decreased levels of neurogenesis and dendritic branching (the generation of new neurons and the generation of new dendrites in existing neurons, respectively) (Wang et al., 2010). While it is impossible to make a causal claim with correlational research like this, studies have demonstrated behavioral improvements and hippocampal volume increases following either pharmacological or cognitive-behavioral therapy in individuals suffering from PTSD (Bremner & Vermetten, 2004; Levy-Gigi, Szabó, Kelemen, & Kéri, 2013).

Watch this video about research that demonstrates how the volume of the hippocampus can vary as a function of traumatic experiences: BYU study finds brain shrinkage after traumatic events.

TEST YOUR UNDERSTANDING



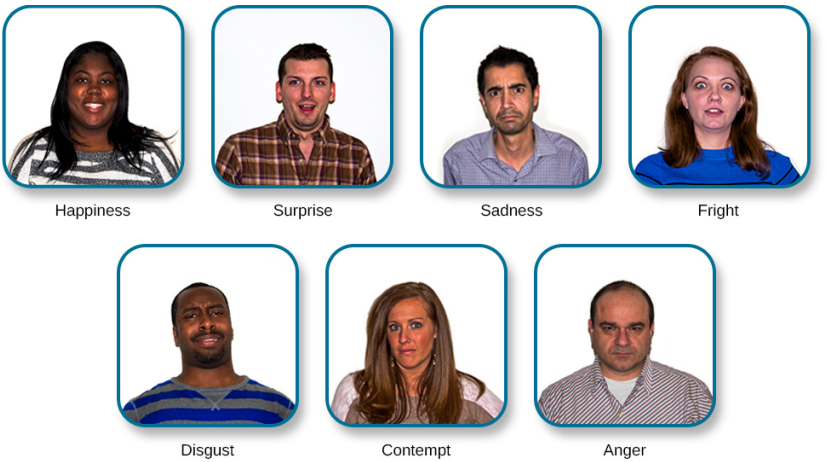
FACIAL EXPRESSION AND RECOGNITION OF EMOTIONS

Culture can impact the way in which people display emotion. A cultural display rule is one of a collection of culturally specific standards that govern the types and frequencies of displays of emotions that are acceptable (Malatesta & Haviland, 1982). Therefore, people from varying cultural backgrounds can have very different cultural display rules of emotion. For example, research has shown that individuals from the United States express negative emotions like fear, anger, and disgust both alone and in the

presence of others, while Japanese individuals only do so while alone (Matsumoto, 1990). Furthermore, individuals from cultures that tend to emphasize social cohesion are more likely to engage in suppression of emotional reaction so they can evaluate which response is most appropriate in a given context (Matsumoto, Yoo, & Nakagawa, 2008).

Other distinct cultural characteristics might be involved in emotionality. For instance, there may be gender differences involved in emotional processing. While research into gender differences in emotional display is equivocal, there is some evidence that men and women may differ in regulation of emotions (McRae, Ochsner, Mauss, Gabrieli, & Gross, 2008).

Despite different emotional display rules, our ability to recognize and produce facial expressions of emotion appears to be universal. In fact, even congenitally blind individuals produce the same facial expression of emotions, despite their never having the opportunity to observe these facial displays of emotion in other people. This would seem to suggest that the pattern of activity in facial muscles involved in generating emotional expressions is universal, and indeed, this idea was suggested in the late 19th century in Charles Darwin's book *The Expression of Emotions in Man and Animals* (1872). In fact, there is substantial evidence for seven universal emotions that are each associated with distinct facial expressions. These include: happiness, surprise, sadness, fright, disgust, contempt, and anger (Ekman & Keltner, 1997).



The seven universal facial expressions of emotion are shown. (credit: modification of work by Cory Zanker)

Does smiling make you happy? Or does being happy make you smile? The facial feedback hypothesis asserts that facial expressions are capable of influencing our emotions, meaning that smiling can make you feel happier (Buck, 1980; Soussignan, 2001; Strack, Martin, & Stepper, 1988). Recent research explored how Botox, which paralyzes facial muscles and limits facial expression, might affect emotion. Havas, Glenberg, Gutowski, Lucarelli, and Davidson (2010) discovered that depressed individuals reported less depression after paralysis of their frowning muscles with Botox injections.

Of course, emotion is not only displayed through facial expression. We also use the tone of our voices, various behaviors, and body language to communicate information about our emotional states. Body language is the expression of emotion in terms of body position or movement. Research suggests that we are quite sensitive to the emotional information communicated through body language, even if we're not consciously aware of it (de Gelder, 2006; Tamietto et al., 2009).

Watch this short History Channel video about body language to see how it plays out in the tense situation of a political debate: Decoding debate body language.

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=141#oembed-1

To apply these same concepts to the more everyday situations most of us face, check out these tips from an interview on the show Today with body language expert Janine Driver.

Autism Spectrum Disorder and Expression of Emotions

Autism spectrum disorder (ASD) is a set of neurodevelopmental disorders characterized by repetitive behaviors and communication and social problems. Children who have autism spectrum disorders have difficulty recognizing the emotional states of others, and research has shown that this may stem from an inability to distinguish various nonverbal expressions of emotion (i.e., facial expressions) from one another (Hobson, 1986). In addition, there is evidence to suggest that autistic individuals also have difficulty expressing emotion through tone of voice and by producing facial expressions (Macdonald et al., 1989). Difficulties with emotional recognition and expression may contribute to the impaired social interaction and communication that characterize autism; therefore, various therapeutic approaches have been explored to address these difficulties. Various educational curricula, cognitive-behavioral therapies, and pharmacological therapies have shown some promise in helping autistic individuals process emotionally relevant information (Bauminger, 2002; Golan & Baron-Cohen, 2006; Guastella et al., 2010).

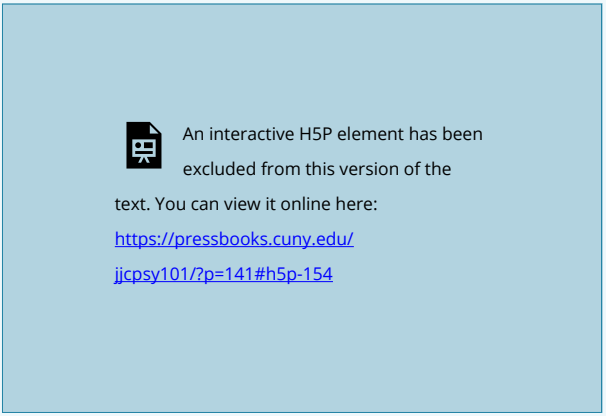
SUMMARY

Emotions are subjective experiences that consist of physiological arousal and cognitive appraisal. Various theories have been put forward to explain our emotional experiences. The James-Lange theory asserts that emotions arise as a function of physiological arousal. The Cannon-Bard theory maintains that emotional experience occurs simultaneous to and independent of physiological arousal. The Schachter-Singer two-factor theory suggests that physiological arousal receives cognitive labels as a function of the relevant context and that these two factors together result in an emotional experience.

The limbic system is the brain's emotional circuit, which includes the amygdala and the hippocampus. Both of these structures are implicated in playing a role in normal emotional processing as well as in psychological mood and anxiety disorders. Increased amygdala activity is associated with learning to fear, and it is seen in individuals who are at risk for or suffering from mood disorders. The volume of the hippocampus has been shown to be reduced in individuals suffering from posttraumatic stress disorder.

The ability to produce and recognize facial expressions of emotions seems to be universal regardless of cultural background. However, there are cultural display rules which influence how often and under what circumstances various emotions can be expressed. Tone of voice and body language also serve as a means by which we communicate information about our emotional states.

REVIEW QUESTIONS



CRITICAL THINKING QUESTIONS

Imagine you find a venomous snake crawling up your leg just after taking a drug that prevented sympathetic nervous system activation. What would

the James-Lange theory predict about your experience?

The James-Lange theory would predict that I would not feel fear because I haven't had the physiological arousal necessary to induce that emotional state.

Why can we not make causal claims regarding the relationship between the volume of the hippocampus and PTSD?

The research that exists is correlational in nature. It could be the case that reduced hippocampal volume predisposes people to develop PTSD or the decreased volume could result from PTSD. Causal claims can only be made when performing an experiment.

PERSONAL APPLICATION QUESTION

Think about times in your life when you have been absolutely elated (e.g., perhaps your school's basketball team just won a closely contested ballgame for the national championship) and very fearful (e.g., you are about to give a speech in your public speaking class to a roomful of 100 strangers). How would you describe how your arousal manifested itself physically? Were there marked differences in physiological arousal associated with each emotional state?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

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Media Attributions

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PERSONALITY

What makes two individuals have different personalities? (credit: modification of work by Nicolas Alejandro)

Three months before William Jefferson Blythe III was born, his father died in a car accident. He was raised by his mother, Virginia Dell, and grandparents, in Hope, Arkansas. When he turned 4, his mother married Roger Clinton, Jr., an alcoholic who was physically abusive to William's mother. Six years later, Virginia gave birth to another son, Roger. William, who later took the last name Clinton from his stepfather, became the 42nd president of the United States. While Bill Clinton was making his political ascendance, his half-brother, Roger Clinton, was arrested numerous times for drug

charges, including possession, conspiracy to distribute cocaine, and driving under the influence, serving time in jail. Two brothers, raised by the same people, took radically different paths in their lives. Why did they make the choices they did? What internal forces shaped their decisions? Personality psychology can help us answer these questions and more.

Learning Objectives

By the end of this section, you will be able to:

- Define personality
- Describe early theories about personality development

Personality refers to the long-standing traits and patterns that propel individuals to consistently think, feel, and behave in specific ways. Our personality is what makes us unique individuals. Each person has an idiosyncratic pattern of enduring, long-term characteristics and a manner in which he or she interacts with other individuals and the world around them. Our personalities are thought to be long term, stable, and not easily changed. The word *personality* comes from the Latin word *persona*. In the ancient world, a persona was a mask worn by an actor. While we tend to think of a mask as being worn to conceal one's identity, the theatrical mask was originally used to either represent or project a specific personality trait of a character.

Happy, sad, impatient, shy, fearful, curious, helpful. What characteristics describe your personality?

HISTORICAL PERSPECTIVES

The concept of personality has been studied for at least 2,000 years, beginning with Hippocrates in 370 BCE (Fazeli, 2012). Hippocrates theorized that personality traits and human behaviors are based on four separate temperaments associated with four fluids ("humors") of the body: choleric temperament (yellow bile from the liver), melancholic temperament (black bile from the kidneys), sanguine temperament (red blood from the heart), and phlegmatic temperament (white phlegm from the lungs) (Clark & Watson, 2008; Eysenck & Eysenck, 1985; Lecci & Magnavita, 2013; Noga, 2007). Centuries later, the influential Greek physician and philosopher Galen built on Hippocrates's theory, suggesting that both diseases and personality differences could be explained by imbalances in the humors and that each person exhibits one of the four temperaments. For example, the choleric person is passionate, ambitious, and bold; the melancholic person is reserved, anxious, and unhappy; the sanguine person is joyful,

eager, and optimistic; and the phlegmatic person is calm, reliable, and thoughtful (Clark & Watson, 2008; Stelmack & Stalikas, 1991). Galen's theory was prevalent for over 1,000 years and continued to be popular through the Middle Ages.

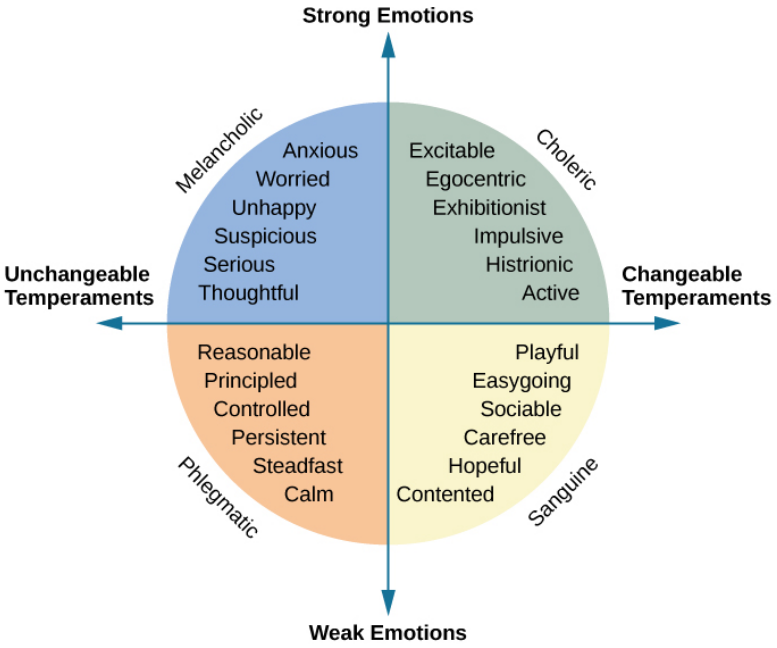
In 1780, Franz Gall, a German physician, proposed that the distances between bumps on the skull reveal a person's personality traits, character, and mental abilities. According to Gall, measuring these distances revealed the sizes of the brain areas underneath, providing information that could be used to determine whether a person was friendly, prideful, murderous, kind, good with languages, and so on. Initially, phrenology was very popular; however, it was soon discredited for lack of empirical support and has long been relegated to the status of pseudoscience (Fancher, 1979).

(a) (b)

The pseudoscience of measuring the areas of a person's skull is known as phrenology. (a) Gall developed a chart that depicted which areas of the skull corresponded to particular personality traits or characteristics (Hothersall, 1995). (b) An 1825 lithograph depicts Gall examining the skull of a young woman. (credit b: modification of work by Wellcome Library, London)

In the centuries after Galen, other researchers contributed to the development of his four primary temperament types, most

prominently Immanuel Kant (in the 18th century) and psychologist Wilhelm Wundt (in the 19th century) (Eysenck, 2009; Stelmack & Stalikas, 1991; Wundt, 1874/1886). Kant agreed with Galen that everyone could be sorted into one of the four temperaments and that there was no overlap between the four categories (Eysenck, 2009). He developed a list of traits that could be used to describe the personality of a person from each of the four temperaments. However, Wundt suggested that a better description of personality could be achieved using two major axes: emotional/nonemotional and changeable/unchangeable. The first axis separated strong from weak emotions (the melancholic and choleric temperaments from the phlegmatic and sanguine). The second axis divided the changeable temperaments (choleric and sanguine) from the unchangeable ones (melancholic and phlegmatic) (Eysenck, 2009).



Developed from Galen's theory of the four temperaments, Kant proposed trait words to describe each temperament. Wundt later suggested the arrangement of the traits on two major axes.

Sigmund Freud's psychodynamic perspective of personality was the first comprehensive theory of personality, explaining a wide variety of both normal and abnormal behaviors. According to Freud, unconscious drives influenced by sex and aggression, along with childhood sexuality, are the forces that influence our personality. Freud attracted many followers who modified his ideas to create new theories about personality. These theorists, referred to as neo-Freudians, generally agreed with Freud that childhood experiences matter, but they reduced the emphasis on sex and focused more on the social environment and effects of culture on personality. The perspective of personality proposed by Freud and his followers was the dominant theory of personality for the first half of the 20th century.

Other major theories then emerged, including the learning, humanistic, biological, evolutionary, trait, and cultural perspectives. In this chapter, we will explore these various perspectives on personality in depth.

View this video for a brief overview of some of the psychological perspectives on personality: Video Lecture: Explain the Major Perspectives on Personality.

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TEST YOUR UNDERSTANDING

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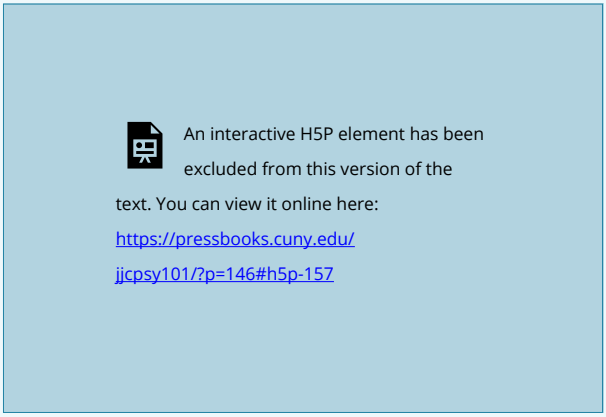
SUMMARY

Personality has been studied for over 2,000 years, beginning with Hippocrates. More recent theories of personality have been proposed, including Freud's psychodynamic perspective, which holds that personality is formed through early childhood experiences. Other perspectives then emerged in reaction to the psychodynamic perspective, including the learning, humanistic, biological, trait, and cultural perspectives.

REVIEW QUESTIONS



CRITICAL THINKING QUESTIONS

What makes a personal quality part of someone's personality?

The particular quality or trait must be part of an

enduring behavior pattern, so that it is a consistent or predictable quality.

PERSONAL APPLICATION QUESTIONS

1. How would you describe your own personality? Do you think that friends and family would describe you in much the same way? Why or why not?
2. How would you describe your personality in an online dating profile?
3. What are some of your positive and negative personality qualities? How do you think these qualities will affect your choice of career?

GLOSSARY

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Media Attributions

- "Video Lecture: Explain the Major Perspectives on Personality" by wkeenecsu. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Describe the assumptions of the psychodynamic perspective on personality development
- Define and describe the nature and function of the id, ego, and superego
- Define and describe the defense mechanisms
- Define and describe the psychosexual stages of personality development

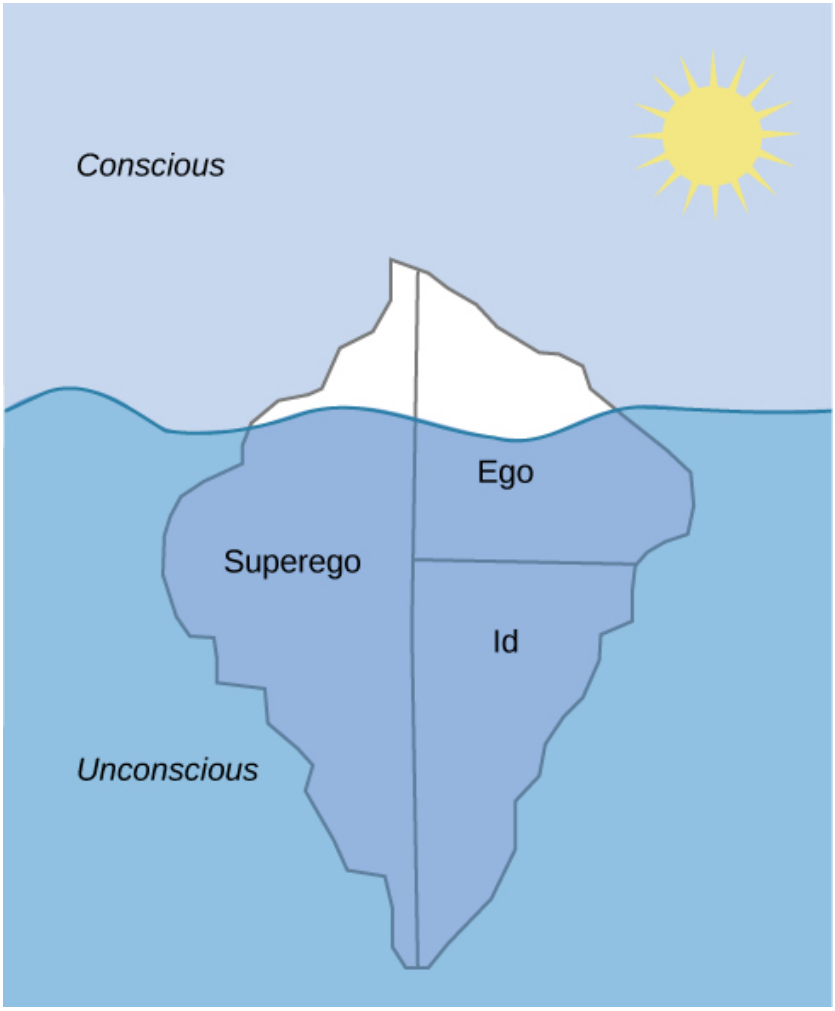
Sigmund Freud (1856–1939) is probably the most controversial and misunderstood psychological theorist. When reading Freud's theories, it is important to remember that he was a medical doctor, not a psychologist. There was no such thing as a degree in psychology at the time that he received his education, which can help us understand some of the controversy over his theories today. However, Freud was the first to systematically study and theorize the workings of the unconscious mind in the manner that we associate with modern psychology.

In the early years of his career, Freud worked with Josef Breuer, a Viennese physician. During this time, Freud became intrigued by the story of one of Breuer's patients, Bertha Pappenheim, who was referred to by the pseudonym Anna O. (Launer, 2005). Anna O. had been caring for her dying father when she began to experience symptoms such as partial paralysis, headaches, blurred vision, amnesia, and hallucinations (Launer, 2005). In Freud's day, these

symptoms were commonly referred to as hysteria. Anna O. turned to Breuer for help. He spent 2 years (1880–1882) treating Anna O. and discovered that allowing her to talk about her experiences seemed to bring some relief of her symptoms. Anna O. called his treatment the "talking cure" (Launer, 2005). Despite the fact the Freud never met Anna O., her story served as the basis for the 1895 book, *Studies on Hysteria*, which he co-authored with Breuer. Based on Breuer's description of Anna O.'s treatment, Freud concluded that hysteria was the result of sexual abuse in childhood and that these traumatic experiences had been hidden from consciousness. Breuer disagreed with Freud, which soon ended their work together. However, Freud continued to work to refine talk therapy and build his theory on personality.

LEVELS OF CONSCIOUSNESS

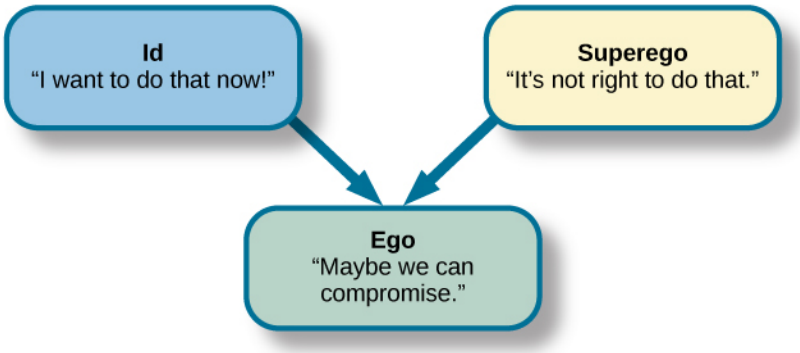
To explain the concept of conscious versus unconscious experience, Freud compared the mind to an iceberg. He said that only about one-tenth of our mind is conscious, and the rest of our mind is unconscious. Our unconscious refers to that mental activity of which we are unaware and are unable to access (Freud, 1923). According to Freud, unacceptable urges and desires are kept in our unconscious through a process called repression. For example, we sometimes say things that we don't intend to say by unintentionally substituting another word for the one we meant. You've probably heard of a Freudian slip, the term used to describe this. Freud suggested that slips of the tongue are actually sexual or aggressive urges, accidentally slipping out of our unconscious. Speech errors such as this are quite common. Seeing them as a reflection of unconscious desires, linguists today have found that slips of the tongue tend to occur when we are tired, nervous, or not at our optimal level of cognitive functioning (Motley, 2002).



Freud believed that we are only aware of a small amount of our mind's activities and that most of it remains hidden from us in our unconscious. The information in our unconscious affects our behavior, although we are unaware of it.

According to Freud, our personality develops from a conflict between two forces: our biological aggressive and pleasure-seeking

drives versus our internal (socialized) control over these drives. Our personality is the result of our efforts to balance these two competing forces. Freud suggested that we can understand this by imagining three interacting systems within our minds. He called them the id, ego, and superego.



The job of the ego, or self, is to balance the aggressive/pleasure-seeking drives of the id with the moral control of the superego.

The unconscious id contains our most primitive drives or urges, and is present from birth. It directs impulses for hunger, thirst, and sex. Freud believed that the id operates on what he called the "pleasure principle," in which the id seeks immediate gratification. Through social interactions with parents and others in a child's environment, the ego and superego develop to help control the id. The superego develops as a child interacts with others, learning the social rules for right and wrong. The superego acts as our conscience; it is our moral compass that tells us how we should behave. It strives for perfection and judges our behavior, leading to feelings of pride or—when we fall short of the ideal—feelings of guilt. In contrast to the instinctual id and the rule-based superego, the ego is the rational part of our personality. It's what Freud considered to be the self, and it is the part of our personality that is seen by others. Its job is to balance the demands of the id and superego in the context of reality; thus, it operates on what Freud

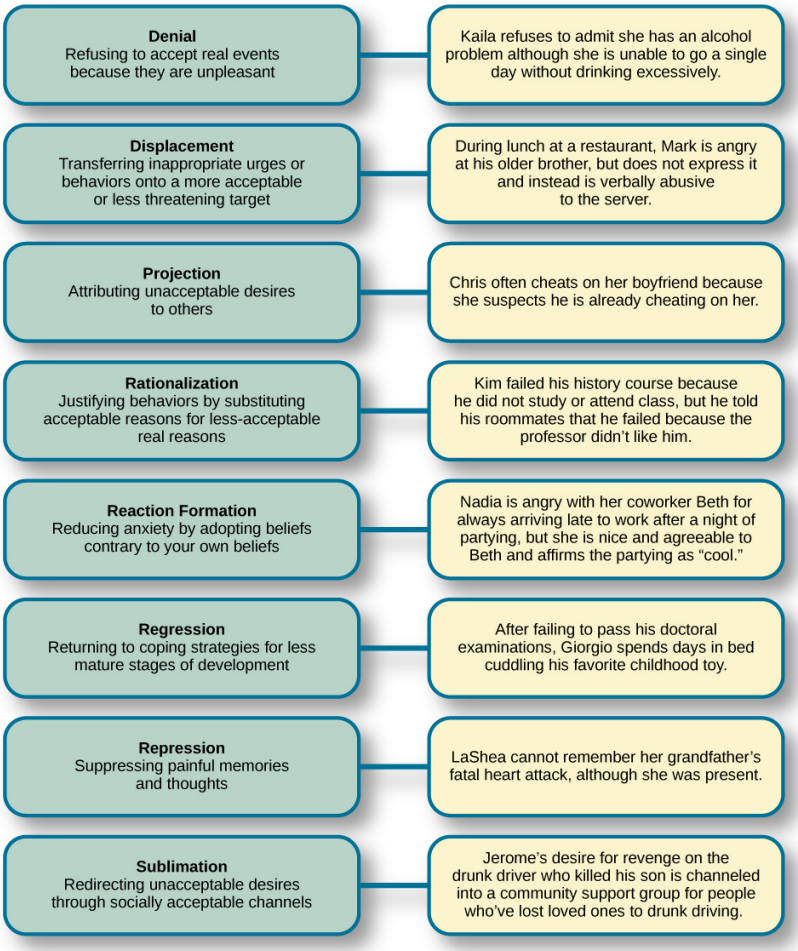
called the "reality principle." The ego helps the id satisfy its desires in a realistic way.

The id and superego are in constant conflict, because the id wants instant gratification regardless of the consequences, but the superego tells us that we must behave in socially acceptable ways. Thus, the ego's job is to find the middle ground. It helps satisfy the id's desires in a rational way that will not lead us to feelings of guilt. According to Freud, a person who has a strong ego, which can balance the demands of the id and the superego, has a healthy personality. Freud maintained that imbalances in the system can lead to neurosis (a tendency to experience negative emotions), anxiety disorders, or unhealthy behaviors. For example, a person who is dominated by their id might be narcissistic and impulsive. A person with a dominant superego might be controlled by feelings of guilt and deny themselves even socially acceptable pleasures; conversely, if the superego is weak or absent, a person might become a psychopath. An overly dominant superego might be seen in an over-controlled individual whose rational grasp on reality is so strong that they are unaware of their emotional needs, or, in a neurotic who is overly defensive (overusing ego defense mechanisms).

DEFENSE MECHANISMS

Freud believed that feelings of anxiety result from the ego's inability to mediate the conflict between the id and superego. When this happens, Freud believed that the ego seeks to restore balance through various protective measures known as defense mechanisms. When certain events, feelings, or yearnings cause an individual anxiety, the individual wishes to reduce that anxiety. To do that, the individual's unconscious mind uses ego defense mechanisms, unconscious protective behaviors that aim to reduce anxiety. The ego, usually conscious, resorts to unconscious

strivings to protect the ego from being overwhelmed by anxiety. When we use defense mechanisms, we are unaware that we are using them. Further, they operate in various ways that distort reality. According to Freud, we all use ego defense mechanisms.



Defense mechanisms are unconscious protective behaviors that work to reduce anxiety.

While everyone uses defense mechanisms, Freud believed that overuse of them may be problematic. For example, let's say Joe Smith is a high school football player. Deep down, Joe feels sexually attracted to males. His conscious belief is that being gay is immoral and that if he were gay, his family would disown him and he would be ostracized by his peers. Therefore, there is a conflict between his conscious beliefs (being gay is wrong and will result in being ostracized) and his unconscious urges (attraction to males). The idea that he might be gay causes Joe to have feelings of anxiety. How can he decrease his anxiety? Joe may find himself acting very "macho," making gay jokes, and picking on a school peer who is gay. This way, Joe's unconscious impulses are further submerged.

There are several different types of defense mechanisms. For instance, in repression, anxiety-causing memories from consciousness are blocked. As an analogy, let's say your car is making a strange noise, but because you do not have the money to get it fixed, you just turn up the radio so that you no longer hear the strange noise. Eventually you forget about it. Similarly, in the human psyche, if a memory is too overwhelming to deal with, it might be repressed and thus removed from conscious awareness (Freud, 1920). This repressed memory might cause symptoms in other areas.

Another defense mechanism is reaction formation, in which someone expresses feelings, thoughts, and behaviors opposite to their inclinations. In the above example, Joe made fun of a homosexual peer while himself being attracted to males. In regression, an individual acts much younger than their age. For example, a four-year-old child who resents the arrival of a newborn sibling may act like a baby and revert to drinking out of a bottle. In projection, a person refuses to acknowledge her own unconscious feelings and instead sees those feelings in someone else. Other defense mechanisms include rationalization, displacement, and sublimation.

Watch this video for a review of Freud's defense mechanisms: Freudian Defense Mechanisms.

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STAGES OF PSYCHOSEXUAL DEVELOPMENT

Freud believed that personality develops during early childhood: Childhood experiences shape our personalities as well as our behavior as adults. He asserted that we develop via a series of stages during childhood. Each of us must pass through these childhood stages, and if we do not have the proper nurturing and parenting during a stage, we will be stuck, or fixated, in that stage, even as adults.

In each psychosexual stage of development, the child's pleasure-seeking urges, coming from the id, are focused on a different area

of the body, called an erogenous zone. The stages are oral, anal, phallic, latency, and genital.

Freud's psychosexual development theory is quite controversial. To understand the origins of the theory, it is helpful to be familiar with the political, social, and cultural influences of Freud's day in Vienna at the turn of the 20th century. During this era, a climate of sexual repression, combined with limited understanding and education surrounding human sexuality, heavily influenced Freud's perspective. Given that sex was a taboo topic, Freud assumed that negative emotional states (neuroses) stemmed from suppression of unconscious sexual and aggressive urges. For Freud, his own recollections and interpretations of patients' experiences and dreams were sufficient proof that psychosexual stages were universal events in early childhood.

Freud's Stages of Psychosexual Development

Stage	Age (years)	Erogenous Zone	Major Conflict	Adult Fixation Example
Oral	0–1	Mouth	Weaning off breast or bottle	Smoking, overeating
Anal	1–3	Anus	Toilet training	Neatness, messiness
Phallic	3–6	Genitals	Oedipus/Electra complex	Vanity, overambition
Latency	6–12	None	None	None
Genital	12+	Genitals	None	None

ORAL STAGE

In the oral stage (birth to 1 year), pleasure is focused on the mouth. Eating and the pleasure derived from sucking (nipples, pacifiers, and thumbs) play a large part in a baby's first year of life. At around 1 year of age, babies are weaned from the bottle or breast, and this process can create conflict if not handled properly by caregivers. According to Freud, an adult who smokes, drinks, overeats, or bites her nails is fixated in the oral stage of her psychosexual

development; she may have been weaned too early or too late, resulting in these fixation tendencies, all of which seek to ease anxiety.

ANAL STAGE

After passing through the oral stage, children enter what Freud termed the anal stage (1–3 years). In this stage, children experience pleasure in their bowel and bladder movements, so it makes sense that the conflict in this stage is over toilet training. Freud suggested that success at the anal stage depended on how parents handled toilet training. Parents who offer praise and rewards encourage positive results and can help children feel competent. Parents who are harsh in toilet training can cause a child to become fixated at the anal stage, leading to the development of an anal-retentive personality. The anal-retentive personality is stingy and stubborn, has a compulsive need for order and neatness, and might be considered a perfectionist. If parents are too lenient in toilet training, the child might also become fixated and display an anal-expulsive personality. The anal-expulsive personality is messy, careless, disorganized, and prone to emotional outbursts.

PHALLIC STAGE

Freud's third stage of psychosexual development is the phallic stage (3–6 years), corresponding to the age when children become aware of their bodies and recognize the differences between boys and girls. The erogenous zone in this stage is the genitals. Conflict arises when the child feels a desire for the opposite-sex parent, and jealousy and hatred toward the same-sex parent. For boys, this is called the Oedipus complex, involving a boy's desire for his mother and his urge to replace his father who is seen as a rival for the mother's attention. At the same time, the boy is afraid his father will punish him for his feelings, so he experiences *castration anxiety*.

The Oedipus complex is successfully resolved when the boy begins to identify with his father as an indirect way to have the mother. Failure to resolve the Oedipus complex may result in fixation and development of a personality that might be described as vain and overly ambitious.

Girls experience a comparable conflict in the phallic stage—the Electra complex. The Electra complex, while often attributed to Freud, was actually proposed by Freud's protégé, Carl Jung (Jung & Kerenyi, 1963). A girl desires the attention of her father and wishes to take her mother's place. Jung also said that girls are angry with the mother for not providing them with a penis—hence the term *penis envy*. While Freud initially embraced the Electra complex as a parallel to the Oedipus complex, he later rejected it, yet it remains as a cornerstone of Freudian theory, thanks in part to academics in the field (Freud, 1931/1968; Scott, 2005).

LATENCY PERIOD

Following the phallic stage of psychosexual development is a period known as the latency period (6 years to puberty). This period is not considered a stage, because sexual feelings are dormant as children focus on other pursuits, such as school, friendships, hobbies, and sports. Children generally engage in activities with peers of the same sex, which serves to consolidate a child's gender-role identity.

GENITAL STAGE

The final stage is the genital stage (from puberty on). In this stage, there is a sexual reawakening as the incestuous urges resurface. The young person redirects these urges to other, more socially acceptable partners (who often resemble the other-sex parent). People in this stage have mature sexual interests, which for Freud meant a strong desire for the opposite sex. Individuals who

successfully completed the previous stages, reaching the genital stage with no fixations, are said to be well-balanced, healthy adults.

While most of Freud's ideas have not found support in modern research, we cannot discount the contributions that Freud has made to the field of psychology. It was Freud who pointed out that a large part of our mental life is influenced by the experiences of early childhood and takes place outside of our conscious awareness; his theories paved the way for others.

TEST YOUR UNDERSTANDING

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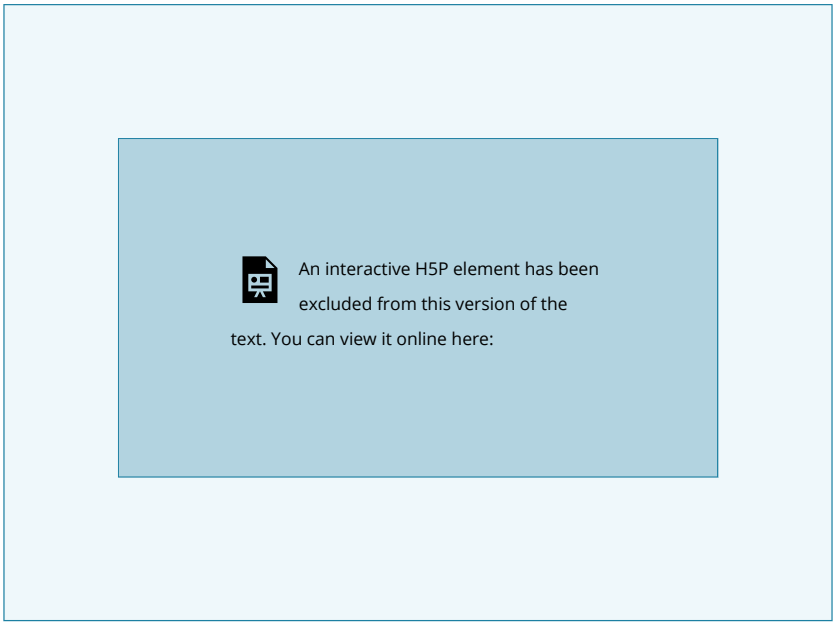
SUMMARY

Sigmund Freud presented the first comprehensive theory of personality. He was also the first to recognize that much of our mental life takes place outside of our conscious awareness. Freud also proposed three components to our personality: the id, ego, and superego. The job of the ego is to balance the sexual and aggressive drives of the id with the moral ideal of the superego. Freud also said that personality develops through a series of psychosexual stages. In each stage, pleasure focuses on a specific erogenous zone. Failure to resolve a stage can lead one to become fixated in that stage, leading to unhealthy personality traits. Successful resolution of the stages leads to a healthy adult.

REVIEW QUESTIONS



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CRITICAL THINKING QUESTIONS

How might the common expression "daddy's girl" be rooted in the idea of the Electra complex?

Since the idea behind the Electra complex is that the daughter competes with her same-sex parent for the attention of her opposite-sex parent, the term "daddy's girl" might suggest that the daughter has an overly close relationship with her father and a more distant—or even antagonistic—relationship with her mother.

Describe the personality of someone who is fixated at the anal stage.

If parents are too harsh during potty training, a person could become fixated at this stage and would be called anal retentive. The anal-retentive personality is stingy, stubborn, has a compulsive need for order and neatness, and might be considered a perfectionist. On the other hand, some parents may be too soft when it comes to potty training. In this case, Freud said that children could also become fixated and display an anal-expulsive personality. As an adult, an anal-expulsive personality is messy, careless, disorganized, and prone to emotional outbursts.

PERSONAL APPLICATION QUESTIONS

What are some examples of defense mechanisms that you have used yourself or have witnessed others using?

GLOSSARY

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Media Attributions

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Learning Objectives

By the end of this section, you will be able to:

- Discuss the concept of the inferiority complex
- Discuss the core differences between Erikson's and Freud's views on personality
- Discuss Jung's ideas of the collective unconscious and archetypes
- Discuss the work of Karen Horney, including her revision of Freud's "penis envy"

Freud attracted many followers who modified his ideas to create new theories about personality. These theorists, referred to as neo-Freudians, generally agreed with Freud that childhood experiences matter, but deemphasized sex, focusing more on the social environment and effects of culture on personality. Four notable neo-Freudians include Alfred Adler, Erik Erikson, Carl Jung (pronounced "Yoong"), and Karen Horney (pronounced "HORN-eye").

ALFRED ADLER

Alfred Adler, a colleague of Freud's and the first president of the Vienna Psychoanalytical Society (Freud's inner circle of colleagues), was the first major theorist to break away from Freud. He subsequently founded a school of psychology called individual

psychology, which focuses on our drive to compensate for feelings of inferiority. Adler (1937, 1956) proposed the concept of the inferiority complex. An inferiority complex refers to a person's feelings that they lack worth and don't measure up to the standards of others or of society. Adler's ideas about inferiority represent a major difference between his thinking and Freud's. Freud believed that we are motivated by sexual and aggressive urges, but Adler (1930, 1961) believed that feelings of inferiority in childhood are what drive people to attempt to gain superiority and that this striving is the force behind all of our thoughts, emotions, and behaviors.

Alfred Adler proposed the concept of the inferiority complex.

Adler also believed in the importance of social connections, seeing childhood development emerging through social development rather than the sexual stages Freud outlined. Adler noted the interrelatedness of humanity and the need to work together for the betterment of all. He said, "The happiness of mankind lies in

working together, in living as if each individual had set himself the task of contributing to the common welfare" (Adler, 1964, p. 255) with the main goal of psychology being "to recognize the equal rights and equality of others" (Adler, 1961, p. 691).

With these ideas, Adler identified three fundamental social tasks that all of us must experience: occupational tasks (careers), societal tasks (friendship), and love tasks (finding an intimate partner for a long-term relationship). Rather than focus on sexual or aggressive motives for behavior as Freud did, Adler focused on social motives. He also emphasized conscious rather than unconscious motivation, since he believed that the three fundamental social tasks are explicitly known and pursued. That is not to say that Adler did not also believe in unconscious processes—he did—but he felt that conscious processes were more important.

One of Adler's major contributions to personality psychology was the idea that our birth order shapes our personality. He proposed that older siblings, who start out as the focus of their parents' attention but must share that attention once a new child joins the family, compensate by becoming overachievers. The youngest children, according to Adler, may be spoiled, leaving the middle child with the opportunity to minimize the negative dynamics of the youngest and oldest children. Despite popular attention, research has not conclusively confirmed Adler's hypotheses about birth order.

One of Adler's major contributions to personality psychology was the idea that our birth order shapes our personality. Follow this link to view a summary of

birth order theory: The Birth Order Effect.

ERIK ERIKSON

As an art school dropout with an uncertain future, young Erik Erikson met Freud's daughter, Anna Freud, while he was tutoring the children of an American couple undergoing psychoanalysis in Vienna. It was Anna Freud who encouraged Erikson to study psychoanalysis. Erikson received his diploma from the Vienna Psychoanalytic Institute in 1933, and as Nazism spread across Europe, he fled the country and immigrated to the United States that same year. As you learned when you studied lifespan development, Erikson later proposed a psychosocial theory of development, suggesting that an individual's personality develops throughout the lifespan—a departure from Freud's view that personality is fixed in early life. In his theory, Erikson emphasized the social relationships that are important at each stage of personality development, in contrast to Freud's emphasis on sex. Erikson identified eight stages, each of which represents a conflict or developmental task. The development of a healthy personality and a sense of competence depend on the successful completion of each task.

Erikson's Psychosocial Stages of Development

Stage	Age (years)	Developmental Task	Description
1	0–1	Trust vs. mistrust	Trust (or mistrust) that basic needs, such as nourishment and affection, will be met
2	1–3	Autonomy vs. shame/doubt	Sense of independence in many tasks develops
3	3–6	Initiative vs. guilt	Take initiative on some activities, may develop guilt when success not met or boundaries overstepped
4	7–11	Industry vs. inferiority	Develop self-confidence in abilities when competent or sense of inferiority when not
5	12–18	Identity vs. confusion	Experiment with and develop identity and roles
6	19–29	Intimacy vs. isolation	Establish intimacy and relationships with others
7	30–64	Generativity vs. stagnation	Contribute to society and be part of a family
8	65–	Integrity vs. despair	Assess and make sense of life and meaning of contributions

CARL JUNG

Carl Jung was a Swiss psychiatrist and protégé of Freud, who later split off from Freud and developed his own theory, which he called analytical psychology. The focus of analytical psychology is on working to balance opposing forces of conscious and unconscious thought, and experience within one's personality. According to Jung, this work is a continuous learning process—mainly occurring in the second half of life—of becoming aware of unconscious elements and integrating them into consciousness.

Carl Jung was interested in exploring the collective unconscious.

Jung's split from Freud was based on two major disagreements. First, Jung, like Adler and Erikson, did not accept that sexual drive was the primary motivator in a person's mental life. Second, although Jung agreed with Freud's concept of a personal unconscious, he thought it to be incomplete. In addition to the personal unconscious, Jung focused on the collective unconscious.

The collective unconscious is a universal version of the personal unconscious, holding mental patterns, or memory traces, which are common to all of us (Jung, 1928). These ancestral memories, which Jung called archetypes, are represented by universal themes in various cultures, as expressed through literature, art, and dreams (Jung). Jung said that these themes reflect common experiences of people the world over, such as facing death, becoming independent, and striving for mastery. Jung (1964) believed that through biology, each person is handed down the same themes and that the same types of symbols—such as the hero, the maiden, the sage, and the trickster—are present in the folklore and fairy tales of every culture. In Jung's view, the task of integrating these unconscious archetypal aspects of the self is part of the self-realization process in the second half of life. With this orientation toward self-realization, Jung parted ways with Freud's belief that personality is determined solely by past events and anticipated the humanistic movement with its emphasis on self-actualization and orientation toward the future.

Jung also proposed two attitudes or approaches toward life: extroversion and introversion (Jung, 1923). These ideas are considered Jung's most important contributions to the field of personality psychology, as almost all models of personality now include these concepts. If you are an extrovert, then you are a person who is energized by being outgoing and socially oriented: You derive your energy from being around others. If you are an introvert, then you are a person who may be quiet and reserved, or you may be social, but your energy is derived from your inner psychic activity. Jung believed a balance between extroversion and introversion best served the goal of self-realization.

Introverts and Extroverts

Introvert	Extrovert
Energized by being alone	Energized by being with others
Avoids attention	Seeks attention
Speaks slowly and softly	Speaks quickly and loudly
Thinks before speaking	Thinks out loud
Stays on one topic	Jumps from topic to topic
Prefers written communication	Prefers verbal communication
Pays attention easily	Distractible
Cautious	Acts first, thinks later

Another concept proposed by Jung was the persona, which he referred to as a mask that we adopt. According to Jung, we consciously create this persona; however, it is derived from both our conscious experiences and our collective unconscious. What is the purpose of the persona? Jung believed that it is a compromise between who we really are (our true self) and what society expects us to be. We hide those parts of ourselves that are not aligned with society's expectations.

Jung's view of extroverted and introverted types serves as a basis of the Myers-Briggs Type Indicator (MBTI). This questionnaire describes a person's degree of introversion versus extroversion, thinking versus feeling, intuition versus sensation, and judging versus perceiving. This site provides a modified questionnaire based on the MBTI: Jung Typology Test.

Are Archetypes Genetically Based?

Jung proposed that human responses to archetypes are similar to instinctual responses in animals. One criticism of Jung is that there is no evidence that archetypes are biologically based or similar to animal instincts (Roesler, 2012). Jung formulated his ideas about 100 years ago, and great advances have been made in the field of genetics since that time. We've found that human babies are born with certain capacities, including the ability to acquire language. However, we've also found that symbolic information (such as archetypes) is not encoded on the genome and that babies cannot decode symbolism, refuting the idea of a biological basis to archetypes. Rather than being seen as purely biological, more recent research suggests that archetypes emerge directly from our experiences and are reflections of linguistic or cultural characteristics (Young-Eisendrath, 1995). Today, most Jungian scholars believe that the collective unconscious and archetypes are based on both innate and environmental influences, with the differences being in the role and degree of each (Sotirova-Kohli et al., 2013).

KAREN HORNEY

Karen Horney was one of the first women trained as a Freudian psychoanalyst. During the Great Depression, Horney moved from Germany to the United States, and subsequently moved away from

Freud's teachings. Like Jung, Horney believed that each individual has the potential for self-realization and that the goal of psychoanalysis should be moving toward a healthy self rather than exploring early childhood patterns of dysfunction. Horney also disagreed with the Freudian idea that girls have penis envy and are jealous of male biological features. According to Horney, any jealousy is most likely culturally based, due to the greater privileges that males often have, meaning that the differences between men's and women's personalities are culturally based, not biologically based. She further suggested that men have womb envy, because they cannot give birth.

Horney's theories focused on the role of unconscious anxiety. She suggested that normal growth can be blocked by basic anxiety stemming from needs not being met, such as childhood experiences of loneliness and/or isolation. How do children learn to handle this anxiety? Horney suggested three styles of coping. The first coping style, *moving toward people*, relies on affiliation and dependence. These children become dependent on their parents and other caregivers in an effort to receive attention and affection, which provides relief from anxiety (Burger, 2008). When these children grow up, they tend to use this same coping strategy to deal with relationships, expressing an intense need for love and acceptance (Burger, 2008). The second coping style, *moving against people*, relies on aggression and assertiveness. Children with this coping style find that fighting is the best way to deal with an unhappy home situation, and they deal with their feelings of insecurity by bullying other children (Burger, 2008). As adults, people with this coping style tend to lash out with hurtful comments and exploit others (Burger, 2008). The third coping style, *moving away from people*, centers on detachment and isolation. These children handle their anxiety by withdrawing from the world. They need privacy and tend to be self-sufficient. When these children are adults, they continue to avoid such things as love

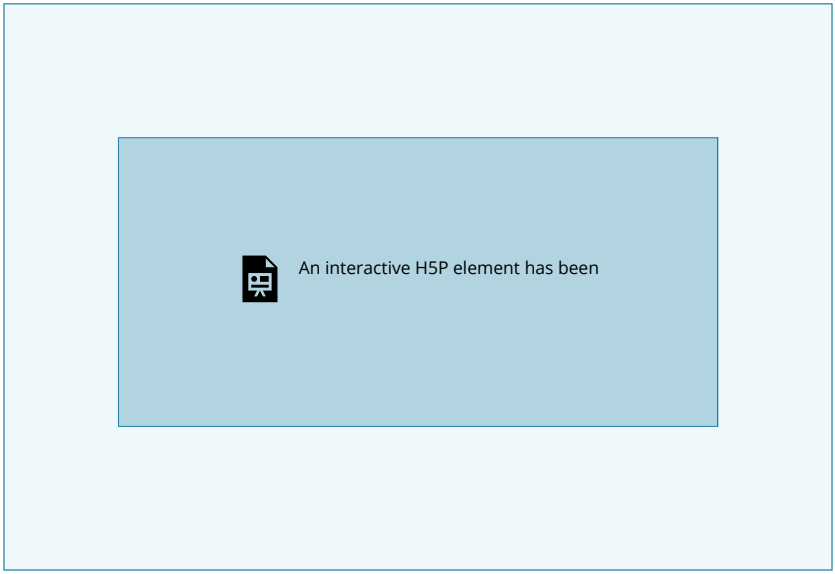
and friendship, and they also tend to gravitate toward careers that require little interaction with others (Burger, 2008).

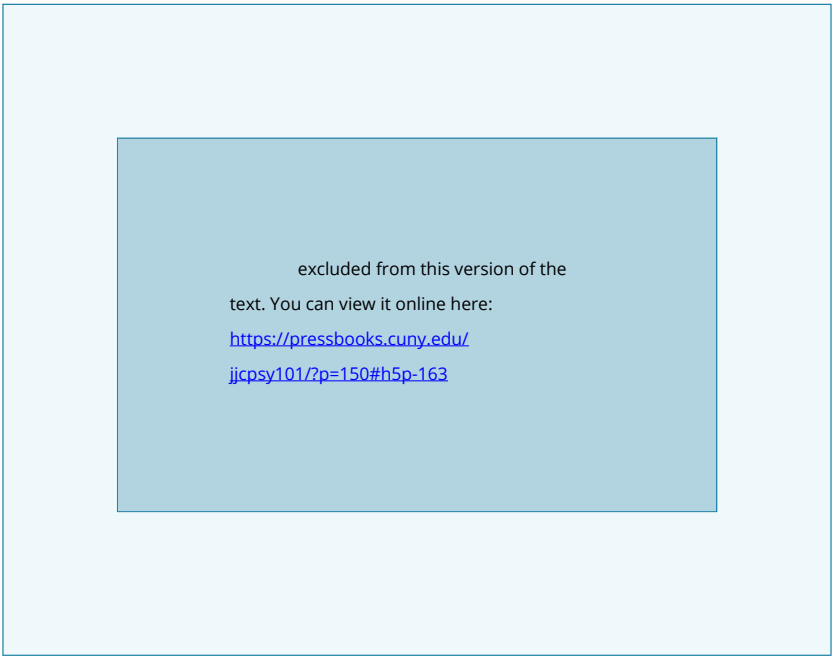
Horney's Coping Styles

Coping Style	Description	Example
Moving toward people	Affiliation and dependence	Child seeking positive attention and affection from parent; adult needing love
Moving against people	Aggression and manipulation	Child fighting or bullying other children; adult who is abrasive and verbally hurtful, or who exploits others
Moving away from people	Detachment and isolation	Child withdrawn from the world and isolated; adult loner

Horney believed these three styles are ways in which people typically cope with day-to-day problems; however, the three coping styles can become neurotic strategies if they are used rigidly and compulsively, leading a person to become alienated from others.

TEST YOUR UNDERSTANDING





SUMMARY

The neo-Freudians were psychologists whose work followed from Freud's. They generally agreed with Freud that childhood experiences matter, but they decreased the emphasis on sex and focused more on the social environment and effects of culture on personality. Some of the notable neo-Freudians are Alfred Adler, Carl Jung, Erik Erikson, and Karen Horney. The neo-Freudian approaches have been criticized, because they tend to be philosophical rather than based on sound scientific research. For example, Jung's conclusions about the existence of the collective unconscious are based on myths, legends, dreams, and art. In addition, as with Freud's psychoanalytic theory, the neo-Freudians

based much of their theories of personality on information from their patients.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Describe the difference between extroverts and introverts in terms of what is energizing to each.

Extroverts are energized by social engagement. Introverts are recharged by solitary time.

Discuss Horney's perspective on Freud's concept of penis envy.

Horney disagreed with the Freudian idea that women had penis envy and were jealous of a man's biological features. Horney discussed that the jealousy was more likely culturally based, due to the greater privileges that males often have, and that differences between men and women's personalities were cultural, not biologically based. Horney also suggested that men may have womb envy, because men cannot give birth.

PERSONAL APPLICATION QUESTIONS

What is your birth order? Do you agree or disagree with Adler's description of your personality based on his birth order theory, as described in the Link to Learning? Provide examples for support.

Would you describe yourself as an extrovert or an introvert? Does this vary based on the situation? Provide examples to support your points.

Select an epic story that is popular in contemporary society (such as *Harry Potter* or *Star Wars*) and explain it terms of Jung's concept of archetypes.

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Describe the behaviorist perspective on personality
- Describe the cognitive perspective on personality
- Describe the social cognitive perspective on personality

In contrast to the psychodynamic approaches of Freud and the neo-Freudians, which relate personality to inner (and hidden) processes, the learning approaches focus only on observable behavior. This illustrates one significant advantage of the learning approaches over psychodynamics: Because learning approaches involve observable, measurable phenomena, they can be scientifically tested.

THE BEHAVIORAL PERSPECTIVE

Behaviorists do not believe in biological determinism: They do not see personality traits as inborn. Instead, they view personality as significantly shaped by the reinforcements and consequences outside of the organism. In other words, people behave in a consistent manner based on prior learning. B. F. Skinner, a strict behaviorist, believed that environment was solely responsible for all behavior, including the enduring, consistent behavior patterns studied by personality theorists.

As you may recall from your study on the psychology of learning,

Skinner proposed that we demonstrate consistent behavior patterns because we have developed certain response tendencies (Skinner, 1953). In other words, we *learn* to behave in particular ways. We increase the behaviors that lead to positive consequences, and we decrease the behaviors that lead to negative consequences. Skinner disagreed with Freud's idea that personality is fixed in childhood. He argued that personality develops over our entire life, not only in the first few years. Our responses can change as we come across new situations; therefore, we can expect more variability over time in personality than Freud would anticipate. For example, consider a young woman, Greta, a risk taker. She drives fast and participates in dangerous sports such as hang gliding and kiteboarding. But after she gets married and has children, the system of reinforcements and punishments in her environment changes. Speeding and extreme sports are no longer reinforced, so she no longer engages in those behaviors. In fact, Greta now describes herself as a cautious person.

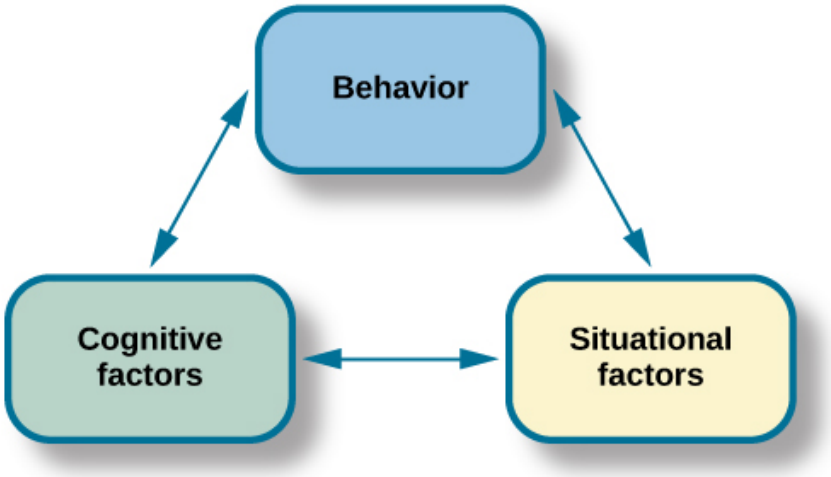
THE SOCIAL-COGNITIVE PERSPECTIVE

Albert Bandura agreed with Skinner that personality develops through learning. He disagreed, however, with Skinner's strict behaviorist approach to personality development, because he felt that thinking and reasoning are important components of learning. He presented a social-cognitive theory of personality that emphasizes both learning and cognition as sources of individual differences in personality. In social-cognitive theory, the concepts of reciprocal determinism, observational learning, and self-efficacy all play a part in personality development.

RECIPROCAL DETERMINISM

In contrast to Skinner's idea that the environment alone

determines behavior, Bandura (1990) proposed the concept of reciprocal determinism, in which cognitive processes, behavior, and context all interact, each factor influencing and being influenced by the others simultaneously. *Cognitive processes* refer to all characteristics previously learned, including beliefs, expectations, and personality characteristics. *Behavior* refers to anything that we do that may be rewarded or punished. Finally, the *context* in which the behavior occurs refers to the environment or situation, which includes rewarding/punishing stimuli.



Bandura proposed the idea of reciprocal determinism: Our behavior, cognitive processes, and situational context all influence each other.

Consider, for example, that you're at a festival and one of the attractions is bungee jumping from a bridge. Do you do it? In this example, the behavior is bungee jumping. Cognitive factors that might influence this behavior include your beliefs and values, and your past experiences with similar behaviors. Finally, context refers to the reward structure for the behavior. According to reciprocal determinism, all of these factors are in play.

OBSERVATIONAL LEARNING

Bandura's key contribution to learning theory was the idea that much learning is vicarious. We learn by observing someone else's behavior and its consequences, which Bandura called observational learning. He felt that this type of learning also plays a part in the development of our personality. Just as we learn individual behaviors, we learn new behavior patterns when we see them performed by other people or models. Drawing on the behaviorists' ideas about reinforcement, Bandura suggested that whether we choose to imitate a model's behavior depends on whether we see the model reinforced or punished. Through observational learning, we come to learn what behaviors are acceptable and rewarded in our culture, and we also learn to inhibit deviant or socially unacceptable behaviors by seeing what behaviors are punished.

We can see the principles of reciprocal determinism at work in observational learning. For example, personal factors determine which behaviors in the environment a person chooses to imitate, and those environmental events in turn are processed cognitively according to other personal factors.

SELF-EFFICACY

Bandura (1977, 1995) has studied a number of cognitive and personal factors that affect learning and personality development, and most recently has focused on the concept of self-efficacy. Self-efficacy is our level of confidence in our own abilities, developed through our social experiences. Self-efficacy affects how we approach challenges and reach goals. In observational learning, self-efficacy is a cognitive factor that affects which behaviors we choose to imitate as well as our success in performing those behaviors.

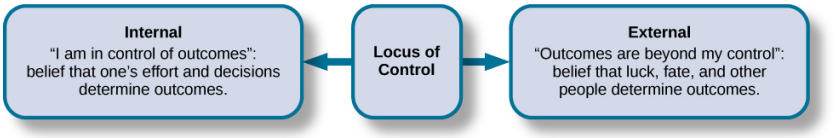
People who have high self-efficacy believe that their goals are

within reach, have a positive view of challenges seeing them as tasks to be mastered, develop a deep interest in and strong commitment to the activities in which they are involved, and quickly recover from setbacks. Conversely, people with low self-efficacy avoid challenging tasks because they doubt their ability to be successful, tend to focus on failure and negative outcomes, and lose confidence in their abilities if they experience setbacks. Feelings of self-efficacy can be specific to certain situations. For instance, a student might feel confident in her ability in English class but much less so in math class.

JULIAN ROTTER AND LOCUS OF CONTROL

Julian Rotter (1966) proposed the concept of locus of control, another cognitive factor that affects learning and personality development. Distinct from self-efficacy, which involves our belief in our own abilities, locus of control refers to our beliefs about the power we have over our lives. In Rotter's view, people possess either an internal or an external locus of control. Those of us with an internal locus of control ("internals") tend to believe that most of our outcomes are the direct result of our efforts. Those of us with an external locus of control ("externals") tend to believe that our outcomes are outside of our control. Externals see their lives as being controlled by other people, luck, or chance. For example, say you didn't spend much time studying for your psychology test and went out to dinner with friends instead. When you receive your test score, you see that you earned a D. If you possess an internal locus of control, you would most likely admit that you failed because you didn't spend enough time studying and decide to study more for the next test. On the other hand, if you possess an external locus of control, you might conclude that the test was too hard and not bother studying for the next test, because you figure you will fail it anyway. Researchers have found that people with an internal

locus of control perform better academically, achieve more in their careers, are more independent, are healthier, are better able to cope, and are less depressed than people who have an external locus of control (Benassi, Sweeney, & Durfour, 1988; Lefcourt, 1982; Maltby, Day, & Macaskill, 2007; Whyte, 1977, 1978, 1980).



Locus of control occurs on a continuum from internal to external.

WALTER MISCHEL AND THE PERSON-SITUATION DEBATE

Walter Mischel was a student of Julian Rotter and taught for years at Stanford, where he was a colleague of Albert Bandura. Mischel surveyed several decades of empirical psychological literature regarding trait prediction of behavior, and his conclusion shook the foundations of personality psychology. Mischel found that the data did not support the central principle of the field—that a person's personality traits are consistent across situations. His report triggered a decades-long period of self-examination, known as the person-situation debate, among personality psychologists.

Mischel suggested that perhaps we were looking for consistency in the wrong places. He found that although behavior was inconsistent across different situations, it was much more consistent within situations—so that a person's behavior in one situation would likely be repeated in a similar one. And as you will see next regarding his famous "marshmallow test," Mischel also found that behavior is consistent in equivalent situations across time.

One of Mischel's most notable contributions to personality

psychology was his ideas on self-regulation. According to Lecci & Magnavita (2013), "Self-regulation is the process of identifying a goal or set of goals and, in pursuing these goals, using both internal (e.g., thoughts and affect) and external (e.g., responses of anything or anyone in the environment) feedback to maximize goal attainment" (p. 6.3). Self-regulation is also known as will power. When we talk about will power, we tend to think of it as the ability to delay gratification. For example, Bettina's teenage daughter made strawberry cupcakes, and they looked delicious. However, Bettina forfeited the pleasure of eating one, because she is training for a 5K race and wants to be fit and do well in the race. Would you be able to resist getting a small reward now in order to get a larger reward later? This is the question Mischel investigated in his now-classic marshmallow test.

Mischel designed a study to assess self-regulation in young children. In the marshmallow study, Mischel and his colleagues placed a preschool child in a room with one marshmallow on the table. The child was told that he could either eat the marshmallow now, or wait until the researcher returned to the room and then he could have two marshmallows (Mischel, Ebbesen & Raskoff, 1972). This was repeated with hundreds of preschoolers. What Mischel and his team found was that young children differ in their degree of self-control. Mischel and his colleagues continued to follow this group of preschoolers through high school, and what do you think they discovered? The children who had more self-control in preschool (the ones who waited for the bigger reward) were more successful in high school. They had higher SAT scores, had positive peer relationships, and were less likely to have substance abuse issues; as adults, they also had more stable marriages (Mischel, Shoda, & Rodriguez, 1989; Mischel et al., 2010). On the other hand, those children who had poor self-control in preschool (the ones who grabbed the one marshmallow) were not as successful in high school, and they were found to have academic and behavioral problems.

To learn more about the marshmallow test and view the test given to children in Columbia, follow the link below to Joachim de Posada's TEDTalks video: Don't eat the marshmallow! | Joachim de Posada.

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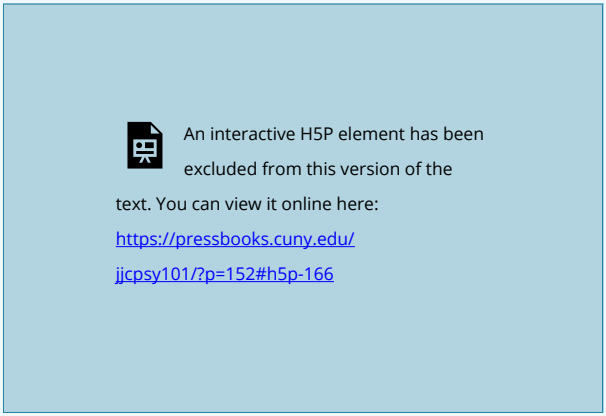
Today, the debate is mostly resolved, and most psychologists consider both the situation and personal factors in understanding behavior. For Mischel (1993), people are situation processors. The children in the marshmallow test each processed, or interpreted, the rewards structure of that situation in their own way. Mischel's approach to personality stresses the importance of both the situation and the way the person perceives the situation. Instead of behavior being determined by the situation, people use cognitive processes to interpret the situation and then behave in accordance with that interpretation.

TEST YOUR UNDERSTANDING



SUMMARY

Behavioral theorists view personality as significantly shaped and impacted by the reinforcements and consequences outside of the organism. People behave in a consistent manner based on prior learning. B. F. Skinner, a prominent behaviorist, said that we demonstrate consistent behavior patterns, because we have developed certain response tendencies. Mischel focused on how personal goals play a role in the self-regulation process. Albert Bandura said that one's environment can determine behavior, but at the same time, people can influence the environment with both

their thoughts and behaviors, which is known as reciprocal determinism. Bandura also emphasized how we learn from watching others. He felt that this type of learning also plays a part in the development of our personality. Bandura discussed the concept of self-efficacy, which is our level of confidence in our own abilities. Finally, Rotter proposed the concept of locus of control, which refers to our beliefs about the power we have over our lives. He said that people fall along a continuum between a purely internal and a purely external locus of control.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Compare the personalities of someone who has high self-efficacy to someone who has low self-efficacy.

People who have high self-efficacy believe that their efforts matter. They perceive their goals as being within reach; have a positive view of challenges, seeing them as tasks to be mastered; develop a deep interest in and strong commitment to the activities in which they are involved; and quickly recover from setbacks. Conversely, people with low self-efficacy believe their efforts have little or no effect, and that outcomes are beyond their control. They avoid challenging tasks because they doubt their abilities to be successful; tend to focus on failure and negative outcomes; and lose confidence in their abilities if they experience setbacks.

Compare and contrast Skinner's perspective on personality development to Freud's.

Skinner disagreed with Freud's idea that childhood plays an important role in shaping our personality. He argued that personality develops over our entire life, rather than in the first few years of life as Freud suggested. Skinner said that our responses can change as we come across new situations; therefore, we can see more variability over time in personality.

PERSONAL APPLICATION QUESTIONS

Do you have an internal or an external locus of control? Provide examples to support your answer.

GLOSSARY

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Media Attributions

- "Don't eat the marshmallow! | Joachim de Posada" by TED. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Discuss the contributions of Abraham Maslow and Carl Rogers to personality development

As the "third force" in psychology, humanism is touted as a reaction both to the pessimistic determinism of psychoanalysis, with its emphasis on psychological disturbance, and to the behaviorists' view of humans passively reacting to the environment, which has been criticized as making people out to be personality-less robots. It does not suggest that psychoanalytic, behaviorist, and other points of view are incorrect but argues that these perspectives do not recognize the depth and meaning of human experience, and fail to recognize the innate capacity for self-directed change and transforming personal experiences. This perspective focuses on how healthy people develop. One pioneering humanist, Abraham Maslow, studied people who he considered to be healthy, creative, and productive, including Albert Einstein, Eleanor Roosevelt, Thomas Jefferson, Abraham Lincoln, and others. Maslow (1950, 1970) found that such people share similar characteristics, such as being open, creative, loving, spontaneous, compassionate, concerned for others, and accepting of themselves. When you studied motivation, you learned about one of the best-known humanistic theories, Maslow's hierarchy of needs theory, in which Maslow proposes that human beings have certain needs in common and that these needs must be met in a certain order.

The highest need is the need for self-actualization, which is the achievement of our fullest potential.

Another humanistic theorist was Carl Rogers. One of Rogers's main ideas about personality regards self-concept, our thoughts and feelings about ourselves. How would you respond to the question, "Who am I?" Your answer can show how you see yourself. If your response is primarily positive, then you tend to feel good about who you are, and you see the world as a safe and positive place. If your response is mainly negative, then you may feel unhappy with who you are. Rogers further divided the self into two categories: the ideal self and the real self. The ideal self is the person that you would like to be; the real self is the person you actually are. Rogers focused on the idea that we need to achieve consistency between these two selves. We experience congruence when our thoughts about our real self and ideal self are very similar—in other words, when our self-concept is accurate**.** High congruence leads to a greater sense of self-worth and a healthy, productive life. Parents can help their children achieve this by giving them unconditional positive regard, or unconditional love. According to Rogers (1980), "As persons are accepted and prized, they tend to develop a more caring attitude towards themselves" (p. 116). Conversely, when there is a great discrepancy between our ideal and actual selves, we experience a state Rogers called incongruence, which can lead to maladjustment. Both Rogers's and Maslow's theories focus on individual choices and do not believe that biology is deterministic.

TEST YOUR UNDERSTANDING

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SUMMARY

Humanistic psychologists Abraham Maslow and Carl Rogers focused on the growth potential of healthy individuals. They believed that people strive to become self-actualized. Both Rogers's and Maslow's theories greatly contributed to our understanding of the self. They emphasized free will and self-determination, with each individual desiring to become the best person they can become.

REVIEW QUESTIONS

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PERSONAL APPLICATION QUESTIONS

Respond to the question, “Who am I?” Based on your response, do you have a negative or a positive self-concept? What are some experiences that led you to develop this particular self-concept?

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Discuss the findings of the Minnesota Study of Twins Reared Apart as they relate to personality and genetics
- Discuss temperament and describe the three infant temperaments identified by Thomas and Chess
- Discuss the evolutionary perspective on personality development

How much of our personality is in-born and biological, and how much is influenced by the environment and culture we are raised in? Psychologists who favor the biological approach believe that inherited predispositions as well as physiological processes can be used to explain differences in our personalities (Burger, 2008).

In the field of behavioral genetics, the Minnesota Study of Twins Reared Apart—a well-known study of the genetic basis for personality—conducted research with twins from 1979 to 1999. In studying 350 pairs of twins, including pairs of identical and fraternal twins reared together and apart, researchers found that identical twins, whether raised together or apart, have very similar personalities (Bouchard, 1994; Bouchard, Lykken, McGue, Segal, & Tellegen, 1990; Segal, 2012). These findings suggest the heritability of some personality traits. Heritability refers to the proportion of difference among people that is attributed to genetics. Some of the traits that the study reported as having more than a 0.50 heritability ratio include leadership, obedience to authority, a sense

of well-being, alienation, resistance to stress, and fearfulness. The implication is that some aspects of our personalities are largely controlled by genetics; however, it's important to point out that traits are not determined by a single gene, but by a combination of many genes, as well as by epigenetic factors that control whether the genes are expressed.

To what extent is our personality dictated by our genetic makeup? View this video to learn more: Is Personality Genetic? – DNA's Influence on Temperament

TEMPERAMENT

Most contemporary psychologists believe temperament has a biological basis due to its appearance very early in our lives (Rothbart, 2011). As you learned when you studied lifespan development, Thomas and Chess (1977) found that babies could be categorized into one of three temperaments: easy, difficult, or slow to warm up. However, environmental factors (family interactions, for example) and maturation can affect the ways in which children's personalities are expressed (Carter et al., 2008).

Research suggests that there are two dimensions of our temperament that are important parts of our adult personality—reactivity and self-regulation (Rothbart, Ahadi, & Evans, 2000). Reactivity refers to how we respond to new or challenging environmental stimuli; self-regulation refers to our ability to control that response (Rothbart & Derryberry, 1981;

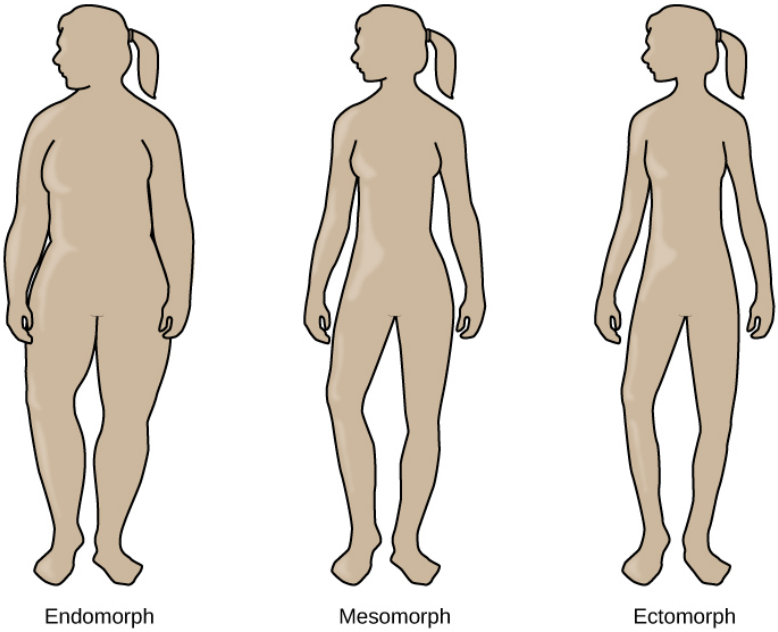
Rothbart, Sheese, Rueda, & Posner, 2011). For example, one person may immediately respond to new stimuli with a high level of anxiety, while another barely notices it.

Body Type and Temperament

Is there an association between your body type and your temperament? The constitutional perspective, which examines the relationship between the structure of the human body and behavior, seeks to answer this question (Genovese, 2008). The first comprehensive system of constitutional psychology was proposed by American psychologist William H. Sheldon (1940, 1942). He believed that your body type can be linked to your personality. Sheldon's life's work was spent observing human bodies and temperaments. Based on his observations and interviews of hundreds of people, he proposed three body/personality types, which he called somatotypes.

The three somatotypes are ectomorphs, endomorphs, and mesomorphs. Ectomorphs are thin with a small bone structure and very little fat on their bodies. According to Sheldon, the ectomorph personality is anxious, self-conscious, artistic, thoughtful, quiet, and private. They enjoy intellectual stimulation and feel uncomfortable in social situations. Actors Adrien Brody and Nicole Kidman would be characterized as ectomorphs. Endomorphs are the opposite of ectomorphs. Endomorphs have narrow shoulders and wide hips, and carry extra fat on their round bodies. Sheldon described

endomorphs as being relaxed, comfortable, good-humored, even-tempered, sociable, and tolerant. Endomorphs enjoy affection and detest disapproval. Queen Latifah and Jack Black would be considered endomorphs. The third somatotype is the mesomorph. This body type falls between the ectomorph and the endomorph. Mesomorphs have large bone structure, well-defined muscles, broad shoulders, narrow waists, and attractive, strong bodies. According to Sheldon, mesomorphs are adventurous, assertive, competitive, and fearless. They are curious and enjoy trying new things, but can also be obnoxious and aggressive. Channing Tatum and Scarlett Johannson would likely be mesomorphs.



Sheldon proposed three somatotypes: endomorphs, mesomorphs, and ectomorphs. Do you think Sheldon's ideas about somatotypes are generally accurate about most people?

Sheldon (1949) also conducted further research into somatotypes and criminality. He measured the physical proportions of hundreds

of juvenile delinquent boys in comparison to male college students, and found that problem youth were primarily mesomorphs. Why might this be? Perhaps it's because they are quick to anger and don't have the restraint demonstrated by ectomorphs. Maybe it's because a person with a mesomorphic body type reflects high levels of testosterone, which may lead to more aggressive behavior. Can you think of other explanations for Sheldon's findings?

Sheldon's method of somatotyping is not without criticism, as it has been considered largely subjective (Carter & Heath, 1990; Cortés & Gatti, 1972; Parnell, 1958). More systematic and controlled research methods did not support his findings (Eysenck, 1970). Consequently, it's not uncommon to see his theory labeled as pseudoscience, much like Gall's theory of phrenology (Rafter, 2007; Rosenbaum, 1995). However, studies involving correlations between somatotype, temperament, and children's school performance (Sanford et al., 1943; Parnell); somatotype and performance of pilots during wartime (Damon, 1955); and somatotype and temperament (Peterson, Liivamagi, & Koskel, 2006) did support his theory.

TEST YOUR UNDERSTANDING

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SUMMARY

Some aspects of our personalities are largely controlled by genetics; however, environmental factors (such as family interactions) and maturation can affect the ways in which children's personalities are expressed.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=156#h5p-173

CRITICAL THINKING QUESTIONS

How might a temperament mix between parent and child affect family life?

An easygoing parent may be irritated by a difficult child. If both parent and child have difficult temperaments, then conflicts in the parent-child relationship might result quite often.

PERSONAL APPLICATION QUESTIONS

Research suggests that many of our personality characteristics have a genetic component. What traits do you think you inherited from your parents? Provide examples. How might modeling (environment) influenced your characteristics as well?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:
https://pressbooks.cuny.edu/jjcpsy101/?p=156#h5p-174

Learning Objectives

By the end of this section, you will be able to:

- Discuss early trait theories of Cattell and Eysenck
- Discuss the Big Five factors and describe someone who is high and low on each of the five traits

Trait theorists believe personality can be understood via the approach that all people have certain traits, or characteristic ways of behaving. Do you tend to be sociable or shy? Passive or aggressive? Optimistic or pessimistic? Moody or even-tempered? Early trait theorists tried to describe all human personality traits. For example, one trait theorist, Gordon Allport (Allport & Odbert, 1936), found 4,500 words in the English language that could describe people. He organized these personality traits into three categories: cardinal traits, central traits, and secondary traits. A cardinal trait is one that dominates your entire personality, and hence your life—such as Ebenezer Scrooge's greed and Mother Theresa's altruism. Cardinal traits are not very common: Few people have personalities dominated by a single trait. Instead, our personalities typically are composed of multiple traits. Central traits are those that make up our personalities (such as loyal, kind, agreeable, friendly, sneaky, wild, and grouchy). Secondary traits are those that are not quite as obvious or as consistent as central traits. They are present under specific circumstances and include preferences and attitudes. For example, one person gets angry when people try to tickle him; another can only sleep on the left

side of the bed; and yet another always orders her salad dressing on the side. And you—although not normally an anxious person—feel nervous before making a speech in front of your English class.

In an effort to make the list of traits more manageable, Raymond Cattell (1946, 1957) narrowed down the list to about 171 traits. However, saying that a trait is either present or absent does not accurately reflect a person's uniqueness, because all of our personalities are actually made up of the same traits; we differ only in the degree to which each trait is expressed. Cattell (1957) identified 16 factors or dimensions of personality: warmth, reasoning, emotional stability, dominance, liveliness, rule-consciousness, social boldness, sensitivity, vigilance, abstractedness, privateness, apprehension, openness to change, self-reliance, perfectionism, and tension ([link]). He developed a personality assessment based on these 16 factors, called the 16PF. Instead of a trait being present or absent, each dimension is scored over a continuum, from high to low. For example, your level of warmth describes how warm, caring, and nice to others you are. If you score low on this index, you tend to be more distant and cold. A high score on this index signifies you are supportive and comforting.

Personality Factors Measured by the 16PF Questionnaire

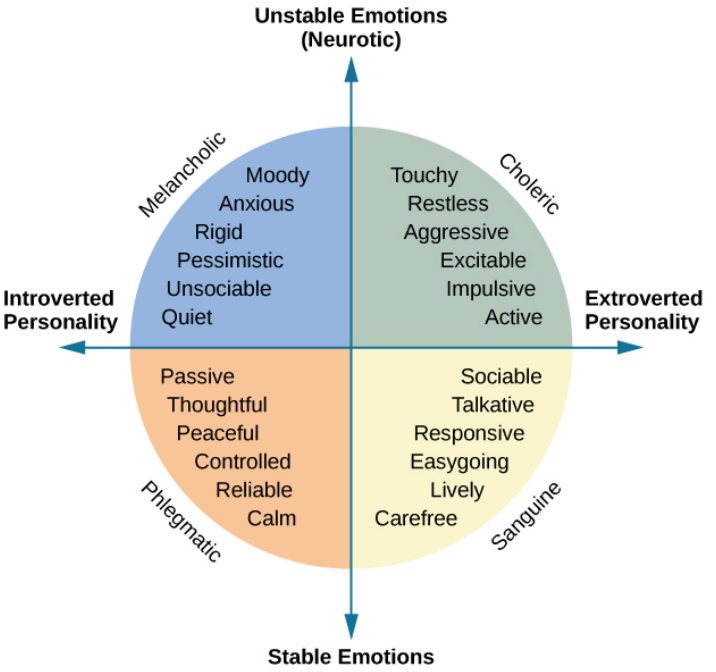
Factor	Low Score	High Score
Warmth	Reserved, detached	Outgoing, supportive
Intellect	Concrete thinker	Analytical
Emotional stability	Moody, irritable	Stable, calm
Aggressiveness	Docile, submissive	Controlling, dominant
Liveliness	Somber, prudent	Adventurous, spontaneous
Dutifulness	Unreliable	Conscientious
Social assertiveness	Shy, restrained	Uninhibited, bold
Sensitivity	Tough-minded	Sensitive, caring
Paranoia	Trusting	Suspicious
Abstractness	Conventional	Imaginative
Introversion	Open, straightforward	Private, shrewd
Anxiety	Confident	Apprehensive
Openmindedness	Closeminded, traditional	Curious, experimental
Independence	Outgoing, social	Self-sufficient
Perfectionism	Disorganized, casual	Organized, precise
Tension	Relaxed	Stressed

Psychologists Hans and Sybil Eysenck were personality theorists who focused on temperament, the inborn, genetically based personality differences that you studied earlier in the chapter. They believed personality is largely governed by biology. The Eysencks (Eysenck, 1990, 1992; Eysenck & Eysenck, 1963) viewed people as having two specific personality dimensions: extroversion/introversion and neuroticism/stability.

Hans and Sybil Eysenck believed that our personality traits are influenced by our genetic inheritance. (credit: "Sirswindon"/Wikimedia Commons)

According to their theory, people high on the trait of extroversion are sociable and outgoing, and readily connect with others, whereas people high on the trait of introversion have a higher need to be alone, engage in solitary behaviors, and limit their interactions with others. In the neuroticism/stability dimension, people high on neuroticism tend to be anxious; they tend to have

an overactive sympathetic nervous system and, even with low stress, their bodies and emotional state tend to go into a flight-or-fight reaction. In contrast, people high on stability tend to need more stimulation to activate their flight-or-fight reaction and are considered more emotionally stable. Based on these two dimensions, the Eysencks' theory divides people into four quadrants. These quadrants are sometimes compared with the four temperaments described by the Greeks: melancholic, choleric, phlegmatic, and sanguine.



The Eysencks described two factors to account for variations in our personalities: extroversion/introversion and emotional stability/instability.

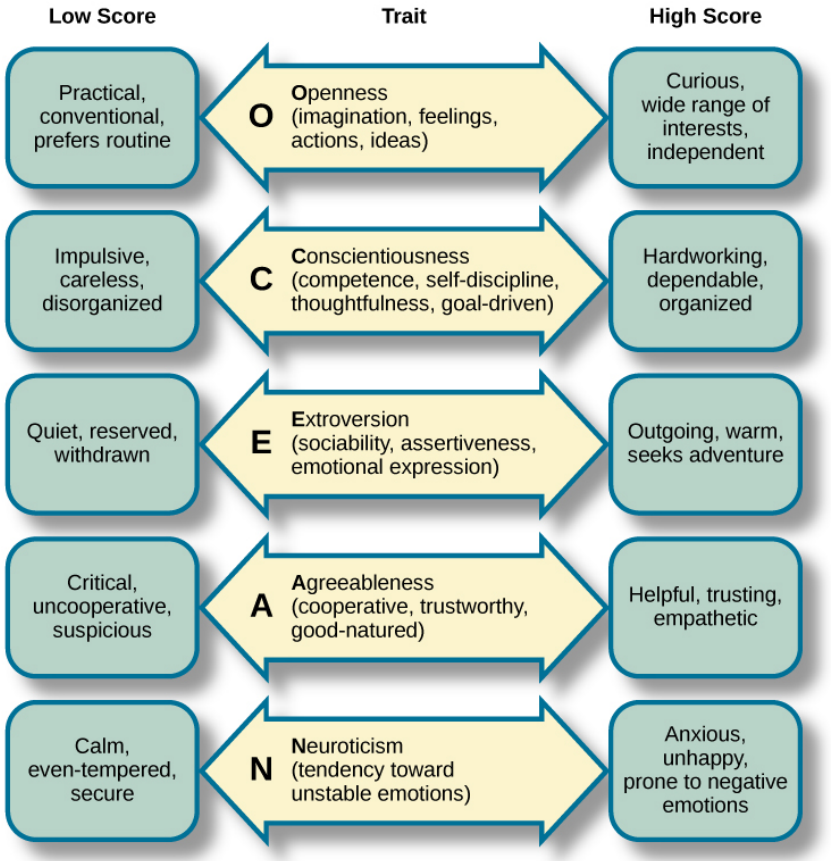
Later, the Eysencks added a third dimension: psychoticism versus superego control (Eysenck, Eysenck & Barrett, 1985). In this dimension, people who are high on psychoticism tend to be

independent thinkers, cold, nonconformists, impulsive, antisocial, and hostile, whereas people who are high on superego control tend to have high impulse control—they are more altruistic, empathetic, cooperative, and conventional (Eysenck, Eysenck & Barrett, 1985).

While Cattell's 16 factors may be too broad, the Eysenck's two-factor system has been criticized for being too narrow. Another personality theory, called the Five Factor Model, effectively hits a middle ground, with its five factors referred to as the Big Five personality traits. It is the most popular theory in personality psychology today and the most accurate approximation of the basic trait dimensions (Funder, 2001). The five traits are openness to experience, conscientiousness, extroversion, agreeableness, and neuroticism. A helpful way to remember the traits is by using the mnemonic OCEAN.

In the Five Factor Model, each person has each trait, but they occur along a spectrum. Openness to experience is characterized by imagination, feelings, actions, and ideas. People who score high on this trait tend to be curious and have a wide range of interests. Conscientiousness is characterized by competence, self-discipline, thoughtfulness, and achievement-striving (goal-directed behavior). People who score high on this trait are hardworking and dependable. Numerous studies have found a positive correlation between conscientiousness and academic success (Akomolafe, 2013; Chamorro-Premuzic & Furnham, 2008; Conrad & Patry, 2012; Noftle & Robins, 2007; Wagerman & Funder, 2007). Extroversion is characterized by sociability, assertiveness, excitement-seeking, and emotional expression. People who score high on this trait are usually described as outgoing and warm. Not surprisingly, people who score high on both extroversion and openness are more likely to participate in adventure and risky sports due to their curious and excitement-seeking nature (Tok, 2011). The fourth trait is agreeableness, which is the tendency to be pleasant, cooperative, trustworthy, and good-natured. People who score low on agreeableness tend to be described as rude and uncooperative,

yet one recent study reported that men who scored low on this trait actually earned more money than men who were considered more agreeable (Judge, Livingston, & Hurst, 2012). The last of the Big Five traits is neuroticism, which is the tendency to experience negative emotions. People high on neuroticism tend to experience emotional instability and are characterized as angry, impulsive, and hostile. Watson and Clark (1984) found that people reporting high levels of neuroticism also tend to report feeling anxious and unhappy. In contrast, people who score low in neuroticism tend to be calm and even-tempered.



In the Five Factor Model, each person has five traits, each scored on a continuum from high to low. In the center column, notice that the first letter of each trait spells the mnemonic OCEAN.

The Big Five personality factors each represent a range between two extremes. In reality, most of us tend to lie somewhere midway along the continuum of each factor, rather than at polar ends. It's important to note that the Big Five traits are relatively stable over our lifespan, with some tendency for the traits to increase or decrease slightly. Researchers have found that conscientiousness increases through young adulthood into middle age, as we become

better able to manage our personal relationships and careers (Donnellan & Lucas, 2008). Agreeableness also increases with age, peaking between 50 to 70 years (Terracciano, McCrae, Brant, & Costa, 2005). Neuroticism and extroversion tend to decline slightly with age (Donnellan & Lucas; Terracciano et al.). Additionally, The Big Five traits have been shown to exist across ethnicities, cultures, and ages, and may have substantial biological and genetic components (Jang, Livesley, & Vernon, 1996; Jang et al., 2006; McCrae & Costa, 1997; Schmitt et al., 2007).

To find out about your personality and where you fall on the Big Five traits, follow this link to take the Big Five personality test: The Big Five Project – Personality Test.

TEST YOUR UNDERSTANDING

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SUMMARY

Trait theorists attempt to explain our personality by identifying our stable characteristics and ways of behaving. They have identified important dimensions of personality. The Five Factor Model is the most widely accepted trait theory today. The five factors are openness, conscientiousness, extroversion, agreeableness, and neuroticism. These traits occur along a continuum.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

How stable are the Big Five traits over one's lifespan?

The Big Five traits are relatively stable over our lifespan with a tendency for the traits to increase or decrease slightly. Researchers have found that conscientiousness increases through young adulthood into middle age, as we become better able to manage our personal relationships and careers. Agreeableness also increases with age, peaking between 50 to 70 years. However, neuroticism and extroversion tend to decline slightly with age.

Compare the personality of someone who scores high on agreeableness to someone who scores low on agreeableness.

A person with a high score on agreeableness is typically pleasant, cooperative, trustworthy and good-natured. People who score low on agreeableness tend to be described as rude and uncooperative. They may be difficult with which to work.

PERSONAL APPLICATION QUESTIONS

Review the Big Five personality traits. On which areas would you expect you'd score high? In which areas does the low score more accurately describe you?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=158#h5p-177

Learning Objectives

By the end of this section you should be able to:

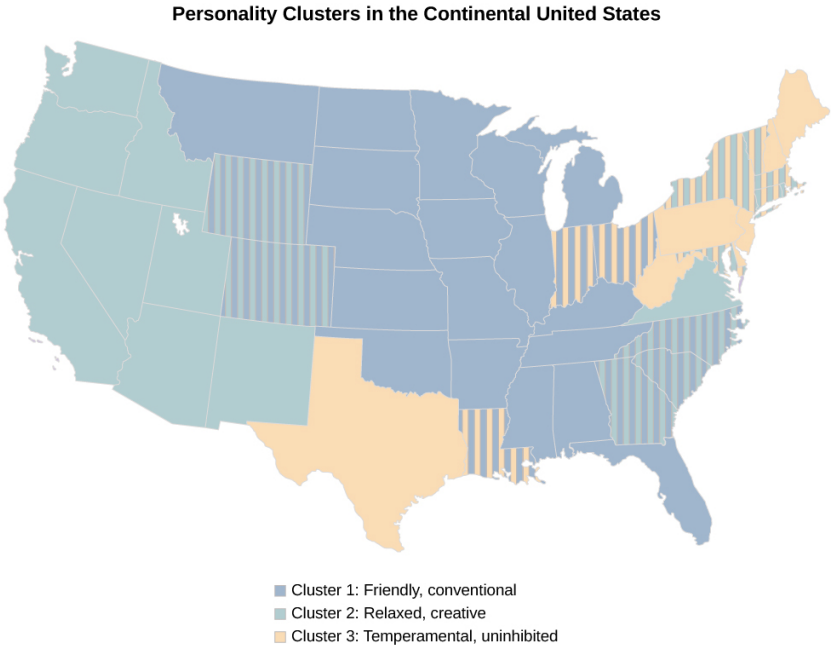
- Discuss personality differences of people from collectivist and individualist cultures
- Discuss the three approaches to studying personality in a cultural context

As you have learned in this chapter, personality is shaped by both genetic and environmental factors. The culture in which you live is one of the most important environmental factors that shapes your personality (Triandis & Suh, 2002). The term culture refers to all of the beliefs, customs, art, and traditions of a particular society. Culture is transmitted to people through language as well as through the modeling of culturally acceptable and nonacceptable behaviors that are either rewarded or punished (Triandis & Suh, 2002). With these ideas in mind, personality psychologists have become interested in the role of culture in understanding personality. They ask whether personality traits are the same across cultures or if there are variations. It appears that there are both universal and culture-specific aspects that account for variation in people's personalities.

Why might it be important to consider cultural influences on personality? Western ideas about personality may not be applicable to other cultures (Benet-Martinez & Oishi, 2008). In fact, there is evidence that the strength of personality traits varies across cultures. Let's take a look at some of the Big Five factors

(conscientiousness, neuroticism, openness, and extroversion) across cultures. As you will learn when you study social psychology, Asian cultures are more collectivist, and people in these cultures tend to be less extroverted. People in Central and South American cultures tend to score higher on openness to experience, whereas Europeans score higher on neuroticism (Benet-Martinez & Karakitapoglu-Aygun, 2003).

According to this study, there also seem to be regional personality differences within the United States. Researchers analyzed responses from over 1.5 million individuals in the United States and found that there are three distinct regional personality clusters: Cluster 1, which is in the Upper Midwest and Deep South, is dominated by people who fall into the "friendly and conventional" personality; Cluster 2, which includes the West, is dominated by people who are more relaxed, emotionally stable, calm, and creative; and Cluster 3, which includes the Northeast, has more people who are stressed, irritable, and depressed. People who live in Clusters 2 and 3 are also generally more open (Rentfrow et al., 2013).



Researchers found three distinct regional personality clusters in the United States. People tend to be friendly and conventional in the Upper Midwest and Deep South; relaxed, emotionally stable, and creative in the West; and stressed, irritable, and depressed in the Northeast (Rentfrow et al., 2013).

One explanation for the regional differences is selective migration (Rentfrow et al., 2013). Selective migration is the concept that people choose to move to places that are compatible with their personalities and needs. For example, a person high on the agreeable scale would likely want to live near family and friends, and would choose to settle or remain in such an area. In contrast, someone high on openness would prefer to settle in a place that is recognized as diverse and innovative (such as California).

PERSONALITY IN INDIVIDUALIST AND COLLECTIVIST CULTURES

Individualist cultures and collectivist cultures place emphasis on different basic values. People who live in individualist cultures tend to believe that independence, competition, and personal achievement are important. Individuals in Western nations such as the United States, England, and Australia score high on individualism (Oyserman, Coon, & Kemmelmier, 2002). People who live in collectivist cultures value social harmony, respectfulness, and group needs over individual needs. Individuals who live in countries in Asia, Africa, and South America score high on collectivism (Hofstede, 2001; Triandis, 1995). These values influence personality. For example, Yang (2006) found that people in individualist cultures displayed more personally oriented personality traits, whereas people in collectivist cultures displayed more socially oriented personality traits.

APPROACHES TO STUDYING PERSONALITY IN A CULTURAL CONTEXT

There are three approaches that can be used to study personality in a cultural context, the *cultural-comparative approach*; the *indigenous approach*; and the *combined approach*, which incorporates elements of both views. Since ideas about personality have a Western basis, the cultural-comparative approach seeks to test Western ideas about personality in other cultures to determine whether they can be generalized and if they have cultural validity (Cheung van de Vijver, & Leong, 2011). For example, recall from the previous section on the trait perspective that researchers used the cultural-comparative approach to test the universality of McCrae and Costa's Five Factor Model. They found applicability in

numerous cultures around the world, with the Big Five traits being stable in many cultures (McCrae & Costa, 1997; McCrae et al., 2005). The indigenous approach came about in reaction to the dominance of Western approaches to the study of personality in non-Western settings (Cheung et al., 2011). Because Western-based personality assessments cannot fully capture the personality constructs of other cultures, the indigenous model has led to the development of personality assessment instruments that are based on constructs relevant to the culture being studied (Cheung et al., 2011). The third approach to cross-cultural studies of personality is the combined approach, which serves as a bridge between Western and indigenous psychology as a way of understanding both universal and cultural variations in personality (Cheung et al., 2011).

TEST YOUR UNDERSTANDING

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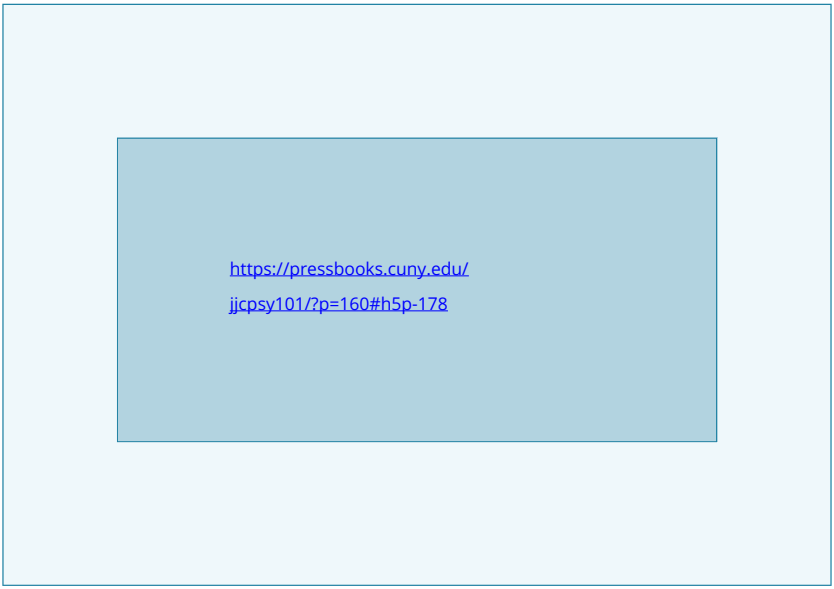
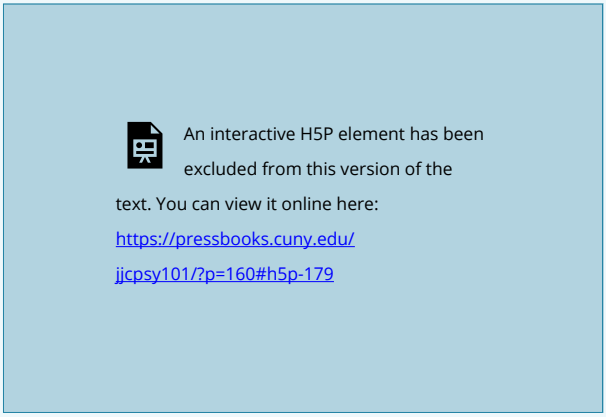


SUMMARY

The culture in which you live is one of the most important environmental factors that shapes your personality. Western ideas about personality may not be applicable to other cultures. In fact, there is evidence that the strength of personality traits varies across cultures. Individualist cultures and collectivist cultures place emphasis on different basic values. People who live in individualist cultures tend to believe that independence, competition, and personal achievement are important. People who live in collectivist cultures value social harmony, respectfulness, and group needs over individual needs. There are three approaches that can be used to study personality in a cultural context: the cultural-comparative approach, the indigenous approach, and the combined approach, which incorporates both elements of both views.

REVIEW QUESTIONS



CRITICAL THINKING QUESTIONS

Why might it be important to consider cultural influences on personality?

Since culture influences one's personality, then Western ideas about personality may not be applicable to people of other cultures. In addition, Western-based measures of personality assessment may not be valid when used to collect data on people from other cultures.

PERSONAL APPLICATION QUESTIONS

According to the work of Rentfrow and colleagues, personalities are not randomly distributed. Instead they fit into distinct geographic clusters. Based on where you live, do you agree or disagree with the traits associated with yourself and the residents of your area of the country? Why or why not?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=160#h5p-180

Learning Objectives

By the end of this section, you will be able to:

- Discuss the Minnesota Multiphasic Personality Inventory
- Recognize and describe common projective tests used in personality assessment

Roberto, Mikhail, and Nat are college friends and all want to be police officers. Roberto is quiet and shy, lacks self-confidence, and usually follows others. He is a kind person, but lacks motivation. Mikhail is loud and boisterous, a leader. He works hard, but is impulsive and drinks too much on the weekends. Nat is thoughtful and well liked. He is trustworthy, but sometimes he has difficulty making quick decisions. Of these three men, who would make the best police officer? What qualities and personality factors make someone a good police officer? What makes someone a bad or dangerous police officer?

A police officer's job is very high in stress, and law enforcement agencies want to make sure they hire the right people. Personality testing is often used for this purpose—to screen applicants for employment and job training. Personality tests are also used in criminal cases and custody battles, and to assess psychological disorders. This section explores the best known among the many different types of personality tests.

SELF-REPORT INVENTORIES

Self-report inventories are a kind of objective test used to assess personality. They typically use multiple-choice items or numbered scales, which represent a range from 1 (strongly disagree) to 5 (strongly agree). They often are called Likert scales after their developer, Rensis Likert (1932).

	Strongly Disagree	Somewhat Disagree	No Opinion	Somewhat Agree	Strongly Agree
I am easygoing.	O	O	O	O	O
I have high standards.	O	O	O	O	O
I enjoy time alone.	O	O	O	O	O
I work well with others.	O	O	O	O	O
I dislike confrontation.	O	O	O	O	O
I prefer crowds over intimacy.	O	O	O	O	O

If you've ever taken a survey, you are probably familiar with Likert-type scale questions. Most personality inventories employ these types of response scales.

One of the most widely used personality inventories is the Minnesota Multiphasic Personality Inventory (MMPI), first published in 1943, with 504 true/false questions, and updated to the MMPI-2 in 1989, with 567 questions. The original MMPI was based on a small, limited sample, composed mostly of Minnesota farmers and psychiatric patients; the revised inventory was based on a more representative, national sample to allow for better standardization. The MMPI-2 takes 1–2 hours to complete. Responses are scored to produce a clinical profile composed of 10 scales: hypochondriasis, depression, hysteria, psychopathic deviance (social deviance), masculinity versus femininity, paranoia,

psychasthenia (obsessive/compulsive qualities), schizophrenia, hypomania, and social introversion. There is also a scale to ascertain risk factors for alcohol abuse. In 2008, the test was again revised, using more advanced methods, to the MMPI-2-RF. This version takes about one-half the time to complete and has only 338 questions. Despite the new test's advantages, the MMPI-2 is more established and is still more widely used. Typically, the tests are administered by computer. Although the MMPI was originally developed to assist in the clinical diagnosis of psychological disorders, it is now also used for occupational screening, such as in law enforcement, and in college, career, and marital counseling (Ben-Porath & Tellegen, 2008).

	True	False
1. I like gardening magazines.	O	O
2. I am unhappy with my sex life.	O	O
3. I feel like no one understands me.	O	O
4. I think I would enjoy the work of a teacher.	O	O
5. I am not easily awakened by noise.	O	O

These true/false questions resemble the kinds of questions you would find on the MMPI.

In addition to clinical scales, the tests also have validity and reliability scales. (Recall the concepts of reliability and validity from your study of psychological research.) One of the validity scales, the Lie Scale (or "L" Scale), consists of 15 items and is used to ascertain whether the respondent is "faking good" (underreporting psychological problems to appear healthier). For example, if someone responds "yes" to a number of unrealistically positive

items such as "I have never told a lie," they may be trying to "fake good" or appear better than they actually are.

Reliability scales test an instrument's consistency over time, assuring that if you take the MMPI-2-RF today and then again 5 years later, your two scores will be similar. Beutler, Nussbaum, and Meredith (1988) gave the MMPI to newly recruited police officers and then to the same police officers 2 years later. After 2 years on the job, police officers' responses indicated an increased vulnerability to alcoholism, somatic symptoms (vague, unexplained physical complaints), and anxiety. When the test was given an additional 2 years later (4 years after starting on the job), the results suggested high risk for alcohol-related difficulties.

PROJECTIVE TESTS

Another method for assessment of personality is projective testing. This kind of test relies on one of the defense mechanisms proposed by Freud—projection—as a way to assess unconscious processes. During this type of testing, a series of ambiguous cards is shown to the person being tested, who then is encouraged to project his feelings, impulses, and desires onto the cards—by telling a story, interpreting an image, or completing a sentence. Many projective tests have undergone standardization procedures (for example, Exner, 2002) and can be used to access whether someone has unusual thoughts or a high level of anxiety, or is likely to become volatile. Some examples of projective tests are the Rorschach Inkblot Test, the Thematic Apperception Test (TAT), the Contemporized-Themes Concerning Blacks test, the TEMAS (Tell-Me-A-Story), and the Rotter Incomplete Sentence Blank (RISB).

The Rorschach Inkblot Test was developed in 1921 by a Swiss psychologist named Hermann Rorschach (pronounced "ROAR-shock"). It is a series of symmetrical inkblot cards that are presented to a client by a psychologist. Upon presentation of each

card, the psychologist asks the client, "What might this be?" What the test-taker sees reveals unconscious feelings and struggles (Piotrowski, 1987; Weiner, 2003). The Rorschach has been standardized using the Exner system and is effective in measuring depression, psychosis, and anxiety.

A second projective test is the Thematic Apperception Test (TAT), created in the 1930s by Henry Murray, an American psychologist, and a psychoanalyst named Christiana Morgan. A person taking the TAT is shown 8–12 ambiguous pictures and is asked to tell a story about each picture. The stories give insight into their social world, revealing hopes, fears, interests, and goals. The storytelling format helps to lower a person's resistance divulging unconscious personal details (Cramer, 2004). The TAT has been used in clinical settings to evaluate psychological disorders; more recently, it has been used in counseling settings to help clients gain a better understanding of themselves and achieve personal growth. Standardization of test administration is virtually nonexistent among clinicians, and the test tends to be modest to low on validity and reliability (Aronow, Weiss, & Rezinkoff, 2001; Lilienfeld, Wood, & Garb, 2000). Despite these shortcomings, the TAT has been one of the most widely used projective tests.

A third projective test is the Rotter Incomplete Sentence Blank (RISB) developed by Julian Rotter in 1950 (recall his theory of locus of control, covered earlier in this chapter). There are three forms of this test for use with different age groups: the school form, the college form, and the adult form. The tests include 40 incomplete sentences that people are asked to complete as quickly as possible. The average time for completing the test is approximately 20 minutes, as responses are only 1–2 words in length. This test is similar to a word association test, and like other types of projective tests, it is presumed that responses will reveal desires, fears, and struggles. The RISB is used in screening college students for adjustment problems and in career counseling (Holaday, Smith, & Sherry, 2010; Rotter & Rafferty 1950).

1. I feel...	
2. I regret...	
3. At home...	
4. My mother...	
5. My greatest worry...	

These incomplete sentences resemble the types of questions on the RISB. How would you complete these sentences?

For many decades, these traditional projective tests have been used in cross-cultural personality assessments. However, it was found that test bias limited their usefulness (Hoy-Watkins & Jenkins-Moore, 2008). It is difficult to assess the personalities and lifestyles of members of widely divergent ethnic/cultural groups using personality instruments based on data from a single culture or race (Hoy-Watkins & Jenkins-Moore, 2008). For example, when the TAT was used with African-American test takers, the result was often shorter story length and low levels of cultural identification (Duzant, 2005). Therefore, it was vital to develop other personality assessments that explored factors such as race, language, and level of acculturation (Hoy-Watkins & Jenkins-Moore, 2008). To address this need, Robert Williams developed the first culturally specific projective test designed to reflect the everyday life experiences of African Americans (Hoy-Watkins & Jenkins-Moore, 2008). The updated version of the instrument is the Contemporized-Themes Concerning Blacks Test (C-TCB) (Williams, 1972). The C-TCB contains 20 color images that show scenes of African-American lifestyles. When the C-TCB was compared with the TAT for African Americans, it was found that use of the C-TCB led to increased story length, higher degrees of positive feelings, and stronger identification with the C-TCB (Hoy, 1997; Hoy-Watkins & Jenkins-Moore, 2008).

The TEMAS Multicultural Thematic Apperception Test is another tool designed to be culturally relevant to minority groups, especially Hispanic youths. TEMAS—standing for "Tell Me a Story" but also a play on the Spanish word *temas* (themes)—uses images and storytelling cues that relate to minority culture (Constantino, 1982).

TEST YOUR UNDERSTANDING

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=162#h5p-181

SUMMARY

Personality tests are techniques designed to measure one's personality. They are used to diagnose psychological problems as well as to screen candidates for college and employment. There are two types of personality tests: self-report inventories and projective tests. The MMPI is one of the most common self-report inventories. It asks a series of true/false questions that are designed to provide a clinical profile of an individual. Projective tests use ambiguous images or other ambiguous stimuli to assess an individual's unconscious fears, desires, and challenges. The Rorschach Inkblot Test, the TAT, the RISB, and the C-TCB are all forms of projective tests.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here:

CRITICAL THINKING QUESTIONS

Why might a prospective employer screen applicants using personality assessments?

They can help an employer predict a candidate's reactions and attitudes to various situations they might encounter on the job, thus helping choose the right person for the job. This is particularly important in hiring for a high-risk job such as law enforcement.

Personality tests can also reveal a potential employee's desirable qualities such as honesty, motivation, and conscientiousness.

Why would a clinician give someone a projective test?

A projective test could give the clinician clues about dreams, fears, and personal struggles of which the client may be unaware, since these tests are designed to reveal unconscious motivations and attitudes. They can also help clinicians diagnose psychological disorders.

PERSONAL APPLICATION QUESTIONS

How objective do you think you can be about yourself in answering questions on self-report personality assessment measures? What implications might this have for the validity of the personality test?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

https://pressbooks.cuny.edu/jjcpsy101/?p=162#h5p-183

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SOCIAL PSYCHOLOGY

Trayvon Martin, 17, was shot to death at the hands of George Zimmerman, a volunteer neighborhood watchman, in 2012. Was his death the result of self-defense or racial bias? That question drew hundreds of people to rally on each side of this heated debate. (credit "signs": modification of work by David Shankbone; credit "walk": modification of work by "Fibonacci Blue"/Flickr)

Humans are diverse, and sometimes our differences make it challenging for us to get along with one another. A poignant example is that of Trayvon Martin, a 17-year-old African American who was shot by a neighborhood watch volunteer, George Zimmerman, in a predominantly White neighborhood in 2012. Zimmerman grew suspicious of the boy dressed in a hoodie and

pursued Martin. A physical altercation ended with Zimmerman fatally shooting Martin. Zimmerman claimed that he acted in self-defense; Martin was unarmed. A Florida jury found Zimmerman not guilty of second degree murder nor of manslaughter.

Several groups protested what they deemed racial profiling and brutality against an unarmed Black male. Zimmerman, who has a Peruvian mother and a German father, was accused of being racist. Some media coverage was criticized for inflaming racial politics in their coverage.

In spite of conflicts such as these, people also to work together to create positive change. For example, after the 9/11 terrorist attacks, people rallied together and charitable donations skyrocket (Brown & Minty, 2006). This chapter explores how the presence of other people influences the behavior of individuals, dyads, and groups. Social factors can determine whether human behavior tends toward conflict or harmony.

Learning Objectives

By the end of this section, you will be able to:

- Define social psychology
- Describe situational versus dispositional influences on behavior
- Describe the fundamental attribution error

Social psychology examines how people affect one another, and it looks at the power of the situation. Social psychologists assert that an individual's thoughts, feelings, and behaviors are very much influenced by social situations. Essentially, people will change their behavior to align with the social situation at hand. If we are in a new situation or are unsure how to behave, we will take our cues from other individuals.

The field of social psychology studies topics at both the intra- and interpersonal levels. Intrapersonal topics (those that pertain to the individual) include emotions and attitudes, the self, and social cognition (the ways in which we think about ourselves and others). Interpersonal topics (those that pertain to dyads and groups) include helping behavior, aggression, prejudice and discrimination, attraction and close relationships, and group processes and intergroup relationships.

Social psychology deals with all kinds of interactions between people, spanning a wide range of how we connect: from moments of confrontation to moments of working together and helping others, as shown here. (credit: Sgt. Derec Pierson, U.S. Army)

Social psychologists focus on how people construe or interpret situations and how these interpretations influence their thoughts, feelings, and behaviors (Ross & Nisbett, 1991). Thus, social psychology studies individuals in a social context and how situational variables interact to influence behavior. In this chapter, we discuss the intrapersonal processes of self-presentation, cognitive dissonance and attitude change, and the interpersonal processes of conformity and obedience, aggression and altruism, and, finally, love and attraction.

SITUATIONAL AND DISPOSITIONAL INFLUENCES ON BEHAVIOR

Behavior is a product of both the situation (e.g., cultural influences, social roles, and the presence of bystanders) and of the person (e.g., personality characteristics). Subfields of psychology tend to focus on one influence or behavior over others. Situationism is the view that our behavior and actions are determined by our immediate environment and surroundings. In contrast, dispositionism holds that our behavior is determined by internal factors (Heider, 1958). An internal factor is an attribute of a person and includes personality traits and temperament. Social psychologists have tended to take the situationist perspective, whereas personality psychologists have promoted the dispositionist perspective. Modern approaches to social psychology, however, take both the situation and the individual into account when studying human behavior (Fiske, Gilbert, & Lindzey, 2010). In fact, the field of social-personality psychology has emerged to study the complex interaction of internal and situational factors that affect human behavior (Mischel, 1977; Richard, Bond, & Stokes-Zoota, 2003).

FUNDAMENTAL ATTRIBUTION ERROR

In the United States, the predominant culture tends to favor a dispositional approach in explaining human behavior. Why do you think this is? We tend to think that people are in control of their own behaviors, and, therefore, any behavior change must be due to something internal, such as their personality, habits, or temperament. According to some social psychologists, people tend to overemphasize internal factors as explanations—or attributions—for the behavior of other people. They tend to assume that the behavior of another person is a *trait* of that

person, and to underestimate the power of the situation on the behavior of others. They tend to fail to recognize when the behavior of another is due to situational variables, and thus to the person's *state*. This erroneous assumption is called the fundamental attribution error (Ross, 1977; Riggio & Garcia, 2009). To better understand, imagine this scenario: Greg returns home from work, and upon opening the front door his wife happily greets him and inquires about his day. Instead of greeting his wife, Greg yells at her, "Leave me alone!" Why did Greg yell at his wife? How would someone committing the fundamental attribution error explain Greg's behavior? The most common response is that Greg is a mean, angry, or unfriendly person (his traits). This is an internal or dispositional explanation. However, imagine that Greg was just laid off from his job due to company downsizing. Would your explanation for Greg's behavior change? Your revised explanation might be that Greg was frustrated and disappointed for losing his job; therefore, he was in a bad mood (his state). This is now an external or situational explanation for Greg's behavior.

The fundamental attribution error is so powerful that people often overlook obvious situational influences on behavior. A classic example was demonstrated in a series of experiments known as the quizmaster study (Ross, Amabile, & Steinmetz, 1977). Student participants were randomly assigned to play the role of a questioner (the quizmaster) or a contestant in a quiz game. Questioners developed difficult questions to which they knew the answers, and they presented these questions to the contestants. The contestants answered the questions correctly only 4 out of 10 times. After the task, the questioners and contestants were asked to rate their own general knowledge compared to the average student. Questioners did not rate their general knowledge higher than the contestants, but the contestants rated the questioners' intelligence higher than their own. In a second study, observers of the interaction also rated the questioner as having more general knowledge than the contestant. The obvious influence on

performance is the situation. The questioners wrote the questions, so of course they had an advantage. Both the contestants and observers made an internal attribution for the performance. They concluded that the questioners must be more intelligent than the contestants.

In the quizmaster study, people tended to disregard the influence of the situation and wrongly concluded that a questioner's knowledge was greater than their own. (credit: Steve Jurvetson)

As demonstrated in the example above, the fundamental attribution error is considered a powerful influence in how we explain the behaviors of others. However, it should be noted that some researchers have suggested that the fundamental attribution error may not be as powerful as it is often portrayed. In fact, a recent review of more than 173 published studies suggests that several factors (e.g., high levels of idiosyncrasy of the character and how well hypothetical events are explained) play a role in determining just how influential the fundamental attribution error is (Malle, 2006).

IS THE FUNDAMENTAL ATTRIBUTION ERROR A UNIVERSAL PHENOMENON?

You may be able to think of examples of the fundamental attribution error in your life. Do people in all cultures commit the fundamental attribution error? Research suggests that they do not. People from an individualistic culture, that is, a culture that focuses on individual achievement and autonomy, have the greatest tendency to commit the fundamental attribution error. Individualistic cultures, which tend to be found in western countries such as the United States, Canada, and the United Kingdom, promote a focus on the individual. Therefore, a person's disposition is thought to be the primary explanation for her behavior. In contrast, people from a collectivistic culture, that is, a culture that focuses on communal relationships with others, such as family, friends, and community, are less likely to commit the fundamental attribution error (Markus & Kitayama, 1991; Triandis, 2001).

(a) (b) (c)

People from collectivistic cultures, such as some Asian cultures, are more likely to emphasize relationships with others than to focus primarily on the individual. Activities such as (a) preparing a meal, (b) hanging out, and (c) playing a game engage people in a group. (credit a: modification of work by Arian Zwegers; credit b: modification of work by "conbon33"/Flickr; credit c: modification of work by Anja Disseldorp)

Why do you think this is the case? Collectivistic cultures, which tend to be found in east Asian countries and in Latin American and African countries, focus on the group more than on the individual

(Nisbett, Peng, Choi, & Norenzayan, 2001). This focus on others provides a broader perspective that takes into account both situational and cultural influences on behavior; thus, a more nuanced explanation of the causes of others' behavior becomes more likely.

Characteristics of Individualistic and Collectivistic Cultures

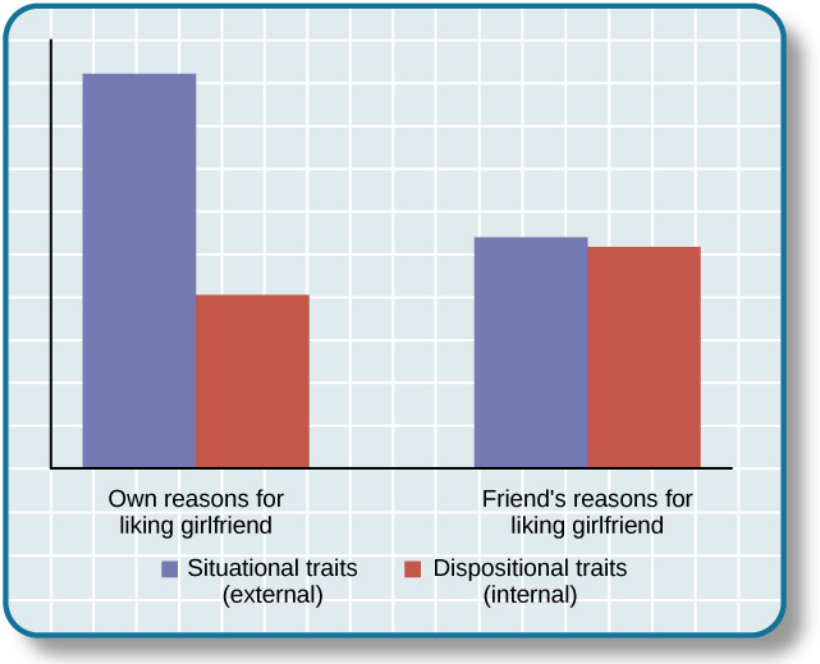
Individualistic Culture	Collectivistic Culture
Achievement oriented	Relationship oriented
Focus on autonomy	Focus on group autonomy
Dispositional perspective	Situational perspective
Independent	Interdependent
Analytic thinking style	Holistic thinking style

ACTOR-OBSERVER BIAS

Returning to our earlier example, Greg knew that he lost his job, but an observer would not know. So a naïve observer would tend to attribute Greg's hostile behavior to Greg's disposition rather than to the true, situational cause. Why do you think we underestimate the influence of the situation on the behaviors of others? One reason is that we often don't have all the information we need to make a situational explanation for another person's behavior. The only information we might have is what is observable. Due to this lack of information we have a tendency to assume the behavior is due to a dispositional, or internal, factor. When it comes to explaining our own behaviors, however, we have much more information available to us. If you came home from school or work angry and yelled at your dog or a loved one, what would your explanation be? You might say you were very tired or feeling unwell and needed quiet time—a situational explanation. The actor-observer bias is the phenomenon of attributing other people's behavior to internal factors (fundamental attribution error) while attributing our own

behavior to situational forces (Jones & Nisbett, 1971; Nisbett, Caputo, Legant, & Marecek, 1973; Choi & Nisbett, 1998). As actors of behavior, we have more information available to explain our own behavior. However as observers, we have less information available; therefore, we tend to default to a dispositionist perspective.

One study on the actor-observer bias investigated reasons male participants gave for why they liked their girlfriend (Nisbett et al., 1973). When asked why participants liked their own girlfriend, participants focused on internal, dispositional qualities of their girlfriends (for example, her pleasant personality). The participants' explanations rarely included causes internal to themselves, such as dispositional traits (for example, "I need companionship."). In contrast, when speculating why a male friend likes his girlfriend, participants were equally likely to give dispositional and external explanations. This supports the idea that actors tend to provide few internal explanations but many situational explanations for their own behavior. In contrast, observers tend to provide more dispositional explanations for a friend's behavior.



Actor-observer bias is evident when subjects explain their own reasons for liking a girlfriend versus their impressions of others' reasons for liking a girlfriend.

SELF-SERVING BIAS

Following an outcome, self-serving bias are those attributions that enable us to see ourselves in favorable light (for example, making internal attributions for success and external attributions for failures). When you do well at a task, for example acing an exam, it is in your best interest to make a dispositional attribution for your behavior ("I'm smart,") instead of a situational one ("The exam was easy,"). The tendency of an individual to take credit by making dispositional or internal attributions for positive outcomes but

situational or external attributions for negative outcomes is known as the self-serving bias (Miller & Ross, 1975). This bias serves to protect self-esteem. You can imagine that if people always made situational attributions for their behavior, they would never be able to take credit and feel good about their accomplishments.

We can understand self-serving bias by digging more deeply into attribution, a belief about the cause of a result. One model of attribution proposes three main dimensions: locus of control (internal versus external), stability (stable versus unstable), and controllability (controllable versus uncontrollable). In this context, stability refers the extent to which the circumstances that result in a given outcome are changeable. The circumstances are considered stable if they are unlikely to change. Controllability refers to the extent to which the circumstances that are associated with a given outcome can be controlled. Obviously, those things that we have the power to control would be labeled controllable (Weiner, 1979).

Consider the example of how we explain our favorite sports team's wins. Research shows that we make internal, stable, and controllable attributions for our team's victory (Grove, Hanrahan, & McInman, 1991). For example, we might tell ourselves that our team is talented (internal), consistently works hard (stable), and uses effective strategies (controllable). In contrast, we are more likely to make external, unstable, and uncontrollable attributions when our favorite team loses. For example, we might tell ourselves that the other team has more experienced players or that the referees were unfair (external), the other team played at home (unstable), and the cold weather affected our team's performance (uncontrollable).

We tend to believe that our team wins because it's better, but loses for reasons it cannot control (Roesch & Amirkham, 1997). (credit: "TheAHL"/Flickr)

JUST-WORLD HYPOTHESIS

One consequence of westerners' tendency to provide dispositional explanations for behavior is victim blame (Jost & Major, 2001). When people experience bad fortune, others tend to assume that they somehow are responsible for their own fate. A common ideology, or worldview, in the United States is the just-world hypothesis. The just-world hypothesis is the belief that people get the outcomes they deserve (Lerner & Miller, 1978). In order to maintain the belief that the world is a fair place, people tend to think that good people experience positive outcomes, and bad people experience negative outcomes (Jost, Banaji, & Nosek, 2004; Jost & Major, 2001). The ability to think of the world as a fair place, where people get what they deserve, allows us to feel that the world is predictable and that we have some control over our life outcomes (Jost et al., 2004; Jost & Major, 2001). For example, if you

want to experience positive outcomes, you just need to work hard to get ahead in life.

Can you think of a negative consequence of the just-world hypothesis? One negative consequence is people's tendency to blame poor individuals for their plight. What common explanations are given for why people live in poverty? Have you heard statements such as, "The poor are lazy and just don't want to work" or "Poor people just want to live off the government"? What types of explanations are these, dispositional or situational? These dispositional explanations are clear examples of the fundamental attribution error. Blaming poor people for their poverty ignores situational factors that impact them, such as high unemployment rates, recession, poor educational opportunities, and the familial cycle of poverty. Other research shows that people who hold just-world beliefs have negative attitudes toward people who are unemployed and people living with AIDS (Sutton & Douglas, 2005). In the United States and other countries, victims of sexual assault may find themselves blamed for their abuse. Victim advocacy groups, such as Domestic Violence Ended (DOVE), attend court in support of victims to ensure that blame is directed at the perpetrators of sexual violence, not the victims.

People who hold just-world beliefs tend to blame the people in poverty for their circumstances, ignoring situational and cultural causes of poverty. (credit: Adrian Miles)

TEST YOUR UNDERSTANDING

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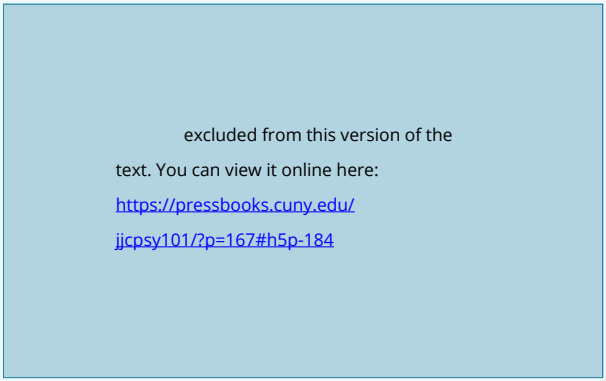
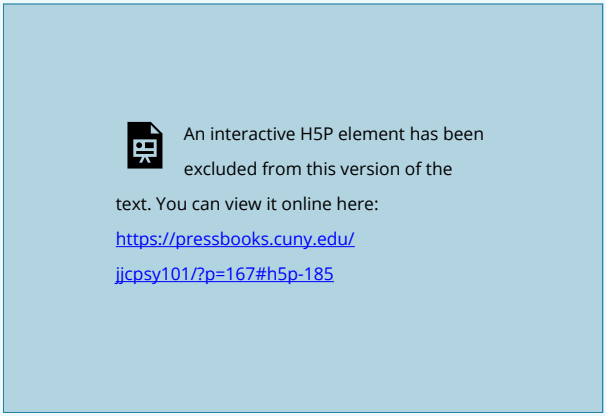


SUMMARY

Social psychology is the subfield of psychology that studies the power of the situation to influence individuals' thoughts, feelings, and behaviors. Psychologists categorize the causes of human behavior as those due to internal factors, such as personality, or those due to external factors, such as cultural and other social influences. Behavior is better explained, however, by using both approaches. Lay people tend to over-rely on dispositional explanations for behavior and ignore the power of situational influences, a perspective called the fundamental attribution error. People from individualistic cultures are more likely to display this bias versus people from collectivistic cultures. Our explanations for

our own and others behaviors can be biased due to not having enough information about others' motivations for behaviors and by providing explanations that bolster our self-esteem.

REVIEW QUESTIONS



CRITICAL THINKING QUESTIONS

Compare and contrast situational influences and dispositional influences and give an example of each. Explain how situational influences and dispositional influences might explain inappropriate behavior.

A situationism view is that our behaviors are determined by the situation—for example, a person who is late for work claims that heavy traffic caused the delay. A dispositional view is that our behaviors are determined by personality traits—for example, a driver in a road rage incident claims the driver who cut her off is an aggressive person. Thus, a situational view tends to provide an excuse for inappropriate behavior, and a dispositional view tends to lay blame for inappropriate behavior.

Provide an example of how people from individualistic and collectivistic cultures would differ in explaining why they won an important sporting event.

People from individualistic cultures would tend to attribute athletic success to individual hard work and ability. People from collectivistic cultures would tend attribute athletic success to the team working together and the support and encouragement of the coach.

PERSONAL APPLICATION QUESTIONS

Provide a personal example of an experience in which your behavior was influenced by the power of the situation.

Think of an example in the media of a sports figure—player or coach—who gives a self-serving attribution for winning or losing. Examples might include accusing the referee of incorrect calls, in the case of losing, or citing their own hard work and talent, in the case of winning.

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=167#h5p-186

Learning Objectives

By the end of this section, you will be able to:

- Describe social roles and how they influence behavior
- Explain what social norms are and how they influence behavior
- Define script
- Describe the findings of Zimbardo's Stanford prison experiment

As you've learned, social psychology is the study of how people affect one another's thoughts, feelings, and behaviors. We have discussed situational perspectives and social psychology's emphasis on the ways in which a person's environment, including culture and other social influences, affect behavior. In this section, we examine situational forces that have a strong influence on human behavior including social roles, social norms, and scripts. We discuss how humans use the social environment as a source of information, or cues, on how to behave. Situational influences on our behavior have important consequences, such as whether we will help a stranger in an emergency or how we would behave in an unfamiliar environment.

SOCIAL ROLES

One major social determinant of human behavior is our social

roles. A social role is a pattern of behavior that is expected of a person in a given setting or group (Hare, 2003). Each one of us has several social roles. You may be, at the same time, a student, a parent, an aspiring teacher, a son or daughter, a spouse, and a lifeguard. How do these social roles influence your behavior? Social roles are defined by culturally shared knowledge. That is, nearly everyone in a given culture knows what behavior is expected of a person in a given role. For example, what is the social role for a student? If you look around a college classroom you will likely see students engaging in studious behavior, taking notes, listening to the professor, reading the textbook, and sitting quietly at their desks. Of course you may see students deviating from the expected studious behavior such as texting on their phones or using Facebook on their laptops, but in all cases, the students that you observe are attending class—a part of the social role of students.

Being a student is just one of the many social roles you have. (credit: "University of Michigan MSIS"/Flickr)

Social roles, and our related behavior, can vary across different settings. How do you behave when you are engaging in the role

of son or daughter and attending a family function? Now imagine how you behave when you are engaged in the role of employee at your workplace. It is very likely that your behavior will be different. Perhaps you are more relaxed and outgoing with your family, making jokes and doing silly things. But at your workplace you might speak more professionally, and although you may be friendly, you are also serious and focused on getting the work completed. These are examples of how our social roles influence and often dictate our behavior to the extent that identity and personality can vary with context (that is, in different social groups) (Malloy, Albright, Kenny, Agatstein & Winquist, 1997).

SOCIAL NORMS

As discussed previously, social roles are defined by a culture's shared knowledge of what is expected behavior of an individual in a specific role. This shared knowledge comes from social norms. A social norm is a group's expectation of what is appropriate and acceptable behavior for its members—how they are supposed to behave and think (Deutsch & Gerard, 1955; Berkowitz, 2004). How are we expected to act? What are we expected to talk about? What are we expected to wear? In our discussion of social roles we noted that colleges have social norms for students' behavior in the role of student and workplaces have social norms for employees' behaviors in the role of employee. Social norms are everywhere including in families, gangs, and on social media outlets. What are some social norms on Facebook?

Tweens, Teens, and Social Norms

My 11-year-old daughter, Jessica, recently told me she needed shorts and shirts for the summer, and that she wanted me to take her to a store at the mall that is popular with preteens and teens to buy them. I have noticed that many girls have clothes from that store, so I tried teasing her. I said, "All the shirts say 'Aero' on the front. If you are wearing a shirt like that and you have a substitute teacher, and the other girls are all wearing that type of shirt, won't the substitute teacher think you are all named 'Aero'?"

My daughter replied, in typical 11-year-old fashion, "Mom, you are not funny. Can we please go shopping?"

I tried a different tactic. I asked Jessica if having clothing from that particular store will make her popular. She replied, "No, it will not make me popular. It is what the popular kids wear. It will make me feel happier." How can a label or name brand make someone feel happier?

Think back to what you've learned about lifespan development. What is it about pre-teens and young teens that make them want to fit in? Does this change over time? Think back to your high school experience, or look around your college campus. What is the main name brand clothing you see? What messages do we get from the media about how to fit in?

Young people struggle to become independent at the same time they are desperately trying to fit in with their peers. (credit: Monica Arellano-Ongpin)

SCRIPTS

Because of social roles, people tend to know what behavior is expected of them in specific, familiar settings. A script is a person's knowledge about the sequence of events expected in a specific setting (Schank & Abelson, 1977). How do you act on the first day of school, when you walk into an elevator, or are at a restaurant? For example, at a restaurant in the United States, if we want the server's attention, we try to make eye contact. In Brazil, you would make the sound "psst" to get the server's attention. You can see the cultural differences in scripts. To an American, saying "psst" to a server might seem rude, yet to a Brazilian, trying to make eye contact might not seem an effective strategy. Scripts are important sources of information to guide behavior in given situations. Can you imagine being in an unfamiliar situation and not having a script

for how to behave? This could be uncomfortable and confusing. How could you find out about social norms in an unfamiliar culture?

ZIMBARDO'S STANFORD PRISON EXPERIMENT

The famous Stanford prison experiment, conducted by social psychologist Philip Zimbardo and his colleagues at Stanford University, demonstrated the power of social roles, social norms, and scripts. In the summer of 1971, an advertisement was placed in a California newspaper asking for male volunteers to participate in a study about the psychological effects of prison life. More than 70 men volunteered, and these volunteers then underwent psychological testing to eliminate candidates who had underlying psychiatric issues, medical issues, or a history of crime or drug abuse. The pool of volunteers was whittled down to 24 healthy male college students. Each student was paid $15 per day and was randomly assigned to play the role of either a prisoner or a guard in the study. Based on what you have learned about research methods, why is it important that participants were randomly assigned?

A mock prison was constructed in the basement of the psychology building at Stanford. Participants assigned to play the role of prisoners were "arrested" at their homes by Palo Alto police officers, booked at a police station, and subsequently taken to the mock prison. The experiment was scheduled to run for several weeks. To the surprise of the researchers, both the "prisoners" and "guards" assumed their roles with zeal. In fact, on day 2, some of the prisoners revolted, and the guards quelled the rebellion by threatening the prisoners with night sticks. In a relatively short time, the guards came to harass the prisoners in an increasingly sadistic manner, through a complete lack of privacy, lack of basic comforts such as mattresses to sleep on, and through degrading chores and late-night counts.

The prisoners, in turn, began to show signs of severe anxiety and hopelessness—they began tolerating the guards' abuse. Even the Stanford professor who designed the study and was the head researcher, Philip Zimbardo, found himself acting as if the prison was real and his role, as prison supervisor, was real as well. After only six days, the experiment had to be ended due to the participants' deteriorating behavior. Zimbardo explained,

> At this point it became clear that we had to end the study. We had created an overwhelmingly powerful situation—a situation in which prisoners were withdrawing and behaving in pathological ways, and in which some of the guards were behaving sadistically. Even the "good" guards felt helpless to intervene, and none of the guards quit while the study was in progress. Indeed, it should be noted that no guard ever came late for his shift, called in sick, left early, or demanded extra pay for overtime work. (Zimbardo, 2013)

The Stanford prison experiment demonstrated the power of social roles, norms, and scripts in affecting human behavior. The guards and prisoners enacted their social roles by engaging in behaviors appropriate to the roles: The guards gave orders and the prisoners followed orders. Social norms require guards to be authoritarian and prisoners to be submissive. When prisoners rebelled, they violated these social norms, which led to upheaval. The specific acts engaged by the guards and the prisoners derived from scripts. For example, guards degraded the prisoners by forcing them do push-ups and by removing all privacy. Prisoners rebelled by throwing pillows and trashing their cells. Some prisoners became so immersed in their roles that they exhibited symptoms of mental breakdown; however, according to Zimbardo, none of the participants suffered long term harm (Alexander, 2001).

The Stanford Prison Experiment has some parallels with the abuse of prisoners of war by U.S. Army troops and CIA personnel at the Abu Ghraib prison in 2003 and 2004. The offenses at Abu Ghraib were documented by photographs of the abuse, some taken by the abusers themselves.

Iraqi prisoners of war were abused by their American captors in Abu Ghraib prison, during the second Iraq war. (credit: United States Department of Defense)

Visit this website to hear an NPR interview with Philip Zimbardo where he discusses the parallels between the Stanford prison experiment and the Abu Ghraib prison in Iraq: Prison Psychology and the Stanford Prison Experiment.

TEST YOUR UNDERSTANDING

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SUMMARY

Human behavior is largely influenced by our social roles, norms, and scripts. In order to know how to act in a given situation, we have shared cultural knowledge of how to behave depending on our role in society. Social norms dictate the behavior that is appropriate or inappropriate for each role. Each social role has scripts that help humans learn the sequence of appropriate behaviors in a given setting. The famous Stanford prison experiment is an example of how the power of the situation can

dictate the social roles, norms, and scripts we follow in a given situation, even if this behavior is contrary to our typical behavior.

REVIEW QUESTIONS

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https://pressbooks.cuny.edu/jjcpsy101/?p=169#h5p-188

CRITICAL THINKING QUESTIONS

Why didn't the "good" guards in the Stanford prison experiment object to other guards' abusive behavior? Were the student prisoners simply weak people? Why didn't they object to being abused?

The good guards were fulfilling their social roles and they did not object to other guards' abusive behavior because of the power of the situation. In addition, the prison supervisor's behavior sanctioned the guards' negative treatment of prisoners. The prisoners were not weak people; they were recruited because they were healthy, mentally stable adults. The power of their social role influenced them to engage in subservient prisoner behavior. The script for prisoners is to accept abusive behavior from authority figures, especially for punishment, when they do not follow the rules.

Describe how social roles, social norms, and scripts were evident in the Stanford prison experiment. How can this experiment be applied to everyday life? Are there any more recent examples where people started fulfilling a role and became abusive?

Social roles were in play as each participant acted out behaviors appropriate to his role as prisoner, guard, or supervisor. Scripts determined the specific

behaviors the guards and prisoners displayed, such as humiliation and passivity. The social norms of a prison environment sanctions abuse of prisoners since they have lost many of their human rights and became the property of the government. This experiment can be applied to other situations in which social norms, roles, and scripts dictate our behavior, such as in mob behavior. A more recent example of similar behavior was the abuse of prisoners by American soldiers who were working as prison guards at the Abu Ghraib prison in Iraq.

PERSONAL APPLICATION QUESTIONS

Try attending a religious service very different from your own and see how you feel and behave without knowing the appropriate script. Or, try attending an important, personal event that you have never attended before, such as a bar mitzvah (a coming-of-age ritual in Jewish culture), a quinceañera (in some Latin American cultures a party is given to a girl who is turning 15 years old), a wedding, a funeral, or a sporting event new to you, such as horse racing or bull riding. Observe and record your feelings and behaviors in this unfamiliar setting for which you lack the appropriate script. Do you silently observe the action, or do you ask another person for help interpreting the behaviors of people at the

event? Describe in what ways your behavior would change if you were to attend a similar event in the future?

Name and describe at least three social roles you have adopted for yourself. Why did you adopt these roles? What are some roles that are expected of you, but that you try to resist?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=169#h5p-189

Learning Objectives

By the end of this section, you will be able to:

- Define attitude
- Describe how people's attitudes are internally changed through cognitive dissonance
- Explain how people's attitudes are externally changed through persuasion
- Describe the peripheral and central routes to persuasion

Social psychologists have documented how the power of the situation can influence our behaviors. Now we turn to how the power of the situation can influence our attitudes and beliefs. Attitude is our evaluation of a person, an idea, or an object. We have attitudes for many things ranging from products that we might pick up in the supermarket to people around the world to political policies. Typically, attitudes are favorable or unfavorable: positive or negative (Eagly & Chaiken, 1993). And, they have three components: an affective component (feelings), a behavioral component (the effect of the attitude on behavior), and a cognitive component (belief and knowledge) (Rosenberg & Hovland, 1960).

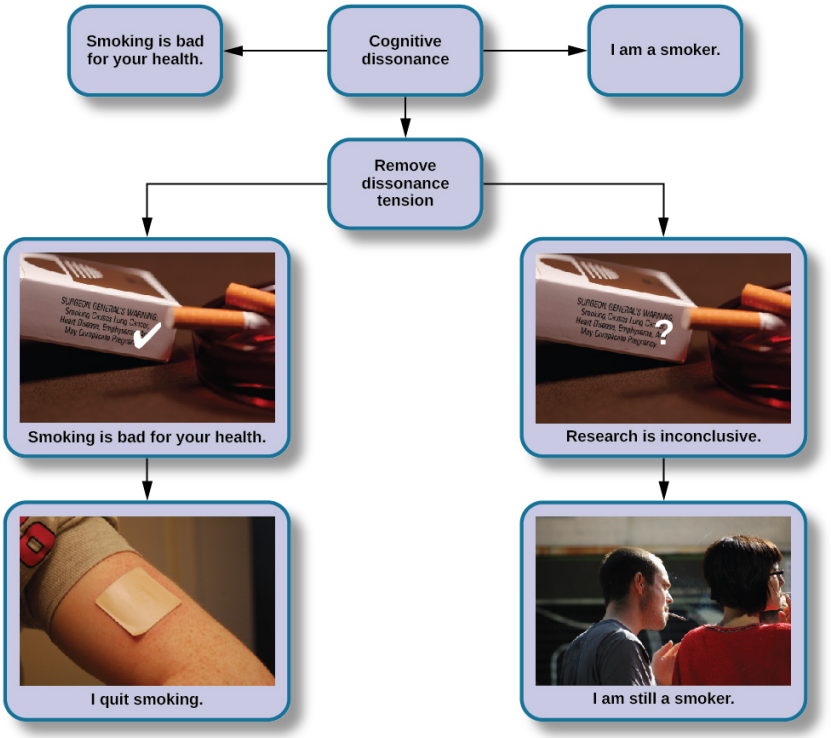
For example, you may hold a positive attitude toward recycling. This attitude should result in positive feelings toward recycling (such as "It makes me feel good to recycle" or "I enjoy knowing that I make a small difference in reducing the amount of waste that ends up in landfills"). Certainly, this attitude should be reflected in our behavior: You actually recycle as often as you can. Finally,

this attitude will be reflected in favorable thoughts (for example, "Recycling is good for the environment" or "Recycling is the responsible thing to do").

Our attitudes and beliefs are not only influenced by external forces, but also by internal influences that we control. Like our behavior, our attitudes and thoughts are not always changed by situational pressures, but they can be consciously changed by our own free will. In this section we discuss the conditions under which we would want to change our own attitudes and beliefs.

WHAT IS COGNITIVE DISSONANCE?

Social psychologists have documented that feeling good about ourselves and maintaining positive self-esteem is a powerful motivator of human behavior (Tavris & Aronson, 2008). In the United States, members of the predominant culture typically think very highly of themselves and view themselves as good people who are above average on many desirable traits (Ehrlinger, Gilovich, & Ross, 2005). Often, our behavior, attitudes, and beliefs are affected when we experience a threat to our self-esteem or positive self-image. Psychologist Leon Festinger (1957) defined cognitive dissonance as psychological discomfort arising from holding two or more inconsistent attitudes, behaviors, or cognitions (thoughts, beliefs, or opinions). Festinger's theory of cognitive dissonance states that when we experience a conflict in our behaviors, attitudes, or beliefs that runs counter to our positive self-perceptions, we experience psychological discomfort (dissonance). For example, if you believe smoking is bad for your health but you continue to smoke, you experience conflict between your belief and behavior.



Cognitive dissonance is aroused by inconsistent beliefs and behaviors. Believing cigarettes are bad for your health, but smoking cigarettes anyway, can cause cognitive dissonance. To reduce cognitive dissonance, individuals can change their behavior, as in quitting smoking, or change their belief, such as discounting the evidence that smoking is harmful. (credit "cigarettes": modification of work by CDC/Debora Cartagena; "patch": modification of "RegBarc"/Wikimedia Commons; "smoking": modification of work by Tim Parkinson)

Later research documented that only conflicting cognitions that threaten individuals' positive self-image cause dissonance (Greenwald & Ronis, 1978). Additional research found that dissonance is not only psychologically uncomfortable but also can cause physiological arousal (Croyle & Cooper, 1983) and activate

regions of the brain important in emotions and cognitive functioning (van Veen, Krug, Schooler, & Carter, 2009). When we experience cognitive dissonance, we are motivated to decrease it because it is psychologically, physically, and mentally uncomfortable. We can reduce cognitive dissonance by bringing our cognitions, attitudes, and behaviors in line—that is, making them harmonious. This can be done in different ways, such as:

- changing our discrepant behavior (e.g., stop smoking),
- changing our cognitions through rationalization or denial (e.g., telling ourselves that health risks can be reduced by smoking filtered cigarettes),
- adding a new cognition (e.g., "Smoking suppresses my appetite so I don't become overweight, which is good for my health.").

A classic example of cognitive dissonance is John, a 20-year-old who enlists in the military. During boot camp he is awakened at 5:00 a.m., is chronically sleep deprived, yelled at, covered in sand flea bites, physically bruised and battered, and mentally exhausted. It gets worse. Recruits that make it to week 11 of boot camp have to do 54 hours of continuous training.

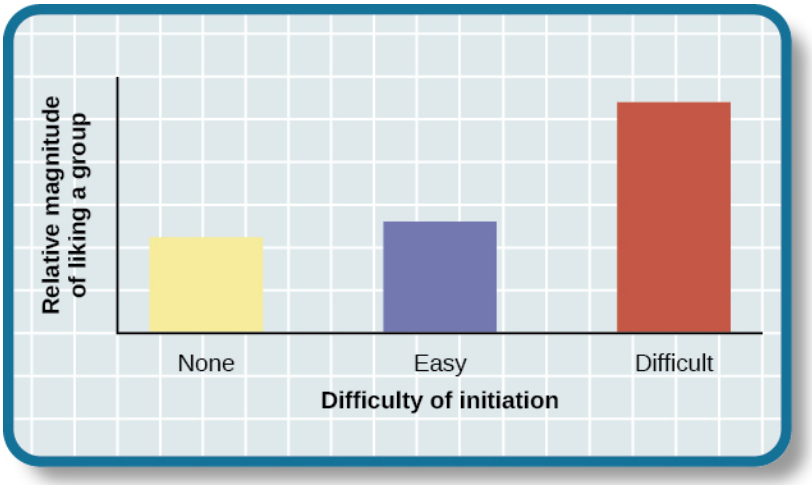
A person who has chosen a difficult path must deal with cognitive dissonance in addition to many other discomforts. (credit: Tyler J. Bolken)

Not surprisingly, John is miserable. No one likes to be miserable. In this type of situation, people can change their beliefs, their attitudes, or their behaviors. The last option, a change of behaviors, is not available to John. He has signed on to the military for four years, and he cannot legally leave.

If John keeps thinking about how miserable he is, it is going to be a very long four years. He will be in a constant state of cognitive dissonance. As an alternative to this misery, John can change his beliefs or attitudes. He can tell himself, "I am becoming stronger, healthier, and sharper. I am learning discipline and how to defend myself and my country. What I am doing is really important." If this is his belief, he will realize that he is becoming stronger through his challenges. He then will feel better and not experience cognitive dissonance, which is an uncomfortable state.

THE EFFECT OF INITIATION

The military example demonstrates the observation that a difficult initiation into a group influences us to like the group *more*, due to the justification of effort. We do not want to have wasted time and effort to join a group that we eventually leave. A classic experiment by Aronson and Mills (1959) demonstrated this justification of effort effect. College students volunteered to join a campus group that would meet regularly to discuss the psychology of sex. Participants were randomly assigned to one of three conditions: no initiation, an easy initiation, and a difficult initiation into the group. After participating in the first discussion, which was deliberately made very boring, participants rated how much they liked the group. Participants who underwent a difficult initiation process to join the group rated the group more favorably than did participants with an easy initiation or no initiation.



Justification of effort has a distinct effect on a person liking a group. Students in the difficult initiation condition liked the group more than students in other conditions due to the justification of effort.

Similar effects can be seen in a more recent study of how student effort affects course evaluations. Heckert, Latier, Ringwald-Burton, and Drazen (2006) surveyed 463 undergraduates enrolled in courses at a midwestern university about the amount of effort that their courses required of them. In addition, the students were also asked to evaluate various aspects of the course. Given what you've just read, it will come as no surprise that those courses that were associated with the highest level of effort were evaluated as being more valuable than those that did not. Furthermore, students indicated that they learned more in courses that required more effort, regardless of the grades that they received in those courses (Heckert et al., 2006).

Besides the classic military example and group initiation, can you think of other examples of cognitive dissonance? Here is one: Marco and Maria live in Fairfield County, Connecticut, which is one of the wealthiest areas in the United States and has a very high cost of living. Marco telecommutes from home and Maria does not work outside of the home. They rent a very small house for more than $3000 a month. Maria shops at consignment stores for clothes and economizes where she can. They complain that they never have any money and that they cannot buy anything new. When asked why they do not move to a less expensive location, since Marco telecommutes, they respond that Fairfield County is beautiful, they love the beaches, and they feel comfortable there. How does the theory of cognitive dissonance apply to Marco and Maria's choices?

PERSUASION

In the previous section we discussed that the motivation to reduce cognitive dissonance leads us to change our attitudes, behaviors, and/or cognitions to make them consonant. Persuasion is the process of changing our attitude toward something based on some kind of communication. Much of the persuasion we experience

comes from outside forces. How do people convince others to change their attitudes, beliefs, and behaviors? What communications do you receive that attempt to persuade you to change your attitudes, beliefs, and behaviors?

We encounter attempts at persuasion attempts everywhere. Persuasion is not limited to formal advertising; we are confronted with it throughout our everyday world. (credit: Robert Couse-Baker)

A subfield of social psychology studies persuasion and social influence, providing us with a plethora of information on how humans can be persuaded by others.

YALE ATTITUDE CHANGE APPROACH

The topic of persuasion has been one of the most extensively researched areas in social psychology (Fiske et al., 2010). During the Second World War, Carl Hovland extensively researched

persuasion for the U.S. Army. After the war, Hovland continued his exploration of persuasion at Yale University. Out of this work came a model called the Yale attitude change approach, which describes the conditions under which people tend to change their attitudes. Hovland demonstrated that certain features of the source of a persuasive message, the content of the message, and the characteristics of the audience will influence the persuasiveness of a message (Hovland, Janis, & Kelley, 1953).

Features of the source of the persuasive message include the credibility of the speaker (Hovland & Weiss, 1951) and the physical attractiveness of the speaker (Eagly & Chaiken, 1975; Petty, Wegener, & Fabrigar, 1997). Thus, speakers who are credible, or have expertise on the topic, and who are deemed as trustworthy are more persuasive than less credible speakers. Similarly, more attractive speakers are more persuasive than less attractive speakers. The use of famous actors and athletes to advertise products on television and in print relies on this principle. The immediate and long term impact of the persuasion also depends, however, on the credibility of the messenger (Kumkale & Albarracín, 2004).

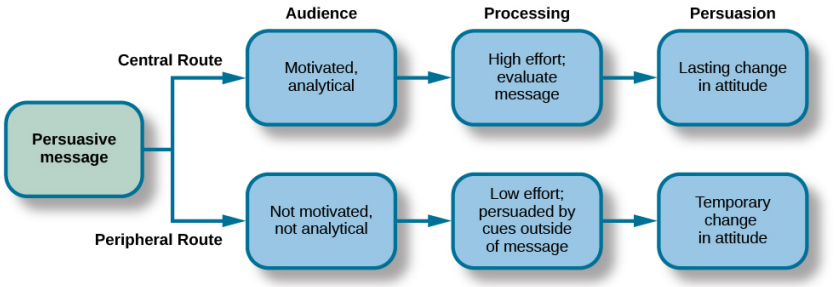
Features of the message itself that affect persuasion include subtlety (the quality of being important, but not obvious) (Petty & Cacioppo, 1986; Walster & Festinger, 1962); sidedness (that is, having more than one side) (Crowley & Hoyer, 1994; Igou & Bless, 2003; Lumsdaine & Janis, 1953); timing (Haugtvedt & Wegener, 1994; Miller & Campbell, 1959), and whether both sides are presented. Messages that are more subtle are more persuasive than direct messages. Arguments that occur first, such as in a debate, are more influential if messages are given back-to-back. However, if there is a delay after the first message, and before the audience needs to make a decision, the last message presented will tend to be more persuasive (Miller & Campbell, 1959).

Features of the audience that affect persuasion are attention (Albarracín & Wyer, 2001; Festinger & Maccoby, 1964), intelligence,

self-esteem (Rhodes & Wood, 1992), and age (Krosnick & Alwin, 1989). In order to be persuaded, audience members must be paying attention. People with lower intelligence are more easily persuaded than people with higher intelligence; whereas people with moderate self-esteem are more easily persuaded than people with higher or lower self-esteem (Rhodes & Wood, 1992). Finally, younger adults aged 18–25 are more persuadable than older adults.

ELABORATION LIKELIHOOD MODEL

An especially popular model that describes the dynamics of persuasion is the elaboration likelihood model of persuasion (Petty & Cacioppo, 1986). The elaboration likelihood model considers the variables of the attitude change approach—that is, features of the source of the persuasive message, contents of the message, and characteristics of the audience are used to determine when attitude change will occur. According to the elaboration likelihood model of persuasion, there are two main routes that play a role in delivering a persuasive message: central and peripheral.



Persuasion can take one of two paths, and the durability of the end result depends on the path.

The central route is logic driven and uses data and facts to convince people of an argument's worthiness. For example, a car company seeking to persuade you to purchase their model will emphasize

the car's safety features and fuel economy. This is a direct route to persuasion that focuses on the quality of the information. In order for the central route of persuasion to be effective in changing attitudes, thoughts, and behaviors, the argument must be strong and, if successful, will result in lasting attitude change.

The central route to persuasion works best when the target of persuasion, or the audience, is analytical and willing to engage in processing of the information. From an advertiser's perspective, what products would be best sold using the central route to persuasion? What audience would most likely be influenced to buy the product? One example is buying a computer. It is likely, for example, that small business owners might be especially influenced by the focus on the computer's quality and features such as processing speed and memory capacity.

The peripheral route is an indirect route that uses peripheral cues to associate positivity with the message (Petty & Cacioppo, 1986). Instead of focusing on the facts and a product's quality, the peripheral route relies on association with positive characteristics such as positive emotions and celebrity endorsement. For example, having a popular athlete advertise athletic shoes is a common method used to encourage young adults to purchase the shoes. This route to attitude change does not require much effort or information processing. This method of persuasion may promote positivity toward the message or product, but it typically results in less permanent attitude or behavior change. The audience does not need to be analytical or motivated to process the message. In fact, a peripheral route to persuasion may not even be noticed by the audience, for example in the strategy of product placement. Product placement refers to putting a product with a clear brand name or brand identity in a TV show or movie to promote the product (Gupta & Lord, 1998). For example, one season of the reality series *American Idol* prominently showed the panel of judges drinking out of cups that displayed the Coca-Cola logo. What other products would be best sold using the peripheral route to

persuasion? Another example is clothing: A retailer may focus on celebrities that are wearing the same style of clothing.

FOOT-IN-THE-DOOR TECHNIQUE

Researchers have tested many persuasion strategies that are effective in selling products and changing people's attitude, ideas, and behaviors. One effective strategy is the foot-in-the-door technique (Cialdini, 2001; Pliner, Hart, Kohl, & Saari, 1974). Using the foot-in-the-door technique, the persuader gets a person to agree to bestow a small favor or to buy a small item, only to later request a larger favor or purchase of a bigger item. The foot-in-the-door technique was demonstrated in a study by Freedman and Fraser (1966) in which participants who agreed to post small sign in their yard or sign a petition were more likely to agree to put a large sign in their yard than people who declined the first request. Research on this technique also illustrates the principle of consistency (Cialdini, 2001): Our past behavior often directs our future behavior, and we have a desire to maintain consistency once we have a committed to a behavior.

(a)

(b)

With the foot-in-the-door technique, a small request such as (a) wearing a campaign button can turn into a large request, such as (b) putting campaigns signs in your yard. (credit a: modification of work by Joe Crawford; credit b: modification of work by "shutterblog"/Flickr)

A common application of foot-in-the-door is when teens ask their parents for a small permission (for example, extending curfew by a half hour) and then asking them for something larger. Having granted the smaller request increases the likelihood that parents will acquiesce with the later, larger request.

How would a store owner use the foot-in-the-door technique to sell you an expensive product? For example, say that you are buying the latest model smartphone, and the salesperson suggests you purchase the best data plan. You agree to this. The salesperson then suggests a bigger purchase—the three-year extended warranty. After agreeing to the smaller request, you are more likely to also agree to the larger request. You may have encountered this if you have bought a car. When salespeople realize that a buyer intends to purchase a certain model, they might try to get the customer to pay for many or most available options on the car.

TEST YOUR UNDERSTANDING

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SUMMARY

Attitudes are our evaluations or feelings toward a person, idea, or object and typically are positive or negative. Our attitudes and beliefs are influenced not only by external forces, but also by internal influences that we control. An internal form of attitude change is cognitive dissonance or the tension we experience when our thoughts, feelings, and behaviors are in conflict. In order to reduce dissonance, individuals can change their behavior, attitudes, or cognitions, or add a new cognition. External forces

of persuasion include advertising; the features of advertising that influence our behaviors include the source, message, and audience. There are two primary routes to persuasion. The central route to persuasion uses facts and information to persuade potential consumers. The peripheral route uses positive association with cues such as beauty, fame, and positive emotions.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=171#h5p-191

CRITICAL THINKING QUESTIONS

Give an example (one not used in class or your text) of cognitive dissonance and how an individual might resolve this.

One example is choosing which college to attend—the public school close to home or the Ivy League school out of state. Since both schools are desirable, the student is likely to experience cognitive dissonance in making this decision. In order to justify choosing the public school close to home, the student could change her cognition about Ivy League school, asserting that it is too expensive and the quality of education at the public school is just as good. She could change her attitude toward the Ivy League school and determine that the students there are too stuffy and wouldn't make good classmates.

Imagine that you work for an advertising agency, and you've been tasked with developing an advertising campaign to increase sales of Bliss Soda. How would you develop an advertisement for this product that uses a central route of persuasion? How would you develop an ad using a peripheral route of persuasion?

Although potential answers will vary, advertisements using the central route of persuasion might involve a doctor listing logical reasons for

drinking this product. For example, the doctor might cite research suggesting that the soda is better than alternatives because of its reduced calorie content, lack of adverse health consequences, etc. An advertisement using a peripheral route of persuasion might show very attractive people consuming the product while spending time on a beautiful, sunny beach..

PERSONAL APPLICATION QUESTIONS

Cognitive dissonance often arises after making an important decision, called post-decision dissonance (or in popular terms, buyer's remorse). Describe a recent decision you made that caused dissonance and describe how you resolved it.

Describe a time when you or someone you know used the foot-in-the-door technique to gain someone's compliance.

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

https://pressbooks.cuny.edu/jjcpsy101/?p=171#h5p-192

Learning Objectives

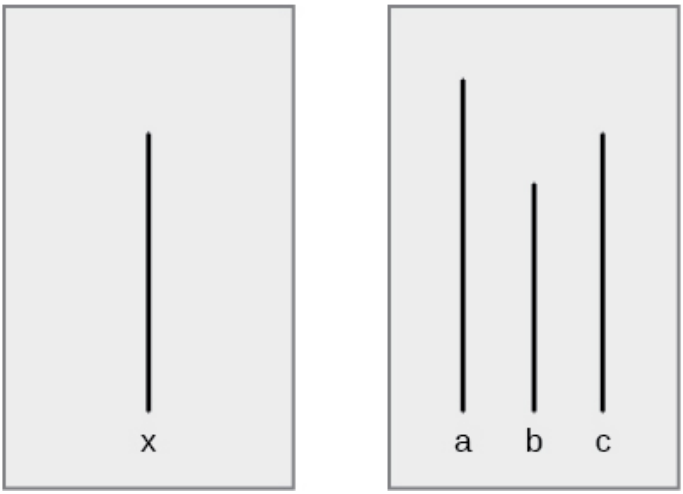
By the end of this section, you will be able to:

- Explain the Asch effect
- Define conformity and types of social influence
- Describe Stanley Milgram's experiment and its implications
- Define groupthink, social facilitation, and social loafing

In this section, we discuss additional ways in which people influence others. The topics of conformity, social influence, obedience, and group processes demonstrate the power of the social situation to change our thoughts, feelings, and behaviors. We begin this section with a discussion of a famous social psychology experiment that demonstrated how susceptible humans are to outside social pressures.

CONFORMITY

Solomon Asch conducted several experiments in the 1950s to determine how people are affected by the thoughts and behaviors of other people. In one study, a group of participants was shown a series of printed line segments of different lengths: a, b, and c. Participants were then shown a fourth line segment: x. They were asked to identify which line segment from the first group (a, b, or c) most closely resembled the fourth line segment in length.



These line segments illustrate the judgment task in Asch's conformity study. Which line on the right—a, b, or c—is the same length as line x on the left?

Each group of participants had only one true, naïve subject. The remaining members of the group were confederates of the researcher. A confederate is a person who is aware of the experiment and works for the researcher. Confederates are used to manipulate social situations as part of the research design, and the true, naïve participants believe that confederates are, like them, uninformed participants in the experiment. In Asch's study, the confederates identified a line segment that was obviously shorter than the target line—a wrong answer. The naïve participant then had to identify aloud the line segment that best matched the target line segment.

How often do you think the true participant aligned with the confederates' response? That is, how often do you think the group influenced the participant, and the participant gave the wrong answer? Asch (1955) found that 76% of participants conformed to group pressure at least once by indicating the incorrect line. Conformity is the change in a person's behavior to go along with the group, even if he does not agree with the group. Why would

people give the wrong answer? What factors would increase or decrease someone giving in or conforming to group pressure?

The Asch effect is the influence of the group majority on an individual's judgment.

What factors make a person more likely to yield to group pressure? Research shows that the size of the majority, the presence of another dissenter, and the public or relatively private nature of responses are key influences on conformity.

- The size of the majority: The greater the number of people in the majority, the more likely an individual will conform. There is, however, an upper limit: a point where adding more members does not increase conformity. In Asch's study, conformity increased with the number of people in the majority—up to seven individuals. At numbers beyond seven, conformity leveled off and decreased slightly (Asch, 1955).
- The presence of another dissenter: If there is at least one dissenter, conformity rates drop to near zero (Asch, 1955).
- The public or private nature of the responses: When responses are made publicly (in front of others), conformity is more likely; however, when responses are made privately (e.g., writing down the response), conformity is less likely (Deutsch & Gerard, 1955).

The finding that conformity is more likely to occur when responses are public than when they are private is the reason government elections require voting in secret, so we are not coerced by others. The Asch effect can be easily seen in children when they have to publicly vote for something. For example, if the teacher asks whether the children would rather have extra recess, no homework, or candy, once a few children vote, the rest will comply and go with the majority. In a different classroom, the majority might vote differently, and most of the children would comply with

that majority. When someone's vote changes if it is made in public versus private, this is known as compliance. Compliance can be a form of conformity. Compliance is going along with a request or demand, even if you do not agree with the request. In Asch's studies, the participants complied by giving the wrong answers, but privately did not accept that the obvious wrong answers were correct.

Voting for government officials in the United States is private to reduce the pressure of conformity. (credit: Nicole Klauss)

Now that you have learned about the Asch line experiments, why do you think the participants conformed? The correct answer to the line segment question was obvious, and it was an easy task. Researchers have categorized the motivation to conform into two types: normative social influence and informational social influence (Deutsch & Gerard, 1955).

In normative social influence, people conform to the group norm to fit in, to feel good, and to be accepted by the group. However, with informational social influence, people conform because they

believe the group is competent and has the correct information, particularly when the task or situation is ambiguous. What type of social influence was operating in the Asch conformity studies? Since the line judgment task was unambiguous, participants did not need to rely on the group for information. Instead, participants complied to fit in and avoid ridicule, an instance of normative social influence.

An example of informational social influence may be what to do in an emergency situation. Imagine that you are in a movie theater watching a film and what seems to be smoke comes in the theater from under the emergency exit door. You are not certain that it is smoke—it might be a special effect for the movie, such as a fog machine. When you are uncertain you will tend to look at the behavior of others in the theater. If other people show concern and get up to leave, you are likely to do the same. However, if others seem unconcerned, you are likely to stay put and continue watching the movie.

(a)

(b)

People in crowds tend to take cues from others and act accordingly. (a) An audience is listening to a lecture and people are relatively quiet, still, and attentive to the speaker on the stage. (b) An audience is at a rock concert where people are dancing, singing, and possibly engaging in activities like crowd surfing. (credit a: modification of work by Matt Brown; credit b: modification of work by Christian Holmér)

How would you have behaved if you were a participant in Asch's study? Many students say they would not conform, that the study is outdated, and that people nowadays are more independent. To some extent this may be true. Research suggests that overall rates of conformity may have reduced since the time of Asch's research. Furthermore, efforts to replicate Asch's study have made it clear that many factors determine how likely it is that someone will demonstrate conformity to the group. These factors include the participant's age, gender, and socio-cultural background (Bond & Smith, 1996; Larsen, 1990; Walker & Andrade, 1996).

Watch this video to see a replication of the Asch experiment: conformity.

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=173#oembed-1

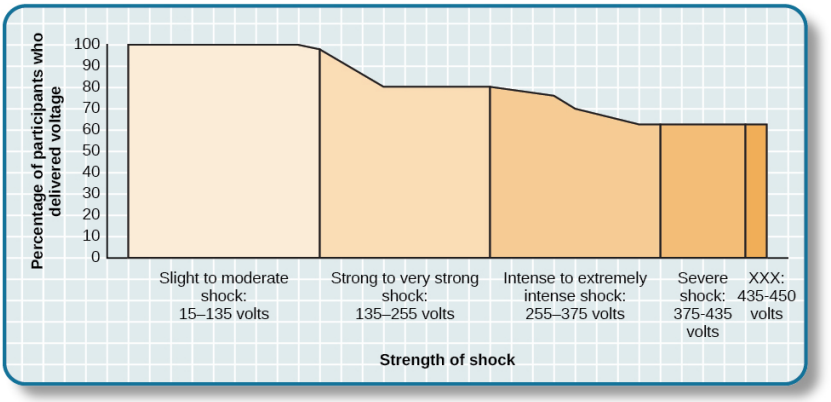
STANLEY MILGRAM'S EXPERIMENT

Conformity is one effect of the influence of others on our thoughts, feelings, and behaviors. Another form of social influence is obedience to authority. Obedience is the change of an individual's behavior to comply with a demand by an authority figure. People often comply with the request because they are concerned about a consequence if they do not comply. To demonstrate this phenomenon, we review another classic social psychology experiment.

Stanley Milgram was a social psychology professor at Yale who was influenced by the trial of Adolf Eichmann, a Nazi war criminal. Eichmann's defense for the atrocities he committed was that he was "just following orders." Milgram (1963) wanted to test the validity of this defense, so he designed an experiment and initially recruited 40 men for his experiment. The volunteer participants were led to believe that they were participating in a study to improve learning and memory. The participants were told that they were to teach other students (learners) correct answers to a series of test items. The participants were shown how to use a device that they were told delivered electric shocks of different intensities to the learners. The participants were told to shock the learners if they gave a wrong answer to a test item—that the shock would help them to learn. The participants gave (or believed they gave) the learners shocks, which increased in 15-volt increments, all the way up to 450 volts. The participants did not know that the learners were confederates and that the confederates did not actually receive shocks.

In response to a string of incorrect answers from the learners, the participants obediently and repeatedly shocked them. The confederate learners cried out for help, begged the participant teachers to stop, and even complained of heart trouble. Yet, when the researcher told the participant-teachers to continue the shock, 65% of the participants continued the shock to the maximum

voltage and to the point that the learner became unresponsive. What makes someone obey authority to the point of potentially causing serious harm to another person?



The Milgram experiment showed the surprising degree to which people obey authority. Two out of three (65%) participants continued to administer shocks to an unresponsive learner.

Several variations of the original Milgram experiment were conducted to test the boundaries of obedience. When certain features of the situation were changed, participants were less likely to continue to deliver shocks (Milgram, 1965). For example, when the setting of the experiment was moved to an office building, the percentage of participants who delivered the highest shock dropped to 48%. When the learner was in the same room as the teacher, the highest shock rate dropped to 40%. When the teachers' and learners' hands were touching, the highest shock rate dropped to 30%. When the researcher gave the orders by phone, the rate dropped to 23%. These variations show that when the humanity of the person being shocked was increased, obedience decreased. Similarly, when the authority of the experimenter decreased, so did obedience.

This case is still very applicable today. What does a person do if an authority figure orders something done? What if the person

believes it is incorrect, or worse, unethical? In a study by Martin and Bull (2008), midwives privately filled out a questionnaire regarding best practices and expectations in delivering a baby. Then, a more senior midwife and supervisor asked the junior midwives to do something they had previously stated they were opposed to. Most of the junior midwives were obedient to authority, going against their own beliefs.

GROUPTHINK

When in group settings, we are often influenced by the thoughts, feelings, and behaviors around us. Whether it is due to normative or informational social influence, groups have power to influence individuals. Another phenomenon of group conformity is groupthink. Groupthink is the modification of the opinions of members of a group to align with what they believe is the group consensus (Janis, 1972). In group situations, the group often takes action that individuals would not perform outside the group setting because groups make more extreme decisions than individuals do. Moreover, groupthink can hinder opposing trains of thought. This elimination of diverse opinions contributes to faulty decision by the group.

GROUPTHINK IN THE U.S. GOVERNMENT

There have been several instances of groupthink in the U.S. government. One example occurred when the United States led a small coalition of nations to invade Iraq in March 2003. This invasion occurred because a small group of advisors and former President George W. Bush were convinced that Iraq represented a significant terrorism threat with a large stockpile of weapons of mass destruction at its disposal. Although some of these individuals may have had some doubts about the credibility of the

information available to them at the time, in the end, the group arrived at a consensus that Iraq had weapons of mass destruction and represented a significant threat to national security. It later came to light that Iraq did not have weapons of mass destruction, but not until the invasion was well underway. As a result, 6000 American soldiers were killed and many more civilians died. How did the Bush administration arrive at their conclusions?

Why does groupthink occur? There are several causes of groupthink, which makes it preventable. When the group is highly cohesive, or has a strong sense of connection, maintaining group harmony may become more important to the group than making sound decisions. If the group leader is directive and makes his opinions known, this may discourage group members from disagreeing with the leader. If the group is isolated from hearing alternative or new viewpoints, groupthink may be more likely. How do you know when groupthink is occurring?

There are several symptoms of groupthink including the following:

- perceiving the group as invulnerable or invincible—believing it can do no wrong
- believing the group is morally correct
- self-censorship by group members, such as withholding information to avoid disrupting the group consensus
- the quashing of dissenting group members' opinions
- the shielding of the group leader from dissenting views
- perceiving an illusion of unanimity among group members
- holding stereotypes or negative attitudes toward the out-group or others' with differing viewpoints (Janis, 1972)

Given the causes and symptoms of groupthink, how can it be avoided? There are several strategies that can improve group

decision making including seeking outside opinions, voting in private, having the leader withhold position statements until all group members have voiced their views, conducting research on all viewpoints, weighing the costs and benefits of all options, and developing a contingency plan (Janis, 1972; Mitchell & Eckstein, 2009).

GROUP POLARIZATION

Another phenomenon that occurs within group settings is group polarization. Group polarization (Teger & Pruitt, 1967) is the strengthening of an original group attitude after the discussion of views within a group. That is, if a group initially favors a viewpoint, after discussion the group consensus is likely a stronger endorsement of the viewpoint. Conversely, if the group was initially opposed to a viewpoint, group discussion would likely lead to stronger opposition. Group polarization explains many actions taken by groups that would not be undertaken by individuals. Group polarization can be observed at political conventions, when platforms of the party are supported by individuals who, when not in a group, would decline to support them. A more everyday example is a group's discussion of how attractive someone is. Does your opinion change if you find someone attractive, but your friends do not agree? If your friends vociferously agree, might you then find this person even more attractive?

SOCIAL FACILITATION

Not all intergroup interactions lead to the negative outcomes we have described. Sometimes being in a group situation can improve performance. Social facilitation occurs when an individual performs better when an audience is watching than when the individual performs the behavior alone. This typically occurs when people are

performing a task for which they are skilled. Can you think of an example in which having an audience could improve performance? One common example is sports. Skilled basketball players will be more likely to make a free throw basket when surrounded by a cheering audience than when playing alone in the gym. However, there are instances when even skilled athletes can have difficulty under pressure. For example, if an athlete is less skilled or nervous about making a free throw, having an audience may actually hinder rather than help. In sum, social facilitation is likely to occur for easy tasks, or tasks at which we are skilled, but worse performance may occur when performing in front of others, depending on the task.

The attention of the crowd can motivate a skilled athlete. (credit: Tommy Gilligan/USMA)

SOCIAL LOAFING

Another way in which a group presence can affect our performance is social loafing. Social loafing is the exertion of less effort by a person working together with a group. Social loafing occurs when our individual performance cannot be evaluated separately from the group. Thus, group performance declines on easy tasks (Karau & Williams, 1993). Essentially individual group members loaf and let other group members pick up the slack. Because each individual's efforts cannot be evaluated, individuals become less motivated to perform well. For example, consider a group of people cooperating to clean litter from the roadside. Some people will exert a great amount of effort, while others will exert little effort. Yet the entire job gets done, and it may not be obvious who worked hard and who didn't.

As a college student you may have experienced social loafing while working on a group project. Have you ever had to contribute more than your fair share because your fellow group members weren't putting in the work? This may happen when a professor assigns a group grade instead of individual grades. If the professor doesn't know how much effort each student contributed to a project, some students may be inclined to let more conscientious students do more of the work. The chance of social loafing in student work groups increases as the size of the group increases (Shepperd & Taylor, 1999).

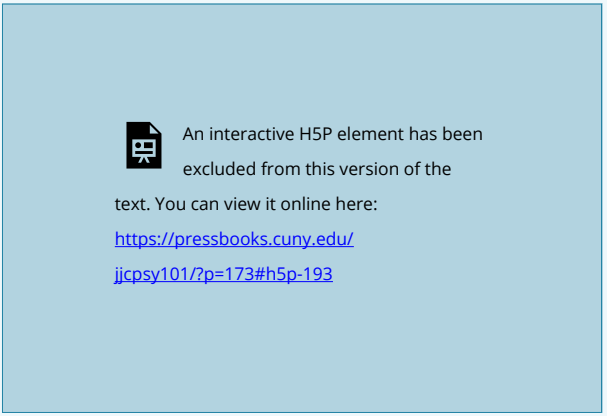
Interestingly, the opposite of social loafing occurs when the task is complex and difficult (Bond & Titus, 1983; Geen, 1989). Remember the previous discussion of choking under pressure? This happens when you perform a difficult task and your individual performance can be evaluated. In a group setting, such as the student work group, if your individual performance cannot be evaluated, there is less pressure for you to do well, and thus less anxiety or physiological arousal (Latané, Williams, & Harkens, 1979). This puts you in a relaxed state in which you can perform

your best, if you choose (Zajonc, 1965). If the task is a difficult one, many people feel motivated and believe that their group needs their input to do well on a challenging project (Jackson & Williams, 1985). Given what you learned about social loafing, what advice would you give a new professor about how to design group projects? If you suggested that individuals' efforts should not be evaluated, to prevent the anxiety of choking under pressure, but that the task must be challenging, you have a good understanding of the concepts discussed in this section. Alternatively, you can suggest that individuals' efforts should be evaluated, but the task should be easy so as to facilitate performance. Good luck trying to convince your professor to only assign easy projects.

Types of Social Influence

Type of Social Influence	Description
Conformity	Changing your behavior to go along with the group even if you do not agree with the group
Compliance	Going along with a request or demand
Normative social influence	Conformity to a group norm to fit in, feel good, and be accepted by the group
Informational social influence	Conformity to a group norm prompted by the belief that the group is competent and has the correct information
Obedience	Changing your behavior to please an authority figure or to avoid aversive consequences
Groupthink	Group members modify their opinions to match what they believe is the group consensus
Group polarization	Strengthening of the original group attitude after discussing views within a group
Social facilitation	Improved performance when an audience is watching versus when the individual performs the behavior alone
Social loafing	Exertion of less effort by a person working in a group because individual performance cannot be evaluated separately from the group, thus causing performance decline on easy tasks

TEST YOUR UNDERSTANDING

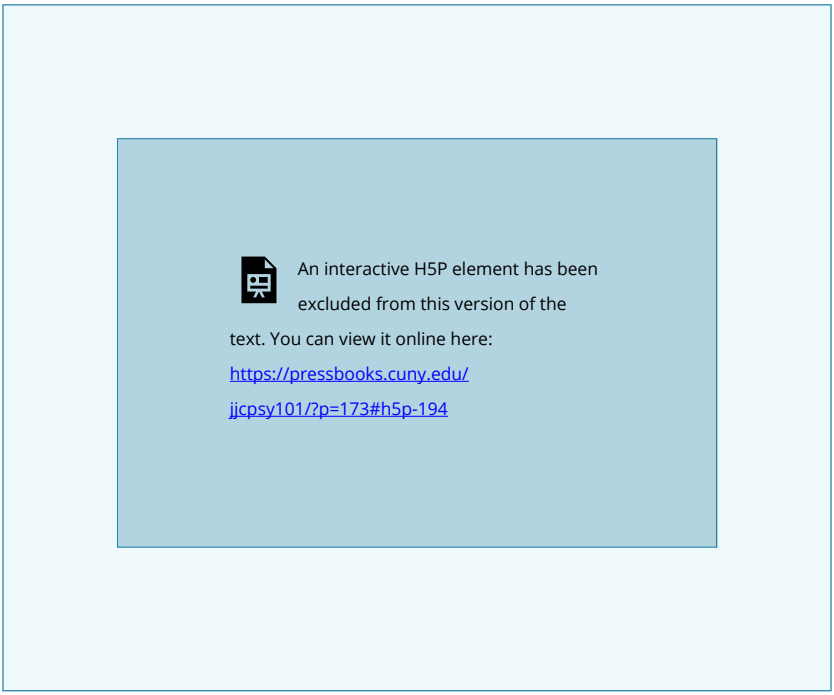


SUMMARY

The power of the situation can lead people to conform, or go along with the group, even in the face of inaccurate information. Conformity to group norms is driven by two motivations, the desire to fit in and be liked and the desire to be accurate and gain information from the group. Authority figures also have influence over our behaviors, and many people become obedient and follow orders even if the orders are contrary to their personal values. Conformity to group pressures can also result in groupthink, or the faulty decision-making process that results from cohesive group

members trying to maintain group harmony. Group situations can improve human behavior through facilitating performance on easy tasks, but inhibiting performance on difficult tasks. The presence of others can also lead to social loafing when individual efforts cannot be evaluated.

REVIEW QUESTIONS



CRITICAL THINKING QUESTIONS

Describe how seeking outside opinions can prevent groupthink.

Outsiders can serve as a quality control by offering diverse views and views that may differ from the leader's opinion. The outsider can also remove the illusion of invincibility by having the group's action held up to outside scrutiny. An outsider may offer additional information and uncover information that group members withheld.

Compare and contrast social loafing and social facilitation.

In social loafing individual performance cannot be evaluated; however, in social facilitation individual performance can be evaluated. Social loafing and social facilitation both occur for easy or well-known tasks and when individuals are relaxed.

PERSONAL APPLICATION QUESTIONS

1. Conduct a conformity study the next time you are in an elevator. After you enter the elevator, stand with your back toward the door. See if others conform to your behavior. Did your results turn out as expected?

2. Most students adamantly state that they would never have turned up the voltage in the Milligram experiment. Do you think you would have refused to shock the learner? Looking at your own past behavior, what evidence suggests that you would go along with the order to increase the voltage?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=173#h5p-195

Media Attributions

- "conformity" by freescoring. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Define and distinguish among prejudice, stereotypes, and discrimination
- Provide examples of prejudice, stereotypes, and discrimination
- Explain why prejudice and discrimination exist

Human conflict can result in crime, war, and mass murder, such as genocide. Prejudice and discrimination often are root causes of human conflict, which explains how strangers come to hate one another to the extreme of causing others harm. Prejudice and discrimination affect everyone. In this section we will examine the definitions of prejudice and discrimination, examples of these concepts, and causes of these biases.

(a)

(b)

(c)

Prejudice and discrimination occur across the globe. (a) A 1939 sign in German-occupied Poland warns "No Entrance for Poles!" (b) An African-American male drinks from a designated "colored" water fountain in Oklahoma in 1939 during the era of racial segregation as a practice of discrimination. (c) A member of the Westboro Baptist Church, widely identified as a hate group, engages in discrimination based on religion and sexual orientation. (credit b: modification of work by United States Farm Security Administration; credit c: modification of work by "JCWilmore"/Wikimedia Commons)

UNDERSTANDING PREJUDICE AND DISCRIMINATION

As we discussed in the opening story of Trayvon Martin, humans are very diverse and although we share many similarities, we also have many differences. The social groups we belong to help form our identities (Tajfel, 1974). These differences may be difficult for some people to reconcile, which may lead to prejudice toward people who are different. Prejudice is a negative attitude and feeling toward an individual based solely on one's membership in a particular social group (Allport, 1954; Brown, 2010). Prejudice is common against people who are members of an unfamiliar cultural group. Thus, certain types of education, contact, interactions, and building relationships with members of different cultural groups can reduce the tendency toward prejudice. In fact, simply imagining interacting with members of different cultural groups might affect prejudice. Indeed, when experimental participants were asked to

imagine themselves positively interacting with someone from a different group, this led to an increased positive attitude toward the other group and an increase in positive traits associated with the other group. Furthermore, imagined social interaction can reduce anxiety associated with inter-group interactions (Crisp & Turner, 2009). What are some examples of social groups that you belong to that contribute to your identity? Social groups can include gender, race, ethnicity, nationality, social class, religion, sexual orientation, profession, and many more. And, as is true for social roles, you can simultaneously be a member of more than one social group. An example of prejudice is having a negative attitude toward people who are not born in the United States. Although people holding this prejudiced attitude do not know all people who were not born in the United States, they dislike them due to their status as foreigners.

Can you think of a prejudiced attitude you have held toward a group of people? How did your prejudice develop? Prejudice often begins in the form of a stereotype—that is, a specific belief or assumption about individuals based solely on their membership in a group, regardless of their individual characteristics. Stereotypes become overgeneralized and applied to all members of a group. For example, someone holding prejudiced attitudes toward older adults, may believe that older adults are slow and incompetent (Cuddy, Norton, & Fiske, 2005; Nelson, 2004). We cannot possibly know each individual person of advanced age to know that all older adults are slow and incompetent. Therefore, this negative belief is overgeneralized to all members of the group, even though many of the individual group members may in fact be spry and intelligent.

Another example of a well-known stereotype involves beliefs about racial differences among athletes. As Hodge, Burden, Robinson, and Bennett (2008) point out, Black male athletes are often believed to be more athletic, yet less intelligent, than their White male counterparts. These beliefs persist despite a number of high profile examples to the contrary. Sadly, such beliefs often

influence how these athletes are treated by others and how they view themselves and their own capabilities. Whether or not you agree with a stereotype, stereotypes are generally well-known within in a given culture (Devine, 1989).

Sometimes people will act on their prejudiced attitudes toward a group of people, and this behavior is known as discrimination. Discrimination is negative action toward an individual as a result of one's membership in a particular group (Allport, 1954; Dovidio & Gaertner, 2004). As a result of holding negative beliefs (stereotypes) and negative attitudes (prejudice) about a particular group, people often treat the target of prejudice poorly, such as excluding older adults from their circle of friends. Have you ever been the target of discrimination? If so, how did this negative treatment make you feel?

Connecting Stereotypes, Prejudice, and Discrimination

Item	Function	Connection	Example
Stereotype	Cognitive; thoughts about people	Overgeneralized beliefs about people may lead to prejudice.	"Yankees fans are arrogant and obnoxious."
Prejudice	Affective; feelings about people, both positive and negative	Feelings may influence treatment of others, leading to discrimination.	"I hate Yankees fans; they make me angry."
Discrimination	Behavior; positive or negative treatment of others	Holding stereotypes and harboring prejudice may lead to excluding, avoiding, and biased treatment of group members.	"I would never hire nor become friends with a person if I knew he or she were a Yankees fan."

So far, we've discussed stereotypes, prejudice, and discrimination as negative thoughts, feelings, and behaviors because these are typically the most problematic. However, it is important to also point out that people can hold positive thoughts, feelings, and behaviors toward individuals based on group membership; for example, they would show preferential treatment for people who are like themselves—that is, who share the same gender, race, or favorite sports team.

This video demonstrates the concepts of prejudice, stereotypes, and discrimination. In the video, a social experiment is conducted in a park where three people try to steal a bike out in the open. The race and gender of the thief is varied: a White male teenager, a Black male teenager, and a White female. Does anyone try to stop them? The treatment of the teenagers in the video demonstrates the concept of racism: What Would You Do? Bike Theft (White Guy, Black Guy, Pretty Girl).

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=175#oembed-1

TYPES OF PREJUDICE AND DISCRIMINATION

When we meet strangers we automatically process three pieces of information about them: their race, gender, and age (Ito & Urland,

2003). Why are these aspects of an unfamiliar person so important? Why don't we instead notice whether their eyes are friendly, whether they are smiling, their height, the type of clothes they are wearing? Although these secondary characteristics are important in forming a first impression of a stranger, the social categories of race, gender, and age provide a wealth of information about an individual. This information, however, often is based on stereotypes. We may have different expectations of strangers depending on their race, gender, and age. What stereotypes and prejudices do you hold about people who are from a race, gender, and age group different from your own?

RACISM

Racism is prejudice and discrimination against an individual based solely on one's membership in a specific racial group (such as toward African Americans, Asian Americans, Latinos, Native Americans, European Americans). What are some stereotypes of various racial or ethnic groups? Research suggests cultural stereotypes for Asian Americans include cold, sly, and intelligent; for Latinos, cold and unintelligent; for European Americans, cold and intelligent; and for African Americans, aggressive, athletic, and more likely to be law breakers (Devine & Elliot, 1995; Fiske, Cuddy, Glick, & Xu, 2002; Sommers & Ellsworth, 2000; Dixon & Linz, 2000).

Racism exists for many racial and ethnic groups. For example, Blacks are significantly more likely to have their vehicles searched during traffic stops than Whites, particularly when Blacks are driving in predominately White neighborhoods, (a phenomenon often termed "DWB," or "driving while Black.") (Rojek, Rosenfeld, & Decker, 2012)

Mexican Americans and other Latino groups also are targets of racism from the police and other members of the community. For example, when purchasing items with a personal check, Latino

shoppers are more likely than White shoppers to be asked to show formal identification (Dovidio et al., 2010).

In one case of alleged harassment by the police, several East Haven, Connecticut, police officers were arrested on federal charges due to reportedly continued harassment and brutalization of Latinos. When the accusations came out, the mayor of East Haven was asked, "What are you doing for the Latino community today?" The Mayor responded, "I might have tacos when I go home, I'm not quite sure yet" ("East Haven Mayor," 2012) This statement undermines the important issue of racial profiling and police harassment of Latinos, while belittling Latino culture by emphasizing an interest in a food product stereotypically associated with Latinos.

Racism is prevalent toward many other groups in the United States including Native Americans, Arab Americans, Jewish Americans, and Asian Americans. Have you witnessed racism toward any of these racial or ethnic groups? Are you aware of racism in your community?

One reason modern forms of racism, and prejudice in general, are hard to detect is related to the dual attitudes model (Wilson, Lindsey, & Schooler, 2000). Humans have two forms of attitudes: explicit attitudes, which are conscious and controllable, and implicit attitudes, which are unconscious and uncontrollable (Devine, 1989; Olson & Fazio, 2003). Because holding egalitarian views is socially desirable (Plant & Devine, 1998), most people do not show extreme racial bias or other prejudices on measures of their explicit attitudes. However, measures of implicit attitudes often show evidence of mild to strong racial bias or other prejudices (Greenwald, McGee, & Schwartz, 1998; Olson & Fazio, 2003).

SEXISM

Sexism is prejudice and discrimination toward individuals based on their sex. Typically, sexism takes the form of men holding biases

against women, but either sex can show sexism toward their own or their opposite sex. Like racism, sexism may be subtle and difficult to detect. Common forms of sexism in modern society include gender role expectations, such as expecting women to be the caretakers of the household. Sexism also includes people's expectations for how members of a gender group should behave. For example, women are expected to be friendly, passive, and nurturing, and when women behave in an unfriendly, assertive, or neglectful manner they often are disliked for violating their gender role (Rudman, 1998). Research by Laurie Rudman (1998) finds that when female job applicants self-promote, they are likely to be viewed as competent, but they may be disliked and are less likely to be hired because they violated gender expectations for modesty. Sexism can exist on a societal level such as in hiring, employment opportunities, and education. Women are less likely to be hired or promoted in male-dominated professions such as engineering, aviation, and construction (Blau, Ferber, & Winkler, 2010; Ceci & Williams, 2011). Have you ever experienced or witnessed sexism? Think about your family members' jobs or careers. Why do you think there are differences in the jobs women and men have, such as more women nurses but more male surgeons (Betz, 2008)?

Women now have many jobs previously closed to them, though they still face challenges in male-dominated occupations. (credit: "Alex"/Flickr)

AGEISM

People often form judgments and hold expectations about people based on their age. These judgments and expectations can lead to ageism, or prejudice and discrimination toward individuals based solely on their age. Typically, ageism occurs against older adults, but ageism also can occur toward younger adults. Think of expectations you hold for older adults. How could someone's expectations influence the feelings they hold toward individuals from older age groups? Ageism is widespread in U.S. culture (Nosek, 2005), and a common ageist attitude toward older adults is that they are incompetent, physically weak, and slow (Greenberg, Schimel, & Martens, 2002) and some people consider older adults less attractive. Some cultures, however, including some Asian, Latino, and African American cultures, both outside and within the United States afford older adults respect and honor.

Ageism can also occur toward younger adults. What expectations do you hold toward younger people? Does society expect younger

adults to be immature and irresponsible? How might these two forms of ageism affect a younger and older adult who are applying for a sales clerk position?

HOMOPHOBIA

Another form of prejudice is homophobia: prejudice and discrimination of individuals based solely on their sexual orientation. Like ageism, homophobia is a widespread prejudice in U.S. society that is tolerated by many people (Herek & McLemore, 2013; Nosek, 2005). Negative feelings often result in discrimination, such as the exclusion of lesbian, gay, bisexual, and transgender (LGBT) people from social groups and the avoidance of LGBT neighbors and co-workers. This discrimination also extends to employers deliberately declining to hire qualified LGBT job applicants. Have you experienced or witnessed homophobia? If so, what stereotypes, prejudiced attitudes, and discrimination were evident?

Research into Homophobia

Some people are quite passionate in their hatred for nonheterosexuals in our society. In some cases, people have been tortured and/or murdered simply because they were not heterosexual. This passionate response has led some researchers to question what motives might exist for homophobic people.

Adams, Wright, & Lohr (1996) conducted a study investigating this issue and their results were quite an eye-opener.

In this experiment, male college students were given a scale that assessed how homophobic they were; those with extreme scores were recruited to participate in the experiment. In the end, 64 men agreed to participate and were split into 2 groups: homophobic men and nonhomophobic men. Both groups of men were fitted with a penile plethysmograph, an instrument that measures changes in blood flow to the penis and serves as an objective measurement of sexual arousal.

All men were shown segments of sexually explicit videos. One of these videos involved a sexual interaction between a man and a woman (heterosexual clip). One video displayed two females engaged in a sexual interaction (homosexual female clip), and the final video displayed two men engaged in a sexual interaction (homosexual male clip). Changes in penile tumescence were recorded during all three clips, and a subjective measurement of sexual arousal was also obtained. While both groups of men became sexually aroused to the heterosexual and female homosexual video clips, only those men who were identified as homophobic showed sexual arousal to the homosexual male video clip. While all men reported that their erections indicated arousal for the heterosexual and female homosexual clips, the homophobic men indicated that they were not sexually aroused (despite their erections) to the male homosexual clips. Adams et al. (1996) suggest that these findings may indicate that homophobia is related to homosexual arousal that the homophobic individuals either deny or are unaware.

WHY DO PREJUDICE AND DISCRIMINATION EXIST?

Prejudice and discrimination persist in society due to social

learning and conformity to social norms. Children learn prejudiced attitudes and beliefs from society: their parents, teachers, friends, the media, and other sources of socialization, such as Facebook (O'Keeffe & Clarke-Pearson, 2011). If certain types of prejudice and discrimination are acceptable in a society, there may be normative pressures to conform and share those prejudiced beliefs, attitudes, and behaviors. For example, public and private schools are still somewhat segregated by social class. Historically, only children from wealthy families could afford to attend private schools, whereas children from middle- and low-income families typically attended public schools. If a child from a low-income family received a merit scholarship to attend a private school, how might the child be treated by classmates? Can you recall a time when you held prejudiced attitudes or beliefs or acted in a discriminatory manner because your group of friends expected you to?

STEREOTYPES AND SELF-FULFILLING PROPHECY

When we hold a stereotype about a person, we have expectations that he or she will fulfill that stereotype. A self-fulfilling prophecy is an expectation held by a person that alters his or her behavior in a way that tends to make it true. When we hold stereotypes about a person, we tend to treat the person according to our expectations. This treatment can influence the person to act according to our stereotypic expectations, thus confirming our stereotypic beliefs. Research by Rosenthal and Jacobson (1968) found that disadvantaged students whose teachers expected them to perform well had higher grades than disadvantaged students whose teachers expected them to do poorly.

Consider this example of cause and effect in a self-fulfilling prophecy: If an employer expects an openly gay male job applicant to be incompetent, the potential employer might treat the

applicant negatively during the interview by engaging in less conversation, making little eye contact, and generally behaving coldly toward the applicant (Hebl, Foster, Mannix, & Dovidio, 2002). In turn, the job applicant will perceive that the potential employer dislikes him, and he will respond by giving shorter responses to interview questions, making less eye contact, and generally disengaging from the interview. After the interview, the employer will reflect on the applicant's behavior, which seemed cold and distant, and the employer will conclude, based on the applicant's poor performance during the interview, that the applicant was in fact incompetent. Thus, the employer's stereotype—gay men are incompetent and do not make good employees—is reinforced. Do you think this job applicant is likely to be hired? Treating individuals according to stereotypic beliefs can lead to prejudice and discrimination.

Another dynamic that can reinforce stereotypes is confirmation bias. When interacting with the target of our prejudice, we tend to pay attention to information that is consistent with our stereotypic expectations and ignore information that is inconsistent with our expectations. In this process, known as confirmation bias, we seek out information that supports our stereotypes and ignore information that is inconsistent with our stereotypes (Wason & Johnson-Laird, 1972). In the job interview example, the employer may not have noticed that the job applicant was friendly and engaging, and that he provided competent responses to the interview questions in the beginning of the interview. Instead, the employer focused on the job applicant's performance in the later part of the interview, after the applicant changed his demeanor and behavior to match the interviewer's negative treatment.

Have you ever fallen prey to the self-fulfilling prophecy or confirmation bias, either as the source or target of such bias? How might we stop the cycle of the self-fulfilling prophecy? Social class stereotypes of individuals tend to arise when information about

the individual is ambiguous. If information is unambiguous, stereotypes do not tend to arise (Baron et al., 1995).

IN-GROUPS AND OUT-GROUPS

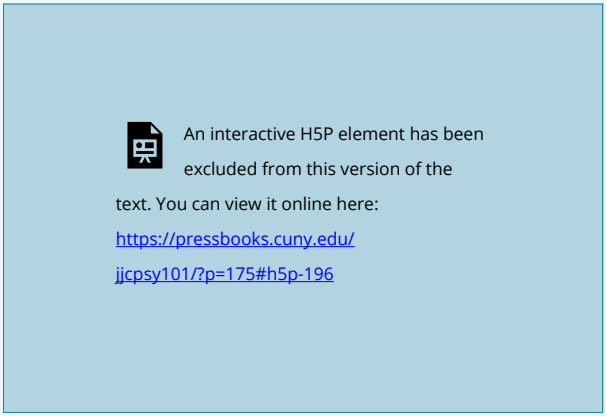
As discussed previously in this section, we all belong to a gender, race, age, and social economic group. These groups provide a powerful source of our identity and self-esteem (Tajfel & Turner, 1979). These groups serve as our in-groups. An in-group is a group that we identify with or see ourselves as belonging to. A group that we don't belong to, or an out-group, is a group that we view as fundamentally different from us. For example, if you are female, your gender in-group includes all females, and your gender out-group includes all males. People often view gender groups as being fundamentally different from each other in personality traits, characteristics, social roles, and interests. Because we often feel a strong sense of belonging and emotional connection to our in-groups, we develop in-group bias: a preference for our own group over other groups. This in-group bias can result in prejudice and discrimination because the out-group is perceived as different and is less preferred than our in-group.

These children are very young, but they are already aware of their gender in-group and out-group. (credit: modification of work by Simone Ramella)

Despite the group dynamics that seem only to push groups toward conflict, there are forces that promote reconciliation between groups: the expression of empathy, of acknowledgment of past suffering on both sides, and the halt of destructive behaviors.

One function of prejudice is to help us feel good about ourselves and maintain a positive self-concept. This need to feel good about ourselves extends to our in-groups: We want to feel good and protect our in-groups. We seek to resolve threats individually and at the in-group level. This often happens by blaming an out-group for the problem. Scapegoating is the act of blaming an out-group when the in-group experiences frustration or is blocked from obtaining a goal (Allport, 1954).

TEST YOUR UNDERSTANDING



SUMMARY

As diverse individuals, humans can experience conflict when interacting with people who are different from each other. Prejudice, or negative feelings and evaluations, is common when people are from a different social group (i.e., out-group). Negative attitudes toward out-groups can lead to discrimination. Prejudice and discrimination against others can be based on gender, race, ethnicity, social class, sexual orientation, or a variety of other social identities. In-group's who feel threatened may blame the out-

groups for their plight, thus using the out-group as a scapegoat for their frustration.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Some people seem more willing to openly display prejudice regarding sexual orientation than prejudice regarding race and gender. Speculate on why this might be.

In the United States, many people believe that sexual orientation is a choice, and there is some debate in the research literature as to the extent sexual orientation is biological or influenced by social factors. Because race and gender are not chosen, many Americans believe it is unfair to negatively judge women or racial minority groups for a characteristic that is determined by genetics. In addition, many people in the United States practice religions that believe homosexuality is wrong.

When people blame a scapegoat, how do you think they choose evidence to support the blame?

One way in which they might do this is to selectively attend to information that would bolster their argument. Furthermore, they may actively seek out information to confirm their assertions.

PERSONAL APPLICATION QUESTIONS

1. Give an example when you felt that someone was

prejudiced against you. What do you think caused this attitude? Did this person display any discrimination behaviors and, if so, how?

2. Give an example when you felt prejudiced against someone else. How did you discriminate against them? Why do you think you did this?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=175#h5p-198

Media Attributions

- "What Would You Do? Bike Theft (White Guy, Black Guy, Pretty Girl)" by VladCantSleep. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Define aggression
- Define cyberbullying
- Describe the bystander effect

Throughout this chapter we have discussed how people interact and influence one another's thoughts, feelings, and behaviors in both positive and negative ways. People can work together to achieve great things, such as helping each other in emergencies: recall the heroism displayed during the 9/11 terrorist attacks. People also can do great harm to one another, such as conforming to group norms that are immoral and obeying authority to the point of murder: consider the mass conformity of Nazis during WWII. In this section we will discuss a negative side of human behavior—aggression.

AGGRESSION

Humans engage in aggression when they seek to cause harm or pain to another person. Aggression takes two forms depending on one's motives: hostile or instrumental. Hostile aggression is motivated by feelings of anger with intent to cause pain; a fight in a bar with a stranger is an example of hostile aggression. In contrast, instrumental aggression is motivated by achieving a goal

and does not necessarily involve intent to cause pain (Berkowitz, 1993); a contract killer who murders for hire displays instrumental aggression.

There are many different theories as to why aggression exists. Some researchers argue that aggression serves an evolutionary function (Buss, 2004). Men are more likely than women to show aggression (Wilson & Daly, 1985). From the perspective of evolutionary psychology, human male aggression, like that in nonhuman primates, likely serves to display dominance over other males, both to protect a mate and to perpetuate the male's genes. Sexual jealousy is part of male aggression; males endeavor to make sure their mates are not copulating with other males, thus ensuring their own paternity of the female's offspring. Although aggression provides an obvious evolutionary advantage for men, women also engage in aggression. Women typically display instrumental forms of aggression, with their aggression serving as a means to an end (Dodge & Schwartz, 1997). For example, women may express their aggression covertly, for example, by communication that impairs the social standing of another person. Another theory that explains one of the functions of human aggression is frustration aggression theory (Dollard, Doob, Miller, Mowrer, & Sears, 1939). This theory states that when humans are prevented from achieving an important goal, they become frustrated and aggressive.

Human males and nonhuman male primates endeavor to gain and display dominance over other males, as demonstrated in the behavior of these monkeys. (credit: "Arcadiuš"/Flickr)

BULLYING

A modern form of aggression is bullying. As you learn in your study of child development, socializing and playing with other children is beneficial for children's psychological development. However, as you may have experienced as a child, not all play behavior has positive outcomes. Some children are aggressive and want to play roughly. Other children are selfish and do not want to share toys. One form of negative social interactions among children that has become a national concern is bullying. Bullying is repeated negative treatment of another person, often an adolescent, over time (Olweus, 1993). A one-time incident in which one child hits another child on the playground would not be considered bullying: Bullying is repeated behavior. The negative treatment typical in bullying is the attempt to inflict harm, injury, or humiliation, and bullying can include physical or verbal attacks. However, bullying doesn't have to be physical or verbal, it can be psychological.

Research finds gender differences in how girls and boys bully others (American Psychological Association, 2010; Olweus, 1993). Boys tend to engage in direct, physical aggression such as physically harming others. Girls tend to engage in indirect, social forms of aggression such as spreading rumors, ignoring, or socially isolating others. Based on what you have learned about child development and social roles, why do you think boys and girls display different types of bullying behavior?

Bullying involves three parties: the bully, the victim, and witnesses or bystanders. The act of bullying involves an imbalance of power with the bully holding more power—physically, emotionally, and/or socially over the victim. The experience of bullying can be positive for the bully, who may enjoy a boost to self-esteem. However, there are several negative consequences of bullying for the victim, and also for the bystanders. How do you think bullying negatively impacts adolescents? Being the victim of bullying is associated with decreased mental health, including experiencing anxiety and depression (APA, 2010). Victims of bullying may underperform in schoolwork (Bowen, 2011). Bullying also can result in the victim committing suicide (APA, 2010). How might bullying negatively affect witnesses?

Although there is not one single personality profile for who becomes a bully and who becomes a victim of bullying (APA, 2010), researchers have identified some patterns in children who are at a greater risk of being bullied (Olweus, 1993):

Children who are emotionally reactive are at a greater risk for being bullied. Bullies may be attracted to children who get upset easily because the bully can quickly get an emotional reaction from them.

Children who are different from others are likely to be targeted for bullying. Children who are overweight, cognitively impaired, or racially or ethnically different from their peer group may be at higher risk.

Gay, lesbian, bisexual, and transgender teens are at very high risk

of being bullied and hurt due to their sexual orientation.

CYBERBULLYING

With the rapid growth of technology, and widely available mobile technology and social networking media, a new form of bullying has emerged: cyberbullying (Hoff & Mitchell, 2009). Cyberbullying, like bullying, is repeated behavior that is intended to cause psychological or emotional harm to another person. What is unique about cyberbullying is that it is typically covert, concealed, done in private, and the bully can remain anonymous. This anonymity gives the bully power, and the victim may feel helpless, unable to escape the harassment, and unable to retaliate (Spears, Slee, Owens, & Johnson, 2009).

Cyberbullying can take many forms, including harassing a victim by spreading rumors, creating a website defaming the victim, and ignoring, insulting, laughing at, or teasing the victim (Spears et al., 2009). In cyberbullying, it is more common for girls to be the bullies and victims because cyberbullying is nonphysical and is a less direct form of bullying (Hoff & Mitchell, 2009). Interestingly, girls who become cyberbullies often have been the victims of cyberbullying at one time (Vandebosch & Van Cleemput, 2009). The effects of cyberbullying are just as harmful as traditional bullying and include the victim feeling frustration, anger, sadness, helplessness, powerlessness, and fear. Victims will also experience lower self-esteem (Hoff & Mitchell, 2009; Spears et al., 2009). Furthermore, recent research suggests that both cyberbullying victims and perpetrators are more likely to experience suicidal ideation, and they are more likely to attempt suicide than individuals who have no experience with cyberbullying (Hinduja & Patchin, 2010). What features of technology make cyberbullying easier and perhaps more accessible to young adults? What can parents, teachers, and social networking websites, like Facebook, do to prevent cyberbullying?

Because cyberbullying is not physical in nature, cyberbullies and their victims are most often female; however, there is much evidence that male homosexuals are frequently victims of cyberbullying as well (Hinduja & Patchin, 2011). (credit: Steven Depolo)

THE BYSTANDER EFFECT

The discussion of bullying highlights the problem of witnesses not intervening to help a victim. This is a common occurrence, as the following well-publicized event demonstrates. In 1964, in Queens, New York, a 19-year-old woman named Kitty Genovese was attacked by a person with a knife near the back entrance to her apartment building and again in the hallway inside her apartment building. When the attack occurred, she screamed for help numerous times and eventually died from her stab wounds. This story became famous because reportedly numerous residents in the apartment building heard her cries for help and did nothing—neither helping her nor summoning the police—though these have facts been disputed.

Based on this case, researchers Latané and Darley (1968)

described a phenomenon called the bystander effect. The bystander effect is a phenomenon in which a witness or bystander does not volunteer to help a victim or person in distress. Instead, they just watch what is happening. Social psychologists hold that we make these decisions based on the social situation, not our own personality variables. Why do you think the bystanders didn't help Genovese? What are the benefits to helping her? What are the risks? It is very likely you listed more costs than benefits to helping. In this situation, bystanders likely feared for their own lives—if they went to her aid the attacker might harm them. However, how difficult would it have been to make a phone call to the police from the safety of their apartments? Why do you think no one helped in any way? Social psychologists claim that diffusion of responsibility is the likely explanation. Diffusion of responsibility is the tendency for no one in a group to help because the responsibility to help is spread throughout the group (Bandura, 1999). Because there were many witnesses to the attack on Genovese, as evidenced by the number of lit apartment windows in the building, individuals assumed someone else must have already called the police. The responsibility to call the police was diffused across the number of witnesses to the crime. Have you ever passed an accident on the freeway and assumed that a victim or certainly another motorist has already reported the accident? In general, the greater the number of bystanders, the less likely any one person will help.

TEST YOUR UNDERSTANDING

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SUMMARY

Aggression is seeking to cause another person harm or pain. Hostile aggression is motivated by feelings of anger with intent to cause pain, and instrumental aggression is motivated by achieving a goal and does not necessarily involve intent to cause pain Bullying is an international public health concern that largely affects the adolescent population. Bullying is repeated behaviors that are intended to inflict harm on the victim and can take the form of physical, psychological, emotional, or social abuse. Bullying has

negative mental health consequences for youth including suicide. Cyberbullying is a newer form of bullying that takes place in an online environment where bullies can remain anonymous and victims are helpless to address the harassment. Despite the social norm of helping others in need, when there are many bystanders witnessing an emergency, diffusion of responsibility will lead to a lower likelihood of any one person helping.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=177#h5p-200

CRITICAL THINKING QUESTIONS

Compare and contrast hostile and instrumental aggression.

Hostile aggression is intentional with the purpose to inflict pain. Hostile aggression is often motivated by anger. In contrast, instrumental aggression is not motivated by anger or the intention to cause pain. Instrumental aggression serves as a means to reach a goal. In a sense it is a more practical or functional form of aggression, whereas hostile aggression is more emotion-driven and less functional and rational.

What evidence discussed in the previous section suggests that cyberbullying is difficult to detect and prevent?

Cyberbullying is difficult to prevent because there are so many forms of media that adolescents use and are exposed to. The Internet is virtually everywhere: computers, phones, tablets, TVs, gaming systems, and so on. Parents likely do not monitor all of their children's use of the Internet, thus their children could be exposed to cyberbullying without their knowledge. Cyberbullying is difficult to detect because it can be done anonymously. Cyberbullies can use pseudonyms and can attack victims in untraceable ways, such as hacking into Facebook accounts or making Twitter posts on their behalf.

PERSONAL APPLICATION QUESTIONS

Have you ever experienced or witnessed bullying or cyberbullying? How did it make you feel? What did you do about it? After reading this section would you have done anything differently?
The next time you see someone needing help, observe your surroundings. Look to see if the bystander effect is in action and take measures to make sure the person gets help. If you aren't able to help, notify an adult or authority figure that can.

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:
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Learning Objectives

By the end of this section, you will be able to:

- Describe altruism
- Describe conditions that influence the formation of relationships
- Identify what attracts people to each other
- Describe the triangular theory of love
- Explain social exchange theory in relationships

You've learned about many of the negative behaviors of social psychology, but the field also studies many positive social interactions and behaviors. What makes people like each other? With whom are we friends? Whom do we date? Researchers have documented several features of the situation that influence whether we form relationships with others. There are also universal traits that humans find attractive in others. In this section we discuss conditions that make forming relationships more likely, what we look for in friendships and romantic relationships, the different types of love, and a theory explaining how our relationships are formed, maintained, and terminated.

PROSOCIAL BEHAVIOR AND ALTRUISM

Do you voluntarily help others? Voluntary behavior with the intent

to help other people is called prosocial behavior. Why do people help other people? Is personal benefit such as feeling good about oneself the only reason people help one another? Research suggests there are many other reasons. Altruism is people's desire to help others even if the costs outweigh the benefits of helping. In fact, people acting in altruistic ways may disregard the personal costs associated with helping. For example, news accounts of the 9/11 terrorist attacks on the World Trade Center in New York reported an employee in the first tower helped his co-workers make it to the exit stairwell. After helping a co-worker to safety he went back in the burning building to help additional co-workers. In this case the costs of helping were great, and the hero lost his life in the destruction (Stewart, 2002).

The events of 9/11 unleashed an enormous show of altruism and heroism on the parts of first responders and many ordinary people. (credit: Don Halasy)

Some researchers suggest that altruism operates on empathy. Empathy is the capacity to understand another person's perspective, to feel what he or she feels. An empathetic person makes an emotional connection with others and feels compelled to help (Batson, 1991). Other researchers argue that altruism is a form of selfless helping that is not motivated by benefits or feeling good about oneself. Certainly, after helping, people feel good about themselves, but some researchers argue that this is a consequence of altruism, not a cause. Other researchers argue that helping is always self-serving because our egos are involved, and we receive benefits from helping (Cialdini, Brown, Lewis, Luce, & Neuberg 1997). It is challenging to determine experimentally

the true motivation for helping, whether is it largely self-serving (egoism) or selfless (altruism). Thus, a debate on whether pure altruism exists continues.

See this excerpt from the popular TV series *Friends* episode for a discussion of the egoism versus altruism debate: Good Deeds Do NOT exist | Friends | 77Math. https://youtube.com/watch?v=ahDxg3hc5pM

FORMING RELATIONSHIPS

What do you think is the single most influential factor in determining with whom you become friends and whom you form romantic relationships? You might be surprised to learn that the answer is simple: the people with whom you have the most contact. This most important factor is proximity. You are more likely to be friends with people you have regular contact with. For example, there are decades of research that shows that you are more likely to become friends with people who live in your dorm, your apartment building, or your immediate neighborhood than with people who live farther away (Festinger, Schachler, & Back, 1950). It is simply easier to form relationships with people you see often because you have the opportunity to get to know them.

Similarity is another factor that influences who we form relationships with. We are more likely to become friends or lovers with someone who is similar to us in background, attitudes, and lifestyle. In fact, there is no evidence that opposites attract. Rather, we are attracted to people who are most like us (McPherson, Smith-

Lovin, & Cook, 2001). Why do you think we are attracted to people who are similar to us? Sharing things in common will certainly make it easy to get along with others and form connections. When you and another person share similar music taste, hobbies, food preferences, and so on, deciding what to do with your time together might be easy. Homophily is the tendency for people to form social networks, including friendships, marriage, business relationships, and many other types of relationships, with others who are similar (McPherson et al., 2001).

People tend to be attracted to similar people. Many couples share a cultural background. This can be quite obvious in a ceremony such as a wedding, and more subtle (but no less significant) in the day-to-day workings of a relationship. (credit: modification of work by Shiraz Chanawala)

But, homophily limits our exposure to diversity (McPherson et al., 2001). By forming relationships only with people who are similar to us, we will have homogenous groups and will not be exposed to different points of view. In other words, because we are likely to spend time with those who are most like ourselves, we will have limited exposure to those who are different than ourselves,

including people of different races, ethnicities, social-economic status, and life situations.

Once we form relationships with people, we desire reciprocity. Reciprocity is the give and take in relationships. We contribute to relationships, but we expect to receive benefits as well. That is, we want our relationships to be a two way street. We are more likely to like and engage with people who like us back. Self-disclosure is part of the two way street. Self-disclosure is the sharing of personal information (Laurenceau, Barrett, & Pietromonaco, 1998). We form more intimate connections with people with whom we disclose important information about ourselves. Indeed, self-disclosure is a characteristic of healthy intimate relationships, as long as the information disclosed is consistent with our own views (Cozby, 1973).

ATTRACTION

We have discussed how proximity and similarity lead to the formation of relationships, and that reciprocity and self-disclosure are important for relationship maintenance. But, what features of a person do we find attractive? We don't form relationships with everyone that lives or works near us, so how is it that we decide which specific individuals we will select as friends and lovers?

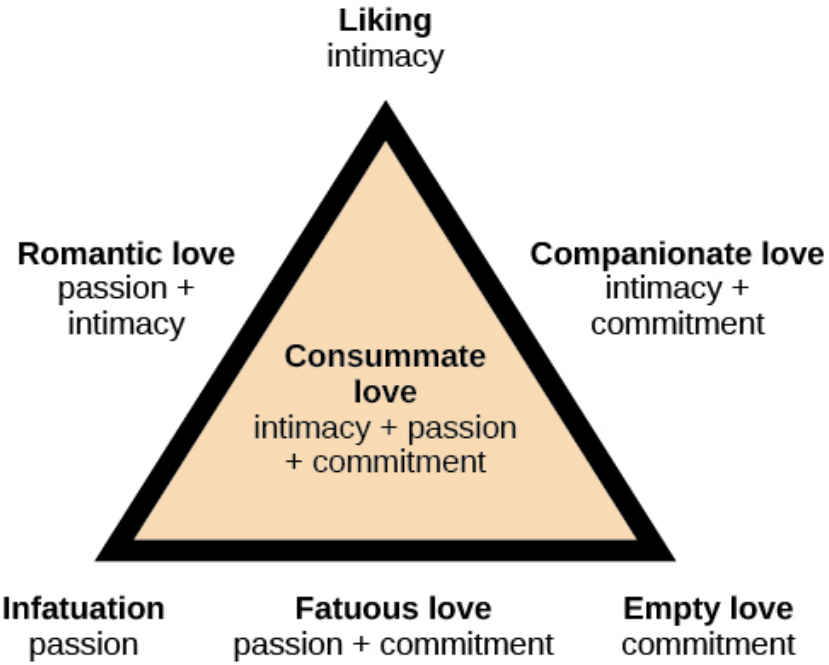
Researchers have documented several characteristics in men and women that humans find attractive. First we look for friends and lovers who are physically attractive. People differ in what they consider attractive, and attractiveness is culturally influenced. Research, however, suggests that some universally attractive features in women include large eyes, high cheekbones, a narrow jaw line, a slender build (Buss, 1989), and a lower waist-to-hip ratio (Singh, 1993). For men, attractive traits include being tall, having broad shoulders, and a narrow waist (Buss, 1989). Both men and women with high levels of facial and body symmetry are

generally considered more attractive than asymmetric individuals (Fink, Neave, Manning, & Grammer, 2006; Penton-Voak et al., 2001; Rikowski & Grammer, 1999). Social traits that people find attractive in potential female mates include warmth, affection, and social skills; in males, the attractive traits include achievement, leadership qualities, and job skills (Regan & Berscheid, 1997). Although humans want mates who are physically attractive, this does not mean that we look for the most attractive person possible. In fact, this observation has led some to propose what is known as the matching hypothesis which asserts that people tend to pick someone they view as their equal in physical attractiveness and social desirability (Taylor, Fiore, Mendelsohn, & Cheshire, 2011). For example, you and most people you know likely would say that a very attractive movie star is out of your league. So, even if you had proximity to that person, you likely would not ask them out on a date because you believe you likely would be rejected. People weigh a potential partner's attractiveness against the likelihood of success with that person. If you think you are particularly unattractive (even if you are not), you likely will seek partners that are fairly unattractive (that is, unattractive in physical appearance or in behavior).

STERNBERG'S TRIANGULAR THEORY OF LOVE

We typically love the people with whom we form relationships, but the type of love we have for our family, friends, and lovers differs. Robert Sternberg (1986) proposed that there are three components of love: intimacy, passion, and commitment. These three components form a triangle that defines multiple types of love: this is known as Sternberg's triangular theory of love. Intimacy is the sharing of details and intimate thoughts and emotions. Passion is the physical attraction—the flame in the fire.

Commitment is standing by the person—the "in sickness and health" part of the relationship.



According to Sternberg's triangular theory of love, seven types of love can be described from combinations of three components: intimacy, passion, and commitment. (credit: modification of work by "Lnesa"/Wikimedia Commons)

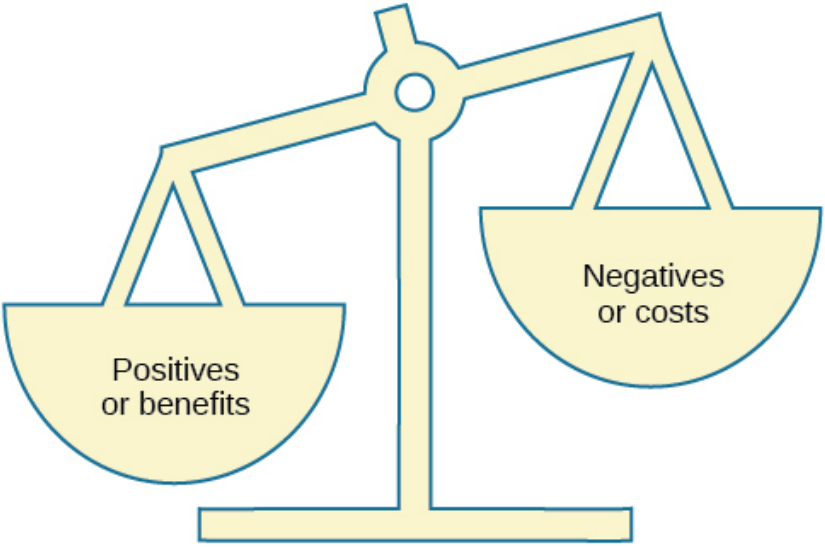
Sternberg (1986) states that a healthy relationship will have all three components of love—intimacy, passion, and commitment—which is described as consummate love. However, different aspects of love might be more prevalent at different life stages. Other forms of love include liking, which is defined as having intimacy but no passion or commitment. Infatuation is the presence of passion without intimacy or commitment. Empty love is having commitment without intimacy or passion. Companionate love, which is characteristic of close friendships and family relationships, consists of intimacy and commitment but no passion.

Romantic love is defined by having passion and intimacy, but no commitment. Finally, fatuous love is defined by having passion and commitment, but no intimacy, such as a long term sexual love affair. Can you describe other examples of relationships that fit these different types of love?

According to Sternberg, consummate love describes a healthy relationship containing intimacy, passion, and commitment. (credit: Kerry Ceszyk)

SOCIAL EXCHANGE THEORY

We have discussed why we form relationships, what attracts us to others, and different types of love. But what determines whether we are satisfied with and stay in a relationship? One theory that provides an explanation is social exchange theory. According to social exchange theory, we act as naïve economists in keeping a tally of the ratio of costs and benefits of forming and maintaining a relationship with others (Rusbult & Van Lange, 2003).



Acting like naïve economists, people may keep track of the costs and benefits of maintaining a relationship. Typically, only those relationships in which the benefits outweigh the costs will be maintained.

People are motivated to maximize the benefits of social exchanges, or relationships, and minimize the costs. People prefer to have more benefits than costs, or to have nearly equal costs and benefits, but most people are dissatisfied if their social exchanges create more costs than benefits. Let's discuss an example. If you have ever decided to commit to a romantic relationship, you probably considered the advantages and disadvantages of your decision. What are the benefits of being in a committed romantic relationship? You may have considered having companionship, intimacy, and passion, but also being comfortable with a person you know well. What are the costs of being in a committed romantic relationship? You may think that over time boredom from being with only one person may set in; moreover, it may be expensive to share activities such as attending movies and going to dinner.

However, the benefits of dating your romantic partner presumably outweigh the costs, or you wouldn't continue the relationship.

TEST YOUR UNDERSTANDING

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SUMMARY

Altruism is a pure form of helping others out of empathy, which can be contrasted with egoistic motivations for helping. Forming relationships with others is a necessity for social beings. We typically form relationships with people who are close to us in proximity and people with whom we share similarities. We expect reciprocity and self-disclosure in our relationships. We also want

to form relationships with people who are physically attractive, though standards for attractiveness vary by culture and gender. There are many types of love that are determined by various combinations of intimacy, passion, and commitment; consummate love, which is the ideal form of love, contains all three components. When determining satisfaction and whether to maintain a relationship, individuals often use a social exchange approach and weigh the costs and benefits of forming and maintaining a relationship.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Describe what influences whether relationships will be formed.

Proximity is a major situational factor in relationship formation; people who have frequent contact are more likely to form relationships. Whether or not individuals will form a relationship is based on non-situational factors such as similarity, reciprocity, self-disclosure, and physical attractiveness. In relationships, people seek reciprocity (i.e., a give and take in costs and benefits), self-disclosure of intimate information, and physically attractive partners.

The evolutionary theory argues that humans are motivated to perpetuate their genes and reproduce. Using an evolutionary perspective, describe traits in men and women that humans find attractive.

Traits that promote reproduction in females warmth, affection, and social skills; women with these traits are presumably better able to care for children. Traits that are desired in males include achievement, leadership qualities, and job skills; men with these traits are thought to be better able to financially provide for their families.

PERSONAL APPLICATION QUESTIONS

1. Think about your recent friendships and romantic relationship(s). What factors do you think influenced the development of these relationships? What attracted you to becoming friends or romantic partners?
2. Have you ever used a social exchange theory approach to determine how satisfied you were in a relationship, either a friendship or romantic relationship? Have you ever had the costs outweigh the benefits of a relationship? If so, how did you address this imbalance?

GLOSSARY

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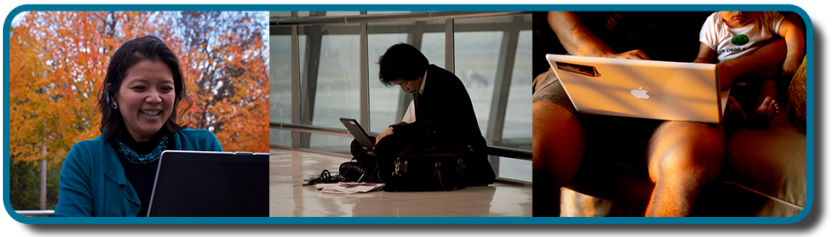
INDUSTRIAL-ORGANIZATIONAL PSYCHOLOGY



What does an office look like? For people who telecommute, their workspace may be adapted to fit their lifestyle. (credit: "left": modification of work by Cory Zanker; credit "center": modification of work by "@Saigon"/Flickr; credit "right": modification of work by Daniel Lobo)

In July 2012, Yahoo!, one of the largest and oldest web companies, announced the appointment of Marissa Mayer as CEO. Yahoo! had struggled to define itself and excel in the industry for several years, and the appointment of Mayer, a top Google executive, made big news. Among her many decisions, in February 2013, Mayer announced that employees would no longer be allowed to telecommute. Telecommuting is representative of many management innovations that have been made in recent years,

largely by tech companies. Telecommuting reflects a belief on the part of companies that employees are responsible, self-motivating, and perhaps work best when they are left alone. It also has an impact on work–family balance, though which way is yet unclear. And telecommuting reflects the more general trend of increasing overlap between workers' time spent *on the job* and time spent *off the job*.

The reversal of this policy at Yahoo! brought controversy and a lot of questions about what it meant. Mayer has stayed largely quiet on her reasoning behind the decision, except to say that it was meant to better the company. She finally addressed her decision briefly at the 2013 Great Place to Work conference (Tkaczyk, 2013) by saying, among other things, that while "people are more productive when they're alone, they're more collaborative and innovative when they're together." Interestingly, shortly after the Yahoo! change, consumer electronics retailer Best Buy also eliminated telecommuting as an option for their employees. Will the change make Yahoo! more innovative or more productive? How has the change affected employees at the company, particularly working parents and those taking care of elderly relatives? Was the change introduced in the most effective way? These are questions that are commonly studied by a branch of psychology called industrial and organizational psychology.

Learning Objectives

By the end of this section, you will be able to:

- Understand the scope of study in the field of industrial and organizational psychology
- Describe the history of industrial and organizational psychology

In 2012, people who worked in the United States spent an average of 56.4 hours per week working (Bureau of Labor Statistics—U.S. Department of Labor, 2013). Sleeping was the only other activity they spent more time on with an average of 61.2 hours per week. The workday is a significant portion of workers' time and energy. It impacts their lives and their family's lives in positive and negative physical and psychological ways. Industrial and organizational (I-O) psychology is a branch of psychology that studies how human behavior and psychology affect work and how they are affected by work.

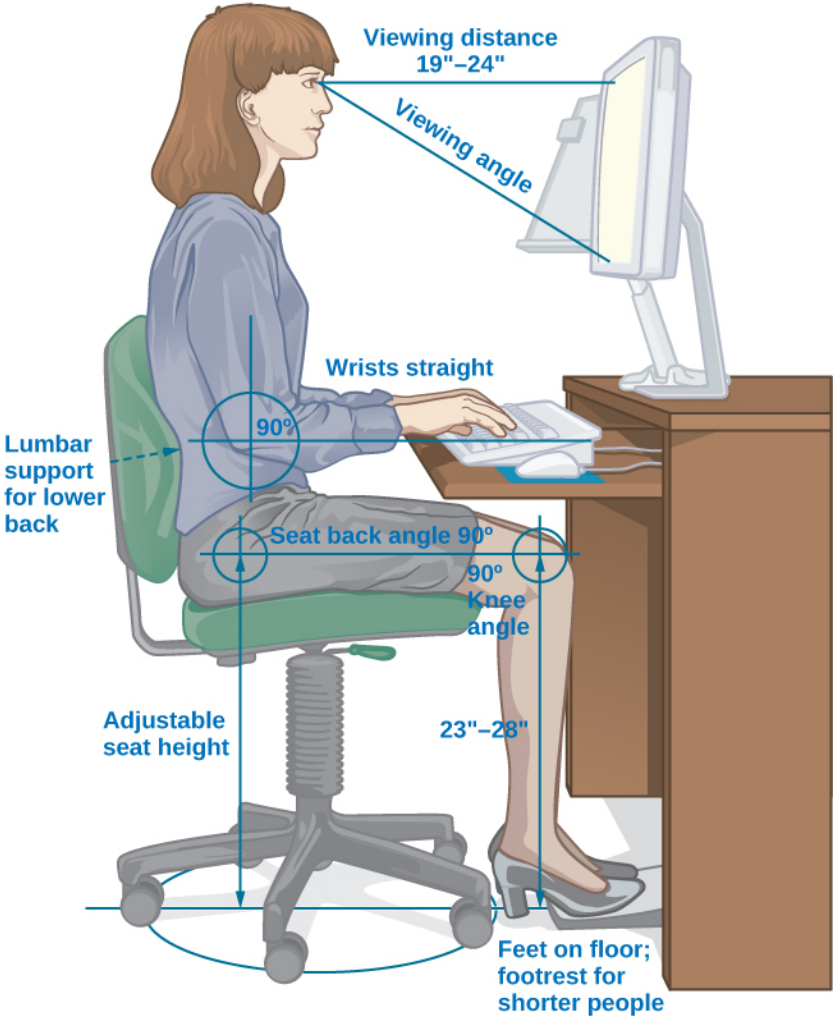
Industrial and organizational psychologists work in four main contexts: academia, government, consulting firms, and business. Most I-O psychologists have a master's or doctorate degree. The field of I-O psychology can be divided into three broad areas: industrial, organizational, and human factors. Industrial psychology is concerned with describing job requirements and assessing individuals for their ability to meet those requirements. In addition, once employees are hired, industrial psychology studies and develops ways to train, evaluate, and respond to those

evaluations. As a consequence of its concern for candidate characteristics, industrial psychology must also consider issues of legality regarding discrimination in hiring. Organizational psychology is a discipline interested in how the relationships among employees affect those employees and the performance of a business. This includes studying worker satisfaction, motivation, and commitment. This field also studies management, leadership, and organizational culture, as well as how an organization's structures, management and leadership styles, social norms, and role expectations affect individual behavior. As a result of its interest in worker wellbeing and relationships, organizational psychology also considers the subjects of harassment, including sexual harassment, and workplace violence. Human factors psychology is the study of how workers interact with the tools of work and how to design those tools to optimize workers' productivity, safety, and health. These studies can involve interactions as straightforward as the fit of a desk, chair, and computer to a human having to sit on the chair at the desk using the computer for several hours each day. They can also include the examination of how humans interact with complex displays and their ability to interpret them accurately and quickly. In Europe, this field is referred to as ergonomics.

(a)

(b)

(a) Industrial psychology focuses on hiring and maintaining employees. (b) Organizational psychology is interested in employee relationships and organizational culture. (credit a: modification of work by Cory Zanker; credit b: modification of work by Vitor Lima)



Human factors psychology is the study of interactions between humans, tools, and work systems.

Find out what I-O psychologists do on the Society for Industrial and Organizational Psychology (SIOP) website—a professional organization for people working in the discipline. This site also offers several I-O psychologist profiles: Society for Industrial and Organizational Psychology (SIOP).

THE HISTORICAL DEVELOPMENT OF INDUSTRIAL AND ORGANIZATIONAL PSYCHOLOGY

Industrial and organizational psychology had its origins in the early 20th century. Several influential early psychologists studied issues that today would be categorized as industrial psychology: James Cattell (1860–1944) at Columbia, Hugo Münsterberg (1863–1916) at Harvard, Walter Dill Scott (1869–1955) at Northwestern, Robert Yerkes (1876–1956) and Walter Bingham (1880–1952) at Dartmouth, and Lillian Gilbreth (1878–1972) at Purdue. Cattell, Münsterberg, and Scott had been students of Wilhelm Wundt, the father of experimental psychology. Some of these researchers had been involved in work in the area of industrial psychology before World War I. Cattell's contribution to industrial psychology is largely reflected in his founding of a psychological consulting company, which is still operating today called the Psychological Corporation, and in the accomplishments of students at Columbia in the area of industrial psychology. In 1913, Münsterberg published *Psychology*

and Industrial Efficiency, which covered topics such as employee selection, employee training, and effective advertising.

Scott was one of the first psychologists to apply psychology to advertising, management, and personnel selection. In 1903, Scott published two books: *The Theory of Advertising* and *Psychology of Advertising*. They are the first books to describe the use of psychology in the business world. By 1911 he published two more books, *Influencing Men in Business* and *Increasing Human Efficiency in Business*. In 1916 a newly formed division in the Carnegie Institute of Technology hired Scott to conduct applied research on employee selection (Katzell & Austin, 1992).

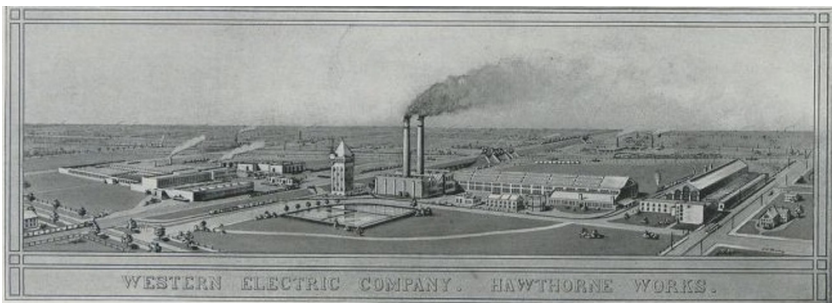
The focus of all this research was in what we now know as industrial psychology; it was only later in the century that the field of organizational psychology developed as an experimental science (Katzell & Austin, 1992). In addition to their academic positions, these researchers also worked directly for businesses as consultants.

The involvement of the United States in World War I in April 1917 catalyzed the participation in the military effort of psychologists working in this area. At that time Yerkes was the president of the 25-year-old American Psychological Association (APA). The APA is a professional association in the United States for clinical and research psychologists. Today the APA performs a number of functions including holding conferences, accrediting university degree programs, and publishing scientific journals. Yerkes organized a group under the Surgeon General's Office (SGO) that developed methods for screening and selecting enlisted men. They developed the Army Alpha test to measure mental abilities. The Army Beta test was a non-verbal form of the test that was administered to illiterate and non-English-speaking draftees. Scott and Bingham organized a group under the Adjutant General's Office (AGO) with the goal to develop selection methods for officers. They created a catalogue of occupational needs for the Army, essentially a job-description system and a system of

performance ratings and occupational skill tests for officers (Katzell & Austin, 1992).

After the war, work on personnel selection continued. For example, Millicent Pond, who received a PhD from Yale University, worked at several businesses and was director of employment test research at Scoville Manufacturing Company. She researched the selection of factory workers, comparing the results of pre-employment tests with various indicators of job performance. These studies were published in a series of research articles in the *Journal of Personnel Research* in the late 1920s (Vinchur & Koppes, 2014).

From 1929 to 1932 Elton Mayo (1880–1949) and his colleagues began a series of studies at a plant near Chicago, Western Electric's Hawthorne Works. This long-term project took industrial psychology beyond just employee selection and placement to a study of more complex problems of interpersonal relations, motivation, and organizational dynamics. These studies mark the origin of organizational psychology. They began as research into the effects of the physical work environment (e.g., level of lighting in a factory), but the researchers found that the psychological and social factors in the factory were of more interest than the physical factors. These studies also examined how human interaction factors, such as supervisorial style, enhanced or decreased productivity.



Hawthorne Works provided the setting for several early I-O studies.

Analysis of the findings by later researchers led to the term the Hawthorne effect, which describes the increase in performance of individuals who are noticed, watched, and paid attention to by researchers or supervisors. What the original researchers found was that any change in a variable, such as lighting levels, led to an improvement in productivity; this was true even when the change was negative, such as a return to poor lighting. The effect faded when the attention faded (Roethlisberg & Dickson, 1939). The Hawthorne-effect concept endures today as an important experimental consideration in many fields and a factor that has to be controlled for in an experiment. In other words, an experimental treatment of some kind may produce an effect simply because it involves greater attention of the researchers on the participants (McCarney et al., 2007).

Researchers discovered that employees performed better when researchers or supervisors observed and interacted with them, a dynamic termed the Hawthorne effect.

Watch this video to hear first-hand accounts of the original Hawthorne studies from those who participated in the research: AT&T Archives: The Year They Discovered People.

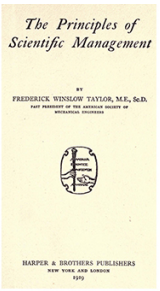
One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=184#oembed-1

In the 1930s, researchers began to study employees' feelings about their jobs. Kurt Lewin also conducted research on the effects of various leadership styles, team structure, and team dynamics (Katzell & Austin, 1992). Lewin is considered the founder of social psychology and much of his work and that of his students produced results that had important influences in organizational psychology. Lewin and his students' research included an important early study that used children to study the effect of leadership style on aggression, group dynamics, and satisfaction (Lewin, Lippitt, & White, 1939). Lewin was also responsible for coining the term *group dynamics*, and he was involved in studies of

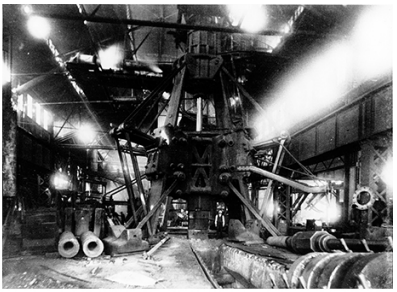
group interactions, cooperation, competition, and communication that bear on organizational psychology.

Parallel to these studies in industrial and organizational psychology, the field of human factors psychology was also developing. Frederick Taylor was an engineer who saw that if one could redesign the workplace there would be an increase in both output for the company and wages for the workers. In 1911 he put forward his theory in a book titled, *The Principles of Scientific Management*. His book examines management styles, personnel selection and training, as well as the work itself, using time and motion studies.

(a)



(b)



(c)

(a) Frederick Taylor (1911) strived to engineer workplaces to increase productivity, based on the ideas he set forth in (b) his book, The Principles of Scientific Management. (c) Taylor designed this steam hammer at the Midvale Steel Company. (credit c: modification of work by "Kheel Center, Cornell University"/Flickr)

One of the examples of Taylor's theory in action involved workers handling heavy iron ingots. Taylor showed that the workers could be more productive by taking work rests. This method of rest increased worker productivity from 12.5 to 47.0 tons moved per day with less reported fatigue as well as increased wages for the workers who were paid by the ton. At the same time, the company's cost was reduced from 9.2 cents to 3.9 cents per ton. Despite these increases in productivity, Taylor's theory received a great

deal of criticism at the time because it was believed that it would exploit workers and reduce the number of workers needed. Also controversial was the underlying concept that only a manager could determine the most efficient method of working, and that while at work, a worker was incapable of this. Taylor's theory was underpinned by the notion that a worker was fundamentally lazy and the goal of Taylor's scientific management approach was to maximize productivity without much concern for worker well-being. His approach was criticized by unions and those sympathetic to workers (Van De Water, 1997).

Gilbreth was another influential I-O psychologist who strove to find ways to increase productivity. Using time and motion studies, Gilbreth and her husband, Frank, worked to make workers more efficient by reducing the number of motions required to perform a task. She not only applied these methods to industry but also to the home, office, shops, and other areas. She investigated employee fatigue and time management stress and found many employees were motivated by money and job satisfaction. In 1914, Gilbreth wrote the book title, *The Psychology of Management: The Function of the Mind in Determining, Teaching, and Installing Methods of Least Waste*, and she is known as the mother of modern management. Some of Gilbreth's contributions are still in use today: you can thank her for the idea to put shelves inside on refrigerator doors, and she also came up with the concept of using a foot pedal to operate the lid of trash can (Gilbreth, 1914, 1998; Koppes, 1997; Lancaster, 2004). Gilbreth was the first woman to join the American Society of Mechanical Engineers in 1926, and in 1966 she was awarded the Hoover Medal of the American Society of Civil Engineers.

Taylor and Gilbreth's work improved productivity, but these innovations also improved the fit between technology and the human using it. The study of machine–human fit is known as ergonomics or human factors psychology.

(a)

(b)

(c)

(a) Lillian Gilbreth studied efficiency improvements that were applicable in the workplace, home, and other areas. She is credited with the idea of (b) putting shelves on the inside of refrigerator doors and (c) foot-pedal-operated garbage cans. (credit b: modification of work by "Goedeker's"/Flickr; credit c: modification of work by Kerry Ceszyk)

FROM WWII TO TODAY

World War II also drove the expansion of industrial psychology. Bingham was hired as the chief psychologist for the War Department (now the Department of Defense) and developed new systems for job selection, classification, training, ad performance review, plus methods for team development, morale change, and attitude change (Katzell & Austin, 1992). Other countries, such as Canada and the United Kingdom, likewise saw growth in I-O psychology during World War II (McMillan, Stevens, & Kelloway, 2009). In the years after the war, both industrial psychology and organizational psychology became areas of significant research effort. Concerns about the fairness of employment tests arose, and the ethnic and gender biases in various tests were evaluated with mixed results. In addition, a great deal of research went into studying job satisfaction and employee motivation (Katzell &

Austin, 1992). Today, I-O psychology is a diverse and deep field of research and practice, as you will learn about in the rest of this chapter. The Society for Industrial and Organizational Psychology (SIOP), a division of the APA, lists 8,000 members (SIOP, 2014) and the Bureau of Labor Statistics—U.S. Department of Labor (2013) has projected this profession will have the greatest growth of all job classifications in the 20 years following 2012. On average, a person with a master's degree in industrial-organizational psychology will earn over $80,000 a year, while someone with a doctorate will earn over $110,000 a year (Khanna, Medsker, & Ginter, 2012).

TEST YOUR UNDERSTANDING

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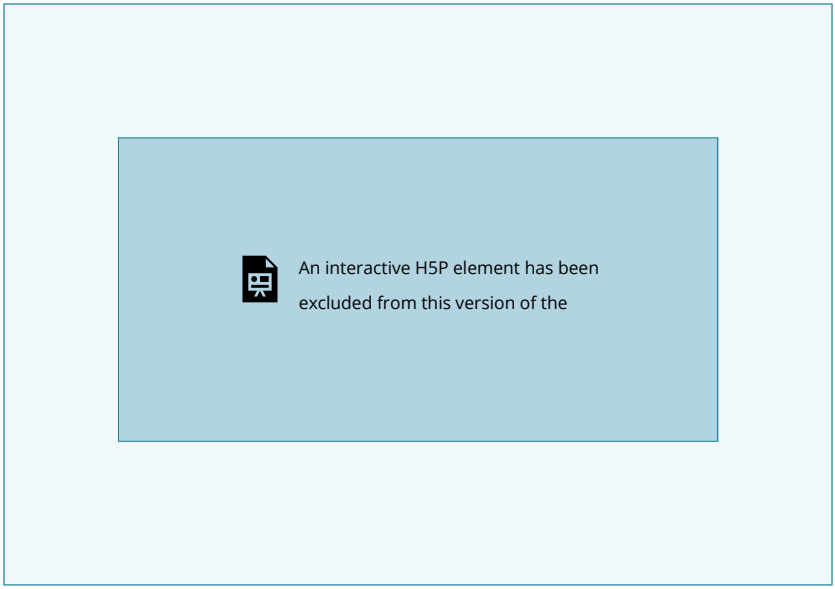
SUMMARY

The field of I-O psychology had its birth in industrial psychology and the use of psychological concepts to aid in personnel selection. However, with research such as the Hawthorne study, it was found that productivity was affected more by human interaction and not physical factors; the field of industrial psychology expanded to include organizational psychology. Both WWI and WWII had a strong influence on the development of an expansion of industrial psychology in the United States and elsewhere: The tasks the psychologists were assigned led to development of tests and research in how the psychological concepts could assist industry and other areas. This movement aided in expanding industrial psychology to include organizational psychology.

REVIEW QUESTIONS



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CRITICAL THINKING QUESTIONS

What societal and management attitudes might have caused organizational psychology to develop later than industrial psychology?

Answers will vary. The student should allude to the fact that organizational psychology finds social relations of employees important and that historically workers were thought of more as individual machines rather than a social group.

Many of the examples of I-O psychology are applications to businesses. Name four different non-business contexts that I-O psychology could impact?

Answers will vary. The student should identify organizations that are dedicated to accomplishing specific tasks, in the general sense. Examples are hospitals, non-profit organizations, government agencies (including the military), law enforcement, universities, schools, and so on.

PERSONAL APPLICATION QUESTION

Which of the broad areas of I-O psychology interests you the most and why?

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Explain the aspects of employee selection
- Describe the kinds of job training
- Describe the approaches to and issues surrounding performance assessment

The branch of I-O psychology known as industrial psychology focuses on identifying and matching persons to tasks within an organization. This involves job analysis, which means accurately describing the task or job. Then, organizations must identify the characteristics of applicants for a match to the job analysis. It also involves training employees from their first day on the job throughout their tenure within the organization, and appraising their performance along the way.

SELECTING EMPLOYEES

When you read job advertisements, do you ever wonder how the company comes up with the job description? Often, this is done with the help of I-O psychologists. There are two related but different approaches to job analysis—you may be familiar with the results of each as they often appear on the same job advertisement. The first approach is task-oriented and lists in detail the tasks that will be performed for the job. Each task is typically

rated on scales for how frequently it is performed, how difficult it is, and how important it is to the job. The second approach is worker-oriented. This approach describes the characteristics required of the worker to successfully perform the job. This second approach has been called job specification (Dierdorff & Wilson, 2003). For job specification, the knowledge, skills, and abilities (KSAs) that the job requires are identified.

Observation, surveys, and interviews are used to obtain the information required for both types of job analysis. It is possible to observe someone who is proficient in a position and analyze what skills are apparent. Another approach used is to interview people presently holding that position, their peers, and their supervisors to get a consensus of what they believe are the requirements of the job.

How accurate and reliable is a job analysis? Research suggests that it can depend on the nature of the descriptions and the source for the job analysis. For example, Dierdorff & Wilson (2003) found that job analyses developed from descriptions provided by people holding the job themselves were the least reliable; however, they did not study or speculate why this was the case.

The United States Department of Labor maintains a database of previously compiled job analyses for different jobs and occupations. This allows the I-O psychologist to access previous analyses for nearly any type of occupation. This system is called O*Net (accessible at www.online.onetcenter.org). The site is open and you can see the KSAs that are listed for your own position or one you might be curious about. Each occupation lists the tasks, knowledge, skills, abilities, work context, work activities, education requirements, interests, personality requirements, and work styles that are deemed necessary for success in that position. You can also see data on average earnings and projected job growth in that industry.

The O*Net database describes the skills, knowledge, and education required for occupations, as well as what personality types and work styles are best suited to the role. See what it has to say about being:

- a food server in a restaurant
- an elementary school teacher
- an industrial-organizational psychologist.

CANDIDATE ANALYSIS AND TESTING

Once a company identifies potential candidates for a position, the candidates' knowledge, skills, and other abilities must be evaluated and compared with the job description. These evaluations can involve testing, an interview, and work samples or exercises. You learned about personality tests in the chapter on personality; in the I-O context, they are used to identify the personality characteristics of the candidate in an effort to match those to personality characteristics that would ensure good performance on the job. For example, a high rating of agreeableness might be desirable in a customer support position. However, it is not always clear how best to correlate personality characteristics with predictions of job performance. It might be that too high of a score on agreeableness is actually a hindrance in the customer support position. For example, if a customer has a misperception about a product or service, agreeing with their misperception will not ultimately lead

to resolution of their complaint. Any use of personality tests should be accompanied by a verified assessment of what scores on the test correlate with good performance (Arthur, Woehr, & Graziano, 2001). Other types of tests that may be given to candidates include IQ tests, integrity tests, and physical tests, such as drug tests or physical fitness tests.

USING CUTOFF SCORES TO DETERMINE JOB SELECTION

Many positions require applicants to take tests as part of the selection process. These can include IQ tests, job-specific skills tests, or personality tests. The organization may set cutoff scores (i.e., a score below which a candidate will not move forward) for each test to determine whether the applicant moves on to the next stage. For example, there was a case of Robert Jordan, a 49-year-old college graduate who applied for a position with the police force in New London, Connecticut. As part of the selection process, Jordan took the Wonderlic Personnel Test (WPT), a test designed to measure cognitive ability.

Jordan did not make it to the interview stage because his WPT score of 33, equivalent to an IQ score of 125 (100 is the average IQ score), was too high. The New London Police department policy is to not interview anyone who has a WPT score over 27 (equivalent to an IQ score over 104) because they believe anyone who scores higher would be bored with police work. The average score for police officers nationwide is the equivalent of an IQ score of 104 (*Jordan v. New London*, 2000; ABC News, 2000).

Jordan sued the police department alleging that his rejection was discrimination and his civil rights were violated because he was denied equal protection under the law. The 2nd U.S. Circuit Court of Appeals upheld a lower court's decision that the city of New London did not discriminate against him because the same standards were applied to everyone who took the exam (The New York Times, 1999).

What do you think? When might universal cutoff points make sense in a hiring decision, and when might they eliminate otherwise potentially strong employees?

INTERVIEWS

Most jobs for mid-size to large-size businesses in the United States require a personal interview as a step in the selection process. Because interviews are commonly used, they have been the subject of considerable research by industrial psychologists. Information derived from job analysis usually forms the basis for the types of questions asked. Interviews can provide a more dynamic source of information about the candidate than standard testing measures. Importantly, social factors and body language can influence the outcome of the interview. These include influences, such as the degree of similarity of the applicant to the interviewer and nonverbal behaviors, such as hand gestures, head nodding, and smiling (Bye, Horverak, Sandal, Sam, & Vivjer, 2014; Rakić, Steffens, & Mummendey, 2011).

There are two types of interviews: unstructured and structured. In an unstructured interview, the interviewer may ask different questions of each different candidate. One candidate might be asked about her career goals, and another might be asked about his previous work experience. In an unstructured interview, the questions are often, though not always, unspecified beforehand. And in an unstructured interview the responses to questions asked are generally not scored using a standard system. In a structured interview, the interviewer asks the same questions of every candidate, the questions are prepared in advance, and the interviewer uses a standardized rating system for each response. With this approach, the interviewer can accurately compare two candidates' interviews. In a meta-analysis of studies examining the effectiveness of various types of job interviews, McDaniel, Whetzel, Schmidt & Maurer (1994) found that structured interviews were

more effective at predicting subsequent job performance of the job candidate.

PREPARING FOR THE JOB INTERVIEW

You might be wondering if psychology research can tell you how to succeed in a job interview. As you can imagine, most research is concerned with the employer's interest in choosing the most appropriate candidate for the job, a goal that makes sense for the candidate too. But suppose you are not the only qualified candidate for the job; is there a way to increase your chances of being hired? A limited amount of research has addressed this question.

As you might expect, nonverbal cues are important in an interview. Liden, Martin, & Parsons (1993) found that lack of eye contact and smiling on the part of the applicant led to lower applicant ratings. Studies of impression management on the part of an applicant have shown that self-promotion behaviors generally have a positive impact on interviewers (Gilmore & Ferris, 1989). Different personality types use different forms of impression management, for example extroverts use verbal self-promotion, and applicants high in agreeableness use non-verbal methods such as smiling and eye contact. Self-promotion was most consistently related with a positive outcome for the interview, particularly if it was related to the candidate's person–job fit. However, it is possible to overdo self-promotion with experienced interviewers (Howard & Ferris, 1996). Barrick, Swider & Stewart (2010) examined the effect of first impressions during the rapport building that typically occurs before an interview begins. They found that initial judgments by interviewers during this period were related to job offers and that the judgments were about the candidate's competence and not just likability. Levine and Feldman (2002) looked at the influence of several nonverbal behaviors in mock interviews on candidates' likability and projections of competence. Likability was affected

positively by greater smiling behavior. Interestingly, other behaviors affected likability differently depending on the gender of the applicant. Men who displayed higher eye contact were less likable; women were more likable when they made greater eye contact. However, for this study male applicants were interviewed by men and female applicants were interviewed by women. In a study carried out in a real setting, DeGroot & Gooty (2009) found that nonverbal cues affected interviewers' assessments about candidates. They looked at visual cues, which can often be modified by the candidate and vocal (nonverbal) cues, which are more difficult to modify. They found that interviewer judgment was positively affected by visual and vocal cues of conscientiousness, visual and vocal cues of openness to experience, and vocal cues of extroversion.

What is the take home message from the limited research that has been done? Learn to be aware of your behavior during an interview. You can do this by practicing and soliciting feedback from mock interviews. Pay attention to any nonverbal cues you are projecting and work at presenting nonverbal cures that project confidence and positive personality traits. And finally, pay attention to the first impression you are making as it may also have an impact in the interview.

TRAINING

Training is an important element of success and performance in many jobs. Most jobs begin with an orientation period during which the new employee is provided information regarding the company history, policies, and administrative protocols such as time tracking, benefits, and reporting requirements. An important goal of orientation training is to educate the new employee about the organizational culture, the values, visions, hierarchies, norms and ways the company's employees interact—essentially how the organization is run, how it operates, and how it makes decisions.

There will also be training that is specific to the job the individual was hired to do, or training during the individual's period of employment that teaches aspects of new duties, or how to use new physical or software tools. Much of these kinds of training will be formalized for the employee; for example, orientation training is often accomplished using software presentations, group presentations by members of the human resources department or with people in the new hire's department.

Training usually begins with an orientation period during which a new employee learns about company policies, practices, and culture. (credit: Cory Zanker)

Mentoring is a form of informal training in which an experienced employee guides the work of a new employee. In some situations, mentors will be formally assigned to a new employee, while in others a mentoring relationship may develop informally.

Mentoring effects on the mentor and the employee being mentored, the protégé, have been studied in recent years. In a review of mentoring studies, Eby, Allen, Evans, Ng, & DuBois (2008)

found significant but small effects of mentoring on performance (i.e., behavioral outcomes), motivation and satisfaction, and actual career outcomes. In a more detailed review, Allen, Eby, Poteet, Lentz, & Lima (2004) found that mentoring positively affected a protégé's compensation and number of promotions compared with non-mentored employees. In addition, protégés were more satisfied with their careers and had greater job satisfaction. All of the effects were small but significant. Eby, Durley, Evans, & Ragins (2006) examined mentoring effects on the mentor and found that mentoring was associated with greater job satisfaction and organizational commitment. Gentry, Weber, & Sadri (2008) found that mentoring was positively related with performance ratings by supervisors. Allen, Lentz, & Day (2006) found in a comparison of mentors and non-mentors that mentoring led to greater reported salaries and promotions.

Mentoring is recognized to be particularly important to the career success of women (McKeen & Bujaki, 2007) by creating connections to informal networks, adopting a style of interaction that male managers are comfortable with, and with overcoming discrimination in job promotions.

Gender combinations in mentoring relationships are also an area of active study. Ragins & Cotton (1999) studied the effects of gender on the outcomes of mentoring relationships and found that protégés with a history of male mentors had significantly higher compensation especially for male protégés. The study found that female mentor–male protégé relationships were considerably rarer than the other gender combinations.

In an examination of a large number of studies on the effectiveness of organizational training to meet its goals, Arthur, Bennett, Edens, and Bell (2003) found that training was, in fact, effective when measured by the immediate response of the employee to the training effort, evaluation of learning outcomes (e.g., a test at the end of the training), behavioral measurements of job activities by a supervisor, and results-based criteria (e.g.,

productivity or profits). The examined studies represented diverse forms of training including self-instruction, lecture and discussion, and computer assisted training.

EVALUATING EMPLOYEES

Industrial and organizational psychologists are typically involved in designing performance-appraisal systems for organizations. These systems are designed to evaluate whether each employee is performing her job satisfactorily. Industrial and organizational psychologists study, research, and implement ways to make work evaluations as fair and positive as possible; they also work to decrease the subjectivity involved with performance ratings. Fairly evaluated work helps employees do their jobs better, improves the likelihood of people being in the right jobs for their talents, maintains fairness, and identifies company and individual training needs.

Performance appraisals are typically documented several times a year, often with a formal process and an annual face-to-face brief meeting between an employee and his supervisor. It is important that the original job analysis play a role in performance appraisal as well as any goals that have been set by the employee or by the employee and supervisor. The meeting is often used for the supervisor to communicate specific concerns about the employee's performance and to positively reinforce elements of good performance. It may also be used to discuss specific performance rewards, such as a pay increase, or consequences of poor performance, such as a probationary period. Part of the function of performance appraisals for the organization is to document poor performance to bolster decisions to terminate an employee.

Performance appraisals are becoming more complex processes within organizations and are often used to motivate employees to improve performance and expand their areas of competence,

in addition to assessing their job performance. In this capacity, performance appraisals can be used to identify opportunities for training or whether a particular training program has been successful. One approach to performance appraisal is called 360-degree feedback appraisal. In this system, the employee's appraisal derives from a combination of ratings by supervisors, peers, employees supervised by the employee, and from the employee herself. Occasionally, outside observers may be used as well, such as customers. The purpose of 360-degree system is to give the employee (who may be a manager) and supervisor different perspectives of the employee's job performance; the system should help employees make improvements through their own efforts or through training. The system is also used in a traditional performance-appraisal context, providing the supervisor with more information with which to make decisions about the employee's position and compensation (Tornow, 1993a).

In a 360-degree performance appraisal, supervisors, customers, direct reports, peers, and the employee himself rate an employee's performance.

Few studies have assessed the effectiveness of 360-degree methods, but Atkins and Wood (2002) found that the self and peer ratings were unreliable as an assessment of an employee's performance and that even supervisors tended to underrate employees that gave themselves modest feedback ratings. However, a different perspective sees this variability in ratings as a positive in that it provides for greater learning on the part of the employees as they and their supervisor discuss the reasons for the discrepancies (Tornow, 1993b).

In theory, performance appraisals should be an asset for an organization wishing to achieve its goals, and most employees will actually solicit feedback regarding their jobs if it is not offered

(DeNisi & Kluger, 2000). However, in practice, many performance evaluations are disliked by organizations, employees, or both (Fletcher, 2001), and few of them have been adequately tested to see if they do in fact improve performance or motivate employees (DeNisi & Kluger, 2000). One of the reasons evaluations fail to accomplish their purpose in an organization is that performance appraisal systems are often used incorrectly or are of an inappropriate type for an organization's particular culture (Schraeder, Becton, & Portis, 2007). An organization's culture is how the organization is run, how it operates, and how it makes decisions. It is based on the collective values, hierarchies, and how individuals within the organization interact. Examining the effectiveness of performance appraisal systems in particular organizations and the effectiveness of training for the implementation of the performance appraisal system is an active area of research in industrial psychology (Fletcher, 2001).

BIAS AND PROTECTIONS IN HIRING

In an ideal hiring process, an organization would generate a job analysis that accurately reflects the requirements of the position, and it would accurately assess candidates' KSAs to determine who the best individual is to carry out the job's requirements. For many reasons, hiring decisions in the real world are often made based on factors other than matching a job analysis to KSAs. As mentioned earlier, interview rankings can be influenced by other factors: similarity to the interviewer (Bye, Horverak, Sandal, Sam, & Vijver, 2014) and the regional accent of the interviewee (Rakić, Steffens, & Mummendey 2011). A study by Agerström & Rooth (2011) examined hiring managers' decisions to invite equally qualified normal-weight and obese job applicants to an interview. The decisions of the hiring managers were based on photographs of the two applicants. The study found that hiring managers that

scored high on a test of negative associations with overweight people displayed a bias in favor of inviting the equally qualified normal-weight applicant but not inviting the obese applicant. The association test measures automatic or subconscious associations between an individual's negative or positive values and, in this case, the body-weight attribute. A meta-analysis of experimental studies found that physical attractiveness benefited individuals in various job-related outcomes such as hiring, promotion, and performance review (Hosoda, Stone-Romero, & Coats, 2003). They also found that the strength of the benefit appeared to be decreasing with time between the late 1970s and the late 1990s.

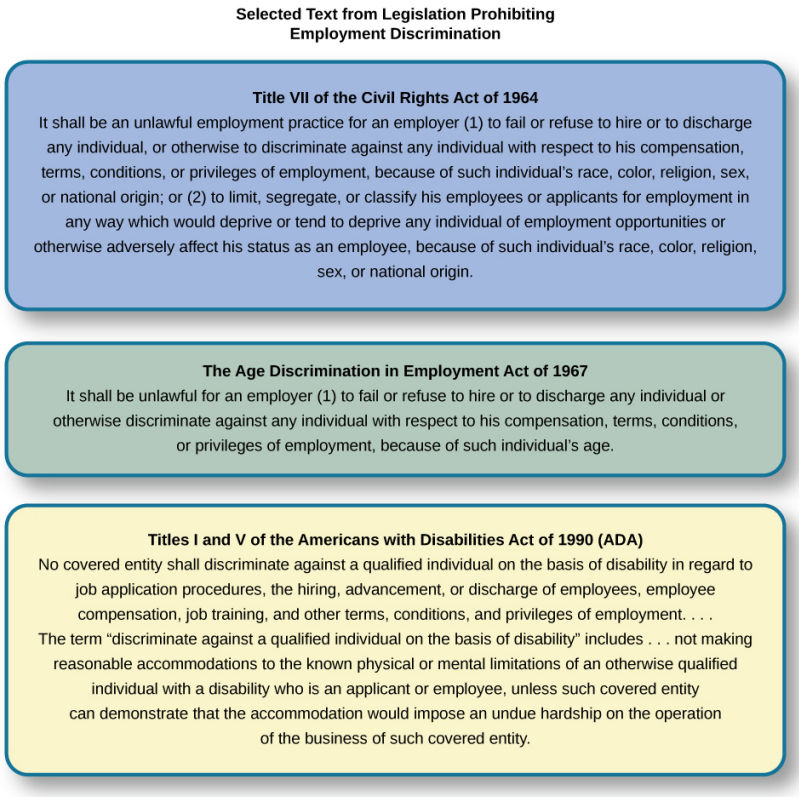
Some hiring criteria may be related to a particular group an applicant belongs to and not individual abilities. Unless membership in that group directly affects potential job performance, a decision based on group membership is discriminatory. To combat hiring discrimination, in the United States there are numerous city, state, and federal laws that prevent hiring based on various group-membership criteria. For example, did you know it is illegal for a potential employer to ask your age in an interview? Did you know that an employer cannot ask you whether you are married, a U.S. citizen, have disabilities, or what your race or religion is? They cannot even ask questions that might shed some light on these attributes, such as where you were born or who you live with. These are only a few of the restrictions that are in place to prevent discrimination in hiring. In the United States, federal anti-discrimination laws are administered by the U.S. Equal Employment Opportunity Commission (EEOC).

(a) (b) (c)

(a) Pregnancy, (b) religion, and (c) age are some of the criteria on which hiring decisions cannot legally be made. (credit a: modification of work by Sean McGrath; credit b: modification of work by Ze'ev Barkan; credit c: modification of work by David Hodgson)

THE U.S. EQUAL EMPLOYMENT OPPORTUNITY COMMISSION (EEOC)

The U.S. Equal Employment Opportunity Commission (EEOC) is responsible for enforcing federal laws that make it illegal to discriminate against a job applicant or an employee because of the person's race, color, religion, sex (including pregnancy), national origin, age (40 or older), disability, or genetic information.



The laws shown here protect employees in the U.S. from discriminatory practices.

The United States has several specific laws regarding fairness and avoidance of discrimination. The Equal Pay Act requires that equal pay for men and women in the same workplace who are performing equal work. Despite the law, persistent inequities in earnings between men and women exist. Corbett & Hill (2012) studied one facet of the gender gap by looking at earnings in the first year after college in the United States. Just comparing the earnings of women to men, women earn about 82 cents for every dollar a man earns in their first year out of college. However, some

of this difference can be explained by education, career, and life choices, such as choosing majors with lower earning potential or specific jobs within a field that have less responsibility. When these factors were corrected the study found an unexplained seven-cents-on-the-dollar gap in the first year after college that can be attributed to gender discrimination in pay. This approach to analysis of the gender pay gap, called the human capital model, has been criticized. Lips (2013) argues that the education, career, and life choices can, in fact, be constrained by necessities imposed by gender discrimination. This suggests that removing these factors entirely from the gender gap equation leads to an estimate of the size of the pay gap that is too small.

Title VII of the Civil Rights Act of 1964 makes it illegal to treat individuals unfavorably because of their race or color of their skin: An employer cannot discriminate based on skin color, hair texture, or other immutable characteristics, which are traits of an individual that are fundamental to her identity, in hiring, benefits, promotions, or termination of employees. The Pregnancy Discrimination Act of 1978 amends the Civil Rights Act; it prohibits job (e.g., employment, pay, and termination) discrimination of a woman because she is pregnant as long as she can perform the work required.

The Supreme Court ruling in *Griggs v. Duke Power Co.* made it illegal under Title VII of the Civil Rights Act to include educational requirements in a job description (e.g., high school diploma) that negatively impacts one race over another if the requirement cannot be shown to be directly related to job performance. The EEOC (2014) received more than 94,000 charges of various kinds of employment discrimination in 2013. Many of the filings are for multiple forms of discrimination and include charges of retaliation for making a claim, which itself is illegal. Only a small fraction of these claims become suits filed in a federal court, although the suits may represent the claims of more than one person. In 2013, there were 148 suits filed in federal courts.

In 2011, the U.S. Supreme Court decided a case in which women plaintiffs were attempting to group together in a class-action suit against Walmart for gender discrimination in promotion and pay. The case was important because it was the only practical way for individual women who felt they had been discriminated against to sustain a court battle for redress of their claims. The Court ultimately decided against the plaintiffs, and the right to a class-action suit was denied. However, the case itself effectively publicized the issue of gender discrimination in employment.

This video discusses the case history and issues: DUKES v. WAL-MART (short).

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This PBS NewsHour presents the arguments in the court case: Supreme Court Hears Wal-Mart Gender Bias Discrimination Case.

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Federal legislation does not protect employees in the private sector from discrimination related to sexual orientation and gender identity. These groups include lesbian, gay, bisexual, and transgender individuals. There is evidence of discrimination derived from surveys of workers, studies of complaint filings, wage comparison studies, and controlled job-interview studies (Badgett, Sears, Lau, & Ho, 2009). Federal legislation protects federal employees from such discrimination; the District of Columbia and 20 states have laws protecting public and private employees from discrimination for sexual orientation (American Civil Liberties Union, n.d). Most of the states with these laws also protect against discrimination based on gender identity. Gender identity, as

discussed when you learned about sexual behavior, refers to one's sense of being male or female.

Many cities and counties have adopted local legislation preventing discrimination based on sexual orientation or gender identity (Human Rights Campaign, 2013a), and some companies have recognized a benefit to explicitly stating that their hiring must not discriminate on these bases (Human Rights Campaign, 2013b).

AMERICANS WITH DISABILITIES ACT (ADA)

The Americans with Disabilities Act (ADA) of 1990 states people may not be discriminated against due to the nature of their disability. A disability is defined as a physical or mental impairment that limits one or more major life activities such as hearing, walking, and breathing. An employer must make reasonable accommodations for the performance of a disabled employee's job. This might include making the work facility handicapped accessible with ramps, providing readers for blind personnel, or allowing for more frequent breaks. The ADA has now been expanded to include individuals with alcoholism, former drug use, obesity, or psychiatric disabilities. The premise of the law is that disabled individuals can contribute to an organization and they cannot be discriminated against because of their disabilities (O'Keefe & Bruyere, 1994).

The Civil Rights Act and the Age Discrimination in Employment Act make provisions for bona fide occupational qualifications (BFOQs), which are requirements of certain occupations for which denying an individual employment would otherwise violate the law. For example, there may be cases in which religion, national origin, age, and sex are bona fide occupational qualifications. There are no BFOQ exceptions that apply to race, although the first amendment protects artistic expressions, such as films, in making race a requirement of a role. Clearcut examples of BFOQs would be hiring someone of a specific religion for a leadership position in a worship

facility, or for an executive position in religiously affiliated institutions, such as the president of a university with religious ties. Age has been determined to be a BFOQ for airline pilots; hence, there are mandatory retirement ages for safety reasons. Sex has been determined as a BFOQ for guards in male prisons.

Sex (gender) is the most common reason for invoking a BFOQ as a defense against accusing an employer of discrimination (Manley, 2009). Courts have established a three-part test for sex-related BFOQs that are often used in other types of legal cases for determining whether a BFOQ exists. The first of these is whether all or substantially all women would be unable to perform a job. This is the reason most physical limitations, such as "able to lift 30 pounds," fail as reasons to discriminate because most women are able to lift this weight. The second test is the "essence of the business" test, in which having to choose the other gender would undermine the essence of the business operation. This test was the reason the now defunct Pan American World Airways (i.e., Pan Am) was told it could not hire only female flight attendants. Hiring men would not have undermined the essense of this business. On a deeper level, this means that hiring cannot be made purely on customers' or others' preferences. The third and final test is whether the employer cannot make reasonable alternative accomodations, such as reassigning staff so that a woman does not have to work in a male-only part of a jail or other gender-specific facility. Privacy concerns are a major reason why discrimination based on gender is upheld by the courts, for example in situations such as hires for nursing or custodial staff (Manley, 2009). Most cases of BFOQs are decided on a case-by-case basis and these court decisions inform policy and future case decisions.

Hooters and BFOQ Laws Hooters restaurants only hire female wait staff. (credit: "BemLoira BemDavassa"/Flickr)

The restaurant chain Hooters, which hires only female wait staff and has them dress in a sexually provocative manner, is commonly cited as a discriminatory employer. The chain would argue that the female employees are an essential part of their business in that they market through sex appeal and the wait staff attract customers. Men have filed discrimination charges against Hooters in the past for not hiring them as wait staff simply because they are men. The chain has avoided a court decision on their hiring practices by settling out of court with the plaintiffs in each case. Do you think their practices violate the Civil Rights Act? See if you can apply the three court tests to this case and make a decision about whether a case that went to trial would find in favor of the plaintiff or the chain.

TEST YOUR UNDERSTANDING

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SUMMARY

Industrial psychology studies the attributes of jobs, applicants of those jobs, and methods for assessing fit to a job. These procedures include job analysis, applicant testing, and interviews. It also studies and puts into place procedures for the orientation of new employees and ongoing training of employees. The process of hiring employees can be vulnerable to bias, which is illegal, and industrial psychologists must develop methods for adhering to the law in hiring. Performance appraisal systems are an active area of research and practice in industrial psychology.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Construct a good interview question for a position of your choosing. The question should relate to a specific skill requirement for the position and you will

need to include the criteria for rating the applicants answer.

Answers will vary depending on the occupation and question. The question should relate to a specific skill for the job and the rating should relate to how the answer demonstrates the skill.

What might be useful mechanisms for avoiding bias during employment interviews?

Answers will vary, but they could include clear measurement standards for answer and applicant quality, ensuring diversity in interviewers or multiple interviews with different diverse interviewers, and clear education about the nature of bias for interviewers and those who make hiring decisions.

PERSONAL APPLICATION QUESTION

What are some of the KSAs (knowledge, skills, and abilities) that are required for your current position or a position you wish to have in the future?

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Define organizational psychology
- Explain the measurement and determinants of job satisfaction
- Describe key elements of management and leadership
- Explain the significance of organizational culture

Organizational psychology is the second major branch of study and practice within the discipline of industrial and organizational psychology. In organizational psychology, the focus is on social interactions and their effect on the individual and on the functioning of the organization. In this section, you will learn about the work organizational psychologists have done to understand job satisfaction, different styles of management, different styles of leadership, organizational culture, and teamwork.

JOB SATISFACTION

Some people love their jobs, some people tolerate their jobs, and some people cannot stand their jobs. Job satisfaction describes the degree to which individuals enjoy their job. It was described by Edwin Locke (1976) as the state of feeling resulting from appraising one's job experiences. While job satisfaction results from both how we think about our work (our cognition) and how we feel about

our work (our affect) (Saari & Judge, 2004), it is described in terms of affect. Job satisfaction is impacted by the work itself, our personality, and the culture we come from and live in (Saari & Judge, 2004).

Job satisfaction is typically measured after a change in an organization, such as a shift in the management model, to assess how the change affects employees. It may also be routinely measured by an organization to assess one of many factors expected to affect the organization's performance. In addition, polling companies like Gallup regularly measure job satisfaction on a national scale to gather broad information on the state of the economy and the workforce (Saad, 2012).

Job satisfaction is measured using questionnaires that employees complete. Sometimes a single question might be asked in a very straightforward way to which employees respond using a rating scale, such as a Likert scale, which was discussed in the chapter on personality. A Likert scale (typically) provides five possible answers to a statement or question that allows respondents to indicate their positive-to-negative strength of agreement or strength of feeling regarding the question or statement. Thus the possible responses to a question such as "How satisfied are you with your job today?" might be "Very satisfied," "Somewhat satisfied," "Neither satisfied, nor dissatisfied," "Somewhat dissatisfied," and "Very dissatisfied." More commonly the survey will ask a number of questions about the employee's satisfaction to determine more precisely why he is satisfied or dissatisfied. Sometimes these surveys are created for specific jobs; at other times, they are designed to apply to any job. Job satisfaction can be measured at a global level, meaning how satisfied in general the employee is with work, or at the level of specific factors intended to measure which aspects of the job lead to satisfaction.

Factors Involved in Job Satisfaction–Dissatisfaction

Factor	Description
Autonomy	Individual responsibility, control over decisions
Work content	Variety, challenge, role clarity
Communication	Feedback
Financial rewards	Salary and benefits
Growth and development	Personal growth, training, education
Promotion	Career advancement opportunity
Coworkers	Professional relations or adequacy
Supervision and feedback	Support, recognition, fairness
Workload	Time pressure, tedium
Work demands	Extra work requirements, insecurity of position

Research has suggested that the work-content factor, which includes variety, difficulty level, and role clarity of the job, is the most strongly predictive factor of overall job satisfaction (Saari & Judge, 2004). In contrast, there is only a weak correlation between pay level and job satisfaction (Judge, Piccolo, Podsakoff, Shaw, & Rich, 2010). Judge et al. (2010) suggest that individuals adjust or adapt to higher pay levels: Higher pay no longer provides the satisfaction the individual may have initially felt when her salary increased.

Why should we care about job satisfaction? Or more specifically, why should an employer care about job satisfaction? Measures of job satisfaction are somewhat correlated with job performance; in particular, they appear to relate to organizational citizenship or discretionary behaviors on the part of an employee that further the goals of the organization (Judge & Kammeyer-Mueller, 2012). Job satisfaction is related to general life satisfaction, although there has been limited research on how the two influence each other or whether personality and cultural factors affect both job and general life satisfaction. One carefully controlled study suggested that the relationship is reciprocal: Job satisfaction affects life

satisfaction positively, and vice versa (Judge & Watanabe, 1993). Of course, organizations cannot control life satisfaction's influence on job satisfaction. Job satisfaction, specifically low job satisfaction, is also related to withdrawal behaviors, such as leaving a job or absenteeism (Judge & Kammeyer-Mueller, 2012). The relationship with turnover itself, however, is weak (Judge & Kammeyer-Mueller, 2012). Finally, it appears that job satisfaction is related to organizational performance, which suggests that implementing organizational changes to improve employee job satisfaction will improve organizational performance (Judge & Kammeyer-Mueller, 2012).

There is opportunity for more research in the area of job satisfaction. For example, Weiss (2002) suggests that the concept of job satisfaction measurements have combined both emotional and cognitive concepts, and measurements would be more reliable and show better relationships with outcomes like performance if the measurement of job satisfaction separated these two possible elements of job satisfaction.

JOB SATISFACTION IN FEDERAL GOVERNMENT AGENCIES

A 2013 study of job satisfaction in the U.S. federal government found indexes of job satisfaction plummeting compared to the private sector. The largest factor in the decline was satisfaction with pay, followed by training and development opportunities. The Partnership for Public Service, a nonprofit, nonpartisan organization, has conducted research on federal employee job satisfaction since 2003. Its primary goal is to improve the federal government's management. However, the results also provide information to those interested in obtaining employment with the federal government.

Among large agencies, the highest job satisfaction ranking went to NASA, followed by the Department of Commerce and the

intelligence community. The lowest scores went to the Department of Homeland Security.

The data used to derive the job satisfaction score come from three questions on the Federal Employee Viewpoint Survey. The questions are:

I recommend my organization as a good place to work.
Considering everything, how satisfied are you with your job?
Considering everything, how satisfied are you with your organization?

The questions have a range of six possible answers, spanning a range of strong agreement or satisfaction to strong disagreement or dissatisfaction. How would you answer these questions with regard to your own job? Would these questions adequately assess your job satisfaction?

You can explore the Best Places To Work In The Federal Government study at their Web site: www.bestplacestowork.org. The Office of Personnel Management also produces a report based on their survey: www.fedview.opm.gov.

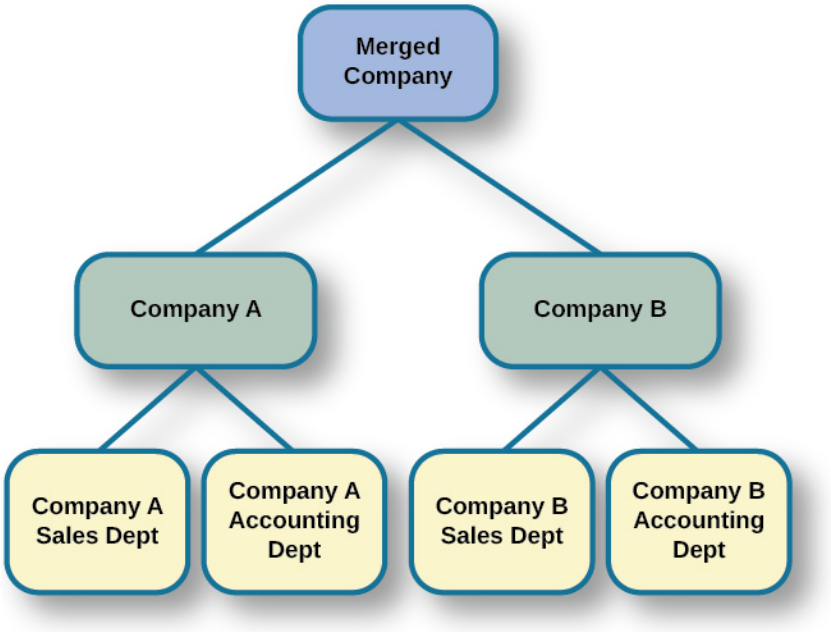
Job stress affects job satisfaction. Job stress, or job strain, is caused by specific stressors in an occupation. Stress can be an ambigious term as it is used in common language. Stress is the perception and response of an individual to events judged as ovewhelming or threatening to the individual's well-being (Gyllensten & Palmer, 2005). The events themselves are the stressors. Stress is a result of an employee's perception that the demands placed on them exceed their ability to meet them (Gyllensten & Palmer, 2005), such as having to fill multiple roles in a job or life in general, workplace role ambiguity, lack of career progress, lack of job security, lack of control over work outcomes, isolation, work overload, discrimination, harrassment, and bullying (Colligan & Higgins, 2005). The stressors are different for women than men and these differences are a significant area of research (Gyllensten & Palmer, 2005). Job stress leads to poor employee health, job performance, and family life (Colligan & Higgins, 2005).

As already mentioned, job insecurity contributes significantly to job stress. Two increasing threats to job security are downsizing events and corporate mergers. Businesses typically involve I-O psychologists in planning for, implementing, and managing these types of organizational change.

Downsizing is an increasingly common response to a business's pronounced failure to achieve profit goals, and it involves laying off a significant percentage of the company's employees. Industrial-organizational psychologists may be involved in all aspects of downsizing: how the news is delivered to employees (both those being let go and those staying), how laid-off employees are supported (e.g., separation packages), and how retained employees are supported. The latter is important for the organization because downsizing events affect the retained employee's intent to quit, organizational commitment, and job insecurity (Ugboro, 2006).

In addition to downsizing as a way of responding to outside strains on a business, corporations often grow larger by combining with other businesses. This can be accomplished through a merger (i.e., the joining of two organizations of equal power and status) or an acquisition (i.e., one organization purchases the other). In an acquisition, the purchasing organization is usually the more powerful or dominant partner. In both cases, there is usually a duplication of services between the two companies, such as two accounting departments and two sales forces. Both departments must be merged, which commonly involves a reduction of staff. This leads to organizational processes and stresses similar to those that occur in downsizing events. Mergers require determining how the organizational culture will change, to which employees also must adjust (van Knippenberg, van Knippenberg, Monden, & de Lima, 2002). There can be additional stress on workers as they lose their connection to the old organization and try to make connections with the new combined group (Amiot, Terry, Jimmieson, & Callan, 2006). Research in this area focuses on

understanding employee reactions and making practical recommendations for managing these organizational changes.



When companies are combined through a merger (or acquisition), there are often cuts due to duplication of core functions, like sales and accounting, at each company.

WORK-FAMILY BALANCE

Many people juggle the demands of work life with the demands of their home life, whether it be caring for children or taking care of an elderly parent; this is known as work-family balance. We might commonly think about work interfering with family, but it is also the case that family responsibilities may conflict with work obligations (Carlson, Kacmar, & Williams, 2000). Greenhaus and Beutell (1985) first identified three sources of work–family conflicts:

time devoted to work makes it difficult to fulfill requirements of family, or vice versa,
strain from participation in work makes it difficult to fulfill requirements of family, or vice versa, and
specific behaviors required by work make it difficult to fulfill the requirements of family, or vice versa.

Women often have greater responsibility for family demands, including home care, child care, and caring for aging parents, yet men in the United States are increasingly assuming a greater share of domestic responsibilities. However, research has documented that women report greater levels of stress from work–family conflict (Gyllensten & Palmer, 2005).

There are many ways to decrease work–family conflict and improve people's job satisfaction (Posig & Kickul, 2004). These include support in the home, which can take various forms: emotional (listening), practical (help with chores). Workplace support can include understanding supervisors, flextime, leave with pay, and telecommuting. Flextime usually involves a requirement of core hours spent in the workplace around which the employee may schedule his arrival and departure from work to meet family demands. Telecommuting involves employees working at home and setting their own hours, which allows them to work during different parts of the day, and to spend part of the day with their family. Recall that Yahoo! had a policy of allowing employees to telecommute and then rescinded the policy. There are also organizations that have onsite daycare centers, and some companies even have onsite fitness centers and health clinics. In a study of the effectiveness of different coping methods, Lapierre & Allen (2006) found practical support from home more important than emotional support. They also found that immediate-supervisor support for a worker significantly reduced work–family conflict through such mechanisms as allowing an employee the flexibility needed to fulfill family obligations. In contrast, flextime did not help with coping and telecommuting actually made things

worse, perhaps reflecting the fact that being at home intensifies the conflict between work and family because with the employee in the home, the demands of family are more evident.

Posig & Kickul (2004) identify exemplar corporations with policies designed to reduce work–family conflict. Examples include IBM's policy of three years of job-guaranteed leave after the birth of a child, Lucent Technologies offer of one year's childbirth leave at half pay, and SC Johnson's program of concierge services for daytime errands.

Glassdoor is a website that posts job satisfaction reviews for different careers and organizations. Use this site to research possible careers and/or organizations that interest you: Glassdoor.

MANAGEMENT AND ORGANIZATIONAL STRUCTURE

A significant portion of I-O research focuses on management and human relations. Douglas McGregor (1960) combined scientific management (a theory of management that analyzes and synthesizes workflows with the main objective of improving economic efficiency, especially labor productivity) and human relations into the notion of leadership behavior. His theory lays out two different styles called Theory X and Theory Y. In the Theory X approach to management, managers assume that most people

dislike work and are not innately self-directed. Theory X managers perceive employees as people who prefer to be led and told which tasks to perform and when. Their employees have to be watched carefully to be sure that they work hard enough to fulfill the organization's goals. Theory X workplaces will often have employees punch a clock when arriving and leaving the workplace: Tardiness is punished. Supervisors, not employees, determine whether an employee needs to stay late, and even this decision would require someone higher up in the command chain to approve the extra hours. Theory X supervisors will ignore employees' suggestions for improved efficiency and reprimand employees for speaking out of order. These supervisors blame efficiency failures on individual employees rather than the systems or policies in place. Managerial goals are achieved through a system of punishments and threats rather than enticements and rewards. Managers are suspicious of employees' motivations and always suspect selfish motivations for their behavior at work (e.g., being paid is their sole motivation for working).

In the Theory Y approach, on the other hand, managers assume that most people seek inner satisfaction and fulfillment from their work. Employees function better under leadership that allows them to participate in, and provide input about, setting their personal and work goals. In Theory Y workplaces, employees participate in decisions about prioritizing tasks; they may belong to teams that, once given a goal, decide themselves how it will be accomplished. In such a workplace, employees are able to provide input on matters of efficiency and safety. One example of Theroy Y in action is the policy of Toyota production lines that allows any employee to stop the entire line if a defect or other issue appears, so that the defect can be fixed and its cause remedied (Toyota Motor Manufacturing, 2013). A Theory Y workplace will also meaningfully consult employees on any changes to the work process or management system. In addition, the organization will encourage employees to contribute their own ideas. McGregor (1960)

characterized Theory X as the traditional method of management used in the United States. He agued that a Theory Y approach was needed to improve organizational output and the wellbeing of individuals.

Theory X and Theory Y Management Styles

Theory X	Theory Y
People dislike work and avoid it.	People enjoy work and find it natural.
People avoid responsibility.	People are more satisified when given responsibility.
People want to be told what to do.	People want to take part in setting their own work goals.
Goals are achieved through rules and punishments.	Goals are achieved through enticements and rewards.

Another management style was described by Donald Clifton, who focused his research on how an organization can best use an individual's strengths, an approach he called strengths-based management. He and his colleagues interviewed 8,000 managers and concluded that it is important to focus on a person's strengths, not their weaknesses. A strength is a particular enduring talent possessed by an individual that allows her to provide consistent, near-perfect performance in tasks involving that talent. Clifton argued that our strengths provide the greatest opportunity for growth (Buckingham & Clifton, 2001). An example of a strength is public speaking or the ability to plan a successful event. The strengths-based approach is very popular although its effect on organization performance is not well-studied. However, Kaiser & Overfield (2011) found that managers often neglected improving their weaknesses and overused their strengths, both of which interfered with performance.

Leadership is an important element of management. Leadership styles have been of major interest within I-O research, and researchers have proposed numerous theories of leadership. Bass (1985) popularized and developed the concepts of transactional

leadership versus transformational leadership styles. In transactional leadership, the focus is on supervision and organizational goals, which are achieved through a system of rewards and punishments (i.e., transactions). Transactional leaders maintain the status quo: They are managers. This is in contrast to the transformational leader. People who have transformational leadership possess four attributes to varying degrees: They are charismatic (highly liked role models), inspirational (optimistic about goal attainment), intellectually stimulating (encourage critical thinking and problem solving), and considerate (Bass, Avolio, & Atwater, 1996).

As women increasingly take on leadership roles in corporations, questions have arisen as to whether there are differences in leadership styles between men and women (Eagly, Johannesen-Schmidt, & van Engen, 2003). Eagly & Johnson (1990) conducted a meta-analysis to examine gender and leadership style. They found, to a slight but significant degree, that women tend to practice an interpersonal style of leadership (i.e., she focuses on the morale and welfare of the employees) and men practice a task-oriented style (i.e., he focuses on accomplishing tasks). However, the differences were less pronounced when one looked only at organizational studies and excluded laboratory experiments or surveys that did not involve actual organizational leaders. Larger sex-related differences were observed when leadership style was categorized as democratic or autocratic, and these differences were consistent across all types of studies. The authors suggest that similarities between the sexes in leadership styles are attributable to both sexes needing to conform the organization's culture; additionally, they propose that sex-related differences reflect inherent differences in the strengths each sex brings to bear on leadership practice. In another meta-analysis of leadership style, Eagly, Johannesen-Schmidt, & van Engen (2003) found that women tended to exhibit the characteristics of transformational leaders, while men were more likely to be transactional leaders. However,

the differences are not absolute; for example, women were found to use methods of reward for performance more often than men, which is a component of transactional leadership. The differences they found were relatively small. As Eagly, Johannesen-Schmidt, & van Engen (2003) point out, research shows that transformational leadership approaches are more effective than transactional approaches, although individual leaders typically exhibit elements of both approaches.

GOALS, TEAMWORK AND WORK TEAMS

The workplace today is rapidly changing due to a variety of factors, such as shifts in technology, economics, foreign competition, globalization, and workplace demographics. Organizations need to respond quickly to changes in these factors. Many companies are responding to these changes by structuring their organizations so that work can be delegated to work teams, which bring together diverse skills, experience, and expertise. This is in contrast to organizational structures that have individuals at their base (Naquin & Tynan, 2003). In the team-based approach, teams are brought together and given a specific task or goal to accomplish. Despite their burgeoning popularity, team structures do not always deliver greater productivity—the work of teams is an active area of research (Naquin & Tynan, 2003).

Why do some teams work well while others do not? There are many contributing factors. For example, teams can mask team members that are not working (i.e., social loafing). Teams can be inefficient due to poor communication; they can have poor decision-making skills due to conformity effects; and, they can have conflict within the group. The popularity of teams may in part result from the team halo effect: Teams are given credit for their successes. but individuals within a team are blamed for team failures (Naquin & Tynan, 2003). One aspect of team diversity is

their gender mix. Researchers have explored whether gender mix has an effect on team performance. On the one hand, diversity can introduce communication and interpersonal-relationship problems that hinder performance, but on the other hand diversity can also increase the team's skill set, which may include skills that can actually improve team member interactions. Hoogendoorn, Oosterbeek, & van Praag (2013) studied project teams in a university business school in which the gender mix of the teams was manipulated. They found that gender-balanced teams (i.e., nearly equal numbers of men and women) performed better, as measured by sales and profits, than predominantly male teams. The study did not have enough data to determine the relative performance of female dominated teams. The study was unsuccessful in identifying which mechanism (interpersonal relationships, learning, or skills mixes) accounted for performance improvement.

There are three basic types of teams: problem resolution teams, creative teams, and tactical teams. Problem resolution teams are created for the purpose of solving a particular problem or issue; for example, the diagnostic teams at the Centers for Disease Control. Creative teams are used to develop innovative possibilities or solutions; for example, design teams for car manufacturers create new vehicle models. Tactical teams are used to execute a well-defined plan or objective, such as a police or FBI SWAT team handling a hostage situation (Larson & LaFasto, 1989). One area of active research involves a fourth kind of team—the virtual team; these studies examine how groups of geographically disparate people brought together using digital communications technology function (Powell, Piccoli, & Ives, 2004). Virtual teams are more common due to the growing globalization of organizations and the use of consulting and partnerships facilitated by digital communication.

ORGANIZATION CULTURE

Each company and organization has an organizational culture. Organizational culture encompasses the values, visions, hierarchies, norms, and interactions among its employees. It is how an organization is run, how it operates, and how it makes decisions—the industry in which the organization participates may have an influence. Different departments within one company can develop their own subculture within the organization's culture. Ostroff, Kinicki, and Tamkins (2003) identify three layers in organizational culture: observable artifacts, espoused values, and basic assumptions. Observable artifacts are the symbols, language (jargon, slang, and humor), narratives (stories and legends), and practices (rituals) that represent the underlying cultural assumptions. Espoused values are concepts or beliefs that the management or the entire organization endorses. They are the rules that allow employees to know which actions they should take in different situations and which information they should adhere to. These basic assumptions generally are unobservable and unquestioned. Researchers have developed survey instruments to measure organizational culture.

With the workforce being a global marketplace, your company may have a supplier in Korea and another in Honduras and have employees in the United States, China, and South Africa. You may have coworkers of different religious, ethnic, or racial backgrounds than yourself. Your coworkers may be from different places around the globe. Many workplaces offer diversity training to help everyone involved bridge and understand cultural differences. Diversity training educates participants about cultural differences with the goal of improving teamwork. There is always the potential for prejudice between members of two groups, but the evidence suggests that simply working together, particularly if the conditions of work are set carefully that such prejudice can be reduced or eliminated. Pettigrew and Tropp (2006) conducted a meta-analysis

to examine the question of whether contact between groups reduced prejudice between those groups. They found that there was a moderate but significant effect. They also found that, as previously theorized, the effect was enhanced when the two groups met under conditions in which they have equal standing, common goals, cooperation between the groups, and especially support on the part of the institution or authorities for the contact.

Managing Generational Differences

An important consideration in managing employees is age. Workers' expectations and attitudes are developed in part by experience in particular cultural time periods. Generational constructs are somewhat arbitrary, yet they may be helpful in setting broad directions to organizational management as one generation leaves the workforce and another enters it. The baby boomer generation (born between 1946 and 1964) is in the process of leaving the workforce and will continue to depart it for a decade or more. Generation X (born between the early 1960s and the 1980s) are now in the middle of their careers. Millennials (born from 1979 to the early 1994) began to come of age at the turn of the century, and are early in their careers.

Today, as these three different generations work side by side

in the workplace, employers and managers need to be able to identify their unique characteristics. Each generation has distinctive expectations, habits, attitudes, and motivations (Elmore, 2010). One of the major differences among these generations is knowledge of the use of technology in the workplace. Millennials are technologically sophisticated and believe their use of technology sets them apart from other generations. They have also been characterized as self-centered and overly self-confident. Their attitudinal differences have raised concerns for managers about maintaining their motivation as employees and their ability to integrate into organizational culture created by baby boomers (Myers & Sadaghiani, 2010). For example, millennials may expect to hear that they need to pay their dues in their jobs from baby boomers who believe they paid their dues in their time. Yet millennials may resist doing so because they value life outside of work to a greater degree (Myers & Sadaghiani, 2010). Meister & Willyerd (2010) suggest alternative approaches to training and mentoring that will engage millennials and adapt to their need for feedback from supervisors: reverse mentoring, in which a younger employee educates a senior employee in social media or other digital resources. The senior employee then has the opportunity to provide useful guidance within a less demanding role.

Recruiting and retaining millennials and Generation X employees poses challenges that did not exist in previous generations. The concept of building a career with the company is not relatable to most Generation X employees, who do not expect to stay with one employer for their career. This expectation arises from of a reduced sense of loyalty because they do not expect their employer to be loyal to them (Gibson, Greenwood, & Murphy, 2009). Retaining Generation X workers thus relies on motivating them by making their work meaningful (Gibson, Greenwood, & Murphy, 2009). Since millennials lack an inherent loyalty to the company, retaining them also requires effort in the form of nurturing through frequent rewards, praise, and feedback.

Millennials are also interested in having many choices, including options in work scheduling, choice of job duties, and so on. They also expect more training and education from their employers. Companies that offer the best benefit package and brand attract millennials (Myers & Sadaghiani, 2010).

One well-recognized negative aspect of organizational culture is a culture of harassment, including sexual harassment. Most organizations of any size have developed sexual harassment policies that define sexual harassment (or harassment in general) and the procedures the organization has set in place to prevent and address it when it does occur. Thus, in most jobs you have held, you were probably made aware of the company's sexual harassment policy and procedures, and may have received training related to the policy. The U.S. Equal Employment Opportunity Commission (n.d.) provides the following description of sexual harassment:

> Unwelcome sexual advances, requests for sexual favors, and other verbal or physical conduct of a sexual nature constitute sexual harassment when this conduct explicitly or implicitly affects an individual's employment, unreasonably interferes with an individual's work performance, or creates an intimidating, hostile, or offensive work environment. (par. 2)

One form of sexual harassment is called quid pro quo. Quid pro quo means you give something to get something, and it refers to a situation in which organizational rewards are offered in exchange for sexual favors. Quid pro quo harassment is often between an employee and a person with greater power in the organization. For example, a supervisor might request an action, such as a kiss or a touch, in exchange for a promotion, a positive performance review, or a pay raise. Another form of sexual harassment is the threat of withholding a reward if a sexual request is refused. Hostile environment sexual harassment is another type of workplace harassment. In this situation, an employee experiences conditions in the workplace that are considered hostile or intimidating. For

example, a work environment that allows offensive language or jokes or displays sexually explicit images. Isolated occurrences of these events do not constitute harassment, but a pattern of repeated occurrences does. In addition to violating organizational policies against sexual harassment, these forms of harassment are illegal.

Harassment does not have to be sexual; it may be related to any of the protected classes in the statutes regulated by the EEOC: race, national origin, religion, or age.

VIOLENCE IN THE WORKPLACE

In the summer of August 1986, a part-time postal worker with a troubled work history walked into the Edmond, Oklahoma, post office and shot and killed 15 people, including himself. From his action, the term "going postal" was coined, describing a troubled employee who engages in extreme violence.

Workplace violence is one aspect of workplace safety that I-O psychologists study. Workplace violence is any act or threat of physical violence, harassment, intimidation, or other threatening, disruptive behavior that occurs at the workplace. It ranges from threats and verbal abuse to physical assaults and even homicide (Occupational Safety & Health Administration, 2014).

There are different targets of workplace violence: a person could commit violence against coworkers, supervisors, or property. Warning signs often precede such actions: intimidating behavior, threats, sabotaging equipment, or radical changes in a coworker's behavior. Often there is intimidation and then escalation that leads to even further escalation. It is important for employees to involve their immediate supervisor if they ever feel intimidated or unsafe.

Murder is the second leading cause of death in the workplace. It is also the primary cause of death for women in the workplace. Every year there are nearly two million workers who are physically

assaulted or threatened with assault. Many are murdered in domestic violence situations by boyfriends or husbands who chose the woman's workplace to commit their crimes.

There are many triggers for workplace violence. A significant trigger is the feeling of being treated unfairly, unjustly, or disrespectfully. In a research experiment, Greenberg (1993) examined the reactions of students who were given pay for a task. In one group, the students were given extensive explanations for the pay rate. In the second group, the students were given a curt uninformative explanation. The students were made to believe the supervisor would not know how much money the student withdrew for payment. The rate of stealing (taking more pay than they were told they deserved) was higher in the group who had been given the limited explanation. This is a demonstration of the importance of procedural justice in organizations. Procedural justice refers to the fairness of the processes by which outcomes are determined in conflicts with or among employees.

In another study by Greenberg & Barling (1999), they found a history of aggression and amount of alcohol consumed to be accurate predictors of workplace violence against a coworker. Aggression against a supervisor was predicted if a worker felt unfairly treated or untrusted. Job security and alcohol consumption predicted aggression against a subordinate. To understand and predict workplace violence, Greenberg & Barling (1999) emphasize the importance of considering the employee target of aggression or violence and characteristics of both the workplace characteristics and the aggressive or violent person.

TEST YOUR UNDERSTANDING

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=188#h5p-211

SUMMARY

Organizational psychology is concerned with the effects of interactions among people in the workplace on the employees themselves and on organizational productivity. Job satisfaction and its determinants and outcomes are a major focus of organizational psychology research and practice. Organizational psychologists have also studied the effects of management styles and leadership styles on productivity. In addition to the employees and management, organizational psychology also looks at the

organizational culture and how that might affect productivity. One aspect of organization culture is the prevention and addressing of sexual and other forms of harassment in the workplace. Sexual harassment includes language, behavior, or displays that create a hostile environment; it also includes sexual favors requested in exchange for workplace rewards (i.e., quid pro quo). Industrial-organizational psychology has conducted extensive research on the triggers and causes of workplace violence and safety. This enables the organization to establish procedures that can identify these triggers before they become a problem.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=188#h5p-212

CRITICAL THINKING QUESTIONS

If you designed an assessment of job satisfaction, what elements would it include?

Answers may vary, but they should include that the assessment would include more than one question to try to understand the reasons for the level of job satisfaction. It may also include questions that assess the importance of emotional and cognitive job satisfaction factors.

Downsizing has commonly shown to result in a period of lowered productivity for the organizations experiencing it. What might be some of the reasons for this observation?

Answers may vary, but they should include factors like lower job satisfaction, higher job stress, disruption of organizational culture, and other factors related to the concepts covered.

PERSONAL APPLICATION QUESTIONS

How would you handle the situation if you were being sexually harassed? What would you consider sexual harassment?

Answers may vary, but they should include telling the person that you are not comfortable with these actions and then reporting it to human resources. The definition of sexual harassment may discuss the sexual nature of the event, feelings of discomfort, fear, or anxiety, and recurrences of events.

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=188#h5p-213

Learning Objectives

By the end of this section, you will be able to:

- Describe the field of human factors psychology
- Explain the role of human factors psychology in safety, productivity, and job satisfaction

Human factors psychology (or ergonomics, a term that is favored in Europe) is the third subject area within industrial and organizational psychology. This field is concerned with the integration of the human-machine interface in the workplace, through design, and specifically with researching and designing machines that fit human requirements. The integration may be physical or cognitive, or a combination of both. Anyone who needs to be convinced that the field is necessary need only try to operate an unfamiliar television remote control or use a new piece of software for the first time. Whereas the two other areas of I-O psychology focus on the interface between the worker and team, group, or organization, human factors psychology focuses on the individual worker's interaction with a machine, work station, information displays, and the local environment, such as lighting. In the United States, human factors psychology has origins in both psychology and engineering; this is reflected in the early contributions of Lillian Gilbreth (psychologist and engineer) and her husband Frank Gilbreth (engineer).

Human factor professionals are involved in design from the beginning of a project, as is more common in software design

projects, or toward the end in testing and evaluation, as is more common in traditional industries (Howell, 2003). Another important role of human factor professionals is in the development of regulations and principles of best design. These regulations and principles are often related to work safety. For example, the Three Mile Island nuclear accident lead to Nuclear Regulatory Commission (NRC) requirements for additional instrumentation in nuclear facilities to provide operators with more critical information and increased operator training (United States Nuclear Regulatory Commission, 2013). The American National Standards Institute (ANSI, 2000), an independent developer of industrial standards, develops many standards related to ergonomic design, such as the design of control-center workstations that are used for transportation control or industrial process control.

Many of the concerns of human factors psychology are related to workplace safety. These concerns can be studied to help prevent work-related injuries of individual workers or those around them. Safety protocols may also be related to activities, such as commercial driving or flying, medical procedures, and law enforcement, that have the potential to impact the public.

One of the methods used to reduce accidents in the workplace is a checklist. The airline industry is one industry that uses checklists. Pilots are required to go through a detailed checklist of the different parts of the aircraft before takeoff to ensure that all essential equipment is working correctly. Astronauts also go through checklists before takeoff. The surgical safety checklist was developed by the World Health Organization (WHO) and serves as the basis for many checklists at medical facilities.

Sign in

Before induction of anesthesia, members of the team (at least the nurse and an anesthesia professional) orally confirm that:

- ☐ The patient has verified his or her identity, the surgical site and procedure, and consent. The surgical site is marked or site marking is not applicable. The pulse oximeter is on the patient and functioning
- ☐ All members of the team are aware of whether the patient has a known allergy
- ☐ The patient's airway and risk of aspiration have been evaluated and appropriate equipment and assistance are available
- ☐ If there is a risk of blood loss of at least 500 ml (or 7 ml/kg of body weight, in children), appropriate access and fluids are available

Time out

Before skin incision, the entire team (nurses, surgeons, anesthesia professionals, and any others participating in the care of the patient) orally:

- ☐ Confirms that all team members have been introduced by name and role. Confirms the patient's identity, surgical site, and procedure. Reviews the anticipated critical events:
  - ☐ Surgeon reviews critical and unexpected steps, operative duration, and anticipated blood loss
  - ☐ Anesthesia staff review concerns specific to the patient
  - ☐ Nursing staff review confirmation of sterility, equipment availability, and other concerns
- ☐ Confirms that prophylactic antibiotics have been administered = 60 min before incision is made or that antibiotics are not indicated
- ☐ Confirms that all essential imaging results for the correct patient are displayed in the operating room

Sign out

Before the patient leaves the operating room:

- ☐ Nurse reviews items aloud with the team:
  - ☐ Name of the procedure as recorded
  - ☐ That the needle, sponge, and instrument counts are complete (or not applicable)
  - ☐ That the specimen (if any) is correctly labeled, including with the patient's name
  - ☐ Whether there are any issues with equipment to be addressed
- ☐ The surgeon, nurse, and anesthesia professional review aloud the key concerns for the recovery and care of the patient

Checklists, such as the WHO surgical checklist shown here, help reduce workplace accidents.

Safety concerns also lead to limits to how long an operator, such as a pilot or truck driver, is allowed to operate the equipment. Recently the Federal Aviation Administration (FAA) introduced limits for how long a pilot is allowed to fly without an overnight break.

Howell (2003) outlines some important areas of research and practice in the field of human factors.

Areas of Study in Human Factors Psychology

Area	Description	I-O Questions
Attention	Includes vigilance and monitoring, recognizing signals in noise, mental resources, and divided attention	How is attention maintained? What about tasks maintains attention? How to design systems to support attention?
Cognitive engineering	Includes human software interactions in complex automated systems, especially the decision-making processes of workers as they are supported by the software system	How do workers use and obtain information provided by software?
Task analysis	Breaking down the elements of a task	How can a task be performed more efficiently? How can a task be performed more safely?
Cognitive task analysis	Breaking down the elements of a cognitive task	How are decisions made?

As an example of research in human factors psychology Bruno & Abrahão (2012) examined the impact of the volume of operator decisions on the accuracy of decisions made within an information security center at a banking institution in Brazil. The study examined a total of about 45,000 decisions made by 35 operators and 4 managers over a period of 60 days. Their study found that as the number of decisions made per day by the operators climbed, that is, as their cognitive effort increased, the operators made more mistakes in falsely identifying incidents as real security breaches (when, in reality, they were not). Interestingly, the opposite mistake of identifying real intrusions as false alarms did not increase with increased cognitive demand. This appears to be good news for the bank, since false alarms are not as costly as incorrectly rejecting a genuine threat. These kinds of studies combine research on attention, perception, teamwork, and human–computer interactions in a field of considerable societal and business significance. This is exactly the context of the events that led to the massive data breach for Target in the fall of 2013. Indications are that security personnel received signals of a security breach but did not interpret them correctly, thus allowing the breach to

continue for two weeks until an outside agency, the FBI, informed the company (Riley, Elgin, Lawrence, & Matlack, 2014).

TEST YOUR UNDERSTANDING

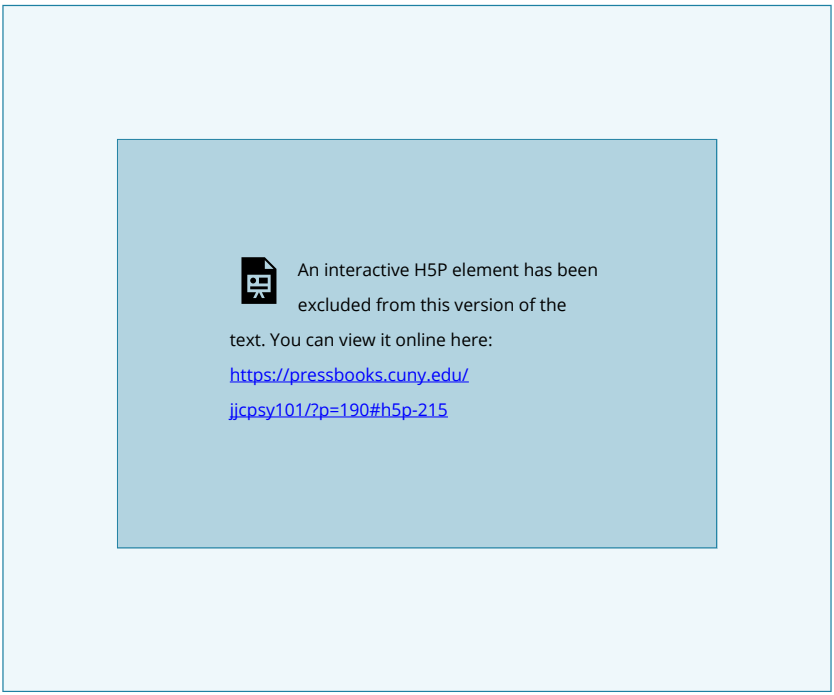
An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=190#h5p-214

SUMMARY

Human factors psychology, or ergonomics, studies the interface between workers and their machines and physical environments. Human factors psychologists specifically seek to design machines to better support the workers using them. Psychologists may be

involved in design of work tools such as software, displays, or machines from the beginning of the design process or during the testing an already developed product. Human factor psychologists are also involved in the development of best design recommendations and regulations. One important aspect of human factors psychology is enhancing worker safety. Human factors research involves efforts to understand and improve interactions between technology systems and their human operators. Human–software interactions are a large sector of this research.

REVIEW QUESTIONS



CRITICAL THINKING

What role could a flight simulator play in the design of a new aircraft?

Answers will vary, but they should include that the simulator would be used to determine how pilots interact with the controls and displays within the cockpit, including under conditions of simulated emergencies.

PERSONAL APPLICATION

Describe an example of a technology or team and technology interaction that you have had in the context of school or work that could have benefited from better design. What were the effects of the poor design? Make one suggestion for its improvement.

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

https://pressbooks.cuny.edu/jjcpsy101/?p=190#h5p-216

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STRESS, LIFESTYLE, AND HEALTH

Exams are a stressful, but unavoidable, element of college life. (credit "left": modification of work by Travis K. Mendoza; credit "center": modification of work by "albertogp123"/Flickr; credit "right": modification of work by Jeffrey Pioquinto, SJ)

Few would deny that today's college students are under a lot of pressure. In addition to many usual stresses and strains incidental to the college experience (e.g., exams, term papers, and the dreaded freshman 15), students today are faced with increased college tuitions, burdensome debt, and difficulty finding employment after graduation. A significant population of non-traditional college students may face additional stressors, such as raising children or holding down a full-time job while working toward a degree.

Of course, life is filled with many additional challenges beyond those incurred in college or the workplace. We might have concerns with financial security, difficulties with friends or neighbors, family responsibilities, and we may not have enough time to do the things we want to do. Even minor hassles—losing things, traffic jams, and loss of internet service—all involve pressure and demands that can make life seem like a struggle and that can compromise our sense of well-being. That is, all can be stressful in some way.

Scientific interest in stress, including how we adapt and cope, has been longstanding in psychology; indeed, after nearly a century of research on the topic, much has been learned and many insights have been developed. This chapter examines stress and highlights our current understanding of the phenomenon, including its psychological and physiological natures, its causes and consequences, and the steps we can take to master stress rather than become its victim.

Learning Objectives

By the end of this section, you will be able to:

- Differentiate between stimulus-based and response-based definitions of stress
- Define stress as a process
- Differentiate between good stress and bad stress
- Describe the early contributions of Walter Cannon and Hans Selye to the stress research field
- Understand the physiological basis of stress and describe the general adaptation syndrome

The term stress as it relates to the human condition first emerged in scientific literature in the 1930s, but it did not enter the popular vernacular until the 1970s (Lyon, 2012). Today, we often use the term loosely in describing a variety of unpleasant feeling states; for example, we often say we are stressed out when we feel frustrated, angry, conflicted, overwhelmed, or fatigued. Despite the widespread use of the term, stress is a fairly vague concept that is difficult to define with precision.

Researchers have had a difficult time agreeing on an acceptable definition of stress. Some have conceptualized stress as a demanding or threatening event or situation (e.g., a high-stress job, overcrowding, and long commutes to work). Such conceptualizations are known as stimulus-based definitions because they characterize stress as a stimulus that causes certain

reactions. Stimulus-based definitions of stress are problematic, however, because they fail to recognize that people differ in how they view and react to challenging life events and situations. For example, a conscientious student who has studied diligently all semester would likely experience less stress during final exams week than would a less responsible, unprepared student.

Others have conceptualized stress in ways that emphasize the physiological responses that occur when faced with demanding or threatening situations (e.g., increased arousal). These conceptualizations are referred to as response-based definitions because they describe stress as a response to environmental conditions. For example, the endocrinologist Hans Selye, a famous stress researcher, once defined stress as the "response of the body to any demand, whether it is caused by, or results in, pleasant or unpleasant conditions" (Selye, 1976, p. 74). Selye's definition of stress is response-based in that it conceptualizes stress chiefly in terms of the body's physiological reaction to any demand that is placed on it. Neither stimulus-based nor response-based definitions provide a complete definition of stress. Many of the physiological reactions that occur when faced with demanding situations (e.g., accelerated heart rate) can also occur in response to things that most people would not consider to be genuinely stressful, such as receiving unanticipated good news: an unexpected promotion or raise.

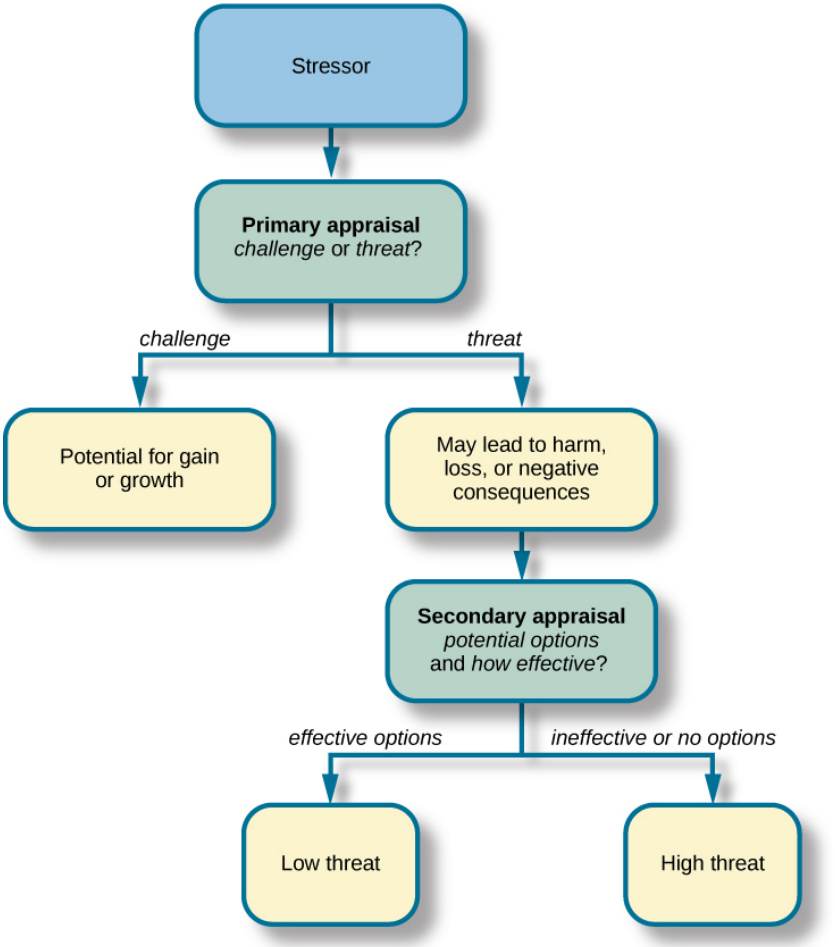
A useful way to conceptualize stress is to view it as a process whereby an individual perceives and responds to events that he appraises as overwhelming or threatening to his well-being (Lazarus & Folkman, 1984). A critical element of this definition is that it emphasizes the importance of how we appraise—that is, judge—demanding or threatening events (often referred to as stressors); these appraisals, in turn, influence our reactions to such events. Two kinds of appraisals of a stressor are especially important in this regard: primary and secondary appraisals. A primary appraisal involves judgment about the degree of potential

harm or threat to well-being that a stressor might entail. A stressor would likely be appraised as a threat if one anticipates that it could lead to some kind of harm, loss, or other negative consequence; conversely, a stressor would likely be appraised as a challenge if one believes that it carries the potential for gain or personal growth. For example, an employee who is promoted to a leadership position would likely perceive the promotion as a much greater threat if she believed the promotion would lead to excessive work demands than if she viewed it as an opportunity to gain new skills and grow professionally. Similarly, a college student on the cusp of graduation may face the change as a threat or a challenge.

Graduating from college and entering the workforce can be viewed as either a threat (loss of financial support) or a challenge (opportunity for independence and growth). (credit: Timothy Zanker)

The perception of a threat triggers a secondary appraisal: judgment of the options available to cope with a stressor, as well as perceptions of how effective such options will be (Lyon, 2012). As you may recall from what you learned about self-efficacy, an

individual's belief in his ability to complete a task is important (Bandura, 1994). A threat tends to be viewed as less catastrophic if one believes something can be done about it (Lazarus & Folkman, 1984). Imagine that two middle-aged women, Robin and Maria, perform breast self-examinations one morning and each woman notices a lump on the lower region of her left breast. Although both women view the breast lump as a potential threat (primary appraisal), their secondary appraisals differ considerably. In considering the breast lump, some of the thoughts racing through Robin's mind are, "Oh my God, I could have breast cancer! What if the cancer has spread to the rest of my body and I cannot recover? What if I have to go through chemotherapy? I've heard that experience is awful! What if I have to quit my job? My husband and I won't have enough money to pay the mortgage. Oh, this is just horrible…I can't deal with it!" On the other hand, Maria thinks, "Hmm, this may not be good. Although most times these things turn out to be benign, I need to have it checked out. If it turns out to be breast cancer, there are doctors who can take care of it because the medical technology today is quite advanced. I'll have a lot of different options, and I'll be just fine." Clearly, Robin and Maria have different outlooks on what might turn out to be a very serious situation: Robin seems to think that little could be done about it, whereas Maria believes that, worst case scenario, a number of options that are likely to be effective would be available. As such, Robin would clearly experience greater stress than would Maria.



When encountering a stressor, a person judges its potential threat (primary appraisal) and then determines if effective options are available to manage the situation. Stress is likely to result if a stressor is perceived as extremely threatening or threatening with few or no effective coping options available.

To be sure, some stressors are inherently more stressful than others in that they are more threatening and leave less potential for variation in cognitive appraisals (e.g., objective threats to one's

health or safety). Nevertheless, appraisal will still play a role in augmenting or diminishing our reactions to such events (Everly & Lating, 2002).

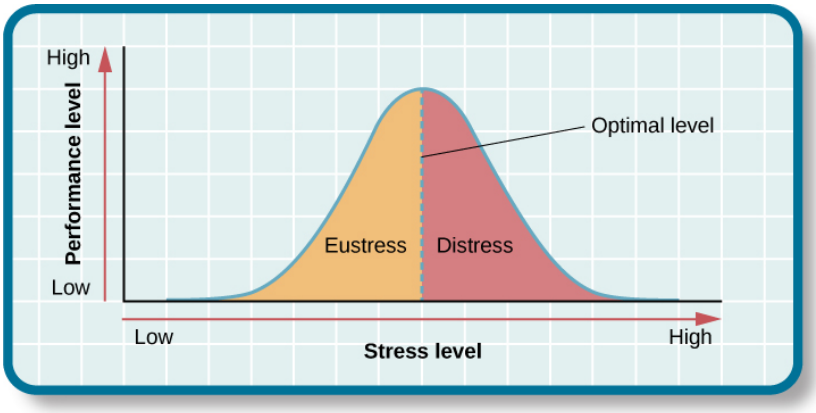
If a person appraises an event as harmful and believes that the demands imposed by the event exceed the available resources to manage or adapt to it, the person will subjectively experience a state of stress. In contrast, if one does not appraise the same event as harmful or threatening, she is unlikely to experience stress. According to this definition, environmental events trigger stress reactions by the way they are interpreted and the meanings they are assigned. In short, stress is largely in the eye of the beholder: it's not so much what happens to you as it is how you respond (Selye, 1976).

GOOD STRESS?

Although stress carries a negative connotation, at times it may be of some benefit. Stress can motivate us to do things in our best interests, such as study for exams, visit the doctor regularly, exercise, and perform to the best of our ability at work. Indeed, Selye (1974) pointed out that not all stress is harmful. He argued that stress can sometimes be a positive, motivating force that can improve the quality of our lives. This kind of stress, which Selye called eustress (from the Greek *eu* = "good"), is a good kind of stress associated with positive feelings, optimal health, and performance. A moderate amount of stress can be beneficial in challenging situations. For example, athletes may be motivated and energized by pregame stress, and students may experience similar beneficial stress before a major exam. Indeed, research shows that moderate stress can enhance both immediate and delayed recall of educational material. Male participants in one study who memorized a scientific text passage showed improved memory of the passage immediately after exposure to a mild stressor as well

as one day following exposure to the stressor (Hupbach & Fieman, 2012).

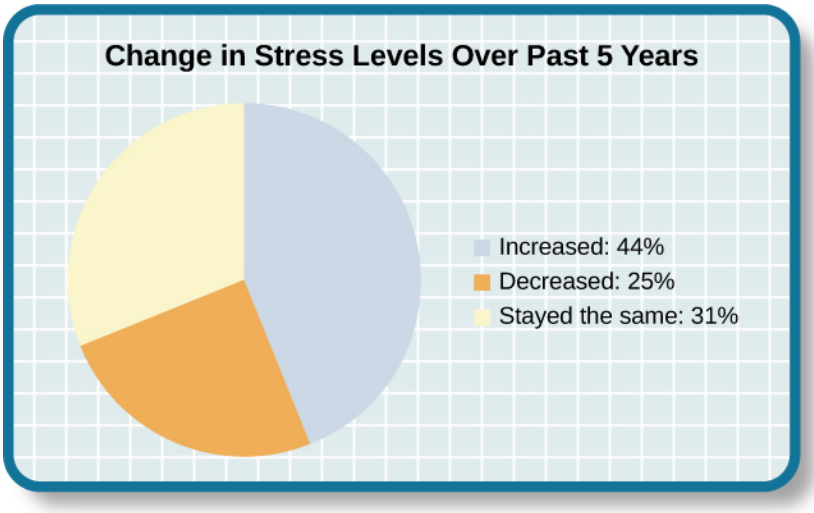
Increasing one's level of stress will cause performance to change in a predictable way. As stress increases, so do performance and general well-being (eustress); when stress levels reach an optimal level (the highest point of the curve), performance reaches its peak. A person at this stress level is colloquially at the top of his game, meaning he feels fully energized, focused, and can work with minimal effort and maximum efficiency. But when stress exceeds this optimal level, it is no longer a positive force—it becomes excessive and debilitating, or what Selye termed distress (from the Latin *dis* = "bad"). People who reach this level of stress feel burned out; they are fatigued, exhausted, and their performance begins to decline. If the stress remains excessive, health may begin to erode as well (Everly & Lating, 2002).



As the stress level increases from low to moderate, so does performance (eustress). At the optimal level (the peak of the curve), performance has reached its peak. If stress exceeds the optimal level, it will reach the distress region, where it will become excessive and debilitating, and performance will decline (Everly & Lating, 2002).

THE PREVALENCE OF STRESS

Stress is everywhere and, it has been on the rise over the last several years. Each of us is acquainted with stress—some are more familiar than others. In many ways, stress feels like a load you just can't carry—a feeling you experience when, for example, you have to drive somewhere in a crippling blizzard, when you wake up late the morning of an important job interview, when you run out of money before the next pay period, and before taking an important exam for which you realize you are not fully prepared.



Nearly half of U.S. adults indicated that their stress levels have increased over the last five years (Neelakantan, 2013).

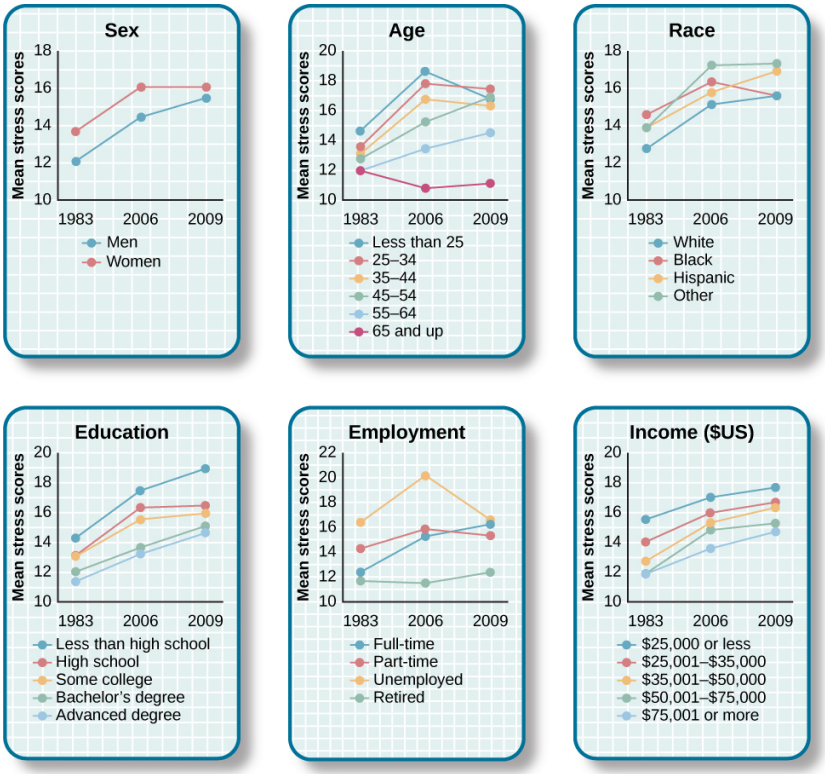
Stress is an experience that evokes a variety of responses, including those that are physiological (e.g., accelerated heart rate, headaches, or gastrointestinal problems), cognitive (e.g., difficulty concentrating or making decisions), and behavioral (e.g., drinking alcohol, smoking, or taking actions directed at eliminating the cause of the stress). Although stress can be positive at times, it can have

deleterious health implications, contributing to the onset and progression of a variety of physical illnesses and diseases (Cohen & Herbert, 1996).

The scientific study of how stress and other psychological factors impact health falls within the realm of health psychology, a subfield of psychology devoted to understanding the importance of psychological influences on health, illness, and how people respond when they become ill (Taylor, 1999). Health psychology emerged as a discipline in the 1970s, a time during which there was increasing awareness of the role behavioral and lifestyle factors play in the development of illnesses and diseases (Straub, 2007). In addition to studying the connection between stress and illness, health psychologists investigate issues such as why people make certain lifestyle choices (e.g., smoking or eating unhealthy food despite knowing the potential adverse health implications of such behaviors). Health psychologists also design and investigate the effectiveness of interventions aimed at changing unhealthy behaviors. Perhaps one of the more fundamental tasks of health psychologists is to identify which groups of people are especially at risk for negative health outcomes, based on psychological or behavioral factors. For example, measuring differences in stress levels among demographic groups and how these levels change over time can help identify populations who may have an increased risk for illness or disease.

The results of three national surveys in which several thousand individuals from different demographic groups completed a brief stress questionnaire; the surveys were administered in 1983, 2006, and 2009 (Cohen & Janicki-Deverts, 2012). All three surveys demonstrated higher stress in women than in men. Unemployed individuals reported high levels of stress in all three surveys, as did those with less education and income; retired persons reported the lowest stress levels. However, from 2006 to 2009 the greatest increase in stress levels occurred among men, Whites, people aged 45–64, college graduates, and those with full-time employment.

One interpretation of these findings is that concerns surrounding the 2008–2009 economic downturn (e.g., threat of or actual job loss and substantial loss of retirement savings) may have been especially stressful to White, college-educated, employed men with limited time remaining in their working careers.



The charts above, adapted from Cohen & Janicki-Deverts (2012), depict the mean stress level scores among different demographic groups during the years 1983, 2006, and 2009. Across categories of sex, age, race, education level, employment status, and income, stress levels generally show a marked increase over this quarter-century time span.

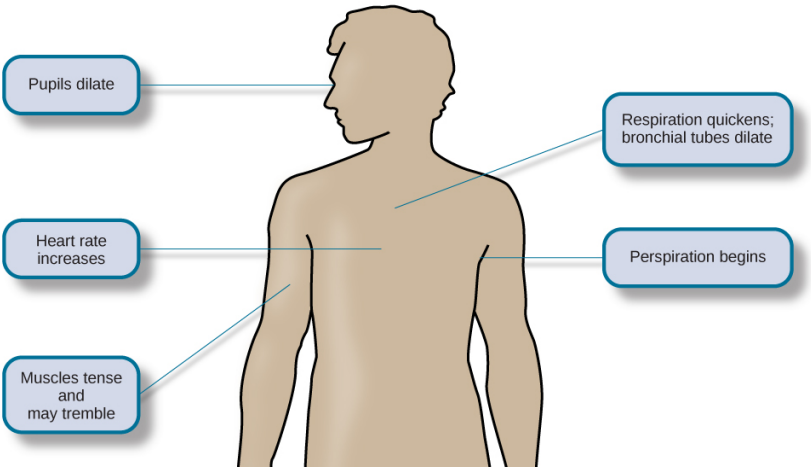
EARLY CONTRIBUTIONS TO THE STUDY OF STRESS

As previously stated, scientific interest in stress goes back nearly a century. One of the early pioneers in the study of stress was Walter Cannon, an eminent American physiologist at Harvard Medical School. In the early part of the 20th century, Cannon was the first to identify the body's physiological reactions to stress.

Harvard physiologist Walter Cannon first articulated and named the fight-or-flight response, the nervous system's sympathetic response to a significant stressor.

CANNON AND THE FIGHT-OR-FLIGHT RESPONSE

Imagine that you are hiking in the beautiful mountains of Colorado on a warm and sunny spring day. At one point during your hike, a large, frightening-looking black bear appears from behind a stand of trees and sits about 50 yards from you. The bear notices you, sits up, and begins to lumber in your direction. In addition to thinking, "This is definitely not good," a constellation of physiological reactions begins to take place inside you. Prompted by a deluge of epinephrine (adrenaline) and norepinephrine (noradrenaline) from your adrenal glands, your pupils begin to dilate. Your heart starts to pound and speeds up, you begin to breathe heavily and perspire, you get butterflies in your stomach, and your muscles become tense, preparing you to take some kind of direct action. Cannon proposed that this reaction, which he called the fight-or-flight response, occurs when a person experiences very strong emotions—especially those associated with a perceived threat (Cannon, 1932). During the fight-or-flight response, the body is rapidly aroused by activation of both the sympathetic nervous system and the endocrine system. This arousal helps prepare the person to either fight or flee from a perceived threat.



Fight or flight is a physiological response to a stressor.

According to Cannon, the fight-or-flight response is a built-in mechanism that assists in maintaining homeostasis—an internal environment in which physiological variables such as blood pressure, respiration, digestion, and temperature are stabilized at levels optimal for survival. Thus, Cannon viewed the fight-or-flight response as adaptive because it enables us to adjust internally and externally to changes in our surroundings, which is helpful in species survival.

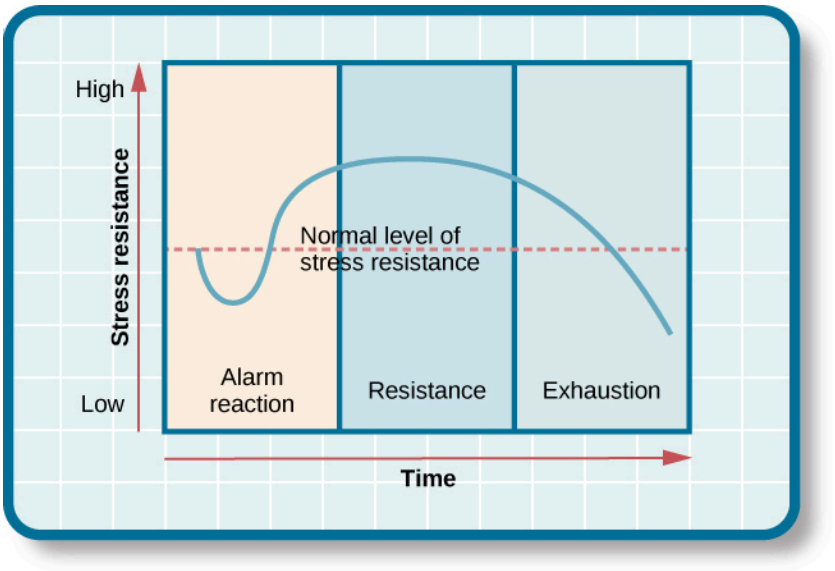
SELYE AND THE GENERAL ADAPTATION SYNDROME

Another important early contributor to the stress field was Hans Selye, mentioned earlier. He would eventually become one of the world's foremost experts in the study of stress. As a young assistant in the biochemistry department at McGill University in the 1930s, Selye was engaged in research involving sex hormones in rats. Although he was unable to find an answer for what he was initially researching, he incidentally discovered that when exposed to prolonged negative stimulation (stressors)—such as extreme cold,

surgical injury, excessive muscular exercise, and shock—the rats showed signs of adrenal enlargement, thymus and lymph node shrinkage, and stomach ulceration. Selye realized that these responses were triggered by a coordinated series of physiological reactions that unfold over time during continued exposure to a stressor. These physiological reactions were nonspecific, which means that regardless of the type of stressor, the same pattern of reactions would occur. What Selye discovered was the general adaptation syndrome, the body's nonspecific physiological response to stress.

Hans Selye specialized in research about stress. In 2009, his native Hungary honored his work with this stamp, released in conjunction with the 2nd annual World Conference on Stress.

The general adaptation syndrome, consists of three stages: (1) alarm reaction, (2) stage of resistance, and (3) stage of exhaustion (Selye, 1936; 1976). Alarm reaction describes the body's immediate reaction upon facing a threatening situation or emergency, and it is roughly analogous to the fight-or-flight response described by Cannon. During an alarm reaction, you are alerted to a stressor, and your body alarms you with a cascade of physiological reactions that provide you with the energy to manage the situation. A person who wakes up in the middle of the night to discover her house is on fire, for example, is experiencing an alarm reaction.



The three stages of Selye's general adaptation syndrome are shown in this graph. Prolonged stress ultimately results in exhaustion.

If exposure to a stressor is prolonged, the organism will enter the stage of resistance. During this stage, the initial shock of alarm reaction has worn off and the body has adapted to the stressor. Nevertheless, the body also remains on alert and is prepared to respond as it did during the alarm reaction, although with less intensity. For example, suppose a child who went missing is still missing 72 hours later. Although the parents would obviously remain extremely disturbed, the magnitude of physiological reactions would likely have diminished over the 72 intervening hours due to some adaptation to this event.

If exposure to a stressor continues over a longer period of time, the stage of exhaustion ensues. At this stage, the person is no longer able to adapt to the stressor: the body's ability to resist becomes depleted as physical wear takes its toll on the body's tissues and organs. As a result, illness, disease, and other permanent damage to the body—even death—may occur. If a

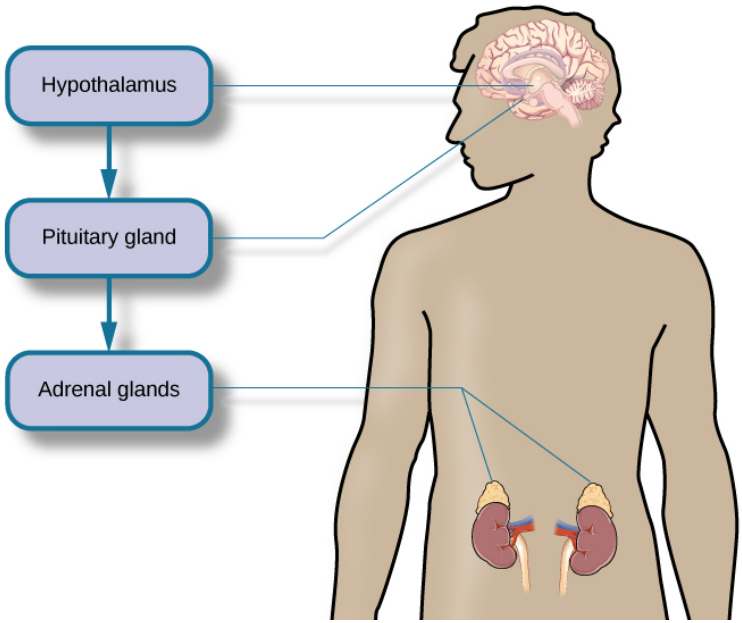
missing child still remained missing after three months, the long-term stress associated with this situation may cause a parent to literally faint with exhaustion at some point or even to develop a serious and irreversible illness.

In short, Selye's general adaptation syndrome suggests that stressors tax the body via a three-phase process—an initial jolt, subsequent readjustment, and a later depletion of all physical resources—that ultimately lays the groundwork for serious health problems and even death. It should be pointed out, however, that this model is a response-based conceptualization of stress, focusing exclusively on the body's physical responses while largely ignoring psychological factors such as appraisal and interpretation of threats. Nevertheless, Selye's model has had an enormous impact on the field of stress because it offers a general explanation for how stress can lead to physical damage and, thus, disease. As we shall discuss later, prolonged or repeated stress has been implicated in development of a number of disorders such as hypertension and coronary artery disease.

THE PHYSIOLOGICAL BASIS OF STRESS

What goes on inside our bodies when we experience stress? The physiological mechanisms of stress are extremely complex, but they generally involve the work of two systems—the sympathetic nervous system and the hypothalamic-pituitary-adrenal (HPA) axis. When a person first perceives something as stressful (Selye's alarm reaction), the sympathetic nervous system triggers arousal via the release of adrenaline from the adrenal glands. Release of these hormones activates the fight-or-flight responses to stress, such as accelerated heart rate and respiration. At the same time, the HPA axis, which is primarily endocrine in nature, becomes especially active, although it works much more slowly than the sympathetic nervous system. In response to stress, the hypothalamus (one of

the limbic structures in the brain) releases corticotrophin-releasing factor, a hormone that causes the pituitary gland to release adrenocorticotropic hormone (ACTH). The ACTH then activates the adrenal glands to secrete a number of hormones into the bloodstream; an important one is cortisol, which can affect virtually every organ within the body. Cortisol is commonly known as a stress hormone and helps provide that boost of energy when we first encounter a stressor, preparing us to run away or fight. However, sustained elevated levels of cortisol weaken the immune system.



This diagram shows the functioning of the hypothalamic-pituitary-adrenal (HPA) axis. The hypothalamus activates the pituitary gland, which in turn activates the adrenal glands, increasing their secretion of cortisol.

In short bursts, this process can have some favorable effects, such as providing extra energy, improving immune system functioning temporarily, and decreasing pain sensitivity. However, extended

release of cortisol—as would happen with prolonged or chronic stress—often comes at a high price. High levels of cortisol have been shown to produce a number of harmful effects. For example, increases in cortisol can significantly weaken our immune system (Glaser & Kiecolt-Glaser, 2005), and high levels are frequently observed among depressed individuals (Geoffroy, Hertzman, Li, & Power, 2013). In summary, a stressful event causes a variety of physiological reactions that activate the adrenal glands, which in turn release epinephrine, norepinephrine, and cortisol. These hormones affect a number of bodily processes in ways that prepare the stressed person to take direct action, but also in ways that may heighten the potential for illness.

When stress is extreme or chronic, it can have profoundly negative consequences. For example, stress often contributes to the development of certain psychological disorders, including post-traumatic stress disorder, major depressive disorder, and other serious psychiatric conditions. Additionally, we noted earlier that stress is linked to the development and progression of a variety of physical illnesses and diseases. For example, researchers in one study found that people injured during the September 11, 2001, World Trade Center disaster or who developed post-traumatic stress symptoms afterward later suffered significantly elevated rates of heart disease (Jordan, Miller-Archie, Cone, Morabia, & Stellman, 2011). Another investigation yielded that self-reported stress symptoms among aging and retired Finnish food industry workers were associated with morbidity 11 years later. This study also predicted the onset of musculoskeletal, nervous system, and endocrine and metabolic disorders (Salonen, Arola, Nygård, & Huhtala, 2008). Another study reported that male South Korean manufacturing employees who reported high levels of work-related stress were more likely to catch the common cold over the next several months than were those employees who reported lower work-related stress levels (Park et al., 2011). Later, you will explore

the mechanisms through which stress can produce physical illness and disease.

TEST YOUR UNDERSTANDING

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=195#h5p-217

SUMMARY

Stress is a process whereby an individual perceives and responds to events appraised as overwhelming or threatening to one's well-being. The scientific study of how stress and emotional factors impact health and well-being is called health psychology, a field

devoted to studying the general impact of psychological factors on health. The body's primary physiological response during stress, the fight-or-flight response, was first identified in the early 20th century by Walter Cannon. The fight-or-flight response involves the coordinated activity of both the sympathetic nervous system and the hypothalamic-pituitary-adrenal (HPA) axis. Hans Selye, a noted endocrinologist, referred to these physiological reactions to stress as part of general adaptation syndrome, which occurs in three stages: alarm reaction (fight-or-flight reactions begin), resistance (the body begins to adapt to continuing stress), and exhaustion (adaptive energy is depleted, and stress begins to take a physical toll).

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=195#h5p-218

CRITICAL THINKING QUESTIONS

Provide an example (other than the one described earlier) of a situation or event that could be appraised as either threatening or challenging.

Answers will vary. One example is divorce. People may perceive a divorce as a threat if they believe it will result in loneliness, change of lifestyle (due to loss of additional income), or humiliation in the eyes of their family. However, divorce may be perceived as a challenge if they view it as an opportunity to find somebody more compatible, and if they consider the process of finding a new partner a pleasant one, perhaps involving mystery and excitement.

Provide an example of a stressful situation that may cause a person to become seriously ill. How would Selye's general adaptation syndrome explain this occurrence?

Answers will vary. One example is when somebody's spouse dies or is unexpectedly diagnosed with a fatal disease. In both cases, the stress experienced by the surviving spouse would be intense, continuous, and—according the general

adaptation syndrome—would eventually increase vulnerability to illness or disease (exhaustion stage).

PERSONAL APPLICATION QUESTION

Think of a time in which you and others you know (family members, friends, and classmates) experienced an event that some viewed as threatening and others viewed as challenging. What were some of the differences in the reactions of those who experienced the event as threatening compared to those who viewed the event as challenging? Why do you think there were differences in how these individuals judged the same event?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

https://pressbooks.cuny.edu/jjcpsy101/?p=195#h5p-219

Learning Objectives

By the end of this section, you will be able to:

- Describe different types of possible stressors
- Explain the importance of life changes as potential stressors
- Describe the Social Readjustment Rating Scale
- Understand the concepts of job strain and job burnout

For an individual to experience stress, he must first encounter a potential stressor. In general, stressors can be placed into one of two broad categories: chronic and acute. Chronic stressors include events that persist over an extended period of time, such as caring for a parent with dementia, long-term unemployment, or imprisonment. Acute stressors involve brief focal events that sometimes continue to be experienced as overwhelming well after the event has ended, such as falling on an icy sidewalk and breaking your leg (Cohen, Janicki-Deverts, & Miller, 2007). Whether chronic or acute, potential stressors come in many shapes and sizes. They can include major traumatic events, significant life changes, daily hassles, as well as other situations in which a person is regularly exposed to threat, challenge, or danger.

TRAUMATIC EVENTS

Some stressors involve traumatic events or situations in which a

person is exposed to actual or threatened death or serious injury. Stressors in this category include exposure to military combat, threatened or actual physical assaults (e.g., physical attacks, sexual assault, robbery, childhood abuse), terrorist attacks, natural disasters (e.g., earthquakes, floods, hurricanes), and automobile accidents. Men, non-Whites, and individuals in lower socioeconomic status (SES) groups report experiencing a greater number of traumatic events than do women, Whites, and individuals in higher SES groups (Hatch & Dohrenwend, 2007). Some individuals who are exposed to stressors of extreme magnitude develop post-traumatic stress disorder (PTSD): a chronic stress reaction characterized by experiences and behaviors that may include intrusive and painful memories of the stressor event, jumpiness, persistent negative emotional states, detachment from others, angry outbursts, and avoidance of reminders of the event (American Psychiatric Association [APA], 2013).

LIFE CHANGES

Most stressors that we encounter are not nearly as intense as the ones described above. Many potential stressors we face involve events or situations that require us to make changes in our ongoing lives and require time as we adjust to those changes. Examples include death of a close family member, marriage, divorce, and moving.

Some fairly typical life events, such as moving, can be significant stressors. Even when the move is intentional and positive, the amount of resulting change in daily life can cause stress. (credit: "Jellaluna"/Flickr)

In the 1960s, psychiatrists Thomas Holmes and Richard Rahe wanted to examine the link between life stressors and physical illness, based on the hypothesis that life events requiring significant changes in a person's normal life routines are stressful, whether these events are desirable or undesirable. They developed the Social Readjustment Rating Scale (SRRS), consisting of 43 life events that require varying degrees of personal readjustment (Holmes & Rahe, 1967). Many life events that most people would consider pleasant (e.g., holidays, retirement, marriage) are among those listed on the SRRS; these are examples of eustress. Holmes and Rahe also proposed that life events can add up over time, and that experiencing a cluster of stressful events increases one's risk of developing physical illnesses.

In developing their scale, Holmes and Rahe asked 394 participants to provide a numerical estimate for each of the 43 items; each estimate corresponded to how much readjustment

participants felt each event would require. These estimates resulted in mean value scores for each event—often called life change units (LCUs) (Rahe, McKeen, & Arthur, 1967). The numerical scores ranged from 11 to 100, representing the perceived magnitude of life change each event entails. Death of a spouse ranked highest on the scale with 100 LCUs, and divorce ranked second highest with 73 LCUs. In addition, personal injury or illness, marriage, and job termination also ranked highly on the scale with 53, 50, and 47 LCUs, respectively. Conversely, change in residence (20 LCUs), change in eating habits (15 LCUs), and vacation (13 LCUs) ranked low on the scale. Minor violations of the law ranked the lowest with 11 LCUs. To complete the scale, participants checked yes for events experienced within the last 12 months. LCUs for each checked item are totaled for a score quantifying the amount of life change. Agreement on the amount of adjustment required by the various life events on the SRRS is highly consistent, even cross-culturally (Holmes & Masuda, 1974).

Some Stressors on the Social Readjustment Rating Scale (Holmes & Rahe, 1967)

Life event	Life change units
Death of a close family member	63
Personal injury or illness	53
Dismissal from work	47
Change in financial state	38
Change to different line of work	36
Outstanding personal achievement	28
Beginning or ending school	26
Change in living conditions	25
Change in working hours or conditions	20
Change in residence	20
Change in schools	20
Change in social activities	18
Change in sleeping habits	16
Change in eating habits	15
Minor violation of the law	11

Extensive research has demonstrated that accumulating a high number of life change units within a brief period of time (one or two years) is related to a wide range of physical illnesses (even accidents and athletic injuries) and mental health problems (Monat & Lazarus, 1991; Scully, Tosi, & Banning, 2000). In an early demonstration, researchers obtained LCU scores for U.S. and Norwegian Navy personnel who were about to embark on a six-month voyage. A later examination of medical records revealed positive (but small) correlations between LCU scores prior to the voyage and subsequent illness symptoms during the ensuing six-month journey (Rahe, 1974). In addition, people tend to experience more physical symptoms, such as backache, upset stomach, diarrhea, and acne, on specific days in which self-reported LCU

values are considerably higher than normal, such as the day of a family member's wedding (Holmes & Holmes, 1970).

The Social Readjustment Rating Scale (SRRS) provides researchers a simple, easy-to-administer way of assessing the amount of stress in people's lives, and it has been used in hundreds of studies (Thoits, 2010). Despite its widespread use, the scale has been subject to criticism. First, many of the items on the SRRS are vague; for example, death of a close friend could involve the death of a long-absent childhood friend that requires little social readjustment (Dohrenwend, 2006). In addition, some have challenged its assumption that undesirable life events are no more stressful than desirable ones (Derogatis & Coons, 1993). However, most of the available evidence suggests that, at least as far as mental health is concerned, undesirable or negative events are more strongly associated with poor outcomes (such as depression) than are desirable, positive events (Hatch & Dohrenwend, 2007). Perhaps the most serious criticism is that the scale does not take into consideration respondents' appraisals of the life events it contains. As you recall, appraisal of a stressor is a key element in the conceptualization and overall experience of stress. Being fired from work may be devastating to some but a welcome opportunity to obtain a better job for others. The SRRS remains one of the most well-known instruments in the study of stress, and it is a useful tool for identifying potential stress-related health outcomes (Scully et al., 2000).

Go to this site to complete the SRRS scale and determine the total number of LCUs you have

experienced over the last year: The Holmes and Rahe Stress Scale.

Correlational Research

The Holmes and Rahe Social Readjustment Rating Scale (SRRS) uses the correlational research method to identify the connection between stress and health. That is, respondents' LCU scores are correlated with the number or frequency of self-reported symptoms indicating health problems. These correlations are typically positive—as LCU scores increase, the number of symptoms increase. Consider all the thousands of studies that have used this scale to correlate stress and illness symptoms: If you were to assign an average correlation coefficient to this body of research, what would be your best guess? How strong do you think the correlation coefficient would be? Why can't the SRRS show a causal relationship between stress and illness? If it were possible to

show causation, do you think stress causes illness or illness causes stress?

HASSLES

Potential stressors do not always involve major life events. Daily hassles—the minor irritations and annoyances that are part of our everyday lives (e.g., rush hour traffic, lost keys, obnoxious coworkers, inclement weather, arguments with friends or family)—can build on one another and leave us just as stressed as life change events (Kanner, Coyne, Schaefer, & Lazarus, 1981).

(a)

(b)

Daily commutes, whether (a) on the road or (b) via public transportation, can be hassles that contribute to our feelings of everyday stress. (credit a: modification of work by Jeff Turner; credit b: modification of work by "epSos.de"/Flickr)

Researchers have demonstrated that the frequency of daily hassles is actually a better predictor of both physical and psychological health than are life change units. In a well-known study of San Francisco residents, the frequency of daily hassles was found to be more strongly associated with physical health problems than were life change events (DeLongis, Coyne, Dakof, Folkman, & Lazarus, 1982). In addition, daily minor hassles, especially interpersonal conflicts, often lead to negative and distressed mood states (Bolger, DeLongis, Kessler, & Schilling, 1989). Cyber hassles that occur on

social media may represent a new source of stress. In one investigation, undergraduates who, over a 10-week period, reported greater Facebook-induced stress (e.g., guilt or discomfort over rejecting friend requests and anger or sadness over being unfriended by another) experienced increased rates of upper respiratory infections, especially if they had larger social networks (Campisi et al., 2012). Clearly, daily hassles can add up and take a toll on us both emotionally and physically.

OTHER STRESSORS

Stressors can include situations in which one is frequently exposed to challenging and unpleasant events, such as difficult, demanding, or unsafe working conditions. Although most jobs and occupations can at times be demanding, some are clearly more stressful than others. For example, most people would likely agree that a firefighter's work is inherently more stressful than that of a florist. Equally likely, most would agree that jobs containing various unpleasant elements, such as those requiring exposure to loud noise (heavy equipment operator), constant harassment and threats of physical violence (prison guard), perpetual frustration (bus driver in a major city), or those mandating that an employee work alternating day and night shifts (hotel desk clerk), are much more demanding—and thus, more stressful—than those that do not contain such elements.

(a)

(b)

(a) Police officers and (b) firefighters hold high stress occupations. (credit a: modification of work by Australian Civil-Military Centre; credit b: modification of work by Andrew Magill)

Occupations and Their Related Stressors

Occupation	Stressors Specific to Occupation (Sulsky & Smith, 2005)
Police officer	physical dangers, excessive paperwork, red tape, dealing with court system, coworker and supervisor conflict, lack of support from the public
Firefighter	uncertainty over whether a serious fire or hazard awaits after an alarm
Social worker	little positive feedback from jobs or from the public, unsafe work environments, frustration in dealing with bureaucracy, excessive paperwork, sense of personal responsibility for clients, work overload
Teacher	Excessive paperwork, lack of adequate supplies or facilities, work overload, lack of positive feedback, vandalism, threat of physical violence
Nurse	Work overload, heavy physical work, patient concerns (dealing with death and medical concerns), interpersonal problems with other medical staff (especially physicians)
Emergency medical worker	Unpredictable and extreme nature of the job, inexperience
Air traffic controller	Little control over potential crisis situations and workload, fear of causing an accident, peak traffic situations, general work environment
Clerical and secretarial work	Little control over job mobility, unsupportive supervisors, work overload, lack of perceived control
Managerial work	Work overload, conflict and ambiguity in defining the managerial role, difficult work relationships

Although the specific stressors for these occupations are diverse,

they seem to share two common denominators: heavy workload and uncertainty about and lack of control over certain aspects of a job. Both of these factors contribute to job strain, a work situation that combines excessive job demands and workload with little discretion in decision making or job control (Karasek & Theorell, 1990). Clearly, many occupations other than the ones listed involve at least a moderate amount of job strain in that they often involve heavy workloads and little job control (e.g., inability to decide when to take breaks). Such jobs are often low-status and include those of factory workers, postal clerks, supermarket cashiers, taxi drivers, and short-order cooks. Job strain can have adverse consequences on both physical and mental health; it has been shown to be associated with increased risk of hypertension (Schnall & Landsbergis, 1994), heart attacks (Theorell et al., 1998), recurrence of heart disease after a first heart attack (Aboa-Éboulé et al., 2007), significant weight loss or gain (Kivimäki et al., 2006), and major depressive disorder (Stansfeld, Shipley, Head, & Fuhrer, 2012). A longitudinal study of over 10,000 British civil servants reported that workers under 50 years old who earlier had reported high job strain were 68% more likely to later develop heart disease than were those workers under 50 years old who reported little job strain (Chandola et al., 2008).

Some people who are exposed to chronically stressful work conditions can experience job burnout, which is a general sense of emotional exhaustion and cynicism in relation to one's job (Maslach & Jackson, 1981). Job burnout occurs frequently among those in human service jobs (e.g., social workers, teachers, therapists, and police officers). Job burnout consists of three dimensions. The first dimension is exhaustion—a sense that one's emotional resources are drained or that one is at the end of her rope and has nothing more to give at a psychological level. Second, job burnout is characterized by depersonalization: a sense of emotional detachment between the worker and the recipients of his services, often resulting in callous, cynical, or indifferent attitudes toward

these individuals. Third, job burnout is characterized by diminished personal accomplishment, which is the tendency to evaluate one's work negatively by, for example, experiencing dissatisfaction with one's job-related accomplishments or feeling as though one has categorically failed to influence others' lives through one's work.

Job strain appears to be one of the greatest risk factors leading to job burnout, which is most commonly observed in workers who are older (ages 55–64), unmarried, and whose jobs involve manual labor. Heavy alcohol consumption, physical inactivity, being overweight, and having a physical or lifetime mental disorder are also associated with job burnout (Ahola, et al., 2006). In addition, depression often co-occurs with job burnout. One large-scale study of over 3,000 Finnish employees reported that half of the participants with severe job burnout had some form of depressive disorder (Ahola et al., 2005). Job burnout is often precipitated by feelings of having invested considerable energy, effort, and time into one's work while receiving little in return (e.g., little respect or support from others or low pay) (Tatris, Peeters, Le Blanc, Schreurs, & Schaufeli, 2001).

As an illustration, consider CharlieAnn, a nursing assistant who worked in a nursing home. CharlieAnn worked long hours for little pay in a difficult facility. Her supervisor was domineering, unpleasant, and unsupportive; he was disrespectful of CharlieAnn's personal time, frequently informing her at the last minute she must work several additional hours after her shift ended or that she must report to work on weekends. CharlieAnn had very little autonomy at her job. She had little say in her day-to-day duties and how to perform them, and she was not permitted to take breaks unless her supervisor explicitly told her that she could. CharlieAnn did not feel as though her hard work was appreciated, either by supervisory staff or by the residents of the home. She was very unhappy over her low pay, and she felt that many of the residents treated her disrespectfully.

After several years, CharlieAnn began to hate her job. She

dreaded going to work in the morning, and she gradually developed a callous, hostile attitude toward many of the residents. Eventually, she began to feel as though she could no longer help the nursing home residents. CharlieAnn's absenteeism from work increased, and one day she decided that she had had enough and quit. She now has a job in sales, vowing never to work in nursing again.

A humorous example illustrating lack of supervisory support can be found in the 1999 comedy *Office Space*. Follow this link to view a brief excerpt in which a sympathetic character's insufferable boss makes a last-minute demand that he "go ahead and come in" to the office on both Saturday and Sunday: Office Space – Working Tomorrow.

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=197#oembed-1

Finally, our close relationships with friends and family—particularly the negative aspects of these relationships—can be a potent source of stress. Negative aspects of close relationships can include adverse exchanges and conflicts, lack of emotional support or confiding, and lack of reciprocity. All of these can be overwhelming, threatening to the relationship, and thus stressful. Such stressors can take a toll both emotionally and physically. A longitudinal investigation of over 9,000 British civil servants found that those who at one point had reported the highest levels of negative interactions in their closest relationship were 34% more likely to experience serious heart problems (fatal or nonfatal heart attacks) over a 13–15 year period, compared to those who experienced the lowest levels of negative interaction (De Vogli, Chandola & Marmot, 2007).

TEST YOUR UNDERSTANDING

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SUMMARY

Stressors can be chronic (long term) or acute (short term), and can include traumatic events, significant life changes, daily hassles, and situations in which people are frequently exposed to challenging and unpleasant events. Many potential stressors include events or situations that require us to make changes in our lives, such as a divorce or moving to a new residence. Thomas Holmes and Richard Rahe developed the Social Readjustment Rating Scale (SRRS) to measure stress by assigning a number of life change units to life events that typically require some adjustment, including positive events. Although the SRRS has been criticized on a number of grounds, extensive research has shown that the accumulation of many LCUs is associated with increased risk of illness. Many potential stressors also include daily hassles, which are minor irritations and annoyances that can build up over time. In addition, jobs that are especially demanding, offer little control over one's

working environment, or involve unfavorable working conditions can lead to job strain, thereby setting the stage for job burnout.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Review the items on the Social Readjustment Rating Scale. Select one of the items and discuss how it might bring about distress and eustress.

Answers will vary. For example, many people look forward to celebrating the Christmas holiday, but it can be stressful in that it requires some degree of readjustment. Getting together with family may bring eustress, while the schedule and travel demands of may bring distress. Giving gifts to others and seeing their enjoyment may bring eustress, but the financial burden associated with buying presents could produce distress. Each of these things requires making some minor adjustments to one's life, and thus is considered somewhat stressful.

Job burnout tends to be high in people who work in human service jobs. Considering the three dimensions of job burnout, explain how various job aspects unique to being a police officer might lead to job burnout in that line of work.

Answers will vary. Many calls that police officers make can be emotionally draining (e.g., tragic deaths, suicides, and children who live in squalid conditions), which might eventually lead to feelings of exhaustion that one can no longer deal with such things. Depersonalization may occur if a police officer works

in an environment in which she feels disrespected and unappreciated, which may lead to cynical and callous feelings toward the public. Constant disrespect from others may diminish a police officer's sense of personal accomplishment.

PERSONAL APPLICATION QUESTION

Suppose you want to design a study to examine the relationship between stress and illness, but you cannot use the Social Readjustment Rating Scale. How would you go about measuring stress? How would you measure illness? What would you need to do in order to tell if there is a cause-effect relationship between stress and illness?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

https://pressbooks.cuny.edu/jjcpsy101/?p=197#h5p-222

Media Attributions

- "Office Space – Working Tomorrow" by ResoluteProductions. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Explain the nature of psychophysiological disorders
- Describe the immune system and how stress impacts its functioning
- Describe how stress and emotional factors can lead to the development and exacerbation of cardiovascular disorders, asthma, and tension headaches

In this section, we will discuss stress and illness. As stress researcher Robert Sapolsky (1998) describes,

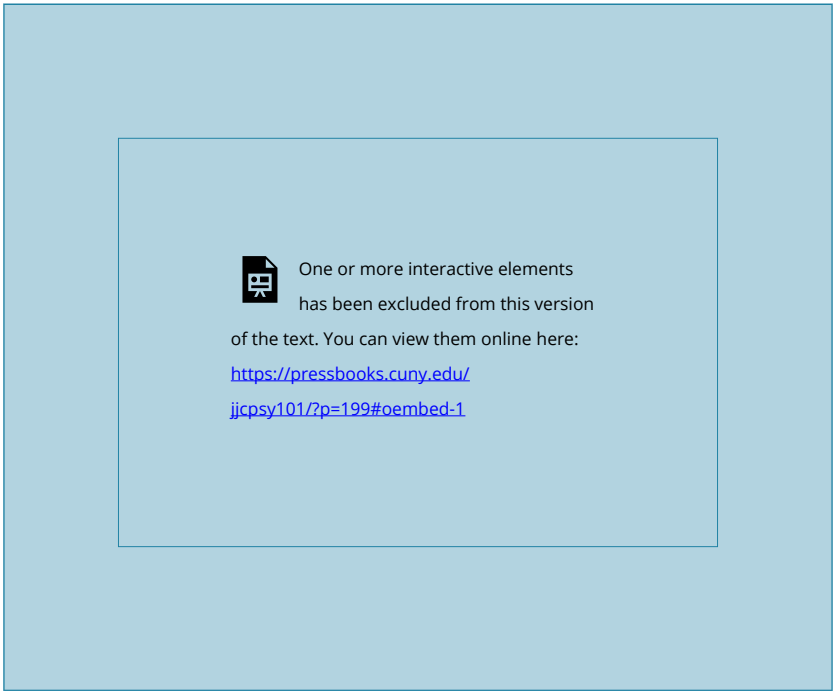
> stress-related disease emerges, predominantly, out of the fact that we so often activate a physiological system that has evolved for responding to acute physical emergencies, but we turn it on for months on end, worrying about mortgages, relationships, and promotions. (p. 6)

The stress response, as noted earlier, consists of a coordinated but complex system of physiological reactions that are called upon as needed. These reactions are beneficial at times because they prepare us to deal with potentially dangerous or threatening situations (for example, recall our old friend, the fearsome bear on the trail). However, health is affected when physiological reactions are sustained, as can happen in response to ongoing stress.

PSYCHOPHYSIOLOGICAL DISORDERS

If the reactions that compose the stress response are chronic or if they frequently exceed normal ranges, they can lead to cumulative wear and tear on the body, in much the same way that running your air conditioner on full blast all summer will eventually cause wear and tear on it. For example, the high blood pressure that a person under considerable job strain experiences might eventually take a toll on his heart and set the stage for a heart attack or heart failure. Also, someone exposed to high levels of the stress hormone cortisol might become vulnerable to infection or disease because of weakened immune system functioning (McEwen, 1998).

Robert Sapolsky, a noted Stanford University neurobiologist and professor, has for over 30 years conducted extensive research on stress, its impact on our bodies, and how psychological tumult can escalate stress—even in baboons. Here is a video featuring Dr. Sapolsky, an excellent in-depth documentary from *National Geographic*: Stress, Portrait of a Killer – Full Documentary (2008).



Physical disorders or diseases whose symptoms are brought about or worsened by stress and emotional factors are called psychophysiological disorders. The physical symptoms of psychophysiological disorders are real and they can be produced or exacerbated by psychological factors (hence the *psycho* and *physiological* in psychophysiological).

Types of Psychophysiological Disorders (adapted from Everly & Lating, 2002)

Type of Psychophysiological Disorder	Examples
Cardiovascular	hypertension, coronary heart disease
Gastrointestinal	irritable bowel syndrome
Respiratory	asthma, allergy
Musculoskeletal	low back pain, tension headaches
Skin	acne, eczema, psoriasis

In addition to stress itself, emotional upset and certain stressful personality traits have been proposed as potential contributors to ill health. Franz Alexander (1950), an early-20th-century psychoanalyst and physician, once postulated that various diseases are caused by specific unconscious conflicts. For example, he linked hypertension to repressed anger, asthma to separation anxiety, and ulcers to an unconscious desire to "remain in the dependent infantile situation—to be loved and cared for" (Alexander, 1950, p. 102). Although hypertension does appear to be linked to anger (as you will learn below), Alexander's assertions have not been supported by research. Years later, Friedman and Booth-Kewley (1987), after statistically reviewing 101 studies examining the link between personality and illness, proposed the existence of disease-prone personality characteristics, including depression, anger/hostility, and anxiety. Indeed, a study of over 61,000 Norwegians identified depression as a risk factor for all major disease-related causes of death (Mykletun et al., 2007). In addition, neuroticism—a personality trait that reflects how anxious, moody, and sad one is—has been identified as a risk factor for chronic health problems and mortality (Ploubidis & Grundy, 2009).

Below, we discuss two kinds of psychophysiological disorders about which a great deal is known: cardiovascular disorders and asthma. First, however, it is necessary to turn our attention to a discussion of the immune system—one of the major pathways through which stress and emotional factors can lead to illness and disease.

STRESS THE IMMUNE SYSTEM

In a sense, the immune system is the body's surveillance system. It consists of a variety of structures, cells, and mechanisms that serve to protect the body from invading toxins and microorganisms that can harm or damage the body's tissues and organs. When

the immune system is working as it should, it keeps us healthy and disease free by eliminating bacteria, viruses, and other foreign substances that have entered the body (Everly & Lating, 2002).

IMMUNE SYSTEM ERRORS

Sometimes, the immune system will function erroneously. For example, sometimes it can go awry by mistaking your body's own healthy cells for invaders and repeatedly attacking them. When this happens, the person is said to have an autoimmune disease, which can affect almost any part of the body. How an autoimmune disease affects a person depends on what part of the body is targeted. For instance, rheumatoid arthritis, an autoimmune disease that affects the joints, results in joint pain, stiffness, and loss of function. Systemic lupus erythematosus, an autoimmune disease that affects the skin, can result in rashes and swelling of the skin. Grave's disease, an autoimmune disease that affects the thyroid gland, can result in fatigue, weight gain, and muscle aches (National Institute of Arthritis and Musculoskeletal and Skin Diseases [NIAMS], 2012).

In addition, the immune system may sometimes break down and be unable to do its job. This situation is referred to as immunosuppression, the decreased effectiveness of the immune system. When people experience immunosuppression, they become susceptible to any number of infections, illness, and diseases. For example, acquired immune deficiency syndrome (AIDS) is a serious and lethal disease that is caused by human immunodeficiency virus (HIV), which greatly weakens the immune system by infecting and destroying antibody-producing cells, thus rendering a person vulnerable to any of a number of opportunistic infections (Powell, 1996).

STRESSORS AND IMMUNE FUNCTION

The question of whether stress and negative emotional states can influence immune function has captivated researchers for over three decades, and discoveries made over that time have dramatically changed the face of health psychology (Kiecolt-Glaser, 2009). Psychoneuroimmunology is the field that studies how psychological factors such as stress influence the immune system and immune functioning. The term psychoneuroimmunology was first coined in 1981, when it appeared as the title of a book that reviewed available evidence for associations between the brain, endocrine system, and immune system (Zacharie, 2009). To a large extent, this field evolved from the discovery that there is a connection between the central nervous system and the immune system.

Some of the most compelling evidence for a connection between the brain and the immune system comes from studies in which researchers demonstrated that immune responses in animals could be classically conditioned (Everly & Lating, 2002). For example, Ader and Cohen (1975) paired flavored water (the conditioned stimulus) with the presentation of an immunosuppressive drug (the unconditioned stimulus), causing sickness (an unconditioned response). Not surprisingly, rats exposed to this pairing developed a conditioned aversion to the flavored water. However, the taste of the water itself later produced immunosuppression (a conditioned response), indicating that the immune system itself had been conditioned. Many subsequent studies over the years have further demonstrated that immune responses can be classically conditioned in both animals and humans (Ader & Cohen, 2001). Thus, if classical conditioning can alter immunity, other psychological factors should be capable of altering it as well.

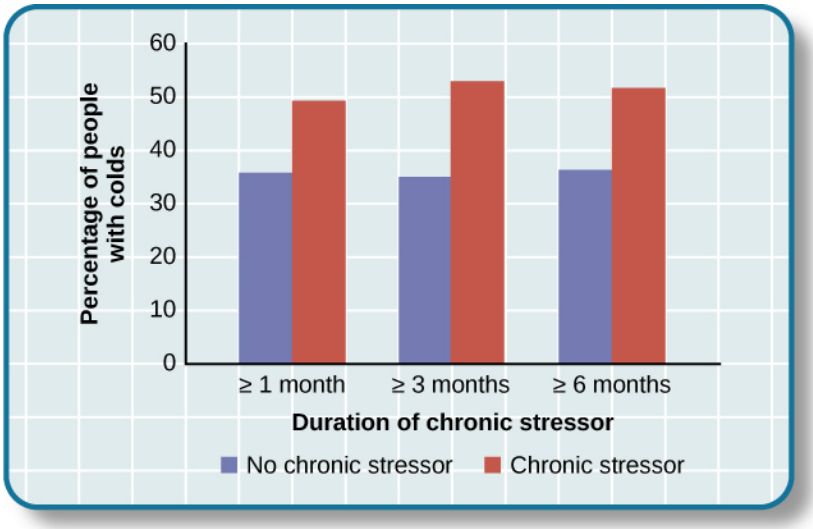
Hundreds of studies involving tens of thousands of participants have tested many kinds of brief and chronic stressors and their

effect on the immune system (e.g., public speaking, medical school examinations, unemployment, marital discord, divorce, death of spouse, burnout and job strain, caring for a relative with Alzheimer's disease, and exposure to the harsh climate of Antarctica). It has been repeatedly demonstrated that many kinds of stressors are associated with poor or weakened immune functioning (Glaser & Kiecolt-Glaser, 2005; Kiecolt-Glaser, McGuire, Robles, & Glaser, 2002; Segerstrom & Miller, 2004).

When evaluating these findings, it is important to remember that there is a tangible physiological connection between the brain and the immune system. For example, the sympathetic nervous system innervates immune organs such as the thymus, bone marrow, spleen, and even lymph nodes (Maier, Watkins, & Fleshner, 1994). Also, we noted earlier that stress hormones released during hypothalamic-pituitary-adrenal (HPA) axis activation can adversely impact immune function. One way they do this is by inhibiting the production of lymphocytes, white blood cells that circulate in the body's fluids that are important in the immune response (Everly & Lating, 2002).

Some of the more dramatic examples demonstrating the link between stress and impaired immune function involve studies in which volunteers were exposed to viruses. The rationale behind this research is that because stress weakens the immune system, people with high stress levels should be more likely to develop an illness compared to those under little stress. In one memorable experiment using this method, researchers interviewed 276 healthy volunteers about recent stressful experiences (Cohen et al., 1998). Following the interview, these participants were given nasal drops containing the cold virus (in case you are wondering why anybody would ever want to participate in a study in which they are subjected to such treatment, the participants were paid $800 for their trouble). When examined later, participants who reported experiencing chronic stressors for more than one month—especially enduring difficulties involving work or

relationships—were considerably more likely to have developed colds than were participants who reported no chronic stressors.



This graph shows the percentages of participants who developed colds (after receiving the cold virus) after reporting having experienced chronic stressors lasting at least one month, three months, and six months (adapted from Cohen et al., 1998).

In another study, older volunteers were given an influenza virus vaccination. Compared to controls, those who were caring for a spouse with Alzheimer's disease (and thus were under chronic stress) showed poorer antibody response following the vaccination (Kiecolt-Glaser, Glaser, Gravenstein, Malarkey, & Sheridan, 1996).

Other studies have demonstrated that stress slows down wound healing by impairing immune responses important to wound repair (Glaser & Kiecolt-Glaser, 2005). In one study, for example, skin blisters were induced on the forearm. Subjects who reported higher levels of stress produced lower levels of immune proteins necessary for wound healing (Glaser et al., 1999). Stress, then, is not so much the sword that kills the knight, so to speak; rather, it's

the sword that breaks the knight's shield, and your immune system is that shield.

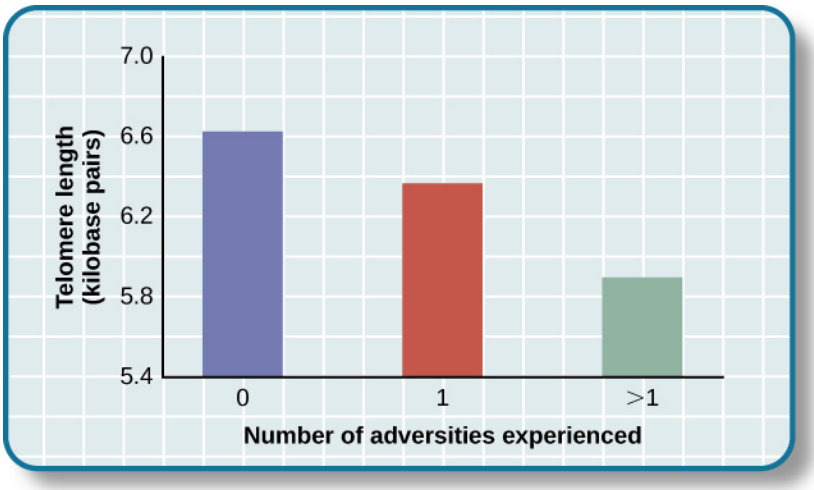
STRESS AND AGING: A TALE OF TELOMERES

Have you ever wondered why people who are stressed often seem to have a haggard look about them? A pioneering study from 2004 suggests that the reason is because stress can actually accelerate the cell biology of aging.

Stress, it seems, can shorten telomeres, which are segments of DNA that protect the ends of chromosomes. Shortened telomeres can inhibit or block cell division, which includes growth and proliferation of new cells, thereby leading to more rapid aging (Sapolsky, 2004). In the study, researchers compared telomere lengths in the white blood cells in mothers of chronically ill children to those of mothers of healthy children (Epel et al., 2004). Mothers of chronically ill children would be expected to experience more stress than would mothers of healthy children. The longer a mother had spent caring for her ill child, the shorter her telomeres (the correlation between years of caregiving and telomere length was r = -.40). In addition, higher levels of perceived stress were negatively correlated with telomere size (r = -.31). These researchers also found that the average telomere length of the most stressed mothers, compared to the least stressed, was similar to what you would find in people who were 9–17 years older than they were on average.

Numerous other studies since have continued to find associations between stress and eroded telomeres (Blackburn & Epel, 2012). Some studies have even demonstrated that stress can begin to erode telomeres in childhood and perhaps even before children are born. For example, childhood exposure to violence (e.g., maternal domestic violence, bullying victimization, and physical maltreatment) was found in one study to accelerate telomere erosion from ages 5 to 10 (Shalev et al., 2013). Another

study reported that young adults whose mothers had experienced severe stress during their pregnancy had shorter telomeres than did those whose mothers had stress-free and uneventful pregnancies (Entringer et al., 2011). Further, the corrosive effects of childhood stress on telomeres can extend into young adulthood. In an investigation of over 4,000 U.K. women ages 41–80, adverse experiences during childhood (e.g., physical abuse, being sent away from home, and parent divorce) were associated with shortened telomere length (Surtees et al., 2010), and telomere size decreased as the amount of experienced adversity increased.



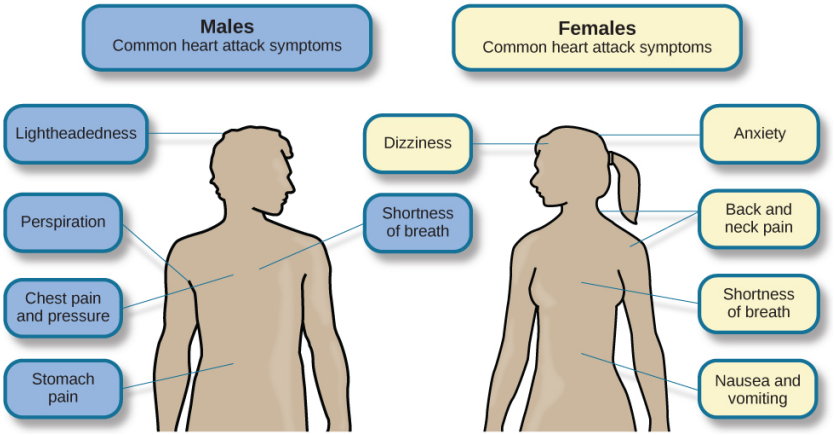
Telomeres are shorter in adults who experienced more trauma as children (adapted from Blackburn & Epel, 2012).

Efforts to dissect the precise cellular and physiological mechanisms linking short telomeres to stress and disease are currently underway. For the time being, telomeres provide us with yet another reminder that stress, especially during early life, can be just as harmful to our health as smoking or fast food (Blackburn & Epel, 2012).

CARDIOVASCULAR DISORDERS

The cardiovascular system is composed of the heart and blood circulation system. For many years, disorders that involve the cardiovascular system—known as cardiovascular disorders—have been a major focal point in the study of psychophysiological disorders because of the cardiovascular system's centrality in the stress response (Everly & Lating, 2002). **Heart disease** is one such condition. Each year, heart disease causes approximately one in three deaths in the United States, and it is the leading cause of death in the developed world (Centers for Disease Control and Prevention [CDC], 2011; Shapiro, 2005).

The symptoms of heart disease vary somewhat depending on the specific kind of heart disease one has, but they generally involve angina—chest pains or discomfort that occur when the heart does not receive enough blood (Office on Women's Health, 2009). The pain often feels like the chest is being pressed or squeezed; burning sensations in the chest and shortness of breath are also commonly reported. Such pain and discomfort can spread to the arms, neck, jaws, stomach (as nausea), and back (American Heart Association [AHA], 2012a).



Males and females often experience different symptoms of a heart attack.

A major risk factor for heart disease is hypertension, which is high blood pressure. Hypertension forces a person's heart to pump harder, thus putting more physical strain on the heart. If left unchecked, hypertension can lead to a heart attack, stroke, or heart failure; it can also lead to kidney failure and blindness. Hypertension is a serious cardiovascular disorder, and it is sometimes called the silent killer because it has no symptoms—one who has high blood pressure may not even be aware of it (AHA, 2012b).

Many risk factors contributing to cardiovascular disorders have been identified. These risk factors include social determinants such as aging, income, education, and employment status, as well as behavioral risk factors that include unhealthy diet, tobacco use, physical inactivity, and excessive alcohol consumption; obesity and diabetes are additional risk factors (World Health Organization [WHO], 2013).

Over the past few decades, there has been much greater recognition and awareness of the importance of stress and other psychological factors in cardiovascular health (Nusair, Al-dadah, & Kumar, 2012). Indeed, exposure to stressors of many kinds has

also been linked to cardiovascular problems; in the case of hypertension, some of these stressors include job strain (Trudel, Brisson, & Milot, 2010), natural disasters (Saito, Kim, Maekawa, Ikeda, & Yokoyama, 1997), marital conflict (Nealey-Moore, Smith, Uchino, Hawkins, & Olson-Cerny, 2007), and exposure to high traffic noise levels at one's home (de Kluizenaar, Gansevoort, Miedema, & de Jong, 2007). Perceived discrimination appears to be associated with hypertension among African Americans (Sims et al., 2012). In addition, laboratory-based stress tasks, such as performing mental arithmetic under time pressure, immersing one's hand into ice water (known as the cold pressor test), mirror tracing, and public speaking have all been shown to elevate blood pressure (Phillips, 2011).

ARE YOU TYPE A OR TYPE B?

Sometimes research ideas and theories emerge from seemingly trivial observations. In the 1950s, cardiologist Meyer Friedman was looking over his waiting room furniture, which consisted of upholstered chairs with armrests. Friedman decided to have these chairs reupholstered. When the man doing the reupholstering came to the office to do the work, he commented on how the chairs were worn in a unique manner—the front edges of the cushions were worn down, as were the front tips of the arm rests. It seemed like the cardiology patients were tapping or squeezing the front of the armrests, as well as literally sitting on the edge of their seats (Friedman & Rosenman, 1974). Were cardiology patients somehow different than other types of patients? If so, how?

After researching this matter, Friedman and his colleague, Ray Rosenman, came to understand that people who are prone to heart disease tend to think, feel, and act differently than those who are not. These individuals tend to be intensively driven workaholics who are preoccupied with deadlines and always seem to be in

a rush. According to Friedman and Rosenman, these individuals exhibit Type A behavior pattern; those who are more relaxed and laid-back were characterized as Type B. In a sample of Type As and Type Bs, Friedman and Rosenman were startled to discover that heart disease was over seven times more frequent among the Type As than the Type Bs (Friedman & Rosenman, 1959).

(a)

(b)

(a) Type A individuals are characterized as intensely driven, (b) while Type B people are characterized as laid-back and relaxed. (credit a: modification of work by Greg Hernandez; credit b: modification of work by Elvert Barnes)

The major components of the Type A pattern include an aggressive and chronic struggle to achieve more and more in less and less time (Friedman & Rosenman, 1974). Specific characteristics of the Type A pattern include an excessive competitive drive, chronic sense of time urgency, impatience, and hostility toward others (particularly those who get in the person's way).

An example of a person who exhibits Type A behavior pattern is Jeffrey. Even as a child, Jeffrey was intense and driven. He excelled at school, was captain of the swim team, and graduated with honors from an Ivy League college. Jeffrey never seems able to relax; he is always working on something, even on the weekends. However, Jeffrey always seems to feel as though there are not enough hours in the day to accomplish all he feels he should. He volunteers to take on extra tasks at work and often brings his work home with him; he often goes to bed angry late at night because he feels that he has not done enough. Jeffrey is quick tempered with

his coworkers; he often becomes noticeably agitated when dealing with those coworkers he feels work too slowly or whose work does not meet his standards. He typically reacts with hostility when interrupted at work. He has experienced problems in his marriage over his lack of time spent with family. When caught in traffic during his commute to and from work, Jeffrey incessantly pounds on his horn and swears loudly at other drivers. When Jeffrey was 52, he suffered his first heart attack.

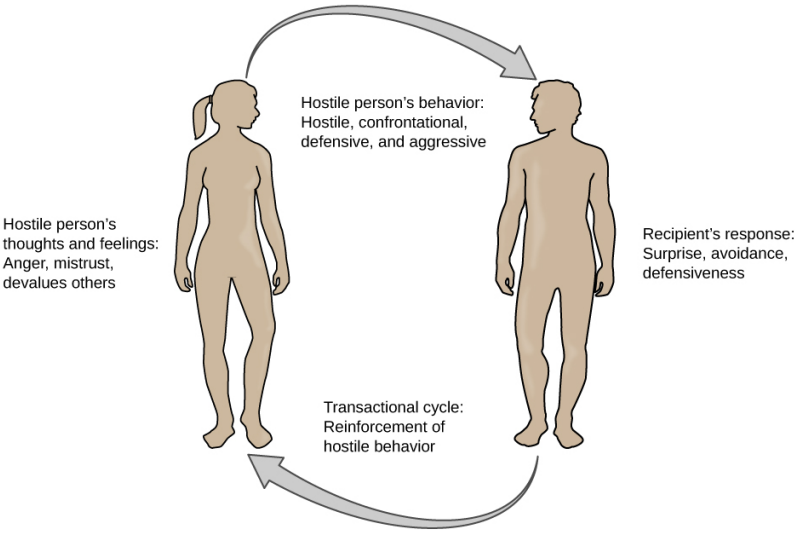
By the 1970s, a majority of practicing cardiologists believed that Type A behavior pattern was a significant risk factor for heart disease (Friedman, 1977). Indeed, a number of early longitudinal investigations demonstrated a link between Type A behavior pattern and later development of heart disease (Rosenman et al., 1975; Haynes, Feinleib, & Kannel, 1980).

Subsequent research examining the association between Type A and heart disease, however, failed to replicate these earlier findings (Glassman, 2007; Myrtek, 2001). Because Type A theory did not pan out as well as they had hoped, researchers shifted their attention toward determining if any of the specific elements of Type A predict heart disease.

Extensive research clearly suggests that the anger/hostility dimension of Type A behavior pattern may be one of the most important factors in the development of heart disease. This relationship was initially described in the Haynes et al. (1980) study mentioned above: Suppressed hostility was found to substantially elevate the risk of heart disease for both men and women. Also, one investigation followed over 1,000 male medical students from 32 to 48 years. At the beginning of the study, these men completed a questionnaire assessing how they react to pressure; some indicated that they respond with high levels of anger, whereas others indicated that they respond with less anger. Decades later, researchers found that those who earlier had indicated the highest levels of anger were over 6 times more likely than those who indicated less anger to have had a heart attack by age 55, and they

were 3.5 times more likely to have experienced heart disease by the same age (Chang, Ford, Meoni, Wang, & Klag, 2002). From a health standpoint, it clearly does not pay to be an angry young person.

After reviewing and statistically summarizing 35 studies from 1983 to 2006, Chida and Steptoe (2009) concluded that the bulk of the evidence suggests that anger and hostility constitute serious long-term risk factors for adverse cardiovascular outcomes among both healthy individuals and those already suffering from heart disease. One reason angry and hostile moods might contribute to cardiovascular diseases is that such moods can create social strain, mainly in the form of antagonistic social encounters with others. This strain could then lay the foundation for disease-promoting cardiovascular responses among hostile individuals (Vella, Kamarck, Flory, & Manuck, 2012). In this transactional model, hostility and social strain form a cycle.



According to the transactional model of hostility for predicting social interactions (Vella et al., 2012), the thoughts and feelings of a hostile person promote antagonistic behavior toward others, which in turn reinforces complimentary reactions from others, thereby intensifying ones' hostile disposition and intensifying the cyclical nature of this relationship.

For example, suppose Kaitlin has a hostile disposition; she has a cynical, distrustful attitude toward others and often thinks that other people are out to get her. She is very defensive around people, even those she has known for years, and she is always looking for signs that others are either disrespecting or belittling her. In the shower each morning before work, she often mentally rehearses what she would say to someone who said or did something that angered her, such as making a political statement that was counter to her own ideology. As Kaitlin goes through these mental rehearsals, she often grins and thinks about the retaliation on anyone who will irk her that day.

Socially, she is confrontational and tends to use a harsh tone with people, which often leads to very disagreeable and sometimes argumentative social interactions. As you might imagine, Kaitlin is

not especially popular with others, including coworkers, neighbors, and even members of her own family. They either avoid her at all costs or snap back at her, which causes Kaitlin to become even more cynical and distrustful of others, making her disposition even more hostile. Kaitlin's hostility—through her own doing—has created an antagonistic environment that cyclically causes her to become even more hostile and angry, thereby potentially setting the stage for cardiovascular problems.

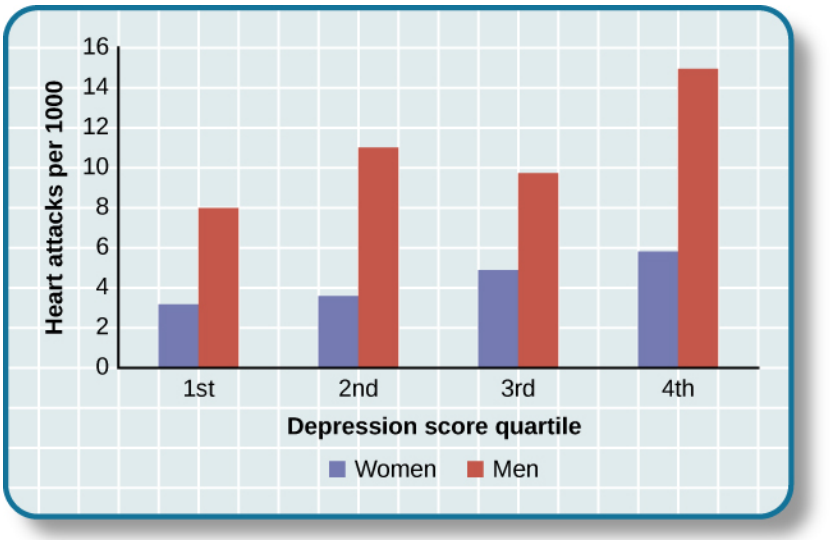
In addition to anger and hostility, a number of other negative emotional states have been linked with heart disease, including negative affectivity and depression (Suls & Bunde, 2005). Negative affectivity is a tendency to experience distressed emotional states involving anger, contempt, disgust, guilt, fear, and nervousness (Watson, Clark, & Tellegen, 1988). It has been linked with the development of both hypertension and heart disease. For example, over 3,000 initially healthy participants in one study were tracked longitudinally, up to 22 years. Those with higher levels of negative affectivity at the time the study began were substantially more likely to develop and be treated for hypertension during the ensuing years than were those with lower levels of negative affectivity (Jonas & Lando, 2000). In addition, a study of over 10,000 middle-aged London-based civil servants who were followed an average of 12.5 years revealed that those who earlier had scored in the upper third on a test of negative affectivity were 32% more likely to have experienced heart disease, heart attack, or angina over a period of years than were those who scored in the lowest third (Nabi, Kivimaki, De Vogli, Marmot, & Singh-Manoux, 2008). Hence, negative affectivity appears to be a potentially vital risk factor for the development of cardiovascular disorders.

DEPRESSION AND THE HEART

For centuries, poets and folklore have asserted that there is a

connection between moods and the heart (Glassman & Shapiro, 1998). You are no doubt familiar with the notion of a broken heart following a disappointing or depressing event and have encountered that notion in songs, films, and literature.

Perhaps the first to recognize the link between depression and heart disease was Benjamin Malzberg (1937), who found that the death rate among institutionalized patients with melancholia (an archaic term for depression) was six times higher than that of the population. A classic study in the late 1970s looked at over 8,000 manic-depressive persons in Denmark, finding a nearly 50% increase in deaths from heart disease among these patients compared with the general Danish population (Weeke, 1979). By the early 1990s, evidence began to accumulate showing that depressed individuals who were followed for long periods of time were at increased risk for heart disease and cardiac death (Glassman, 2007). In one investigation of over 700 Denmark residents, those with the highest depression scores were 71% more likely to have experienced a heart attack than were those with lower depression scores (Barefoot & Schroll, 1996)



This graph shows the incidence of heart attacks among men and women by depression score quartile (adapted from Barefoot & Schroll, 1996).

After more than two decades of research, it is now clear that a relationship exists: Patients with heart disease have more depression than the general population, and people with depression are more likely to eventually develop heart disease and experience higher mortality than those who do not have depression (Hare, Toukhsati, Johansson, & Jaarsma, 2013); the more severe the depression, the higher the risk (Glassman, 2007). Consider the following:

In one study, death rates from cardiovascular problems was substantially higher in depressed people; depressed men were 50% more likely to have died from cardiovascular problems, and depressed women were 70% more likely (Ösby, Brandt, Correia, Ekbom, & Sparén, 2001).

A statistical review of 10 longitudinal studies involving initially healthy individuals revealed that those with elevated depressive symptoms have, on average, a 64% greater risk of developing

heart disease than do those with fewer symptoms (Wulsin & Singal, 2003).

A study of over 63,000 registered nurses found that those with more depressed symptoms when the study began were 49% more likely to experience fatal heart disease over a 12-year period (Whang et al., 2009).

The American Heart Association, fully aware of the established importance of depression in cardiovascular diseases, several years ago recommended routine depression screening for all heart disease patients (Lichtman et al., 2008). Recently, they have recommended including depression as a risk factor for heart disease patients (AHA, 2014).

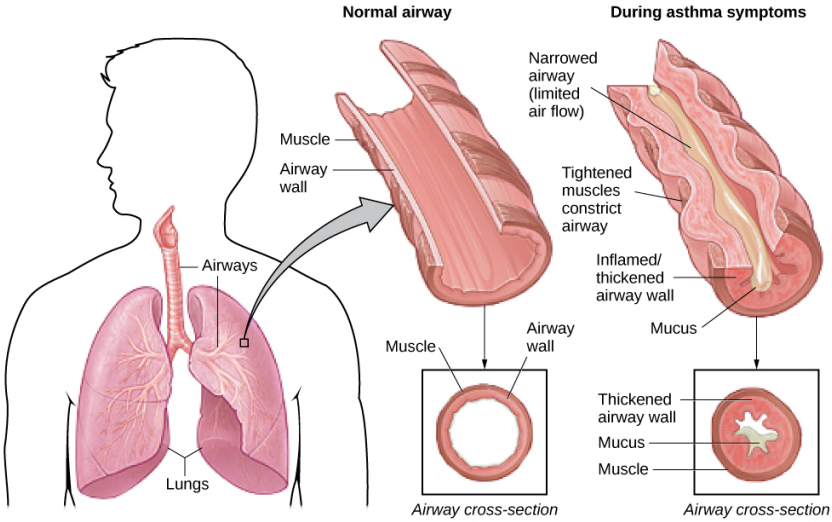
Although the exact mechanisms through which depression might produce heart problems have not been fully clarified, a recent investigation examining this connection in early life has shed some light. In an ongoing study of childhood depression, adolescents who had been diagnosed with depression as children were more likely to be obese, smoke, and be physically inactive than were those who had not received this diagnosis (Rottenberg et al., 2014). One implication of this study is that depression, especially if it occurs early in life, may increase the likelihood of living an unhealthy lifestyle, thereby predisposing people to an unfavorable cardiovascular disease risk profile.

It is important to point out that depression may be just one piece of the emotional puzzle in elevating the risk for heart disease, and that chronically experiencing several negative emotional states may be especially important. A longitudinal investigation of Vietnam War veterans found that depression, anxiety, hostility, and trait anger each independently predicted the onset of heart disease (Boyle, Michalek, & Suarez, 2006). However, when each of these negative psychological attributes was combined into a single variable, this new variable (which researchers called psychological risk factor) predicted heart disease more strongly than any of the individual variables. Thus, rather than examining the predictive

power of isolated psychological risk factors, it seems crucial for future researchers to examine the effects of combined and more general negative emotional and psychological traits in the development of cardiovascular illnesses.

ASTHMA

Asthma is a chronic and serious disease in which the airways of the respiratory system become obstructed, leading to great difficulty expelling air from the lungs. The airway obstruction is caused by inflammation of the airways (leading to thickening of the airway walls) and a tightening of the muscles around them, resulting in a narrowing of the airways (American Lung Association, 2010). Because airways become obstructed, a person with asthma will sometimes have great difficulty breathing and will experience repeated episodes of wheezing, chest tightness, shortness of breath, and coughing, the latter occurring mostly during the morning and night (CDC, 2006).



In asthma, the airways become inflamed and narrowed.

According to the Centers for Disease Control and Prevention (CDC), around 4,000 people die each year from asthma-related causes, and asthma is a contributing factor to another 7,000 deaths each year (CDC, 2013a). The CDC has revealed that asthma affects 18.7 million U.S. adults and is more common among people with lower education and income levels (CDC, 2013b). Especially concerning is that asthma is on the rise, with rates of asthma increasing 157% between 2000 and 2010 (CDC, 2013b).

Asthma attacks are acute episodes in which an asthma sufferer experiences the full range of symptoms. Asthma exacerbation is often triggered by environmental factors, such as air pollution, allergens (e.g., pollen, mold, and pet hairs), cigarette smoke, airway infections, cold air or a sudden change in temperature, and exercise (CDC, 2013b).

Psychological factors appear to play an important role in asthma (Wright, Rodriguez, & Cohen, 1998), although some believe that psychological factors serve as potential triggers in only a subset of asthma patients (Ritz, Steptoe, Bobb, Harris, & Edwards, 2006).

Many studies over the years have demonstrated that some people with asthma will experience asthma-like symptoms if they expect to experience such symptoms, such as when breathing an inert substance that they (falsely) believe will lead to airway obstruction (Sodergren & Hyland, 1999). As stress and emotions directly affect immune and respiratory functions, psychological factors likely serve as one of the most common triggers of asthma exacerbation (Trueba & Ritz, 2013).

People with asthma tend to report and display a high level of negative emotions such as anxiety, and asthma attacks have been linked to periods of high emotionality (Lehrer, Isenberg, & Hochron, 1993). In addition, high levels of emotional distress during both laboratory tasks and daily life have been found to negatively affect airway function and can produce asthma-like symptoms in people with asthma (von Leupoldt, Ehnes, & Dahme, 2006). In one investigation, 20 adults with asthma wore preprogrammed wristwatches that signaled them to breathe into a portable device that measures airway function. Results showed that higher levels of negative emotions and stress were associated with increased airway obstruction and self-reported asthma symptoms (Smyth, Soefer, Hurewitz, Kliment, & Stone, 1999). In addition, D'Amato, Liccardi, Cecchi, Pellegrino, & D'Amato (2010) described a case study of an 18-year-old man with asthma whose girlfriend had broken up with him, leaving him in a depressed state. She had also unfriended him on Facebook , while friending other young males. Eventually, the young man was able to "friend" her once again and could monitor her activity through Facebook. Subsequently, he would experience asthma symptoms whenever he logged on and accessed her profile. When he later resigned not to use Facebook any longer, the asthma attacks stopped. This case suggests that the use of Facebook and other forms of social media may represent a new source of stress—it may be a triggering factor for asthma attacks, especially in depressed asthmatic individuals.

Exposure to stressful experiences, particularly those that involve

parental or interpersonal conflicts, has been linked to the development of asthma throughout the lifespan. A longitudinal study of 145 children found that parenting difficulties during the first year of life increased the chances that the child developed asthma by 107% (Klinnert et al., 2001). In addition, a cross-sectional study of over 10,000 Finnish college students found that high rates of parent or personal conflicts (e.g., parental divorce, separation from spouse, or severe conflicts in other long-term relationships) increased the risk of asthma onset (Kilpeläinen, Koskenvuo, Helenius, & Terho, 2002). Further, a study of over 4,000 middle-aged men who were interviewed in the early 1990s and again a decade later found that breaking off an important life partnership (e.g., divorce or breaking off relationship from parents) increased the risk of developing asthma by 124% over the time of the study (Loerbroks, Apfelbacher, Thayer, Debling, & Stürmer, 2009).

TENSION HEADACHES

A headache is a continuous pain anywhere in the head and neck region. Migraine headaches are a type of headache thought to be caused by blood vessel swelling and increased blood flow (McIntosh, 2013). Migraines are characterized by severe pain on one or both sides of the head, an upset stomach, and disturbed vision. They are more frequently experienced by women than by men (American Academy of Neurology, 2014). Tension headaches are triggered by tightening/tensing of facial and neck muscles; they are the most commonly experienced kind of headache, accounting for about 42% of all headaches worldwide (Stovner et al., 2007). In the United States, well over one-third of the population experiences tension headaches each year, and 2–3% of the population suffers from chronic tension headaches (Schwartz, Stewart, Simon, & Lipton, 1998).

A number of factors can contribute to tension headaches,

including sleep deprivation, skipping meals, eye strain, overexertion, muscular tension caused by poor posture, and stress (MedicineNet, 2013). Although there is uncertainty regarding the exact mechanisms through which stress can produce tension headaches, stress has been demonstrated to increase sensitivity to pain (Caceres & Burns, 1997; Logan et al., 2001). In general, tension headache sufferers, compared to non-sufferers, have a lower threshold for and greater sensitivity to pain (Ukestad & Wittrock, 1996), and they report greater levels of subjective stress when faced with a stressor (Myers, Wittrock, & Foreman, 1998). Thus, stress may contribute to tension headaches by increasing pain sensitivity in already-sensitive pain pathways in tension headache sufferers (Cathcart, Petkov, & Pritchard, 2008).

TEST YOUR UNDERSTANDING

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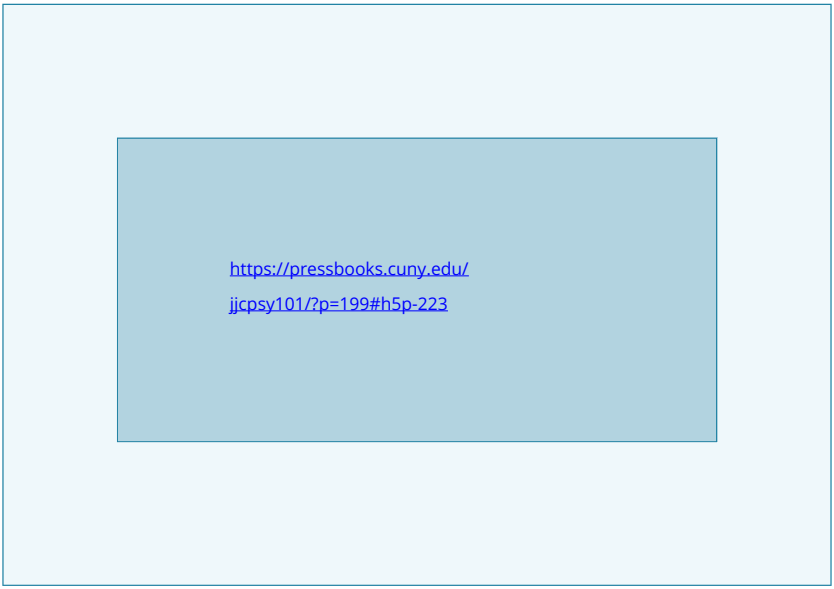

SUMMARY

Psychophysiological disorders are physical diseases that are either brought about or worsened by stress and other emotional factors. One of the mechanisms through which stress and emotional factors can influence the development of these diseases is by adversely affecting the body's immune system. A number of studies have demonstrated that stress weakens the functioning of the immune system. Cardiovascular disorders are serious medical conditions that have been consistently shown to be influenced by stress and negative emotions, such as anger, negative affectivity, and depression. Other psychophysiological disorders that are known to be influenced by stress and emotional factors include asthma and tension headaches.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Discuss the concept of Type A behavior pattern, its history, and what we now know concerning its role in heart disease.

Type A was conceptualized as a behavioral style characterized by competitiveness, time urgency, impatience, and anger/hostility. It was later discovered, however, that anger/hostility seems to be the dimension that most clearly predicts heart disease.

Consider the study in which volunteers were given nasal drops containing the cold virus to examine the relationship between stress and immune function (Cohen et al., 1998). How might this finding explain how people seem to become sick during stressful times in their lives (e.g., final exam week)?

The results of the study showed that people exposed to the virus were more likely to develop a cold if they had high stress scores. The implication of this finding is that during stressful times, like final exam weeks, the immune system becomes compromised. Thus, it's much easier to get sick during these periods because the immune system is not working at full capacity.

PERSONAL APPLICATION QUESTION

If a family member or friend of yours has asthma, talk to that person (if he or she is willing) about their symptom triggers. Does

this person mention stress or emotional states? If so, are there any commonalities in these asthma triggers?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:
https://pressbooks.cuny.edu/jjcpsy101/?p=199#h5p-225

Media Attributions

- "Stress, Portrait of a Killer – Full Documentary (2008)" by interface. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Define coping and differentiate between problem-focused and emotion-focused coping
- Describe the importance of perceived control in our reactions to stress
- Explain how social support is vital in health and longevity

As we learned in the previous section, stress—especially if it is chronic—takes a toll on our bodies and can have enormously negative health implications. When we experience events in our lives that we appraise as stressful, it is essential that we use effective coping strategies to manage our stress. Coping refers to mental and behavioral efforts that we use to deal with problems relating to stress, including its presumed cause and the unpleasant feelings and emotions it produces.

COPING STYLES

Lazarus and Folkman (1984) distinguished two fundamental kinds of coping: problem-focused coping and emotion-focused coping. In problem-focused coping, one attempts to manage or alter the problem that is causing one to experience stress (i.e., the stressor). Problem-focused coping strategies are similar to strategies used in everyday problem-solving: they typically involve identifying the

problem, considering possible solutions, weighing the costs and benefits of these solutions, and then selecting an alternative (Lazarus & Folkman, 1984). As an example, suppose Bradford receives a midterm notice that he is failing statistics class. If Bradford adopts a problem-focused coping approach to managing his stress, he would be proactive in trying to alleviate the source of the stress. He might contact his professor to discuss what must be done to raise his grade, he might also decide to set aside two hours daily to study statistics assignments, and he may seek tutoring assistance. A problem-focused approach to managing stress means we actively try to do things to address the problem.

Emotion-focused coping, in contrast, consists of efforts to change or reduce the negative emotions associated with stress. These efforts may include avoiding, minimizing, or distancing oneself from the problem, or positive comparisons with others ("I'm not as bad off as she is"), or seeking something positive in a negative event ("Now that I've been fired, I can sleep in for a few days"). In some cases, emotion-focused coping strategies involve reappraisal, whereby the stressor is construed differently (and somewhat self-deceptively) without changing its objective level of threat (Lazarus & Folkman, 1984). For example, a person sentenced to federal prison who thinks, "This will give me a great chance to network with others," is using reappraisal. If Bradford adopted an emotion-focused approach to managing his midterm deficiency stress, he might watch a comedy movie, play video games, or spend hours on Twitter to take his mind off the situation. In a certain sense, emotion-focused coping can be thought of as treating the symptoms rather than the actual cause.

While many stressors elicit both kinds of coping strategies, problem-focused coping is more likely to occur when encountering stressors we perceive as controllable, while emotion-focused coping is more likely to predominate when faced with stressors that we believe we are powerless to change (Folkman & Lazarus, 1980). Clearly, emotion-focused coping is more effective in dealing

with uncontrollable stressors. For example, if at midnight you are stressing over a 40-page paper due in the morning that you have not yet started, you are probably better off recognizing the hopelessness of the situation and doing something to take your mind off it; taking a problem-focused approach by trying to accomplish this task would only lead to frustration, anxiety, and even more stress.

Fortunately, most stressors we encounter can be modified and are, to varying degrees, controllable. A person who cannot stand her job can quit and look for work elsewhere; a middle-aged divorcee can find another potential partner; the freshman who fails an exam can study harder next time, and a breast lump does not necessarily mean that one is fated to die of breast cancer.

CONTROL AND STRESS

The desire and ability to predict events, make decisions, and affect outcomes—that is, to enact control in our lives—is a basic tenet of human behavior (Everly & Lating, 2002). Albert Bandura (1997) stated that "the intensity and chronicity of human stress is governed largely by perceived control over the demands of one's life" (p. 262). As cogently described in his statement, our reaction to potential stressors depends to a large extent on how much control we feel we have over such things. Perceived control is our beliefs about our personal capacity to exert influence over and shape outcomes, and it has major implications for our health and happiness (Infurna & Gerstorf, 2014). Extensive research has demonstrated that perceptions of personal control are associated with a variety of favorable outcomes, such as better physical and mental health and greater psychological well-being (Diehl & Hay, 2010). Greater personal control is also associated with lower reactivity to stressors in daily life. For example, researchers in one investigation found that higher levels of perceived control at one

point in time were later associated with lower emotional and physical reactivity to interpersonal stressors (Neupert, Almeida, & Charles, 2007). Further, a daily diary study with 34 older widows found that their stress and anxiety levels were significantly reduced on days during which the widows felt greater perceived control (Ong, Bergeman, & Bisconti, 2005).

Learned Helplessness

When we lack a sense of control over the events in our lives, particularly when those events are threatening, harmful, or noxious, the psychological consequences can be profound. In one of the better illustrations of this concept, psychologist Martin Seligman conducted a series of classic experiments in the 1960s (Seligman & Maier, 1967) in which dogs were placed in a chamber where they received electric shocks from which they could not escape. Later, when these dogs were given the opportunity to escape the shocks by jumping across a partition, most failed to even try; they seemed to just give up and passively accept any shocks the experimenters chose to administer. In comparison, dogs who were previously allowed to escape the shocks tended to jump the partition and escape the pain.

Seligman's learned helplessness experiments with dogs used an apparatus that measured when the animals would move from a floor delivering shocks to one without.

Seligman believed that the dogs who failed to try to escape the later shocks were demonstrating learned helplessness: They had acquired a belief that they were powerless to do anything about the noxious stimulation they were receiving. Seligman also believed that the passivity and lack of initiative these dogs demonstrated was similar to that observed in human depression. Therefore, Seligman speculated that acquiring a sense of learned helplessness might be an important cause of depression in humans: Humans who experience negative life events that they believe they are unable to control may become helpless. As a result, they give up trying to control or change the situation and some may become depressed and show lack of initiative in future situations in which they can control the outcomes (Seligman, Maier, & Geer, 1968).

Seligman and colleagues later reformulated the original learned helplessness model of depression (Abramson, Seligman, & Teasdale, 1978). In their reformulation, they emphasized

attributions (i.e., a mental explanation for why something occurred) that lead to the perception that one lacks control over negative outcomes are important in fostering a sense of learned helplessness. For example, suppose a coworker shows up late to work; your belief as to what caused the coworker's tardiness would be an attribution (e.g., too much traffic, slept too late, or just doesn't care about being on time).

The reformulated version of Seligman's study holds that the attributions made for negative life events contribute to depression. Consider the example of a student who performs poorly on a midterm exam. This model suggests that the student will make three kinds of attributions for this outcome: internal vs. external (believing the outcome was caused by his own personal inadequacies or by environmental factors), stable vs. unstable (believing the cause can be changed or is permanent), and global vs. specific (believing the outcome is a sign of inadequacy in most everything versus just this area). Assume that the student makes an internal ("I'm just not smart"), stable ("Nothing can be done to change the fact that I'm not smart") and global ("This is another example of how lousy I am at everything") attribution for the poor performance. The reformulated theory predicts that the student would perceive a lack of control over this stressful event and thus be especially prone to developing depression. Indeed, research has demonstrated that people who have a tendency to make internal, global, and stable attributions for bad outcomes tend to develop symptoms of depression when faced with negative life experiences (Peterson & Seligman, 1984).

Seligman's learned helplessness model has emerged over the years as a leading theoretical explanation for the onset of major depressive disorder. When you study psychological disorders, you will learn more about the latest reformulation of this model—now called hopelessness theory.

People who report higher levels of perceived control view their health as controllable, thereby making it more likely that they will

better manage their health and engage in behaviors conducive to good health (Bandura, 2004). Not surprisingly, greater perceived control has been linked to lower risk of physical health problems, including declines in physical functioning (Infurna, Gerstorf, Ram, Schupp, & Wagner, 2011), heart attacks (Rosengren et al., 2004), and both cardiovascular disease incidence (Stürmer, Hasselbach, & Amelang, 2006) and mortality from cardiac disease (Surtees et al., 2010). In addition, longitudinal studies of British civil servants have found that those in low-status jobs (e.g., clerical and office support staff) in which the degree of control over the job is minimal are considerably more likely to develop heart disease than those with high-status jobs or considerable control over their jobs (Marmot, Bosma, Hemingway, & Stansfeld, 1997).

The link between perceived control and health may provide an explanation for the frequently observed relationship between social class and health outcomes (Kraus, Piff, Mendoza-Denton, Rheinschmidt, & Keltner, 2012). In general, research has found that more affluent individuals experience better health mainly because they tend to believe that they can personally control and manage their reactions to life's stressors (Johnson & Krueger, 2006). Perhaps buoyed by the perceived level of control, individuals of higher social class may be prone to overestimating the degree of influence they have over particular outcomes. For example, those of higher social class tend to believe that their votes have greater sway on election outcomes than do those of lower social class, which may explain higher rates of voting in more affluent communities (Krosnick, 1990). Other research has found that a sense of perceived control can protect less affluent individuals from poorer health, depression, and reduced life-satisfaction—all of which tend to accompany lower social standing (Lachman & Weaver, 1998).

Taken together, findings from these and many other studies clearly suggest that perceptions of control and coping abilities are

important in managing and coping with the stressors we encounter throughout life.

SOCIAL SUPPORT

The need to form and maintain strong, stable relationships with others is a powerful, pervasive, and fundamental human motive (Baumeister & Leary, 1995). Building strong interpersonal relationships with others helps us establish a network of close, caring individuals who can provide social support in times of distress, sorrow, and fear. Social support can be thought of as the soothing impact of friends, family, and acquaintances (Baron & Kerr, 2003). Social support can take many forms, including advice, guidance, encouragement, acceptance, emotional comfort, and tangible assistance (such as financial help). Thus, other people can be very comforting to us when we are faced with a wide range of life stressors, and they can be extremely helpful in our efforts to manage these challenges. Even in nonhuman animals, species mates can offer social support during times of stress. For example, elephants seem to be able to sense when other elephants are stressed and will often comfort them with physical contact—such as a trunk touch—or an empathetic vocal response (Krumboltz, 2014).

Scientific interest in the importance of social support first emerged in the 1970s when health researchers developed an interest in the health consequences of being socially integrated (Stroebe & Stroebe, 1996). Interest was further fueled by longitudinal studies showing that social connectedness reduced mortality. In one classic study, nearly 7,000 Alameda County, California, residents were followed over 9 years. Those who had previously indicated that they lacked social and community ties were more likely to die during the follow-up period than those with more extensive social networks. Compared to those with the

most social contacts, isolated men and women were, respectively, 2.3 and 2.8 times more likely to die. These trends persisted even after controlling for a variety of health-related variables, such as smoking, alcohol consumption, self-reported health at the beginning of the study, and physical activity (Berkman & Syme, 1979).

Since the time of that study, social support has emerged as one of the well-documented psychosocial factors affecting health outcomes (Uchino, 2009). A statistical review of 148 studies conducted between 1982 and 2007 involving over 300,000 participants concluded that individuals with stronger social relationships have a 50% greater likelihood of survival compared to those with weak or insufficient social relationships (Holt-Lunstad, Smith, & Layton, 2010). According to the researchers, the magnitude of the effect of social support observed in this study is comparable with quitting smoking and exceeded many well-known risk factors for mortality, such as obesity and physical inactivity.

(a) (b)

Close relationships with others, whether (a) a group of friends or (b) a family circle, provide more than happiness and fulfillment—they can help foster good health. (credit a: modification of work by Nattachai Noogure; credit b: modification of work by Christian Haugen)

A number of large-scale studies have found that individuals with low levels of social support are at greater risk of mortality, especially from cardiovascular disorders (Brummett et al., 2001). Further, higher levels of social supported have been linked to

better survival rates following breast cancer (Falagas et al., 2007) and infectious diseases, especially HIV infection (Lee & Rotheram-Borus, 2001). In fact, a person with high levels of social support is less likely to contract a common cold. In one study, 334 participants completed questionnaires assessing their sociability; these individuals were subsequently exposed to a virus that causes a common cold and monitored for several weeks to see who became ill. Results showed that increased sociability was linearly associated with a decreased probability of developing a cold (Cohen, Doyle, Turner, Alper, & Skoner, 2003).

For many of us, friends are a vital source of social support. But what if you found yourself in a situation in which you lacked friends or companions? For example, suppose a popular high school student attends a far-away college, does not know anyone, and has trouble making friends and meaningful connections with others during the first semester. What can be done? If real life social support is lacking, access to distant friends via social media may help compensate. In a study of college freshmen, those with few face-to-face friends on campus but who communicated electronically with distant friends were less distressed that those who did not (Raney & Troop-Gordon, 2012). Also, for some people, our families—especially our parents—are a major source of social support.

Social support appears to work by boosting the immune system, especially among people who are experiencing stress (Uchino, Vaughn, Carlisle, & Birmingham, 2012). In a pioneering study, spouses of cancer patients who reported high levels of social support showed indications of better immune functioning on two out of three immune functioning measures, compared to spouses who were below the median on reported social support (Baron, Cutrona, Hicklin, Russell, & Lubaroff, 1990). Studies of other populations have produced similar results, including those of spousal caregivers of dementia sufferers, medical students, elderly

adults, and cancer patients (Cohen & Herbert, 1996; Kiecolt-Glaser, McGuire, Robles, & Glaser, 2002).

In addition, social support has been shown to reduce blood pressure for people performing stressful tasks, such as giving a speech or performing mental arithmetic (Lepore, 1998). In these kinds of studies, participants are usually asked to perform a stressful task either alone, with a stranger present (who may be either supportive or unsupportive), or with a friend present. Those tested with a friend present generally exhibit lower blood pressure than those tested alone or with a stranger (Fontana, Diegnan, Villeneuve, & Lepore, 1999). In one study, 112 female participants who performed stressful mental arithmetic exhibited lower blood pressure when they received support from a friend rather than a stranger, but only if the friend was a male (Phillips, Gallagher, & Carroll, 2009). Although these findings are somewhat difficult to interpret, the authors mention that it is possible that females feel less supported and more evaluated by other females, particularly females whose opinions they value.

Taken together, the findings above suggest one of the reasons social support is connected to favorable health outcomes is because it has several beneficial physiological effects in stressful situations. However, it is also important to consider the possibility that social support may lead to better health behaviors, such as a healthy diet, exercising, smoking cessation, and cooperation with medical regimens (Uchino, 2009).

COPING WITH PREJUDICE AND DISCRIMINATION

While having social support is quite beneficial, being the recipient of prejudicial attitudes and discriminatory behaviors is associated with a number of negative outcomes. In their literature review, Brondolo, Brady, Pencille, Beatty, and Contrada (2009) describe how racial prejudice and discrimination serve as unique, significant stressors for those who are the targets of such attitudes and

behavior. Being the target of racism is associated with increased rates of depression, lowered self-esteem, hypertension, and cardiovascular disease.

Given the complex and pervasive nature of racism as a stressor, Brondolo et al. (2009) point out the importance of coping with this specific stressor. Their review is aimed at determining which coping strategies are most effective at offsetting negative health outcomes associated with racism-related stress. The authors examine the effectiveness of three coping strategies: focusing on racial identity to handle race-related stress, anger expression/suppression, and seeking social support. You've learned a bit about social support, so we'll focus the remainder of this discussion on the potential coping strategies of focusing on racial identity and anger expression/suppression.

Focusing on racial identity refers to the process by which a person comes to feel as if he belongs to a given racial group; this may increase a sense of pride associated with group membership. Brondolo et al. (2009) suggest that a strong sense of racial identity might help an individual who is the target of racism differentiate between prejudicial attitudes/behaviors that are directed toward his group as a whole rather than at him as a person. Furthermore, the sense of belonging to his group might alleviate the distress of being ostracized by others. However, the research literature on the effectiveness of this technique has produced mixed results.

Anger expression/suppression refers to the options available as a function of the anger evoked by racial prejudice and discrimination. Put simply, a target of racist attitudes and behaviors can act upon her anger or suppress her anger. As discussed by Brondolo et al. (2009), there has been very little research on the effectiveness of either approach; the results are quite mixed with some showing anger expression and others showing anger suppression as the healthier option.

In the end, racism-related stress is a complex issue and each of the coping strategies discussed here has strengths and

weaknesses. Brondolo et al. (2009) argue that it is imperative that additional research be conducted to ascertain the most effective strategies for coping with the negative outcomes that are experienced by the targets of racism.

STRESS REDUCTION TECHNIQUES

Beyond having a sense of control and establishing social support networks, there are numerous other means by which we can manage stress. A common technique people use to combat stress is exercise (Salmon, 2001). It is well-established that exercise, both of long (aerobic) and short (anaerobic) duration, is beneficial for both physical and mental health (Everly & Lating, 2002). There is considerable evidence that physically fit individuals are more resistant to the adverse effects of stress and recover more quickly from stress than less physically fit individuals (Cotton, 1990). In a study of more than 500 Swiss police officers and emergency service personnel, increased physical fitness was associated with reduced stress, and regular exercise was reported to protect against stress-related health problems (Gerber, Kellman, Hartman, & Pühse, 2010).

(a)

(b)

(c)

Stress reduction techniques may include (a) exercise, (b) meditation and relaxation, or (c) biofeedback. (credit a: modification of work by "UNE Photos"/Flickr; credit b: modification of work by Caleb Roenigk; credit c: modification of work by Dr. Carmen Russoniello)

One reason exercise may be beneficial is because it might buffer

some of the deleterious physiological mechanisms of stress. One study found rats that exercised for six weeks showed a decrease in hypothalamic-pituitary-adrenal responsiveness to mild stressors (Campeau et al., 2010). In high-stress humans, exercise has been shown to prevent telomere shortening, which may explain the common observation of a youthful appearance among those who exercise regularly (Puterman et al., 2010). Further, exercise in later adulthood appears to minimize the detrimental effects of stress on the hippocampus and memory (Head, Singh, & Bugg, 2012). Among cancer survivors, exercise has been shown to reduce anxiety (Speck, Courneya, Masse, Duval, & Schmitz, 2010) and depressive symptoms (Craft, VanIterson, Helenowski, Rademaker, & Courneya, 2012). Clearly, exercise is a highly effective tool for regulating stress.

In the 1970s, Herbert Benson, a cardiologist, developed a stress reduction method called the relaxation response technique (Greenberg, 2006). The relaxation response technique combines relaxation with transcendental meditation, and consists of four components (Stein, 2001):

- sitting upright on a comfortable chair with feet on the ground and body in a relaxed position,
- a quiet environment with eyes closed,
- repeating a word or a phrase—a mantra—to oneself, such as "alert mind, calm body,"
- passively allowing the mind to focus on pleasant thoughts, such as nature or the warmth of your blood nourishing your body.

The relaxation response approach is conceptualized as a general approach to stress reduction that reduces sympathetic arousal, and it has been used effectively to treat people with high blood pressure (Benson & Proctor, 1994).

Another technique to combat stress, biofeedback, was

developed by Gary Schwartz at Harvard University in the early 1970s. Biofeedback is a technique that uses electronic equipment to accurately measure a person's neuromuscular and autonomic activity—feedback is provided in the form of visual or auditory signals. The main assumption of this approach is that providing somebody biofeedback will enable the individual to develop strategies that help gain some level of voluntary control over what are normally involuntary bodily processes (Schwartz & Schwartz, 1995). A number of different bodily measures have been used in biofeedback research, including facial muscle movement, brain activity, and skin temperature, and it has been applied successfully with individuals experiencing tension headaches, high blood pressure, asthma, and phobias (Stein, 2001).

TEST YOUR UNDERSTANDING

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SUMMARY

When faced with stress, people must attempt to manage or cope with it. In general, there are two basic forms of coping: problem-focused coping and emotion-focused coping. Those who use problem-focused coping strategies tend to cope better with stress because these strategies address the source of stress rather than the resulting symptoms. To a large extent, perceived control greatly impacts reaction to stressors and is associated with greater physical and mental well-being. Social support has been demonstrated to be a highly effective buffer against the adverse effects of stress. Extensive research has shown that social support has beneficial physiological effects for people, and it seems to influence immune functioning. However, the beneficial effects of

social support may be related to its influence on promoting healthy behaviors.

REVIEW QUESTIONS

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https://pressbooks.cuny.edu/jjcpsy101/?p=201#h5p-227

CRITICAL THINKING QUESTIONS

Although problem-focused coping seems to be a more effective strategy when dealing with stressors, do you think there are any kinds of stressful situations in which emotion-focused coping might be a better strategy?

Emotion-focused coping would likely be a better coping strategy in situations in which a stressor is uncontrollable, or in which nothing could otherwise be done about it, such as a fatal illness.

Describe how social support can affect health both directly and indirectly.

Social support seems to have a direct effect on immune system functioning. Social support can affect health indirectly by influencing health-related behaviors, such as exercise and eating properly..

PERSONAL APPLICATION QUESTION

Try to think of an example in which you coped with a particular stressor by using problem-focused coping. What was the stressor? What did your problem-focused efforts involve? Were they effective?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=201#h5p-228

Learning Objectives

By the end of this section, you will be able to:

- Define and discuss happiness, including its determinants
- Describe the field of positive psychology and identify the kinds of problems it addresses
- Explain the meaning of positive affect and discuss its importance in health outcomes
- Describe the concept of flow and its relationship to happiness and fulfillment

Although the study of stress and how it affects us physically and psychologically is fascinating, it is—admittedly—somewhat of a grim topic. Psychology is also interested in the study of a more upbeat and encouraging approach to human affairs—the quest for happiness.

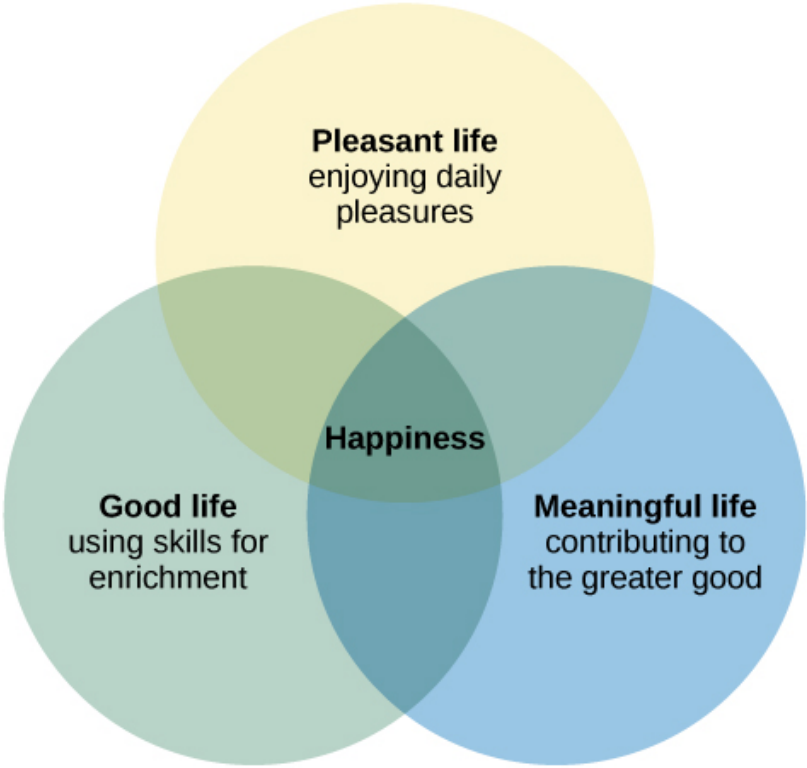
HAPPINESS

America's founders declared that its citizens have an unalienable right to pursue happiness. But what is happiness? When asked to define the term, people emphasize different aspects of this elusive state. Indeed, happiness is somewhat ambiguous and can be defined from different perspectives (Martin, 2012). Some people, especially those who are highly committed to their religious faith, view happiness in ways that emphasize virtuosity, reverence, and

enlightened spirituality. Others see happiness as primarily contentment—the inner peace and joy that come from deep satisfaction with one's surroundings, relationships with others, accomplishments, and oneself. Still others view happiness mainly as pleasurable engagement with their personal environment—having a career and hobbies that are engaging, meaningful, rewarding, and exciting. These differences, of course, are merely differences in emphasis. Most people would probably agree that each of these views, in some respects, captures the essence of happiness.

ELEMENTS OF HAPPINESS

Some psychologists have suggested that happiness consists of three distinct elements: the pleasant life, the good life, and the meaningful life (Seligman, 2002; Seligman, Steen, Park, & Peterson, 2005). The pleasant life is realized through the attainment of day-to-day pleasures that add fun, joy, and excitement to our lives. For example, evening walks along the beach and a fulfilling sex life can enhance our daily pleasure and contribute to the pleasant life. The good life is achieved through identifying our unique skills and abilities and engaging these talents to enrich our lives; those who achieve the good life often find themselves absorbed in their work or their recreational pursuits. The meaningful life involves a deep sense of fulfillment that comes from using our talents in the service of the greater good: in ways that benefit the lives of others or that make the world a better place. In general, the happiest people tend to be those who pursue the full life—they orient their pursuits toward all three elements (Seligman et al., 2005).



Happiness is an enduring state of well-being involving satisfaction in the pleasant, good, and meaningful aspects of life.

For practical purposes, a precise definition of happiness might incorporate each of these elements: an enduring state of mind consisting of joy, contentment, and other positive emotions, plus the sense that one's life has meaning and value (Lyubomirsky, 2001). The definition implies that happiness is a long-term state—what is often characterized as subjective well-being—rather than merely a transient positive mood we all experience from time to time. It is this enduring happiness that has captured the interests of psychologists and other social scientists.

The study of happiness has grown dramatically in the last three

decades (Diener, 2013). One of the most basic questions that happiness investigators routinely examine is this: How happy are people in general? The average person in the world tends to be relatively happy and tends to indicate experiencing more positive feelings than negative feelings (Diener, Ng, Harter, & Arora, 2010). When asked to evaluate their current lives on a scale ranging from 0 to 10 (with 0 representing "worst possible life" and 10 representing "best possible life"), people in more than 150 countries surveyed from 2010–2012 reported an average score of 5.2. People who live in North America, Australia, and New Zealand reported the highest average score at 7.1, whereas those living Sub-Saharan Africa reported the lowest average score at 4.6 (Helliwell, Layard, & Sachs, 2013). Worldwide, the five happiest countries are Denmark, Norway, Switzerland, the Netherlands, and Sweden; the United States is ranked 17th happiest (Helliwell et al., 2013).

(a) (b)

(a) Surveys of residents in over 150 countries indicate that Denmark has the happiest citizens in the world. (b) Americans ranked the United States as the 17th happiest country in which to live. (credit a: modification of work by "JamesZ_Flickr"/Flickr; credit b: modification of work by Ryan Swindell)

Several years ago, a Gallup survey of more than 1,000 U.S. adults found that 52% reported that they were "very happy." In addition, more than 8 in 10 indicated that they were "very satisfied" with their lives (Carroll, 2007). However, a recent poll of 2,345 U.S. adults

surprisingly revealed that only one-third reported they are "very happy." The poll also revealed that the happiness levels of certain groups, including minorities, recent college graduates, and the disabled, have trended downward in recent years (Gregoire, 2013). Although it is difficult to explain this apparent decline in happiness, it may be connected to the challenging economic conditions the United States has endured over the last several years. Of course, this presumption would imply that happiness is closely tied to one's finances. But, is it? This question brings us to the next important issue: What factors influence happiness?

FACTORS CONNECTED TO HAPPINESS

What really makes people happy? What factors contribute to sustained joy and contentment? Is it money, attractiveness, material possessions, a rewarding occupation, a satisfying relationship? Extensive research over the years has examined this question. One finding is that age is related to happiness: Life satisfaction usually increases the older people get, but there do not appear to be gender differences in happiness (Diener, Suh, Lucas, & Smith, 1999). Although it is important to point out that much of this work has been correlational, many of the key findings (some of which may surprise you) are summarized below.

Family and other social relationships appear to be key factors correlated with happiness. Studies show that married people report being happier than those who are single, divorced, or widowed (Diener et al., 1999). Happy individuals also report that their marriages are fulfilling (Lyubomirsky, King, & Diener, 2005). In fact, some have suggested that satisfaction with marriage and family life is the strongest predictor of happiness (Myers, 2000). Happy people tend to have more friends, more high-quality social relationships, and stronger social support networks than less happy people (Lyubomirsky et al., 2005). Happy people also have a high frequency of contact with friends (Pinquart & Sörensen, 2000).

Can money buy happiness? In general, extensive research suggests that the answer is yes, but with several caveats. While a nation's per capita gross domestic product (GDP) is associated with happiness levels (Helliwell et al., 2013), changes in GDP (which is a less certain index of household income) bear little relationship to changes in happiness (Diener, Tay, & Oishi, 2013). On the whole, residents of affluent countries tend to be happier than residents of poor countries; within countries, wealthy individuals are happier than poor individuals, but the association is much weaker (Diener & Biswas-Diener, 2002). To the extent that it leads to increases in purchasing power, increases in income are associated with increases in happiness (Diener, Oishi, & Ryan, 2013). However, income within societies appears to correlate with happiness only up to a point. In a study of over 450,000 U.S. residents surveyed by the Gallup Organization, Kahneman and Deaton (2010) found that well-being rises with annual income, but only up to $75,000. The average increase in reported well-being for people with incomes greater than $75,000 was null. As implausible as these findings might seem—after all, higher incomes would enable people to indulge in Hawaiian vacations, prime seats as sporting events, expensive automobiles, and expansive new homes—higher incomes may impair people's ability to savor and enjoy the small pleasures of life (Kahneman, 2011). Indeed, researchers in one study found that participants exposed to a subliminal reminder of wealth spent less time savoring a chocolate candy bar and exhibited less enjoyment of this experience than did participants who were not reminded of wealth (Quoidbach, Dunn, Petrides, & Mikolajczak, 2010).

What about education and employment? Happy people, compared to those who are less happy, are more likely to graduate from college and secure more meaningful and engaging jobs. Once they obtain a job, they are also more likely to succeed (Lyubomirsky et al., 2005). While education shows a positive (but weak)

correlation with happiness, intelligence is not appreciably related to happiness (Diener et al., 1999).

Does religiosity correlate with happiness? In general, the answer is yes (Hackney & Sanders, 2003). However, the relationship between religiosity and happiness depends on societal circumstances. Nations and states with more difficult living conditions (e.g., widespread hunger and low life expectancy) tend to be more highly religious than societies with more favorable living conditions. Among those who live in nations with difficult living conditions, religiosity is associated with greater well-being; in nations with more favorable living conditions, religious and nonreligious individuals report similar levels of well-being (Diener, Tay, & Myers, 2011).

Clearly the living conditions of one's nation can influence factors related to happiness. What about the influence of one's culture? To the extent that people possess characteristics that are highly valued by their culture, they tend to be happier (Diener, 2012). For example, self-esteem is a stronger predictor of life satisfaction in individualistic cultures than in collectivistic cultures (Diener, Diener, & Diener, 1995), and extraverted people tend to be happier in extraverted cultures than in introverted cultures (Fulmer et al., 2010).

So we've identified many factors that exhibit some correlation to happiness. What factors don't show a correlation? Researchers have studied both parenthood and physical attractiveness as potential contributors to happiness, but no link has been identified. Although people tend to believe that parenthood is central to a meaningful and fulfilling life, aggregate findings from a range of countries indicate that people who do not have children are generally happier than those who do (Hansen, 2012). And although one's perceived level of attractiveness seems to predict happiness, a person's objective physical attractiveness is only weakly correlated with her happiness (Diener, Wolsic, & Fujita, 1995).

LIFE EVENTS AND HAPPINESS

An important point should be considered regarding happiness. People are often poor at affective forecasting: predicting the intensity and duration of their future emotions (Wilson & Gilbert, 2003). In one study, nearly all newlywed spouses predicted their marital satisfaction would remain stable or improve over the following four years; despite this high level of initial optimism, their marital satisfaction actually declined during this period (Lavner, Karner, & Bradbury, 2013). In addition, we are often incorrect when estimating how our long-term happiness would change for the better or worse in response to certain life events. For example, it is easy for many of us to imagine how euphoric we would feel if we won the lottery, were asked on a date by an attractive celebrity, or were offered our dream job. It is also easy to understand how long-suffering fans of the Chicago Cubs baseball team, which has not won a World Series championship since 1908, think they would feel permanently elated if their team would finally win another World Series. Likewise, it easy to predict that we would feel permanently miserable if we suffered a crippling accident or if a romantic relationship ended.

However, something similar to sensory adaptation often occurs when people experience emotional reactions to life events. In much the same way our senses adapt to changes in stimulation (e.g., our eyes adapting to bright light after walking out of the darkness of a movie theater into the bright afternoon sun), we eventually adapt to changing emotional circumstances in our lives (Brickman & Campbell, 1971; Helson, 1964). When an event that provokes positive or negative emotions occurs, at first we tend to experience its emotional impact at full intensity. We feel a burst of pleasure following such things as a marriage proposal, birth of a child, acceptance to law school, an inheritance, and the like; as you might imagine, lottery winners experience a surge of happiness after hitting the jackpot (Lutter, 2007). Likewise, we experience a

surge of misery following widowhood, a divorce, or a layoff from work. In the long run, however, we eventually adjust to the emotional new normal; the emotional impact of the event tends to erode, and we eventually revert to our original baseline happiness levels. Thus, what was at first a thrilling lottery windfall or World Series championship eventually loses its luster and becomes the status quo. Indeed, dramatic life events have much less long-lasting impact on happiness than might be expected (Brickman, Coats, & Janoff-Bulman, 1978).

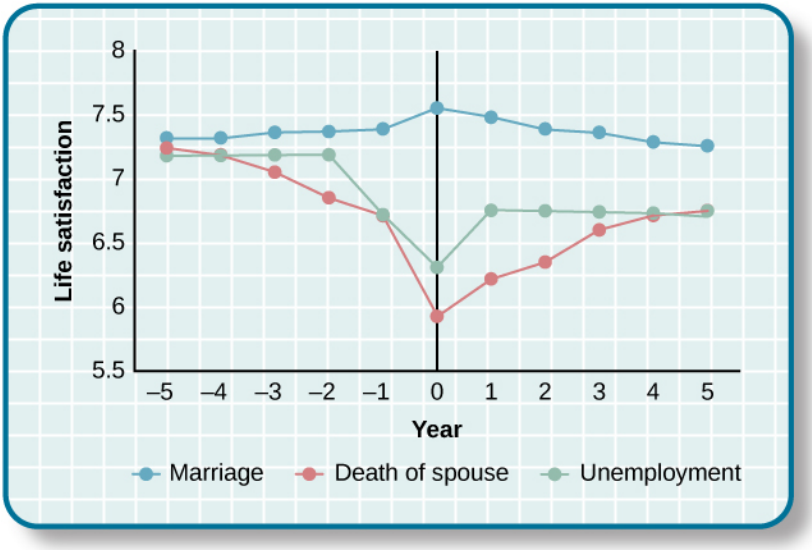
(a)

(b)

(a) Long-suffering Chicago Cub fans would no doubt feel elated if their team won a World Series championship, a feat that has not been accomplished by that franchise in over a century. (b) In ways that are similar, those who play the lottery rightfully think that choosing the correct numbers and winning millions would lead to a surge in happiness. However, the initial burst of elation following such elusive events would most likely erode with time. (credit a: modification of work by Phil Roeder; credit b: modification of work by Robert S. Donovan)

Recently, some have raised questions concerning the extent to which important life events can permanently alter people's happiness set points (Diener, Lucas, & Scollon, 2006). Evidence from a number of investigations suggests that, in some circumstances, happiness levels do not revert to their original positions. For example, although people generally tend to adapt to

marriage so that it no longer makes them happier or unhappier than before, they often do not fully adapt to unemployment or severe disabilities (Diener, 2012). This is based on longitudinal data from a sample of over 3,000 German respondents, shows life satisfaction scores several years before, during, and after various life events, and it illustrates how people adapt (or fail to adapt) to these events. German respondents did not get lasting emotional boosts from marriage; instead, they reported brief increases in happiness, followed by quick adaptation. In contrast, widows and those who had been laid off experienced sizeable decreases in happiness that appeared to result in long-term changes in life satisfaction (Diener et al., 2006). Further, longitudinal data from the same sample showed that happiness levels changed significantly over time for nearly a quarter of respondents, with 9% showing major changes (Fujita & Diener, 2005). Thus, long-term happiness levels can and do change for some people.



This graphs shows life satisfaction scores several years before and after three significant life events (0 represents the year the event happened) (Diener et al., 2006).

INCREASING HAPPINESS

Some recent findings about happiness provide an optimistic picture, suggesting that real changes in happiness are possible. For example, thoughtfully developed well-being interventions designed to augment people's baseline levels of happiness may increase happiness in ways that are permanent and long-lasting, not just temporary. These changes in happiness may be targeted at individual, organizational, and societal levels (Diener et al., 2006). Researchers in one study found that a series of happiness interventions involving such exercises as writing down three good things that occurred each day led to increases in happiness that lasted over six months (Seligman et al., 2005).

Measuring happiness and well-being at the societal level over

time may assist policy makers in determining if people are generally happy or miserable, as well as when and why they might feel the way they do. Studies show that average national happiness scores (over time and across countries) relate strongly to six key variables: per capita gross domestic product (GDP, which reflects a nation's economic standard of living), social support, freedom to make important life choices, healthy life expectancy, freedom from perceived corruption in government and business, and generosity (Helliwell et al., 2013). Investigating why people are happy or unhappy might help policymakers develop programs that increase happiness and well-being within a society (Diener et al., 2006). Resolutions about contemporary political and social issues that are frequent topics of debate—such as poverty, taxation, affordable health care and housing, clean air and water, and income inequality—might be best considered with people's happiness in mind.

POSITIVE PSYCHOLOGY

In 1998, Seligman (the same person who conducted the learned helplessness experiments mentioned earlier), who was then president of the American Psychological Association, urged psychologists to focus more on understanding how to build human strength and psychological well-being. In deliberately setting out to create a new direction and new orientation for psychology, Seligman helped establish a growing movement and field of research called positive psychology (Compton, 2005). In a very general sense, positive psychology can be thought of as the science of happiness; it is an area of study that seeks to identify and promote those qualities that lead to greater fulfillment in our lives. This field looks at people's strengths and what helps individuals to lead happy, contented lives, and it moves away from focusing

on people's pathology, faults, and problems. According to Seligman and Csikszentmihalyi (2000), positive psychology,

> at the subjective level is about valued subjective experiences: well-being, contentment, and satisfaction (in the past); hope and optimism (for the future); and... happiness (in the present). At the individual level, it is about positive individual traits: the capacity for love and vocation, courage, interpersonal skill, aesthetic sensibility, perseverance, forgiveness, originality, future mindedness, spirituality, high talent, and wisdom. (p. 5)

Some of the topics studied by positive psychologists include altruism and empathy, creativity, forgiveness and compassion, the importance of positive emotions, enhancement of immune system functioning, savoring the fleeting moments of life, and strengthening virtues as a way to increase authentic happiness (Compton, 2005). Recent efforts in the field of positive psychology have focused on extending its principles toward peace and well-being at the level of the global community. In a war-torn world in which conflict, hatred, and distrust are common, such an extended "positive peace psychology" could have important implications for understanding how to overcome oppression and work toward global peace (Cohrs, Christie, White, & Das, 2013).

The Center for Investigating

Healthy Minds

On the campus of the University of Wisconsin–Madison, the Center for Investigating Healthy Minds at the Waisman Center conducts rigorous scientific research on healthy aspects of the mind, such as kindness, forgiveness, compassion, and mindfulness. Established in 2008 and led by renowned neuroscientist Dr. Richard J. Davidson, the Center examines a wide range of ideas, including such things as a kindness curriculum in schools, neural correlates of prosocial behavior, psychological effects of Tai Chi training, digital games to foster prosocial behavior in children, and the effectiveness of yoga and breathing exercises in reducing symptoms of post-traumatic stress disorder.

According to its website, the Center was founded after Dr. Davidson was challenged by His Holiness, the 14th Dalai Lama, "to apply the rigors of science to study positive qualities of mind" (Center for Investigating Health Minds, 2013). The Center continues to conduct scientific research with the aim of developing mental health training approaches that help people to live happier, healthier lives).

POSITIVE AFFECT AND OPTIMISM

Taking a cue from positive psychology, extensive research over the last 10-15 years has examined the importance of positive psychological attributes in physical well-being. Qualities that help promote psychological well-being (e.g., having meaning and purpose in life, a sense of autonomy, positive emotions, and satisfaction with life) are linked with a range of favorable health outcomes (especially improved cardiovascular health) mainly through their relationships with biological functions and health behaviors (such as diet, physical activity, and sleep quality) (Boehm

& Kubzansky, 2012). The quality that has received attention is positive affect, which refers to pleasurable engagement with the environment, such as happiness, joy, enthusiasm, alertness, and excitement (Watson, Clark, & Tellegen, 1988). The characteristics of positive affect, as with negative affect (discussed earlier), can be brief, long-lasting, or trait-like (Pressman & Cohen, 2005). Independent of age, gender, and income, positive affect is associated with greater social connectedness, emotional and practical support, adaptive coping efforts, and lower depression; it is also associated with longevity and favorable physiological functioning (Steptoe, O'Donnell, Marmot, & Wardle, 2008).

Positive affect also serves as a protective factor against heart disease. In a 10-year study of Nova Scotians, the rate of heart disease was 22% lower for each one-point increase on the measure of positive affect, from 1 (no positive affect expressed) to 5 (extreme positive affect) (Davidson, Mostofsky, & Whang, 2010). In terms of our health, the expression, "don't worry, be happy" is helpful advice indeed. There has also been much work suggesting that optimism—the general tendency to look on the bright side of things—is also a significant predictor of positive health outcomes.

Although positive affect and optimism are related in some ways, they are not the same (Pressman & Cohen, 2005). Whereas positive affect is mostly concerned with positive feeling states, optimism has been regarded as a generalized tendency to expect that good things will happen (Chang, 2001). It has also been conceptualized as a tendency to view life's stressors and difficulties as temporary and external to oneself (Peterson & Steen, 2002). Numerous studies over the years have consistently shown that optimism is linked to longevity, healthier behaviors, fewer postsurgical complications, better immune functioning among men with prostate cancer, and better treatment adherence (Rasmussen & Wallio, 2008). Further, optimistic people report fewer physical symptoms, less pain, better physical functioning, and are less likely to be rehospitalized following heart surgery (Rasmussen, Scheier, & Greenhouse, 2009).

FLOW

Another factor that seems to be important in fostering a deep sense of well-being is the ability to derive flow from the things we do in life. Flow is described as a particular experience that is so engaging and engrossing that it becomes worth doing for its own sake (Csikszentmihalyi, 1997). It is usually related to creative endeavors and leisure activities, but it can also be experienced by workers who like their jobs or students who love studying (Csikszentmihalyi, 1999). Many of us instantly recognize the notion of flow. In fact, the term derived from respondents' spontaneous use of the term when asked to describe how it felt when what they were doing was going well. When people experience flow, they become involved in an activity to the point where they feel they lose themselves in the activity. They effortlessly maintain their concentration and focus, they feel as though they have complete control of their actions, and time seems to pass more quickly than usual (Csikszentmihalyi, 1997). Flow is considered a pleasurable experience, and it typically occurs when people are engaged in challenging activities that require skills and knowledge they know they possess. For example, people would be more likely report flow experiences in relation to their work or hobbies than in relation to eating. When asked the question, "Do you ever get involved in something so deeply that nothing else seems to matter, and you lose track of time?" about 20% of Americans and Europeans report having these flow-like experiences regularly (Csikszentmihalyi, 1997).

Although wealth and material possessions are nice to have, the notion of flow suggests that neither are prerequisites for a happy and fulfilling life. Finding an activity that you are truly enthusiastic about, something so absorbing that doing it is reward itself (whether it be playing tennis, studying Arabic, writing children's novels, or cooking lavish meals) is perhaps the real key. According to Csikszentmihalyi (1999), creating conditions that make flow

experiences possible should be a top social and political priority. How might this goal be achieved? How might flow be promoted in school systems? In the workplace? What potential benefits might be accrued from such efforts?

In an ideal world, scientific research endeavors should inform us on how to bring about a better world for all people. The field of positive psychology promises to be instrumental in helping us understand what truly builds hope, optimism, happiness, healthy relationships, flow, and genuine personal fulfillment.

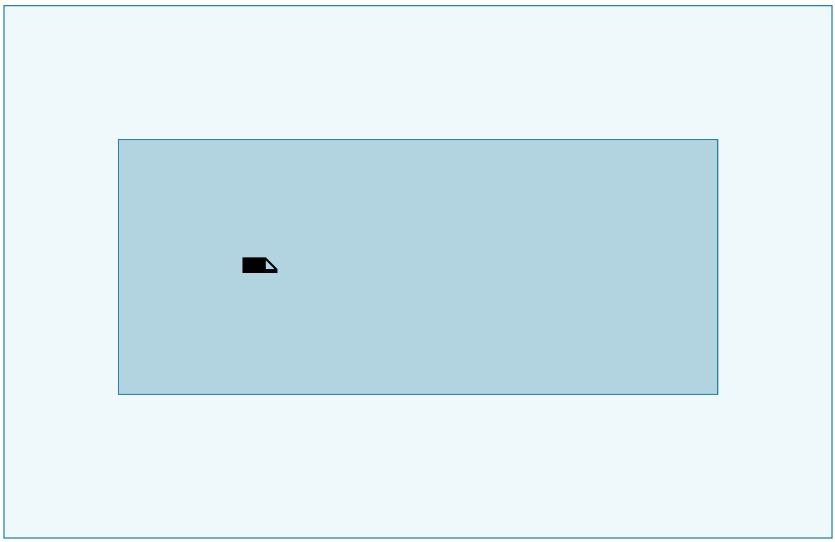
TEST YOUR UNDERSTANDING

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=203#h5p-229

SUMMARY

Happiness is conceptualized as an enduring state of mind that consists of the capacity to experience pleasure in daily life, as well as the ability to engage one's skills and talents to enrich one's life and the lives of others. Although people around the world generally report that they are happy, there are differences in average happiness levels across nations. Although people have a tendency to overestimate the extent to which their happiness set points would change for the better or for the worse following certain life events, researchers have identified a number of factors that are consistently related to happiness. In recent years, positive psychology has emerged as an area of study seeking to identify and promote qualities that lead to greater happiness and fulfillment in our lives. These components include positive affect, optimism, and flow.

REVIEW QUESTIONS



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CRITICAL THINKING QUESTIONS

In considering the three dimensions of happiness discussed in this section (the pleasant life, the good life, and the meaningful life), what are some steps you could take to improve your personal level of

happiness?

Answers will vary, but may include mentioning things that boost positive emotions (the pleasant life), developing and using skills and talents (the good life), and using one's talents to help others (the meaningful life).

The day before the drawing of a $300 million Powerball lottery, you notice that a line of people waiting to buy their Powerball tickets is stretched outside the door of a nearby convenience store. Based on what you've learned, provide some perspective on why these people are doing this, and what would likely happen if one of these individuals happened to pick the right numbers.

These individuals' affective forecasting is such that they believe their lives would be immeasurably happier if they won the lottery. Although winning would certainly lead to a surge of euphoria in the short term, long term they would likely adjust, and their happiness levels would likely return to normal. This fact is lost on most people, especially when considering the intensity and duration of their emotions following a major life event.

PERSONAL APPLICATION QUESTION

Think of an activity you participate in that you find engaging and absorbing. For example, this might be something like playing video games, reading, or a hobby. What are your experiences typically like while engaging in this activity? Do your experiences conform to the notion of flow? If so, how? Do you think these experiences have enriched your life? Why or why not?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

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PSYCHOLOGICAL DISORDERS

A wreath is laid in memoriam to victims of the Washington Navy Yard shooting. (credit: modification of work by D. Myles Cullen, US Department of Defense)

On Monday, September 16, 2013, a gunman killed 12 people as the workday began at the Washington Navy Yard in Washington, DC. Aaron Alexis, 34, had a troubled history: he thought that he was being controlled by radio waves. He called the police to complain about voices in his head and being under surveillance by "shadowy forces" (Thomas, Levine, Date, & Cloherty, 2013). While Alexis's

actions cannot be excused, it is clear that he had some form of mental illness. Mental illness is not necessarily a cause of violence; it is far more likely that the mentally ill will be victims rather than perpetrators of violence (Stuart, 2003). If, however, Alexis had received the help he needed, this tragedy might have been averted.

Learning Objectives

By the end of this section, you will be able to:

- Understand the problems inherent in defining the concept of psychological disorder
- Describe what is meant by harmful dysfunction
- Identify the formal criteria that thoughts, feelings, and behaviors must meet to be considered abnormal and, thus, symptomatic of a psychological disorder

A psychological disorder is a condition characterized by abnormal thoughts, feelings, and behaviors. Psychopathology is the study of psychological disorders, including their symptoms, etiology (i.e., their causes), and treatment. The term *psychopathology* can also refer to the manifestation of a psychological disorder. Although consensus can be difficult, it is extremely important for mental health professionals to agree on what kinds of thoughts, feelings, and behaviors are truly abnormal in the sense that they genuinely indicate the presence of psychopathology. Certain patterns of behavior and inner experience can easily be labeled as abnormal and clearly signify some kind of psychological disturbance. The person who washes his hands 40 times per day and the person who claims to hear the voices of demons exhibit behaviors and inner experiences that most would regard as abnormal: beliefs and behaviors that suggest the existence of a psychological disorder. But, consider the nervousness a young man feels when talking to attractive women or the loneliness and longing for home a

freshman experiences during her first semester of college—these feelings may not be regularly present, but they fall in the range of normal. So, what kinds of thoughts, feelings, and behaviors represent a true psychological disorder? Psychologists work to distinguish psychological disorders from inner experiences and behaviors that are merely situational, idiosyncratic, or unconventional.

DEFINITION OF A PSYCHOLOGICAL DISORDER

Perhaps the simplest approach to conceptualizing psychological disorders is to label behaviors, thoughts, and inner experiences that are atypical, distressful, dysfunctional, and sometimes even dangerous, as signs of a disorder. For example, if you ask a classmate for a date and you are rejected, you probably would feel a little dejected. Such feelings would be normal. If you felt extremely depressed—so much so that you lost interest in activities, had difficulty eating or sleeping, felt utterly worthless, and contemplated suicide—your feelings would be atypical, would deviate from the norm, and could signify the presence of a psychological disorder. Just because something is atypical, however, does not necessarily mean it is disordered.

For example, only about 4% of people in the United States have red hair, so red hair is considered an atypical characteristic, but it is not considered disordered, it's just unusual. And it is less unusual in Scotland, where approximately 13% of the population has red hair ("DNA Project Aims," 2012). As you will learn, some disorders, although not exactly typical, are far from atypical, and the rates in which they appear in the population are surprisingly high.

(a)

(b)

(c)

Red hair is considered unusual, but not abnormal. (a) Isla Fischer, (b) Prince Harry, and (c) Marcia Cross are three natural redheads. (credit a: modification of work by Richard Goldschmidt; credit b: modification of work by Glyn Lowe; credit c: modification of work by Kirk Weaver)

If we can agree that merely being atypical is an insufficient criterion for a having a psychological disorder, is it reasonable to consider behavior or inner experiences that differ from widely expected cultural values or expectations as disordered? Using this criterion, a woman who walks around a subway platform wearing a heavy winter coat in July while screaming obscenities at strangers may be considered as exhibiting symptoms of a psychological disorder. Her actions and clothes violate socially accepted rules governing appropriate dress and behavior; these characteristics are atypical.

CULTURAL EXPECTATIONS

Violating cultural expectations is not, in and of itself, a satisfactory means of identifying the presence of a psychological disorder. Since behavior varies from one culture to another, what may be expected and considered appropriate in one culture may not be viewed as such in other cultures. For example, returning a stranger's smile is expected in the United States because a pervasive social norm dictates that we reciprocate friendly

gestures. A person who refuses to acknowledge such gestures might be considered socially awkward—perhaps even disordered—for violating this expectation. However, such expectations are not universally shared. Cultural expectations in Japan involve showing reserve, restraint, and a concern for maintaining privacy around strangers. Japanese people are generally unresponsive to smiles from strangers (Patterson et al., 2007). Eye contact provides another example. In the United States and Europe, eye contact with others typically signifies honesty and attention. However, most Latin-American, Asian, and African cultures interpret direct eye contact as rude, confrontational, and aggressive (Pazain, 2010). Thus, someone who makes eye contact with you could be considered appropriate and respectful or brazen and offensive, depending on your culture.

Eye contact is one of many social gestures that vary from culture to culture. (credit: Joi Ito)

Hallucinations (seeing or hearing things that are not physically present) in Western societies is a violation of cultural expectations,

and a person who reports such inner experiences is readily labeled as psychologically disordered. In other cultures, visions that, for example, pertain to future events may be regarded as normal experiences that are positively valued (Bourguignon, 1970). Finally, it is important to recognize that cultural norms change over time: what might be considered typical in a society at one time may no longer be viewed this way later, similar to how fashion trends from one era may elicit quizzical looks decades later—imagine how a headband, legwarmers, and the big hair of the 1980s would go over on your campus today.

The Myth of Mental Illness

In the 1950s and 1960s, the concept of mental illness was widely criticized. One of the major criticisms focused on the notion that mental illness was a "myth that justifies psychiatric intervention in socially disapproved behavior" (Wakefield, 1992). Thomas Szasz (1960), a noted psychiatrist, was perhaps the biggest proponent of this view. Szasz argued that the notion of mental illness was invented by society (and the mental health establishment) to stigmatize and subjugate people whose behavior violates accepted social and legal norms. Indeed, Szasz suggested that what appear to be symptoms of mental illness are more appropriately characterized as "problems in living" (Szasz, 1960).

In his 1961 book, *The Myth of Mental Illness: Foundations of a Theory of Personal Conduct*, Szasz expressed his disdain for the

concept of mental illness and for the field of psychiatry in general (Oliver, 2006). The basis for Szasz's attack was his contention that detectable abnormalities in bodily structures and functions (e.g., infections and organ damage or dysfunction) represent the defining features of genuine illness or disease, and because symptoms of purported mental illness are not accompanied by such detectable abnormalities, so-called psychological disorders are not disorders at all. Szasz (1961/2010) proclaimed that "disease or illness can only affect the body; hence, there can be no mental illness" (p. 267).

Today, we recognize the extreme level of psychological suffering experienced by people with psychological disorders: the painful thoughts and feelings they experience, the disordered behavior they demonstrate, and the levels of distress and impairment they exhibit. This makes it very difficult to deny the reality of mental illness.

However controversial Szasz's views and those of his supporters might have been, they have influenced the mental health community and society in several ways. First, lay people, politicians, and professionals now often refer to mental illness as mental health "problems," implicitly acknowledging the "problems in living" perspective Szasz described (Buchanan-Barker & Barker, 2009). Also influential was Szasz's view of homosexuality. Szasz was perhaps the first psychiatrist to openly challenge the idea that homosexuality represented a form of mental illness or disease (Szasz, 1965). By challenging the idea that homosexuality represented a form a mental illness, Szasz helped pave the way for the social and civil rights that gay and lesbian people now have (Barker, 2010). His work also inspired legal changes that protect the rights of people in psychiatric institutions and allow such individuals a greater degree of influence and responsibility over their lives (Buchanan-Barker & Barker, 2009).

HARMFUL DYSFUNCTION

If none of the criteria discussed so far is adequate by itself to define the presence of a psychological disorder, how can a disorder be conceptualized? Many efforts have been made to identify the specific dimensions of psychological disorders, yet none is entirely satisfactory. No universal definition of psychological disorder exists that can apply to all situations in which a disorder is thought to be present (Zachar & Kendler, 2007). However, one of the more influential conceptualizations was proposed by Wakefield (1992), who defined psychological disorder as a harmful dysfunction. Wakefield argued that natural internal mechanisms—that is, psychological processes honed by evolution, such as cognition, perception, and learning—have important functions, such as enabling us to experience the world the way others do and to engage in rational thought, problem solving, and communication. For example, learning allows us to associate a fear with a potential danger in such a way that the intensity of fear is roughly equal to the degree of actual danger. Dysfunction occurs when an internal mechanism breaks down and can no longer perform its normal function. But, the presence of a dysfunction by itself does not determine a disorder. The dysfunction must be harmful in that it leads to negative consequences for the individual or for others, as judged by the standards of the individual's culture. The harm may include significant internal anguish (e.g., high levels of anxiety or depression) or problems in day-to-day living (e.g., in one's social or work life).

To illustrate, Janet has an extreme fear of spiders. Janet's fear might be considered a dysfunction in that it signals that the internal mechanism of learning is not working correctly (i.e., a faulty process prevents Janet from appropriately associating the magnitude of her fear with the actual threat posed by spiders). Janet's fear of spiders has a significant negative influence on her life: she avoids all situations in which she suspects spiders to be present (e.g., the

basement or a friend's home), and she quit her job last month because she saw a spider in the restroom at work and is now unemployed. According to the harmful dysfunction model, Janet's condition would signify a disorder because (a) there is a dysfunction in an internal mechanism, and (b) the dysfunction has resulted in harmful consequences. Similar to how the symptoms of physical illness reflect dysfunctions in biological processes, the symptoms of psychological disorders presumably reflect dysfunctions in mental processes. The internal mechanism component of this model is especially appealing because it implies that disorders may occur through a breakdown of biological functions that govern various psychological processes, thus supporting contemporary neurobiological models of psychological disorders (Fabrega, 2007).

THE AMERICAN PSYCHIATRIC ASSOCIATION (APA) DEFINITION

Many of the features of the harmful dysfunction model are incorporated in a formal definition of psychological disorder developed by the American Psychiatric Association (APA). According to the APA (2013), a psychological disorder is a condition that is said to consist of the following:

- **There are significant disturbances in thoughts, feelings, and behaviors**. A person must experience inner states (e.g., thoughts and/or feelings) and exhibit behaviors that are clearly disturbed—that is, unusual, but in a negative, self-defeating way. Often, such disturbances are troubling to those around the individual who experiences them. For example, an individual who is uncontrollably preoccupied by thoughts of germs spends hours each day bathing, has inner experiences, and

displays behaviors that most would consider atypical and negative (disturbed) and that would likely be troubling to family members.

- **The disturbances reflect some kind of biological, psychological, or developmental dysfunction**. Disturbed patterns of inner experiences and behaviors should reflect some flaw (dysfunction) in the internal biological, psychological, and developmental mechanisms that lead to normal, healthy psychological functioning. For example, the hallucinations observed in schizophrenia could be a sign of brain abnormalities.
- **The disturbances lead to significant distress or disability in one's life**. A person's inner experiences and behaviors are considered to reflect a psychological disorder if they cause the person considerable distress, or greatly impair his ability to function as a normal individual (often referred to as functional impairment, or occupational and social impairment). As an illustration, a person's fear of social situations might be so distressing that it causes the person to avoid all social situations (e.g., preventing that person from being able to attend class or apply for a job).
- **The disturbances do not reflect expected or culturally approved responses to certain events**. Disturbances in thoughts, feelings, and behaviors must be socially unacceptable responses to certain events that often happen in life. For example, it is perfectly natural (and expected) that a person would experience great sadness and might wish to be left alone following the death of a close family member. Because such reactions are in some ways culturally expected, the individual would not be assumed to signify a mental disorder.

Some believe that there is no essential criterion or set of criteria that can definitively distinguish all cases of disorder from nondisorder (Lilienfeld & Marino, 1999). In truth, no single approach to defining a psychological disorder is adequate by itself, nor is there universal agreement on where the boundary is between disordered and not disordered. From time to time we all experience anxiety, unwanted thoughts, and moments of sadness; our behavior at other times may not make much sense to ourselves or to others. These inner experiences and behaviors can vary in their intensity, but are only considered disordered when they are highly disturbing to us and/or others, suggest a dysfunction in normal mental functioning, and are associated with significant distress or disability in social or occupational activities.

TEST YOUR UNDERSTANDING

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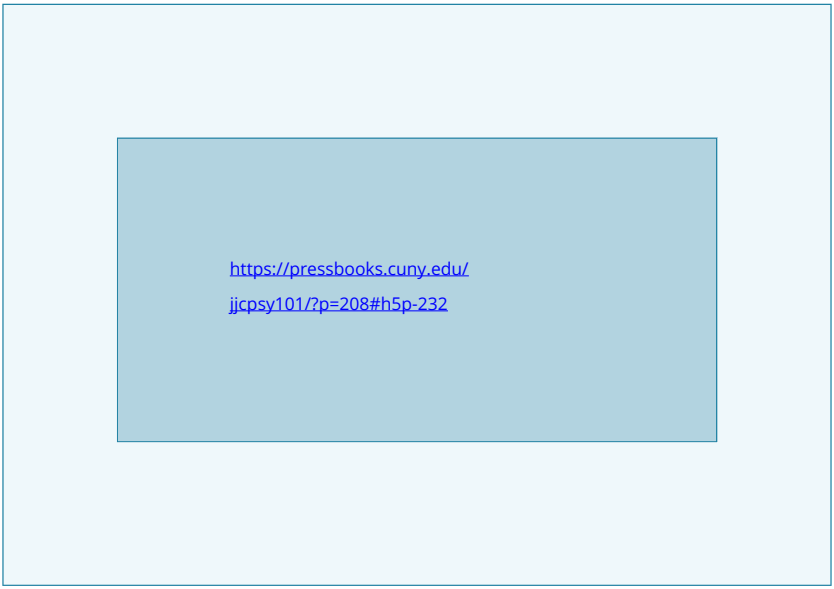

SUMMARY

Psychological disorders are conditions characterized by abnormal thoughts, feelings, and behaviors. Although challenging, it is essential for psychologists and mental health professionals to agree on what kinds of inner experiences and behaviors constitute the presence of a psychological disorder. Inner experiences and behaviors that are atypical or violate social norms could signify the presence of a disorder; however, each of these criteria alone is inadequate. Harmful dysfunction describes the view that psychological disorders result from the inability of an internal mechanism to perform its natural function. Many of the features of harmful dysfunction conceptualization have been incorporated in the APA's formal definition of psychological disorders. According to this definition, the presence of a psychological disorder is signaled

by significant disturbances in thoughts, feelings, and behaviors; these disturbances must reflect some kind of dysfunction (biological, psychological, or developmental), must cause significant impairment in one's life, and must not reflect culturally expected reactions to certain life events.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTION

Discuss why thoughts, feelings, or behaviors that are merely atypical or unusual would not necessarily signify the presence of a psychological disorder. Provide an example.

Just because something is atypical or unusual does not mean it is disordered. A person may experience atypical inner experiences or exhibit unusual behaviors, but she would not be considered disordered if they are not distressing, disturbing, or reflecting a dysfunction. For example, a classmate might stay up all night studying before exams; although atypical, this behavior is unlikely to possess any of the other criteria for psychological disorder mentioned previously.

PERSONAL APPLICATION QUESTION

Identify a behavior that is considered unusual or abnormal in your own culture; however, it would be considered normal and expected in another culture.

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

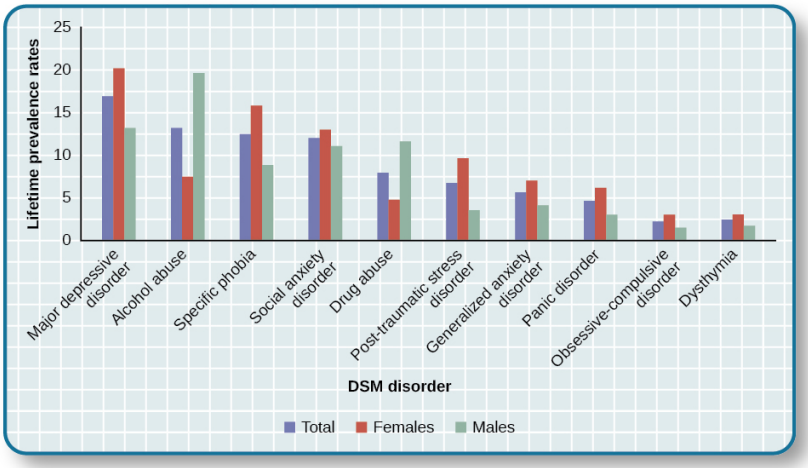
- Explain why classification systems are necessary in the study of psychopathology
- Describe the basic features of the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)
- Discuss changes in the DSM over time, including criticisms of the current edition
- Identify which disorders are generally the most common

A first step in the study of psychological disorders is carefully and systematically discerning significant signs and symptoms. How do mental health professionals ascertain whether or not a person's inner states and behaviors truly represent a psychological disorder? Arriving at a proper diagnosis—that is, appropriately identifying and labeling a set of defined symptoms—is absolutely crucial. This process enables professionals to use a common language with others in the field and aids in communication about the disorder with the patient, colleagues and the public. A proper diagnosis is an essential element to guide proper and successful treatment. For these reasons, classification systems that organize psychological disorders systematically are necessary.

THE DIAGNOSTIC AND STATISTICAL MANUAL

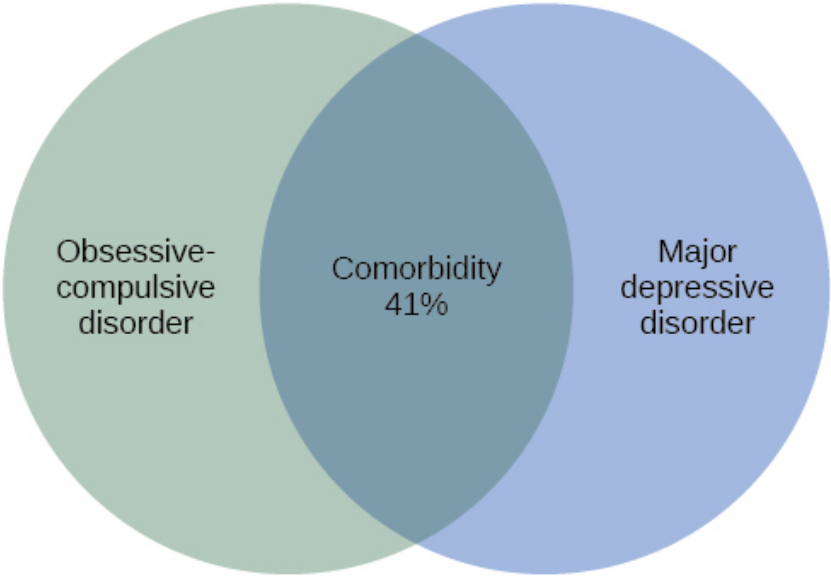
OF MENTAL DISORDERS (DSM)

Although a number of classification systems have been developed over time, the one that is used by most mental health professionals in the United States is the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5), published by the American Psychiatric Association (2013). (Note that the American Psychiatric Association differs from the American Psychological Association; both are abbreviated APA.) The first edition of the DSM, published in 1952, classified psychological disorders according to a format developed by the U.S. Army during World War II (Clegg, 2012). In the years since, the DSM has undergone numerous revisions and editions. The most recent edition, published in 2013, is the DSM-5 (APA, 2013). The DSM-5 includes many categories of disorders (e.g., anxiety disorders, depressive disorders, and dissociative disorders). Each disorder is described in detail, including an overview of the disorder (diagnostic features), specific symptoms required for diagnosis (diagnostic criteria), prevalence information (what percent of the population is thought to be afflicted with the disorder), and risk factors associated with the disorder. The chart below shows lifetime prevalence rates—the percentage of people in a population who develop a disorder in their lifetime—of various psychological disorders among U.S. adults. These data were based on a national sample of 9,282 U.S. residents (National Comorbidity Survey, 2007).



The graph shows the breakdown of psychological disorders, comparing the percentage prevalence among adult males and adult females in the United States. Because the data is from 2007, the categories shown here are from the DSM-IV, which has been supplanted by the DSM-5. Most categories remain the same; however, alcohol abuse now falls under a broader Alcohol Use Disorder category.

The DSM-5 also provides information about comorbidity; the co-occurrence of two disorders. For example, the DSM-5 mentions that 41% of people with obsessive-compulsive disorder (OCD) also meet the diagnostic criteria for major depressive disorder. Drug use is highly comorbid with other mental illnesses; 6 out of 10 people who have a substance use disorder also suffer from another form of mental illness (National Institute on Drug Abuse [NIDA], 2007).



Obsessive-compulsive disorder and major depressive disorder frequently occur in the same person.

The DSM has changed considerably in the half-century since it was originally published. The first two editions of the DSM, for example, listed homosexuality as a disorder; however, in 1973, the APA voted to remove it from the manual (Silverstein, 2009). Additionally, beginning with the DSM-III in 1980, mental disorders have been described in much greater detail, and the number of diagnosable conditions has grown steadily, as has the size of the manual itself. DSM-I included 106 diagnoses and was 130 total pages, whereas DSM-III included more than 2 times as many diagnoses (265) and was nearly seven times its size (886 total pages) (Mayes & Horowitz, 2005). Although DSM-5 is longer than DSM-IV, the volume includes only 237 disorders, a decrease from the 297 disorders that were listed in DSM-IV. The latest edition, DSM-5, includes revisions in the organization and naming of categories and in the diagnostic criteria for various disorders (Regier, Kuhl, & Kupfer, 2012), while emphasizing careful

consideration of the importance of gender and cultural difference in the expression of various symptoms (Fisher, 2010).

Some believe that establishing new diagnoses might overpathologize the human condition by turning common human problems into mental illnesses (The Associated Press, 2013). Indeed, the finding that nearly half of all Americans will meet the criteria for a DSM disorder at some point in their life (Kessler et al., 2005) likely fuels much of this skepticism. The DSM-5 is also criticized on the grounds that its diagnostic criteria have been loosened, thereby threatening to "turn our current diagnostic inflation into diagnostic hyperinflation" (Frances, 2012, para. 22). For example, DSM-IV specified that the symptoms of major depressive disorder must not be attributable to normal bereavement (loss of a loved one). The DSM-5, however, has removed this bereavement exclusion, essentially meaning that grief and sadness after a loved one's death can constitute major depressive disorder.

THE INTERNATIONAL CLASSIFICATION OF DISEASES

A second classification system, the *International Classification of Diseases* (ICD), is also widely recognized. Published by the World Health Organization (WHO), the ICD was developed in Europe shortly after World War II and, like the DSM, has been revised several times. The categories of psychological disorders in both the DSM and ICD are similar, as are the criteria for specific disorders; however, some differences exist. Although the ICD is used for clinical purposes, this tool is also used to examine the general health of populations and to monitor the prevalence of diseases and other health problems internationally (WHO, 2013). The ICD is in its 10th edition (ICD-10); however, efforts are now underway to develop a new edition (ICD-11) that, in conjunction with the

changes in DSM-5, will help harmonize the two classification systems as much as possible (APA, 2013).

A study that compared the use of the two classification systems found that worldwide the ICD is more frequently used for clinical diagnosis, whereas the DSM is more valued for research (Mezzich, 2002). Most research findings concerning the etiology and treatment of psychological disorders are based on criteria set forth in the DSM (Oltmanns & Castonguay, 2013). The DSM also includes more explicit disorder criteria, along with an extensive and helpful explanatory text (Regier et al., 2012). The DSM is the classification system of choice among U.S. mental health professionals, and this chapter is based on the DSM paradigm.

THE COMPASSIONATE VIEW OF PSYCHOLOGICAL DISORDERS

As these disorders are outlined, please bear two things in mind. First, remember that psychological disorders represent *extremes* of inner experience and behavior. If, while reading about these disorders, you feel that these descriptions begin to personally characterize you, do not worry—this moment of enlightenment probably means nothing more than you are normal. Each of us experiences episodes of sadness, anxiety, and preoccupation with certain thoughts—times when we do not quite feel ourselves. These episodes should not be considered problematic unless the accompanying thoughts and behaviors become extreme and have a disruptive effect on one's life. Second, understand that people with psychological disorders are far more than just embodiments of their disorders. We do not use terms such as schizophrenics, depressives, or phobics because they are labels that objectify people who suffer from these conditions, thus promoting biased and disparaging assumptions about them. It is important to remember that a psychological disorder is not what a person *is*; it

is something that a person *has*—through no fault of his or her own. As is the case with cancer or diabetes, those with psychological disorders suffer debilitating, often painful conditions that are not of their own choosing. These individuals deserve to be viewed and treated with compassion, understanding, and dignity.

TEST YOUR UNDERSTANDING

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SUMMARY

The diagnosis and classification of psychological disorders is

essential in studying and treating psychopathology. The classification system used by most U.S. professionals is the DSM-5. The first edition of the DSM was published in 1952, and has undergone numerous revisions. The 5th and most recent edition, the DSM-5, was published in 2013. The diagnostic manual includes a total of 237 specific diagnosable disorders, each described in detail, including its symptoms, prevalence, risk factors, and comorbidity. Over time, the number of diagnosable conditions listed in the DSM has grown steadily, prompting criticism from some. Nevertheless, the diagnostic criteria in the DSM are more explicit than that of any other system, which makes the DSM system highly desirable for both clinical diagnosis and research.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Describe the DSM-5. What is it, what kind of information does it contain, and why is it important to the study and treatment of psychological disorders?

The DSM-5 is the classification system of psychological disorders preferred by most U.S. mental health professionals, and it is published by the American Psychiatric Association (APA). It consists of broad categories of disorders and specific disorders that fall within each category. Each disorder has an explicit description of its symptoms, as well as information concerning prevalence, risk factors, and comorbidity. The DSM-5 provides a common language that enables mental health professionals to communicate effectively about sets of symptoms.

The International Classification of Diseases (ICD) and the DSM differ in various ways. What are some of the differences in these two classification systems?

The ICD is used primarily for making clinical diagnoses and more broadly for examining the general health of populations and monitoring the international prevalence of diseases and other health problems. While the DSM is also used for diagnostic

purposes, it is also highly valued as a research tool. For example, much of the data regarding the etiology and treatment of psychological disorders are based on diagnostic criteria set forth in the DSM.

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=210#h5p-237

Learning Objectives

By the end of this section, you will be able to:

- Discuss supernatural perspectives on the origin of psychological disorders, in their historical context
- Describe modern biological and psychological perspectives on the origin of psychological disorders
- Identify which disorders generally show the highest degree of heritability
- Describe the diathesis-stress model and its importance to the study of psychopathology

Scientists and mental health professionals may adopt different perspectives in attempting to understand or explain the underlying mechanisms that contribute to the development of a psychological disorder. The perspective used in explaining a psychological disorder is extremely important, in that it will consist of explicit assumptions regarding how best to study the disorder, its etiology, and what kinds of therapies or treatments are most beneficial. Different perspectives provide alternate ways for how to think about the nature of psychopathology.

SUPERNATURAL PERSPECTIVES OF PSYCHOLOGICAL DISORDERS

For centuries, psychological disorders were viewed from a

supernatural perspective: attributed to a force beyond scientific understanding. Those afflicted were thought to be practitioners of black magic or possessed by spirits (Maher & Maher, 1985). For example, convents throughout Europe in the 16th and 17th centuries reported hundreds of nuns falling into a state of frenzy in which the afflicted foamed at the mouth, screamed and convulsed, sexually propositioned priests, and confessed to having carnal relations with devils or Christ. Although, today, these cases would suggest serious mental illness; at the time, these events were routinely explained as possession by devilish forces (Waller, 2009a). Similarly, grievous fits by young girls are believed to have precipitated the witch panic in New England late in the 17th century (Demos, 1983). Such beliefs in supernatural causes of mental illness are still held in some societies today; for example, beliefs that supernatural forces cause mental illness are common in some cultures in modern-day Nigeria (Aghukwa, 2012).

In The Extraction of the Stone of Madness, a 15th century painting by Hieronymus Bosch, a practitioner is using a tool to extract an object (the supposed "stone of madness") from the head of an afflicted person.

Dancing Mania

Between the 11th and 17th centuries, a curious epidemic swept across Western Europe. Groups of people would suddenly begin to dance with wild abandon. This compulsion to dance—referred to as dancing mania—sometimes gripped thousands of people at a time. Historical accounts indicate that those afflicted would sometimes dance with bruised and bloody feet for days or weeks, screaming of terrible visions and begging priests and monks to save their souls (Waller, 2009b). What caused dancing mania is not known, but several explanations have been proposed, including spider venom and ergot poisoning ("Dancing Mania," 2011).

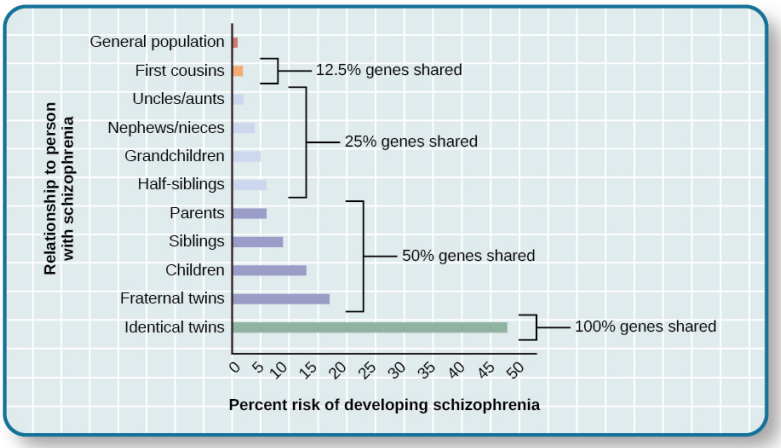
Although the cause of dancing mania, depicted in this painting, was unclear, the behavior was attributed to supernatural forces.

Historian John Waller (2009a, 2009b) has provided a comprehensive and convincing explanation of dancing mania that suggests the phenomenon was attributable to a combination of

three factors: psychological distress, social contagion, and belief in supernatural forces. Waller argued that various disasters of the time (such as famine, plagues, and floods) produced high levels of psychological distress that could increase the likelihood of succumbing to an involuntary trance state. Waller indicated that anthropological studies and accounts of possession rituals show that people are more likely to enter a trance state if they expect it to happen, and that entranced individuals behave in a ritualistic manner, their thoughts and behavior shaped by the spiritual beliefs of their culture. Thus, during periods of extreme physical and mental distress, all it took were a few people—believing themselves to have been afflicted with a dancing curse—to slip into a spontaneous trance and then act out the part of one who is cursed by dancing for days on end.

BIOLOGICAL PERSPECTIVES OF PSYCHOLOGICAL DISORDERS

The biological perspective views psychological disorders as linked to biological phenomena, such as genetic factors, chemical imbalances, and brain abnormalities; it has gained considerable attention and acceptance in recent decades (Wyatt & Midkiff, 2006). Evidence from many sources indicates that most psychological disorders have a genetic component; in fact, there is little dispute that some disorders are largely due to genetic factors.



A person's risk of developing schizophrenia increases if a relative has schizophrenia. The closer the genetic relationship, the higher the risk.

Findings such as these have led many of today's researchers to search for specific genes and genetic mutations that contribute to mental disorders. Also, sophisticated neural imaging technology in recent decades has revealed how abnormalities in brain structure and function might be directly involved in many disorders, and advances in our understanding of neurotransmitters and hormones have yielded insights into their possible connections. The biological perspective is currently thriving in the study of psychological disorders.

THE DIATHESIS-STRESS MODEL OF PSYCHOLOGICAL DISORDERS

Despite advances in understanding the biological basis of psychological disorders, the psychosocial perspective is still very important. This perspective emphasizes the importance of learning, stress, faulty and self-defeating thinking patterns, and

environmental factors. Perhaps the best way to think about psychological disorders, then, is to view them as originating from a combination of biological and psychological processes. Many develop not from a single cause, but from a delicate fusion between partly biological and partly psychosocial factors.

The diathesis-stress model (Zuckerman, 1999) integrates biological and psychosocial factors to predict the likelihood of a disorder. This diathesis-stress model suggests that people with an underlying predisposition for a disorder (i.e., a diathesis) are more likely than others to develop a disorder when faced with adverse environmental or psychological events (i.e., stress), such as childhood maltreatment, negative life events, trauma, and so on. A diathesis is not always a biological vulnerability to an illness; some diatheses may be psychological (e.g., a tendency to think about life events in a pessimistic, self-defeating way).

The key assumption of the diathesis-stress model is that both factors, diathesis and stress, are necessary in the development of a disorder. Different models explore the relationship between the two factors: the level of stress needed to produce the disorder is inversely proportional to the level of diathesis.

TEST YOUR UNDERSTANDING

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SUMMARY

Psychopathology is very complex, involving a plethora of etiological theories and perspectives. For centuries, psychological disorders were viewed primarily from a supernatural perspective and thought to arise from divine forces or possession from spirits. Some cultures continue to hold this supernatural belief. Today, many who study psychopathology view mental illness from a biological perspective, whereby psychological disorders are thought to result largely from faulty biological processes. Indeed,

scientific advances over the last several decades have provided a better understanding of the genetic, neurological, hormonal, and biochemical bases of psychopathology. The psychological perspective, in contrast, emphasizes the importance of psychological factors (e.g., stress and thoughts) and environmental factors in the development of psychological disorders. A contemporary, promising approach is to view disorders as originating from an integration of biological and psychosocial factors. The diathesis-stress model suggests that people with an underlying diathesis, or vulnerability, for a psychological disorder are more likely than those without the diathesis to develop the disorder when faced with stressful events.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTION

Why is the perspective one uses in explaining a psychological disorder important?

The perspective one uses in explaining a psychological disorder consists of assumptions that will guide how to best study and understand the nature of a disorder, including its causes, and how to most effectively treat the disorder.

PERSONAL APPLICATION QUESTION

Even today, some believe that certain occurrences have supernatural causes. Think of an event, recent or historical, for which others have provided supernatural explanation.

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=212#h5p-240

Learning Objectives

By the end of this section, you will be able to:

- Distinguish normal anxiety from pathological anxiety
- List and describe the major anxiety disorders, including their main features and prevalence
- Describe basic psychological and biological factors that are suspected to be important in the etiology of anxiety disorder

Everybody experiences anxiety from time to time. Although anxiety is closely related to fear, the two states possess important differences. Fear involves an instantaneous reaction to an imminent threat, whereas anxiety involves apprehension, avoidance, and cautiousness regarding a potential threat, danger, or other negative event (Craske, 1999). While anxiety is unpleasant to most people, it is important to our health, safety, and well-being. Anxiety motivates us to take actions—such as preparing for exams, watching our weight, showing up to work on time—that enable us to avert potential future problems. Anxiety also motivates us to avoid certain things—such as running up debts and engaging in illegal activities—that could lead to future trouble. Most individuals' level and duration of anxiety approximates the magnitude of the potential threat they face. For example, suppose a single woman in her late 30s who wishes to marry is concerned about the possibility of having to settle for a spouse who is less attractive and educated than desired. This woman likely would experience anxiety of

greater intensity and duration than would a 21-year-old college junior who is having trouble finding a date for the annual social. Some people, however, experience anxiety that is excessive, persistent, and greatly out of proportion to the actual threat; if one's anxiety has a disruptive influence on one's live, this is a strong indicator that the individual is experiencing an anxiety disorder.

Anxiety disorders are characterized by excessive and persistent fear and anxiety, and by related disturbances in behavior (APA, 2013). Although anxiety is universally experienced, anxiety disorders cause considerable distress. As a group, anxiety disorders are common: approximately 25%–30% of the U.S. population meets the criteria for at least one anxiety disorder during their lifetime (Kessler et al., 2005). Also, these disorders appear to be much more common in women than they are in men; within a 12-month period, around 23% of women and 14% of men will experience at least one anxiety disorder (National Comorbidity Survey, 2007). Anxiety disorders are the most frequently occurring class of mental disorders and are often comorbid with each other and with other mental disorders (Kessler, Ruscio, Shear, & Wittchen, 2009).

SPECIFIC PHOBIA

Phobia is a Greek word that means fear. A person diagnosed with a specific phobia (formerly known as simple phobia) experiences excessive, distressing, and persistent fear or anxiety about a specific object or situation (such as animals, enclosed spaces, elevators, or flying) (APA, 2013). Even though people realize their level of fear and anxiety in relation to the phobic stimulus is irrational, some people with a specific phobia may go to great lengths to avoid the phobic stimulus (the object or situation that triggers the fear and anxiety). Typically, the fear and anxiety a phobic stimulus elicits is disruptive to the person's life. For

example, a man with a phobia of flying might refuse to accept a job that requires frequent air travel, thus negatively affecting his career. Clinicians who have worked with people who have specific phobias have encountered many kinds of phobias.

Specific Phobias

Phobia	Feared Object or Situation
Acrophobia	heights
Aerophobia	flying
Arachnophobia	spiders
Claustrophobia	enclosed spaces
Cynophobia	dogs
Hematophobia	blood
Ophidiophobia	snakes
Taphophobia	being buried alive
Trypanophobia	injections
Xenophobia	strangers

Specific phobias are common; in the United States, around 12.5% of the population will meet the criteria for a specific phobia at some point in their lifetime (Kessler et al., 2005). One type of phobia, agoraphobia, is listed in the DSM-5 as a separate anxiety disorder. Agoraphobia, which literally means "fear of the marketplace," is characterized by intense fear, anxiety, and avoidance of situations in which it might be difficult to escape or receive help if one experiences symptoms of a panic attack (a state of extreme anxiety that we will discuss shortly). These situations include public transportation, open spaces (parking lots), enclosed spaces (stores), crowds, or being outside the home alone (APA, 2013). About 1.4% of Americans experience agoraphobia during their lifetime (Kessler et al., 2005).

ACQUISITION OF PHOBIAS THROUGH LEARNING

Many theories suggest that phobias develop through learning. Rachman (1977) proposed that phobias can be acquired through three major learning pathways. The first pathway is through classical conditioning. As you may recall, classical conditioning is a form of learning in which a previously neutral stimulus is paired with an unconditioned stimulus (UCS) that reflexively elicits an unconditioned response (UCR), eliciting the same response through its association with the unconditioned stimulus. The response is called a conditioned response (CR). For example, a child who has been bitten by a dog may come to fear dogs because of her past association with pain. In this case, the dog bite is the UCS and the fear it elicits is the UCR. Because a dog was associated with the bite, any dog may come to serve as a conditioned stimulus, thereby eliciting fear; the fear the child experiences around dogs, then, becomes a CR.

The second pathway of phobia acquisition is through vicarious learning, such as modeling. For example, a child who observes his cousin react fearfully to spiders may later express the same fears, even though spiders have never presented any danger to him. This phenomenon has been observed in both humans and nonhuman primates (Olsson & Phelps, 2007). A study of laboratory-reared monkeys readily acquired a fear of snakes after observing wild-reared monkeys react fearfully to snakes (Mineka & Cook, 1993).

The third pathway is through verbal transmission or information. For example, a child whose parents, siblings, friends, and classmates constantly tell her how disgusting and dangerous snakes are may come to acquire a fear of snakes.

Interestingly, people are more likely to develop phobias of things that do not represent much actual danger to themselves, such as animals and heights, and are less likely to develop phobias toward

things that present legitimate danger in contemporary society, such as motorcycles and weapons (Öhman & Mineka, 2001). Why might this be so? One theory suggests that the human brain is evolutionarily predisposed to more readily associate certain objects or situations with fear (Seligman, 1971). This theory argues that throughout our evolutionary history, our ancestors associated certain stimuli (e.g., snakes, spiders, heights, and thunder) with potential danger. As time progressed, the mind has become adapted to more readily develop fears of these things than of others. Experimental evidence has consistently demonstrated that conditioned fears develop more readily to fear-relevant stimuli (images of snakes and spiders) than to fear-irrelevant stimuli (images of flowers and berries) (Öhman & Mineka, 2001). Such prepared learning has also been shown to occur in monkeys. In one study (Cook & Mineka, 1989), monkeys watched videotapes of model monkeys reacting fearfully to either fear-relevant stimuli (toy snakes or a toy crocodile) or fear-irrelevant stimuli (flowers or a toy rabbit). The observer monkeys developed fears of the fear-relevant stimuli but not the fear-irrelevant stimuli.

SOCIAL ANXIETY DISORDER

Social anxiety disorder (formerly called social phobia) is characterized by extreme and persistent fear or anxiety and avoidance of social situations in which the person could potentially be evaluated negatively by others (APA, 2013). As with specific phobias, social anxiety disorder is common in the United States; a little over 12% of all Americans experience social anxiety disorder during their lifetime (Kessler et al., 2005).

The heart of the fear and anxiety in social anxiety disorder is the person's concern that he may act in a humiliating or embarrassing way, such as appearing foolish, showing symptoms of anxiety (blushing), or doing or saying something that might lead to

rejection (such as offending others). The kinds of social situations in which individuals with social anxiety disorder usually have problems include public speaking, having a conversation, meeting strangers, eating in restaurants, and, in some cases, using public restrooms. Although many people become anxious in social situations like public speaking, the fear, anxiety, and avoidance experienced in social anxiety disorder are highly distressing and lead to serious impairments in life. Adults with this disorder are more likely to experience lower educational attainment and lower earnings (Katzelnick et al., 2001), perform more poorly at work and are more likely to be unemployed (Moitra, Beard, Weisberg, & Keller, 2011), and report greater dissatisfaction with their family lives, friends, leisure activities, and income (Stein & Kean, 2000).

When people with social anxiety disorder are unable to avoid situations that provoke anxiety, they typically perform safety behaviors: mental or behavioral acts that reduce anxiety in social situations by reducing the chance of negative social outcomes. Safety behaviors include avoiding eye contact, rehearsing sentences before speaking, talking only briefly, and not talking about oneself (Alden & Bieling, 1998). Other examples of safety behaviors include the following (Marker, 2013):

- assuming roles in social situations that minimize interaction with others (e.g., taking pictures, setting up equipment, or helping prepare food)
- asking people many questions to keep the focus off of oneself
- selecting a position to avoid scrutiny or contact with others (sitting in the back of the room)
- wearing bland, neutral clothes to avoid drawing attention to oneself
- avoiding substances or activities that might cause anxiety symptoms (such as caffeine, warm clothing, and physical

exercise)

Although these behaviors are intended to prevent the person with social anxiety disorder from doing something awkward that might draw criticism, these actions usually exacerbate the problem because they do not allow the individual to disconfirm his negative beliefs, often eliciting rejection and other negative reactions from others (Alden & Bieling, 1998).

People with social anxiety disorder may resort to self-medication, such as drinking alcohol, as a means to avert the anxiety symptoms they experience in social situations (Battista & Kocovski, 2010). The use of alcohol when faced with such situations may become negatively reinforcing: encouraging individuals with social anxiety disorder to turn to the substance whenever they experience anxiety symptoms. The tendency to use alcohol as a coping mechanism for social anxiety, however, can come with a hefty price tag: a number of large scale studies have reported a high rate of comorbidity between social anxiety disorder and alcohol use disorder (Morris, Stewart, & Ham, 2005).

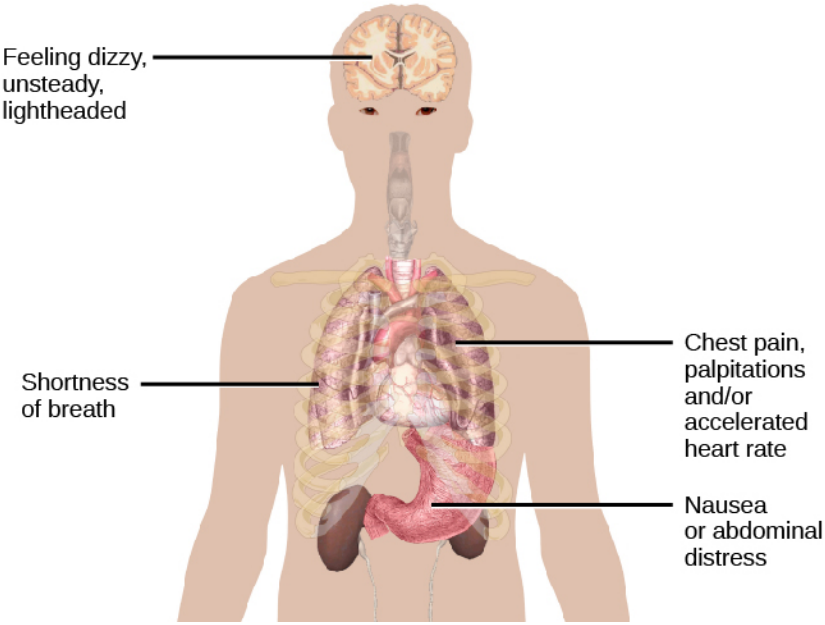
As with specific phobias, it is highly probable that the fears inherent to social anxiety disorder can develop through conditioning experiences. For example, a child who is subjected to early unpleasant social experiences (e.g., bullying at school) may develop negative social images of herself that become activated later in anxiety-provoking situations (Hackmann, Clark, & McManus, 2000). Indeed, one study reported that 92% of a sample of adults with social anxiety disorder reported a history of severe teasing in childhood, compared to only 35% of a sample of adults with panic disorder (McCabe, Antony, Summerfeldt, Liss, & Swinson, 2003).

One of the most well-established risk factors for developing social anxiety disorder is behavioral inhibition (Clauss & Blackford, 2012). Behavioral inhibition is thought to be an inherited trait, and it is characterized by a consistent tendency to show fear and

restraint when presented with unfamiliar people or situations (Kagan, Reznick, & Snidman, 1988). Behavioral inhibition is displayed very early in life; behaviorally inhibited toddlers and children respond with great caution and restraint in unfamiliar situations, and they are often timid, fearful, and shy around unfamiliar people (Fox, Henderson, Marshall, Nichols, & Ghera, 2005). A recent statistical review of studies demonstrated that behavioral inhibition was associated with more than a sevenfold increase in the risk of development of social anxiety disorder, demonstrating that behavioral inhibition is a major risk factor for the disorder (Clauss & Blackford, 2012).

PANIC DISORDER

Imagine that you are at the mall one day with your friends and—suddenly and inexplicably—you begin sweating and trembling, your heart starts pounding, you have trouble breathing, and you start to feel dizzy and nauseous. This episode lasts for 10 minutes and is terrifying because you start to think that you are going to die. When you visit your doctor the following morning and describe what happened, she tells you that you have experienced a panic attack. If you experience another one of these episodes two weeks later and worry for a month or more that similar episodes will occur in the future, it is likely that you have developed panic disorder.



Some of the physical manifestations of a panic attack are shown. People may also experience sweating, trembling, feelings of faintness, or a fear of losing control, among other symptoms.

People with panic disorder experience recurrent (more than one) and unexpected panic attacks, along with at least one month of persistent concern about additional panic attacks, worry over the consequences of the attacks, or self-defeating changes in behavior related to the attacks (e.g., avoidance of exercise or unfamiliar situations) (APA, 2013). As is the case with other anxiety disorders, the panic attacks cannot result from the physiological effects of drugs and other substances, a medical condition, or another mental disorder. A panic attack is defined as a period of extreme fear or discomfort that develops abruptly and reaches a peak within 10 minutes. Its symptoms include accelerated heart rate, sweating, trembling, choking sensations, hot flashes or chills, dizziness or lightheadedness, fears of losing control or going crazy,

and fears of dying (APA, 2013). Sometimes panic attacks are expected, occurring in response to specific environmental triggers (such as being in a tunnel); other times, these episodes are unexpected and emerge randomly (such as when relaxing). According to the DSM-5, the person must experience unexpected panic attacks to qualify for a diagnosis of panic disorder.

Experiencing a panic attack is often terrifying. Rather than recognizing the symptoms of a panic attack merely as signs of intense anxiety, individuals with panic disorder often misinterpret them as a sign that something is intensely wrong internally (thinking, for example, that the pounding heart represents an impending heart attack). Panic attacks can occasionally precipitate trips to the emergency room because several symptoms of panic attacks are, in fact, similar to those associated with heart problems (e.g., palpitations, racing pulse, and a pounding sensation in the chest) (Root, 2000). Unsurprisingly, those with panic disorder fear future attacks and may become preoccupied with modifying their behavior in an effort to avoid future panic attacks. For this reason, panic disorder is often characterized as fear of fear (Goldstein & Chambless, 1978).

Panic attacks themselves are not mental disorders. Indeed, around 23% of Americans experience isolated panic attacks in their lives without meeting the criteria for panic disorder (Kessler et al., 2006), indicating that panic attacks are fairly common. Panic disorder is, of course, much less common, afflicting 4.7% of Americans during their lifetime (Kessler et al., 2005). Many people with panic disorder develop agoraphobia, which is marked by fear and avoidance of situations in which escape might be difficult or help might not be available if one were to develop symptoms of a panic attack. People with panic disorder often experience a comorbid disorder, such as other anxiety disorders or major depressive disorder (APA, 2013).

Researchers are not entirely sure what causes panic disorder. Children are at a higher risk of developing panic disorder if their

parents have the disorder (Biederman et al., 2001), and family and twins studies indicate that the heritability of panic disorder is around 43% (Hettema, Neale, & Kendler, 2001). The exact genes and gene functions involved in this disorder, however, are not well-understood (APA, 2013). Neurobiological theories of panic disorder suggest that a region of the brain called the locus coeruleus may play a role in this disorder. Located in the brainstem, the locus coeruleus is the brain's major source of norepinephrine, a neurotransmitter that triggers the body's fight-or-flight response. Activation of the locus coeruleus is associated with anxiety and fear, and research with nonhuman primates has shown that stimulating the locus coeruleus either electrically or through drugs produces panic-like symptoms (Charney et al., 1990). Such findings have led to the theory that panic disorder may be caused by abnormal norepinephrine activity in the locus coeruleus (Bremner, Krystal, Southwick, & Charney, 1996).

Conditioning theories of panic disorder propose that panic attacks are classical conditioning responses to subtle bodily sensations resembling those normally occurring when one is anxious or frightened (Bouton, Mineka, & Barlow, 2001). For example, consider a child who has asthma. An acute asthma attack produces sensations, such as shortness of breath, coughing, and chest tightness, that typically elicit fear and anxiety. Later, when the child experiences subtle symptoms that resemble the frightening symptoms of earlier asthma attacks (such as shortness of breath after climbing stairs), he may become anxious, fearful, and then experience a panic attack. In this situation, the subtle symptoms would represent a conditioned stimulus, and the panic attack would be a conditioned response. The finding that panic disorder is nearly three times as frequent among people with asthma as it is among people without asthma (Weiser, 2007) supports the possibility that panic disorder has the potential to develop through classical conditioning.

Cognitive factors may play an integral part in panic disorder.

Generally, cognitive theories (Clark, 1996) argue that those with panic disorder are prone to interpret ordinary bodily sensations catastrophically, and these fearful interpretations set the stage for panic attacks. For example, a person might detect bodily changes that are routinely triggered by innocuous events such getting up from a seated position (dizziness), exercising (increased heart rate, shortness of breath), or drinking a large cup of coffee (increased heart rate, trembling). The individual interprets these subtle bodily changes catastrophically ("Maybe I'm having a heart attack!"). Such interpretations create fear and anxiety, which trigger additional physical symptoms; subsequently, the person experiences a panic attack. Support of this contention rests with findings that people with more severe catastrophic thoughts about sensations have more frequent and severe panic attacks, and among those with panic disorder, reducing catastrophic cognitions about their sensations is as effective as medication in reducing panic attacks (Good & Hinton, 2009).

GENERALIZED ANXIETY DISORDER

Alex was always worried about many things. He worried that his children would drown when they played at the beach. Each time he left the house, he worried that an electrical short circuit would start a fire in his home. He worried that his wife would lose her job at the prestigious law firm. He worried that his daughter's minor staph infection could turn into a massive life-threatening condition. These and other worries constantly weighed heavily on Alex's mind, so much so that they made it difficult for him to make decisions and often left him feeling tense, irritable, and worn out. One night, Alex's wife was to drive their son home from a soccer game. However, his wife stayed after the game and talked with some of the other parents, resulting in her arriving home 45 minutes late. Alex had tried to call his cell phone three or four

times, but he could not get through because the soccer field did not have a signal. Extremely worried, Alex eventually called the police, convinced that his wife and son had not arrived home because they had been in a terrible car accident.

Alex suffers from generalized anxiety disorder: a relatively continuous state of excessive, uncontrollable, and pointless worry and apprehension. People with generalized anxiety disorder often worry about routine, everyday things, even though their concerns are unjustified. For example, an individual may worry about her health and finances, the health of family members, the safety of her children, or minor matters (e.g., being late for an appointment) without having any legitimate reason for doing so (APA, 2013). A diagnosis of generalized anxiety disorder requires that the diffuse worrying and apprehension characteristic of this disorder—what Sigmund Freud referred to as free-floating anxiety—is not part of another disorder, occurs more days than not for at least six months, and is accompanied by any three of the following symptoms: restlessness, difficulty concentrating, being easily fatigued, muscle tension, irritability, and sleep difficulties.

Worry is a defining feature of generalized anxiety disorder. (credit: Freddie Peña)

About 5.7% of the U.S. population will develop symptoms of generalized anxiety disorder during their lifetime (Kessler et al., 2005), and females are 2 times as likely as males to experience the disorder (APA, 2013). Generalized anxiety disorder is highly comorbid with mood disorders and other anxiety disorders (Noyes, 2001), and it tends to be chronic. Also, generalized anxiety disorder appears to increase the risk for heart attacks and strokes, especially in people with preexisting heart conditions (Martens et al., 2010).

Although there have been few investigations aimed at determining the heritability of generalized anxiety disorder, a summary of available family and twin studies suggests that genetic factors play a modest role in the disorder (Hettema et al., 2001). Cognitive theories of generalized anxiety disorder suggest that worry represents a mental strategy to avoid more powerful negative emotions (Aikins & Craske, 2001), perhaps stemming from earlier unpleasant or traumatic experiences. Indeed, one longitudinal study found that childhood maltreatment was strongly related to the development of this disorder during adulthood (Moffitt et al., 2007); worrying might distract people from remembering painful childhood experiences.

TEST YOUR UNDERSTANDING

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SUMMARY

Anxiety disorders are a group of disorders in which a person experiences excessive, persistent, and distressing fear and anxiety that interferes with normal functioning. Anxiety disorders include specific phobia: a specific unrealistic fear; social anxiety disorder: extreme fear and avoidance of social situations; panic disorder: suddenly overwhelmed by panic even though there is no apparent reason to be frightened; agoraphobia: an intense fear and avoidance of situations in which it might be difficult to escape;

and generalized anxiety disorder: a relatively continuous state of tension, apprehension, and dread.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTION

Describe how cognitive theories of the etiology of anxiety disorders differ from learning theories.

Learning theories suggest that some anxiety disorders, especially specific phobia, can develop through a number of learning mechanisms. These mechanisms can include classical and operant conditioning, modeling, or vicarious learning. Cognitive theories, in contrast, assume that some anxiety disorder, especially panic disorder, develop through cognitive misinterpretations of anxiety and other symptoms.

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=214#h5p-243

Learning Objectives

By the end of this section, you will be able to:

- Describe the main features and prevalence of obsessive-compulsive disorder, body dysmorphic disorder, and hoarding disorder
- Understand some of the factors in the development of obsessive-compulsive disorder

Obsessive-compulsive and related disorders are a group of overlapping disorders that generally involve intrusive, unpleasant thoughts and repetitive behaviors. Many of us experience unwanted thoughts from time to time (e.g., craving double cheeseburgers when dieting), and many of us engage in repetitive behaviors on occasion (e.g., pacing when nervous). However, obsessive-compulsive and related disorders elevate the unwanted thoughts and repetitive behaviors to a status so intense that these cognitions and activities disrupt daily life. Included in this category are obsessive-compulsive disorder (OCD), body dysmorphic disorder, and hoarding disorder.

OBSESSIVE-COMPULSIVE DISORDER

People with obsessive-compulsive disorder (OCD) experience thoughts and urges that are intrusive and unwanted (obsessions) and/or the need to engage in repetitive behaviors or mental acts

(compulsions). A person with this disorder might, for example, spend hours each day washing his hands or constantly checking and rechecking to make sure that a stove, faucet, or light has been turned off.

Obsessions are more than just unwanted thoughts that seem to randomly jump into our head from time to time, such as recalling an insensitive remark a coworker made recently, and they are more significant than day-to-day worries we might have, such as justifiable concerns about being laid off from a job. Rather, obsessions are characterized as persistent, unintentional, and unwanted thoughts and urges that are highly intrusive, unpleasant, and distressing (APA, 2013). Common obsessions include concerns about germs and contamination, doubts ("Did I turn the water off?"), order and symmetry ("I need all the spoons in the tray to be arranged a certain way"), and urges that are aggressive or lustful. Usually, the person knows that such thoughts and urges are irrational and thus tries to suppress or ignore them, but has an extremely difficult time doing so. These obsessive symptoms sometimes overlap, such that someone might have both contamination and aggressive obsessions (Abramowitz & Siqueland, 2013).

Compulsions are repetitive and ritualistic acts that are typically carried out primarily as a means to minimize the distress that obsessions trigger or to reduce the likelihood of a feared event (APA, 2013). Compulsions often include such behaviors as repeated and extensive hand washing, cleaning, checking (e.g., that a door is locked), and ordering (e.g., lining up all the pencils in a particular way), and they also include such mental acts as counting, praying, or reciting something to oneself. Compulsions characteristic of OCD are not performed out of pleasure, nor are they connected in a realistic way to the source of the distress or feared event. Approximately 2.3% of the U.S. population will experience OCD in their lifetime (Ruscio, Stein, Chiu, & Kessler, 2010) and, if left untreated, OCD tends to be a chronic condition creating lifelong

interpersonal and psychological problems (Norberg, Calamari, Cohen, & Riemann, 2008).

(a)

(b)

(a) Repetitive hand washing and (b) checking (e.g., that a door is locked) are common compulsions among those with obsessive-compulsive disorder. (credit a: modification of work by the USDA; credit b: modification of work by Bradley Gordon)

BODY DYSMORPHIC DISORDER

An individual with body dysmorphic disorder is preoccupied with a perceived flaw in her physical appearance that is either nonexistent or barely noticeable to other people (APA, 2013). These perceived physical defects cause the person to think she is unattractive, ugly, hideous, or deformed. These preoccupations can focus on any bodily area, but they typically involve the skin, face, or hair. The preoccupation with imagined physical flaws drives the person to engage in repetitive and ritualistic behavioral and mental acts, such as constantly looking in the mirror, trying to hide the offending body part, comparisons with others, and, in some extreme cases, cosmetic surgery (Phillips, 2005). An estimated 2.4% of the adults in the United States meet the criteria for body dysmorphic disorder, with slightly higher rates in women than in men (APA, 2013).

HOARDING DISORDER

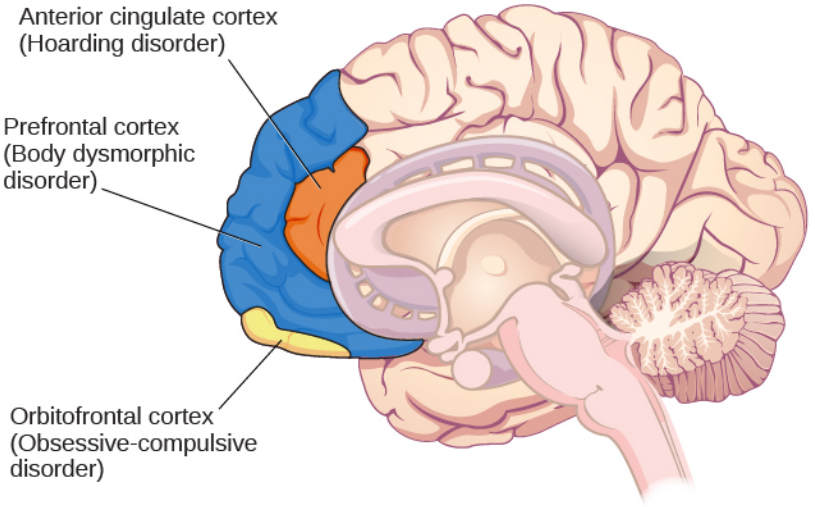
Although hoarding was traditionally considered to be a symptom of OCD, considerable evidence suggests that hoarding represents an entirely different disorder (Mataix-Cols et al., 2010). People with hoarding disorder cannot bear to part with personal possessions, regardless of how valueless or useless these possessions are. As a result, these individuals accumulate excessive amounts of usually worthless items that clutter their living areas. Often, the quantity of cluttered items is so excessive that the person is unable use his kitchen, or sleep in his bed. People who suffer from this disorder have great difficulty parting with items because they believe the items might be of some later use, or because they form a sentimental attachment to the items (APA, 2013). Importantly, a diagnosis of hoarding disorder is made only if the hoarding is not caused by another medical condition and if the hoarding is not a symptom of another disorder (e.g., schizophrenia) (APA, 2013).

Those who suffer from hoarding disorder have great difficulty in discarding possessions, usually resulting in an accumulation of items that clutter living or work areas. (credit: "puuikibeach"/Flickr)

CAUSES OF OCD

The results of family and twin studies suggest that OCD has a moderate genetic component. The disorder is five times more frequent in the first-degree relatives of people with OCD than in people without the disorder (Nestadt et al., 2000). Additionally, the concordance rate of OCD among identical twins is around 57%; however, the concordance rate for fraternal twins is 22% (Bolton, Rijsdijk, O'Connor, Perrin, & Eley, 2007). Studies have implicated about two dozen potential genes that may be involved in OCD; these genes regulate the function of three neurotransmitters: serotonin, dopamine, and glutamate (Pauls, 2010). Many of these studies included small sample sizes and have yet to be replicated. Thus, additional research needs to be done in this area.

A brain region that is believed to play a critical role in OCD is the orbitofrontal cortex (Kopell & Greenberg, 2008), an area of the frontal lobe involved in learning and decision-making (Rushworth, Noonan, Boorman, Walton, & Behrens, 2011). In people with OCD, the orbitofrontal cortex becomes especially hyperactive when they are provoked with tasks in which, for example, they are asked to look at a photo of a toilet or of pictures hanging crookedly on a wall (Simon, Kaufmann, Müsch, Kischkel, & Kathmann, 2010). The orbitofrontal cortex is part of a series of brain regions that, collectively, is called the OCD circuit; this circuit consists of several interconnected regions that influence the perceived emotional value of stimuli and the selection of both behavioral and cognitive responses (Graybiel & Rauch, 2000). As with the orbitofrontal cortex, other regions of the OCD circuit show heightened activity during symptom provocation (Rotge et al., 2008), which suggests that abnormalities in these regions may produce the symptoms of OCD (Saxena, Bota, & Brody, 2001). Consistent with this explanation, people with OCD show a substantially higher degree of connectivity of the orbitofrontal cortex and other regions of the OCD circuit than do those without OCD (Beucke et al., 2013).



Different regions of the brain may be associated with different psychological disorders.

The findings discussed above were based on imaging studies, and they highlight the potential importance of brain dysfunction in OCD. However, one important limitation of these findings is the inability to explain differences in obsessions and compulsions. Another limitation is that the correlational relationship between neurological abnormalities and OCD symptoms cannot imply causation (Abramowitz & Siqueland, 2013).

CONDITIONING AND OCD

The symptoms of OCD have been theorized to be learned responses, acquired and sustained as the result of a combination of two forms of learning: classical conditioning and operant conditioning (Mowrer, 1960; Steinmetz, Tracy, & Green, 2001). Specifically, the acquisition of OCD may occur first as the result of classical conditioning, whereby a neutral stimulus becomes

associated with an unconditioned stimulus that provokes anxiety or distress. When an individual has acquired this association, subsequent encounters with the neutral stimulus trigger anxiety, including obsessive thoughts; the anxiety and obsessive thoughts (which are now a conditioned response) may persist until she identifies some strategy to relieve it. Relief may take the form of a ritualistic behavior or mental activity that, when enacted repeatedly, reduces the anxiety. Such efforts to relieve anxiety constitute an example of negative reinforcement (a form of operant conditioning). Recall from the chapter on learning that negative reinforcement involves the strengthening of behavior through its ability to remove something unpleasant or aversive. Hence, compulsive acts observed in OCD may be sustained because they are negatively reinforcing, in the sense that they reduce anxiety triggered by a conditioned stimulus.

Suppose an individual with OCD experiences obsessive thoughts about germs, contamination, and disease whenever she encounters a doorknob. What might have constituted a viable unconditioned stimulus? Also, what would constitute the conditioned stimulus, unconditioned response, and conditioned response? What kinds of compulsive behaviors might we expect, and how do they reinforce themselves? What is decreased? Additionally, and from the standpoint of learning theory, how might the symptoms of OCD be treated successfully?

CHECK YOUR UNDERSTANDING

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SUMMARY

Obsessive-compulsive and related disorders are a group of DSM-5 disorders that overlap somewhat in that they each involve intrusive thoughts and/or repetitive behaviors. Perhaps the most recognized of these disorders is obsessive-compulsive disorder, in which a person is obsessed with unwanted, unpleasant thoughts and/or compulsively engages in repetitive behaviors or mental acts, perhaps as a way of coping with the obsessions. Body dysmorphic disorder is characterized by the individual becoming excessively

preoccupied with one or more perceived flaws in his physical appearance that are either nonexistent or unnoticeable to others. Preoccupation with the perceived physical defects causes the person to experience significant anxiety regarding how he appears to others. Hoarding disorder is characterized by persistent difficulty in discarding or parting with objects, regardless of their actual value, often resulting in the accumulation of items that clutter and congest her living area.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTIONS

Discuss the common elements of each of the three disorders covered in this section: obsessive-compulsive disorder, body dysmorphic disorder, and hoarding disorder.

Each of the three disorders is characterized by repetitive thoughts and urges, as well as an uncontrollable need to engage in repetitive behavior and mental acts. For example, repetitive thoughts include concerns over contamination (OCD), imaged physical defects (body dysmorphic disorder), and over discarding one's possessions (hoarding disorder). An uncontrollable need to engage in repetitive behaviors and mental acts include persistent hand-washing (OCD), constantly looking in the mirror (body dysmorphic disorder), and engaging in efforts to acquire new possessions (hoarding disorder).

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Describe the nature and symptoms of posttraumatic stress disorder
- Identify the risk factors associated with this disorder
- Understand the role of learning and cognitive factors in its development

Extremely stressful or traumatic events, such as combat, natural disasters, and terrorist attacks, place the people who experience them at an increased risk for developing psychological disorders such as posttraumatic stress disorder (PTSD). Throughout much of the 20th century, this disorder was called *shell shock* and *combat neurosis* because its symptoms were observed in soldiers who had engaged in wartime combat. By the late 1970s it had become clear that women who had experienced sexual traumas (e.g., rape, domestic battery, and incest) often experienced the same set of symptoms as did soldiers (Herman, 1997). The term *posttraumatic stress disorder* was developed given that these symptoms could happen to anyone who experienced psychological trauma.

A BROADER DEFINITION OF PTSD

PTSD was listed among the anxiety disorders in previous DSM editions. In DSM-5, it is now listed among a group called Trauma-

and-Stressor-Related Disorders. For a person to be diagnosed with PTSD, she be must exposed to, witness, or experience the details of a traumatic experience (e.g., a first responder), one that involves "actual or threatened death, serious injury, or sexual violence" (APA, 2013, p. 271). These experiences can include such events as combat, threatened or actual physical attack, sexual assault, natural disasters, terrorist attacks, and automobile accidents. This criterion makes PTSD the only disorder listed in the DSM in which a cause (extreme trauma) is explicitly specified.

Symptoms of PTSD include intrusive and distressing memories of the event, flashbacks (states that can last from a few seconds to several days, during which the individual relives the event and behaves as if the event were occurring at that moment [APA, 2013]), avoidance of stimuli connected to the event, persistently negative emotional states (e.g., fear, anger, guilt, and shame), feelings of detachment from others, irritability, proneness toward outbursts, and an exaggerated startle response (jumpiness). For PTSD to be diagnosed, these symptoms must occur for at least one month.

Roughly 7% of adults in the United States, including 9.7% of women and 3.6% of men, experience PTSD in their lifetime (National Comorbidity Survey, 2007), with higher rates among people exposed to mass trauma and people whose jobs involve duty-related trauma exposure (e.g., police officers, firefighters, and emergency medical personnel) (APA, 2013). Nearly 21% of residents of areas affected by Hurricane Katrina suffered from PTSD one year following the hurricane (Kessler et al., 2008), and 12.6% of Manhattan residents were observed as having PTSD 2–3 years after the 9/11 terrorist attacks (DiGrande et al., 2008).

RISK FACTORS FOR PTSD

Of course, not everyone who experiences a traumatic event will go on to develop PTSD; several factors strongly predict the

development of PTSD: trauma experience, greater trauma severity, lack of immediate social support, and more subsequent life stress (Brewin, Andrews, & Valentine, 2000). Traumatic events that involve harm by others (e.g., combat, rape, and sexual molestation) carry greater risk than do other traumas (e.g., natural disasters) (Kessler, Sonnega, Bromet, Hughes, & Nelson, 1995). Factors that increase the risk of PTSD include female gender, low socioeconomic status, low intelligence, personal history of mental disorders, history of childhood adversity (abuse or other trauma during childhood), and family history of mental disorders (Brewin et al., 2000). Personality characteristics such as neuroticism and somatization (the tendency to experience physical symptoms when one encounters stress) have been shown to elevate the risk of PTSD (Bramsen, Dirkzwager, & van der Ploeg, 2000). People who experience childhood adversity and/or traumatic experiences during adulthood are at significantly higher risk of developing PTSD if they possess one or two short versions of a gene that regulates the neurotransmitter serotonin (Xie et al., 2009). This suggests a possible diathesis-stress interpretation of PTSD: its development is influenced by the interaction of psychosocial and biological factors.

SUPPORT FOR SUFFERERS OF PTSD

Research has shown that social support following a traumatic event can reduce the likelihood of PTSD (Ozer, Best, Lipsey, & Weiss, 2003). Social support is often defined as the comfort, advice, and assistance received from relatives, friends, and neighbors. Social support can help individuals cope during difficult times by allowing them to discuss feelings and experiences and providing a sense of being loved and appreciated. A 14-year study of 1,377 American Legionnaires who had served in the Vietnam War found that those who perceived less social support when they came home were more likely to develop PTSD than were those who perceived

greater support. In addition, those who became involved in the community were less likely to develop PTSD, and they were more likely to experience a remission of PTSD than were those who were less involved (Koenen, Stellman, Stellman, & Sommer, 2003).

PTSD was first recognized in soldiers who had engaged in combat. Research has shown that strong social support decreases the risk of PTSD. This person stands at the Vietnam Traveling Memorial Wall. (credit: Kevin Stanchfield)

LEARNING AND THE DEVELOPMENT OF PTSD

PTSD learning models suggest that some symptoms are developed and maintained through classical conditioning. The traumatic event may act as an unconditioned stimulus that elicits an unconditioned response characterized by extreme fear and anxiety. Cognitive, emotional, physiological, and environmental cues accompanying or related to the event are conditioned stimuli. These traumatic reminders evoke conditioned responses (extreme fear and anxiety) similar to those caused by the event itself (Nader, 2001). A person who was in the vicinity of the Twin Towers during the 9/11 terrorist attacks and who developed PTSD may display excessive hypervigilance and distress when planes fly overhead; this behavior constitutes a conditioned response to the traumatic reminder (conditioned stimulus of the sight and sound of an airplane). Differences in how conditionable individuals are help to explain differences in the development and maintenance of PTSD symptoms (Pittman, 1988). Conditioning studies demonstrate facilitated acquisition of conditioned responses and delayed extinction of conditioned responses in people with PTSD (Orr et al., 2000).

Cognitive factors are important in the development and maintenance of PTSD. One model suggests that two key processes are crucial: disturbances in memory for the event, and negative appraisals of the trauma and its aftermath (Ehlers & Clark, 2000). According to this theory, some people who experience traumas do not form coherent memories of the trauma; memories of the traumatic event are poorly encoded and, thus, are fragmented, disorganized, and lacking in detail. Therefore, these individuals are unable remember the event in a way that gives it meaning and context. A rape victim who cannot coherently remember the event may remember only bits and pieces (e.g., the attacker repeatedly telling her she is stupid); because she was unable to develop a fully integrated memory, the fragmentary memory tends to stand out.

Although unable to retrieve a complete memory of the event, she may be haunted by intrusive fragments involuntarily triggered by stimuli associated with the event (e.g., memories of the attacker's comments when encountering a person who resembles the attacker). This interpretation fits previously discussed material concerning PTSD and conditioning. The model also proposes that negative appraisals of the event ("I deserved to be raped because I'm stupid") may lead to dysfunctional behavioral strategies (e.g., avoiding social activities where men are likely to be present) that maintain PTSD symptoms by preventing both a change in the nature of the memory and a change in the problematic appraisals.

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SUMMARY

Posttraumatic stress disorder (PTSD) was described through much of the 20th century and was referred to as shell shock and combat neurosis in the belief that its symptoms were thought to emerge from the stress of active combat. Today, PTSD is defined as a disorder in which the experience of a traumatic or profoundly stressful event, such as combat, sexual assault, or natural disaster, produces a constellation of symptoms that must last for one month or more. These symptoms include intrusive and distressing memories of the event, flashbacks, avoidance of stimuli or situations that are connected to the event, persistently negative emotional states, feeling detached from others, irritability, proneness toward outbursts, and a tendency to be easily startled. Not everyone who experiences a traumatic event will develop PTSD; a variety of risk factors associated with its development have been identified.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTION

List some of the risk factors associated with the development of PTSD following a traumatic event.

Risk factors associated with PTSD include gender (female), low socioeconomic status, low intelligence, personal and family history of mental illness, and childhood abuse or trauma. Personality factors, including neuroticism and somatization, may also serve as risk factors. Also, certain versions of a gene that regulates serotonin may constitute a diathesis.

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Distinguish normal states of sadness and euphoria from states of depression and mania
- Describe the symptoms of major depressive disorder and bipolar disorder
- Understand the differences between major depressive disorder and persistent depressive disorder, and identify two subtypes of depression
- Define the criteria for a manic episode
- Understand genetic, biological, and psychological explanations of major depressive disorder
- Discuss the relationship between mood disorders and suicidal ideation, as well as factors associated with suicide

Blake cries all day and feeling that he is worthless and his life is hopeless, he cannot get out of bed. Crystal stays up all night, talks very rapidly, and went on a shopping spree in which she spent $3,000 on furniture, although she cannot afford it. Maria recently had a baby, and she feels overwhelmed, teary, anxious, and panicked, and believes she is a terrible mother—practically every day since the baby was born. All these individuals demonstrate symptoms of a potential mood disorder.

Mood disorders are characterized by severe disturbances in mood and emotions—most often depression, but also mania and

elation (Rothschild, 1999). All of us experience fluctuations in our moods and emotional states, and often these fluctuations are caused by events in our lives. We become elated if our favorite team wins the World Series and dejected if a romantic relationship ends or if we lose our job. At times, we feel fantastic or miserable for no clear reason. People with mood disorders also experience mood fluctuations, but their fluctuations are extreme, distort their outlook on life, and impair their ability to function.

Mood disorders are characterized by massive disruptions in mood. Symptoms can range from the extreme sadness and hopelessness of depression to the extreme elation and irritability of mania. (credit: Kiran Foster)

The DSM-5 lists two general categories of mood disorders. Depressive disorders are a group of disorders in which depression is the main feature. Depression is a vague term that, in everyday language, refers to an intense and persistent sadness. Depression is a heterogeneous mood state—it consists of a broad spectrum of symptoms that range in severity. Depressed people feel sad,

discouraged, and hopeless. These individuals lose interest in activities once enjoyed, often experience a decrease in drives such as hunger and sex, and frequently doubt personal worth. Depressive disorders vary by degree, but this chapter highlights the most well-known: major depressive disorder (sometimes called unipolar depression).

Bipolar and related disorders are a group of disorders in which mania is the defining feature. Mania is a state of extreme elation and agitation. When people experience mania, they may become extremely talkative, behave recklessly, or attempt to take on many tasks simultaneously. The most recognized of these disorders is bipolar disorder.

MAJOR DEPRESSIVE DISORDER

According to the DSM-5, the defining symptoms of major depressive disorder include "depressed mood most of the day, nearly every day" (feeling sad, empty, hopeless, or appearing tearful to others), and loss of interest and pleasure in usual activities (APA, 2013). In addition to feeling overwhelmingly sad most of each day, people with depression will no longer show interest or enjoyment in activities that previously were gratifying, such as hobbies, sports, sex, social events, time spent with family, and so on. Friends and family members may notice that the person has completely abandoned previously enjoyed hobbies; for example, an avid tennis player who develops major depressive disorder no longer plays tennis (Rothschild, 1999).

To receive a diagnosis of major depressive disorder, one must experience a total of five symptoms for at least a two-week period; these symptoms must cause significant distress or impair normal functioning, and they must not be caused by substances or a medical condition. At least one of the two symptoms mentioned

above must be present, plus any combination of the following symptoms (APA, 2013):

- significant weight loss (when not dieting) or weight gain and/or significant decrease or increase in appetite;
- difficulty falling asleep or sleeping too much;
- psychomotor agitation (the person is noticeably fidgety and jittery, demonstrated by behaviors like the inability to sit, pacing, hand-wringing, pulling or rubbing of the skin, clothing, or other objects) or psychomotor retardation (the person talks and moves slowly, for example, talking softly, very little, or in a monotone);
- fatigue or loss of energy;
- feelings of worthlessness or guilt;
- difficulty concentrating and indecisiveness; and
- suicidal ideation: thoughts of death (not just fear of dying), thinking about or planning suicide, or making an actual suicide attempt.

Major depressive disorder is considered episodic: its symptoms are typically present at their full magnitude for a certain period of time and then gradually abate. Approximately 50%–60% of people who experience an episode of major depressive disorder will have a second episode at some point in the future; those who have had two episodes have a 70% chance of having a third episode, and those who have had three episodes have a 90% chance of having a fourth episode (Rothschild, 1999). Although the episodes can last for months, a majority a people diagnosed with this condition (around 70%) recover within a year. However, a substantial number do not recover; around 12% show serious signs of impairment associated with major depressive disorder after 5 years (Boland & Keller, 2009). In the long-term, many who do recover will still show minor symptoms that fluctuate in their severity (Judd, 2012).

RESULTS OF MAJOR DEPRESSIVE DISORDER

Major depressive disorder is a serious and incapacitating condition that can have a devastating effect on the quality of one's life. The person suffering from this disorder lives a profoundly miserable existence that often results in unavailability for work or education, abandonment of promising careers, and lost wages; occasionally, the condition requires hospitalization. The majority of those with major depressive disorder report having faced some kind of discrimination, and many report that having received such treatment has stopped them from initiating close relationships, applying for jobs for which they are qualified, and applying for education or training (Lasalvia et al., 2013). Major depressive disorder also takes a toll on health. Depression is a risk factor for the development of heart disease in healthy patients, as well as adverse cardiovascular outcomes in patients with preexisting heart disease (Whooley, 2006).

RISK FACTORS FOR MAJOR DEPRESSIVE DISORDER

Major depressive disorder is often referred to as the common cold of psychiatric disorders. Around 6.6% of the U.S. population experiences major depressive disorder each year; 16.9% will experience the disorder during their lifetime (Kessler & Wang, 2009). It is more common among women than among men, affecting approximately 20% of women and 13% of men at some point in their life (National Comorbidity Survey, 2007). The greater risk among women is not accounted for by a tendency to report symptoms or to seek help more readily, suggesting that gender differences in the rates of major depressive disorder may reflect biological and gender-related environmental experiences (Kessler, 2003).

Lifetime rates of major depressive disorder tend to be highest in North and South America, Europe, and Australia; they are

considerably lower in Asian countries (Hasin, Fenton, & Weissman, 2011). The rates of major depressive disorder are higher among younger age cohorts than among older cohorts, perhaps because people in younger age cohorts are more willing to admit depression (Kessler & Wang, 2009).

A number of risk factors are associated with major depressive disorder: unemployment (including homemakers); earning less than $20,000 per year; living in urban areas; or being separated, divorced, or widowed (Hasin et al., 2011). Comorbid disorders include anxiety disorders and substance abuse disorders (Kessler & Wang, 2009).

SUBTYPES OF DEPRESSION

The DSM-5 lists several different subtypes of depression. These subtypes—what the DSM-5 refer to as specifiers—are not specific disorders; rather, they are labels used to indicate specific patterns of symptoms or to specify certain periods of time in which the symptoms may be present. One subtype, seasonal pattern, applies to situations in which a person experiences the symptoms of major depressive disorder only during a particular time of year (e.g., fall or winter). In everyday language, people often refer to this subtype as the winter blues.

Another subtype, peripartum onset (commonly referred to as postpartum depression), applies to women who experience major depression during pregnancy or in the four weeks following the birth of their child (APA, 2013). These women often feel very anxious and may even have panic attacks. They may feel guilty, agitated, and be weepy. They may not want to hold or care for their newborn, even in cases in which the pregnancy was desired and intended. In extreme cases, the mother may have feelings of wanting to harm her child or herself. In a horrific illustration, a woman named Andrea Yates, who suffered from extreme

peripartum-onset depression (as well as other mental illnesses), drowned her five children in a bathtub (Roche, 2002). Most women with peripartum-onset depression do not physically harm their children, but most do have difficulty being adequate caregivers (Fields, 2010). A surprisingly high number of women experience symptoms of peripartum-onset depression. A study of 10,000 women who had recently given birth found that 14% screened positive for peripartum-onset depression, and that nearly 20% reported having thoughts of wanting to harm themselves (Wisner et al., 2013).

People with persistent depressive disorder (previously known as dysthymia) experience depressed moods most of the day nearly every day for at least two years, as well as at least two of the other symptoms of major depressive disorder. People with persistent depressive disorder are chronically sad and melancholy, but do not meet all the criteria for major depression. However, episodes of full-blown major depressive disorder can occur during persistent depressive disorder (APA, 2013).

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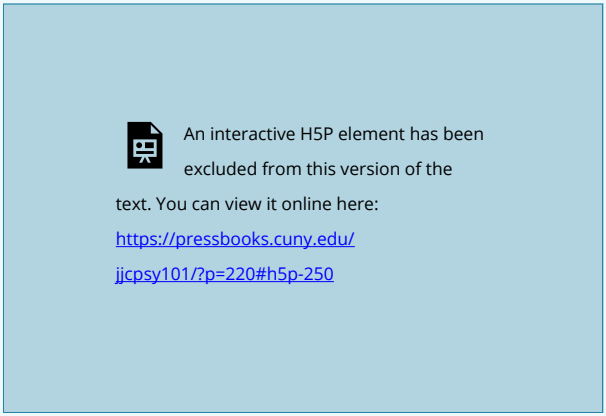

BIPOLAR DISORDER

A person with bipolar disorder (commonly known as manic depression) often experiences mood states that vacillate between depression and mania; that is, the person's mood is said to alternate from one emotional extreme to the other (in contrast to unipolar, which indicates a persistently sad mood).

To be diagnosed with bipolar disorder, a person must have experienced a manic episode at least once in his life; although major depressive episodes are common in bipolar disorder, they are not required for a diagnosis (APA, 2013). According to the

DSM-5, a manic episode is characterized as a "distinct period of abnormally and persistently elevated, expansive, or irritable mood and abnormally and persistently increased activity or energy lasting at least one week," that lasts most of the time each day (APA, 2013, p. 124). During a manic episode, some experience a mood that is almost euphoric and become excessively talkative, sometimes spontaneously starting conversations with strangers; others become excessively irritable and complain or make hostile comments. The person may talk loudly and rapidly, exhibiting flight of ideas, abruptly switching from one topic to another. These individuals are easily distracted, which can make a conversation very difficult. They may exhibit grandiosity, in which they experience inflated but unjustified self-esteem and self-confidence. For example, they might quit a job in order to "strike it rich" in the stock market, despite lacking the knowledge, experience, and capital for such an endeavor. They may take on several tasks at the same time (e.g., several time-consuming projects at work) and yet show little, if any, need for sleep; some may go for days without sleep. Patients may also recklessly engage in pleasurable activities that could have harmful consequences, including spending sprees, reckless driving, making foolish investments, excessive gambling, or engaging in sexual encounters with strangers (APA, 2013).

During a manic episode, individuals usually feel as though they are not ill and do not need treatment. However, the reckless behaviors that often accompany these episodes—which can be antisocial, illegal, or physically threatening to others—may require involuntary hospitalization (APA, 2013). Some patients with bipolar disorder will experience a rapid-cycling subtype, which is characterized by at least four manic episodes (or some combination of at least four manic and major depressive episodes) within one year.

In the 1997 independent film *Sweetheart*, actress Janeane Garofalo plays the part of Jasmine, a young woman with bipolar disorder. Watch this video to see a portion of this film in which Jasmine experiences a manic episode: The Other Side of Me – inside my bipolar mind.

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RISK FACTORS FOR BIPOLAR DISORDER

Bipolar disorder is considerably less frequent than major depressive disorder. In the United States, 1 out of every 167 people meets the criteria for bipolar disorder each year, and 1 out of 100 meet the criteria within their lifetime (Merikangas et al., 2011). The rates are higher in men than in women, and about half of those with this disorder report onset before the age of 25 (Merikangas et al., 2011). Around 90% of those with bipolar disorder have a

comorbid disorder, most often an anxiety disorder or a substance abuse problem. Unfortunately, close to half of the people suffering from bipolar disorder do not receive treatment (Merikangas & Tohen, 2011). Suicide rates are extremely high among those with bipolar disorder: around 36% of individuals with this disorder attempt suicide at least once in their lifetime (Novick, Swartz, & Frank, 2010), and between 15%–19% complete suicide (Newman, 2004).

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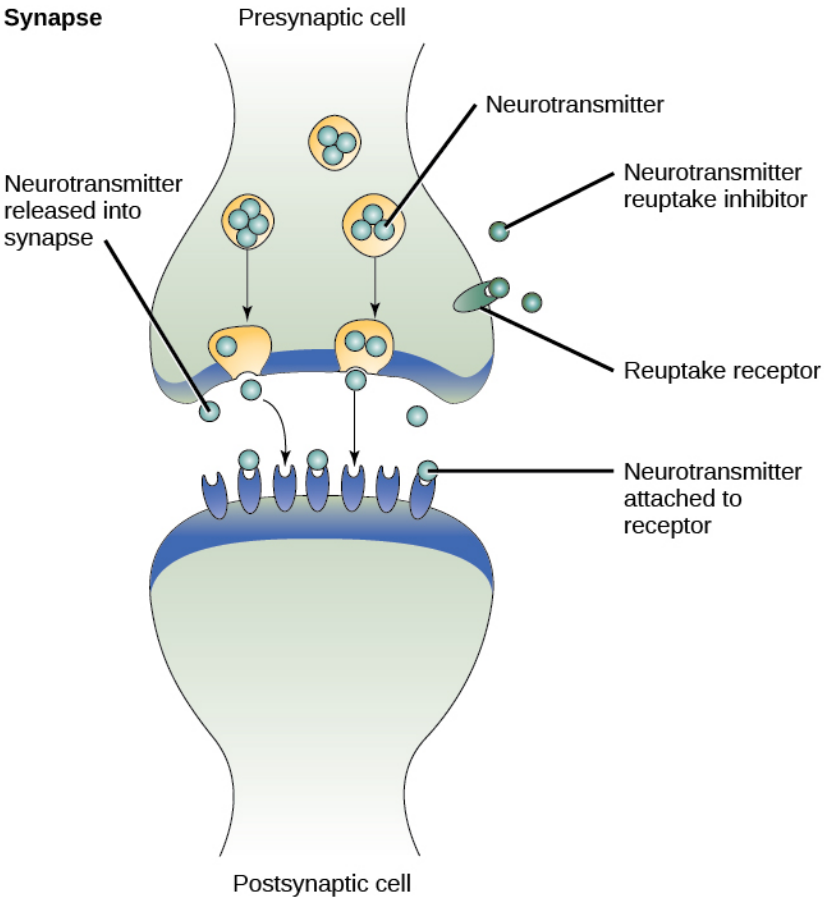
THE BIOLOGICAL BASIS OF MOOD DISORDERS

Mood disorders have been shown to have a strong genetic and biological basis. Relatives of those with major depressive disorder have double the risk of developing major depressive disorder, whereas relatives of patients with bipolar disorder have over nine times the risk (Merikangas et al., 2011). The rate of concordance for major depressive disorder is higher among identical twins than fraternal twins (50% vs. 38%, respectively), as is that of bipolar disorder (67% vs. 16%, respectively), suggesting that genetic factors play a stronger role in bipolar disorder than in major depressive disorder (Merikangas et al. 2011).

People with mood disorders often have imbalances in certain neurotransmitters, particularly norepinephrine and serotonin (Thase, 2009). These neurotransmitters are important regulators of the bodily functions that are disrupted in mood disorders, including appetite, sex drive, sleep, arousal, and mood. Medications that are used to treat major depressive disorder typically boost serotonin and norepinephrine activity, whereas lithium—used in the treatment of bipolar disorder—blocks norepinephrine activity at the synapses.



Many medications designed to treat mood disorders work by altering neurotransmitter activity in the neural synapse.

Depression is linked to abnormal activity in several regions of the brain (Fitzgerald, Laird, Maller, & Daskalakis, 2008) including those important in assessing the emotional significance of stimuli and experiencing emotions (amygdala), and in regulating and controlling emotions (like the prefrontal cortex, or PFC) (LeMoult, Castonguay, Joormann, & McAleavey, 2013). Depressed individuals show elevated amygdala activity (Drevets, Bogers, & Raichle, 2002),

especially when presented with negative emotional stimuli, such as photos of sad faces (Surguladze et al., 2005). Interestingly, heightened amygdala activation to negative emotional stimuli among depressed persons occurs even when stimuli are presented outside of conscious awareness (Victor, Furey, Fromm, Öhman, & Drevets, 2010), and it persists even after the negative emotional stimuli are no longer present (Siegle, Thompson, Carter, Steinhauer, & Thase, 2007). Additionally, depressed individuals exhibit less activation in the prefrontal, particularly on the left side (Davidson, Pizzagalli, & Nitschke, 2009). Because the PFC can dampen amygdala activation, thereby enabling one to suppress negative emotions (Phan et al., 2005), decreased activation in certain regions of the PFC may inhibit its ability to override negative emotions that might then lead to more negative mood states (Davidson et al., 2009). These findings suggest that depressed persons are more prone to react to emotionally negative stimuli, yet have greater difficulty controlling these reactions.

Depressed individuals react to negative emotional stimuli, such as sad faces, with greater amygdala activation than do non-depressed individuals. (credit: Ian Munroe)

Since the 1950s, researchers have noted that depressed individuals

have abnormal levels of cortisol, a stress hormone released into the blood by the neuroendocrine system during times of stress (Mackin & Young, 2004). When cortisol is released, the body initiates a fight-or-flight response in reaction to a threat or danger. Many people with depression show elevated cortisol levels (Holsboer & Ising, 2010), especially those reporting a history of early life trauma such as the loss of a parent or abuse during childhood (Baes, Tofoli, Martins, & Juruena, 2012). Such findings raise the question of whether high cortisol levels are a cause or a consequence of depression. High levels of cortisol are a risk factor for future depression (Halligan, Herbert, Goodyer, & Murray, 2007), and cortisol activates activity in the amygdala while deactivating activity in the PFC (McEwen, 2005)—both brain disturbances are connected to depression. Thus, high cortisol levels may have a causal effect on depression, as well as on its brain function abnormalities (van Praag, 2005). Also, because stress results in increased cortisol release (Michaud, Matheson, Kelly, Anisman, 2008), it is equally reasonable to assume that stress may precipitate depression.

A DIATHESIS-STRESS MODEL AND MAJOR DEPRESSIVE DISORDERS

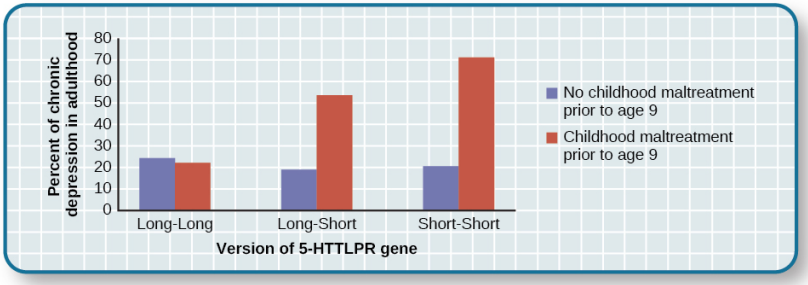
Indeed, it has long been believed that stressful life events can trigger depression, and research has consistently supported this conclusion (Mazure, 1998). Stressful life events include significant losses, such as death of a loved one, divorce or separation, and serious health and money problems; life events such as these often precede the onset of depressive episodes (Brown & Harris, 1989). In particular, exit events—instances in which an important person departs (e.g., a death, divorce or separation, or a family member leaving home)—often occur prior to an episode (Paykel, 2003). Exit events are especially likely to trigger depression if these happenings occur in a way that humiliates or devalues the

individual. For example, people who experience the breakup of a relationship initiated by the other person develop major depressive disorder at a rate more than 2 times that of people who experience the death of a loved one (Kendler, Hettema, Butera, Gardner, & Prescott, 2003).

Likewise, individuals who are exposed to traumatic stress during childhood—such as separation from a parent, family turmoil, and maltreatment (physical or sexual abuse)—are at a heightened risk of developing depression at any point in their lives (Kessler, 1997). A recent review of 16 studies involving over 23,000 subjects concluded that those who experience childhood maltreatment are more than 2 times as likely to develop recurring and persistent depression (Nanni, Uher, & Danese, 2012).

Of course, not everyone who experiences stressful life events or childhood adversities succumbs to depression—indeed, most do not. Clearly, a diathesis-stress interpretation of major depressive disorder, in which certain predispositions or vulnerability factors influence one's reaction to stress, would seem logical. If so, what might such predispositions be? A study by Caspi and others (2003) suggests that an alteration in a specific gene that regulates serotonin (the 5-HTTLPR gene) might be one culprit. These investigators found that people who experienced several stressful life events were significantly more likely to experience episodes of major depression if they carried one or two short versions of this gene than if they carried two long versions. Those who carried one or two short versions of the 5-HTTLPR gene were unlikely to experience an episode, however, if they had experienced few or no stressful life events. Numerous studies have replicated these findings, including studies of people who experienced maltreatment during childhood (Goodman & Brand, 2009). In a recent investigation conducted in the United Kingdom (Brown & Harris, 2013), researchers found that childhood maltreatment before age 9 elevated the risk of chronic adult depression (a depression episode lasting for at least 12 months) among those

individuals having one (LS) or two (SS) short versions of the 5-HTTLPR gene. Childhood maltreatment did not increase the risk for chronic depression for those have two long (LL) versions of this gene. Thus, genetic vulnerability may be one mechanism through which stress potentially leads to depression.



A study on gene-environment interaction in people experiencing chronic depression in adulthood suggests a much higher incidence in individuals with a short version of the gene in combination with childhood maltreatment (Brown & Harris, 2013).

COGNITIVE THEORIES OF DEPRESSION

Cognitive theories of depression take the view that depression is triggered by negative thoughts, interpretations, self-evaluations, and expectations (Joormann, 2009). These diathesis-stress models propose that depression is triggered by a "cognitive vulnerability" (negative and maladaptive thinking) and by precipitating stressful life events (Gotlib & Joormann, 2010). Perhaps the most well-known cognitive theory of depression was developed in the 1960s by psychiatrist Aaron Beck, based on clinical observations and supported by research (Beck, 2008). Beck theorized that depression-prone people possess depressive schemas, or mental predispositions to think about most things in a negative way (Beck, 1976). Depressive schemas contain themes of loss, failure,

rejection, worthlessness, and inadequacy, and may develop early in childhood in response to adverse experiences, then remain dormant until they are activated by stressful or negative life events. Depressive schemas prompt dysfunctional and pessimistic thoughts about the self, the world, and the future. Beck believed that this dysfunctional style of thinking is maintained by cognitive biases, or errors in how we process information about ourselves, which lead us to focus on negative aspects of experiences, interpret things negatively, and block positive memories (Beck, 2008). A person whose depressive schema consists of a theme of rejection might be overly attentive to social cues of rejection (more likely to notice another's frown), and he might interpret this cue as a sign of rejection and automatically remember past incidents of rejection. Longitudinal studies have supported Beck's theory, in showing that a preexisting tendency to engage in this negative, self-defeating style of thinking—when combined with life stress—over time predicts the onset of depression (Dozois & Beck, 2008). Cognitive therapies for depression, aimed at changing a depressed person's negative thinking, were developed as an expansion of this theory (Beck, 1976).

Another cognitive theory of depression, hopelessness theory, postulates that a particular style of negative thinking leads to a sense of hopelessness, which then leads to depression (Abramson, Metalsky, & Alloy, 1989). According to this theory, hopelessness is an expectation that unpleasant outcomes will occur or that desired outcomes will not occur, and there is nothing one can do to prevent such outcomes. A key assumption of this theory is that hopelessness stems from a tendency to perceive negative life events as having stable ("It's never going to change") and global ("It's going to affect my whole life") causes, in contrast to unstable ("It's fixable") and specific ("It applies only to this particular situation") causes, especially if these negative life events occur in important life realms, such as relationships, academic achievement, and the like. Suppose a student who wishes to go to law school does poorly

on an admissions test. If the student infers negative life events as having stable and global causes, she may believe that her poor performance has a stable and global cause ("I lack intelligence, and it's going to prevent me from ever finding a meaningful career"), as opposed to an unstable and specific cause ("I was sick the day of the exam, so my low score was a fluke"). Hopelessness theory predicts that people who exhibit this cognitive style in response to undesirable life events will view such events as having negative implications for their future and self-worth, thereby increasing the likelihood of hopelessness—the primary cause of depression (Abramson et al., 1989). One study testing hopelessness theory measured the tendency to make negative inferences for bad life effects in participants who were experiencing uncontrollable stressors. Over the ensuing six months, those with scores reflecting high cognitive vulnerability were 7 times more likely to develop depression compared to those with lower scores (Kleim, Gonzalo, & Ehlers, 2011).

A third cognitive theory of depression focuses on how people's thoughts about their distressed moods—depressed symptoms in particular—can increase the risk and duration of depression. This theory, which focuses on rumination in the development of depression, was first described in the late 1980s to explain the higher rates of depression in women than in men (Nolen-Hoeksema, 1987). Rumination is the repetitive and passive focus on the fact that one is depressed and dwelling on depressed symptoms, rather that distracting one's self from the symptoms or attempting to address them in an active, problem-solving manner (Nolen-Hoeksema, 1991). When people ruminate, they have thoughts such as "Why am I so unmotivated? I just can't get going. I'm never going to get my work done feeling this way" (Nolen-Hoeksema & Hilt, 2009, p. 393). Women are more likely than men to ruminate when they are sad or depressed (Butler & Nolen-Hoeksema, 1994), and the tendency to ruminate is associated with increases in depression symptoms (Nolen-Hoeksema, Larson, &

Grayson, 1999), heightened risk of major depressive episodes (Abela & Hankin, 2011), and chronicity of such episodes (Robinson & Alloy, 2003)

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SUICIDE

For some people with mood disorders, the extreme emotional pain they experience becomes unendurable. Overwhelmed by hopelessness, devastated by incapacitating feelings of worthlessness, and burdened with the inability to adequately cope with such feelings, they may consider suicide to be a reasonable

way out. Suicide, defined by the CDC as "death caused by self-directed injurious behavior with any intent to die as the result of the behavior" (CDC, 2013a), in a sense represents an outcome of several things going wrong all at the same time Crosby, Ortega, & Melanson, 2011). Not only must the person be biologically or psychologically vulnerable, but he must also have the means to perform the suicidal act, and he must lack the necessary protective factors (e.g., social support from friends and family, religion, coping skills, and problem-solving skills) that provide comfort and enable one to cope during times of crisis or great psychological pain (Berman, 2009).

Suicide is not listed as a disorder in the DSM-5; however, suffering from a mental disorder—especially a mood disorder—poses the greatest risk for suicide. Around 90% of those who complete suicides have a diagnosis of at least one mental disorder, with mood disorders being the most frequent (Fleischman, Bertolote, Belfer, & Beautrais, 2005). In fact, the association between major depressive disorder and suicide is so strong that one of the criteria for the disorder is thoughts of suicide, as discussed above (APA, 2013).

Suicide rates can be difficult to interpret because some deaths that appear to be accidental may in fact be acts of suicide (e.g., automobile crash). Nevertheless, investigations into U.S. suicide rates have uncovered these facts:

- Suicide was the 10th leading cause of death for all ages in 2010 (Centers for Disease Control and Prevention [CDC], 2012).
- There were 38,364 suicides in 2010 in the United States—an average of 105 each day (CDC, 2012).
- Suicide among males is 4 times higher than among females and accounts for 79% of all suicides; firearms are the most commonly used method of suicide for males, whereas poisoning is the most commonly used method

for females (CDC, 2012).

- From 1991 to 2003, suicide rates were consistently higher among those 65 years and older. Since 2001, however, suicide rates among those ages 25–64 have risen consistently, and, since 2006, suicide rates have been greater for those ages 65 and older (CDC, 2013b). This increase in suicide rates among middle-aged Americans has prompted concern in some quarters that baby boomers (individuals born between 1946–1964) who face economic worry and easy access to prescription medication may be particularly vulnerable to suicide (Parker-Pope, 2013).
- The highest rates of suicide within the United States are among American Indians/Alaskan natives and Non-Hispanic Whites (CDC, 2013b).
- Suicide rates vary across the United States, with the highest rates consistently found in the mountain states of the west (Alaska, Montana, Nevada, Wyoming, Colorado, and Idaho) (Berman, 2009).

Contrary to popular belief, suicide rates peak during the springtime (April and May), not during the holiday season or winter. In fact, suicide rates are generally lowest during the winter months (Postolache et al., 2010).

RISK FACTORS FOR SUICIDE

Suicidal risk is especially high among people with substance abuse problems. Individuals with alcohol dependence are at 10 times greater risk for suicide than the general population (Wilcox, Conner, & Caine, 2004). The risk of suicidal behavior is especially high among those who have made a prior suicide attempt. Among those who attempt suicide, 16% make another attempt within a

year and over 21% make another attempt within four years (Owens, Horrocks, & House, 2002). Suicidal individuals may be at high risk for terminating their life if they have a lethal means in which to act, such as a firearm in the home (Brent & Bridge, 2003). Withdrawal from social relationships, feeling as though one is a burden to others, and engaging in reckless and risk-taking behaviors may be precursors to suicidal behavior (Berman, 2009). A sense of entrapment or feeling unable to escape one's miserable feelings or external circumstances (e.g., an abusive relationship with no perceived way out) predicts suicidal behavior (O'Connor, Smyth, Ferguson, Ryan, & Williams, 2013). Tragically, reports of suicides among adolescents following instances of cyberbullying have emerged in recent years. In one widely-publicized case a few years ago, Phoebe Prince, a 15-year-old Massachusetts high school student, committed suicide following incessant harassment and taunting from her classmates via texting and Facebook (McCabe, 2010).

Suicides can have a contagious effect on people. For example, another's suicide, especially that of a family member, heightens one's risk of suicide (Agerbo, Nordentoft, & Mortensen, 2002). Additionally, widely-publicized suicides tend to trigger copycat suicides in some individuals. One study examining suicide statistics in the United States from 1947–1967 found that the rates of suicide skyrocketed for the first month after a suicide story was printed on the front page of the *New York Times* (Phillips, 1974). Austrian researchers found a significant increase in the number of suicides by firearms in the three weeks following extensive reports in Austria's largest newspaper of a celebrity suicide by gun (Etzersdorfer, Voracek, & Sonneck, 2004). A review of 42 studies concluded that media coverage of celebrity suicides is more than 14 times more likely to trigger copycat suicides than is coverage of non-celebrity suicides (Stack, 2000). This review also demonstrated that the medium of coverage is important: televised stories are considerably less likely to prompt a surge in suicides than are

newspaper stories. Research suggests that a trend appears to be emerging whereby people use online social media to leave suicide notes, although it is not clear to what extent suicide notes on such media might induce copycat suicides (Ruder, Hatch, Ampanozi, Thali, & Fischer, 2011). Nevertheless, it is reasonable to conjecture that suicide notes left by individuals on social media may influence the decisions of other vulnerable people who encounter them (Luxton, June, & Fairall, 2012).

One possible contributing factor in suicide is brain chemistry. Contemporary neurological research shows that disturbances in the functioning of serotonin are linked to suicidal behavior (Pompili et al., 2010). Low levels of serotonin predict future suicide attempts and suicide completions, and low levels have been observed post-mortem among suicide victims (Mann, 2003). Serotonin dysfunction, as noted earlier, is also known to play an important role in depression; low levels of serotonin have also been linked to aggression and impulsivity (Stanley et al., 2000). The combination of these three characteristics constitutes a potential formula for suicide—especially violent suicide. A classic study conducted during the 1970s found that patients with major depressive disorder who had very low levels of serotonin attempted suicide more frequently and more violently than did patients with higher levels (Asberg, Thorén, Träskman, Bertilsson, & Ringberger, 1976; Mann, 2003).

Suicidal thoughts, plans, and even off-hand remarks ("I might kill myself this afternoon") should always be taken extremely seriously. People who contemplate terminating their life need immediate help. Below are links to two excellent websites that contain resources (including hotlines) for people who are struggling with suicidal ideation, have loved ones who may be suicidal, or who have lost loved ones to suicide: http://www.afsp.org and http://suicidology.org.

SUMMARY

Mood disorders are those in which the person experiences severe disturbances in mood and emotion. They include depressive disorders and bipolar and related disorders. Depressive disorders include major depressive disorder, which is characterized by episodes of profound sadness and loss of interest or pleasure in usual activities and other associated features, and persistent depressive disorder, which marked by a chronic state of sadness. Bipolar disorder is characterized by mood states that vacillate between sadness and euphoria; a diagnosis of bipolar disorder requires experiencing at least one manic episode, which is defined as a period of extreme euphoria, irritability, and increased activity. Mood disorders appear to have a genetic component, with genetic factors playing a more prominent role in bipolar disorder than in depression. Both biological and psychological factors are important in the development of depression. People who suffer from mental health problems, especially mood disorders, are at heightened risk for suicide.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTION

Describe several of the factors associated with suicide.

The risk of suicide is high among people with

mental health problems, including mood disorders and substance abuse problems. The risk is also high among those who have made a prior suicide attempt and who have lethal means to commit suicide. Rates of suicide are higher among men and during the springtime, and they are higher in the mountain states of the west than in other regions of the United States. Research has also shown that suicides can have a "contagious" effect on people, and that it is associated with serotonin dysfunction.

PERSONAL APPLICATION QUESTION

Think of someone you know who seems to have a tendency to make negative, self-defeating explanations for negative life events. How might this tendency lead to future problems? What steps do you think could be taken to change this thinking style?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=220#h5p-254

Media Attributions

- "The Other Side of Me – inside my bipolar mind" by Julie Kraft. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Recognize the essential nature of schizophrenia, avoiding the misconception that it involves a split personality
- Categorize and describe the major symptoms of schizophrenia
- Understand the interplay between genetic, biological, and environmental factors that are associated with the development of schizophrenia
- Discuss the importance of research examining prodromal symptoms of schizophrenia

Schizophrenia is a devastating psychological disorder that is characterized by major disturbances in thought, perception, emotion, and behavior. About 1% of the population experiences schizophrenia in their lifetime, and usually the disorder is first diagnosed during early adulthood (early to mid-20s). Most people with schizophrenia experience significant difficulties in many day-to-day activities, such as holding a job, paying bills, caring for oneself (grooming and hygiene), and maintaining relationships with others. Frequent hospitalizations are more often the rule rather than the exception with schizophrenia. Even when they receive the best treatments available, many with schizophrenia will continue to experience serious social and occupational impairment throughout their lives.

What is schizophrenia? First, schizophrenia is *not* a condition

involving a split personality; that is, schizophrenia is not the same thing as dissociative identity disorder (better known as multiple personality disorder). These disorders are sometimes confused because the word *schizophrenia* first coined by the Swiss psychiatrist Eugen Bleuler in 1911, derives from Greek words that refer to a "splitting" (schizo) of psychic functions (phrene) (Green, 2001).

Schizophrenia is considered a psychotic disorder, or one in which the person's thoughts, perceptions, and behaviors are impaired to the point where she is not able to function normally in life. In informal terms, one who suffers from a psychotic disorder (that is, has a psychosis) is disconnected from the world in which most of us live.

SYMPTOMS OF SCHIZOPHRENIA

The main symptoms of schizophrenia include hallucinations, delusions, disorganized thinking, disorganized or abnormal motor behavior, and negative symptoms (APA, 2013). A hallucination is a perceptual experience that occurs in the absence of external stimulation. Auditory hallucinations (hearing voices) occur in roughly two-thirds of patients with schizophrenia and are by far the most common form of hallucination (Andreasen, 1987). The voices may be familiar or unfamiliar, they may have a conversation or argue, or the voices may provide a running commentary on the person's behavior (Tsuang, Farone, & Green, 1999).

Less common are visual hallucinations (seeing things that are not there) and olfactory hallucinations (smelling odors that are not actually present).

Delusions are beliefs that are contrary to reality and are firmly held even in the face of contradictory evidence. Many of us hold beliefs that some would consider odd, but a delusion is easily identified because it is clearly absurd. A person with schizophrenia

may believe that his mother is plotting with the FBI to poison his coffee, or that his neighbor is an enemy spy who wants to kill him. These kinds of delusions are known as paranoid delusions, which involve the (false) belief that other people or agencies are plotting to harm the person. People with schizophrenia also may hold grandiose delusions, beliefs that one holds special power, unique knowledge, or is extremely important. For example, the person who claims to be Jesus Christ, or who claims to have knowledge going back 5,000 years, or who claims to be a great philosopher is experiencing grandiose delusions. Other delusions include the belief that one's thoughts are being removed (thought withdrawal) or thoughts have been placed inside one's head (thought insertion). Another type of delusion is somatic delusion, which is the belief that something highly abnormal is happening to one's body (e.g., that one's kidneys are being eaten by cockroaches).

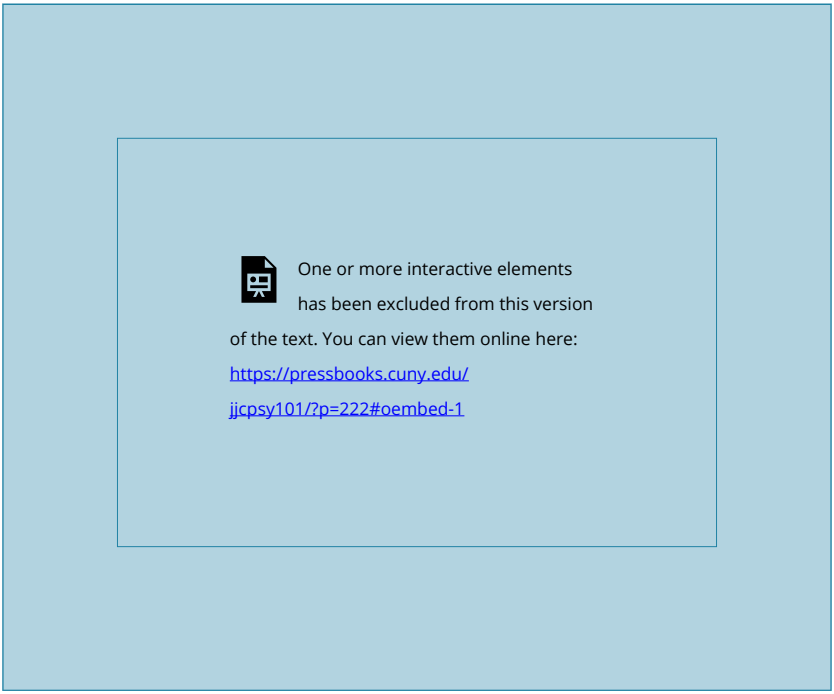
Disorganized thinking refers to disjointed and incoherent thought processes—usually detected by what a person says. The person might ramble, exhibit loose associations (jump from topic to topic), or talk in a way that is so disorganized and incomprehensible that it seems as though the person is randomly combining words. Disorganized thinking is also exhibited by blatantly illogical remarks (e.g., "Fenway Park is in Boston. I live in Boston. Therefore, I live at Fenway Park.") and by tangentiality: responding to others' statements or questions by remarks that are either barely related or unrelated to what was said or asked. For example, if a person diagnosed with schizophrenia is asked if she is interested in receiving special job training, she might state that she once rode on a train somewhere. To a person with schizophrenia, the tangential (slightly related) connection between job *training* and riding a *train* are sufficient enough to cause such a response.

Disorganized or abnormal motor behavior refers to unusual behaviors and movements: becoming unusually active, exhibiting silly child-like behaviors (giggling and self-absorbed smiling),

engaging in repeated and purposeless movements, or displaying odd facial expressions and gestures. In some cases, the person will exhibit catatonic behaviors, which show decreased reactivity to the environment, such as posturing, in which the person maintains a rigid and bizarre posture for long periods of time, or catatonic stupor, a complete lack of movement and verbal behavior.

Negative symptoms are those that reflect noticeable decreases and absences in certain behaviors, emotions, or drives (Green, 2001). A person who exhibits diminished emotional expression shows no emotion in his facial expressions, speech, or movements, even when such expressions are normal or expected. Avolition is characterized by a lack of motivation to engage in self-initiated and meaningful activity, including the most basic of tasks, such as bathing and grooming. Alogia refers to reduced speech output; in simple terms, patients do not say much. Another negative symptom is asociality, or social withdrawal and lack of interest in engaging in social interactions with others. A final negative symptom, anhedonia, refers to an inability to experience pleasure. One who exhibits anhedonia expresses little interest in what most people consider to be pleasurable activities, such as hobbies, recreation, or sexual activity.

Watch this video and try to identify which classic symptoms of schizophrenia are shown: Schizophrenia: Gerald, Part 1.



CAUSES OF SCHIZOPHRENIA

There is considerable evidence suggesting that schizophrenia has a genetic basis. The risk of developing schizophrenia is nearly 6 times greater if one has a parent with schizophrenia than if one does not (Goldstein, Buka, Seidman, & Tsuang, 2010). Additionally, one's risk of developing schizophrenia increases as genetic relatedness to family members diagnosed with schizophrenia increases (Gottesman, 2001).

GENES

When considering the role of genetics in schizophrenia, as in any disorder, conclusions based on family and twin studies are subject

to criticism. This is because family members who are closely related (such as siblings) are more likely to share similar environments than are family members who are less closely related (such as cousins); further, identical twins may be more likely to be treated similarly by others than might fraternal twins. Thus, family and twin studies cannot completely rule out the possible effects of shared environments and experiences. Such problems can be corrected by using adoption studies, in which children are separated from their parents at an early age. One of the first adoption studies of schizophrenia conducted by Heston (1966) followed 97 adoptees, including 47 who were born to mothers with schizophrenia, over a 36-year period. Five of the 47 adoptees (11%) whose mothers had schizophrenia were later diagnosed with schizophrenia, compared to none of the 50 control adoptees. Other adoption studies have consistently reported that for adoptees who are later diagnosed with schizophrenia, their biological relatives have a higher risk of schizophrenia than do adoptive relatives (Shih, Belmonte, & Zandi, 2004).

Although adoption studies have supported the hypothesis that genetic factors contribute to schizophrenia, they have also demonstrated that the disorder most likely arises from a combination of genetic and environmental factors, rather than just genes themselves. For example, investigators in one study examined the rates of schizophrenia among 303 adoptees (Tienari et al., 2004). A total of 145 of the adoptees had biological mothers with schizophrenia; these adoptees constituted the high genetic risk group. The other 158 adoptees had mothers with no psychiatric history; these adoptees composed the low genetic risk group. The researchers managed to determine whether the adoptees' families were either healthy or disturbed. For example, the adoptees were considered to be raised in a disturbed family environment if the family exhibited a lot of criticism, conflict, and a lack of problem-solving skills. The findings revealed that adoptees whose mothers had schizophrenia (high genetic risk) *and* who had

been raised in a disturbed family environment were much more likely to develop schizophrenia or another psychotic disorder (36.8%) than were adoptees whose biological mothers had schizophrenia but who had been raised in a healthy environment (5.8%), or than adoptees with a low genetic risk who were raised in either a disturbed (5.3%) or healthy (4.8%) environment. Because the adoptees who were at high genetic risk were likely to develop schizophrenia *only* if they were raised in a disturbed home environment, this study supports a diathesis-stress interpretation of schizophrenia—both genetic vulnerability and environmental stress are necessary for schizophrenia to develop, genes alone do not show the complete picture.

NEUROTRANSMITTERS

If we accept that schizophrenia is at least partly genetic in origin, as it seems to be, it makes sense that the next step should be to identify biological abnormalities commonly found in people with the disorder. Perhaps not surprisingly, a number of neurobiological factors have indeed been found to be related to schizophrenia. One such factor that has received considerable attention for many years is the neurotransmitter dopamine. Interest in the role of dopamine in schizophrenia was stimulated by two sets of findings: drugs that increase dopamine levels can produce schizophrenia-like symptoms, and medications that block dopamine activity reduce the symptoms (Howes & Kapur, 2009). The dopamine hypothesis of schizophrenia proposed that an overabundance of dopamine or too many dopamine receptors are responsible for the onset and maintenance of schizophrenia (Snyder, 1976). More recent work in this area suggests that abnormalities in dopamine vary by brain region and thus contribute to symptoms in unique ways. In general, this research has suggested that an overabundance of dopamine in the limbic system may be responsible for some symptoms, such as hallucinations and

delusions, whereas low levels of dopamine in the prefrontal cortex might be responsible primarily for the negative symptoms (avolition, alogia, asociality, and anhedonia) (Davis, Kahn, Ko, & Davidson, 1991). In recent years, serotonin has received attention, and newer antipsychotic medications used to treat the disorder work by blocking serotonin receptors (Baumeister & Hawkins, 2004).

BRAIN ANATOMY

Brain imaging studies reveal that people with schizophrenia have enlarged ventricles, the cavities within the brain that contain cerebral spinal fluid (Green, 2001). This finding is important because larger than normal ventricles suggests that various brain regions are reduced in size, thus implying that schizophrenia is associated with a loss of brain tissue. In addition, many people with schizophrenia display a reduction in gray matter (cell bodies of neurons) in the frontal lobes (Lawrie & Abukmeil, 1998), and many show less frontal lobe activity when performing cognitive tasks (Buchsbaum et al., 1990). The frontal lobes are important in a variety of complex cognitive functions, such as planning and executing behavior, attention, speech, movement, and problem solving. Hence, abnormalities in this region provide merit in explaining why people with schizophrenia experience deficits in these of areas.

EVENTS DURING PREGNANCY

Why do people with schizophrenia have these brain abnormalities? A number of environmental factors that could impact normal brain development might be at fault. High rates of obstetric complications in the births of children who later developed schizophrenia have been reported (Cannon, Jones, & Murray, 2002). In addition, people are at an increased risk for developing

schizophrenia if their mother was exposed to influenza during the first trimester of pregnancy (Brown et al., 2004). Research has also suggested that a mother's emotional stress during pregnancy may increase the risk of schizophrenia in offspring. One study reported that the risk of schizophrenia is elevated substantially in offspring whose mothers experienced the death of a relative during the first trimester of pregnancy (Khashan et al., 2008).

MARIJUANA

Another variable that is linked to schizophrenia is marijuana use. Although a number of reports have shown that individuals with schizophrenia are more likely to use marijuana than are individuals without schizophrenia (Thornicroft, 1990), such investigations cannot determine if marijuana use leads to schizophrenia, or vice versa. However, a number of longitudinal studies have suggested that marijuana use is, in fact, a risk factor for schizophrenia. A classic investigation of over 45,000 Swedish conscripts who were followed up after 15 years found that those individuals who had reported using marijuana at least once by the time of conscription were more than 2 times as likely to develop schizophrenia during the ensuing 15 years than were those who reported never using marijuana; those who had indicated using marijuana 50 or more times were 6 times as likely to develop schizophrenia (Andréasson, Allbeck, Engström, & Rydberg, 1987). More recently, a review of 35 longitudinal studies found a substantially increased risk of schizophrenia and other psychotic disorders in people who had used marijuana, with the greatest risk in the most frequent users (Moore et al., 2007). Other work has found that marijuana use is associated with an onset of psychotic disorders at an earlier age (Large, Sharma, Compton, Slade, & Nielssen, 2011). Overall, the available evidence seems to indicate that marijuana use plays a causal role in the development of schizophrenia, although it is important to point out that marijuana use is not an essential or

sufficient risk factor as not all people with schizophrenia have used marijuana and the majority of marijuana users do not develop schizophrenia (Casadio, Fernandes, Murray, & Di Forti, 2011). One plausible interpretation of the data is that early marijuana use may disrupt normal brain development during important early maturation periods in adolescence (Trezza, Cuomo, & Vanderschuren, 2008). Thus, early marijuana use may set the stage for the development of schizophrenia and other psychotic disorders, especially among individuals with an established vulnerability (Casadio et al., 2011).

TEST YOUR UNDERSTANDING

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SCHIZOPHRENIA: EARLY WARNING SIGNS

Early detection and treatment of conditions such as heart disease and cancer have improved survival rates and quality of life for people who suffer from these conditions. A new approach involves identifying people who show minor symptoms of psychosis, such as unusual thought content, paranoia, odd communication, delusions, problems at school or work, and a decline in social functioning—which are coined prodromal symptoms—and following these individuals over time to determine which of them develop a psychotic disorder and which factors best predict such a disorder. A number of factors have been identified that predict a greater likelihood that prodromal individuals will develop a psychotic disorder: genetic risk (a family history of psychosis), recent deterioration in functioning, high levels of unusual thought content, high levels of suspicion or paranoia, poor social functioning, and a history of substance abuse (Fusar-Poli et al., 2013). Further research will enable a more accurate prediction of those at greatest risk for developing schizophrenia, and thus to whom early intervention efforts should be directed.

TEST YOUR UNDERSTANDING

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SUMMARY

Schizophrenia is a severe disorder characterized by a complete breakdown in one's ability to function in life; it often requires hospitalization. People with schizophrenia experience hallucinations and delusions, and they have extreme difficulty regulating their emotions and behavior. Thinking is incoherent and disorganized, behavior is extremely bizarre, emotions are flat, and motivation to engage in most basic life activities is lacking. Considerable evidence shows that genetic factors play a central role in schizophrenia; however, adoption studies have highlighted the additional importance of environmental factors. Neurotransmitter and brain abnormalities, which may be linked to environmental factors such as obstetric complications or exposure

to influenza during the gestational period, have also been implicated. A promising new area of schizophrenia research involves identifying individuals who show prodromal symptoms and following them over time to determine which factors best predict the development of schizophrenia. Future research may enable us to pinpoint those especially at risk for developing schizophrenia and who may benefit from early intervention.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTION

Why is research following individuals who show prodromal symptoms of schizophrenia so important?

This kind of research is important because it enables investigators to identify potential warning signs that predict the onset of schizophrenia. Once such factors are identified, interventions may be developed.

GLOSSARY

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Media Attributions

- "Schizophrenia: Gerald, Part 1" by Neuroslicer. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Describe the essential nature of dissociative disorders
- Identify and differentiate the symptoms of dissociative amnesia, depersonalization/ derealization disorder, and dissociative identity disorder
- Discuss the potential role of both social and psychological factors in dissociative identity disorder

Dissociative disorders are characterized by an individual becoming split off, or dissociated, from her core sense of self. Memory and identity become disturbed; these disturbances have a psychological rather than physical cause. Dissociative disorders listed in the DSM-5 include dissociative amnesia, depersonalization/derealization disorder, and dissociative identity disorder.

DISSOCIATIVE AMNESIA

Amnesia refers to the partial or total forgetting of some experience or event. An individual with dissociative amnesia is unable to recall important personal information, usually following an extremely stressful or traumatic experience such as combat, natural disasters, or being the victim of violence. The memory impairments are not caused by ordinary forgetting. Some individuals with

dissociative amnesia will also experience dissociative fugue (from the word "to flee" in French), whereby they suddenly wander away from their home, experience confusion about their identity, and sometimes even adopt a new identity (Cardeña & Gleaves, 2006). Most fugue episodes last only a few hours or days, but some can last longer. One study of residents in communities in upstate New York reported that about 1.8% experienced dissociative amnesia in the previous year (Johnson, Cohen, Kasen, & Brook, 2006).

Some have questioned the validity of dissociative amnesia (Pope, Hudson, Bodkin, & Oliva, 1998); it has even been characterized as a "piece of psychiatric folklore devoid of convincing empirical support" (McNally, 2003, p. 275). Notably, scientific publications regarding dissociative amnesia rose during the 1980s and reached a peak in the mid-1990s, followed by an equally sharp decline by 2003; in fact, only 13 cases of individuals with dissociative amnesia worldwide could be found in the literature that same year (Pope, Barry, Bodkin, & Hudson, 2006). Further, no description of individuals showing dissociative amnesia following a trauma exists in any fictional or nonfictional work prior to 1800 (Pope, Poliakoff, Parker, Boynes, & Hudson, 2006). However, a study of 82 individuals who enrolled for treatment at a psychiatric outpatient hospital found that nearly 10% met the criteria for dissociative amnesia, perhaps suggesting that the condition is underdiagnosed, especially in psychiatric populations (Foote, Smolin, Kaplan, Legatt, & Lipschitz, 2006).

DEPERSONALIZATION/DEREALIZATION DISORDER

Depersonalization/derealization disorder is characterized by recurring episodes of depersonalization, derealization, or both. Depersonalization is defined as feelings of "unreality or detachment from, or unfamiliarity with, one's whole self or from

aspects of the self" (APA, 2013, p. 302). Individuals who experience depersonalization might believe their thoughts and feelings are not their own; they may feel robotic as though they lack control over their movements and speech; they may experience a distorted sense of time and, in extreme cases, they may sense an "out-of-body" experience in which they see themselves from the vantage point of another person. Derealization is conceptualized as a sense of "unreality or detachment from, or unfamiliarity with, the world, be it individuals, inanimate objects, or all surroundings" (APA, 2013, p. 303). A person who experiences derealization might feel as though he is in a fog or a dream, or that the surrounding world is somehow artificial and unreal. Individuals with depersonalization/derealization disorder often have difficulty describing their symptoms and may think they are going crazy (APA, 2013).

DISSOCIATIVE IDENTITY DISORDER

By far, the most well-known dissociative disorder is dissociative identity disorder (formerly called multiple personality disorder). People with dissociative identity disorder exhibit two or more separate personalities or identities, each well-defined and distinct from one another. They also experience memory gaps for the time during which another identity is in charge (e.g., one might find unfamiliar items in her shopping bags or among her possessions), and in some cases may report hearing voices, such as a child's voice or the sound of somebody crying (APA, 2013). The study of upstate New York residents mentioned above (Johnson et al., 2006) reported that 1.5% of their sample experienced symptoms consistent with dissociative identity disorder in the previous year.

Dissociative identity disorder (DID) is highly controversial. Some believe that people fake symptoms to avoid the consequences of illegal actions (e.g., "I am not responsible for shoplifting because it was my other personality"). In fact, it has been demonstrated that

people are generally skilled at adopting the role of a person with different personalities when they believe it might be advantageous to do so. As an example, Kenneth Bianchi was an infamous serial killer who, along with his cousin, murdered over a dozen females around Los Angeles in the late 1970s. Eventually, he and his cousin were apprehended. At Bianchi's trial, he pled not guilty by reason of insanity, presenting himself as though he had DID and claiming that a different personality ("Steve Walker") committed the murders. When these claims were scrutinized, he admitted faking the symptoms and was found guilty (Schwartz, 1981).

A second reason DID is controversial is because rates of the disorder suddenly skyrocketed in the 1980s. More cases of DID were identified during the five years prior to 1986 than in the preceding two centuries (Putnam, Guroff, Silberman, Barban, & Post, 1986). Although this increase may be due to the development of more sophisticated diagnostic techniques, it is also possible that the popularization of DID—helped in part by *Sybil*, a popular 1970s book (and later film) about a woman with 16 different personalities—may have prompted clinicians to overdiagnose the disorder (Piper & Merskey, 2004). Casting further scrutiny on the existence of multiple personalities or identities is the recent suggestion that the story of Sybil was largely fabricated, and the idea for the book might have been exaggerated (Nathan, 2011).

Despite its controversial nature, DID is clearly a legitimate and serious disorder, and although some people may fake symptoms, others suffer their entire lives with it. People with this disorder tend to report a history of childhood trauma, some cases having been corroborated through medical or legal records (Cardeña & Gleaves, 2006). Research by Ross et al. (1990) suggests that in one study about 95% of people with DID were physically and/or sexually abused as children. Of course, not all reports of childhood abuse can be expected to be valid or accurate. However, there is strong evidence that traumatic experiences can cause people to experience states of dissociation, suggesting that dissociative

states—including the adoption of multiple personalities—may serve as a psychologically important coping mechanism for threat and danger (Dalenberg et al., 2012).

TEST YOUR UNDERSTANDING

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SUMMARY

The main characteristic of dissociative disorders is that people become dissociated from their sense of self, resulting in memory and identity disturbances. Dissociative disorders listed in the

DSM-5 include dissociative amnesia, depersonalization/derealization disorder, and dissociative identity disorder. A person with dissociative amnesia is unable to recall important personal information, often after a stressful or traumatic experience.

Depersonalization/derealization disorder is characterized by recurring episodes of depersonalization (i.e., detachment from or unfamiliarity with the self) and/or derealization (i.e., detachment from or unfamiliarity with the world). A person with dissociative identity disorder exhibits two or more well-defined and distinct personalities or identities, as well as memory gaps for the time during which another identity was present.

Dissociative identity disorder has generated controversy, mainly because some believe its symptoms can be faked by patients if presenting its symptoms somehow benefits the patient in avoiding negative consequences or taking responsibility for one's actions. The diagnostic rates of this disorder have increased dramatically following its portrayal in popular culture. However, many people legitimately suffer over the course of a lifetime with this disorder.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTION

The prevalence of most psychological disorders has increased since the 1980s. However, as discussed

in this section, scientific publications regarding dissociative amnesia peaked in the mid-1990s but then declined steeply through 2003. In addition, no fictional or nonfictional description of individuals showing dissociative amnesia following a trauma exists prior to 1800. How would you explain this phenomenon?

Several explanations are possible. One explanation is that perhaps there is little scientific interest in this phenomenon, maybe because it has yet to gain consistent scientific acceptance. Another possible explanation is that perhaps the dissociative amnesia was fashionable at the time publications dealing with this topic peaked (1990s); perhaps since that time it has become less fashionable.

PERSONAL APPLICATION QUESTION

Try to find an example (via a search engine) of a past instance in which a person committed a horrible crime, was apprehended, and later claimed to have dissociative identity disorder during the trial. What was the outcome? Was the person revealed to be faking? If so, how was this determined?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=224#h5p-261

Learning Objectives

By the end of this section, you will be able to:

- Describe the nature of personality disorders and how they differ from other disorders
- List and distinguish between the three clusters of personality disorders
- Identify the basic features of borderline personality disorder and antisocial personality disorder, and the factors that are important in the etiology of both

The term *personality* refers loosely to one's stable, consistent, and distinctive way of thinking about, feeling, acting, and relating to the world. People with personality disorders exhibit a personality style that differs markedly from the expectations of their culture, is pervasive and inflexible, begins in adolescence or early adulthood, and causes distress or impairment (APA, 2013). Generally, individuals with these disorders exhibit enduring personality styles that are extremely troubling and often create problems for them and those with whom they come into contact. Their maladaptive personality styles frequently bring them into conflict with others, disrupt their ability to develop and maintain social relationships, and prevent them from accomplishing realistic life goals.

The DSM-5 recognizes 10 personality disorders, organized into 3 different clusters. Cluster A disorders include paranoid personality disorder, schizoid personality disorder, and schizotypal personality disorder. People with these disorders display a personality style

that is odd or eccentric. Cluster B disorders include antisocial personality disorder, histrionic personality disorder, narcissistic personality disorder, and borderline personality disorder. People with these disorders usually are impulsive, overly dramatic, highly emotional, and erratic. Cluster C disorders include avoidant personality disorder, dependent personality disorder, and obsessive-compulsive personality disorder (which is not the same thing as obsessive-compulsive disorder). People with these disorders often appear to be nervous and fearful.

DSM-5 Personality Disorders

DSM-5 Personality Disorder	Description	Cluster
Paranoid	harbors a pervasive and unjustifiable suspiciousness and mistrust of others; reluctant to confide in or become close to others; reads hidden demeaning or threatening meaning into benign remarks or events; takes offense easily and bears grudges; not due to schizophrenia or other psychotic disorders	A
Schizoid	lacks interest and desire to form relationships with others; aloof and shows emotional coldness and detachment; indifferent to approval or criticism of others; lacks close friends or confidants; not due to schizophrenia or other psychotic disorders, not an autism spectrum disorder	A
Schizotypal	exhibits eccentricities in thought, perception, emotion, speech, and behavior; shows suspiciousness or paranoia; has unusual perceptual experiences; speech is often idiosyncratic; displays inappropriate emotions; lacks friends or confidants; not due to schizophrenia or other psychotic disorder, or to autism spectrum disorder	A
Antisocial	continuously violates the rights of others; history of antisocial tendencies prior to age 15; often lies, fights, and has problems with the law; impulsive and fails to think ahead; can be deceitful and manipulative in order to gain profit or pleasure; irresponsible and often fails to hold down a job or pay financial debts; lacks feelings for others and remorse over misdeeds	B
Histrionic	excessively overdramatic, emotional, and theatrical; feels uncomfortable when not the center of others' attention; behavior is often inappropriately seductive or provocative; speech is highly emotional but often vague and diffuse; emotions are shallow and often shift rapidly; may alienate friends with demands for constant attention	B
Narcissistic	overinflated and unjustified sense of self-importance and preoccupied with fantasies of success; believes he is entitled to special treatment from others; shows arrogant attitudes and behaviors; takes advantage of others; lacks empathy	B
Borderline	unstable in self-image, mood, and behavior; cannot tolerate being alone and experiences chronic feelings of emptiness; unstable and intense relationships with others; behavior is impulsive, unpredictable, and sometimes self-damaging; shows inappropriate and intense anger; makes suicidal gestures	B
Avoidant	socially inhibited and oversensitive to negative evaluation; avoids occupations that involve interpersonal contact because of fears of criticism or rejection; avoids relationships with others unless guaranteed to be accepted unconditionally; feels inadequate and views self as socially inept and unappealing; unwilling to take risks or engage in new activities if they may prove embarrassing	C

DSM-5 Personality Disorder	Description	Cluster
Dependent	allows others to take over and run her life; is submissive, clingy, and fears separation; cannot make decisions without advice and reassurance from others; lacks self-confidence; cannot do things on her own; feels uncomfortable or helpless when alone	C
Obsessive-Compulsive	pervasive need for perfectionism that interferes with the ability to complete tasks; preoccupied with details, rules, order, and schedules; excessively devoted to work at the expense of leisure and friendships; rigid, inflexible, and stubborn; insists things be done his way; miserly with money	C

Slightly over 9% of the U.S. population suffers from a personality disorder, with avoidant and schizoid personality disorders the most frequent (Lezenweger, Lane, Loranger, & Kessler, 2007). Two of these personality disorders, borderline personality disorder and antisocial personality disorder, are regarded by many as especially problematic.

BORDERLINE PERSONALITY DISORDER

The "borderline" in borderline personality disorder was originally coined in the late 1930s in an effort to describe patients who appeared anxious, but were prone to brief psychotic experiences—that is, patients who were thought to be literally on the borderline between anxiety and psychosis (Freeman, Stone, Martin, & Reinecke, 2005). Today, borderline personality disorder has a completely different meaning. Borderline personality disorder is characterized chiefly by instability in interpersonal relationships, self-image, and mood, as well as marked impulsivity (APA, 2013). People with borderline personality disorder cannot tolerate the thought of being alone and will make frantic efforts (including making suicidal gestures and engaging in self-mutilation) to avoid abandonment or separation (whether real or imagined). Their relationships are intense and unstable; for example, a lover may be idealized early in a relationship, but then later vilified at

the slightest sign she appears to no longer show interest. These individuals have an unstable view of self and, thus, might suddenly display a shift in personal attitudes, interests, career plans, and choice of friends. For example, a law school student may, despite having invested tens of thousands of dollars toward earning a law degree and despite having performed well in the program, consider dropping out and pursuing a career in another field. People with borderline personality disorder may be highly impulsive and may engage in reckless and self-destructive behaviors such as excessive gambling, spending money irresponsibly, substance abuse, engaging in unsafe sex, and reckless driving. They sometimes show intense and inappropriate anger that they have difficulty controlling, and they can be moody, sarcastic, bitter, and verbally abusive.

The prevalence of borderline personality disorder in the U.S. population is estimated to be around 1.4% (Lezenweger et al., 2007), but the rates are higher among those who use mental health services; approximately 10% of mental health outpatients and 20% of psychiatric inpatients meet the criteria for diagnosis (APA, 2013). Additionally, borderline personality disorder is comorbid with anxiety, mood, and substance use disorders (Lezenweger et al., 2007).

BIOLOGICAL BASIS FOR BORDERLINE PERSONALITY DISORDER

Genetic factors appear to be important in the development of borderline personality disorder. For example, core personality traits that characterize this disorder, such as impulsivity and emotional instability, show a high degree of heritability (Livesley, 2008). Also, the rates of borderline personality disorder among relatives of people with this disorder have been found to be as high as 24.9% (White, Gunderson, Zanarani, & Hudson, 2003). Individuals with borderline personality disorder report

experiencing childhood physical, sexual, and/or emotional abuse at rates far greater than those observed in the general population (Afifi et al., 2010), indicating that environmental factors are also crucial. These findings would suggest that borderline personality disorder may be determined by an interaction between genetic factors and adverse environmental experiences. Consistent with this hypothesis, one study found that the highest rates of borderline personality disorder were among individuals with a borderline temperament (characterized by high novelty seeking and high harm-avoidance) and those who experienced childhood abuse and/or neglect (Joyce et al., 2003).

ANTISOCIAL PERSONALITY DISORDER

Most human beings live in accordance with a moral compass, a sense of right and wrong. Most individuals learn at a very young age that there are certain things that should not be done. We learn that we should not lie or cheat. We are taught that it is wrong to take things that do not belong to us, and that it is wrong to exploit others for personal gain. We also learn the importance of living up to our responsibilities, of doing what we say we will do. People with antisocial personality disorder, however, do not seem to have a moral compass. These individuals act as though they neither have a sense of nor care about right or wrong. Not surprisingly, these people represent a serious problem for others and for society in general.

According to the DSM-5, the individual with antisocial personality disorder (sometimes referred to as psychopathy) shows no regard at all for other people's rights or feelings. This lack of regard is exhibited a number of ways and can include repeatedly performing illegal acts, lying to or conning others, impulsivity and recklessness, irritability and aggressiveness toward others, and failure to act in a responsible way (e.g., leaving debts unpaid) (APA, 2013). The worst

part about antisocial personality disorder, however, is that people with this disorder have no remorse over one's misdeeds; these people will hurt, manipulate, exploit, and abuse others and not feel any guilt. Signs of this disorder can emerge early in life; however, a person must be at least 18 years old to be diagnosed with antisocial personality disorder.

People with antisocial personality disorder seem to view the world as self-serving and unkind. They seem to think that they should use whatever means necessary to get by in life. They tend to view others not as living, thinking, feeling beings, but rather as pawns to be used or abused for a specific purpose. They often have an over-inflated sense of themselves and can appear extremely arrogant. They frequently display superficial charm; for example, without really meaning it they might say exactly what they think another person wants to hear. They lack empathy: they are incapable of understanding the emotional point-of-view of others. People with this disorder may become involved in illegal enterprises, show cruelty toward others, leave their jobs with no plans to obtain another job, have multiple sexual partners, repeatedly get into fights with others, and show reckless disregard for themselves and others (e.g., repeated arrests for driving while intoxicated) (APA, 2013).

A useful way to conceptualize antisocial personality disorder is boiling the diagnosis down to three major concepts: disinhibition, boldness, and meanness (Patrick, Fowles, & Krueger, 2009). Disinhibition is a propensity toward impulse control problems, lack of planning and forethought, insistence on immediate gratification, and inability to restrain behavior. Boldness describes a tendency to remain calm in threatening situations, high self-assurance, a sense of dominance, and a tendency toward thrill-seeking. Meanness is defined as "aggressive resource seeking without regard for others," and is signaled by a lack of empathy, disdain for and lack of close relationships with others, and a tendency to accomplish goals through cruelty (Patrick et al., 2009, p. 913).

RISK FACTORS FOR ANTISOCIAL PERSONALITY DISORDER

Antisocial personality disorder is observed in about 3.6% of the population; the disorder is much more common among males, with a 3 to 1 ratio of men to women, and it is more likely to occur in men who are younger, widowed, separated, divorced, of lower socioeconomic status, who live in urban areas, and who live in the western United States (Compton, Conway, Stinson, Colliver, & Grant, 2005). Compared to men with antisocial personality disorder, women with the disorder are more likely to have experienced emotional neglect and sexual abuse during childhood, and they are more likely to have had parents who abused substances and who engaged in antisocial behaviors themselves (Alegria et al., 2013).

This table shows some of the differences in the specific types of antisocial behaviors that men and women with antisocial personality disorder exhibit (Alegria et al., 2013).

Gender Differences in Antisocial Personality Disorder

Men with antisocial personality disorder are more likely than women with antisocial personality disorder to	Women with antisocial personality disorder are more likely than men with antisocial personality to
• do things that could easily hurt themselves or others • receive three or more traffic tickets for reckless driving • have their driver's license suspended • destroy others' property • start a fire on purpose • make money illegally • do anything that could lead to arrest • hit someone hard enough to injure them • hurt an animal on purpose	• run away from home overnight • frequently miss school or work • lie frequently • forge someone's signature • get into a fight that comes to blows with an intimate partner • live with others besides the family for at least one month • harass, threaten, or blackmail someone

Family, twin, and adoption studies suggest that both genetic and environmental factors influence the development of antisocial

personality disorder, as well as general antisocial behavior (criminality, violence, aggressiveness) (Baker, Bezdjian, & Raine, 2006). Personality and temperament dimensions that are related to this disorder, including fearlessness, impulsive antisociality, and callousness, have a substantial genetic influence (Livesley & Jang, 2008). Adoption studies clearly demonstrate that the development of antisocial behavior is determined by the interaction of genetic factors and adverse environmental circumstances (Rhee & Waldman, 2002). For example, one investigation found that adoptees of biological parents with antisocial personality disorder were more likely to exhibit adolescent and adult antisocial behaviors if they were raised in adverse adoptive family environments (e.g., adoptive parents had marital problems, were divorced, used drugs, and had legal problems) than if they were raised in a more normal adoptive environment (Cadoret, Yates, Ed, Woodworth, & Stewart, 1995).

Researchers who are interested in the importance of environment in the development of antisocial personality disorder have directed their attention to such factors as the community, the structure and functioning of the family, and peer groups. Each of these factors influences the likelihood of antisocial behavior. One longitudinal investigation of more than 800 Seattle-area youth measured risk factors for violence at 10, 14, 16, and 18 years of age (Herrenkohl et al., 2000). The risk factors examined included those involving the family, peers, and community.

Risk Factors During Adolescence That Predict Later Violence			
Risk factor	**Age 10 predictor (elementary school)**	**Age 14 predictor (middle school)**	**Age 16 predictor (high school)**
Family			
Parental violence		✕	
Parental criminality		✕	✕
Poof family management		✕	✕
Family conflict		✕	✕
Parental attitudes favorable to violence	✕		
Frequent moves			✕
Peer			
Peer delinquency	✕	✕	✕
Gang membership		✕	✕
Community			
Economic deprivation	✕		✕
Community disorganization		✕	✕
Availability of drugs	✕	✕	✕
Neighborhood adults involved in crime		✕	✕

Longitudinal studies have helped to identify risk factors for predicting violent behavior.

Those with antisocial tendencies do not seem to experience

emotions the way most other people do. These individuals fail to show fear in response to environment cues that signal punishment, pain, or noxious stimulation. For instance, they show less skin conductance (sweatiness on hands) in anticipation of electric shock than do people without antisocial tendencies (Hare, 1965). Skin conductance is controlled by the sympathetic nervous system and is used to assess autonomic nervous system functioning. When the sympathetic nervous system is active, people become aroused and anxious, and sweat gland activity increases. Thus, increased sweat gland activity, as assessed through skin conductance, is taken as a sign of arousal or anxiety. For those with antisocial personality disorder, a lack of skin conductance may indicate the presence of characteristics such as emotional deficits and impulsivity that underlie the propensity for antisocial behavior and negative social relationships (Fung et al., 2005).

While emotional deficits may contribute to antisocial personality disorder, so too might an inability to relate to others' pain. In a recent study, 80 prisoners were shown photos of people being intentionally hurt by others (e.g., someone crushing a person's hand in an automobile door) while undergoing brain imaging (Decety, Skelly, & Kiehl, 2013). Prisoners who scored high on a test of antisocial tendencies showed significantly less activation in brain regions involved in the experience of empathy and feeling concerned for others than did prisoners with low scores on the antisocial test. Notably, the prisoners who scored high on the antisocial test showed greater activation in a brain area involved self-awareness, cognitive function, and interpersonal experience. The investigators suggested that the heightened activation in this region when watching social interactions involving one person harming another may reflect a propensity or desire for this kind of behavior.

TEST YOUR UNDERSTANDING

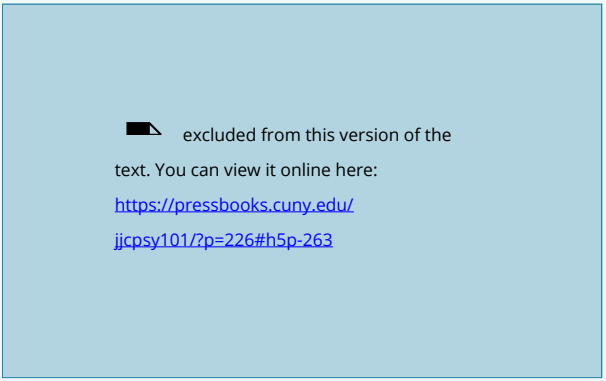
An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=226#h5p-262

SUMMARY

Individuals with personality disorders exhibit a personality style that is inflexible, causes distress and impairment, and creates problems for themselves and others. The DSM-5 recognizes 10 personality disorders, organized into three clusters. The disorders in Cluster A include those characterized by a personality style that is odd and eccentric. Cluster B includes personality disorders characterized chiefly by a personality style that is impulsive, dramatic, highly emotional, and erratic, and those in Cluster C are

characterized by a nervous and fearful personality style. Two Cluster B personality disorders, borderline personality disorder and antisocial personality disorder, are especially problematic. People with borderline personality disorder show marked instability in mood, behavior, and self-image, as well as impulsivity. They cannot stand to be alone, are unpredictable, have a history of stormy relationships, and frequently display intense and inappropriate anger. Genetic factors and adverse childhood experiences (e.g., sexual abuse) appear to be important in its development. People with antisocial personality display a lack of regard for the rights of others; they are impulsive, deceitful, irresponsible, and unburdened by any sense of guilt. Genetic factors and socialization both appear to be important in the origin of antisocial personality disorder. Research has also shown that those with this disorder do not experience emotions the way most other people do.

REVIEW QUESTIONS



CRITICAL THINKING QUESTION

Imagine that a child has a genetic vulnerability to antisocial personality disorder. How might this child's environment shape the likelihood of developing this personality disorder?

The environment is likely to be very instrumental

in determining the likelihood of developing antisocial personality disorder. Research has shown that adverse family environments (e.g., divorce or marital problems, legal problems, and drug use) are connected to antisocial personality disorder, particularly if one is genetically vulnerable. Beyond one's family environment, peer group delinquency and community variables (e.g., economic deprivation, community disorganization, drug use, and the presence of adult antisocial models) heighten the risk of violent behavior.

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=226#h5p-264

Learning Objectives

By the end of this section, you will be able to:

- Describe the nature and symptoms of attention deficit/ hyperactivity disorder and autism spectrum disorder
- Discuss the prevalence and factors that contribute to the development of these disorders

Most of the disorders we have discussed so far are typically diagnosed in adulthood, although they can and sometimes do occur during childhood. However, there are a group of conditions that, when present, are diagnosed early in childhood, often before the time a child enters school. These conditions are listed in the DSM-5 as neurodevelopmental disorders, and they involve developmental problems in personal, social, academic, and intellectual functioning (APA, 2013). In this section, we will discuss two such disorders: attention deficit/ hyperactivity disorder and autism.

ATTENTION DEFICIT/HYPERACTIVITY DISORDER

Diego is always active, from the time he wakes up in the morning until the time he goes to bed at night. His mother reports that he came out the womb kicking and screaming, and he has not stopped moving since. He has a sweet disposition, but always seems to be in trouble with his teachers, parents, and after-school program

counselors. He seems to accidently break things; he lost his jacket three times last winter, and he never seems to sit still. His teachers believe he is a smart child, but he never finishes anything he starts and is so impulsive that he does not seem to learn much in school.

Diego likely has attention deficit/hyperactivity disorder (ADHD). The symptoms of this disorder were first described by Hans Hoffman in the 1920s. While taking care of his son while his wife was in the hospital giving birth to a second child, Hoffman noticed that the boy had trouble concentrating on his homework, had a short attention span, and had to repeatedly go over easy homework to learn the material (Jellinek & Herzog, 1999). Later, it was discovered that many hyperactive children—those who are fidgety, restless, socially disruptive, and have trouble with impulse control—also display short attention spans, problems with concentration, and distractibility. By the 1970s, it had become clear that many children who display attention problems often also exhibit signs of hyperactivity. In recognition of such findings, the DSM-III (published in 1980) included a new disorder: attention deficit disorder with and without hyperactivity, now known as attention deficit/hyperactivity disorder (ADHD).

A child with ADHD shows a constant pattern of inattention and/or hyperactive and impulsive behavior that interferes with normal functioning (APA, 2013). Some of the signs of inattention include great difficulty with and avoidance of tasks that require sustained attention (such as conversations or reading), failure to follow instructions (often resulting in failure to complete school work and other duties), disorganization (difficulty keeping things in order, poor time management, sloppy and messy work), lack of attention to detail, becoming easily distracted, and forgetfulness. Hyperactivity is characterized by excessive movement, and includes fidgeting or squirming, leaving one's seat in situations when remaining seated is expected, having trouble sitting still (e.g., in a restaurant), running about and climbing on things, blurting out responses before another person's question or statement has

been completed, difficulty waiting one's turn for something, and interrupting and intruding on others. Frequently, the hyperactive child comes across as noisy and boisterous. The child's behavior is hasty, impulsive, and seems to occur without much forethought; these characteristics may explain why adolescents and young adults diagnosed with ADHD receive more traffic tickets and have more automobile accidents than do others (Thompson, Molina, Pelham, & Gnagy, 2007).

ADHD occurs in about 5% of children (APA, 2013). On the average, boys are 3 times more likely to have ADHD than are girls; however, such findings might reflect the greater propensity of boys to engage in aggressive and antisocial behavior and thus incur a greater likelihood of being referred to psychological clinics (Barkley, 2006). Children with ADHD face severe academic and social challenges. Compared to their non-ADHD counterparts, children with ADHD have lower grades and standardized test scores and higher rates of expulsion, grade retention, and dropping out (Loe & Feldman, 2007). they also are less well-liked and more often rejected by their peers (Hoza et al., 2005).

Previously, ADHD was thought to fade away by adolescence. However, longitudinal studies have suggested that ADHD is a chronic problem, one that can persist into adolescence and adulthood (Barkley, Fischer, Smallish, & Fletcher, 2002). A recent study found that 29.3% of adults who had been diagnosed with ADHD decades earlier still showed symptoms (Barbaresi et al., 2013). Somewhat troubling, this study also reported that nearly 81% of those whose ADHD persisted into adulthood had experienced at least one other comorbid disorder, compared to 47% of those whose ADHD did not persist.

LIFE PROBLEMS FROM ADHD

Children diagnosed with ADHD face considerably worse long-term outcomes than do those children who do not receive such a

diagnosis. In one investigation, 135 adults who had been identified as having ADHD symptoms in the 1970s were contacted decades later and interviewed (Klein et al., 2012). Compared to a control sample of 136 participants who had never been diagnosed with ADHD, those who were diagnosed as children:
had worse educational attainment (more likely to have dropped out of high school and less likely to have earned a bachelor's degree);

- had lower socioeconomic status;
- held less prestigious occupational positions;
- were more likely to be unemployed;
- made considerably less in salary;
- scored worse on a measure of occupational functioning (indicating, for example, lower job satisfaction, poorer work relationships, and more firings);
- scored worse on a measure of social functioning (indicating, for example, fewer friendships and less involvement in social activities);
- were more likely to be divorced; and
- were more likely to have non-alcohol-related substance abuse problems. (Klein et al., 2012)

Longitudinal studies also show that children diagnosed with ADHD are at higher risk for substance abuse problems. One study reported that childhood ADHD predicted later drinking problems, daily smoking, and use of marijuana and other illicit drugs (Molina & Pelham, 2003). The risk of substance abuse problems appears to be even greater for those with ADHD who also exhibit antisocial tendencies (Marshal & Molina, 2006).

CAUSES OF ADHD

Family and twin studies indicate that genetics play a significant role in the development of ADHD. Burt (2009), in a review of 26 studies, reported that the median rate of concordance for identical twins was .66 (one study reported a rate of .90), whereas the median concordance rate for fraternal twins was .20. This study also found that the median concordance rate for unrelated (adoptive) siblings was .09; although this number is small, it is greater than 0, thus suggesting that the environment may have at least some influence. Another review of studies concluded that the heritability of inattention and hyperactivity were 71% and 73%, respectively (Nikolas & Burt, 2010).

The specific genes involved in ADHD are thought to include at least two that are important in the regulation of the neurotransmitter dopamine (Gizer, Ficks, & Waldman, 2009), suggesting that dopamine may be important in ADHD. Indeed, medications used in the treatment of ADHD, such as methylphenidate (Ritalin) and amphetamine with dextroamphetamine (Adderall), have stimulant qualities and elevate dopamine activity. People with ADHD show less dopamine activity in key regions of the brain, especially those associated with motivation and reward (Volkow et al., 2009), which provides support to the theory that dopamine deficits may be a vital factor in the development this disorder (Swanson et al., 2007).

Brain imaging studies have shown that children with ADHD exhibit abnormalities in their frontal lobes, an area in which dopamine is in abundance. Compared to children without ADHD, those with ADHD appear to have smaller frontal lobe volume, and they show less frontal lobe activation when performing mental tasks. Recall that one of the functions of the frontal lobes is to inhibit our behavior. Thus, abnormalities in this region may go a long way toward explaining the hyperactive, uncontrolled behavior of ADHD.

By the 1970s, many had become aware of the connection between nutritional factors and childhood behavior. At the time, much of the public believed that hyperactivity was caused by sugar and food additives, such as artificial coloring and flavoring. Undoubtedly, part of the appeal of this hypothesis was that it provided a simple explanation of (and treatment for) behavioral problems in children. A statistical review of 16 studies, however, concluded that sugar consumption has no effect at all on the behavioral and cognitive performance of children (Wolraich, Wilson, & White, 1995). Additionally, although food additives have been shown to increase hyperactivity in non-ADHD children, the effect is rather small (McCann et al., 2007). Numerous studies, however, have shown a significant relationship between exposure to nicotine in cigarette smoke during the prenatal period and ADHD (Linnet et al., 2003). Maternal smoking during pregnancy is associated with the development of more severe symptoms of the disorder (Thakur et al., 2013).

Is ADHD caused by poor parenting? Not likely. Remember, the genetics studies discussed above suggested that the family environment does not seem to play much of a role in the development of this disorder; if it did, we would expect the concordance rates to be higher for fraternal twins and adoptive siblings than has been demonstrated. All things considered, the evidence seems to point to the conclusion that ADHD is triggered more by genetic and neurological factors and less by social or environmental ones.

AUTISM SPECTRUM DISORDER

A seminal paper published in 1943 by psychiatrist Leo Kanner described an unusual neurodevelopmental condition he observed in a group of children. He called this condition early infantile autism, and it was characterized mainly by an inability to form close

emotional ties with others, speech and language abnormalities, repetitive behaviors, and an intolerance of minor changes in the environment and in normal routines (Bregman, 2005). What the DSM-5 refers to as autism spectrum disorder today, is a direct extension of Kanner's work.

Autism spectrum disorder is probably the most misunderstood and puzzling of the neurodevelopmental disorders. Children with this disorder show signs of significant disturbances in three main areas: (a) deficits in social interaction, (b) deficits in communication, and (c) repetitive patterns of behavior or interests. These disturbances appear early in life and cause serious impairments in functioning (APA, 2013). The child with autism spectrum disorder might exhibit deficits in social interaction by not initiating conversations with other children or turning their head away when spoken to. These children do not make eye contact with others and seem to prefer playing alone rather than with others. In a certain sense, it is almost as though these individuals live in a personal and isolated social world others are simply not privy to or able to penetrate. Communication deficits can range from a complete lack of speech, to one word responses (e.g., saying "Yes" or "No" when replying to questions or statements that require additional elaboration), to echoed speech (e.g., parroting what another person says, either immediately or several hours or even days later), to difficulty maintaining a conversation because of an inability to reciprocate others' comments. These deficits can also include problems in using and understanding nonverbal cues (e.g., facial expressions, gestures, and postures) that facilitate normal communication.

Repetitive patterns of behavior or interests can be exhibited a number of ways. The child might engage in stereotyped, repetitive movements (rocking, head-banging, or repeatedly dropping an object and then picking it up), or she might show great distress at small changes in routine or the environment. For example, the child might throw a temper tantrum if an object is not in its proper

place or if a regularly-scheduled activity is rescheduled. In some cases, the person with autism spectrum disorder might show highly restricted and fixated interests that appear to be abnormal in their intensity. For instance, the person might learn and memorize every detail about something even though doing so serves no apparent purpose. Importantly, autism spectrum disorder is not the same thing as intellectual disability, although these two conditions are often comorbid. The DSM-5 specifies that the symptoms of autism spectrum disorder are not caused or explained by intellectual disability.

LIFE PROBLEMS FROM AUTISM SPECTRUM DISORDER

Autism spectrum disorder is referred to in everyday language as autism; in fact, the disorder was termed "autistic disorder" in earlier editions of the DSM, and its diagnostic criteria were much narrower than those of autism spectrum disorder. The qualifier "spectrum" in autism spectrum disorder is used to indicate that individuals with the disorder can show a range, or spectrum, of symptoms that vary in their magnitude and severity: some severe, others less severe. The previous edition of the DSM included a diagnosis of Asperger's disorder, generally recognized as a less severe form of autistic disorder; individuals diagnosed with Asperger's disorder were described as having average or high intelligence and a strong vocabulary, but exhibiting impairments in social interaction and social communication, such as talking only about their special interests (Wing, Gould, & Gillberg, 2011). However, because research has failed to demonstrate that Asperger's disorder differs qualitatively from autistic disorder, the DSM-5 does not include it, which is prompting concerns among some parents that their children may no longer be eligible for special services ("Asperger's Syndrome Dropped," 2012). Some individuals with autism spectrum disorder, particularly those with better language and intellectual skills, can live and work independently as adults.

However, most do not because the symptoms remain sufficient to cause serious impairment in many realms of life (APA, 2013).

Here is an instructive and poignant video highlighting severe autism: 10 Early Signs of Autism (UPDATED).

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=228#oembed-1

Currently, estimates indicate that nearly 1 in 88 children in the United States has autism spectrum disorder; the disorder is 5 times more common in boys (1 out of 54) than girls (1 out of 252) (CDC, 2012). Rates of autistic spectrum disorder have increased dramatically since the 1980s. For example, California saw an increase of 273% in reported cases from 1987 through 1998 (Byrd, 2002); between 2000 and 2008, the rate of autism diagnoses in the United States increased 78% (CDC, 2012). Although it is difficult to interpret this increase, it is possible that the rise in prevalence

is the result of the broadening of the diagnosis, increased efforts to identify cases in the community, and greater awareness and acceptance of the diagnosis. In addition, mental health professionals are now more knowledgeable about autism spectrum disorder and are better equipped to make the diagnosis, even in subtle cases (Novella, 2008).

Why Is the Prevalence Rate of ADHD Increasing?

Many people believe that the rates of ADHD have increased in recent years, and there is evidence to support this contention. In a recent study, investigators found that the parent-reported prevalence of ADHD among children (4–17 years old) in the United States increased by 22% during a 4-year period, from 7.8% in 2003 to 9.5% in 2007 (CDC, 2010). Over time this increase in parent-reported ADHD was observed in all sociodemographic groups and was reflected by substantial increases in 12 states (Indiana, North Carolina, and Colorado were the top three). The increases were greatest for older teens (ages 15–17), multiracial and Hispanic children, and children with a primary language other than English. Another investigation found that from 1998–2000 through 2007–2009 the parent-reported prevalence of ADHD increased

among U.S. children between the ages of 5–17 years old, from 6.9% to 9.0% (Akinbami, Liu, Pastor, & Reuben, 2011).

A major weakness of both studies was that children were not actually given a formal diagnosis. Instead, parents were simply asked whether or not a doctor or other health-care provider had ever told them their child had ADHD; the reported prevalence rates thus may have been affected by the accuracy of parental memory. Nevertheless, the findings from these studies raise important questions concerning what appears to be a demonstrable rise in the prevalence of ADHD. Although the reasons underlying this apparent increase in the rates of ADHD over time are poorly understood and, at best, speculative, several explanations are viable:

ADHD may be over-diagnosed by doctors who are too quick to medicate children as a behavior treatment.

There is greater awareness of ADHD now than in the past. Nearly everyone has heard of ADHD, and most parents and teachers are aware of its key symptoms. Thus, parents may be quick to take their children to a doctor if they believe their child possesses these symptoms, or teachers may be more likely now than in the past to notice the symptoms and refer the child for evaluation.

The use of computers, video games, iPhones, and other electronic devices has become pervasive among children in the early 21st century, and these devices could potentially shorten children's attentions spans. Thus, what might seem like inattention to some parents and teachers could simply reflect exposure to too much technology.

ADHD diagnostic criteria have changed over time.

CAUSES OF AUTISM SPECTRUM DISORDER

Early theories of autism placed the blame squarely on the shoulders of the child's parents, particularly the mother. Bruno Bettelheim (an Austrian-born American child psychologist who was

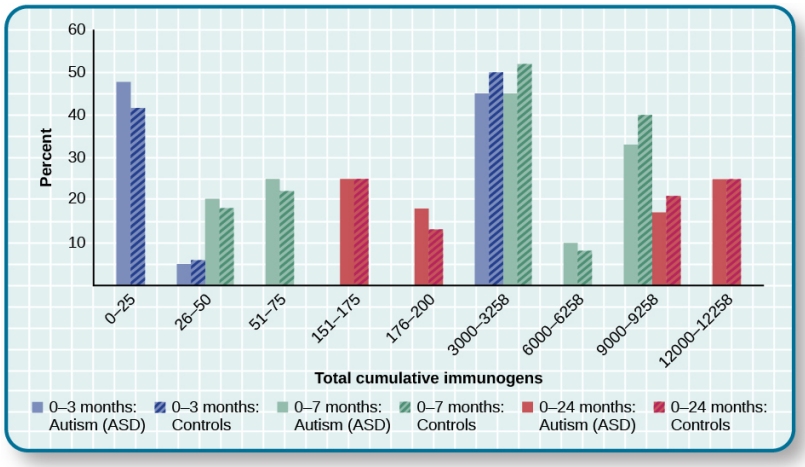
heavily influenced by Sigmund Freud's ideas) suggested that a mother's ambivalent attitudes and her frozen and rigid emotions toward her child were the main causal factors in childhood autism. In what must certainly stand as one of the more controversial assertions in psychology over the last 50 years, he wrote, "I state my belief that the precipitating factor in infantile autism is the parent's wish that his child should not exist" (Bettelheim, 1967, p. 125). As you might imagine, Bettelheim did not endear himself to a lot of people with this position; incidentally, no scientific evidence exists supporting his claims.

The exact causes of autism spectrum disorder remain unknown despite massive research efforts over the last two decades (Meek, Lemery-Chalfant, Jahromi, & Valiente, 2013). Autism appears to be strongly influenced by genetics, as identical twins show concordance rates of 60%–90%, whereas concordance rates for fraternal twins and siblings are 5%–10% (Autism Genome Project Consortium, 2007). Many different genes and gene mutations have been implicated in autism (Meek et al., 2013). Among the genes involved are those important in the formation of synaptic circuits that facilitate communication between different areas of the brain (Gauthier et al., 2011). A number of environmental factors are also thought to be associated with increased risk for autism spectrum disorder, at least in part, because they contribute to new mutations. These factors include exposure to pollutants, such as plant emissions and mercury, urban versus rural residence, and vitamin D deficiency (Kinney, Barch, Chayka, Napoleon, & Munir, 2009).

CHILD VACCINATIONS AND AUTISM SPECTRUM DISORDER

In the late 1990s, a prestigious medical journal published an article purportedly showing that autism is triggered by the MMR (measles, mumps, and rubella) vaccine. These findings were very

controversial and drew a great deal of attention, sparking an international forum on whether children should be vaccinated. In a shocking turn of events, some years later the article was retracted by the journal that had published it after accusations of fraud on the part of the lead researcher. Despite the retraction, the reporting in popular media led to concerns about a possible link between vaccines and autism persisting. A recent survey of parents, for example, found that roughly a third of respondents expressed such a concern (Kennedy, LaVail, Nowak, Basket, & Landry, 2011); and perhaps fearing that their children would develop autism, more than 10% of parents of young children refuse or delay vaccinations (Dempsey et al., 2011). Some parents of children with autism mounted a campaign against scientists who refuted the vaccine-autism link. Even politicians and several well-known celebrities weighed in; for example, actress Jenny McCarthy (who believed that a vaccination caused her son's autism) co-authored a book on the matter. However, there is no scientific evidence that a link exists between autism and vaccinations (Hughes, 2007). Indeed, a recent study compared the vaccination histories of 256 children with autism spectrum disorder with that of 752 control children across three time periods during their first two years of life (birth to 3 months, birth to 7 months, and birth to 2 years) (DeStefano, Price, & Weintraub, 2013). At the time of the study, the children were between 6 and 13 years old, and their prior vaccination records were obtained. Because vaccines contain immunogens (substances that fight infections), the investigators examined medical records to see how many immunogens children received to determine if those children who received more immunogens were at greater risk for developing autism spectrum disorder. The results of this study, clearly demonstrate that the quantity of immunogens from vaccines received during the first two years of life were not at all related to the development of autism spectrum disorder. There is not a relationship between vaccinations and autism spectrum disorders.



In terms of their exposure to immunogens in vaccines, overall, there is not a significant difference between children with autism spectrum disorder and their age-matched controls without the disorder (DeStefano et al., 2013).

Why does concern over vaccines and autism spectrum disorder persist? Since the proliferation of the Internet in the 1990s, parents have been constantly bombarded with online information that can become magnified and take on a life of its own. The enormous volume of electronic information pertaining to autism spectrum disorder, combined with how difficult it can be to grasp complex scientific concepts, can make separating good research from bad challenging (Downs, 2008). Notably, the study that fueled the controversy reported that 8 out of 12 children—according to their parents—developed symptoms consistent with autism spectrum disorder shortly after receiving a vaccination. To conclude that vaccines cause autism spectrum disorder on this basis, as many did, is clearly incorrect for a number of reasons, not the least of which is because correlation does not imply causation, as you've learned.

Additionally, as was the case with diet and ADHD in the 1970s, the notion that autism spectrum disorder is caused by vaccinations

is appealing to some because it provides a simple explanation for this condition. Like all disorders, however, there are no simple explanations for autism spectrum disorder. Although the research discussed above has shed some light on its causes, science is still a long way from complete understanding of the disorder.

TEST YOUR UNDERSTANDING

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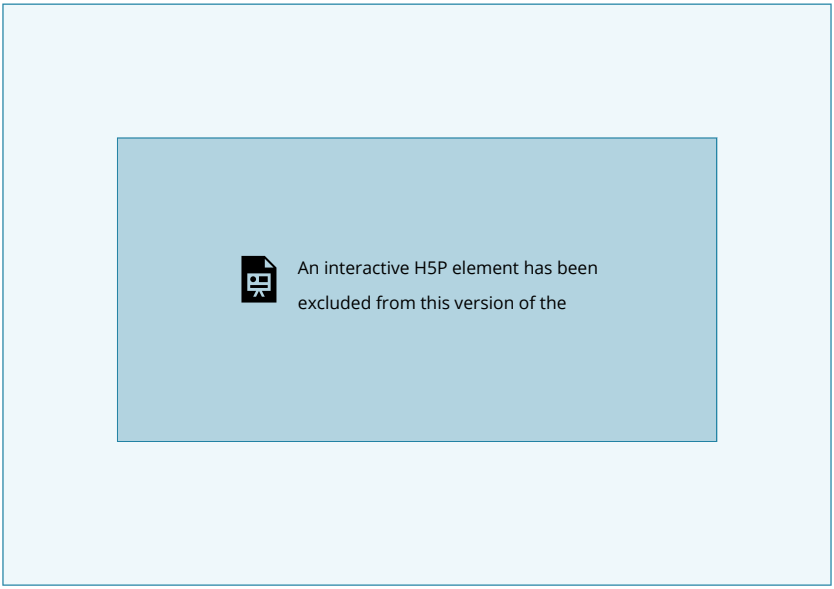
SUMMARY

Neurodevelopmental disorders are a group of disorders that are typically diagnosed during childhood and are characterized by developmental deficits in personal, social, academic, and

intellectual realms; these disorders include attention deficit/hyperactivity disorder (ADHD) and autism spectrum disorder. ADHD is characterized by a pervasive pattern of inattention and/or hyperactive and impulsive behavior that interferes with normal functioning. Genetic and neurobiological factors contribute to the development of ADHD, which can persist well into adulthood and is often associated with poor long-term outcomes. The major features of autism spectrum disorder include deficits in social interaction and communication and repetitive movements or interests. As with ADHD, genetic factors appear to play a prominent role in the development of autism spectrum disorder; exposure to environmental pollutants such as mercury have also been linked to the development of this disorder. Although it is believed by some that autism is triggered by the MMR vaccination, evidence does not support this claim.

REVIEW QUESTIONS



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CRITICAL THINKING QUESTIONS

Compare the factors that are important in the development of ADHD with those that are important in the development of autism spectrum disorder.

Genetic factors appear to play a major role in the development of both ADHD and autism spectrum disorder: studies show higher rates of concordance among identical twins than among fraternal twins for both disorders. In ADHD, genes that regulate

dopamine have been implicated; in autism spectrum disorder, de novo genetic mutations appear to be important. Imaging studies suggest that abnormalities in the frontal lobes may be important in the development of ADHD. Parenting practices are not connected to the development of either disorder. Although environmental toxins are generally unimportant in the development of ADHD, exposure to cigarette smoke during the prenatal period has been linked to the development of the disorder; a number of environmental factors are thought to be associated with an increased risk for autism spectrum disorder: exposure to pollutants, an urban versus rural residence, and vitamin D deficiency. Although some people continue to believe that MMR vaccinations can cause autism spectrum disorder (due to an influential paper that was later retracted), there is no scientific evidence that supports this assertion.

PERSONAL APPLICATION QUESTION

Discuss the characteristics of autism spectrum disorder with a few of your friends or members of your family (choose friends or family members who know little about the disorder) and ask them if they think the cause is due to bad parenting or vaccinations. If they indicate that they believe either to be true, why do you think this might be the case? What would be your response?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:

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Media Attributions

- "10 Early Signs of Autism (UPDATED)" by Autism Family. Standard YouTube License.

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THERAPY AND TREATMENT

Many forms of therapy have been developed to treat a wide array of problems. These marines who served in Iraq and Afghanistan, together with community mental health volunteers, are part of the Ocean Therapy program at Camp Pendleton, a program in which learning to surf is combined with group discussions. The program helps vets recover, especially vets who suffer from post-traumatic stress disorder (PTSD).

What comes to mind when you think about therapy for psychological problems? You might picture someone lying on a

couch talking about his childhood while the therapist sits and takes notes, à la Sigmund Freud. But can you envision a therapy session in which someone is wearing virtual reality headgear to conquer a fear of snakes?

In this chapter, you will see that approaches to therapy include both psychological and biological interventions, all with the goal of alleviating distress. Because psychological problems can originate from various sources—biology, genetics, childhood experiences, conditioning, and sociocultural influences—psychologists have developed many different therapeutic techniques and approaches. The Ocean Therapy program uses multiple approaches to support the mental health of veterans in the group.

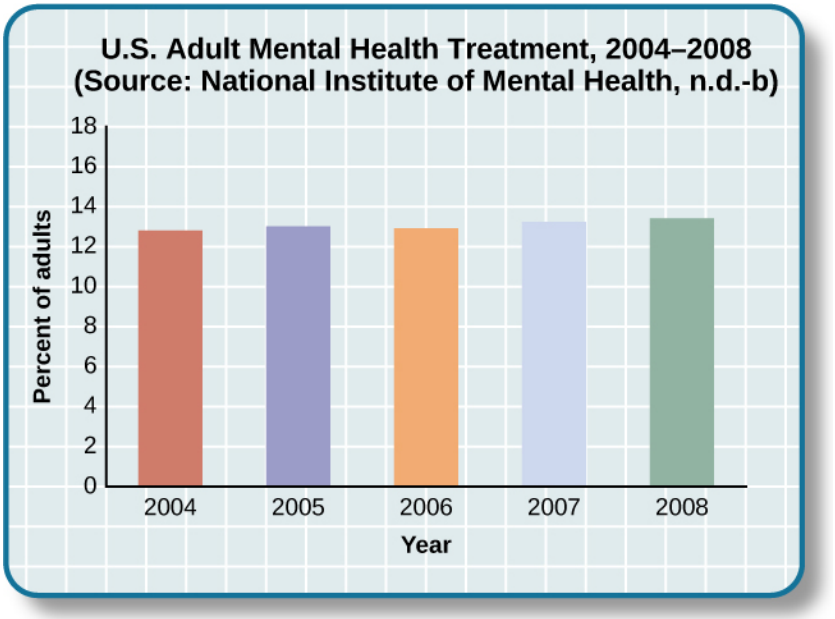
Learning Objectives

By the end of this section, you will be able to:

- Explain how people with psychological disorders have been treated throughout the ages
- Discuss deinstitutionalization
- Discuss the ways in which mental health services are delivered today
- Distinguish between voluntary and involuntary treatment

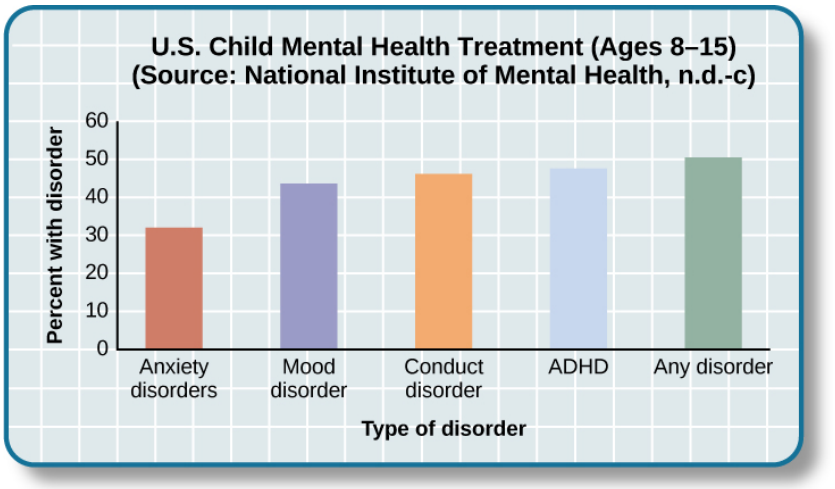
Before we explore the various approaches to therapy used today, let's begin our study of therapy by looking at how many people experience mental illness and how many receive treatment. According to the U.S. Department of Health and Human Services (2013), 19% of U.S. adults experienced mental illness in 2012. For teens (ages 13–18), the rate is similar to that of adults, and for children ages 8–15, current estimates suggest that 13% experience mental illness in a given year (National Institute of Mental Health [NIMH], n.d.-a)

With many different treatment options available, approximately how many people receive mental health treatment per year? According to the Substance Abuse and Mental Health Services Administration (SAMHSA), in 2008, 13.4% of adults received treatment for a mental health issue (NIMH, n.d.-b). These percentages, reflect the number of adults who received care in inpatient and outpatient settings and/or used prescription medication for psychological disorders.



The percentage of adults who received mental health treatment in 2004–2008 is shown. Adults seeking treatment increased slightly from 2004 to 2008.

Children and adolescents also receive mental health services. The Centers for Disease Control and Prevention's National Health and Nutrition Examination Survey (NHANES) found that approximately half (50.6%) of children with mental disorders had received treatment for their disorder within the past year (NIMH, n.d.-c). However, there were some differences between treatment rates by category of disorder. For example, children with anxiety disorders were least likely to have received treatment in the past year, while children with ADHD or a conduct disorder were more likely to receive treatment. Can you think of some possible reasons for these differences in receiving treatment?



About one-third to one-half of U.S. adolescents (ages 8–15) with mental disorders receive treatment, with behavior-related disorders more likely to be treated.

Considering the many forms of treatment for mental health disorders available today, how did these forms of treatment emerge? Let's take a look at the history of mental health treatment from the past (with some questionable approaches in light of modern understanding of mental illness) to where we are today.

TREATMENT IN THE PAST

For much of history, the mentally ill have been treated very poorly. It was believed that mental illness was caused by demonic possession, witchcraft, or an angry god (Szasz, 1960). For example, in medieval times, abnormal behaviors were viewed as a sign that a person was possessed by demons. If someone was considered to be possessed, there were several forms of treatment to release spirits from the individual. The most common treatment was exorcism, often conducted by priests or other religious figures:

Incantations and prayers were said over the person's body, and she may have been given some medicinal drinks. Another form of treatment for extreme cases of mental illness was trephining: A small hole was made in the afflicted individual's skull to release spirits from the body. Most people treated in this manner died. In addition to exorcism and trephining, other practices involved execution or imprisonment of people with psychological disorders. Still others were left to be homeless beggars. Generally speaking, most people who exhibited strange behaviors were greatly misunderstood and treated cruelly. The prevailing theory of psychopathology in earlier history was the idea that mental illness was the result of demonic possession by either an evil spirit or an evil god because early beliefs incorrectly attributed all unexplainable phenomena to deities deemed either good or evil.

From the late 1400s to the late 1600s, a common belief perpetuated by some religious organizations was that some people made pacts with the devil and committed horrible acts, such as eating babies (Blumberg, 2007). These people were considered to be witches and were tried and condemned by courts—they were often burned at the stake. Worldwide, it is estimated that tens of thousands of mentally ill people were killed after being accused of being witches or under the influence of witchcraft (Hemphill, 1966)

By the 18th century, people who were considered odd and unusual were placed in asylums. Asylums were the first institutions created for the specific purpose of housing people with psychological disorders, but the focus was ostracizing them from society rather than treating their disorders. Often these people were kept in windowless dungeons, beaten, chained to their beds, and had little to no contact with caregivers.

This painting by Francisco Goya, called The Madhouse, depicts a mental asylum and its inhabitants in the early 1800s. It portrays those with psychological disorders as victims.

In the late 1700s, a French physician, Philippe Pinel, argued for more humane treatment of the mentally ill. He suggested that they be unchained and talked to, and that's just what he did for patients at La Salpêtrière in Paris in 1795 Patients benefited from this more humane treatment, and many were able to leave the hospital.

This painting by Tony Robert-Fleury depicts Dr. Philippe Pinel ordering the removal of chains from patients at the Salpêtrière asylum in Paris.

In the 19th century, Dorothea Dix led reform efforts for mental health care in the United States. She investigated how those who are mentally ill and poor were cared for, and she discovered an underfunded and unregulated system that perpetuated abuse of this population (Tiffany, 1891). Horrified by her findings, Dix began lobbying various state legislatures and the U.S. Congress for change (Tiffany, 1891). Her efforts led to the creation of the first mental asylums in the United States.

Dorothea Dix was a social reformer who became an advocate for the indigent insane and was instrumental in creating the first American mental asylum. She did this by relentlessly lobbying state legislatures and Congress to set up and fund such institutions.

Despite reformers' efforts, however, a typical asylum was filthy, offered very little treatment, and often kept people for decades. At Willard Psychiatric Center in upstate New York, for example, one treatment was to submerge patients in cold baths for long periods of time. Electroshock treatment was also used, and the way the treatment was administered often broke patients' backs; in 1943, doctors at Willard administered 1,443 shock treatments (Willard Psychiatric Center, 2009). (Electroshock is now called electroconvulsive treatment, and the therapy is still used, but with safeguards and under anesthesia. A brief application of electric stimulus is used to produce a generalized seizure. Controversy

continues over its effectiveness versus the side effects.) Many of the wards and rooms were so cold that a glass of water would be frozen by morning (Willard Psychiatric Center, 2009). Willard's doors were not closed until 1995. Conditions like these remained commonplace until well into the 20th century.

Starting in 1954 and gaining popularity in the 1960s, antipsychotic medications were introduced. These proved a tremendous help in controlling the symptoms of certain psychological disorders, such as psychosis. Psychosis was a common diagnosis of individuals in mental hospitals, and it was often evidenced by symptoms like hallucinations and delusions, indicating a loss of contact with reality. Then in 1963, Congress passed and John F. Kennedy signed the Mental Retardation Facilities and Community Mental Health Centers Construction Act, which provided federal support and funding for community mental health centers (National Institutes of Health, 2013). This legislation changed how mental health services were delivered in the United States. It started the process of deinstitutionalization, the closing of large asylums, by providing for people to stay in their communities and be treated locally. In 1955, there were 558,239 severely mentally ill patients institutionalized at public hospitals (Torrey, 1997). By 1994, by percentage of the population, there were 92% fewer hospitalized individuals (Torrey, 1997).

MENTAL HEALTH TREATMENT TODAY

Today, there are community mental health centers across the nation. They are located in neighborhoods near the homes of clients, and they provide large numbers of people with mental health services of various kinds and for many kinds of problems. Unfortunately, part of what occurred with deinstitutionalization was that those released from institutions were supposed to go to newly created centers, but the system was not set up effectively.

Centers were underfunded, staff was not trained to handle severe illnesses such as schizophrenia, there was high staff burnout, and no provision was made for the other services people needed, such as housing, food, and job training. Without these supports, those people released under deinstitutionalization often ended up homeless. Even today, a large portion of the homeless population is considered to be mentally ill. Statistics show that 26% of homeless adults living in shelters experience mental illness (U.S. Department of Housing and Urban Development [HUD], 2011).

(a)

(b)

(a) Of the homeless individuals in U.S. shelters, about one-quarter have a severe mental illness (HUD, 2011). (b) Correctional institutions also report a high number of individuals living with mental illness. (credit a: modification of work by C.G.P. Grey; credit b: modification of work by Bart Everson)

Another group of the mentally ill population is involved in the corrections system. According to a 2006 special report by the Bureau of Justice Statistics (BJS), approximately 705,600 mentally ill adults were incarcerated in the state prison system, and another 78,800 were incarcerated in the federal prison system. A further 479,000 were in local jails. According to the study, "people with mental illnesses are overrepresented in probation and parole populations at estimated rates ranging from two to four times the

general population" (Prins & Draper, 2009, p. 23). The Treatment Advocacy Center reported that the growing number of mentally ill inmates has placed a burden on the correctional system (Torrey et al., 2014).

Today, instead of asylums, there are psychiatric hospitals run by state governments and local community hospitals focused on short-term care. In all types of hospitals, the emphasis is on short-term stays, with the average length of stay being less than two weeks and often only several days. This is partly due to the very high cost of psychiatric hospitalization, which can be about $800 to $1000 per night (Stensland, Watson, & Grazier, 2012). Therefore, insurance coverage often limits the length of time a person can be hospitalized for treatment. Usually individuals are hospitalized only if they are an imminent threat to themselves or others.

View this timeline showing the history of mental institutions in the United States: A History of Mental Institutions in the United States.

Most people suffering from mental illnesses are not hospitalized. If someone is feeling very depressed, complains of hearing voices, or feels anxious all the time, he or she might seek psychological treatment. A friend, spouse, or parent might refer someone for treatment. The individual might go see his primary care physician first and then be referred to a mental health practitioner.

Some people seek treatment because they are involved with the state's child protective services—that is, their children have been removed from their care due to abuse or neglect. The parents might be referred to psychiatric or substance abuse facilities and

the children would likely receive treatment for trauma. If the parents are interested in and capable of becoming better parents, the goal of treatment might be family reunification. For other children whose parents are unable to change—for example, the parent or parents who are heavily addicted to drugs and refuse to enter treatment—the goal of therapy might be to help the children adjust to foster care and/or adoption.

Therapy with children may involve play. (credit: "LizMarie_AK"/Flick4)

Some people seek therapy because the criminal justice system referred them or required them to go. For some individuals, for example, attending weekly counseling sessions might be a condition of parole. If an individual is mandated to attend therapy, she is seeking services involuntarily. Involuntary treatment refers to therapy that is not the individual's choice. Other individuals might voluntarily seek treatment. Voluntary treatment means the person chooses to attend therapy to obtain relief from symptoms.

Psychological treatment can occur in a variety of places. An individual might go to a community mental health center or a practitioner in private or community practice. A child might see a school counselor, school psychologist, or school social worker. An

incarcerated person might receive group therapy in prison. There are many different types of treatment providers, and licensing requirements vary from state to state. Besides psychologists and psychiatrists, there are clinical social workers, marriage and family therapists, and trained religious personnel who also perform counseling and therapy.

A range of funding sources pay for mental health treatment: health insurance, government, and private pay. In the past, even when people had health insurance, the coverage would not always pay for mental health services. This changed with the Mental Health Parity and Addiction Equity Act of 2008, which requires group health plans and insurers to make sure there is parity of mental health services (U.S. Department of Labor, n.d.). This means that co-pays, total number of visits, and deductibles for mental health and substance abuse treatment need to be equal to and cannot be more restrictive or harsher than those for physical illnesses and medical/surgical problems.

Finding treatment sources is also not always easy: there may be limited options, especially in rural areas and low-income urban areas; waiting lists; poor quality of care available for indigent patients; and financial obstacles such as co-pays, deductibles, and time off from work. Over 85% of the l,669 federally designated mental health professional shortage areas are rural; often primary care physicians and law enforcement are the first-line mental health providers (Ivey, Scheffler, & Zazzali, 1998), although they do not have the specialized training of a mental health professional, who often would be better equipped to provide care. Availability, accessibility, and acceptability (the stigma attached to mental illness) are all problems in rural areas. Approximately two-thirds of those with symptoms receive no care at all (U.S. Department of Health and Human Services, 2005; Wagenfeld, Murray, Mohatt, & DeBruiynb, 1994). At the end of 2013, the U.S. Department of Agriculture announced an investment of $50 million to help improve access and treatment for mental health problems as part

of the Obama administration's effort to strengthen rural communities.

TEST YOUR UNDERSTANDING

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=233#h5p-268

SUMMARY

It was once believed that people with psychological disorders, or those exhibiting strange behavior, were possessed by demons. These people were forced to take part in exorcisms, were imprisoned, or executed. Later, asylums were built to house the

mentally ill, but the patients received little to no treatment, and many of the methods used were cruel. Philippe Pinel and Dorothea Dix argued for more humane treatment of people with psychological disorders. In the mid-1960s, the deinstitutionalization movement gained support and asylums were closed, enabling people with mental illness to return home and receive treatment in their own communities. Some did go to their family homes, but many became homeless due to a lack of resources and support mechanisms.

Today, instead of asylums, there are psychiatric hospitals run by state governments and local community hospitals, with the emphasis on short-term stays. However, most people suffering from mental illness are not hospitalized. A person suffering symptoms could speak with a primary care physician, who most likely would refer him to someone who specializes in therapy. The person can receive outpatient mental health services from a variety of sources, including psychologists, psychiatrists, marriage and family therapists, school counselors, clinical social workers, and religious personnel. These therapy sessions would be covered through insurance, government funds, or private (self) pay.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=233#h5p-269

CRITICAL THINKING QUESTIONS

People with psychological disorders have been treated poorly throughout history. Describe some

efforts to improve treatment, include explanations for the success or lack thereof.

Beginning in the Middle Ages and up until the mid-20th century, the mentally ill were misunderstood and treated cruelly. In the 1700s, Philippe Pinel advocated for patients to be unchained, and he was able to affect this in a Paris hospital. In the 1800s, Dorothea Dix urged the government to provide better funded and regulated care, which led to the creation of asylums, but treatment generally remained quite poor. Federally mandated deinstitutionalization in the 1960s began the elimination of asylums, but it was often inadequate in providing the infrastructure for replacement treatment.

Usually someone is hospitalized only if they are an imminent threat to themselves or others. Describe a situation that might meet these criteria.

Frank is severely depressed. He lost his job one year ago and has not been able to find another one. A few months after losing his job, his home was foreclosed and his wife left him. Lately, he has been thinking that he would be better off dead. He's begun giving his possessions away and has purchased a handgun. He plans to kill himself on what would have been his 20th wedding anniversary, which is coming up in a few weeks.

PERSONAL APPLICATION QUESTIONS

Do you think there is a stigma associated with mentally ill persons today? Why or why not?
What are some places in your community that offer mental health services? Would you feel comfortable seeking assistance at one of these facilities? Why or why not?

GLOSSARY

An interactive H5P element has been excluded from this version of the text. You can view it online here:
https://pressbooks.cuny.edu/jjcpsy101/?p=233#h5p-270

Learning Objectives

By the end of this section, you will be able to:

- Distinguish between psychotherapy and biomedical therapy
- Recognize various orientations to psychotherapy
- Discuss psychotropic medications and recognize which medications are used to treat specific psychological disorders

One of the goals of therapy is to help a person stop repeating and reenacting destructive patterns and to start looking for better solutions to difficult situations. This goal is reflected in the following poem:

> *Autobiography in Five Short Chapters* by Portia Nelson (1993)

> Chapter One
> I walk down the street.
> There is a deep hole in the sidewalk.
> I fall in.
> I am lost. . . . I am helpless.
> It isn't my fault.
> It takes forever to find a way out.

> Chapter Two
> I walk down the same street.
> There is a deep hole in the sidewalk.

I pretend I don't see it.
I fall in again.
I can't believe I am in this same place.
But, it isn't my fault.
It still takes a long time to get out.

Chapter Three
I walk down the same street.
There is a deep hole in the sidewalk.
I *see* it is there.
I still fall in . . . it's a habit . . . but,
my eyes are open.
I know where I am.
It is *my* fault.
I get out immediately.

Chapter Four
I walk down the same street.
There is a deep hole in the sidewalk.
I walk around it.

Chapter Five
I walk down another street.

Two types of therapy are psychotherapy and biomedical therapy. Both types of treatment help people with psychological disorders, such as depression, anxiety, and schizophrenia. Psychotherapy is a psychological treatment that employs various methods to help someone overcome personal problems, or to attain personal growth. In modern practice, it has evolved ino what is known as psychodynamic therapy, which will be discussed later. Biomedical therapy involves medication and/or medical procedures to treat psychological disorders. First, we will explore the various psychotherapeutic orientations, many of these orientations were discussed in the Introduction chapter).

Various Psychotherapy Techniques

Type	Description	Example
Psychodynamic psychotherapy	Talk therapy based on belief that the unconscious and childhood conflicts impact behavior	Patient talks about his past
Play therapy	Psychoanalytical therapy wherein interaction with toys is used instead of talk; used in child therapy	Patient (child) acts out family scenes with dolls
Behavior therapy	Principles of learning applied to change undesirable behaviors	Patient learns to overcome fear of elevators through several stages of relaxation techniques
Cognitive therapy	Awareness of cognitive process helps patients eliminate thought patterns that lead to distress	Patient learns not to overgeneralize failure based on single failure
Cognitive-behavioral therapy	Work to change cognitive distortions and self-defeating behaviors	Patient learns to identify self-defeating behaviors to overcome an eating disorder
Humanistic therapy	Increase self-awareness and acceptance through focus on conscious thoughts	Patient learns to articulate thoughts that keep her from achieving her goals

PSYCHOTHERAPY TECHNIQUES: PSYCHOANALYSIS

Psychoanalysis was developed by Sigmund Freud and was the first form of psychotherapy. It was the dominant therapeutic technique in the early 20th century, but it has since waned significantly in popularity. Freud believed most of our psychological problems are the result of repressed impulses and trauma experienced in childhood, and he believed psychoanalysis would help uncover long-buried feelings. In a psychoanalyst's office, you might see a patient lying on a couch speaking of dreams or childhood memories, and the therapist using various Freudian methods such as free association and dream analysis. In free association, the patient relaxes and then says whatever comes to mind at the moment. However, Freud felt that the ego would at times try to block, or repress, unacceptable urges or painful conflicts during

free association. Consequently, a patient would demonstrate resistance to recalling these thoughts or situations. In dream analysis, a therapist interprets the underlying meaning of dreams.

Psychoanalysis is a therapy approach that typically takes years. Over the course of time, the patient reveals a great deal about himself to the therapist. Freud suggested that during this patient-therapist relationship, the patient comes to develop strong feelings for the therapist—maybe positive feelings, maybe negative feelings. Freud called this transference: the patient transfers all the positive or negative emotions associated with the patient's other relationships to the psychoanalyst. For example, Crystal is seeing a psychoanalyst. During the years of therapy, she comes to see her therapist as a father figure. She transfers her feelings about her father onto her therapist, perhaps in an effort to gain the love and attention she did not receive from her own father.

This is the famous couch in Freud's consulting room. Patients were instructed to lie comfortably on the couch and to face away from Freud in order to feel less inhibited and to help them focus. Today, a psychotherapy patient is not likely to lie on a couch; instead he is more likely to sit facing the therapist (Prochaska & Norcross, 2010). (credit: Robert Huffstutter)

Today, Freud's psychoanalytical perspective has been expanded upon by the developments of subsequent theories and methodologies: the psychodynamic perspective. This approach to therapy remains centered on the role of people's internal drives and forces, but treatment is less intensive than Freud's original model.

View a brief video that presents an overview of

psychoanalysis theory, research, and practice: Psychoanalysis: An Introduction.
https://youtube.com/watch?v=WiLtQATY_1s

PSYCHOTHERAPY: PLAY THERAPY

Play therapy is often used with children since they are not likely to sit on a couch and recall their dreams or engage in traditional talk therapy. This technique uses a therapeutic process of play to "help clients prevent or resolve psychosocial difficulties and achieve optimal growth" (O'Connor, 2000, p. 7). The idea is that children play out their hopes, fantasies, and traumas while using dolls, stuffed animals, and sandbox figurines. Play therapy can also be used to help a therapist make a diagnosis. The therapist observes how the child interacts with toys (e.g., dolls, animals, and home settings) in an effort to understand the roots of the child's disturbed behavior. Play therapy can be nondirective or directive. In nondirective play therapy, children are encouraged to work through their problems by playing freely while the therapist observes (LeBlanc & Ritchie, 2001). In directive play therapy, the therapist provides more structure and guidance in the play session by suggesting topics, asking questions, and even playing with the child (Harter, 1977).

This type of play therapy is known as sandplay or sandtray therapy. Children can set up a three-dimensional world using various figures and objects that correspond to their inner state (Kalff, 1991). (credit: Kristina Walter)

PSYCHOTHERAPY: BEHAVIOR THERAPY

In psychoanalysis, therapists help their patients look into their past to uncover repressed feelings. In behavior therapy, a therapist employs principles of learning to help clients change undesirable behaviors—rather than digging deeply into one's unconscious. Therapists with this orientation believe that dysfunctional behaviors, like phobias and bedwetting, can be changed by teaching clients new, more constructive behaviors. Behavior therapy employs both classical and operant conditioning techniques to change behavior.

One type of behavior therapy utilizes classical conditioning techniques. Therapists using these techniques believe that dysfunctional behaviors are conditioned responses. Applying the conditioning principles developed by Ivan Pavlov, these therapists seek to recondition their clients and thus change their behavior.

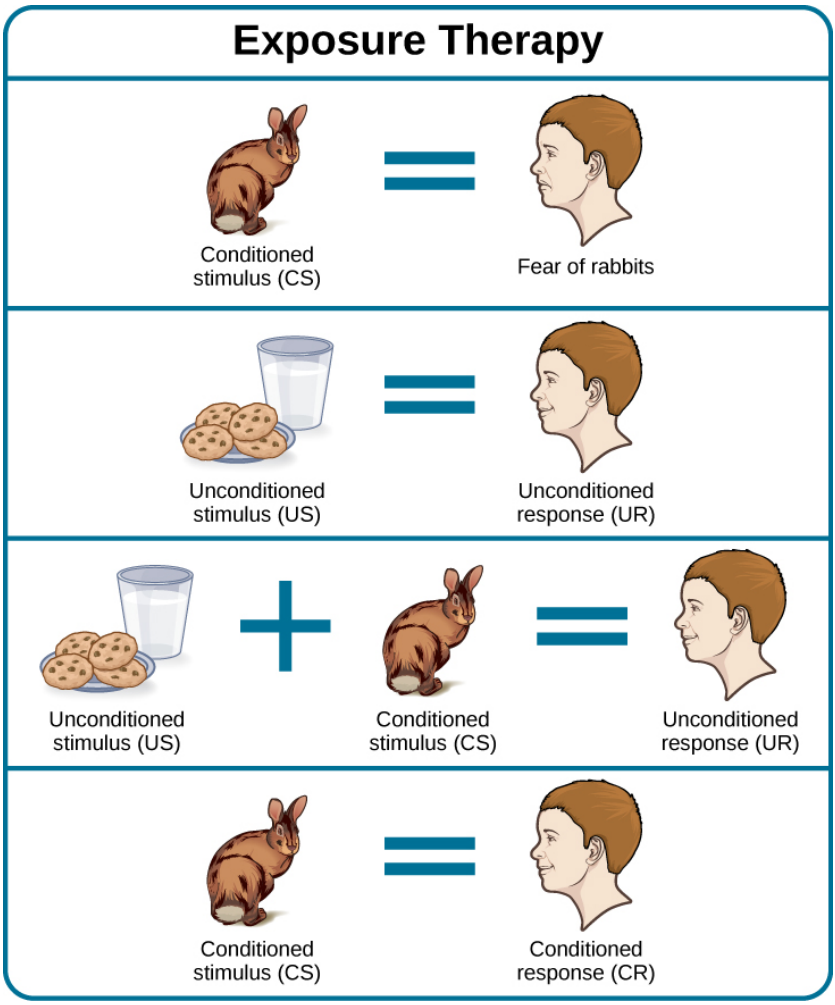
Emmie is eight years old, and frequently wets her bed at night. She's been invited to several sleepovers, but she won't go because of her problem. Using a type of conditioning therapy, Emmie begins to sleep on a liquid-sensitive bed pad that is hooked to an alarm. When moisture touches the pad, it sets off the alarm, waking up Emmie. When this process is repeated enough times, Emmie develops an association between urinary relaxation and waking up, and this stops the bedwetting. Emmie has now gone three weeks without wetting her bed and is looking forward to her first sleepover this weekend.

One commonly used classical conditioning therapeutic technique is counterconditioning: a client learns a new response to a stimulus that has previously elicited an undesirable behavior. Two counterconditioning techniques are aversive conditioning and exposure therapy. Aversive conditioning uses an unpleasant stimulus to stop an undesirable behavior. Therapists apply this technique to eliminate addictive behaviors, such as smoking, nail biting, and drinking. In aversion therapy, clients will typically engage in a specific behavior (such as nail biting) and at the same time are exposed to something unpleasant, such as a mild electric shock or a bad taste. After repeated associations between the unpleasant stimulus and the behavior, the client can learn to stop the unwanted behavior.

Aversion therapy has been used effectively for years in the treatment of alcoholism (Davidson, 1974; Elkins, 1991; Streeton & Whelan, 2001). One common way this occurs is through a chemically based substance known as Antabuse. When a person takes Antabuse and then consumes alcohol, uncomfortable side effects result including nausea, vomiting, increased heart rate, heart palpitations, severe headache, and shortness of breath. Antabuse is repeatedly paired with alcohol until the client associates alcohol with unpleasant feelings, which decreases the client's desire to consume alcohol. Antabuse creates a conditioned

aversion to alcohol because it replaces the original pleasure response with an unpleasant one.

In exposure therapy, a therapist seeks to treat clients' fears or anxiety by presenting them with the object or situation that causes their problem, with the idea that they will eventually get used to it. This can be done via reality, imagination, or virtual reality. Exposure therapy was first reported in 1924 by Mary Cover Jones, who is considered the mother of behavior therapy. Jones worked with a boy named Peter who was afraid of rabbits. Her goal was to replace Peter's fear of rabbits with a conditioned response of relaxation, which is a response that is incompatible with fear. How did she do it? Jones began by placing a caged rabbit on the other side of a room with Peter while he ate his afternoon snack. Over the course of several days, Jones moved the rabbit closer and closer to where Peter was seated with his snack. After two months of being exposed to the rabbit while relaxing with his snack, Peter was able to hold the rabbit and pet it while eating (Jones, 1924).



Exposure therapy seeks to change the response to a conditioned stimulus (CS). An unconditioned stimulus is presented over and over just after the presentation of the conditioned stimulus. This figure shows conditioning as conducted in Mary Cover Jones' 1924 study.

Thirty years later, Joseph Wolpe (1958) refined Jones's techniques, giving us the behavior therapy technique of exposure therapy that

is used today. A popular form of exposure therapy is systematic desensitization, wherein a calm and pleasant state is gradually associated with increasing levels of anxiety-inducing stimuli. The idea is that you can't be nervous and relaxed at the same time. Therefore, if you can learn to relax when you are facing environmental stimuli that make you nervous or fearful, you can eventually eliminate your unwanted fear response (Wolpe, 1958).

This person suffers from arachnophobia (fear of spiders). Through exposure therapy he is learning how to face his fear in a controlled, therapeutic setting. (credit: "GollyGforce – Living My Worst Nightmare"/Flickr)

How does exposure therapy work? Jayden is terrified of elevators. Nothing bad has ever happened to him on an elevator, but he's so afraid of elevators that he will always take the stairs. That wasn't a problem when Jayden worked on the second floor of an office building, but now he has a new job—on the 29th floor of a skyscraper in downtown Los Angeles. Jayden knows he can't climb 29 flights of stairs in order to get to work each day, so he decided to see a behavior therapist for help. The therapist asks Jayden to first construct a hierarchy of elevator-related situations that elicit fear and anxiety. They range from situations of mild anxiety such as

being nervous around the other people in the elevator, to the fear of getting an arm caught in the door, to panic-provoking situations such as getting trapped or the cable snapping. Next, the therapist uses progressive relaxation. She teaches Jayden how to relax each of his muscle groups so that he achieves a drowsy, relaxed, and comfortable state of mind. Once he's in this state, she asks Jayden to imagine a mildly anxiety-provoking situation. Jayden is standing in front of the elevator thinking about pressing the call button.

If this scenario causes Jayden anxiety, he lifts his finger. The therapist would then tell Jayden to forget the scene and return to his relaxed state. She repeats this scenario over and over until Jayden can imagine himself pressing the call button without anxiety. Over time the therapist and Jayden use progressive relaxation and imagination to proceed through all of the situations on Jayden's hierarchy until he becomes desensitized to each one. After this, Jayden and the therapist begin to practice what he only previously envisioned in therapy, gradually going from pressing the button to actually riding an elevator. The goal is that Jayden will soon be able to take the elevator all the way up to the 29th floor of his office without feeling any anxiety.

Sometimes, it's too impractical, expensive, or embarrassing to re-create anxiety- producing situations, so a therapist might employ virtual reality exposure therapy by using a simulation to help conquer fears. Virtual reality exposure therapy has been used effectively to treat numerous anxiety disorders such as the fear of public speaking, claustrophobia (fear of enclosed spaces), aviophobia (fear of flying), and post-traumatic stress disorder (PTSD), a trauma and stressor-related disorder (Gerardi, Cukor, Difede, Rizzo, & Rothbaum, 2010).

A new virtual reality exposure therapy is being used to treat PTSD in soldiers. Virtual Iraq is a simulation that mimics Middle Eastern cities and desert roads with situations similar to those soldiers experienced while deployed in Iraq. This method of virtual reality exposure therapy has been effective in treating PTSD for combat

veterans. Approximately 80% of participants who completed treatment saw clinically significant reduction in their symptoms of PTSD, anxiety, and depression (Rizzo et al., 2010).

Watch this Virtual Iraq video showing soldiers being treated via simulation: Treating PTSD With Virtual Reality Therapy: A Way to Heal Trauma.

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=235#oembed-1

Some behavior therapies employ operant conditioning. Recall what you learned about operant conditioning: We have a tendency to repeat behaviors that are reinforced. What happens to behaviors that are not reinforced? They become extinguished. These principles can be applied to help people with a wide range of psychological problems. For instance, operant conditioning techniques designed to reinforce positive behaviors and punish unwanted behaviors have been an effective tool to help children

with autism (Lovaas, 1987, 2003; Sallows & Graupner, 2005; Wolf & Risley, 1967). This technique is called Applied Behavior Analysis (ABA). In this treatment, child-specific reinforcers (e.g., stickers, praise, candy, bubbles, and extra play time) are used to reward and motivate autistic children when they demonstrate desired behaviors such as sitting on a chair when requested, verbalizing a greeting, or making eye contact. Punishment such as a timeout or a sharp "No!" from the therapist or parent might be used to discourage undesirable behaviors such as pinching, scratching, and pulling hair.

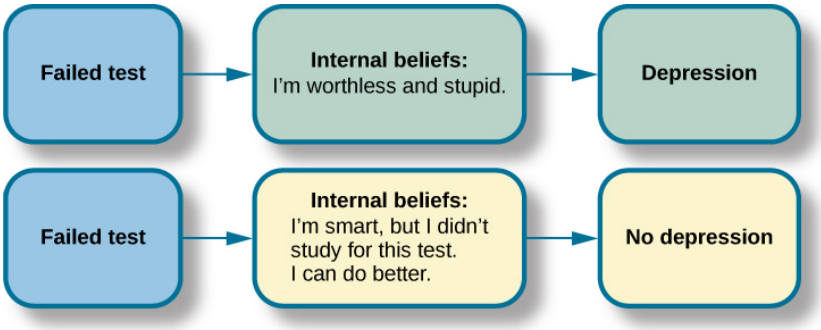
One popular operant conditioning intervention is called the token economy. This involves a controlled setting where individuals are reinforced for desirable behaviors with tokens, such as a poker chip, that can be exchanged for items or privileges. Token economies are often used in psychiatric hospitals to increase patient cooperation and activity levels. Patients are rewarded with tokens when they engage in positive behaviors (e.g., making their beds, brushing their teeth, coming to the cafeteria on time, and socializing with other patients). They can later exchange the tokens for extra TV time, private rooms, visits to the canteen, and so on (Dickerson, Tenhula, & Green-Paden, 2005).

PSYCHOTHERAPY: COGNITIVE THERAPY

Cognitive therapy is a form of psychotherapy that focuses on how a person's thoughts lead to feelings of distress. The idea behind cognitive therapy is that how you think determines how you feel and act. Cognitive therapists help their clients change dysfunctional thoughts in order to relieve distress. They help a client see how they misinterpret a situation (cognitive distortion). For example, a client may overgeneralize. Because Ray failed one test in his Psychology 101 course, he feels he is stupid and worthless. These thoughts then cause his mood to worsen. Therapists also help

clients recognize when they blow things out of proportion. Because Ray failed his Psychology 101 test, he has concluded that he's going to fail the entire course and probably flunk out of college altogether. These errors in thinking have contributed to Ray's feelings of distress. His therapist will help him challenge these irrational beliefs, focus on their illogical basis, and correct them with more logical and rational thoughts and beliefs.

Cognitive therapy was developed by psychiatrist Aaron Beck in the 1960s. His initial focus was on depression and how a client's self-defeating attitude served to maintain a depression despite positive factors in her life (Beck, Rush, Shaw, & Emery, 1979). Through questioning, a cognitive therapist can help a client recognize dysfunctional ideas, challenge catastrophizing thoughts about themselves and their situations, and find a more positive way to view things (Beck, 2011).



Your emotional reactions are the result of your thoughts about the situation rather than the situation itself. For instance, if you consistently interpret events and emotions around the themes of loss and defeat, then you are likely to be depressed. Through therapy, you can learn more logical ways to interpret situations.

View a brief video in which Judith Beck talks about cognitive therapy and conducts a session with a client: Judith Beck Phd talks about Cognitive Therapy.
https://youtube.com/watch?v=45U1F7cDH5k

PSYCHOTHERAPY: COGNITIVE-BEHAVIORAL THERAPY

Cognitive-behavioral therapists focus much more on present issues than on a patient's childhood or past, as in other forms of psychotherapy. One of the first forms of cognitive-behavioral therapy was rational emotive therapy (RET), which was founded by Albert Ellis and grew out of his dislike of Freudian psychoanalysis (Daniel, n.d.). Behaviorists such as Joseph Wolpe also influenced Ellis's therapeutic approach (National Association of Cognitive-Behavioral Therapists, 2009).

Cognitive-behavioral therapy (CBT) helps clients examine how their thoughts affect their behavior. It aims to change cognitive distortions and self-defeating behaviors. In essence, this approach is designed to change the way people think as well as how they act. It is similar to cognitive therapy in that CBT attempts to make individuals aware of their irrational and negative thoughts and helps people replace them with new, more positive ways of thinking. It is also similar to behavior therapies in that CBT teaches people how to practice and engage in more positive and healthy approaches to daily situations. In total, hundreds of studies have shown the effectiveness of cognitive-behavioral therapy in the

treatment of numerous psychological disorders such as depression, PTSD, anxiety disorders, eating disorders, bipolar disorder, and substance abuse (Beck Institute for Cognitive Behavior Therapy, n.d.). For example, CBT has been found to be effective in decreasing levels of hopelessness and suicidal thoughts in previously suicidal teenagers (Alavi, Sharifi, Ghanizadeh, & Dehbozorgi, 2013). Cognitive-behavioral therapy has also been effective in reducing PTSD in specific populations, such as transit workers (Lowinger & Rombom, 2012).

Cognitive-behavioral therapy aims to change cognitive distortions and self-defeating behaviors using techniques like the ABC model. With this model, there is an **A**ction (sometimes called an activating event), the **B**elief about the event, and the **C**onsequences of this belief. Let's say, Jon and Joe both go to a party. Jon and Joe each have met a young woman at the party: Jon is talking with Megan most of the party, and Joe is talking with Amanda. At the end of the party, Jon asks Megan for her phone number and Joe asks Amanda. Megan tells Jon she would rather not give him her number, and Amanda tells Joe the same thing. Both Jon and Joe are surprised, as they thought things were going well. What can Jon and Joe tell themselves about why the women were not interested? Let's say Jon tells himself he is a loser, or is ugly, or "has no game." Jon then gets depressed and decides not to go to another party, which starts a cycle that keeps him depressed. Joe tells himself that he had bad breath, goes out and buys a new toothbrush, goes to another party, and meets someone new.

Jon's belief about what happened results in a consequence of further depression, whereas Joe's belief does not. Jon is internalizing the attribution or reason for the rebuffs, which triggers his depression. On the other hand, Joe is externalizing the cause, so his thinking does not contribute to feelings of depression. Cognitive-behavioral therapy examines specific maladaptive and automatic thoughts and cognitive distortions. Some examples of cognitive distortions are all-or-nothing thinking, overgeneralization,

and jumping to conclusions. In overgeneralization, someone takes a small situation and makes it huge—for example, instead of saying, "This particular woman was not interested in me," the man says, "I am ugly, a loser, and no one is ever going to be interested in me."

All or nothing thinking, which is a common type of cognitive distortion for people suffering from depression, reflects extremes. In other words, everything is black or white. After being turned down for a date, Jon begins to think, "No woman will ever go out with me. I'm going to be alone forever." He begins to feel anxious and sad as he contemplates his future.

The third kind of distortion involves jumping to conclusions—assuming that people are thinking negatively about you or reacting negatively to you, even though there is no evidence. Consider the example of Savannah and Hillaire, who recently met at a party. They have a lot in common, and Savannah thinks they could become friends. She calls Hillaire to invite her for coffee. Since Hillaire doesn't answer, Savannah leaves her a message. Several days go by and Savannah never hears back from her potential new friend. Maybe Hillaire never received the message because she lost her phone or she is too busy to return the phone call. But if Savannah believes that Hillaire didn't like Savannah or didn't want to be her friend, she is demonstrating the cognitive distortion of jumping to conclusions.

How effective is CBT? One client said this about his cognitive-behavioral therapy:

> I have had many painful episodes of depression in my life, and this has had a negative effect on my career and has put considerable strain on my friends and family. The treatments I have received, such as taking antidepressants and psychodynamic counseling, have helped [me] to cope with the symptoms and to get some insights into the roots of my problems. CBT has been by far the most useful approach I have found in tackling these mood problems. It has raised my awareness of how my thoughts impact on my moods. How the way I think about myself, about others

> and about the world can lead me into depression. It is a practical approach, which does not dwell so much on childhood experiences, whilst acknowledging that it was then that these patterns were learned. It looks at what is happening now, and gives tools to manage these moods on a daily basis. (Martin, 2007, n.p.)

PSYCHOTHERAPY: HUMANISTIC THERAPY

Humanistic psychology focuses on helping people achieve their potential. So it makes sense that the goal of humanistic therapy is to help people become more self-aware and accepting of themselves. In contrast to psychoanalysis, humanistic therapists focus on conscious rather than unconscious thoughts. They also emphasize the patient's present and future, as opposed to exploring the patient's past.

Psychologist Carl Rogers developed a therapeutic orientation known as Rogerian, or client-centered therapy. Note the change from *patients* to *clients*. Rogers (1951) felt that the term patient suggested the person seeking help was sick and looking for a cure. Since this is a form of nondirective therapy, a therapeutic approach in which the therapist does not give advice or provide interpretations but helps the person to identify conflicts and understand feelings, Rogers (1951) emphasized the importance of the person taking control of his own life to overcome life's challenges.

In client-centered therapy, the therapist uses the technique of active listening. In active listening, the therapist acknowledges, restates, and clarifies what the client expresses. Therapists also practice what Rogers called unconditional positive regard, which involves not judging clients and simply accepting them for who they are. Rogers (1951) also felt that therapists should demonstrate genuineness, empathy, and acceptance toward their clients because this helps people become more accepting of themselves, which results in personal growth.

EVALUATING VARIOUS FORMS OF PSYCHOTHERAPY

How can we assess the effectiveness of psychotherapy? Is one technique more effective than another? For anyone considering therapy, these are important questions. According to the American Psychological Association, three factors work together to produce successful treatment. The first is the use of evidence-based treatment that is deemed appropriate for your particular issue. The second important factor is the clinical expertise of the psychologist or therapist. The third factor is your own characteristics, values, preferences, and culture. Many people begin psychotherapy feeling like their problem will never be resolved; however, psychotherapy helps people see that they can do things to make their situation better. Psychotherapy can help reduce a person's anxiety, depression, and maladaptive behaviors. Through psychotherapy, individuals can learn to engage in healthy behaviors designed to help them better express emotions, improve relationships, think more positively, and perform more effectively at work or school.

Many studies have explored the effectiveness of psychotherapy. For example, one large-scale study that examined 16 meta-analyses of CBT reported that it was equally effective or more effective than other therapies in treating PTSD, generalized anxiety disorder, depression, and social phobia (Butlera, Chapmanb, Formanc, & Becka, 2006). Another study found that CBT was as effective at treating depression (43% success rate) as prescription medication (50% success rate) compared to the placebo rate of 25% (DeRubeis et al., 2005). Another meta-analysis found that psychodynamic therapy was also as effective at treating these types of psychological issues as CBT (Shedler, 2010). However, no studies have found one psychotherapeutic approach more effective than another (Abbass, Kisely, & Kroenke, 2006; Chorpita et al., 2011), nor have they shown any relationship between a client's

treatment outcome and the level of the clinician's training or experience (Wampold, 2007). Regardless of which type of psychotherapy an individual chooses, one critical factor that determines the success of treatment is the person's relationship with the psychologist or therapist.

BIOMEDICAL THERAPIES

Individuals can be prescribed biologically based treatments or psychotropic medications that are used to treat mental disorders. While these are often used in combination with psychotherapy, they also are taken by individuals not in therapy. This is known as biomedical therapy. Medications used to treat psychological disorders are called psychotropic medications and are prescribed by medical doctors, including psychiatrists. In Louisiana and New Mexico, psychologists are able to prescribe some types of these medications (American Psychological Association, 2014).

Different types and classes of medications are prescribed for different disorders. A depressed person might be given an antidepressant, a bipolar individual might be given a mood stabilizer, and a schizophrenic individual might be given an antipsychotic. These medications treat the symptoms of a psychological disorder. They can help people feel better so that they can function on a daily basis, but they do not cure the disorder. Some people may only need to take a psychotropic medication for a short period of time. Others with severe disorders like bipolar disorder or schizophrenia may need to take psychotropic medication for a long time.

Commonly Prescribed Psychotropic Medications

Type of Medication	Used to Treat	Brand Names of Commonly Prescribed Medications	How They Work	Side Effects
Antipsychotics (developed in the 1950s)	Schizophrenia and other types of severe thought disorders	Haldol, Mellaril, Prolixin, Thorazine	Treat positive psychotic symptoms such as auditory and visual hallucinations, delusions, and paranoia by blocking the neurotransmitter dopamine	Long-term use can lead to tardive dyskinesia, involuntary movements of the arms, legs, tongue and facial muscles, resulting in Parkinson's-like tremors
Atypical Antipsychotics (developed in the late 1980s)	Schizophrenia and other types of severe thought disorders	Abilify, Risperdal, Clozaril	Treat the negative symptoms of schizophrenia, such as withdrawal and apathy, by targeting both dopamine and serotonin receptors; newer medications may treat both positive and negative symptoms	Can increase the risk of obesity and diabetes as well as elevate cholesterol levels; constipation, dry mouth, blurred vision, drowsiness, and dizziness
Anti-depressants	Depression and increasingly for anxiety	Paxil, Prozac, Zoloft (selective serotonin reuptake inhibitors, [SSRIs]); Tofranil and Elavil (tricyclics)	Alter levels of neurotransmitters such as serotonin and norepinephrine	SSRIs: headache, nausea, weight gain, drowsiness, reduced sex drive Tricyclics: dry mouth, constipation, blurred vision, drowsiness, reduced sex drive, increased risk of suicide
Anti-anxiety agents	Anxiety and agitation that occur in OCD, PTSD, panic disorder, and social phobia	Xanax, Valium, Ativan	Depress central nervous system activity	Drowsiness, dizziness, headache, fatigue, lightheadedness

Type of Medication	Used to Treat	Brand Names of Commonly Prescribed Medications	How They Work	Side Effects
Mood Stabilizers	Bipolar disorder	Lithium, Depakote, Lamictal, Tegretol	Treat episodes of mania as well as depression	Excessive thirst, irregular heartbeat, itching/rash, swelling (face, mouth, and extremities), nausea, loss of appetite
Stimulants	ADHD	Adderall, Ritalin	Improve ability to focus on a task and maintain attention	Decreased appetite, difficulty sleeping, stomachache, headache

Another biologically based treatment that continues to be used, although infrequently, is electroconvulsive therapy (ECT) (formerly known by its unscientific name as electroshock therapy). It involves using an electrical current to induce seizures to help alleviate the effects of severe depression. The exact mechanism is unknown, although it does help alleviate symptoms for people with severe depression who have not responded to traditional drug therapy (Pagnin, de Queiroz, Pini, & Cassano, 2004). About 85% of people treated with ECT improve (Reti, n.d.). However, the memory loss associated with repeated administrations has led to it being implemented as a last resort (Donahue, 2000; Prudic, Peyser, & Sackeim, 2000). A more recent alternative is transcranial magnetic stimulation (TMS), a procedure approved by the FDA in 2008 that uses magnetic fields to stimulate nerve cells in the brain to improve depression symptoms; it is used when other treatments have not worked (Mayo Clinic, 2012).

Evidence-based Practice

A buzzword in therapy today is evidence-based practice. However, it's not a novel concept but one that has been used in medicine for at least two decades. Evidence-based practice is used to reduce errors in treatment selection by making clinical decisions based on research (Sackett & Rosenberg, 1995). In any case, evidence-based treatment is on the rise in the field of psychology. So what is it, and why does it matter? In an effort to determine which treatment methodologies are evidenced-based, professional organizations such as the American Psychological Association (APA) have recommended that specific psychological treatments be used to treat certain psychological disorders (Chambless & Ollendick, 2001). According to the APA (2005), "Evidence-based practice in psychology (EBPP) is the integration of the best available research with clinical expertise in the context of patient characteristics, culture, and preferences" (p. 1).

The foundational idea behind evidence based treatment is that best practices are determined by research evidence that has been compiled by comparing various forms of treatment (Charman & Barkham, 2005). These treatments are then operationalized and placed in treatment manuals—trained therapists follow these manuals. The benefits are that evidence-based treatment can reduce variability between therapists to ensure that a specific approach is delivered with integrity (Charman & Barkham, 2005). Therefore, clients have a higher chance of receiving therapeutic interventions that are effective at treating their specific disorder. While EBPP is based on randomized control trials, critics of EBPP reject it stating that the results of trials cannot be applied to

individuals and instead determinations regarding treatment should be based on a therapist's judgment (Mullen & Streiner, 2004).

TEST YOUR UNDERSTANDING

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=235#h5p-271

SUMMARY

Psychoanalysis was developed by Sigmund Freud. Freud's theory is that a person's psychological problems are the result of repressed impulses or childhood trauma. The goal of the therapist is to help a person uncover buried feelings by using techniques such as free association and dream analysis.

Play therapy is a psychodynamic therapy technique often used

with children. The idea is that children play out their hopes, fantasies, and traumas, using dolls, stuffed animals, and sandbox figurines.

In behavior therapy, a therapist employs principles of learning from classical and operant conditioning to help clients change undesirable behaviors. Counterconditioning is a commonly used therapeutic technique in which a client learns a new response to a stimulus that has previously elicited an undesirable behavior via classical conditioning. Principles of operant conditioning can be applied to help people deal with a wide range of psychological problems. Token economy is an example of a popular operant conditioning technique.

Cognitive therapy is a technique that focuses on how thoughts lead to feelings of distress. The idea behind cognitive therapy is that how you think determines how you feel and act. Cognitive therapists help clients change dysfunctional thoughts in order to relieve distress. Cognitive-behavioral therapy explores how our thoughts affect our behavior. Cognitive-behavioral therapy aims to change cognitive distortions and self-defeating behaviors.

Humanistic therapy focuses on helping people achieve their potential. One form of humanistic therapy developed by Carl Rogers is known as client-centered or Rogerian therapy. Client-centered therapists use the techniques of active listening, unconditional positive regard, genuineness, and empathy to help clients become more accepting of themselves.

Often in combination with psychotherapy, people can be prescribed biologically based treatments such as psychotropic medications and/or other medical procedures such as electro-convulsive therapy.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=235#h5p-272

CRITICAL THINKING QUESTION

Imagine that you are a psychiatrist. Your patient, Pat, comes to you with the following symptoms: anxiety and feelings of sadness. Which therapeutic

approach would you recommend and why?

I would recommend psychodynamic talk therapy or cognitive therapy to help the person see how her thoughts and behaviors are having negative effects.

PERSONAL APPLICATION QUESTION

If you were to choose a therapist practicing one of the techniques presented in this section, which kind of therapist would you choose and why?

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Distinguish between the various modalities of treatment
- Discuss benefits of group therapy

Once a person seeks treatment, whether voluntarily or involuntarily, he has an intake done to assess his clinical needs. An intake is the therapist's first meeting with the client. The therapist gathers specific information to address the client's immediate needs, such as the presenting problem, the client's support system, and insurance status. The therapist informs the client about confidentiality, fees, and what to expect in treatment. Confidentiality means the therapist cannot disclose confidential communications to any third party unless mandated or permitted by law to do so. During the intake, the therapist and client will work together to discuss treatment goals. Then a treatment plan will be formulated, usually with specific measurable objectives. Also, the therapist and client will discuss how treatment success will be measured and the estimated length of treatment. There are several different modalities of treatment: Individual therapy, family therapy, couples therapy, and group therapy are the most common.

(a)

(b)

Therapy may occur (a) one-on-one between a therapist and client, or (b) in a group setting. (credit a: modification of work by Connor Ashleigh, AusAID/ Department of Foreign Affairs and Trade)

INDIVIDUAL THERAPY

In individual therapy, also known as individual psychotherapy or individual counseling, the client and clinician meet one-on-one (usually from 45 minutes to 1 hour). These meetings typically occur weekly or every other week, and sessions are conducted in a confidential and caring environment. The clinician will work with clients to help them explore their feelings, work through life challenges, identify aspects of themselves and their lives that they wish to change, and set goals to help them work towards these changes. A client might see a clinician for only a few sessions, or the client may attend individual therapy sessions for a year or longer. The amount of time spent in therapy depends on the needs of the client as well as her personal goals.

In an individual therapy session, a client works one-on-one with a trained therapist. (credit: Alan Cleaver)

GROUP THERAPY

In group therapy, a clinician meets together with several clients

with similar problems. When children are placed in group therapy, it is particularly important to match clients for age and problems. One benefit of group therapy is that it can help decrease a client's shame and isolation about a problem while offering needed support, both from the therapist and other members of the group (American Psychological Association, 2014). A nine-year-old sexual abuse victim, for example, may feel very embarrassed and ashamed. If he is placed in a group with other sexually abused boys, he will realize that he is not alone. A child struggling with poor social skills would likely benefit from a group with a specific curriculum to foster special skills. A woman suffering from post-partum depression could feel less guilty and more supported by being in a group with similar women.

Group therapy also has some specific limitations. Members of the group may be afraid to speak in front of other people because sharing secrets and problems with complete strangers can be stressful and overwhelming. There may be personality clashes and arguments among group members. There could also be concerns about confidentiality: Someone from the group might share what another participant said to people outside of the group.

In group therapy, usually 5–10 people meet with a trained therapist to discuss a common issue such as divorce, grief, an eating disorder, substance abuse, or anger management. (credit: Cory Zanker)

Another benefit of group therapy is that members can confront each other about their patterns. For those with some types of problems, such as sexual abusers, group therapy is the recommended treatment. Group treatment for this population is considered to have several benefits:

> Group treatment is more economical than individual, couples, or family therapy. Sexual abusers often feel more comfortable admitting and discussing their offenses in a treatment group where others are modeling openness. Clients often accept feedback about their behavior more willingly from other group members than from therapists. Finally, clients can practice social skills in group treatment settings. (McGrath, Cumming, Burchard, Zeoli, & Ellerby, 2009)

Groups that have a strong educational component are called psycho-educational groups. For example, a group for children

whose parents have cancer might discuss in depth what cancer is, types of treatment for cancer, and the side effects of treatments, such as hair loss. Often, group therapy sessions with children take place in school. They are led by a school counselor, a school psychologist, or a school social worker. Groups might focus on test anxiety, social isolation, self-esteem, bullying, or school failure (Shechtman, 2002). Whether the group is held in school or in a clinician's office, group therapy has been found to be effective with children facing numerous kinds of challenges (Shechtman, 2002).

During a group session, the entire group could reflect on an individual's problem or difficulties, and others might disclose what they have done in that situation. When a clinician is facilitating a group, the focus is always on making sure that everyone benefits and participates in the group and that no one person is the focus of the entire session. Groups can be organized in various ways: some have an overarching theme or purpose, some are time-limited, some have open membership that allows people to come and go, and some are closed. Some groups are structured with planned activities and goals, while others are unstructured: There is no specific plan, and group members themselves decide how the group will spend its time and on what goals it will focus. This can become a complex and emotionally charged process, but it is also an opportunity for personal growth (Page & Berkow, 1994).

COUPLES THERAPY

Couples therapy involves two people in an intimate relationship who are having difficulties and are trying to resolve them. The couple may be dating, partnered, engaged, or married. The primary therapeutic orientation used in couples counseling is cognitive-behavioral therapy (Rathus & Sanderson, 1999). Couples meet with a therapist to discuss conflicts and/or aspects of their relationship that they want to change. The therapist helps them see how their

individual backgrounds, beliefs, and actions are affecting their relationship. Often, a therapist tries to help the couple resolve these problems, as well as implement strategies that will lead to a healthier and happier relationship, such as how to listen, how to argue, and how to express feelings. However, sometimes, after working with a therapist, a couple will realize that they are too incompatible and will decide to separate. Some couples seek therapy to work out their problems, while others attend therapy to determine whether staying together is the best solution. Counseling couples in a high-conflict and volatile relationship can be difficult. In fact, psychologists Peter Pearson and Ellyn Bader, who founded the Couples Institute in Palo Alto, California, have compared the experience of the clinician in couples' therapy to be like "piloting a helicopter in a hurricane" (Weil, 2012, para. 7).

In couples counseling, a therapist helps people work on their relationship. (credit: Cory Zanker)

FAMILY THERAPY

Family therapy is a special form of group therapy, consisting of one or more families. Although there are many theoretical orientations in family therapy, one of the most predominant is the systems approach. The family is viewed as an organized system, and each individual within the family is a contributing member who creates and maintains processes within the system that shape behavior (Minuchin, 1985). Each member of the family influences and is influenced by the others. The goal of this approach is to enhance the growth of each family member as well as that of the family as a whole.

Often, dysfunctional patterns of communication that develop between family members can lead to conflict. A family with this dynamic might wish to attend therapy together rather than individually. In many cases, one member of the family has problems that detrimentally affect everyone. For example, a mother's depression, teen daughter's eating disorder, or father's alcohol dependence could affect all members of the family. The therapist would work with all members of the family to help them cope with the issue, and to encourage resolution and growth in the case of the individual family member with the problem.

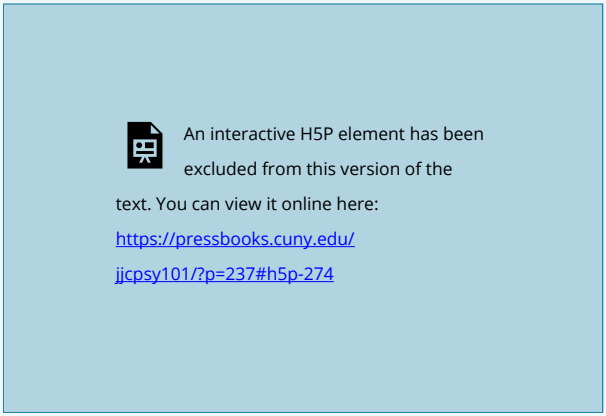
With family therapy, the nuclear family (i.e., parents and children) or the nuclear family plus whoever lives in the household (e.g., grandparent) come into treatment. Family therapists work with the whole family unit to heal the family. There are several different types of family therapy. In structural family therapy, the therapist examines and discusses the boundaries and structure of the family: who makes the rules, who sleeps in the bed with whom, how decisions are made, and what are the boundaries within the family. In some families, the parents do not work together to make rules, or one parent may undermine the other, leading the children to act out. The therapist helps them resolve these issues and learn to communicate more effectively.

Watch this video to view a structural family session: Family Therapy.

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=237#oembed-1

In strategic family therapy, the goal is to address specific problems within the family that can be dealt with in a relatively short amount of time. Typically, the therapist would guide what happens in the therapy session and design a detailed approach to resolving each member's problem (Madanes, 1991).

TEST YOUR UNDERSTANDING



SUMMARY

There are several modalities of treatment: individual therapy, group therapy, couples therapy, and family therapy are the most common. In an individual therapy session, a client works one-on-one with a trained therapist. In group therapy, usually 5–10 people meet with a trained group therapist to discuss a common issue (e.g., divorce, grief, eating disorders, substance abuse, or anger management). Couples therapy involves two people in an intimate relationship who are having difficulties and are trying to resolve them. The couple may be dating, partnered, engaged, or married.

The therapist helps them resolve their problems as well as implement strategies that will lead to a healthier and happier relationship. Family therapy is a special form of group therapy. The therapy group is made up of one or more families. The goal of this approach is to enhance the growth of each individual family member and the family as a whole.

REVIEW QUESTIONS

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CRITICAL THINKING QUESTION

Compare and contrast individual and group therapies.

In an individual therapy session, a client works one-on-one with a trained therapist. In group therapy, usually 5–10 people meet with a trained group therapist to discuss a common issue, such as divorce, grief, eating disorder, substance abuse, or anger management.

PERSONAL APPLICATION

Your best friend tells you that she is concerned about her cousin. The cousin—a teenage girl—is constantly coming home after her curfew, and your friend suspects that she has been drinking. What treatment modality would you recommend to your friend and why?

GLOSSARY

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Media Attributions

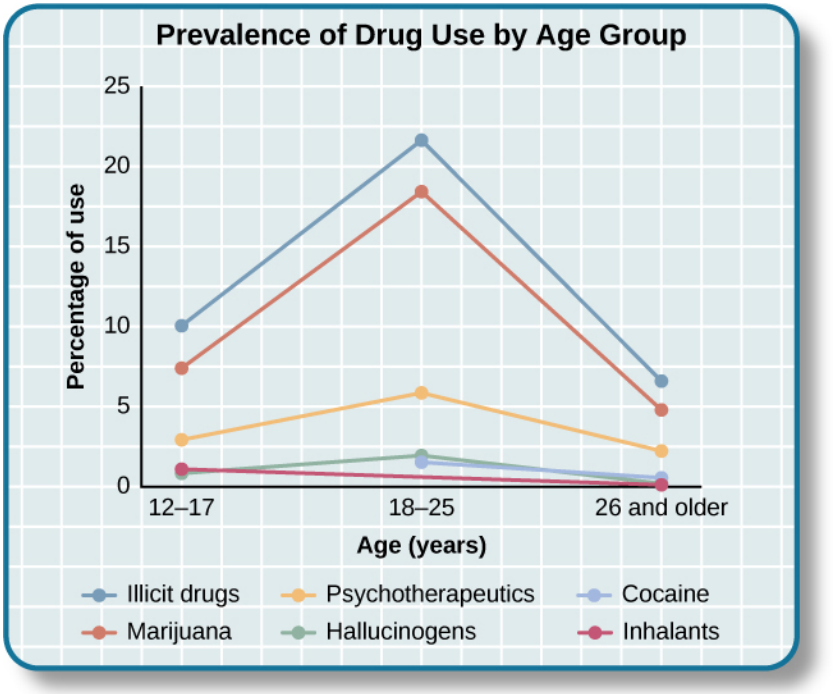
- "Family Therapy" by Andrea Saathoff. Standard YouTube License.

Learning Objectives

By the end of this section, you will be able to:

- Recognize the goal of substance-related and addictive disorders treatment
- Discuss what makes for effective treatment
- Describe how comorbid disorders are treated

Addiction is often viewed as a chronic disease. The choice to use a substance is initially voluntary; however, because chronic substance use can permanently alter the neural structure in the prefrontal cortex, an area of the brain associated with decision-making and judgment, a person becomes driven to use drugs and/or alcohol (Muñoz-Cuevas, Athilingam, Piscopo, & Wilbrecht, 2013). This helps explain why relapse rates tend to be high. About 40%–60% of individuals relapse, which means they return to abusing drugs and/or alcohol after a period of improvement (National Institute on Drug Abuse [NIDA], 2008).



The National Survey on Drug Use and Health shows trends in prevalence of various drugs for ages 12–17, 18–25, and 26 or older.

The goal of substance-related treatment is to help an addicted person stop compulsive drug-seeking behaviors (NIDA, 2012). This means an addicted person will need long-term treatment, similar to a person battling a chronic physical disease such as hypertension or diabetes. Treatment usually includes behavioral therapy and/or medication, depending on the individual (NIDA, 2012). Specialized therapies have also been developed for specific types of substance-related disorders, including alcohol, cocaine, and opioids (McGovern & Carroll, 2003). Substance-related treatment is considered much more cost-effective than incarceration or not treating those with addictions (NIDA, 2012).

Substance use and abuse costs the United States over $600 billion a year (NIDA, 2012). This addict is using heroin. (credit: "jellymc – urbansnaps"/Flickr)

WHAT MAKES TREATMENT EFFECTIVE?

Specific factors make substance-related treatment much more effective. One factor is duration of treatment. Generally, the addict needs to be in treatment for at least three months to achieve a positive outcome (Simpson, 1981; Simpson, Joe, & Bracy, 1982; NIDA, 2012). This is due to the psychological, physiological, behavioral, and social aspects of abuse (Simpson, 1981; Simpson et al., 1982; NIDA, 2012). While in treatment, an addict might receive behavior therapy, which can help motivate the addict to participate in the treatment program and teach strategies for dealing with cravings and how to prevent relapse. Also, treatment needs to be holistic and address multiple needs, not just the drug addiction.

This means that treatment will address factors such as communication, stress management, relationship issues, parenting, vocational concerns, and legal concerns (McGovern & Carroll, 2003; NIDA, 2012).

While individual therapy is used in the treatment of substance-related disorders, group therapy is the most widespread treatment modality (Weiss, Jaffee, de Menil, & Cogley, 2004). The rationale behind using group therapy for addiction treatment is that addicts are much more likely to maintain sobriety in a group format. It has been suggested that this is due to the rewarding and therapeutic benefits of the group, such as support, affiliation, identification, and even confrontation (Center for Substance Abuse Treatment, 2005). For teenagers, the whole family often needs to participate in treatment to address issues such as family dynamics, communication, and relapse prevention. Family involvement in teen drug addiction is vital. Research suggests that greater parental involvement is correlated with a greater reduction in use by teen substance abusers. Also, mothers who participated in treatment displayed better mental health and greater warmth toward their children (Bertrand et al., 2013). However, neither individual nor group therapy has been found to be more effective (Weiss et al., 2004). Regardless of the type of treatment service, the primary focus is on abstinence or at the very least a significant reduction in use (McGovern & Carroll, 2003).

Treatment also usually involves medications to detox the addict safely after an overdose, to prevent seizures and agitation that often occur in detox, to prevent reuse of the drug, and to manage withdrawal symptoms. Getting off drugs often involves the use of drugs—some of which can be just as addictive. Detox can be difficult and dangerous.

Watch this video to find out more about treating substance-related disorders using the biological, behavioral, and psychodynamic approaches: Treating Substance-Related Disorders: Biological, Behavioral and Psychodynamic Approaches.

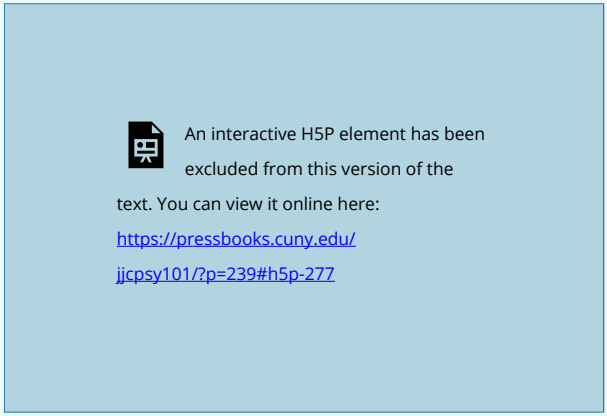
COMORBID DISORDERS

Frequently, a person who is addicted to drugs and/or alcohol has an additional psychological disorder. Saying a person has comorbid disorders means the individual has two or more diagnoses. This can often be a substance-related diagnosis and another psychiatric diagnosis, such as depression, bipolar disorder, or schizophrenia. These individuals fall into the category of mentally ill and chemically addicted (MICA)—their problems are often chronic and expensive to treat, with limited success. Compared with the overall population, substance abusers are twice as likely to have a mood or anxiety disorder. Drug abuse can cause symptoms of mood and anxiety disorders and the reverse is also true—people with debilitating symptoms of a psychiatric disorder may self-medicate and abuse substances.

In cases of comorbidity, the best treatment is thought to address both (or multiple) disorders simultaneously (NIDA, 2012). Behavior therapies are used to treat comorbid conditions, and in many cases, psychotropic medications are used along with psychotherapy. For example, evidence suggests that bupropion (trade names: Wellbutrin and Zyban), approved for treating

depression and nicotine dependence, might also help reduce craving and use of the drug methamphetamine (NIDA, 2011). However, more research is needed to better understand how these medications work—particularly when combined in patients with comorbidities.

TEST YOUR UNDERSTANDING



SUMMARY

Addiction is often viewed as a chronic disease that rewires the

brain. This helps explain why relapse rates tend to be high, around 40%–60% (McLellan, Lewis, & O'Brien, & Kleber, 2000). The goal of treatment is to help an addict stop compulsive drug-seeking behaviors. Treatment usually includes behavioral therapy, which can take place individually or in a group setting. Treatment may also include medication. Sometimes a person has comorbid disorders, which usually means that they have a substance-related disorder diagnosis and another psychiatric diagnosis, such as depression, bipolar disorder, or schizophrenia. The best treatment would address both problems simultaneously.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=239#h5p-278

CRITICAL THINKING QUESTION

You are conducting an intake assessment. Your client is a 45-year-old single, employed male with cocaine dependence. He failed a drug screen at work and is mandated to treatment by his employer if he wants to keep his job. Your client admits that he needs help. Why would you recommend group therapy for him?

The rationale behind using group therapy for addiction treatment is that addicts are much more likely to maintain sobriety when treatment is in a group format. It has been suggested that it's due to the rewarding and therapeutic benefits of the group, such as support, affiliation, identification, and even confrontation. Because this client is single, he may not have family support, so support from the group may be even more important in his ability to recover and maintain his sobriety.

PERSONAL APPLICATION QUESTION

What are some substance-related and addictive disorder treatment facilities in your community, and what types of services do they

provide? Would you recommend any of them to a friend or family member with a substance abuse problem? Why or why not?

GLOSSARY

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Learning Objectives

By the end of this section, you will be able to:

- Explain how the sociocultural model is used in therapy
- Discuss barriers to mental health services among ethnic minorities

The sociocultural perspective looks at you, your behaviors, and your symptoms in the context of your culture and background. For example, José is an 18-year-old Hispanic male from a traditional family. José comes to treatment because of depression. During the intake session, he reveals that he is gay and is nervous about telling his family. He also discloses that he is concerned because his religious background has taught him that homosexuality is wrong. How does his religious and cultural background affect him? How might his cultural background affect how his family reacts if José were to tell them he is gay?

As our society becomes increasingly multiethnic and multiracial, mental health professionals must develop cultural competence, which means they must understand and address issues of race, culture, and ethnicity. They must also develop strategies to effectively address the needs of various populations for which Eurocentric therapies have limited application (Sue, 2004). For example, a counselor whose treatment focuses on individual decision making may be ineffective at helping a Chinese client with a collectivist approach to problem solving (Sue, 2004).

Multicultural counseling and therapy aims to offer both a helping

role and process that uses modalities and defines goals consistent with the life experiences and cultural values of clients. It strives to recognize client identities to include individual, group, and universal dimensions, advocate the use of universal and culture-specific strategies and roles in the healing process, and balancs the importance of individualism and collectivism in the assessment, diagnosis, and treatment of client and client systems (Sue, 2001).

This therapeutic perspective integrates the impact of cultural and social norms, starting at the beginning of treatment. Therapists who use this perspective work with clients to obtain and integrate information about their cultural patterns into a unique treatment approach based on their particular situation (Stewart, Simmons, & Habibpour, 2012). Sociocultural therapy can include individual, group, family, and couples treatment modalities.

How do your cultural and religious beliefs affect your attitude toward mental health treatment? (credit "top-left": modification of work by Staffan Scherz; credit "top-left-middle": modification of work by Alejandra Quintero Sinisterra; credit "top-right-middle": modification of work by Pedro Ribeiro Simões; credit "top-right": modification of work by Agustin Ruiz; credit "bottom-left": modification of work by Czech Provincial Reconstruction Team; credit "bottom-left-middle": modification of work by Arian Zwegers; credit "bottom-right-middle": modification of work by "Wonderlane"/Flickr; credit "bottom-right": modification of work by Shiraz Chanawala)

Watch this short video to learn more about cultural

competence and sociocultural treatments: What is cultural competence and why is it important?

One or more interactive elements has been excluded from this version of the text. You can view them online here: https://pressbooks.cuny.edu/jjcpsy101/?p=241#oembed-1

BARRIERS TO TREATMENT

Statistically, ethnic minorities tend to utilize mental health services less frequently than White, middle-class Americans (Alegría et al., 2008; Richman, Kohn-Wood, & Williams, 2007). Why is this so? Perhaps the reason has to do with access and availability of mental health services. Ethnic minorities and individuals of low socioeconomic status (SES) report that barriers to services include lack of insurance, transportation, and time (Thomas & Snowden, 2002). However, researchers have found that even when income levels and insurance variables are taken into account, ethnic minorities are far less likely to seek out and utilize mental health

services. And when access to mental health services is comparable across ethnic and racial groups, differences in service utilization remain (Richman et al., 2007).

In a study involving thousands of women, it was found that the prevalence rate of anorexia was similar across different races, but that bulimia nervosa was more prevalent among Hispanic and African American women when compared with non-Hispanic whites (Marques et al., 2011). Although they have similar or higher rates of eating disorders, Hispanic and African American women with these disorders tend to seek and engage in treatment far less than Caucasian women. These findings suggest ethnic disparities in access to care, as well as clinical and referral practices that may prevent Hispanic and African American women from receiving care, which could include lack of bilingual treatment, stigma, fear of not being understood, family privacy, and lack of education about eating disorders.

Perceptions and attitudes toward mental health services may also contribute to this imbalance. A recent study at King's College, London, found many complex reasons why people do not seek treatment: self-sufficiency and not seeing the need for help, not seeing therapy as effective, concerns about confidentiality, and the many effects of stigma and shame (Clement et al., 2014). And in another study, African Americans exhibiting depression were less willing to seek treatment due to fear of possible psychiatric hospitalization as well as fear of the treatment itself (Sussman, Robins, & Earls, 1987). Instead of mental health treatment, many African Americans prefer to be self-reliant or use spiritual practices (Snowden, 2001; Belgrave & Allison, 2010). For example, it has been found that the Black church plays a significant role as an alternative to mental health services by providing prevention and treatment-type programs designed to enhance the psychological and physical well-being of its members (Blank, Mahmood, Fox, & Guterbock, 2002).

Additionally, people belonging to ethnic groups that already

report concerns about prejudice and discrimination are less likely to seek services for a mental illness because they view it as an additional stigma (Gary, 2005; Townes, Cunningham, & Chavez-Korell, 2009; Scott, McCoy, Munson, Snowden, & McMillen, 2011). For example, in one recent study of 462 older Korean Americans (over the age of 60) many participants reported suffering from depressive symptoms. However, 71% indicated they thought depression was a sign of personal weakness, and 14% reported that having a mentally ill family member would bring shame to the family (Jang, Chiriboga, & Okazaki, 2009).

Language differences are a further barrier to treatment. In the previous study on Korean Americans' attitudes toward mental health services, it was found that there were no Korean-speaking mental health professionals where the study was conducted (Orlando and Tampa, Florida) (Jang et al., 2009). Because of the growing number of people from ethnically diverse backgrounds, there is a need for therapists and psychologists to develop knowledge and skills to become culturally competent (Ahmed, Wilson, Henriksen, & Jones, 2011). Those providing therapy must approach the process from the context of the unique culture of each client (Sue & Sue, 2007).

TREATMENT PERCEPTIONS

By the time a child is a senior in high school, 20% of his classmates—that is 1 in 5—will have experienced a mental health problem (U.S. Department of Health and Human Services, 1999), and 8%—about 1 in 12—will have attempted suicide (Centers for Disease Control and Prevention, 2014). Of those classmates experiencing mental disorders, only 20% will receive professional help (U.S. Public Health Service, 2000). Why?

It seems that the public has a negative perception of children and teens with mental health disorders. According to researchers from Indiana University, the University of Virginia, and Columbia

University, interviews with over 1,300 U.S. adults show that they believe children with depression are prone to violence and that if a child receives treatment for a psychological disorder, then that child is more likely to be rejected by peers at school.

Bernice Pescosolido, author of the study, asserts that this is a misconception. However, stigmatization of psychological disorders is one of the main reasons why young people do not get the help they need when they are having difficulties. Pescosolido and her colleagues caution that this stigma surrounding mental illness, based on misconceptions rather than facts, can be devastating to the emotional and social well-being of our nation's children.

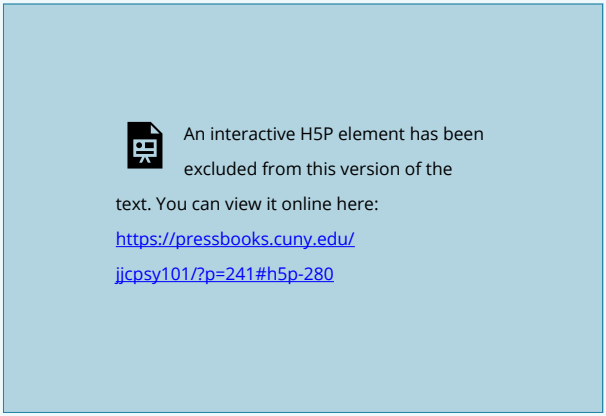
This warning played out as a national tragedy in the 2012 shootings at Sandy Hook Elementary. In her blog, Suzy DeYoung (2013), co-founder of Sandy Hook Promise (the organization parents and concerned others set up in the wake of the school massacre) speaks to treatment perceptions and what happens when children do not receive the mental health treatment they desperately need.

> I've become accustomed to the reaction when I tell people where I'm from.
>
> Eleven months later, it's as consistent as it was back in January.
>
> Just yesterday, inquiring as to the availability of a rental house this holiday season, the gentleman taking my information paused to ask, "Newtown, CT? Isn't that where that…that *thing* happened?
>
> A recent encounter in the Massachusetts Berkshires, however, took me by surprise.
>
> It was in a small, charming art gallery. The proprietor, a woman who looked to be in her 60s, asked where we were from. My response usually depends on my present mood and readiness for the inevitable dialogue. Sometimes it's simply, Connecticut. This time, I replied, Newtown, CT.
>
> The woman's demeanor abruptly shifted from one of amiable

> graciousness to one of visible agitation.
>
> "Oh my god," she said wide eyed and open mouthed. "Did you know her?"
>
> . . . .
>
> "Her?" I inquired
>
> That woman," she replied with disdain, "that woman that raised that monster."
>
> "That woman's" name was Nancy Lanza. Her son, Adam, killed her with a rifle blast to the head before heading out to kill 20 children and six educators at Sandy Hook Elementary School in Newtown, CT last December 14th.
>
> When Nelba Marquez Greene, whose beautiful 6-year-old daughter, Ana, was killed by Adam Lanza, was recently asked how she felt about "that woman," this was her reply:
>
> "She's a victim herself. And it's time in America that we start looking at mental illness with compassion, and helping people who need it.
>
> "This was a family that needed help, an individual that needed help and didn't get it. And what better can come of this, of this time in America, than if we can get help to people who really need it?" (pars. 1–7, 10–15)

Fortunately, we are starting to see campaigns related to the destigmatization of mental illness and an increase in public education and awareness. Join the effort by encouraging and supporting those around you to seek help if they need it. To learn more, visit the National Alliance on Mental Illness (NAMI) website (http://www.nami.org/). The nation's largest nonprofit mental health advocacy and support organization is NAMI.

TEST YOUR UNDERSTANDING



SUMMARY

The sociocultural perspective looks at you, your behaviors, and your symptoms in the context of your culture and background. Clinicians using this approach integrate cultural and religious beliefs into the therapeutic process. Research has shown that ethnic minorities are less likely to access mental health services than their White middle-class American counterparts. Barriers to treatment include lack of insurance, transportation, and time; cultural views that mental illness is a stigma; fears about treatment; and language barriers.

REVIEW QUESTIONS

An interactive H5P element has been excluded from this version of the text. You can view it online here: https://pressbooks.cuny.edu/jjcpsy101/?p=241#h5p-281

CRITICAL THINKING QUESTION

Lashawn is a 24-year-old African American female. For years she has been struggling with bulimia. She knows she has a problem, but she is not

willing to seek mental health services. What are some reasons why she may be hesitant to get help?

One reason may be that her culture views having a mental illness as a stigma. Additionally, perhaps she doesn't have insurance and is worried about the cost of therapy. She could also be afraid that a White counselor would not understand her cultural background, so she would feel uncomfortable sharing things. Also, she may believe she is self-reliant and tell herself that she's a strong woman who can fix this problem on her own without the help of a therapist.

PERSONAL APPLICATION QUESTION

What is your attitude toward mental health treatment? Would you seek treatment if you were experiencing symptoms or having trouble functioning in your life? Why or why not? In what ways do you think your cultural and/or religious beliefs influence your attitude toward psychological intervention?

GLOSSARY

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https://pressbooks.cuny.edu/jjcpsy101/?p=241#h5p-282

Media Attributions

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